

**2 x 660 MW VUNG ANG II
THERMAL POWER PLANT PROJECT**

OWNER'S TECHNICAL REQUIREMENTS

**OWNER
VUNG ANG II THERMAL JOINT STOCK COMPANY
(VAPCO)
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Chapter 1 – General Description

1.1 General

This Owner's Technical Requirements (OTR) is prepared for a 1200 MW net coal fired thermal power plant project (Project) to be developed, constructed, commissioned and operated by the Owner in a Site at Hai Phong, approximately 60 km south of the Ha Tinh City, Vietnam.

The Facility shall consist of two 600 MW nominal, net capacity Units, firing imported coals. The design steam conditions of the Units shall be at enhanced super-critical levels.

The Facility shall be designed to fire a mixture of bituminous and sub-bituminous coals, to be delivered to the Site by sea going vessels. The flue gas leaving the Units shall be cleaned to comply with the specified environmental requirements, by means of adopting NO_x reduction technologies in the boilers, removal of particulates by electrostatic precipitators, and then, further removal of sulphur dioxide (SO₂) by a seawater FGD (SWFGD) system. Following the above treatment, the flue gas from the two Units shall be discharged into the atmosphere via a 210-meter tall chimney, having a common concrete outer windshield with internal individual steel flues for the flue gas, one for each Unit. Selective Catalytic Reduction (SCR) system readiness including ducting, bypass, foundation and piling of SCR for future retrofitting with catalyst and ammonia plant, shall be provided.

Seawater desalination system shall be provided for producing the fresh water for the operational usage of the Facility.

To facilitate the execution of the Works, the Owner will provide the construction power connection at a defined point at the Site boundary. The cost of construction power shall be borne by the Onshore Contractor.

Ash comprising fly ash and bottom ash, as generated from the Units will be removed by ash slurry system or trucking to an ash pond for disposal or to offsite for sales.

The Facility shall deliver electric power to the grid at a voltage level of 500 kV.

1.2 Site Conditions

The Facility shall be designed to deliver the full electrical output (at least 2 x 600 MW net) at TMCR conditions safely and reliably under the following Site conditions as a minimum:

Ambient temperature range

7 – 41 °C

Design wind speed (3-second Gust) at 10 m above ground	54 m/s
Seismic intensity	7 (MKS – 64)
Average annual rainfall	3027 mm
Maximum rainfall rate	519 mm/day 80.9 mm/hr
Atmospheric pressure range	100.02 – 102.09 kPA(A)
Maximum relative humidity	100%
Sea water temperature range ⁽¹⁾	13 – 35 °C
Seawater level at CW intake	
HWL	+1.65 m
MSL	+1.19 m
LWL	- 1.03 m
LLWL	-1.17 m
Sea water quality	
Salinity	23.4 psu
pH	7.94
Total alkalinity (CaCO ₃)	2.46 mmol/L
Elevation (before site formation)	3-5 m
Elevation (after site formation)	8.0 m
Location	Coastal

Note (1) : The cooling water system and associated power output shall be designed according to section 8.1.10.

The Facility shall be designed to meet the requirements of applicable Vietnamese seismic regulations and for the following minimum seismic conditions: the largest horizontal peak ground acceleration (PGA) at the bedrock of the Site for maximum credible earthquake (MCE), maximum design earthquake (MDE) and operating basic earthquake (OBE) are 0.1698 g, 0.0926 g and 0.0418 g respectively.

The Onshore Contractor shall verify the Site conditions, including making all the additional arrangement and measurements (such as information search, surveys and monitoring) as deemed necessary, at their own cost and under their sole responsibility, and shall coordinate with the Offshore Services Contractor and the Offshore Supply Contractor to determine the required design of the Plant so as to ensure that the proposed design of the Facility is compatible with all the anticipated Site conditions.

1.3 Owner Supplied Services

The Owner will provide a number of utilities and services to facilitate the construction, commissioning and testing of the Facility. The major ones are:

- Construction power connection
- Construction water connection
- Raw Water for commissioning ash slurry system will be available at 29 months after Notice to Proceed (NTP)

1.4 Terminal Points

The terminal points for the various connections between the Facility and its immediate external environment are defined as below.

TP1 Ash Disposal

Fly ash and bottom ash extraction systems shall be provided. All ash silos shall be equipped with ash unloading facilities suitable for removal and transportation by trucks. Furthermore, wet ash slurry system shall be provided to transport the fly ash and bottom ash to ash disposal site. Ash disposal site will be provided by the Owner. The Onshore Contractor shall be responsible for the handling and disposal of ash up to Taking Over of the relevant Unit. In the event that the wet ash conveying system is removed, the Owner will be responsible for conveying the ash from the silos to the ash disposal area up to Taking Over of the relevant Unit.

TP2 Construction Power

One single electric feeder of 9.5 MVA, 35kV, will be provided by the Owner to the Site boundary for construction power supplies.

The Onshore Contractor shall pick up from the defined terminal point and provide the necessary transformer, switchgears etc. to receive and distribute the power for his construction use.

TP3 Construction Water

The Owner will provide a terminal point for construction water for supplying treated water to a maximum rate of 1,300 m³ per day at a pressure of 2 bar for the Project. It is the responsibility of Onshore Contractor to get construction water supply from local water company. The Onshore Contractor shall enter into an agreement with the local water company for direct payment of water bills. The Onshore Contractor shall pick up from the defined terminal point and provide all the isolation, metering, distribution pipework and treatment of the water downstream of the terminal point.

TP4 Diesel Oil

Diesel oil shall be used for boiler ignition and low load operation and will be delivered to the Site by road tankers. Suitable diesel unloading facilities with the associated pumping system and pipeline etc. shall be provided at the defined terminal point.

TP5 Effluents Discharge

All effluents shall be collected and treated based on the principle of “minimal discharge” for the Plant. To achieve this, measures shall be employed to maximize the reuse and recycling of all Plant effluents including contaminated storm water within the Site. Contaminated storm water shall be treated to meet the local effluent criteria before routing to the cooling water culverts. Non-contaminated storm water within the Site shall be collected into a rain water pond with overflows before discharging to the sea, while that from the site periphery will be discharged directly to the sea.

During construction and commissioning periods, the Onshore Contractor shall be responsible for the handling, treatment and disposal of all liquid effluents and solid contaminants generated.

TP6 Refer to interfaces for Phase 2 Switchyard in the section 16.2.3 of this Owner's Technical Requirements.

TP7 Access Roads and Entrances

Roads shall be provided within the Site boundary, up to 1m outside the Site boundary.

In addition, the Onshore Contractor shall construct the Temporary Access Road linking the public main road to the main gate of the Site as a temporary road for use during the construction period, after which it shall be repaired and resurfaced as a permanent road, which shall comply with all relevant Vietnamese standards as a condition of handing over to the Owner as part of Section 4 of the Works. The Onshore Contractor shall construct a separate path for people to go to the temple near the Site.

The number of road entrances to the Site, for both construction and permanent use respectively, shall allow for the proper access of vehicles, heavy plant delivery and waste disposals etc. In addition, the number of entrances shall comply with the local fire-fighting requirements for safety access and evacuation. At least two entrances shall be allowed to meet such requirements.

A weighbridge shall be provided for checking the loading of the vehicles going through the entrance near the coal yard.

TP8 Telecommunication, Telemetry and Protection

The provision of telephones lines for external use after Taking Over will be arranged by the Owner. A suitable telephone exchange facility shall be

provided to receive the lines at the defined terminal points located at the Site boundary.

The Onshore Contractor shall apply for the use of the external lines, provide suitable telephone exchange system within the Site area, and pay the telephone charges for his own use during the construction period.

All line relays, communication system and RTUs etc. to interface with the EVN 500 kV substation, the National Load Dispatching Center and Regional Load Dispatching Center shall be provided as required by EVN. The specifications of such relays, RTUs & communication equipment shall match with the equipment to be adopted by EVN.

Systems shall be provided in the station central control room (CCR) to allow the operators to supervise the operation of the Facility, perform switching process and arrange operation for load dispatch. Also, the design of the station distributed control system (DCS) shall meet the requirements of the national grid code regarding any interfaces with the EVN's equipment.

Tariff metering panels for Main, backup no. 1 and backup no. 2 metering systems and power line carrier communication (PLCC) system shall be provided as required by EVN.

TP9 Site Fencing

The Onshore Contractor shall provide both the temporary fencing during construction and permanent boundary wall. The permanent boundary wall shall be made of brick. At each of the entrance gate, suitable security guard houses shall be provided to ensure security controls.

TP10 Landscaping

The Onshore Contractor shall provide all required landscaping work within the station as specified in Chapter 10 to meet local environmental requirements.

TP11 Laydown Area

Owner will provide 3 separate laydown areas of 2.2 ha, 11.5 ha and 23.8 ha. These areas will be provided on an "as is" basis so the Onshore Contractor shall be responsible for any preparatory works to render the areas suitable for use during the construction period. Any additional areas that may be required by the Onshore Contractor for laydown, fabrication yards, housing etc. shall be arranged separately by the Onshore Contractor at its own costs.

TP12 Services to Owner

The equipment for the auxiliary power (2 MW for whole station) and switchgear for Owner's use in ash yard and other facilities outside the Site boundary shall be provided at the defined terminal points.

- TP13 Vung Ang I Outfall Extension*
- The Onshore Contractor shall construct an extension of the Vung Ang I cooling water outfall from existing Vung Ang I diversion chamber.
- TP14 Vung Ang II Cooling Water Outfall*
- A submerged cooling water outfall with discharge diffusers to discharge cooling water to the sea shall be provided.
- TP15 Vung Ang II Cooling Water Intake*
- A cooling water intake including intake head shall be provided.
- TP16 Rain Water Discharge*
- Rainfall shall be collected and discharged into the sea through a rain water discharge.
- TP17 Raw Water Supply for Wet Ash Slurry System*
- The Owner will establish a raw water supply of 1500 m³/day at Site boundary for the use of wet ash slurry system.
- TP18 Vung Ang I Cooling Water Outfall Discharge*
- The submerged Vung Ang I cooling water outfall shall be provided with discharge diffusers to discharge Vung Ang I cooling water to sea.
- TP19 Vung Ang II Jetty*
- A coal unloading jetty with associated coal handling system shall be provided to unload coal from coal colliers.
- TP20 Power Supply to Return Water Pumps at Ash Disposal Site*
- The Owner will provide an electrical connection of 400 VAC at 165kW for the power supplies of two return water pumps at ash pond.

Chapter 2 – General Design Concept

2.1 Facility and Plant Arrangement

The design life of the Facility shall be at least 35 years. Civil works, structures and foundations shall also meet the additional requirements on design life as called for in Chapters 10 and 13.

Unless otherwise agreed, the power block layout for Units 1 and 2 shall be arranged in a “slide along” configuration and not in a mirror image arrangement.

The station layout, operability of the plant and all amenity provisions, such as number and types of auxiliary buildings, building sizing, office and canteen services, effluent and waste treatment facilities etc. shall be designed to suit a team of 300 station operation and maintenance staff, excluding contracted labour. In addition, suitable space and connection facilities shall also be provided to allow for the accommodation of a further temporary labour force of 280 for conducting major overhauls of the Units.

The general arrangement and layout of the Facility shall ensure convenient access to the plant for operation and maintenance. All valves, gates, dampers and other devices shall, wherever possible, be located and oriented in such a way that they are easily accessible from the operating floor level. Where this cannot be achieved, platforms, walkways and access ladders shall be provided to facilitate access for operation and maintenance.

The main station buildings shall be based on an enclosed turbine hall, a deaerator bay, a bunker bay, the boiler house and the associated central control building. Behind the boilers, the Facility shall be arranged sequentially as primary air fan, forced draft fan, electrostatic precipitator, induced draft fan, sea water FGD (SWFGD) and chimney. One central control room (CCR) for two Units shall be provided.

Reliability, availability and maintainability are of prime considerations for the design of the Facility. Hence, the rating and frame size selected for the Facility shall be consistent to fulfill this requirement. Details of operating regime of the Units are as given below. Sufficient redundancies and design margins shall be provided for all Plant in accordance with Best Industry Practice and as specified in this OTR.

2.2 Plant Design and Operating Duties

2.2.1 Operating Modes and Cycle, Availability and Utilization

It is envisaged that the Facility will operate for most of its operating life as a base load power generating station, generating close to its rated output for most of the time. However, in later years of its operating life, it is also envisaged that the Facility may be required to operate on a cycling basis. Hence, the Facility shall also be designed to maximize efficient operation, under both of these operating regimes.

The control and supervision of the Facility shall be carried out in a CCR, from where the generators will normally be started, auto-synchronized, and initially loaded. Provision shall be made to allow the output of the Facility to be controlled remotely by the dispatch centre through Automatic Generation Control (AGC) program. The Facility shall be designed to ensure that automatic hot, warm and cold starts, as well as shut downs, are achieved on a reliable basis throughout its design life. In addition, the Facility shall meet the following defined steady state and transient operating conditions:

- 1) Operation under automatic control for sustained periods at all loads above minimum controllable conditions, for the given ambient range.
- 2) Operation to ensure safe shut down of the Facility.
- 3) Automatic controls shall account for boiler and turbine operating conditions (hot, warm or cold), in setting the ramp rates for start-up and loading.
- 4) Each turbine generator shall be provided with a flexible governing system with characteristics so that it can be readily adjusted as well as any changes in the operating regime that may arise during the life of the Facility. The governing system shall be provided with a set of control algorithms that allow the Facility operators to change governor control parameters, if required. The Facility shall be designed with good capability for load cycling between the minimum load (without oil support for firing) and full load.

The Facility shall be capable to operate on sliding pressure, typically, between 40% and 100% turbine maximum continuous rating (TMCR) while the steam pressure will remain constant below 40% TMCR. The minimum stable load achievable without oil firing shall be 30% boiler maximum continuous rating (BMCR) or below.

The steam evaporation rate of the boiler at BMCR shall be at least equal to the sum of the steam flow required by the turbine at its valve wide open (VWO) rating plus the auxiliary steam consumption requirements.

The Facility shall be designed for the following minimum number of starts and load cycles throughout its design life:

Start Type (Shutdown time t)	No. of starts
Cold Start ($48 \text{ hr} < t$)	280
Warm Start ($8 < t \leq 48 \text{ hr}$)	800
Hot Start ($1 < t \leq 8 \text{ hr}$)	3500
Very Hot Start ($t \leq 1 \text{ hr}$)	360
Operating Hours: 200,000 in 35 Years	

The Facility shall also be designed to be synchronized and run to full load within the following Total Start Time for different start conditions:

Start Types	Time to Synchronisation (Minutes)	Time from Synchronisation to 40% Load (Minutes)	Time from 40% load to Full Load (Minutes)	Total Start Time (Minutes)
Hot Start (Shutdown < 8 hr)	70	40	40	150
Warm Start (Shutdown in 8 – 48 hr)	130	70	110	310
Cold Start (Shutdown > 48 hr)	255	115	155	525

Times to synchronization are per unit and according to PPA requirements.

The Facility shall be designed to vary (including ramp up and ramp down) the loads during normal operation at the following rates:

0-40% load	In accordance with manufacturer's Start curves and manufacturer recommendations;
40-50% load	7 MW per minute
50-100% load	22 MW per minute

All electrical components, including selection of the component design, provision for ventilation, air-conditioning etc., shall be designed to operate at an ambient air temperature of 50°C for outdoor installation and an ambient temperature of 40°C for indoor installation.

The control system of the Unit shall be capable of automatic operation mode above 40% TMCR.

The drain and vent valves which are required to be operated during start-up and shut-down shall be motorized and controllable from the central control room through the DCS.

Plant interlock shall be designed and provided with all required protection devices, logic circuits, relays, instruments and other pertinent equipment.

Under emergency circumstances, the Unit output may be subject to a load rejection from any load to "no load". Re-establishment of full load shall then be achievable as soon as possible after the cause of the load rejection has been cleared. Under such circumstances, the Unit control devices shall keep the Unit in a controlled condition, and maintain the unit within normal parameters for thermal stress, differential expansion, and vibration. In addition to above, the Units shall also comply with the requirements of the load rejection tests as called for in the PPA. Any limitations on rapid re-loading after a load excursion shall be agreed with the Owner during the detailed engineering stage.

In case of blackout, the Facility shall be able to shutdown safely with control systems designed to be "fail safe". The control air receiver capacity, the control electric power system and battery system capacity shall be rated sufficiently for safe shutdown.

With the loss of all AC supplies, the Units shall be capable of being placed onto electric barring gear operation if required. During this period a number of essential services shall be switched to and maintained by DC supply and emergency diesel generator. If required, additional auxiliary pumps shall be provided to maintain essential lube oil flow to the turbine generator bearings in case of loss of station power supplies.

All Unit auxiliary supplies shall be arranged on a unitized basis such that individual Unit is capable of supporting its own auxiliaries. Any failure in one Unit will not affect the operation of another Unit.

2.2.2 Unit Operating Fuels

Coal Specification:

The normal operating fuel for the Facility will be coal. The Units shall be capable of economic and efficient operation over the complete range of the coal as given in this section on coal specification. Specifically, the following parameters of the coal specification shall be used as the design basis, and for testing, commissioning and performance guarantees. The parameters of Coal A and Coal B will form the basis of the specifications that will be used for the sourcing of coals from the market. Subject to provisions in Chapter 3 Section 3.15 of this OTR for the performance of the electrostatic precipitator, the Facility shall be able to perform firing of blends of Coal A and Coal B, mixed-firing of Coal A and Coal B from different mills, and also firing each of Coal A and Coal B on its own.

	Unit	Sub-Bituminous Coal A		Bituminous Coal B		Performance Coal	Coal Blends / Mixed-firing Range
		Typical	Range	Typical	Range		
Total Moisture	% ARB	36.00	< 40.00	12.00	< 21.50	27.40	< 30.00
Proximate Analysis							
Inherent Moisture	% ADB	26.00	< 28.00	8.00	< 13.00	12.90	< 20.00
Ash	% ADB	5.00	< 10.00	22.27	< 28.00	5.88	< 14.00
Volatile Matter	% ADB	40.00	34.00 - 58.00	32.00	21.00 – 45.00	41.16	25.00 – 50.00
Fixed Carbon	% ADB	29.00	By diff	37.73	By diff	40.06	30.00 - 42.50
Total	% ADB	100.00		100.00		100.00	
Total Sulphur	% ADB	0.25	0.10 – 0.70	1.1	0.45 – 1.3	0.85	≤ 1.10
	%ARB	0.22	0.1 – 0.6				
Ultimate Analysis							
Ash	%DB	5.95		24.21		6.75	
Carbon	% DB	68.00		63.90		69.12	
Hydrogen	% DB	5.20		4.19		5.14	
Nitrogen	% DB	1.00		1.47		1.00	
Sulphur	% DB	0.30		1.02		0.98	
Chlorine	% DB	0.01		0.01		0.01	
Oxygen + Others	% DB	19.54		5.20		17.01	

	Unit	Sub-Bituminous Coal A		Bituminous Coal B		Performance Coal	Coal Blends / Mixed-firing Range
		Typical	Range	Typical	Range		
Total	% DB	100.00		100.00		100.00	
Hardness	HGI	50	40 – 65	36	34 – 60	36	34 – 60
Calorific Value							
Gross (ARB)	kcal/kg	4200	> 4000	5450	> 5200	4797	>3844
Net (ARB)	kcal/kg	3850	> 3650	5250	> 5100	4454	>3508
Ash Composition (% in ash, Dry basis)							
Silicon (SiO ₂)	%	24.00		77.50		32.87	5 – 80
Aluminum (Al ₂ O ₃)	%	12.00		15.50		18.85	3 – 35
Iron (Fe ₂ O ₃)	%	16.00		1.00		13.39	1 – 18
Calcium (CaO)	%	30.00		1.00		22.54	0.6 – 32
Magnesium (MgO)	%	6.00		0.80		5.06	0.5 – 15
Titanium (TiO ₂)	%	0.60		1.00		0.96	0.2 – 4
Sodium (Na ₂ O)	%	0.35		0.80		0.48	0.05 – 3
Potassium (K ₂ O)	%	0.40		0.80		0.50	0.2 – 3
Sulphur (SO ₃)	%	9.00		0.80		4.34	0.1 – 15
Phosphorous (P ₂ O ₅)	%	0.25		0.70		0.15	0.04 - 1.25
Manganese (Mn ₃ O ₄)	%	0.50		0.03		0.36	0.01 – 1
Other Metallic Oxide	%	0.90		0.87		0.50	
		100.00		100.00			
Ash Fusion (Reducing)							
Initial Deformation	°C	1150	> 1100	1300	> 1150	1150	>1100
Spherical	°C	1200					
Hemisphere	°C	1250	> 1150	1400	> 1200		
Flow	°C	1300	> 1200	1450	> 1250		>1150

Design criteria:

- Under all firing conditions using the types of coals described above, the Unit shall be capable to deliver 100% BMCR output, except when the total moisture content in the fuel is higher than 38%. The Unit shall be capable of deliver approximately 80% BMCR output when the total moisture content is 40%. Performance Coal shall be used to define the net electrical output, and heat rate tests.
- Performance Coal with ash = 14% (ADB) and gross calorific value of 4797 kcal/kg (ARB) shall be used for performance testing of ESP and ash handling plant. The boiler gas convective passes shall be sized for a design gas velocity selected for the Typical Bituminous Coal B with ash = 17% (ADB) and a minimum Gross Calorific Value of 5600 kcal/kg (ARB).

- Performance Coal with sulfur content of 0.85% (ADB) shall be used for performance testing of SWFGD. The SWFGD shall be designed to operate normally when firing Performance Coal with sulfur content of 1.1% (ADB).
- The results of the Performance Guarantees shall be corrected to the Performance Coal with one mill as standby (not operating).
- The Units shall be designed to deliver 100% BMCR output when firing 100% sub-bituminous with the range as defined above with all mills (including the standby mill) operating, except when the total moisture content in the fuel is higher than 38%. The Unit shall be capable of delivering approximately 80% BMCR output when the total moisture content is 40%.
- The Initial Deformation and Flow Ash Fusion Temperatures of any resulting coal blend of Coal A and Coal B shall not be lower than 1100°C and 1150°C, respectively.
- The Facility shall be designed to produce 100% BMCR on Coal B and meet the NO_x requirement. The range of coal B will be within the following conditions:
 - Dry mineral matter free, Fixed Carbon < 62.7%
 - Nitrogen loading < 1.62 kg/million kcal – as HHV
 - Fuel Ratio < 1.6 which is Fixed carbon/Volatile Matter , where
 - Dry mineral matter free, Fixed Carbon = (% fixed carbon - 0.15 × % sulfur) / (100 - (% fuel H₂O + 1.08 × % ash + 0.55 × % sulfur)) × 100
 - Nitrogen loading = (% nitrogen × 5553.8 / HHV in kcal/kg) where % nitrogen is as received basis
 - Fuel Ratio = Fixed Carbon / Volatile Matter

Specification for Diesel fuel (TCVN 5689:2018)

The Facility shall be designed to use the following diesel oil for startup, shutdown and flame stability for low load operation.

Specification	Limit	Test Methods
Sulphur Content, mg/kg, max.	500	TCVN 6701 (ASTM D 2622) TCVN 7760 (ASTM D 5453) TCVN 3172 (ASTM D 4294)
Cetane , min.		
- Cetane value	46	TCVN 7630 (ASTM D 613)
- Cetane index	46	TCVN 3180 (ASTM D 4737)
Distillation, Deg. C, max.		TCVN 2698 (ASTM D 86)
- 90 %Vol	360	
- 95 %Vol	-	
Flash Point, Deg. C, min.	55	TCVN 6608 (ASTM D 3828)/ TCVN 2693 (ASTM D 93)
Viscosity at 40 Deg. C, mm ² / s	2- 4.5	TCVN 3171 (ASTM D 445)

CCR on 10 % Bottoms, % Wt PCT, max.	0.3	TCVN 6324 (ASTM D 189) TCVN 7865 (ASTM D 4530)
Pour Point, Deg. C - Summer - Winter	+ 12 + 3	TCVN 3753 (ASTM D 97) ASTM D 5950 ASTM D6749
Ash, % Wt PCT, max.	0.01	TCVN 2690 (ASTM D 482)
Water content, mg/kg, max.	200	TCVN 3182 (ASTM D 6304)
Particulate contaminant, mg/l, max.	10	TCVN 2706 (ASTM D 6217)
Copper Strip Corrosion at 50Deg. C, 3 hrs, max.	Type No. 1	TCVN 2694 (ASTM D 130)
Density at 15°C, kg/m ³	820 – 860	TCVN 6594 (ASTM D 1298) TCVN 8314 (ASTM 4052)
Lubricity, µm, maximum	460	TCVN 7758 (ASTM D6079) (ASTM D 7688)
Polycyclic aromatic hydrocarbon (PAH) content, % Wt PCT, max.	-	TCVN 11587 (ASTM D 5186) TCVN 11589 (ASTM D 6591)
Visual Inspection	Clear, visibility of no free water and contaminant	TCVN 7759 (ASTM D4176)

2.2.3 Other Requirements

Local Law "Law 50-2010-QH12 on Efficient and Economical Use of Energy" requires energy users to adopt economical and efficient use of energy. All necessary features shall be incorporated in the design of the Facility to enable the Owner to comply with Law 50 which shall include but not be limited to the following:

- Apply advanced and proven technologies with high energy yield
- Allow the Owner to measure and monitor the energy consumption of major Plant
- Provide a design which is capable of making use of emitted heat and steam of high temperature for the burning and drying of fuels and heating of water supplied to furnaces in order to raise electric generation yield of generating units

2.3 Environmental Emissions Conditions

The following minimum emission levels shall be achieved as set out in the Schedule of Performance Guarantees:

2.3.1 Flue Gas Emissions from Chimney

SO ₂	<	200 mg/Nm ³
NO _x	<	420 mg/Nm ³ (SCR system is not in operation)
	<	84 mg/Nm ³ (SCR system is in operation)
Particulate	<	50 mg/Nm ³ (measured at ESP outlet and at FGD inlet, when firing Performance Coal of ash content of 14% (ADB) and gross calorific value of 4797 kcal/kg (ARB), with one transformer/rectifier (T/R) set of ESP out of service in each train).

Above figures are measured volumetric dry, at 273K, 1 atmosphere and 6% O₂.

Chimney shall be at least of 210m in height and of separate twin flue ducts made of steel, protected by RCC windshield.

2.3.2 Effluent/Water Discharges

The Facility shall adopt a design to achieve minimal discharge of waste water and effluent. To achieve such aims, during normal operation, effluent shall be treated, re-cycled and re-used. Subject to the review and agreement by the Owner, certain excess effluent may be allowed to be treated for discharge to cooling water system.

A waste water treatment plant shall be provided to collect all Facility waste water for centralized treatment. Liquid effluent from demineralization plant, SWFGD, coal yard and oily drains may be treated separately by dedicated effluent management system.

Sludge and water for the centralized treatment plant (clarifier and filtration plants) shall be separated and compacted in sludge presses so that they can be removed for disposal in solid form. The balance water shall be re-cycled to maximize the water recovery.

Effluents from the demineralizer plant resin regeneration circuit, generally acidic effluents from the cation units and alkaline effluents from the anion units shall be neutralized in a neutralizing pit. The neutralized effluents shall be led into the station sump for water recovery.

The boiler drainage shall be controlled to maintain system solids loading within normal limits for proper water chemistry. The effluent shall be led into the station sump mix with other station effluents to reduce temperature.

The coal contaminated water from the coal handling area shall flow through channels around the coal storage area and coal handling buildings into a common settling pond from where the water after settlement shall be treated by waste water treatment plant.

The oil-water mixture shall be collected and led to the various oil-water separators.

The sewage from the various buildings in the Plant shall be treated in a centralized sewage treatment plant.

The discharge quality of effluent from waste water treatment plant and cooling water discharge shall meet the limitation values in the following table.

No.	Parameter		Unit	Limitation Value
1	Temperature		°C	< 40
2	pH			6 – 9
3	Smell			Negligible
4	Color, Co-Pt in pH=7			70
5	BOD 5 (20 °C)		mg/L	50
6	COD		mg/L	100
7	Mineral Oil and Fat		mg/L	5
8	Animal-vegetable Fat and Oil		mg/L	20
9	Total Nitrogen		mg/L	30
10	Total Suspended Solids (TSS)		mg/L	50
11	Total Residual Chlorine		mg/L	< 0.2
12	Pesticides-Organic Chlorine		mg/L	0.1
13	Pesticides-Organic Phosphate		mg/L	1
14	PCBs		mg/L	0.01
15	Arsenic		mg/L	0.1
16	Cadmium		mg/L	0.01
17	Chromium (III)		mg/L	1
18	Chromium	Hexavalent	mg/L	0.1
		Total	mg/L	0.5
19	Copper		mg/L	0.5
20	Iron		mg/L	1.0
21	Lead		mg/L	0.5
22	Mercury		mg/L	0.005
23	Manganese		mg/L	1
24	Nickel		mg/L	0.5
25	Zinc		mg/L	1.0
26	Cyanic		mg/L	0.1
27	Ammonia (as N)		mg/L	10
28	Fluoride		mg/L	10
29	Chloride		mg/L	600
30	Phenols		mg/L	0.5
31	Phosphorus		mg/L	6.0
32	Sulphur		mg/L	0.5
33	Tin		mg/L	1
34	Total Coliform Bacteria		MPN/100ml	5000
35	Gross α Activity		Bq/L	0.1
36	Gross β Activity		Bq/L	1.0

If there is exceedance at cooling water discharge, the causes should be investigated. The contractors are not responsible for the exceedance of the cooling water discharge if it is caused by the inlet cooling water quality and/or the fuel quality.

The temperature of discharged cooling water at the discharge point of the submerged cooling water outfall pipe shall be less than 40°C (absolute) for a sea water temperature at condenser inlet lower than or equal to 33°C and for a Unit net electrical output at lower or equal to 600 MW.

2.3.3 Noise

Noise emission levels shall be in compliance with the following requirements:

- (1) Near field:
 - (a) 85 dB(A), at 1m distance from one source or its acoustical enclosure and 1.5 m above ground level or personnel platforms;
 - (b) 90 dB(A), at 1m distance from any multiple sources or its acoustical enclosure and 1.5 m above ground level or personnel platforms;
- (2) Far field: 70 dB(A) at 1 m outside from the Site boundary fence) and 1.2 m above the ground level;
- (3) Central Control Room: 55 dB(A), when averaged over equally spaced measuring points at a distance of 1 m from the equipment and 1.2 m above the ground level.
- (4) Nearby Residential Area: Compliance with ESHIA and EIA.

All above stated noise requirements shall be met, by adding abatement measures where necessary. The Onshore Contractor shall conduct baseline surveys of the background noise for use in assessment of the compliance mentioned in item (4) above.

2.3.4 Chemical/Solid Waste Disposal

A suitable storage area shall be provided to store various kinds of chemical/solid waste, including but not limited to contaminated oil, degreasing solvent, paint & thinner, sulfuric acid, miscellaneous acids and alkalis. Suitable unloading facilities shall be provided to enable the removal of such waste from the Facility by trucks.

2.3.5 Fugitive Dust Emission

The fugitive dust emission shall comply with EIA/ESHIA requirements. The coal storage shall be based on open coal yard.

2.4 Codes, Standards and Local Regulations

The Facility shall be designed, manufactured, erected and tested in accordance with all relevant codes, standards and regulations. All materials, Plant and services carried out including calculation sheets, drawings, quality and class of goods, methods of inspection, constructional peculiarities of Plant and parts and acceptances of partial plants, as far as these are beyond the special requirements of the OTR, shall comply in every respect with the relevant technical codes.

2.4.1 Codes and Standards

The contractors shall comply with all applicable Laws as defined in the relevant contracts. Each of the Onshore Contractor and the Offshore Services Contractor shall define the standards applying to its Plant design and ensure its design is compliant with local regulations.

The contractors shall comply with all codes and standards described in the OTR. For Key Components, the contractors may in addition select from the codes and standards set out in chapter 3 of Exhibit 4 of the relevant contract. To the extent that the OTR does not prescribe an applicable code, standard or both, subject to the agreement with the Owner, a contractor may adopt codes and standards issued by the institutions listed below.

Reference	
ACI	American Concrete Institute
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
API	American Petroleum Institute
ASHRAE	American Society of Heating, Refrigeration & Air Conditioning Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
AWS	American Welding Society
AWWA	American Water Works Association
BS	British Standards Institution
BIS	Bureau of Indian Standards
IS	Indian Standards
DIN	German Standardization Institute
DTII	Chinese Code for Conveyors
EN	Euro Norm
FEM	European Maintenance Federation
HEI	Heat Exchange Institute
HIS	Hydraulic Institute Standard
IEC	International Electro technical Commission
IEE	Institute of Electrical Engineers
IEEE	Institute of Electrical and Electronics Engineers
IP	Institute of Petroleum
ISA	Instrument Society of America

ISO	International Standards Organization
JIS	Japanese Industrial Standards
MSS	Manufacturer's Standardization Society
NEC	National Electric Code
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
SSPC	Steel Structures Painting Council
TEMA	Tubular Exchanger Manufacturers Association
TRD	Technical Rules for Steam Generators
VDI	Association of German Engineers
VGB	Society of large utility owners
VDEW	Association of German Power Plants
UBC	Uniform Building Codes
GB	Chinese National Standards
JB	Chinese Standards for Industrial Plant and Equipment
DL	Chinese Energy Industry Standards
	Manufacturer Standards

Where the OTR, Exhibit 4 to the relevant contract or a contractor otherwise identifies the adoption of a non-Vietnamese code, standard or both, the Onshore Contractor shall obtain all requisite approvals from local authorities to adopt and apply the non-Vietnamese code and/or standard as required by Circular No. 18/2010/TT-BXD dated October 15, 2010. The Owner shall provide reasonable assistance including documents to the Onshore Contractor at the request of the Onshore Contractor for the Onshore Contractor's application.

2.4.2 Vietnamese Codes, Standards and Regulations

The Offshore Services Contractor shall be fully and exclusively liable and responsible for design of the Facility (and all engineering work) in compliance with all applicable Laws.

A non-exhaustive list of applicable Vietnamese codes, standards and local regulations with which the contractors are required to comply with in their performance of the relevant contracts are described below and elsewhere in this OTR.

2.4.2.1 Environmental Standards

- QCVN 22: 2009/BTNMT National Technical Regulation on Emission of Thermal Power Industry
- QCVN 05: 2013/BTNMT National Technical Regulation on Ambient Air Quality
- QCVN 06: 2009/BTNMT National Technical Regulation on hazardous substances in Ambient Air
- QCVN 19: 2009/BTMNT National Technical Regulation on Industrial Emission of Inorganic Substances and Dusts

- QCVN 20: 2009/BTMNT National Technical Regulation on Industrial Emission of Organic Substances
- QCVN 40: 2011/BTMNT National Technical Regulation on Industrial Wastewater
- QCVN 03-MT:2015/BTNMT National Technical Regulation on the allowable limits of heavy metals in the soils
- QCVN 08-MT:2015/BTNMT National Technical Regulation on Surface Water Quality
- QCVN 09-MT:2015/BTNMT National Technical Regulation on Underground Water Quality
- QCVN 10-MT:2015/BTNMT National Technical Regulation on Coastal Water Quality
- QCVN 14:2008/BTNMT National Technical Regulation on Domestic Waste Water
- TCVN 5525:1995 Water Quality – General Requirements for Ground Water Protection
- TCVN 3985:1999 Acoustics - Allowable Noise Levels at Workplace
- TCVN 5126:1990 Vibration - Permissible value at Workplace
- QCVN 27: 2010/BTNMT National Technical Regulation on Vibration
- QCVN 26: 2010/BTNMT National Technical Regulation on Noise
- QCVN 25: 2009/BTNMT National Technical Regulation on Wastewater of the Solid Waste Landfill Sites
- QCVN 43: 2012/BTNMT National Technical Regulation on Sediment Quality

2.4.2.2 Pressure Vessels and the like

Boiler shall be designed, manufactured and tested in accordance with "ASME Boiler and Pressure Vessel Code ". Listed below are the standards which may be applicable to other parts of the Facility.

- TCVN 7704: 2007 Boiler – Technical Requirements for Design, Construction, Installation, Use and Maintenance.
- TCVN 6413: 1998 Stationary Furnace – Tube & Fire-tube Boilers of Welded Construction Except Water-tube Boilers
- TCVN 6008: 2010 Pressure Plant – Welds – Technical Requirements and Inspection Methods
- TCVN 6158 – 1996 Steam and Hot Water Pipes. Technical Requirements
- TCVN 6159 – 1996 Steam and Hot Water Pipes. Test Method
- TCVN 8366 – 2010 Pressure Vessel Safety Requirements on Design, Construction and Manufacture.

- TCVN 6155 – 96 Pressure Vessel Safety Requirements on Installation, Use and Maintenance
- QCVN:01-2008/BLDTBXH National Codes for Labor Safety with Regards Boilers and Pressure Vessels

2.4.2.3 Lifting Equipment

- TCVN 4244: 2005 Lifting Equipment, Design, Manufacture and Technical Inspections
- TCVN 4755 – 1989 Cranes – Safety Requirements of Hydraulic Equipment
- TCVN 5180 – 1990 Electrical Tackle – General Safety Requirements
- TCVN 3147 – 1990 Safety Codes for loading and unloading works – General Requirements
- TCVN 5179 – 1990 Hoisting Cranes – Test Requirements of hydraulic equipment for safety
- TCVN 5744: 1993 Safety Requirements on Installation and use of Elevators
- TCVN 5866: 1995 Elevator - Safety Mechanism
- TCVN 6904: 2001 Electric Elevator – Test Methods for Construction and Installation Safety Requirements
- TCVN 6905: 2001 Hydraulic Elevator – Test Methods for Construction and Installation Safety Requirements
- TCVN 6906: 2001 Passenger Escalator and Conveyor – Test Methods for Construction and Installation Safety Requirements
- TCVN 5867: 2009 Elevator: Cabinet, Counter Weight and Guide Rail – Safety requirements

2.4.2.4 Electric Safety Codes and Standards

- National codes for electric safety engineering No.QCVN01:2008/BCT published together with Decision No.12/2008/QD-BCT dated 17 June 2008.
- Electrical equipment codes Nos.11 TCN-18-2006, 11 TCN-19-2006, 11 TCN-20-2006, and 11 TCN-21-2006 published in accompany with Decision No.19/2006/QD-BCN.
- National codes for Electrical engineering No. QCVN QTD-5:2009/BCT, QCVN QTD-6: 2009/BCT, QCVN QTD-7: 2009/BCT published in accompany with the Circular 40/2009/TT-BCT by Ministry of Industry and Trade on 31 December 2009.
- TCVN 4756-89 Codes for electric equipment earthing and neutral connections.
- TCVN 8092:2009 Graphical symbols – Safety colours and safety signs – Safe signs used in workplaces and public areas.

2.4.2.5 Explosion and Fire Protection Standards

- TCVN 2622: 1995 Fire prevention and protection for buildings and structures – Design requirements
- TCVN 5760: 1993 Fire fighting system – General requirements for Design, installation and use
- TCVN 5738: 2001 Automatic fire alarm system – Technical requirements
- TCVN 5307: 2009 Oil and petrol storage and oil product – Design requirements
- TCVN 6100: 1996 fire protection - fire extinguishing media – carbon dioxide
- TCVN 4090: 1985 oil products and main transporting pipe – Design standard
- TCVN 9385: 2012 Protection of structures against lightning - Guide for design, inspection and maintenance
- TCVN 6101: 1996 Fire protection and fighting equipment – Carbon Dioxide extinguishing system – Design and Installation
- TCVN 5303: 1990 Fire safety – Terminology and definitions.
- TCVN 3254:1989 Fire safety – General requirement
- TCVN 4878:2009 Fire protection and fighting – Fire classification
- TCVN 4879: 1989 Fire protection – Safety signs
- TCVN 5040: 1990 Fire protection and fighting equipment - Graphical Symbols used for protection schemes - Specifications
- TCVN 6103: 1996 Fire protection and fighting - Terminology – Smoke control.
- TCVN 6305-1: 2007 Fire protection and fighting – Automatic sprinkler system – Part 1: Requirements and test methods for sprinklers
- TCVN 6305-2:2007 Fire protection and fighting – Automatic sprinkler system – Part 2: Requirements and test methods for wet alarm valves, retard chambers and water motor alarms
- TCVN 6305-3:2007 Fire protection and fighting – Automatic sprinkler system – Part 3: Requirements and test methods for dry pipe valves
- TCVN 6305-4:1997 Fire protection and fighting – Automatic sprinkler system – Part 4: Requirements and test methods for quick opening devices
- TCVN 6305-5:2009 Fire protection and fighting – Automatic sprinkler system – Part 5: Requirements and test methods for deluge valves
- TCVN 6305-6:2013 Fire protection and fighting – Automatic sprinkler system – Part 6: Requirements and test methods for check valves
- TCVN 6305-7:2006 Fire protection and fighting – Automatic sprinkler system – Part 7: Requirements and test methods for sprinkler's early suppression fast response (ESFR)
- TCVN 6305-8:2013 Fire protection and fighting – Automatic sprinkler system – Part 8: Requirements and test methods for pre-action dry alarm valves

- TCVN 6305-9:2013 Fire protection and fighting – Automatic sprinkler system – Part 9: Requirements and test methods for water mist nozzles
- TCVN 6305-10:2013 Fire protection and fighting – Automatic sprinkler system – Part 10: Requirements and test methods for domestic sprinklers
- TCVN 6305-11:2006 Fire protection and fighting – Automatic sprinkler system – Part 11: Requirements and test methods for pipe hangers
- TCVN 6305-12:2013 Fire protection and fighting – Automatic sprinkler system – Part 12: Requirements and test methods for grooved-end components for steel pipe systems
- TCVN 7336: 2003 Fire protection and fighting – Automatic sprinkler system - Design and installation requirements.
- TCVN 7161: 2009 Gaseous fire extinguishing system – Physical property and system design.
- TCVN 6379: 1998 Fire fighting equipment – Fire hydrant – Technical requirement.
- TCVN 7435-1: 2004 Fire protection and fighting – Portable and wheeled fire extinguishers – Part 1: Selection and Installation
- TCVN 5279: 1990 Fire and Explosion safety – Combustible dusts General requirement.
- TCVN 3255: 1986 Explosion safety – General requirement.
- QCVN 02:2008/BCT National technical regulation on safety in the storage, transportation, use and disposal of industrial explosive materials.
- TCVN 6174: 1997 Industrial explosive material – Safety requirement for manufacturing, explosion testing and approval.
- TCVN 5334: 2007 Electrical equipment for petroleum storage and products - Safety requirements for design, installation and use.
- TCVN 3991: 2012 Fire protection standard in construction design – Terminology - Definition
- TCVN 6160: 1996 Fire protection and fighting –High rise building – Design requirement.
- TCVN 4317: 1986 Oil and gasoline storage and pipeline - Warehouse – Basic principle for design.
- TCVN 4513: 1988 Internal water supply - Design standard
- TCVN 6102: 1995 Fire protection and fighting - Fire extinguishing media - Power
- TCXDVN 33: 2006 Water supply - Pipeline network and facility– Design standard.
- TCVN 9207: 2012 Electrical line installation in dwellings and public building – Design standard.
- 11TCN-18-2006 Code of electrical installation – General regulation.
- TCXD 218: 1998 Fire detection and alarm system – General regulation.

2.4.2.6 Other Vietnamese Laws, Codes, Standards and Regulations

Transmission system code	National Grid Code
Loading and effect - design standard	TCVN 2737-2006
Seismic design criteria	TCVN 9386-2012
Design of concrete and reinforcement concrete structures	TCVN 5574:2018
Steel structures - design standard	TCVN 5575-2012
Anti-erosion in construction/ Reinforcement concrete & steel structures. Basic Design Principles	TCXDVN 3993-1985
Pile foundation – Design standard	TCVN 10304-2014
Design standard of building and structure foundation	TCVN 9362-2012
Drainage – External networks and facilities – Design standard	TCXDVN 51:2008
Auto roads - design requirements	TCVN 4054- 2005
Soft revetment procedure	22TCN 211-06
Electricity	Law 28/2004/QH11 and 24/2012/QH13
Efficient and Economical Use of Energy	Law 50/2010/QH12
Efficient and Economical Use of Energy	Decree 21/2011/ND-CP
Strengthening Economical Use of Electricity	Direction 34/CT-TTg
On amendments to Decrees on guidelines for the law on environment protection	Decree 40/2019/ND-CP

Chapter 3 – Boiler and Auxiliaries

3.1 General

The boiler and its auxiliaries shall be of the pulverized coal type designed to operate at enhanced super-critical steam conditions.

The proposed Facility shall be sized to operate over the full range of ambient conditions as anticipated for the proposed Site without limiting the electrical output. The boiler proper shall include pressure parts such as the boiler furnace, with the associated economizer, evaporator, superheater and reheater tube bank sections made up of tubing materials, designed to maximize heat transfer.

The thermal cycle of the Facility shall be optimized to suit the specific requirements of the OTR. The boiler and auxiliaries shall be of the standard proven design. Designs incorporating components which may be considered prototype in nature shall not be used.

Pressure parts and materials shall be designed, manufactured and tested in accordance with "ASME Boiler and Pressure Vessel Code ". In addition, the boiler design and materials shall be approved by relevant Vietnamese authorities. The Onshore Contractor shall be responsible for the application for the approval of the boiler design required by the Vietnamese authorities. ASME stamping for boiler is not required for this Project.

Each boiler shall supply steam to a steam turbine of a nominal turbine maximum continuous rating (TMCR) capacity of 600 MW net under the full range of the anticipated Site conditions. The steam evaporation rate of the boiler at boiler maximum continuous rating (BMCR) shall be at least equal to the sum of the steam flow required by the turbine at valve wide open (VWO) rating plus the auxiliary steam consumption requirements of the Unit. The boiler shall be of the single reheat, balanced draft type.

Each of the two boilers shall exhaust through a separate flue inside a common chimney of 210 m in height. The chimney flue shall be made of acid-resistive steel plates with suitable types of protective lining - such as the use of foamed glass tiles. The flue gas exit temperature and velocity shall be sufficient to ensure adequate dispersion of the flue gases to meet the local environmental standards and requirements in EIA/ESHIA.

The boiler shall be based on an semi-outdoor design, top supported, dry bottom, single vortex, single reheat, balanced draft, pulverized coal of the front and rear or tilting tangential firing design. The boiler structure shall include a roof, rain hooded burner level platform and sheltered mill bays at two ends, to top of mill elevation.

The structural steel shall consist of steel columns, lattice trusses or girt & purlin members. It shall be used with hot rolled sections or built-up sections for columns, girder and braces (trusses). To resist walls and roof of structure, it shall be considered

with hot rolled section, cold formed sections or built-up sections for girt and purlins. The Offshore Service Contractor shall state the minimum yield strength of the steel members for Owner's review during design submission stage. All structural steelwork shall be protected against corrosion by use of appropriate surface finish as required to maintain the integrity of the structure over the life of the plant. The steelwork shall be shot blasted to minimum Swedish Standard SA 2.5 in view of the proximity of the plant to the harsh marine environment.

The Offshore Service Contractor shall propose the corrosion protection system for Owner's review during design submission stage.

Each boiler shall be equipped with regenerative type air pre-heaters; two sets of 50% (of BMCR) forced draft (FD), induced draft (ID), primary air (PA) fans. The furnace cross section, volumetric size and heat release rates of the burners shall be selected conservatively to match the characteristics of the fuel. The design of the air heaters, fans and associated boiler auxiliaries shall be provided with adequate margins to avoid limiting the capability of the Facility to achieve full rated output throughout the design life of the Facility. Steam sootblowers shall be provided to support the cleaning operation of the boilers to allow continuous full load operation. The furnace shall also be equipped with wall sootblowers and long retractable sootblowers for the cleaning of the superheater, reheater and economizer areas. The air heaters shall have fixed lance type sootblower units. The BMCR shall be designed with the inclusion of auxiliary steam flow. The boiler shall have a suitable number of duty mill groups to allow for the firing a mixture of bituminous and sub-bituminous coal at BMCR output, with one spare mill group acts as standby for the specified coal ranges. In addition, the boiler shall also allow for the firing of the design bituminous coal alone and sub-bituminous coal alone to deliver full BMCR output.

Firing with oil support shall not be required at or above the 30% of BMCR. All fans shall have at least 21% margin on head corresponding to fan nominal capacity and all mills (together with feeders, PA fans and auxiliaries) shall also have at least 10% throughput margin in pulverizing capability after ageing. Steam flow at BMCR shall be at least 3% more than that at TMCR.

The boiler shall be equipped with three superheaters with interstage spray-type attemperators and two reheater banks. The economizer shall be of a horizontal, multi-loop bank.

The Offshore Supply Contractor shall examine the specified coal specification to ensure the integrity of the boiler.

The boiler shall be suitably designed and constructed for safe, proper and continuous operation under all conditions described or implied in this OTR without undue strain, vibration, corrosion, noise or other operating difficulties.

Plant and components shall be designed and supported to permit free expansion and contraction without causing excessive strains, distortion or leakage.

Parts subject to wear, corrosion or other deterioration, or requiring adjustment, inspection or repair shall be accessible and capable of reasonably convenient removal.

replacement and repair. All such parts shall be chosen from suitable materials to provide a life expectancy as specified.

The boiler and its auxiliaries shall be designed to permit interchangeability of parts and ease of access during inspection, maintenance and repair.

The boilers shall be designed to meet the specified operating regime for base load operation with capability of fulfilling cyclic/two shift duties. Verification of the boiler pressure parts design shall be made available to the Owner at the Offshore Supply Contractor's facility during detailed engineering stage to ensure this design intent is satisfied.

For start-up and low load operations, diesel oil will be used. Fuel oil preparation and firing system shall be designed for startup, coal flame stabilization and low load requirements.

The combustion system shall be designed to restrict the emission from the Unit below acceptable limits as stipulated in the statutory environmental laws and the OTR.

3.2 Furnace

The furnace shall be properly sized in respect of the fuel/coal burner system, in order to meet the requirements of low NO_x emissions in the flue gas whilst maximizing burnout to minimize the amounts of unburned carbon-in-ash. Unburnt carbon-in-ash shall not be more than 5% by weight in fly ash. The furnace design shall be free from 'dead pockets' when prescribed purge procedures are exercised.

Furnace pressure shall be obtained from three redundant measurements to overcome problems of signal noise, derive average pressure and provide signal redundancy. Proper pressure control shall be achieved by modulation of the ID fans. A further three pressure measurements shall be provided to implement furnace cut back control, directional blocking and over-riding of the draught plant control operators on excessive sustained pressure deviation.

The firing system shall be in accordance with the National Fire Protection Association (NFPA) Code.

In the combustion zone, the membrane wall shall be of the spiral wound type, utilizing smooth bore tubes so that the steam/water mixture through each tube is increased. In the upper furnace where the spiral tubes change to vertical tubes, this shall be via a transition header. The mass flow in the furnace-wall tubes shall be sufficiently high to avoid overheating, departure from nucleate boiling (DNB), excessive metal temperatures and uneven steam outlet temperature when operating at supercritical pressure at higher boiler loading. Suitable support measures shall be in place to transfer some of the weight load from the spiral furnace tubes to the support structure.

Observation ports shall be provided at the necessary and accessible locations which enable a wide scope of flame and furnace walls from the bottom to the top of the furnace. Access doors of adequate size shall be provided at suitable location of the

boiler walls to permit access to the furnace bottom, superheaters, reheater and economizer elements. An opening sized approximately 600 mm by 500 mm shall be provided at the bottom of the furnace wall for inspection and maintenance work of the furnace, etc. Cable holes shall be provided on the furnace roof at suitable spacing for slinging, staging and handling material for inspections and maintenance work.

Spacing of the superheater tube bundles shall be designed taking into account the slagging and/or fouling potential of the ash. Detailed design of the furnace and the furnace exit gas temperature profiles shall be presented based on the specified design coal to the Owner for review during the design stage of the Project.

The arrangement of the burners shall be designed so that no flame impingement on the walls would occur when firing the specified fuel under all conditions of load up to BMCR.

Necessary information shall be submitted on the following: furnace design criteria of furnace volume (m^3), effective projected radiant surface areas (EPRS) (m^2), volumetric heat release rate ($\text{kcal}/\text{m}^3\text{h}$) and the EPRS heat release rate ($\text{kcal}/\text{m}^2\text{h}$) based on the specified coal and ash analysis in the detailed engineering stage.

The furnace bottom throat level shall be sufficient so as not to require any excavation below ground floor level for locating furnace bottom hopper and their auxiliary systems.

The furnace coutant bottom shall be designed with steep and self-feeding slopes for gravity feeding of any mass ash or slag that may fall from the boiler. The furnace coutant bottom shall be designed for the maximum ash content of the range of coal specified to freely flow into the submerged scraper conveyor (SSC). The SSC shall be designed for fresh water quenching. Adequate clearance around all parts of hopper shall be given for maintenance purpose.

Suitable devices shall be provided to signal abnormal ash accumulation in the furnace coutant bottom. Access openings shall be provided to allow cleaning of ash in the event ash does not clear the coutant opening.

The bottom ash handling system shall be provided to remove bottom ash from the bottom of the boiler and to transport the bottom ash to the bottom ash discharge bin by SSC. The SSC shall collect ash on a continuous basis whilst the boiler is firing. The SSC shall be designed to be capable to remove bottom ash when burning Coal B with ash content of 28% (ADB) at BMCR output. Each SSC shall have sufficient capacity to contain 8 hours production of ash when burning Performance Coal with ash content of 14% (ADB) at BMCR output without overflowing at any point and without the removal system in operation. This storage of ash shall not result in any bridging of the coutant opening. The system shall also be able to extract a 3-hour backlog of ash within a further 8-hour period with the unit operating on 100% BMCR output. Provision of diverter chute shall be provided upstream of the clinker grinder for emergency discharge of ash. There should be safety provision to allow operation of the SSC alone under emergency mode when ash is being diverted.

Extended seal plates shall be hung off the furnace bottom and extend into the SSC water trough to provide a gas seal that allows for bi-axial expansion of the boiler

bottom. An overflow seal box located at the top of SSC shall collect excess seal and ash water and drain by gravity to the SSC recirculation water sump

The furnace coutant bottom seal shall be stainless steel.

A clinker grinder with chain drives and all other accessories for operation shall be provided downstream of the SSC to limit the size of clinker to maximum 25 mm. Clinker grinder shall be sized to crush and feed ash at a guarantee rate. Heavy-duty roller bearings water sealing arrangements shall be provided as gland seals and arrangement shall be made for free shaft expansion. The clinker grinder shall be suitable for handling the specified ash.

Burners shall be so designed and located that, when burning the specified fuels, furnace deposit and slugging shall be minimized, and if occur, it shall easily be removed by soot-blowing.

During the detailed engineering stage, details of the evaporative water circuit shall be submitted to explain how furnace tube overheating is prevented, particularly during start-up and shut-down.

Where critical piping is located, the temperature, pressure and conditions of the tubes shall be monitored through the DCS. The associated display in the central control room (CCR) shall be provided.

The flue gas velocity shall be chosen based on the proven experience of the Offshore Supply Contractor, taking into account the ash properties of the specified coals. Maximum flue gas velocity in economizer region shall be less than 18 m/s when firing Performance Coal.

The furnace shall be provided with at least two sets of flame monitoring colour CCTV and at least two sets of bottom hopper monitoring color CCTV.

An acoustic boiler tube leak detection system shall be provided to give early warning of steam and water leaks in the boiler.

3.3 Pressure Parts

Boiler pressure parts shall be designed, manufactured and tested in accordance with "ASME Boiler and Pressure Vessel Code, Section 1, Power Boilers". In addition, the boiler design and materials shall be approved by relevant Vietnamese authorities. The scope of the boiler pressure parts shall include, but not be limited to, the following:

- (1) Furnace waterwall tubes, headers, supply tubes and risers
- (2) Steam water separator with all suitable internal fittings
- (2) Superheaters and reheater including attemperators and all associated headers

- and pipe-works
- (3) Economizer including headers and pipe-works
- (4) Valves and pipe-works associated with the above items.

During detailed engineering stage the complete material schedules of all headers, separator, tube banks, integral pipework, main steam pipework, cold and hot reheat pipework, attemperators, flash vessels and other high pressure pipework shall be submitted for the Owner's review.

All boiler pressure parts shall be designed to achieve the required design life. Routine maintenance shall be performed on the boiler and replacement of parts will be installed according to maintenance results. Also, boiler shall be operated and maintained per the Offshore Supply Contractor's instructions and procedures.

The qualification/approval of welding procedures and welders shall be carried out to the standard specified by ASME, and acceptable to local Vietnamese authorities. Welding procedures for pressure part welding, together with their procedure qualification information shall be made available for review by Owner at the factory before welding manufacturing process starts.

Non-destructive examination (NDE) shall be carried out in accordance with written procedures made available to the Owner by the Offshore Supply Contractor together with the defect acceptance standards. The testing and defect acceptance standards shall be identified in the relevant Inspection and Test Plan (ITP) during the detailed engineering stage and procedures shall be made available to the Owner for review at the Offshore Supply Contractor's facility two months before manufacture commences.

3.3.1 General

The supercritical boiler shall be designed for a high degree of operational flexibility in accordance with the defined operating regimes. This flexibility shall be achieved by careful selection of pressure part materials and by design features to avoid the use of thick-section components in critical areas. A study on the cyclic stresses that cause fatigue and high temperature creep shall be made available for the Owner's review during the detailed engineering stage.

Simple carbon manganese steels shall only be used at lower temperatures such as the reheater inlet. As the component temperature increases, the more modern steels that have been referred to as creep-strength enhanced ferritic (CSEF) steels, which exhibit very high creep strength, shall be used. In case of the very high temperature components such as the superheater and reheater, the choice of material shall be dictated by the flue gas corrosiveness as well as stress rupture characteristics.

Membrane panels shall be fabricated either by a fusion welded fin or integral fin method. Resistance welded fin construction shall not be allowed. Main steam pipes, hot and cold reheat steam pipes, and bled steam pipes of all stages shall be manufactured using seamless tubing without longitudinal welds.

A suitable quantity of sootblowers (wall blowers, long blowers, & air heater blowers) shall be provided at appropriate places. Superheater shielding shall include opposite retractable blowers. Any two sootblowers shall be capable of operating simultaneously.

3.3.2 Headers

Header shells shall be manufactured using seamless pipe. Headers shall be provided with access arrangements, allow full internal visual inspection and cleaning. Two access ports per header located at quarter points shall be provided.

The inspection ports shall have an internal diameter of at least 50 mm to allow tele-visual inspection of grade 12 or 22 headers. Grade P91 (Reheater) and P92 (Superheater) headers shall have inspection nipples of at least 50 mm internal diameter to allow cutting off for easy tele-visual inspection of the complete header in the future. The blind cap of these inspection nipples shall be of a permanent type, fixed by means of fully penetrating butt welds. The length of the inspection nipples shall be at least 150 mm for future cutting and rewelding. Where header ends are attached by welding, this shall be achieved by using a fully penetrating plain butt weld joint where possible, with a weld configuration in compliance with the relevant ASME.

All nozzles, branches and tube stubs shall be attached by full penetration welding. Reinforcing pads shall not be used. Nozzle, branch or tube stub weld shall not encroach over the heat-affected-zone of main seam welds unless with prior approval. Stub weld is allowed to encroach over header girth seam which is carbon and low alloy up to grade 22, with a weld configuration in compliance with the relevant ASME.

All nozzles, branches and tube stubs shall be of a sufficient length to ensure adequate access for welding on the adjoining tubes or pipes and to permit effective post weld heat treatment of these butt welds so as not to affect the integrity of the headers.

All drains and vents in pressure part headers shall be self-draining and provided with two drain valves where necessary.

3.3.3 Tubes

These provisions shall be applied to all tubular pressure part that is either exposed over much of its length to hot gases for purposes of heat transfer or is directly butt welded to such a tubular pressure part.

All tubing materials shall comply with the requirements of ASME, and be acceptable to the Vietnamese authorities.

The design basis for controlling mechanical wastage (eg. grit and sootblower erosion) of tubing shall be made available to the Owner during the detailed engineering stage. This shall include a justification of the maximum gas velocities that have been chosen for each tube bank and the measures to be taken to protect exposed tubing including bends and tube sets.

During the design phase, the design basis for controlling chemical wastage (eg: fireside corrosion, dew point corrosion) of tubing shall be made available to the Owner including details of the wastage provision for each tube design.

Membrane panel construction shall be either by a fusion welded fin or integral fin method. Resistance welded fin construction shall not be allowed.

The manufacturing method, the extent and type of testing proposed for each type of tubing and finning and reference experience for the proposed tubing shall be made available to the Owner during the detailed engineering stage.

All tubes shall have a minimum supplied thickness of 3.2 mm (0.125'). Tube bend shall not be circumferentially welded.

Parallel down-flow circuits subject to significant variations in heat absorption and/or resistance to flow between these circuits shall be avoided. If economics dictates that such circuits are unavoidable, then tube materials and tube flexibility shall be determined on the basis that flow stagnation can occur. Such details if applicable shall be made available to the Owner at the Offshore Supply Contractor's facility for the review and agreement by the Owner during detailed engineering stage.

3.3.4 Integral Piping

Where cold bending of boiler integral pipework is anticipated, the Offshore Supply Contractor shall describe the method of testing in design submission.

For carbon steel integral pipes designed to operate at temperatures greater than 300°C, system stresses created at bends due to manipulation during erection shall be minimized. This shall need to take account of poor fit and misalignment of components. Furthermore, dimensional limits for each pipe run or group of runs shall be defined during the design phase.

3.3.5 Vents

The Offshore Services Contractor shall describe the arrangements for complying with the relevant standard and local legislation on noise from escape pipes, safety valves and drains vent systems in design submission.

The Offshore Supply Contractor shall make provision for future retrofit with silencers for boiler safety valves.

3.3.6 Pressure Parts Attachments

Attachment welds to headers shall be at least 200 mm from main seam or other pressure part welds and attachment welds to tubing shall not encroach on any circumferential butt welds except with prior approval. Attachment welds may be fillet welds according to load.

Load carrying attachments to tubes, pipes, and headers shall be welded using fully penetrating welds. Load carrying attachments shall be designated by the designer and indicated on the relevant drawings.

3.3.7 Welding

The qualification/approval of welding procedures and welders shall be carried out as per the standard specified in ASME, and be acceptable to Vietnamese authorities. Welding procedures for pressure part welding, together with their procedure qualification information shall be made available to the Owner for review at the Offshore Supply Contractor's facility prior to manufacture during design stage.

Design of welded joints shall accommodate NDE to be performed. Weld preparations if not formed as an integral part of a casting, shall be formed by machining. All branch, nozzle, stub and butt welds shall have at least the root run made by a tungsten inert gas (TIG) welding plus filler process unless it is not practical. For those welds not practical, these shall be submitted to Owner for review. Tube butt welds shall be made by using TIG welding. Where automatic welding is impractical and manual welding is used, an inspection technique shall be used to demonstrate that acceptable bore protrusion and concavity are maintained over the entire weld length. Procedures for the storage, baking and usage of welding consumables shall be made available to the Owner at the Offshore Supply Contractor's facility before welding commences. Post welding heat treatment procedures shall be made available to the Owner at the Offshore Supply Contractor's facility before production welding commences. Permanent backing rings and Oxy-acetylene welding shall not be used.

3.3.8 Identification of Materials

Raw material incoming inspection shall be conducted and material shall be identified, tagged, marked, or colour coded where appropriate. The Offshore Supply Contractor shall ensure all materials used in the manufacture of equipment comply with the requirements of the relevant specifications. The Offshore Supply Contractor shall establish a Positive Materials Identification program indicating the extent of inspection and testing to be conducted during incoming inspection, throughout the completion of boiler fabrication and submitted for Owner review during design submission stage.

The Offshore Supply Contractor shall establish means by which materials and components are identified through each stage of the supply of the Offshore Equipment to facilitate full traceability of materials. The Offshore Supply Contractor

shall maintain a suitable control of the materials used and keep a record of the material certificates. Material traceability procedure shall be submitted for review.

The material composition of both parent material and weld metal shall be checked and confirmed after welding micro alloyed carbon manganese steels and all ferritic alloy components. The method to be used shall be made available to the Owner at the Offshore Supply Contractor's facility during detailed engineering stage.

3.3.9 Non Destructive Examination

Procedure for NDE shall be made available to the Owner at the Offshore Supply Contractor's facility by the Owner before manufacture commences.

NDE shall be carried out by appropriately qualified personnel as required by a recognized standard or international certification scheme which is acceptable to Vietnamese authorities.

The NDE shall comply with the relevant ASME standard and be acceptable to Vietnamese authorities. In addition, the following inspection requirements shall also be applied to headers and integral pipework:

- Ultrasonic method or radiographic method of testing (RT) shall be used for volumetric examinations.
- Ferritic steel – 100% of welds attaching nozzles, branches and stubs shall be examined over their entire length by magnetic-particle inspection test (MT). In addition, 20% of all nozzle, branch and stub welds shall be inspected over their entire length using an ultrasonic method (UT) or RT. All nozzle, branch and stub welds shall be fully dressed and blended to aid future NDE.
- Austenitic steel – 100% of welds attaching nozzles, branches and stubs shall be examined over their entire length by dye penetration inspection and ultrasonic method. In addition, 20% of all nozzle, branch and stub welds shall be inspected over their entire length using an ultrasonic or RT method. All nozzle, branch and stub welds shall be fully dressed and blended to aid future NDE. RT is preferred where possible.
- Ferritic steel – 100% of butt welds in integral pipework shall be examined over their entire length by magnetic particle inspection and an ultrasonic or RT method.
- Austenitic steel – 100% of butt welds in integral pipework shall be examined over their entire length by dye penetrant inspection and an ultrasonic or RT method.
- Imperfections, such as cracks and lack of fusion, shall be removed. Complete removal shall be verified by either MT or PT examination. When cracks are removed from a weld, the entire weld shall be removed if more than one quarter of its circumference (cumulative) is involved.
- All final NDE inspection and testing shall be performed after any required postweld heat treatment. For shop welding, 100 % RT will be performed for all reasonably accessible tubular butt welds after PWHT for Gr 91, 92, super 304 and

HR3C materials. 20% RT will be performed for other materials after any required PWHT.

For shop welds, all superheater, reheater, economizer, internal surface butt welds shall be tested using X-TV or RT to provide permanent and traceable records. All other tube butt welds that cannot be tested using X-TV shall be examined using radiography. Header tube nipple welds shall be examined by MT. Additional 20 % UT shall be performed (where possible).

For field welds, the following tube butt welds shall be RT with progressive evaluation according to ASME B31.3. RT shall also be performed for welded nozzle connections. Magnetic particle and liquid penetrant test shall be performed for fillet, socket, attachment and seal welds where RT cannot be performed.

100% RT shall be carried out for the following field welds that meet the following conditions:

- Dissimilar metal welds or
- Design pressure ≥ 9.81 MPa or
- Design temperature ≥ 400 deg C

100% UT and 20% RT for the following field welds that meet the following conditions:

- Design pressure < 9.81 MPa or
- Design temperature < 400 deg C

The Onshore Contractor shall establish procedure to ensure the quality of seal fin welds and temporary attachment welds and removal. This procedure shall be submitted to the Owner for review during execution stage.

3.3.10 Main Steam and Reheat Steam System

All main steam, and hot reheat pipe shall be supplied by acceptable international suppliers. The materials of manufacture shall be selected according to the design and operating conditions, and shall be reviewed during detailed design stage. The main steam, cold and hot reheat system piping shall be stress analyzed and routed and supported in such a manner as to provide for thermal expansion and to maintain end reactions and moments within allowable limits.

The Offshore Supply Contractor shall define which austenitic grades of tubes for superheating and reheating surfaces shall be shot-peened to improve high temperature oxidation resistance. Details shall be submitted to the Owner for review during the design submission stage.

The Offshore Supply Contractor shall provide complete details of material of tubes, arrangement of tubes and flow velocities in the tubes and also provisions to prevent high temperature-steam corrosion and steam oxidation of tubes in detailed engineering stage.

The superheaters and reheater shall be designed for maintaining uniform steam temperature distribution in the elements and tubes across the width of the boiler to prevent overheating of tubes and headers. The boiler shall be designed so that the various sections of the superheaters and reheater will not be overheated during the start-up and shut-down, at loss of steam temperature control, boiler trip or sudden load rejection and restoration to full load. Furthermore, the Offshore Supply Contractor shall provide sufficient thermocouples on outlet tube legs for the purpose of monitoring the steam temperature distribution and the Offshore Supply Contractor shall submit the necessary information and back-up data to explain the criteria for monitoring the steam temperature distribution for the approval by the Owner during design submission.

Adequate provision shall be made for thermal expansion of tubes and headers.

Superheaters and reheater tubes shall be designed as widely spaced to avoid bridging by ash in consideration with slagging and fouling characteristics of the ash. All superheaters and reheater parts shall be readily accessible for cleaning and maintenance of the elements.

The superheater and reheater steam temperature control system shall be capable of limiting at plus/minus 5°C to the final steam temperature under steady state load of boiler operation. During the detailed engineering stage, the Offshore Supply Contractor shall submit curves of superheat and reheat temperature plotted against boiler evaporation from no load to BMCR during design stage and start-up characteristics of the boiler for hot, warm and cold starts indicating superheat and reheat steam temperatures, flow and pressure conditions with reference to time.

Superheater and reheater temperature shall be controlled by spray type attemperator. Gas recirculation for steam temperature control is not acceptable.

The steam sampling connections shall be provided on the superheater and reheater outlet headers.

The spray type attemperator for the reheater system shall be installed in the inter stage in the reheater piping and shall be used for emergency operation only, as a safety device and shall be operated automatically when demand arises.

Spray water flow shall not exceed 10% of the BMCR steam flow.

Suitable anchors shall be provided on the terminal headers to withstand thrusts and moments resulting from the thermal expansion of the main steam and hot and cold reheat piping.

Erosion shields of the approved design shall be provided on tubes, where necessary, to prevent erosion by sootblower steam or ash in the flue gas.

3.3.10.1 Main Steam Pipes

From the boiler superheater outlets, main steam header supplies steam to the turbine generator. Steam for steam sampling shall be taken from the boiler top outlet header.

Drip pot drain shall be provided at the low and high points respectively. In general, drains shall discharge to the condenser, contain one motor operated valve and one manual stop valve in series. The turbine stop valve above seat drain shall be routed to suitable flash vessel with desuperheating.

3.3.10.2 Reheat Steam Pipes

The system between the high pressure turbine exhaust and the reheater inlet header is designated as cold reheat. The spray type attenuator for the reheater system shall be installed in the interstage in the reheater piping. The system between the reheater outlet header and the reheat stop valves of the intermediate pressure turbine of the turbine generator is designated as hot reheat. The hot reheat shall connect to the turbine reheat stop valves. Steam for sampling shall be taken from the hot reheat outlet header. The hot reheater outlet header piping shall contain spring loaded safety valves.

3.3.11 Economizer

The Offshore Supply Contractor shall incorporate provisions for preventing thermal fatigue damage to the economizer inlet manifolds, headers and tubing due to feed temperature cycling during start-up. Cold end corrosion shall be avoided.

The tubes shall be seamless drawn and be arranged to facilitate ease of replacement or repair with minimum disturbance of tubes, casing or settings. The elements shall be divided into several banks sufficiently mustered to enable the flue gas passages can be cleaned by sootblowing. Adequate access shall be arranged for routine cleaning, inspection and maintenance of tube elements.

Tube alignment and spacing shall be designed with allowance for thermal expansion and contraction.

The economizer shall preferably be of plain tube continuous loop and non-steaming type and without intermediate headers.

An economizer recirculation line with associated valves, fittings and hardware may be supplied, if required, for start-up and low load operation.

The economizer inlet header shall be provided with suitable supports to permit the header to withstand the thrusts and moments resulting from thermal expansion of the main feedwater piping.

Erosion shields shall also be provided for the economizer, where necessary.

3.3.12 Metallurgical Information

For purpose of facilitating the future boiler life assessment and to demonstrate achievement of quality welds and compliance with metal specification at critical locations on boiler pressure parts, the Offshore Supply Contractor shall provide the following, by referring to Table 1:

- Two ex-work samples of each type of raw material for listed pressure parts.
- Two samples of weld joints which are prepared in identical way to listed welds.
- Cross weld metallographic replicas for the listed welds upon the completion of manufacturing.

The Onshore Contractor shall provide the following, by referring to Table 1:

- Cross weld metallographic replicas for the listed welds upon the completion of site erection.

The replica for each weld shall cover the parent metal zone- heat affected zone- welds-heat affected zone- parent metal zone, i.e. 5 zones altogether. 4 replicas shall be taken along the circumferential welds at 90 degrees apart. i.e. Replicas on weld taken at 12, 3, 6 and 9 o'clock position on the weld.

The production of the metallographic replicas shall conform to ASTM E 1351 "Standard Practice for Production and Evaluation of Field Metallographic Replicas".

- Metal hardness measurements of parts at locations where replicas have been taken upon completion of manufacturing and site erection. The measurement shall be carried out using the Pin Brinell Hardness Tester. The Offshore Supply Contractor shall remove the dents left by the hardness measurements with minimum grinding at Offshore Supply Contractor's facility. The Onshore Supply Contractor shall remove the dents left by the hardness measurements with minimum grinding at Site.

In addition, the Offshore Supply Contractor shall conduct a metallurgical evaluation according to ASTM E 1351 "Standard Practice for Production and Evaluation of Field Metallographic Replicas" and assess the compliance of materials and welds with relevant parent metal specifications and welding specifications at replica locations as part of inspection and testing plan. The Onshore Contractor shall also perform the above evaluation work for the defined site welds.

Table 1 – List of Replica Locations

Pressure Parts	Locations
Main Steam Pipelines	Boiler side terminal weld, turbine end terminal weld and one weld in-between on each of the steam pipelines
Hot Reheat Pipelines	Boiler side terminal weld, turbine side terminal weld and one weld in-between on each of the steam pipelines
Final Superheater Outlet Headers	One weld on the branch which will be subject to highest operating temperature and one butt weld on each of two randomly selected pipes
Final Reheater Outlets Headers	One weld on the branch which will be subject to highest operating temperature and one body butt weld on each of two randomly selected pipes

3.3.13 Boiler Life Analysis

The Offshore Supply Contractor shall perform boiler life analysis to demonstrate that the boiler is able to operate satisfactorily throughout its design life of 35 years at operating modes and cycle as specified in OTR clause 2.2.1. It shall also be able to serve as baseline for future boiler life assessment.

The boiler life analysis shall include fatigue damage, creep damage and total damage analysis. The Offshore Supply Contractor shall identify the high risk damage components which shall include final superheater outlets headers and final reheater outlets headers. The analysis result shall be provided to the Owner in report format. The report shall contain the following information

- Executive Summary
- Method of Calculations adopted
- Assumptions made in the calculations including
 - Boiler operating regime
 - Critical locations and components chosen for analysis
 - Rationale of critical locations chosen for analysis
 - Transient and steady-state operations considered
 - Temperature and pressure profiles in steady and transient operating conditions considered
 - Operation duration
- Calculation steps
- Calculation result at critical components and locations
 - Creep Damage
 - Fatigue Damage
 - Total Equivalent Damage

3.4 Start Up System

The Units shall be capable of performing two shifting operation, this means that during low load and startup period the boiler operates in sub-critical steam conditions. A suitable start up system shall be designed to provide a lower once-through minimum load. The system shall include a water separator system installed between the waterwalls and the primary superheater, a water storage tank and a drain with drain recovery facility. Both the separator and the storage tank shall be adequately sized to allow prolonged low load operation. The drain recovery shall include a low load recirculation pump to maintain the required minimum flow in the economizer and waterwalls during start up and 'wet condition' of the separator.

3.5 Chemical Cleaning of Boiler

The Onshore Contractor shall supply the material, equipment and chemicals and carry out the pre-operational chemical cleaning of the boiler to ensure that the boiler internal surface can achieve a clean condition with a protective magnetite layer to assist in preventing later on-load corrosion of the boiler.

The Onshore Contractor shall provide all necessary temporary neutralizing and sedimentation facilities including pipework for treatment of effluent from chemical cleaning operations and shall carry out treatment work of the effluent to meet the requirements of the local regulation for discharge. The Onshore Contractor shall be responsible for supplying all chemicals, including the disposal of all waste and effluents. Furthermore, the Onshore Contractor shall provide the necessary pipework for the nitrogen gas supply and chemical cleaning of the Unit.

During detailed engineering stage, the Onshore Contractor shall submit the procedure and equipment required for the chemical cleaning process.

3.6 Boiler Vents, Drains and Start-up Flash Tank

The complete vents, drains and start-up flash tank including the necessary piping, valves, silencers (if applicable) and instrumentation shall be supplied by Offshore Supply Contractor and installed by Onshore Contractor.

Nitrogen blanketing connections shall be provided on the high point vents for inerting during shutdown periods.

All boiler proper drains shall have double isolation valves.

Start-up flash tank with 6 mm wear resistance liner shall be provided.

3.7 Sootblowers

A complete set of automatic steam-operated soot-blowing equipment shall be provided for cleaning the furnace waterwalls, superheaters, reheaters, economizer and air pre-heater elements. Steam for the sootblowers shall be taken from superheated steam line before the attemperator for the secondary superheater.

The soot-blowing system shall include, but not be limited, to the following:

- All steam and drain piping, valves fittings, supports and steam pressure reducing valve
- Motorized isolating valves in the steam supply line in addition to the reducing valve
- Sealing air piping, fittings and valves if necessary
- Shielding of boiler tubes in the vicinity of sootblower head to prevent erosion

The soot-blowing steam supply piping system shall be designed and supported to impose the minimum acceptable reactions and moments at the sootblower heads. Spring hangers shall be provided where necessary to accommodate vertical differential movements.

Attention shall be paid to the particular feature of the specified design coals in respect of slagging and fouling on the tube.

All sootblowers shall be capable of operating manually and sequentially in series from the Central Control Room. Local manual operation shall also be provided for testing.

The sootblower system shall contain all automatic sequential control equipment for the blowers including operation of motor drives, steam isolating valves and drain valves.

The sootblower system shall be fully controllable from the unit DCS.

The air pre-heater soot-blowing system shall also be incorporated in the boiler soot-blowing DCS control system. The Offshore Supply Contractor shall ensure that the steam used is adequate for the purpose.

Sootblowers shall each be provided with a manual isolation valve in upstream of the blower (poppet) valve, in order to facilitate individual blower isolation for maintenance.

The system and blowers shall be designed such that any part may be removed and replaced while the boiler is on load.

The steam piping is to be continuously self-draining with the drain valves located at the operating floor level, these valves shall be proceeded by drain control orifices. The drains shall be collected and run to the startup flash tank.

Adequate walkways and platforms shall be provided for access during operation, repair and maintenance purposes. The arrangement of the platforms, access stairways, ladders etc. shall meet the local Vietnamese codes and standards. The arrangement of walkways, platforms, access stairways, ladders, etc. shall be submitted for the Owner's review during the design stage and a 3D model review shall be conducted with the Owner.

The details of sootblowers provided for various sections including layout showing the position of sootblowers and spare holes, angle of blow, etc. shall be submitted for the Owner's review during detailed engineering stage.

(a) Boiler Pressure Part Surfaces

The Offshore Supply Contractor shall carefully select the number of sootblowers to be strategically located in the superheater, reheater, economizer and furnace of the boiler and the selection and locations. The proposed system shall recognize the operation of the unit and due consideration shall be given to the maintenance requirements and reliability.

(b) Furnace Wall Cleaning

Furnace wall blowers shall be provided on several levels on both sides of the furnace and also above the top row of burners on the front and rear walls.

(c) Pendant Surface

The pendant surfaces of the superheater and reheater shall be provided with fully retractable blowers arranged on both sides of the boiler. The complement comprises of blowers in the cavities of the pendant platen superheater, pendant final superheater and pendant reheater section.

Provision shall be provided for the possible future installation of blowers in the cavities of the pendant platen superheater, pendant final superheater, pendant reheater section, front of the pendant primary reheater section and rear of the pendant primary superheater section.

(d) Horizontal Surface

The horizontal surfaces of the superheater, reheater and economizer banks shall be provided with retractable blowers arranged on both sides of the boiler. The complement shall comprise of blowers above the horizontal primary reheater top bank, in the cavities of the horizontal reheater section, above the horizontal primary superheater top bank, in the cavities of the horizontal primary superheater and above the horizontal economizer.

(e) Flue Gas Air pre-heater Plant

Each gas air pre-heater shall be equipped with two steam sootblowers – one above and one below the heater matrix. In addition, each air pre-heater shall be equipped with a high-pressure (5 bar) water-washing lance for cleaning during maintenance.

3.8 Insulation and Refractory

Insulation and refractory shall be provided for all equipment which requires insulation and/or refractory.

The insulation and refractory shall include brickworks, tiles, fittings, weatherproofing, outer metal lagging and other materials required to ensure a sound, safe and functional installation suitable for the service.

The setting shall permit free expansion of enclosures and differential expansion of component parts.

Thermal insulation for boiler, ductworks and similar surfaces shall consist of slabs firmly fixed by being implanted on studs and secured by binding wire and clips or other suitable means.

The maximum pitch of studs shall be 300 mm on downward-facing surfaces, 460 mm on vertical surfaces and 610 mm on upward-facing surfaces.

In the event of borderline cases arising between the standard commercial thickness of insulating material, the greater thickness shall be applied.

Where the insulation is multi-layered, the application shall be such that all joints in the second layer are staggered from those in the first layer and overlapped.

Provisions shall be made at insulation on expansion joints to do not obstruct movement of the joints.

Outer metal cladding of boiler shall be 1.2 mm thickness aluminum plates to be approved by the Owner. The cladding shall not be used to retain refractory or insulation in place.

Pipework insulation shall be performed sections except for the special locations such as for steam separator, large headers, pipes and tube banks inside the penthouse where slabs may be used.

Where insulation is applied on surfaces operating at high temperatures it is essential that the fixings and attachments shall be fabricated from appropriate materials.

All top surfaces of ducting, flues and boiler roof including inclined runs which are walking surfaces shall be capable of withstanding the maximum movement of people and traffic.

3.9 Auxiliary Steam System

Auxiliary steam shall normally be supplied from the cold reheat steam and / or bled steam.

Auxiliary steam for initial start up of the first Unit shall be supplied from auxiliary boiler. The steam from the auxiliary boiler shall be connected to the auxiliary steam header common to two Units.

The auxiliary boiler shall be of light diesel oil fired and has a capacity enough at least for starting-up of the First Unit. The emission of the auxiliary boiler shall fulfill the local requirements. Steam condition of the auxiliary boiler shall be appropriate to the auxiliary steam header.

Scope of supply shall include, but not limited to the following:

- (a) The auxiliary boiler shall be outdoor type and includes burners, oil pump, complete control and instrumentation.
- (b) Auxiliary steam piping, headers, valves, pressure reducing stations, atomizers, insulation, drains, traps and fittings.
- (c) Interconnection from the auxiliary boiler.
- (d) One spare branch connections with isolating valves.
- (e) Local instrumentation and emission monitoring equipment.

3.10 Pulverized Coal Handling System

3.10.1 Pipework

The pipework system shall be arranged such that an equal quantity of pulverized coal is distributed to each burner from each mill under all loading conditions. The system shall be designed to avoid any areas where pulverised fuel (PF) can fall out of suspension.

The pipework system shall be manufactured from miter elbow type with wear resistant liners at the elbows at all mill outlets and elbows at all burner inlets. To improve the wear life of bends, long radius bends are preferred where practicable.

3.10.2 Feeder and Bunker

The system shall consist of boiler bunkers with a design capacity for 12-hour storage of the coal with the lowest calorific value (CV). Auxiliaries as feeders, mills, primary

air fans, distribution pipework to the burners shall also be provided. Sloped bottom conical portion of the bunker shall be lined with 6mm thick stainless steel plate.

The tripper floor of bunker bay shall be fully enclosed. The shape of the bunkers shall be designed to give mass flow of coal under all conditions particularly when reclaiming wet coal from stock, without material holding up on the bunker sides or in any valley angles.

Powered means of isolation shall be provided between bunkers and feeders. The feeder shall be fitted with liners to combat abrasion and corrosion such that the feeder casing is not sacrificial.

The coal transfer chutes between feeders and mills shall be manufactured from stainless steel or contain stainless steel liners. Bunker emptying facilities shall be provided which shall allow the bunker contents to be discharged in a controlled and safe manner.

The feeders shall be variable speed and be able to supply the required quantity of coal to the mill under all circumstances without interruption. The feeders shall be able to re-start against a full bunker head of compacted coal. The mill is positioned directly below the feeder and the coal chutes shall allow uninterrupted flow of coal from feeder to mill in all circumstances particularly when reclaiming wet coal from stock without coal hanging up on the sides of the chutes or bridging over.

The throughput of the mill and classifier assembly shall be able to meet the boiler loading requirements under all circumstances. This shall be achieved without compromising minimum transport velocities and pulverized coal/air ratio.

3.10.3 Mill

At least six (6 no.) mill groups with associated system such as bunkers, ductworks, primary air fans, burners ... etc. shall be provided.

The mill shall have sufficient drying capacity for the specified coal range when the coal is unloaded from jetty and directly loaded to coal bunker during the period of highest rainfall. Routine maintenance of mills and dynamic classifiers shall be feasible with the boiler on load. Fire detectors and a steam inserting system for the mills shall be provided. Vertical bowl-type mill shall be provided. Mill shall be equipped with dynamic classifiers.

The system shall be designed to be fully automatic under all normal operating conditions including start up and shutdown and require minimum operator intervention.

Under all loading conditions, the mill shall reject all tramp metal, pyrites and other non grindable materials, up to 3% of the coal flow. There shall be no dead areas in the feeder casing, mill, classifier or primary air system where coal can accumulate under any circumstances.

Transport velocity, pulverized coal air ratio, the fineness grading, classifier outlet temperature and so on shall meet requirement of the pulverized coal handling system. The performance shall be obtained throughout the wear life of the grinding elements.

The mill shall have sufficient drying capacity for the specified fuel range when the coal is reclaimed from stock during the period of highest rainfall. This shall be achieved with the minimum hot air temperature at mill inlet.

Routine maintenance of mills and classifiers (and exhausters if proposed) shall normally be undertaken with the boiler on load.

Sufficient openings through the feeder floor to provide access platform for the operation staff to access the top of mill from feeder floor.

The whole milling and pulverizing system shall be designed to be gas and fuel tight, to prevent pulverized coal leakage and to withstand an internal explosion without a breach of containment.

The grinding elements of the mill shall be manufactured from wear resistant high-chromium materials. The system shall be designed to automatically reject up to 3% of mill throughput when handling the range of specified fuel.

Facilities shall be provided to fully isolate each mill to allow safe access for internal maintenance with the remaining mills in service.

A means shall be provided for isolating mill from the furnace to prevent blow back in the event of furnace pressurization. This shall be achieved by means of pneumatic actuated isolation gate valves at the mill outlet and manual burner isolation valves. These valves shall be remain opened for mill steam inerting in case of boiler trip in MFT or mill trip to meet NFPA inerting requirement. Moreover, mill and mill components shall be designed to withstand 50 psi pressure per NFPA requirement for explosion. No explosion vent shall be provided.

The PF pipework system shall be fitted with suitably designed sampling ports in each mill outlet pipe and in each burner leg after the final split. The pipework system shall be adequately supported in accordance with the relevant national standard. Expansion provision shall be provided within the pipework system. The expansion joints shall be fitted with suitable restraints to maintain the integrity of the system in the event of an internal expansion. Three dimensional compensators shall be furnished for the pipes near the burners to compensate the internal expansion.

The auto control system shall allow sequenced starting and shutdown of mill groups from a single initiation. The system logic shall take full account of the potential hazards associated with exceptional conditions such as loss of coal, mill trip etc. The handling system shall be equipped with suitable instrumentation which allows the operator to be aware when a potentially hazardous situation exists. Parameters which shall be monitored include pulverized coal carrying velocity, pulverized coal/air mixture strength, fuel/air temperature at mill outlet, air temperature at mill inlet, loss of coal feed and fire detection.

3.10.4 Inerting and Fire Fighting System

A dedicated fire detection system shall be installed to warn the operator of an ignition source within the system. Each mill shall be furnished with steam for inerting and fire water protection spray nozzles within the body of pulverizers. The spray water for fire protection shall be actuated manually from local and/or remote location. Thermocouples shall be provided in the casing to monitor the temperature. When the casing temperature is higher than normal, alarm shall be provided in DCS and warn the operators.

Pulverizer inerting and fire fighting system shall be provided to dilute the oxygen content of the pulverizer so as to safely control a fire and to reduce the risk of explosions. Of primary concern is assuring the safety of plant personnel during a potentially hazardous situation.

The inserting mediums shall be steam which shall be readily available in sufficient quantity to allow a flow through system for continuous purging of volatile gases. Then, it shall allow a safer restart of the pulverizer.

3.11 Burner System

3.11.1 Description

The coal burners shall provide a stable and efficient combustion of the pulverised fuel and minimize emissions of oxides of nitrogen (NO_x), unburnt carbon in dust and carbon monoxide (CO) for the range of fuels specified. A low NO_x firing system shall be used. The coal burners and firing sequence shall be designed in conjunction with the furnace, coal milling facilities and forced draft fans.

Burner management system shall be in full compliance with NFPA standard.

The coal burners/coal combustion system shall be sized such that it is possible to achieve BMCR with the design number of burners in service.

3.11.2 Combustion System Operational Requirements

The design and arrangement of the coal combustion system shall be such that when operating in conjunction with the ignition and oil support firing equipment, continuous furnace firing can be maintained when bringing the Unit up to full load from cold conditions without damage to any part of the Unit.

The coal burners shall be stable (i.e. operate without oil support) over the normal range of coal mill operation and in any case down to 30% of the BMCR fuel throughput. Each burner shall have 5% capacity margin for BMCR conditions.

There shall be no flame impingement on the furnace walls under any of the defined operating conditions. Where appropriate positive design features shall be incorporated to provide a beneficial (i.e. oxidizing) atmosphere at the furnace walls.

3.11.3 Combustion System Component Life

The major structural components of the coal burners/coal combustion system shall achieve the design overall plant life of 35 years. The Offshore Supply Contractor shall list in the design submission those components which cannot achieve the overall plant life without repair, including the estimated frequency of replacement, extent and cost. The design life of the non-consumable burner components shall be 35 years. The Offshore Supply Contractor shall provide routine maintenance spare parts list and the operating and maintenance procedures showing which part need to replace during the design life of the burner.

The minimum life of any burner component shall ensure that no maintenance or repair is necessary between yearly boiler overhauls.

The material of nozzles tip shall be heat resisting stainless alloy and new rare earth alloy such as 253MA.

3.11.4 Combustion System Performance Requirements

The coal burners/coal combustion system shall be sized such that it is possible to achieve BMCR with the design number of burners in service.

The firing system shall employ advanced low NO_x coal-fired burners which have the great advantages of broad range of fuel flexibility, low NO_x emissions, high combustion efficiency, even temperature field of furnace and flexible and safe operation. Burner wall distance shall be sufficient to ensure that the burner swirl direction, operation with low excess air shall be possible without the risk of wall damage.

The coal combustion system shall be designed to minimize the emissions of oxides of nitrogen (NO_x) over the full boiler operating range above the design minimum load whilst maintaining satisfactory combustion efficiency.

3.11.5 Combustion System Design and Fabrication Requirements

The burner shall be fitted with adequate lifting and handling arrangements for erection and maintenance purposes.

The burner shall incorporate observation and de-slagging ports as appropriate. These ports shall be located to ensure operation under suction conditions over the burner's entire operating range. Observation and de-slagging ports shall be sited to ensure adequate access with the burners installed in the boiler.

Cross correlation type flame monitors shall be provided.

The burner components liable to heat distortion (for example PA/PF front end, swirl generation system, etc.) shall be sited away from the furnace and the main heat source as far as possible. The cooling requirements of out of service burners/air nozzles shall be minimized by positive design features. During Unit shutdown or Unit trip, sufficient cooling air shall be blown in to cool the burner tips.

Details of methods for preventing heat distortion shall be provided during the engineering stage.

All dampers and controls for manually adjusting the burners and regulating the combustion air supply to burners from local or emergency conditions shall be arranged for easy operation at a convenient gallery level.

Means shall be provided for measuring the combustion air flow rate through measuring primary air to each mill by venturi and secondary air at each fan inlet by piezometric ring.

All local and remote instrumentation and controls necessary for the safe and efficient operation of the coal burners shall be supplied and shall be rated to withstand the maximum local ambient temperatures.

The burner control system shall maintain a stable coal flame for a period of at least 3 seconds in the event of the voltage dip to any value between normal and zero at the terminals of the motor drives on the burner coal supplying system and the burner control shall be maintained over this period.

An interlock shall be provided to prevent any burner being lit against a closed flue system.

The Offshore Supply Contractor shall design the interlock system for oil burners and coal milling plant and state the minimum number of oil burners required to safely initiate and establish each mill group of coal burners.

The oil burners shall be automatically operated from the central control room. The mill and the coal burners shall also be controlled from the central control room. The DCS shall control both according to the signals from the furnace and other path. The Offshore Supply Contractor shall provide the minimum number of oil burners required to safely initiate and establish the boiler load in their design submission.

The flame monitoring system shall be provided to detect the presence or absence of a stable flame at each burner under all conditions. The flame monitoring system provided shall be able to detect the presence or absence of a stable flame at each burner position under all conditions, regardless of the number and pattern of burners in service.

The system shall be provided with an automatic checking system capable of checking continuously or at pre-determined time intervals for component and circuitry faults and the integrity of the complete flame failure system with the burner firing.

Means for determining the magnitude of the detected flame signal shall be provided local to the burner. A display of the magnitude of the detected flame signal at all burner positions shall be provided in central control room. The maximum period of time that coal can be injected into the furnace with no flame present shall be less than 2 seconds.

An adequate supply of purge air from a pressurized source shall be provided to cool the flame monitor viewing heads and keep the optical systems clean.

The Offshore Supply Contractor shall design the system to prevent PA, ID and FD fans from reverse rotation at anytime of operation and, in particular, during one fan shutoff condition. Under such condition, the cold/ hot air or flue gas from the running fan shall not be circulated back to the shutoff fan by means of tight-closed dampers. All dampers for PA, ID, FD and hot air dampers to the windboxes shall be designed with

- (a) Fast shutoff of the damper to tight-closed position; the time from fully open to fully closed position shall not be more than 30 seconds;
- (b) Low leakage under the tight-closed position; the leakage rate shall not be more than 5% of the rated flow rate of that damper.
- (c) Easy adjustment of the control mechanism to narrow down the gap between any two adjacent damper blades.

3.11.6 Provision of Space for the Plasma Ignition System

The Offshore Service Contractor shall make provision of space for the optional plasma ignition system. The future electrical supply and control system shall be containerized. The Offshore Service Contractor shall submit layout drawings identifying the provision of space during design submission stage.

3.12 Oil-support Firing System

3.12.1 Support Firing System Description

Diesel oil shall be used for boiler start up, support for low load operation and flame stabilization. The total oil burner capacity (oil firing without coal) shall be able to achieve 30% BMCR.

The system shall be designed for automatic operation of all control sequences. Manual intervention shall only be required for lighting up and extinguishing the oil burner and, where applicable, adjusting the oil burner firing rate.

Each oil burner shall be supplied complete with its own burner specific ignition, oil flame monitoring, and burner management system.

3.12.2 Support Firing System Operational Requirements

The ignition and support firing system shall be designed such that at any time when the boiler is firing on coal, oil burners are available for immediate operation on demand from the DCS or the boiler front local control panels.

The oil firing rate(s) shall be selected taking account of the operational requirements for coal combustion support and part load carrying duties. The range of oil burner firing rates shall operate satisfactorily.

3.12.3 Support Firing System Design and Fabrication Requirements

The oil burner atomizer tips shall be designed to ensure that the installed fuel oil pumping plant has sufficient capacity to meet the maximum site fuel oil flow demand.

Each oil burner shall be equipped with a burner specific oil flame monitoring system. The flame monitoring system shall be able to detect the presence or absence of a stable flame at each burner under all conditions, regardless of the number and pattern of burners in service and the composition of the fuel fired. The system shall be provided with an automatic checking system capable of checking continuously or at predetermined time intervals for component and circuitry faults and the integrity of the complete flame failure system with the burner firing.

Local indication of the magnitude of the detected flame signal shall be provided to facilitate alignment of the viewing head.

The oil burners and associated components (carrier tube, flame stabilizer, etc.) shall be installed in the windbox located between the pulverized fuel elevations.

The oil burner assembly shall be fitted with a positive mechanical interlock to ensure that the burner assembly cannot be disconnected with the oil supply on and that oil cannot flow to the burner unless the burner assembly is in the firing or stand-by positions.

Where flexible hoses are used the design of all connections shall be such that mismatch of connections to the burners cannot occur.

An IP65 rated mechanical-type limit switch shall be fitted to each safety shutoff valve for indication for the burner operating mode, i.e. "firing" or 'not-firing'. Indication of the burner-operating mode shall be provided in DCS in the CCR.

An IP65 rated proximity-type limit switch shall be fitted to each burner to indicate that the burner gun is connected. Indication of the status i.e. connected or otherwise, of each burner shall be provided in DCS in the CCR.

A local oil tray shall be mounted immediately beneath the oil burner coupling assembly to contain any small oil leakage that may occur during burner gun removal.

A complete main fuel ignition system shall be furnished. The system shall consist of electric spark ignition and all required accessories.

All required control devices for the igniters shall be furnished, including high voltage transformers, indicating lights, control switches, and relays.

This system shall be mounted in a cabinet with each cabinet serving a group of igniters. The igniter grouping shall be based on the manufacturer's standard arrangement. Each cabinet shall be provided with local igniter control push buttons to provide capability for maintenance and operational testing of the associated igniter group. All required fuel control valves, interconnecting piping, flexible hoses, piping accessories, and wiring between the cabinet and igniter must be furnished. Igniters control shall be suitable for operation from and monitoring by the burner management system of the DCS. All field installed wiring and raceway shall be provided.

Combustion air supply to each burner shall be from the FD fan discharge.

Oil burners shall be sized to provide sufficient heat input for reliable main fuel ignition and flame stabilization at low firing rates.

Atomizing and purge air shall be supplied from compressed air system. All piping and accessories shall be provided. Compressed air purging shutdown of the oil gun shall be provided in the control system. HEA ignitor system shall be provided to light off the oil burners.

All oil burner control and sequencing shall be controlled by DCS.

The common burner management logic shall be provided for the repeating of all common signals between burners, including the firing trip and starting interlock derived from the boiler protection system. A single fault shall have minimum effect on burner availability and the integrity of plant protection. Therefore, where single component failure may result in major plant loss of generation or plant damage, the common protection circuits shall include adequate levels of redundancy.

All equipment necessary to provide sequence completion check signals shall be supplied, including:

- Oil ring main pressure check
- Oil ring main temperature check
- Oil burner pressure and flow check
- Core air flow/pressure check
- Electric igniter check
- Main oil flame established check
- Purge air pressure and flow check
- Furnace purge check

Circuit for both power and control supplies shall be organized to ensure that any individual burner unit can be isolated and its associated electrical equipment can be worked on without hazard whilst the remainder of the burners remains in service. The isolation shall be by means of an OFF/LOCAL/REMOTE lockable selector switch.

Sequence of events for igniting oil shall be;

1. Open oil valve
2. HEA begins sparking
3. Detectors check for flame
4. If no flame is detected within 30 sec from the initiation of the event, then the DCS signals to close oil trip valves.

Each oil burner shall be equipped with a reliable flame monitoring system which meets NFPA requirements and provides the required input signals to the flame safeguard and safety system.

Oil burner controls shall be provided for local operation in situ and remote control from the CCR. The normal mode of oil burner operation and status monitoring shall be from the Unit Control Desk situated in the central control room. Local stop/start push buttons shall be provided for individual burner control. Oil burner status indicators (FIRING or READY/STANDBY) and ignition detector units shall be located at the boiler front.

Each individual boiler front burner control cubicle shall have a minimum rating of IP65.

The burner management system shall provide comprehensive diagnostic facilities. As a minimum the diagnostic display shall indicate the precise position that has been reached in the oil burner light up sequence in the event of a sequence failure. Additionally a facility for interrogating individual burners local to the boiler front shall be provided.

The Offshore Supply Contractor shall provide the following in their design submission

- a functional specification for the burner management system and diagnostic facilities;
- details of the electrical hardware proposed;
- "Watchdog" and protection (fail-safe) arrangements; and
- details of power and control circuit.

Local panel with indicating lamps to the burner management system shall be provided with separate lamp test facilities.

3.13 Boiler Valves, Casings and Access Gantries

3.13.1 Boiler Valves

The Offshore Supply Contractor shall provide details of all valves to be fitted to the boiler in design submission. Welded valves shall be provided to minimize stem leakage. Flanged design will only be allowed for small bored valves with low pressure rating.

The boiler shall be furnished with sufficient safety valves and electro-relief valve to meet ASME requirement or equivalent standards before the main steam stop valve along the steam flow. Before the combined valve of IP turbine, safety valve shall be fixed according to the requirements of ASME standards.

The selection of the safety valves and other valves shall satisfy ASME.

All safety valves shall be located at the place where enough space can be provided for setting and maintenance work. All ordinary valves shall be located at readily accessible areas as near as possible to the headers. All connections to pressure parts shall not be less than 25 mm nominal bore.

Where two blowdown valves or isolating valves are arranged in series they shall be supplied as two separate units. Twin-type valve in a single body is not acceptable.

All pressure parts shall have double isolation valves for instruments, drains and vents.

Local gauge boards shall be provided for the boiler to indicate major operating parameters. The boards shall include pressure gauges for indication of separator and final superheater and reheater outlet pressures.

The lifting and reseating pressure of safety valves after setting shall be correct and reliable to avoid failure to lift or reseal of them.

The Offshore Supply Contractor shall provide data of setting and checking for lifting pressure of the safety valves according to the requirements of standard and code during Project execution and in the maintenance manual.

The reseating pressure difference of safety valves shall be no more than 7% of lifting pressure.

The safety valves shall have good features of tightness under operation pressure.

The safety valve shall be fully open when the steam pressure is 3% above the lifting pressure of the safety valve. For the boiler, the total pressure rise when all safety valves are fully open shall not be more than 8% of the design working pressure.

The Offshore Supply Contractor shall ensure sufficient discharge capacity of the safety valves at the outlets of the superheater and reheater and also cooling of the superheater and the reheater when the safety valves are lifting.

The safety valves shall be furnished with steam discharge piping extended through to outdoors. Steam discharge piping shall not be rigidly connected with safety valves to avoid reacting force transmitted to the piping. Leakage of steam shall not be allowed when the safety valves are lifting. The piping shall be straightened as much as possible and shall be fixed in such a way that the piping shall not hinder the correct movement of safety valves due to thermal expansion or reacting force of discharge and valve shall not be installed. No foreign force shall be exerted on safety valves either in cold or in hot state. The drain piping with drain valves extended to safe site

shall be installed on bottom of the steam discharge piping. The exhaust pipe shall be extended with a certain height above the boiler roof casing.

All silencers if applicable, atmospheric vents, safety valves vents and other piping penetrations through the boiler roof shall terminate 3,000 mm above the roof structure, suitably supported, sleeved and weatherproofed.

Safety valves shall be equipped with devices or measures to prevent set value from deviation after safety valve have been regulated.

The superheater outlet shall be fitted with safety valves having a capacity of not less than 20% of the maximum evaporation rate of the boiler.

The safety valve(s) of reheater on the outlet side shall have a capacity not less than 20% of the maximum steam flow. The electrically operated ball type relief valves shall be fitted. Each valve shall be 65 mm minimum size and shall be complete with silencer if applicable, controller, solenoid, pilot valves, and interconnecting pressure piping between the superheater header and the valve and a controller with two shutoff valves.

All safety valves shall be provided with silencer if applicable and drain pan. Silencers if applicable shall be installed for the first lifting safety valves and power controlled discharge valves separately of superheater outlet, reheater inlet and outlet.

Headers and connectors for installing safety valves shall be able to withstand reacting force of lifting or reseating of safety valves.

The Onshore Contractor shall provide clamping tools for safety valves during boiler hydrostatic test.

The Onshore Contractor shall provide both tools for calibrating safety valves and those for calibrating safety valves in operation.

High pressure valves shall have seal surface. When the valve is fully opened and closed, the stem packing shall not bear the pressure of the working medium, so that packing replacing is convenient.

On the valve casing, there shall be signs indicating the direction of the flow of working medium, rated pressure, temperature and nominal diameter.

The tightness of the valve shall conform to the requirements of American National Standards Institute (ANSI) or equivalent standard; therein the leakage grade of the common using regulating valves shall be not lower than standard ANSI IV equivalent standard.

The minimum bore of the valve seat of any safety valve connected directly to the boiler, superheater or reheater, shall not be less than 20mm.

Over 100% capacity safety valves shall be provided for superheater system on primary superheater outlet, final superheater outlet.

Every reheater shall have at least two safety valves on cold reheater inlet and hot reheater outlet.

Safety valves shall be mounted without any intervening valve's vertically and separately supported.

3.13.2 Boiler Casings & Access Gantries

A suitable gas tight boiler casing shall be provided. The Offshore Supply Contractor shall indicate the design pressure range. The casing shall withstand taking account of anticipated normal and abnormal operational conditions. The flexibility of the casing shall allow for thermal expansion and shall cater for changes in section at corners and access door.

During the design phase, the Offshore Supply Contractor shall provide details of the gas seals for pipes and hangers which pass through the boiler casing.

The Offshore Supply Contractor shall provide all necessary galleries and access and inspection doors to allow comprehensive visual monitoring of boiler operation and provide ready access for tube failure repairs in any area of the boiler. In particular adequate properly placed observation ports shall be provided to permit inspection of the furnace and windbox. Operating platforms shall be provided to allow operation of all valves and on load maintenance of boiler cleaning equipment. Walkway access shall also be provided for furnace and dead space areas for planned maintenance during overhauls. Width of walkways, ladders and platforms etc. shall comply with the relevant codes and the Vietnamese safety standards.

3.14 Air Pre-Heater

The air pre-heater system shall be tri-sector type regenerative heater for primary as well as secondary air system.

The air pre-heater system shall be tri-sector regenerative type and used to heat the combustion air, provide hot air for fuel drying, pulverized fuel transportation and maximize the heat recovery from the flue gas.

3.14.1 Air Pre-Heater Operational Requirements

The air pre-heater shall be designed to work safely and reliably in conjunction with the pulverized fuel handling system.

The minimum diluted flue gas exit temperature and combined cold end air/gas temperature of the air heater shall reflect proven practice for the range of fuels and operating conditions, to include the effects of cross leakage such that adverse fouling and corrosion effects on downstream equipment are minimized. The Offshore Supply Contractor shall describe the temperature control system and temperature limitations for all load conditions and range of fuels in design submission.

The cold end heating surface of the air pre-heater is made up of the corrosion resistant steel, and the Onshore Contractor shall install steam coil preheater to heat the secondary air before going to air preheater (APH) so that the outlet gas temperature remains above the acid dew point for coal firing and oil firing.

Fouling of the heat transfer surface shall be minimized. The methods for minimizing fouling and corrosion of the air pre-heater shall be described in design submission. The method of start-up control of the combustion air temperature and the air pre-heater metal temperature shall be given in the operation and maintenance manual. The Offshore Supply Contractor shall provide the frequency and type of any necessary off-load cleaning operations required between major overhauls in maintenance manual.

The air pre-heater shall be fitted with any necessary dust hoppers.

System shall be provided for water washing including all the equipment required for this system including pumps tanks, pipework, discharge equipment, necessary heaters and sludge troughs.

The Offshore Supply Contractor shall provide an emergency means of air heater rotation in the event of a failure to the main drive motor.

The radial, circumferential and axial seal shall be safe and reliable and equipped with reliable automatic adjusting devices for regulating the sealing gaps with electronic sensor elements. The precision of regulation shall ensure the gaps not greater than 0.5mm.

In case of failure of any air pre-heater, the boiler shall be capable of one-sided operation and prevent deformation and leaking of flue gas taken for the shut-down air pre-heater.

The air pre-heater shall equip with illuminated inspection holes, temperature transmitters, temperature elements, fire-alarming and fire-fighting system.

The bearings of the air pre-heater shall be installed for easy inspection and renewal with lubrication and cooling system.

The air pre-heater shall be equipped with alarm devices for stalling.

3.14.2 Air Pre-Heater Performance Requirements

Each air pre-heater shall be driven by a main electric motor. It shall also be equipped with an auxiliary motor for maintenance purpose. Each air heater shall be equipped with a manual lock arm for the maintenance operation.

The air heater shall be installed with reliable and effective soot blowers on the inlet and outlet of the gas side and water flushing equipment on the inlet of the gas side. The flushing water shall be provided by a common once-through fresh service water or treated service water wash water system.

The heating surface at the cold end of the air heater shall be made of corrosion-resistant steel and made into small packages to facilitate maintenance and renewal.

The Offshore Supply Contractor shall prove in design stage that the design has considered fuels, thermal performance, pressure drop, fouling and corrosion.

The fine, efficient, mature sealing system shall be provided in air heater, which shall be of adjustable gap type.

The total air leakage where the leakages of all air (primary & secondary) in to the flue gas is measure as L2 (@ BMCR) where

$$\% L2 = (E'' - E') \times 100 / E'$$

where E' = flue gas flow entering to air pre-heater and
 E'' = flue gas flow leaving air pre-heater.

The Offshore Supply Contractor shall design the APH seals so that L2 shall be < 8.0% after one year operation.

The Offshore Supply Contractor shall provide APH seal leakage control system where seal clearances will automatically adjust during normal operation and minimize the air leakages to gas.

The Offshore Supply Contractor shall demonstrate in design stage that the thermal and hydrodynamic experience have been confirmed by testing in full size plant and the proposed heat transfer surface has provided satisfactory non-fouling performance with the full range of fuels and operating conditions. Operational margins shall be as a minimum allowance for fouling, air pre-heater leakage and manufacturing and erection tolerances.

Due allowance shall be made for fouling of the air pre-heater (both temporary and permanent) in the design of draught plant temperature control systems, cleaning systems, drainage and selection of operating conditions. Consideration shall be given to all mechanical loads arising from dust and moisture accumulation.

The method of achieving good flue gas and air flow distribution shall be described in design submission. The Offshore Supply Contractor shall describe in design submission how the effects of mal-distribution in gas flow to be minimized with regard to downstream equipment.

Regenerative secondary air pre-heater leakage of air to the gas path (from the primary air to secondary air path) shall be minimized. Off-load seal adjustment is not acceptable. The leakage rate in the air heater shall be continuously adjusted by the leakage control system and shall not exceed 2% point increase in leakage from first year over 30,000 hours operation with a proper operation and maintenance of sealing surface. The casing and mechanical seal devices shall be designed such that drift shall not occur due to mechanical wear or interference. The Offshore Supply Contractor shall provide details of and demonstrate previous satisfactory and reliable operation of the proposed leakage control system in the anticipated environment in design submission.

The Offshore Supply Contractor shall provide details of the sealing arrangement and fit up tolerances between the removable heat transfer surfaces and the permanent structure.

The Offshore Supply Contractor shall provide detail of temperature limitations of air pre-heater components and the measures provided to ensure these are not exceeded under all operational circumstances.

Ideally the life of components shall be the same as the overall Facility life. However, it is accepted that some components might have a shorter life due to corrosion, fouling and mechanical failure. The Offshore Supply Contractor shall list those items of plant which cannot achieve the overall plant life without replacement or repair in design submission. The frequency, spare price list and time duration of repair/replacement shall also be stated. The design shall provide for easy replacement.

3.14.3 Air pre-Heater Design and Fabrication Requirements

The Offshore Supply Contractor shall provide outline drawings to clearly illustrate the arrangement of air pre-heaters, the split of air and gas flows, control and damper arrangements in design submission. The positioning and details of the bypass dampers and ductwork shall be described.

The heat transfer surface shall consist of sections of major depth together with sections of smaller depth.

Means shall be provided for the easy replacement of cold ends packs without disturbing other layers.

Load bearing welds shall be full penetration.

The methods adopted shall minimize the stresses and distortions due to thermal effects and self-weight. Dust and water loads and pressure (or suction) shall be described together with the anticipated temperature differences in design manual.

Reference shall be made in design stage to the design method/standards used to design against fatigue failure and noting the load cycling requirements.

The design of the heat transfer surface and materials of construction shall be chosen for the range of anticipated fuels, operating regimes, size of air pre-heater, the materials of construction and the design details and fabrication procedures.

The air pre-heater system shall work automatically with no operator intervention for seal adjustment.

Details of the method of support arrangements shall be given in design stage together with normal and upset loads to the steelwork.

The Offshore Supply Contractor shall make available to the Owner at its facility details of the fabrication, welding procedures, lining and coating procedures, inspection procedures and acceptance criteria adopted in respect of the flue gas air

pre-heaters in design stage. Details shall also be provided of any critical areas requiring regular visual and/or NDE.

The sootblower system shall be designed for all air pre-heater to minimize pressure losses and the loss of sootblowing fluid. Consideration shall be given to housing rotating joints outside the air heater casing.

Provision shall be made for draining water from the air pre-heaters arising from steam leaks and/or cleaning operations.

All on-line maintenance shall be undertaken from permanently fixed ladders and walkways. Handling equipment for air pre-heater element removal and runway beams lifting tackle, permanent access facilities and special tools shall be provided.

The Offshore Services Contractor shall include for proven fire detection and firefighting systems and shall provide details of the fire detection and firefighting systems and their operation in design submission. References to existing installations shall also be provided.

The fire detection and firefighting systems shall be commissioned and fully available prior to first gas to the air pre-heaters.

The systems shall work reliably in the anticipated environment and under normal and adverse cleaning cycles.

The effects of dust build-up shall be taken into consideration when designing the water wash system, firefighting system and structure. Water washing system (with permanent water pipework and pump) and drain collection shall be provided to water wash the air heater effectively.

The Offshore Supply Contractor shall describe the electrical equipment provided within the nominated terminal points for the air pre-heaters, to include the level of standby/redundancy provided, particularly with respect to drive motors and reliability/availability requirements in design submission.

The Offshore Supply Contractor shall describe the C&I system and define the scope in the design stage. The extent of local control and instrumentation provided shall be described to include the following:

- Automatic fire detection
- Pressure Differential Indicating Transmitter (PDIT) to check pressure drop across the heaters
- Gas outlet temperature indication
- Thrust bearing oil pressure check transmitter for sequence control
- Main bearing lubrication pump discharge pressure gauge
- Lubricating oil flow sight glass
- Oil temperature high alarm

In addition to drive motor current indication in the central control room, a rotor or hood stop alarm of high reliability shall be provided in the central control room.

Clean contacts from the fire detection equipment shall operate in conjunction with the station alarm system.

The Offshore Supply Contractor shall include all necessary seal leakage control system including fault monitoring, local displays and controls and all necessary interfaces to the control room and to the automated sequence system.

3.15 Electrostatic Precipitator (ESP)

The electrostatic precipitator shall be designed and capable of being operated such that particulate emission is less than 50 mg/Nm³ when firing Performance Coal of ash content of 14% (ADB) and gross calorific value of 4797 kcal/kg (ARB), with the following out of service:

- one transformer/rectifier unit in each electrostatic precipitator casing; and
- SWFGD.

This shall include allowances for furnace and tube cleaning operations and any effects of fouling. Design gas volume shall be at least 115% of BMCR gas flow based on Performance Coal.

The electrostatic precipitator shall satisfy the particulate emission limit of less than 50 mg/Nm³ when firing Coal A on its own with properties exhibited below when all transformer/rectifier units are in service and SWFGD is out of service:

- Ash Content < 10% ADB
- SiO₂ + Al₂O₃ < 61% DB
- Fe₂O₃ > 4% DB

The electrostatic precipitator shall also satisfy the particulate emission limit of less than 50 mg/Nm³ when firing Coal B on its own with properties exhibited below when all transformer/rectifier units are in service and SWFGD is out of service.

- Ash Content < 19% ADB
- SiO₂ + Al₂O₃ < 92% DB
- Fe₂O₃ > 2.5% DB

Coal B with Ash Content above 19% will be mix fired with Coal A to achieve properties similar to Performance Coal.

3.15.1 ESP Operational Requirements

The particulate extraction system shall operate automatically with limited operator intervention.

The precipitator shall operate in a safe and efficient manner taking account of the complete boiler operating regime, specified fuels and ambient conditions, to include boiler start-up, shut-down and anticipated transient operation. The Offshore Supply Contractor shall provide the operation requirements as per the recommendation / guidelines in the respective operation & maintenance manual.

Where to meet availability requirements, individual flows are taken out of service for routine or breakdown maintenance with the unit on load, these shall be safely isolated. Sufficient mechanical and electrical isolation shall be provided to accommodate this mode of operation. For each boiler, two ESP casings shall be provided. Electrically operated leak-proof isolation dampers shall be provided at the inlet & outlet of each ESP casing. This facilitation shall enable mechanical isolation for either of the casings at reduced boiler load for routine / breakdown maintenance requirements or availability requirements. For electrical isolation Mechanical Safety interlock system, earth switch and manual earthing rods shall be provided

For oil firing during boiler startup, flue gas shall be allowed to pass through ESP without the transformer rectifier (T/R) sets in service. Once minimum boiler load (with coal firing) is achieved with temperature of the flue gas well above acid dew point and all control enables ensured, ESP (T/R sets) shall be put in service. However, if ESP operates in oil firing, complete combustion of the oil (max hydrocarbon < 0.03% in the inlet dust) has to be ensured and the flue gas temperature shall be well above the acid dew point at entry to the ESP.

3.15.2 ESP Design and Fabrication Requirements

The ESP shall be located after the air preheaters and upstream of the induced draught fans.

The ESP plant design shall not include unproven features. The design shall be based on plant operating under similar operating conditions and similar fuels.

The methods used for transport and storage of electrostatic precipitator electrodes shall ensure that permanent distortion does not occur. Details shall be provided of erection methods and tolerances, together with details of quality control procedures in design stage.

Test points shall be fitted with covers so arranged that their removal does not cause damage to the insulation and of sufficient numbers and positions to fulfill test requirements. They shall be completely gas-tight and the covers retained by chain or other approved means.

The Offshore Supply Contractor shall ensure that collected dust can be removed from hoppers at all times including arrangements for emergency emptying, the details of which shall be provided during design stage. The hoppers shall be arranged such that the position of the discharge flange allows headroom for the installation of the removal equipment. The insulators shall be configured with protective sleeve and self-act temperature-control unit to prevent them from particulates and condensation.

The inside of access doors and frames shall be flush with the inside face of the casing and shall have equal insulation characteristics to the rest of the casing. Seals shall be readily replaceable and remain gas tight between major overhauls. Casings shall be of steel plate construction.

Each zone of a precipitator shall have an independent drive for the rapping of both the discharge and collecting electrodes.

High voltage support insulators shall be located out of the gas stream. Where possible the housing shall be subdivided to give one section for each zone. Insulators in contact with flue gas shall be fitted with electrical heaters, the heaters shall be controlled by adjustable thermostats.

The ESP dust collection hopper of all stages shall have a minimum storage capacity of eight (8) hours at BMCR for firing Performance Coal with ash content of 14% (ADB). Hoppers shall have heating arrangements to prevent ash sticking to the sloping sides and down pipes. Level switches to indicate and trip the ESP in case of high ash levels in the ash hoppers which shall jeopardize the safety of ESP otherwise. Safe arrangements shall be made, such as poke rods or agitator chains to aid clearing of the hoppers. Provision for a safe means of hopper emergency emptying shall be made.

A minimum of four zones in series in the gas flow shall be provided. There shall be at least one transformer rectifier set per zone.

The gas treatment velocity shall not exceed 1.3 m/s at the precipitator design gas low condition of 115% BMCR

The collecting electrode area of precipitators shall be the projected and not the developed area.

The gas treatment velocity shall be based on the overall cross sectional area within the effective treatment zone.

The total treatment length of precipitators shall not include the interspaces between zones. The treatment time shall be based upon the above velocity and the total treatment length.

The design shall provide for easy replacement of wearing parts. The Offshore Supply Contractor shall identify those items of plant that cannot achieve the overall plant life without replacement or repair in design submission.

The Offshore Supply Contractor shall identify those components that require full or partial shutdown of the process for replacement and/or routine maintenance such as broken discharge wires. The impact on emissions during removal from service of any plant component shall be identified, together with a description of the methods for both component removal and replacement in design manual.

The detail of methods of component protection (for example, special materials, linings, and sacrificial surfaces) shall be provided in design manual.

3.15.3 ESP Safety Requirements

No access shall be possible into the gas passages unless all the zones in that particular flow are safely isolated and earthed. Key exchange boxes shall be provided as necessary. In addition, the Offshore Supply Contractor shall, in its supply, comply with the following:

- 1) Provision of access to each access door and the equipment within the casings for the fitting of earth clamps shall be provided. Portable earthing cables shall have terminations that conform with international standards.
- 2) Measures shall be included in the design to ensure that access to the body of the casing cannot be gained from inlet or outlet flues.
- 3) Where the roofs of gas chambers are used for the installation of electrical equipment immersed in oil, oil retaining walls shall be incorporated and be so designed that oil cannot leak into the gas chamber.
- 4) A key interlocks shall be provided so that it shall be impossible to enter the screened area on a zone until the electrical equipment in that zone is de-energised and earthed.

3.15.4 ESP Electrical Equipment

ESP transformer rectifier sets shall use high flash point silicone oil as the cooling medium. The Offshore Supply Contractor shall provide documentary evidence including references to demonstrate that the design (including protection systems) of the proposed high voltage power supply has operated safely and reliably under similar operating conditions and all anticipated fault conditions during design stage.

The Offshore Supply Contractor shall provide the schematic diagram to show the high voltage connections and rapping systems, to include the number of precipitators in parallel, the number of zones in series, size and connection details of all high voltage sets and all switching and earthing arrangements in their design manual.

The following details of the design shall be approved by the Owner during design stage:

- 1) Mechanical screening of high voltage areas.
- 2) Electrical screening of equipment.
- 3) Definition of rated output.
- 4) Level of sectionalisation of the plant.
- 5) Environmental conditions.
- 6) High voltage equipment including transformer rectifier sets, insulation, cables and switches.
- 7) Insulator and hopper heating.
- 8) Hopper level controls.
- 9) Electrical and mechanical interlocks.
- 10) Type and routing testing and associated standards.
- 11) Design/fabrication standards.

3.15.5 ESP C&I System

ESP shall be provided with microprocessor based programmable type (rapper & transformer/rectifier) control system and ESP management system to ensure the safe and optimum operation of ESP.

The high voltage power supply for precipitators shall be controlled by a solid state system which continuously adjusts the electrical input in order that the optimum voltage is impressed to the discharge electrodes. In the event of any faults, alarms shall be initiated. Control of individual high voltage supplies to each zone shall be independent of each other and based on solid state technology. A micro-processor based system allowing ready and reliable changing of parameters such as intermittent energisation and power down/off rapping shall be incorporated to avoid rapping two zones simultaneously. An integrated transformer/rectifier and rapper control system for ESP shall be provided. The controllers shall be mounted on the T/R control cabinets which would be housed inside a separate electrical building. ESP shall be controlled locally through ESP controllers stated above with overall operation & monitoring from ESP control room and CCR.

The ESP shall be controlled from the CCR. Control and instrumentation systems shall be provided in order to enable the plant operation to be carried out in a safe, effective, efficient and reliable manner. This shall be achieved with a minimum number of operation staff and with minimum reliance on maintenance staff and plant attendants. The design of the control and instrumentation systems and equipment shall include all necessary provision in order to meet the reliability, availability and maintainability requirements, by use of redundancy, diversity and appropriate distribution of control and instrumentation functions.

Remote manual control shall be provided to allow the safe operation of the ESP in the event of failure of the automatic system. The Offshore Supply Contractor shall provide an instrumentation list, identifying each item by type and extent of supply during design stage.

3.16 Draught Plant

The draught plant comprises ductwork dampers, air and gas fans associated with the boiler plant. In addition to the operational requirements of the power island the draught plant has its own particular requirements with regard to maximum number of start-up/shut-down cycles, flexibility (including turndown) and emergency shut-down conditions.

The margin between the fan operating point and the onset of instability (such as stall) shall not be less than 20% of the design flow.

The fans shall operate in a completely aerodynamically stable manner over their entire operating range when operated alone or in parallel with another fan. The operating range includes transient states such as boiler load changes, start-up and shut-down.

Emergency shut-down of the fans, initiated by either the operator or protection system, shall be entirely automatic. A manual method of bringing a decelerating fan to rest and preventing fan impeller rotation during maintenance shall also be provided.

The fans shall be controllable at all loads without significant hysteresis (i.e. hunting due to poor mechanical design or construction) or the need for operator intervention.

The fan design capacity (flow and pressure) shall be selected in accordance with the overall boiler performance requirements taking into account the margins inherent in fan manufacture and margins necessary for long term operation purposes such as increased in-leakage boiler and air pre-heater surface fouling and turbine heat rate deterioration. At least 21% margin on head corresponding to nominal fan capacity shall be provided. These margins are above the 100% BMCR requirements using the coal of lowest CV, which shall include full allowance for air heater air leakage and assume the maximum ambient temperature.

Fan impellers shall have sufficient fatigue strength to prevent failure from repeated application of centrifugal loading, aerodynamic loading, acoustically induced load and surging due to interaction of fans throughout design operating life.

Fan impellers shall not impose aerodynamic pulsations on the gas flow which could induce fatigue failure of adjacent ducts.

The Offshore Supply Contractor shall ensure that suction fans do not create a significant risk of implosion of the boiler under any plant fault conditions.

The Offshore Supply Contractor shall ensure that significant erosion does not occur on fan components.

All air intake ductwork shall have safety grillage suitable to prevent damage or blockage of fans and downstream components.

Automatic drainage provision shall be made for all flues and ducts where significant quantities of water are liable to collect including conditions such as air pre-heater firefighting or boiler tube leakage.

All dampers shall be operable and have position indicators both locally and in control room. A manual method of operation, such as a hand wheel, shall be provided on all dampers.

3.17 On Line Boiler Tube Leakage Detection System

With the objective of boiler tube protection, a set of on-line boiler tube leakage detection system (BTLD) for two units shall be provided by the Offshore Supply Contractor. BTLD shall be equipped with associated hardware and software. The functions of monitoring and alarming functions are implemented in CCR.

BTLD shall comply with the following requirement:

- Monitor the tube leakage of water wall, super-heater, re-heater, economizer, and internal surface at the boiler top.
- Discover steam/water leakage from the pressure heating surface.
- Locate the leakage and give early warning.
- Display the leakage with graphics, photo and alarm
- Follow up on the event development.
- Help monitor the operation status of the soot blower.

BTLD shall include at least following three parts:

- Signal collection system.
- Signal treatment system.
- Display and alarm system.

The BTLD shall have enough measurement points and reasonable arrangement on the boiler proper to ensure effective monitoring to the boiler heating surface. About 40 measurement points are required as a minimum for each of the boiler and necessary spare gate shall be reserved. Sound wave sensor and soot blower shall be high temperature resistant, anticorrosive and able to insulate magnetic and noise interference from outside.

The BTLD shall have redundant interface for standard network and communication agreement to communicate with DCS to ensure that data can be transmitted to DCS correctly, reliably and in time. DCS shall have a function to display the tube leakage and alarming information for the corresponding boiler.

Chapter 4 – Steam Turbine and Auxiliaries

4.1 General

The steam turbines shall be of proven design, directly coupled to a 50 Hz synchronous generator.

Design criteria shall be applied to ensure a long creep life, cyclic duty and fast start-up capability. Last stage blade materials and appropriate exhaust steam moisture for design shall be selected to minimize blade erosion. The last low pressure stage blade design shall be based on a proven arrangement with adequate number of operating units in service.

The steam turbine generators shall be located within an enclosed common turbine building with indoor access to the central control room. Two overhead cranes located within turbine building shall be capable of transporting the heaviest piece of equipment (except generator stator) for maintenance in turbine hall.

The steam turbine shall be sized to pass the entire quantity of steam generated by the boiler. The steam turbine exhaust and condenser configuration shall be to the manufacturer's standard design.

The steam turbine shall be supplied with steam for generation of power and for regenerative feed heating through uncontrolled extraction points. All the auxiliaries for the turbine-generator shall be provided.

The high pressure (HP) turbine bypass shall be capable to pass through the steam of 70% boiler maximum continuous rating (BMCR) flow rate and the low pressure (LP) turbine bypass shall be capable to pass through the steam of 40% BMCR flow rate to ensure the capability of a full load rejection to house loads without tripping of the Unit. The condenser shall be sized to receive the LP bypass steam flow of 40% BMCR as well as the normal steam flow from LP turbine.

Pneumatic non-return valves on the cold reheat pipeline shall be provided to safeguard against the back-flow of steam into HP turbine while HP bypass is in operation. Both the HP and LP bypass control valves shall be designed to take the full temperature difference of the inlet and outlet under all conditions.

Cold reheat steam shall be supplied to auxiliary steam header through auxiliary steam pressure reducing & desuperheating station.

The proposed steam turbine and auxiliaries shall be of proven, reliable design with high availability and efficiency; combined or separated HP and intermediate pressure (IP) cylinder and separate double-flow LP cylinders; condensing extracting and tandem compound for an operating speed of 3000 rpm. Proven electronic governor control shall be provided for start up / shutdown, normal running and to prevent operation of the over-speed trip in the event of a unit trip. Turbine supervisory and

monitor systems shall be provided to allow close monitoring of the turbine conditions in central control room. Protection and interlock system shall be provided to safeguard the equipment during abnormal operating conditions.

The steam turbine shall be designed and tested in accordance with the latest applicable requirements of the relevant international codes and standards. The selection of steam turbine shall be based on a robust design suitable for the specified operating regime including frequent two shifting operations.

Each Unit shall provide the required guaranteed output at the HV side of the generator transformer (after mechanical and electrical losses) while operating with eight stages of feedwater heating.

Control regimes of the steam turbine such as normal starting, loading, load pick-up and shut-down shall be explained with details in the design manual.

Full details of the materials proposed for the casing, rotors, blades, diaphragm and connecting pipework shall be supplied to the Owner for review, at detailed engineering stage, including composition and specified mechanical properties. Details of heat treatment procedures, fabrication and repair welding procedures, inspection procedures and defect acceptance standards shall be made available for inspection by the Owner at the manufacturing works.

Details of all casing joint bolts including materials, nominal sizes and initial tightening strain shall be given. Bolts and studs strain controlled tightening method shall be described. The necessary tightening equipment shall be supplied by the Offshore Supply Contractor. Details for pipe connection to the turbine to accommodate thermal expansion shall also be provided for the Owner's review at detailed engineering stage.

The Offshore Supply Contractor shall provide details of the arrangement of supports and fixture to assure the alignment of the power train and allows for the thermal expansion to the Owner for review at detailed engineering stage.

Atmospheric relief diaphragms shall be provided in the turbine exhaust hood to prevent positive pressurization. Motor operated vacuum breaker valves shall be provided at the condenser.

If required by the turbine design, a turbine exhaust water spray system shall be provided for protection against high exhaust hood temperature when operating transiently or continuously at low load. Condensate from the extraction pump discharge shall be used as spray water.

The steam turbine shall be protected from the effects of water carryover under all normal and abnormal conditions. If special protection equipment is provided, the Offshore Supply Contractor shall provide for the operation of the device(s), the set points and the method of resetting the system for normal operation.

The insulation shall be designed for ease of removal and replacement, using segmental units insofar as possible.

Cladding shall be furnished for the turbine generator unit as aesthetic purpose.

Turbine shall be provided with electro-hydraulic governing system. Turbine governing system shall be capable of controlling with stability the turbine speed between zero to maximum power output when the Unit is operating isolated or in parallel with other units.

4.2 Operating Requirements

The turbine shall be designed to operate under the maximum sustained steam pressure of over 5% of the design turbine inlet steam pressure. The maximum sustained steam pressure and maximum momentary pressure rise that can be tolerated and the aggregate duration of such swings beyond maximum sustained steam pressure shall be indicated by the Offshore Supply Contractor.

The Offshore Supply Contractor shall indicate the maximum inlet main steam temperature and maximum reheat steam temperature that can be permitted together with the duration of each such incidence and the total duration thereof. This shall satisfy the relevant stipulations of IEC-60045 or equivalent standard.

The maximum continuous rating of the turbine with valves wide open (VWO) shall not be less than the BMCR. The Offshore Supply Contractor shall also indicate the maximum flow passing capability of the turbine corresponding to rated steam parameters and at rated condenser vacuum. The unit shall be capable of operating with all the HP heaters cut-out and no extraction steam supply to them. The Offshore Supply Contractor shall indicate the maximum output of the Unit under such conditions.

Steam turbine overspeed protection shall operate at 110% of nominal speed.

The steam turbine and auxiliaries shall be able to withstand, without damage, any transient thermal stresses, acceleration and deceleration to which they shall be subjected within limitations set by proper function of the supervisory and protective devices and within the limitations set forth in the Operation and Maintenance Manuals.

Steam turbine and generator shall be statically and dynamically balanced at factory. Furthermore, the turbine-generator assembly shall be tested on Site by the Onshore Contractor to ensure that it is dynamically balanced. The Offshore Supply Contractor shall identify the maximum vibration level that can be tolerated on the bearing housings and shaft. On line vibration monitoring for all bearings shall be provided.

Operational capabilities of the steam turbine shall be as follows:

4.2.1 Normal Operation

The design of steam turbine shall be suitable for base load operation as well as for cyclic duty and frequent start-up requirements with minimum start-up and shut down time, and capable of achieving the specified service life of the Facility. The steam turbine shall be designed to withstand without signs of strain or leakage for any load up to VWO condition. The steam turbine shall operate smoothly under all operating conditions and shall not exhibit characteristics at any speed or load that restricts the operation of the Facility. Vibratory stresses shall be low and no component may have a resonant frequency within the operating range. Turbine shall also be capable of operating continuously with VWO to swallow 103% of turbine maximum continuous rating (TMCR) steam flow to the turbine at rated main steam and reheat steam parameters.

The steam turbine shall be designed for the maximum removal of water particularly during start-up. The turbine internals shall be self-draining and the design shall include passageways and inter-stage drainage where appropriate. Special attention shall be given to avoid erosion damage of LP blading and in this respect all necessary cylinder or diaphragm water grooves shall be included.

All operating controls, gauges and thermometers mounted on the steam turbine shall be readily accessible, easily operated and observed.

The location and arrangement of all equipment and piping shall minimize the potential for oil fires as a result of leakage onto hot areas of the turbine or associated equipment and piping. Proximity of the oil system to high temperature steam pipes shall be avoided to minimize the fire hazard. Guards and drip trays shall be provided where necessary to prevent contact between oil escapes and hot piping and other hot parts of the machine.

The steam turbine shall be capable of operating under sliding pressure control between 40% and 100 % load and fixed pressure control for below 40% load.

Steam turbine shall be capable of accepting variation in steam temperature and pressure as per IEC 60045. The Offshore Supply Contractor shall indicate the allowable turbine over pressurization (under VWO) and ensure corresponding continuous output of turbine generator set.

The steam turbine control system shall provide the required rapid reaction to generator load loss transients so that the stability and synchronization of the Unit can be maintained without excessive level of stress.

The steam turbine shall be sized to operate up to VWO output for the complete range of stipulated ambient conditions.

In the case of all high pressure feed water heaters being bypassed, the Unit shall be able to maintain the 93% TMCR load operation.

4.2.2 Start Up and Shutdown

The Units shall be capable of being started, warmed, run to speed and fully loaded in a short possible time. The time to achieve full output from a cold, warm and hot initial condition shall be stated in the design submission in the form of schedules and curves.

4.2.3 Plant Interlock and Operation in Abnormal Conditions

(1) Plant interlock

Plant interlock shall be designed and provided with all required protection devices, logic circuits, relays, instruments and other pertinent equipment.

(2) Load rejection

Under certain emergency circumstances, the Unit has to be disconnected from the grid and the output of the steam turbine generator of the Unit has to be reduced suddenly from approximately base load to near house load. Re-establishment of full load shall then be required as soon as possible after the cause of the load rejection has been cleared. Accordingly the Unit shall be capable of sustaining such a severe load rejection. Under such circumstances the plant control devices shall keep the Units in a controlled condition and free from such trouble as thermal stress, differential expansion, vibration and other undesirable effects.

Any limitations on rapid re-loading after such a load excursion shall be stated in the design submission.

(3) Operation in abnormal condition

In case of blackout, the Unit shall be able to shutdown safely; every control system shall be designed to be "Fail Safe".

The control air receiver capacity, the control electric power system and battery capacity shall be safely and sufficiently able to shutdown the Plant in case of an emergency.

4.2.4 Operation on Failure of AC Supplies

Following a severe system disturbance resulting in tripping of the Facility from up to full load with the loss of all AC supplies, the Unit shall be capable of being placed onto turning gear operation if required. During this period a number of essential services shall be maintained by DC supply and emergency diesel generator.

All unit auxiliary supplies shall be arranged that individual turbine generators are capable of supporting their own auxiliaries and continuing to run even in the event of major electrical failure which results in the total or partial loss of MV supply.

4.3 Design Criteria and Materials

The major parameters of steam turbine shall be as (or better than) follows:

- a) Main steam pressure before stop valve: 27.5 MPa(a)
- b) Main steam temperature before stop valve: 600 °C.
- c) Reheat steam temperature at IP turbine inlet: 610 °C.
- d) Dryness fraction of turbine exhaust: Not less than 90% at TMCR at cooling water temperature of 29.3 °C.
- e) Cooling water inlet temperature: 29.3 °C for guarantee, 33 °C for sizing of equipment.
- f) Cooling water temperature rise across condenser : < 6 °C
- g) DM water make - up to heat cycle: 1 % of steam flow at BMCR.

The Facility shall conform to the following design basis as a minimum.

Equipment / System		Design Basis
Type	Installation	Indoor
Cycle	Non-reheat/Reheat	Reheat
Net electrical output under design conditions (MW)		600
Operating modes		Base load and two shift operation; constant pressure and sliding pressure;
Extractions		Uncontrolled with force-closed NRVs
Governing		4% - 6% droop
Exhaust direction		Down Flow
Turbine exhaust pressure (bar (a))		0.067
Condenser Type		Surface type, divided shell, single pass
Cooling water temperature	Condenser inlet (°C)	29.3 (max 33)
	Condenser discharge (°C)	Inlet temperature + 6
Cooling water flow rate (t/h)		Offshore Services Contractor to indicate
Condenser cooling surface (m ²)		Maintain rated vacuum at VWO flow plus 5% plugging margin
Condenser shell		Carbon steel with epoxy coating
Condenser tube sheet material		Titanium clad carbon steel
Condenser tube material		Titanium

Equipment / System	Design Basis
Condensate extraction pumps	- Vertical can centrifugal type; 3 x 50% pumps; - Nominal capacity will be calculated for condensate flow corresponds to VWO load plus 1% make-up plus spray water for auxiliaries or any other system plus valve gland sealing water. Two pumps operate at BMCR case, 10% flow margin and 21% margin on calculated pressure losses in the system from condensate extraction pump discharge through LP heaters to the deaerator.
Boiler feed pumps	- Variable speed multi-stage centrifugal barrel type pumps 3 x 50% motor driven pump with variable speed drive - Nominal capacity will be calculated for feed water flow corresponds to VWO load plus 1% make-up plus spray water for superheater, reheater and auxiliaries. - Discharge pressure calculated based on highest set safety valve pressure plus static head plus system losses through HP heaters, economiser, water walls and superheater 10% flow margin and 21% margin on calculated pressure losses in the piping. - Pump sizing to be finalized during detailed design.
Feed water heaters	LP heaters – tube and shell horizontal type HP heaters – tube and shell horizontal / vertical type
HP / LP bypass system	HP bypass: 70% BMCR capacity LP bypass: 40 % BMCR capacity
Condensate polishing plant	2 x 50% capacity with provision for bypass

The materials of all components of the turbine shall be suitable for the steam conditions and the load cycling nature of the steam turbine. As a minimum, the material used for various critical parts shall satisfy or better than the following requirements for materials:

Main steam and hot reheat steam piping	P92
Cold reheat steam piping	P22, or P11 or A691 or carbon steel

	material
Main steam valve casing	10CrMoWVNbBCo or as per OEM's standard
HP inner shell	DIN GX12CrMoWVNbN or as per OEM's standard
HP rotating blading – inlet stages	11CrMoVNbNW or as per OEM's standard

4.4 Casing

The arrangement of the turbine casings together with their supports and location details shall be described and basic details of the proposed casing materials shall be provided at detailed engineering stage. Cast iron shall not be used for any casing or part of a casing, which may be subjected to steam at a temperature in excess of 320 °C unless fully substantiated by the Offshore Supply Contractor. Full details of the casing materials shall be provided as part of design submission, including composition and specified mechanical properties. In addition, the proposed casing inspection, defect acceptance and weld repair procedures shall also be specified.

The design and construction of the turbine casings shall be such, that growth or distortion of any part, under the conditions to be met in practice, shall not affect the efficiency and reliability of the Plant. All portions of the casing and steam chests shall be in materials carefully chosen to withstand the steam conditions to which they may be subjected.

Casings shall be thermally treated to minimize the possibility of distortion occurring in service.

Multiple inlet and exhaust branches shall as far as possible be symmetrically arranged.

The HP and IP turbine sections shall be of combined or separated HP/IP casing design. The casings shall be supported at the centerline. The outer casing may be either cast steel or steel fabrication. The LP casing shall be divided on the horizontal centerline.

For maintenance and inspection purposes, suitable lifting gear and slings shall be provided for raising and lowering the turbine casings and rotors clear of the blades or remainder of the machines. Guide rods for preventing damage to the blades shall be provided.

Jackscrews, lifting lugs, eyebolts, positioning screws, guide rods and/or other necessary tools shall be provided to facilitate alignment, disassembly and reassembly. When jack screws are used as a means of parting contact on faces, the mating face shall be counter-bored to receive the jack screw in order to prevent subsequent improper fit caused by marking.

In no circumstances shall the casings be subjected to high temperatures without the protection of adequate thermal insulation.

The LP cylinder shall be provided with relief diaphragms which shall be arranged to discharge vertically into the turbine hall and a cooling spray water system initiated by exhaust hood high temperature. Spray water shall be taken from condensate extraction pump discharge to control excessive exhaust heating for low load operating and high back pressure situations.

The LP cylinders shall be designed to minimize the effects of moisture erosion and corrosion.

Casings shall be designed to have openings with covers for ease of insertion of weights for turbine rotor balancing.

The LP cylinder outer casing shall incorporate access doors which enable sighting of the last rows of blades from outside the machine. These access doors shall also allow suitable personnel access. Provision shall be made inside and outside the casing for safe access by use of rungs and handrails. Access to the condenser steam space shall be provided with manholes with hinged covers, one on the exhaust neck and at least one on the hotwell, so it shall not through the LP cylinder outer casing access doors.

The outermost casings of the HP/IP turbine cylinders which operate with internal steam pressures above ambient pressure shall be hydraulically tested in-works, the gauge test pressure exceeding the maximum working pressure difference by 50%.

Instrumentation shall be provided for continuous monitoring of thermal stresses in the HP/IP casing and differential expansion between rotor and casing. Redundant (duplex type) thermocouples shall be provided at recommended points on the cylinder casings to indicate metal and steam temperatures and to protect steam turbine. Control of transient thermal stresses in the HP/IP rotor shall be provided to ensure they are well within design limits.

All the casing joints shall be designed to remain steam-tight for a minimum operation period of 45,000 hours without the need for bolt re-tightening or gasket replacement. The associated high temperature fasteners shall be adequately wasted and controlled tightened during assembly. The Offshore Supply Contractor shall provide details regarding the proposed bolting material and methods of tightening the bolts and studs at detailed engineering stage. The Offshore Supply Contractor shall provide all necessary tightening equipment.

4.5 Rotors and Couplings

The turbine rotors shall be of forged construction and designed to avoid high geometric stress concentration and self weight bending stresses. The use of shrunk on bladed discs is not acceptable.

The construction of the rotor shall be described and submitted as part of design

submission with reference to a sectional arrangement drawing. Details shall also be provided as part of design submission of the proposed rotor materials and their minimum mechanical properties including fracture toughness.

Rotors shall be designed to withstand 20% over-speed. Completed rotors shall be dynamically balanced at high speed and tested to an over-speed of 10% for 5 minutes at factory. The Offshore Supply Contractor shall state whether any in situ testing or balancing of the rotors at Site is necessary at detailed engineering stage.

The rotor shall be designed so that all combined rotor critical speeds on the turbine foundations are adequately clear of running speed to ensure satisfactory operation of the Unit. Adequate provision and access shall also be provided for in situ Site balancing of the rotors without requiring major dismantling. The rotors shall be designed such that combined rotor critical speeds are well apart from the normal running speed to ensure satisfactory operation of the Unit. The rotor shall also be capable of safely withstanding the worst case short circuit of the generator at 100% voltage and current.

The rotors of two Units shall be same design.

The procedures proposed for inspection of the rotors in order to ensure their soundness and homogeneity shall be given during design stage, together with particulars of the thermal treatment proposed in order to minimize the possibility of distortion occurring in service.

The turbine rotor complete with blades shall be subject to a no-load overspeed testing and dynamic balancing in the factory test bench.

Couplings shall be designed to allow for removal of any one rotor without opening adjacent turbine casings. Couplings with radially fitted bolts are acceptable. Special tools for bolt tensioning shall be provided.

Grooves for locating balance weights to allow on site balancing of the rotors at a later date shall be incorporated into the rotor design.

4.6 Vibration

The blading shall be constructed to avoid the possibility of damage from vibration when the Unit is running continuously at any speed between 6% below and 3% above normal running speed.

The Offshore Supply Contractor shall identify any proposed moving blades which are not identical to that proven in service elsewhere. In such cases, the Offshore Supply Contractor shall provide estimates of the blade vibration frequencies and modes, together with the proceeding nozzle rows impulse frequency as part of design submission. The estimated frequencies shall take into account both temperature and rotational stiffening effects.

4.7 Blading

Moving blades constructed with integral roots and tip shrouding shall be provided. Riveted on shrouding and blading requiring through holes for damping lacing wires are not acceptable. The type of blade root fixtures, for both the fixed and moving blades shall be described in design stage and submitted as part of design submission. Where diaphragm fixed blade construction is employed, the method of construction and diaphragm support arrangement shall also be described as part of design submission.

The Offshore Supply Contractor shall provide a statement at detailed engineering stage on the minimum expected life of all blading if less than the design life of the turbo-generator unit, defining the criteria upon which the limitations are based.

The moving blades shall be either forged or machined from solid material. The stationary blades may if necessary be cast, provided every precaution is taken to ensure high quality castings are obtained. All blades and nozzles shall be of proven design. Fixed blades and associated nozzle chambers, diaphragms and carrier rings shall be designed for minimum deflection under maximum steam load conditions commensurate with the clearances in the turbine.

The blades material shall have good creep and fatigue strength properties with small elongation at fracture and resistance to cracking, scaling and erosion at the operating conditions. The blades shall be free from sudden changes of section liable to cause stress concentrations.

The blades shall be designed to minimize blade end leakage. Full details of the blades design shall be submitted in design stage, including details such as blade root fixing and shrouding attachment for each blade row. Details of fixed blades shall also be provided including supporting, alignment and inter-stage sealing arrangement.

The blading shall be of a design which has operated successfully in commercial service for its design life. The natural frequency of all the rows of blades shall be such as to avoid resonant vibration at or near the normal operating speeds. A Campbell diagram shall be provided to indicate the blade vibration characteristics as part of design submission for last stage of LP. Blading requiring through holes for damping or lacing wires are not allowed.

The Offshore Supply Contractor shall demonstrate successful service elsewhere of the latter LP turbine blading stages, indicating where and which units with identical blading have been commissioned. The Offshore Supply Contractor shall demonstrate that all blade and wheel mode frequencies of the last three stages are adequately removed from running speed.

The Offshore Supply Contractor shall state the proposed method of protecting the leading edge of the LP moving blades from erosion in the wet region in design stage. If erosion shields are used, details of their location, material specifications including heat treatment, method of attachment to the blading and details of the inspection requirements with acceptance standards shall be stated in design submission. If flame, induction or laser hardened leading edges are being used, the means by which residual stresses after hardening are monitored and relieved shall be explained in design stage.

4.8 Glands and Gland Sealing System

4.8.1 Shaft Glands

Each cylinder shall be provided with efficient shaft glands of the steam sealed multi-stage labyrinth type.

Glands and inter-stage seals shall be of the spring-back type. The glands shall be capable of maintaining full condenser vacuum with the turbine on turning gear and during start up and shut down in addition to normal operation and shall be self regulating under all conditions. Alternatively, if the amount of steam required for LP gland steam is less than the steam supplied from HP/IP gland, backup steam shall be supplied automatically from cold reheat line and/or auxiliary steam line.

4.8.2 Gland Sealing Steam System

A complete gland sealing steam system shall be provided to interconnect the shaft glands with each other and with appropriate sources of sealing steam, or with appropriate points for the discharge of excess steam. Gland steam condenser shall be provided. The gland steam system shall normally be fully automatic in operation but shall also include manual control.

All necessary steam pipes, vents, desuperheaters and valves shall be included. All isolating valves required to be operated when placing the gland steam system into

service (after shut-down of up to 60 hours duration) and when taking it out of service shall be motorized and arranged to be suitable for remote control.

Where the gland steam supply pressure is dependent on the main steam pressure, the gland steam system shall be designed to operate satisfactorily (on automatic control) with the main steam pressure at the turbine HP stop valves at any value between the auxiliary steam pressure and the maximum pressure which can be attained when the boiler safety valves lift. The gland steam system shall be capable of controlling the wide fluctuations in boiler steam pressure which can occur during start-up and shut-down.

Adequate drainage of the gland steam system under all conditions shall be provided.

Where gland steam temperature control is achieved by use of direct contact spray type desuperheaters, the design shall include features to prevent water reaching the shaft glands in the event of malfunction of the desuperheater spray water control system.

The full description of the gland sealing arrangements shall be given in design stage, together with a diagrammatic arrangement of the system.

4.8.3 Gland Steam Exhaust System

The outermost pockets of all shaft glands shall be connected to an exhaust system maintained at a slight vacuum and arranged to prevent steam leakage to atmosphere. Steam from the exhaust system shall be condensed in a single full-duty gland steam condenser. 2 × 100% duty exhaust fans, ac motor driven, shall be provided to maintain this vacuum and to discharge incondensable gases to atmosphere.

The gland steam condenser tubes shall be of type 304 stainless steel.

The system shall be arranged to operate satisfactorily with the gland steam condenser out of service when necessary for maintenance purposes. Under these circumstances the two exhaust fans may be arranged to operate together.

Atmospheric vent pipes including a silencer from the exhaust fans shall be provided and shall be arranged to discharge outside the turbine house, in a location subject to the agreement of the Owner.

Appropriate arrangements shall be included to return the condensate drainage from the gland steam condenser to the system via the main condenser.

4.8.4 Gland Sealing Control and Instrumentation

A comprehensive control and instrumentation scheme shall be developed, taking into account of the various requirements of the OTR.

Duplex thermocouples shall be fitted to monitor the temperature of the inlet steam to each shaft gland. The thermocouples shall be arranged for remote monitoring as with turbine casing thermocouples.

The desuperheating of sealing steam for the LP glands, and elsewhere as appropriate, shall be fully automatic under all operating conditions. The gland steam temperature downstream of any desuperheater shall be monitored locally and remotely and shall also include abnormal temperature alarm features. Alternatively, desuperheating by routing sealing steam pipe through the condenser as per OEM's standard can be applied. Control and monitoring of LP gland steam temperature downstream of this pipe is not required as appropriate temperature will be achieved by design.

Local indication of pressure in the gland sealing steam supply and exhaust manifold shall be provided, appropriately located in the vicinity of the pressure reducing control valves to facilitate local manual control when this is necessary. Alarms when these pressures are abnormal shall be included.

The measurement of the steam leakage flows associated with each shaft gland shall be provided.

4.9 Emergency Stop Valves, Governing Valves & Steam Chests

4.9.1 General

Complete system of steam chests valving including emergency stop and control valves shall be provided to the HP/IP steam turbine main steam inlets to protect the turbine from over-speeding.

Hydraulic control fluid system including pumps, tanks, filter, heater, coolers, vacuum dehydration unit, auxiliary treatment system piping, accumulator and servo valves shall be supplied for driving the governor as well as the valves in the HP/IP steam chests. Necessity of heater and vacuum dehydration unit will be reviewed during execution stage.

Drains shall be provided at all points of the Unit where water may collect during the period of warming up or during normal operation. Drains from points downstream of the turbine stop valves shall discharge into drains vessels with desuperheating facility which shall be vented and discharged to the condenser or elsewhere.

4.9.2 Arrangement of Valves

Emergency stop valve operating in series with the governing valves shall be provided on each steam pipe for controlling the inlet of main steam to the HP turbine. In addition, reheat emergency stop valves with reheat governing valves shall be provided on each steam pipe for controlling the inlet of reheat steam to the IP turbine.

Governing valves shall be provided at the inlet of the HP cylinder to fulfill all requirements of the governing system. In addition, sufficient reheat governing valves shall be provided at the inlet of the IP cylinder to fulfill all requirements involving modulating control of reheat steam flow.

All valves shall be designed to be tripped by the operation of the emergency governor, hand trip lever, solenoid trip, the low vacuum trip, etc. The stop valves shall incorporate bypass valves for start up purposes and shall be located as close as possible to the turbine inlets to minimize the effects of potential over-speed caused by entrained steam during load rejections. Facilities shall be provided for on-load testing of all the valves, over-speed and all other protection devices.

Duplex thermocouples shall be provided at recommended points on the valve bodies (steam chests) to indicate metal and steam temperatures and to enable thermal stress levels to be evaluated.

Satisfactory access for operational and maintenance purposes by means of permanent platforms and walkways shall be provided to all valves. If any valves are not accessible for maintenance purposes from the turbine house crane, additional lifting and dismantling gear shall be provided such that comparable maintenance facilities are available.

The steam connections and any necessary piping between the main steam and reheat emergency stop valves and the turbine cylinders shall be provided. These connections and piping shall be arranged so as to avoid distortion and to avoid undue expansion thrusts being imparted to the turbine casings. The steam pipes and connections shall be fully welded, except where it is necessary to provide dismantling joints and shall be of proven design, details of which shall be submitted in design stage. The design of the piping and valve chests shall be such as to prevent undue noise being produced by the flow of steam through them at any load.

A power operated drain valve installation (or vent if appropriate) shall be provided (as part of the turbine drains system) arranged to drain and/or allow the flow of warming steam from each emergency stop valve chest just upstream of the valve seat. This shall have an adequate steam flow capacity to facilitate warming. Drain or vent facilities shall be arranged to be connected as appropriate just upstream of each group of governing valves.

All main steam and reheat valve bodies or chests shall be thermally insulated so that their cooling rate after shut-down is closely similar to that of the associated turbine casings.

All main steam and reheat steam pipes shall be cleaned in accordance with the steam blow procedure. For this, process steam will be blown via the respective emergency stop valve chests and temporary piping to atmosphere. All necessary temporary pipe work and steam silencing equipment shall be prepared.

4.9.3 General Design Requirements

In the design of steam chest valves and valve bodies, particular attention shall be paid to the surface finish and to minimizing the variation in wall thickness in the interest of thermal symmetry. The Offshore Supply Contractor shall describe what special provisions have been made in the design of the steam chests and valve bodies to reduce the possibility of cracking during the normal life of the machine.

The turbine steam valves shall be designed to allow full-arc admission.

Stop and governor valve stems shall be designed to prevent sticking in the bushing and to resist deposit build-up.

The emergency stop and reheat emergency stop valves shall be capable of withstanding cold hydrostatic pressure of 1.5 times the maximum continuous steam pressure.

The valves, spindles, bushes and glands shall be of compatible materials entirely suitable for the operating conditions. Valve seats and valve plugs or heads shall be hardened or provided with hard alloy areas to minimize erosion. Spindles, guide bushes and glands shall operate for prolonged periods of operation without pick-up, seizure or permanent change of dimensions.

If the valves provided are subject to valve spindle steam leakage with the turbine out of service, an estimate of the leakage steam under full boiler pressure conditions shall be given. The Offshore Supply Contractor shall also give the location of the leakage and drain destinations of each spindle leak-off. All steam leakage from the valve spindles shall be recovered. The spindle steam leakage from the main and reheat steam stop and governing valves shall be capable of being isolated as required from the main turbine gland steam sealing system and directed to the main condenser, when the starting conditions so dictate.

The servomotors of all turbine (main steam and reheat steam), emergency stop and governing valves shall be operated hydraulically by oil supplied from the medium pressure control oil system. All these valves shall be arranged to be held open by the hydraulic pressure and to be closed (against full steam pressure) by spring action.

An electro-hydraulic converter shall be provided for each servomotor as appropriate to operate in conjunction with the electro-hydraulic turbine governing system.

All necessary equipment and design features shall be provided to ensure the rapid closure of all valves in the event of a turbine trip.

Provision shall be made for remote operation (tripping and re-setting) of the emergency stop, reheat emergency stop, governing and reheat governing valves. These valves shall all close automatically on loss of control oil pressure. Adequate provisions shall also be included to facilitate testing of each valve and serve for satisfactory operation with the turbine off-load.

The Offshore Supply Contractor shall describe fully the arrangement and methods of operation proposed for the reheat emergency stop and governing valves and the protection provided to prevent the turbine speed reaching a dangerous value in the event of the normal governing equipment failing to function on loss of load.

Hydraulic power and control equipment for steam valves shall have a positive means of lubrication for all sliding and rotating bearing mechanisms. Where these mechanisms are exposed to atmospheric conditions they shall be adequately protected by dust covers or seals to prevent dust build-up in bearings and on journals, plungers or similar small clearance parts. Suitable venting shall be provided for all servomotor enclosures to ensure water vapor is removed to prevent corrosion of enclosed elements.

Oil pipe connections to all servomotors shall be arranged to absorb all vibration, shock loading and thermal expansion movements, so as to avoid any risk of fatigue cracks or oversteering at the connections.

Jacking screws or other similar provisions shall be included to aid the dismantling of the valves, where components could become bonded together after a period of time, e.g. flange faces.

Reheat governing valves shall be designed to be compatible with the turbine bypass systems. They shall be arranged to operate satisfactorily on hot reheat steam governing duty for extended periods at low loads with the turbine bypass systems in operation.

4.9.4 Steam Strainers

The emergency stop and reheat emergency stop valves shall be provided with permanent rugged steam strainers to protect the turbine against foreign materials.

The strainer shall be arranged so as to permit reasonably easy inspection and cleaning.

4.10 Turbine Bearings

The bearings of the turbines shall be of horizontally divided type, readily accessible, adjustable and replaceable without damage to the adjacent glands. Provision shall be made for ease of cylinder re-alignment. Tilting or PAD or ellipse type bearing shall be provided. All bearings shall be designed to compensate for limited misalignment to withstand loads transmitted in both normal and unusual operating conditions. All bearings shall be split to permit removal of both halves for inspection. The bearing supports shall be so designed that shims can be fitted to adjust the alignment.

Turbine bearings shall be mounted separately from the casings to minimize rotor misalignment with cylinder mounted diaphragms or stator blades. The thrust bearing shall be of the tilting pad type with thrust and surge pads, and shall be capable of withstanding the maximum loads imposed during any conceivable transient

condition of operation. The thrust bearing shall be fitted with wear monitoring devices that shall raise an alarm to the CCR.

Each main bearing shall be provided with a positive visual checkpoint of oil flow through the bearing incorporating a local dial type thermometer. Thermometer pockets shall also be provided local to each sight glass.

Visible indication of the return oil flow from each bearing shall be provided. A dial type thermometer of an approved type shall also be provided to indicate the return oil temperature from each bearing.

Resistance temperature detectors or alternatively duplex thermocouples shall be installed in the white metal of each bearing of the turbine, and also within two specified pad of the thrust bearing. These thermocouples shall be positioned so as to ensure a quick response in detecting abnormal changes in metal temperature.

A resistance temperature detector and a separate locally mounted temperature indicator shall be provided in each main bearing oil drain. In addition thermocouples in the white metal of the turbine bearings shall be supplied for local or remote indication, recording and remote alarm annunciation.

Under thermally stabilized base load operation at nominal speed, the relative shaft vibration for the steam turbine according to ISO 7919-2 shall be within Zone A. The vibration level shall be verified during the Performance Testing.

Vibration of the steam turbine shall not be greater than the specified requirement, when operating at any load and the rated speed. Approved proximity vibration detectors shall be provided at the X and Y directions and bearing pedestal vibration detectors shall also be provided at each rotor bearing (including generator bearing) for the continuous remote indication, recording and alarm annunciation of the machine vibration.

Provisions shall be made by the fitting of an earthing brush or other approved means for prevention of damage to the bearings, reduction gears and oil pump drives due to any electrostatic charges which may be produced. Each of the generator and exciter bearings and oil pipes shall be insulated from the bedplate and foundations to prevent circulation of possible shaft currents. The generator bearing insulation shall be bridged by a removable copper strap to facilitate the routine testing of the insulation.

During overhaul, the thrust and journal bearing shall be taken out from the steam turbine and can be replaced with new one without opening the turbine casing rotor. The bearing shall be split horizontally at the centre line and can be regulated without lifting the rotors. Under any circumstances when operating, the temperature of the oil discharge from the bearing shall be below 65 °C, the oil flow and the oil temperature shall be monitored online through DCS. The metal temperature shall be below 90 °C and the bearing material shall be designed to the temperature of 112 °C without damage. Temperature values shall be reviewed during detailed design stage.

All the journal and thrust bearings shall be split horizontally at the centre line to facilitate inspection and replacement. The lower halves shall be removable by supporting the rotor weight.

Adjustment of rotor alignment and also axial re-positioning of the thrust bearing housings shall execute during the installation period and special tools shall be provided. The adjustment of the bearing alignment shall be able to perform without opening the turbine casing and lifting the turbine rotors, detailed arrangement of the bearing shall be provided during design phase.

4.11 Shaft Turning Gear

A fully automatic shaft turning gear shall be provided by an AC motor for turning speed. The motor shall have capacity to rotate the turbine generator shaft continuously without undue stress during periods of shutdown. The shaft turning gear shall be arranged for automatic disengagement of the drive when the speed of the turbine generator shaft exceeds the turning gear speed and automatic engagement when the turbine generator speed falls below turning gear speed. Electrical interlocks shall be provided so that the turning gear cannot start until the gear is fully engaged and an adequate supply of lubricating oil is available. In the event of failure of the lubricating oil supply the tuning gear shall become inoperative immediately. A remote zero speed indicator at local and an alarm at CCR shall be provided to indicate that the shaft has stopped. All necessary equipment shall be provided to enable local control of the turning gear, if allowed as per OEM's standard.

Hand turning and also shaft inching facilities shall also be provided. An interlock shall ensure that electrical turning cannot be initiated while hand turning is being attempted. To minimize damage between internal rotating and stationary parts, safeguards shall be included to prevent excessive torques from being transmitted to the shaft in the event of a seized rotor.

The turning gear shall be interlocked to prevent operation of the turning gear motor unless the emergency oil pump is running and the turning gear is full engaged. Alternatively, the turning gear shall be interlocked to prevent operation of the turning gear motor unless turning oil pressure is sufficient and the turning gear is full engaged. The turning gear motor shall be arranged to trip in the event of loss of lubricating oil pressure.

The turning gear shall be capable of accelerating the rotor assembly from rest. It shall be continuously rated to enable uninterrupted turning to be carried out under all appropriate circumstances and shall also be entirely suitable for intermittent use during maintenance.

Limit switches shall be provided for remote indication of turning gear status.

The Offshore Supply Contractor shall state the criteria recommended for determining when, after a shut down, the turning of the turbine generator rotor assembly may cease, also those for determining the minimum pre-service turning times after any

shaft standstill period. Full details of the hand turning mechanism and the procedure to be followed in its use shall be given in design stage.

4.12 High Temperature and Bolting

Particular attention shall be paid to the design of all bolts subject to high temperature, such as in the turbine casing, to ensure that they resist creep under all normal operating conditions. The Offshore Supply Contractor shall supply details of materials and physical specifications of all high temperature bolts, nuts, studs, set bolts, dowels, joint rings and connecting rods at detailed engineering stage.

Bolting for the casing joints will be designed for a minimum interval between bolt tightening of 45,000 hours.

The material of all bolts and studs operating at 480 °C and over shall be such as to give a basic life (i.e. the minimum, possible period at design temperature before a crack could appear in any such bolt or stud) in excess of 200,000 hours.

However, the tightening of turbine casing and steam chest flange shall be heating type and one such equipment shall be supplied. Small bolt shall be tightened using torque wrench and large bolt shall be tightened mechanically by bolt heater as per manufacture's standard. The special tools for bolt tightening shall be supplied.

Provisions shall also be made for direct measurement of imposed tightening strain on bolts mechanically, by means of suitable boreholes and the use of depth gauges.

One complete set of this bolt extension measuring equipment shall be provided. The surface finish of the bore holes shall be smooth and shall be adequate to permit ultrasonic testing for cracks using a bore hole probe. The bore hole shall not terminate near the plane of a change in section of the bolt.

If bolt stretching equipment is provided, one set of all necessary items of stretching equipment shall be included together with one full set of spares for all wearing parts.

For tightening using heating elements, one complete set of all necessary apparatus, including transformer, shall be provided together with one full set of spares for the heating elements and insulators.

4.13 Protection Devices

The Offshore Supply Contractor shall describe the protection and tripping logic of the turbine generator and provide a list of protection and tripping devices at detailed engineering stage. In circumstances where these devices may initiate a trip, the signals shall be hard-wired.

4.13.1 Over-speed Protection

To prevent the inadvertent over-speed of the turbine generator unit, an established system of over-speed protection shall be provided to trip the unit at 10% over-speed.

The Offshore Supply Contractor shall describe the equipment provided for testing the over-speed protection systems and the steam admission valves, preferably with full stroking and the minimum reduction of load.

4.13.2 High Exhaust Temperature Protection

Desuperheat and Spray water system shall be provided to prevent exhaust temperature from exceeding its limited value. Desuperheat and sinker shall be provided to exhaust for LP bypass in condenser. Exhaust temperature shall not exceed its limited value, when LP bypass is in operation. The source of cooling water shall come from condensate.

4.13.3 High Condenser Pressure Protection

Protection against excessive pressure in the LP steam turbine casing shall be provided by approved means such as lifting or relief discs or bursting diaphragms etc. These shall be sized to protect the LP steam turbine from over-pressure condition.

4.14 Pipework and Valves

The HP turbine shall be fitted with two combined stop and control valves for main steam admission. Two combined intercept stop and control valves shall be installed at the IP turbine for hot reheat steam admission.

Each combined valve consists of a stop valve and a control valve that shall be mounted on a common body. Every stop valve and control valve shall be equipped with an electro-hydraulic actuator.

All piping system shall be designed to the relevant Vietnamese regulation and acceptable international standards and codes such as ASME or equivalent. Main steam piping and hot reheat piping shall be of P92 seamless. Cold reheat piping from steam turbine to HP bypass shall be of P11 seamless. Cold reheat piping from HP bypass to boiler shall be of A106 Gr C seamwelded. HP bled steam piping shall be seamless. Feedwater piping shall be made of 15NiCuMoNb5. Alternatively, A106-C for main feedwater pump discharge piping and A106-B for main feedwater pump suction piping will be used according to ASME B31.1.

Pipework supports and hangers shall be allowed for movement due to thermal expansion, hydrostatic testing, water hammer and other possible dynamic impacts. Welded valves shall be provided and flanged valves will only be accepted for low pressure rating.

Suitably designed drains and vents complete with necessary pipework shall be provided for the pipework. Drain valves shall operate in a failsafe (open) mode with limited steam flow and pressure drop across the drain valve. These valves shall either be parallel slide valve or globe valve. All drain and vent valves required operating during start up and shutdown shall be remotely controllable from CCR. Double valving in series shall be arranged for each drain line PN40 (bar) and above.

Steam mains shall be routed and supported in such a manner that the thermal expansion of the pipework can be accommodated without imposing excessive end reaction forces and moments to the pipework and the valve chests. Stress analysis shall be performed during design stage to ensure that the allowable stress and the acceptable end reactions and moment at the steam chests are below the acceptable limits.

Steam chest shall be constructed of steel without abrupt changes of section. They shall be adequately drained and supported in such a manner so as to avoid distortion of the turbine cylinder. The steam stop and throttle valves shall be constructed of suitable materials and shall be of the 'fail safe' type, arranged to close under spring force and to open, under governor control, by hydraulic pressure. The main valves shall be fitted with internal bypass valves to permit them to be opened against full steam pressure. Provisions shall be made for exercising all stop valves and the throttle valves in turn with the turbine on load.

4.15 Lubricating Oil System

4.15.1 General

Lubricating oil system with two 100% AC pump and one 100% DC pump, shall be provided with lube oil purifier and 2 × 100% coolers. Space shall be reserved to allow future additional clean lube oil storage tank and dirty lube oil storage tank of replacement capacity for a whole Unit, when required. The lubricating oil module shall be located inside an oil bund of a size sufficient to hold the total quantity of oil in the system.

The Offshore Supply Contractor shall provide the lubricating oil system for:

- lubrication of all the turbine generator journal and thrust bearings;
- removal of the heat dissipated in the journal bearings; and
- jacking of rotors during start-up or turning at low speed, if required as per OEM's standard.

The lubricating oil system for the turbine generator shall comprise at least the following equipment:

- One shaft driven oil pump
- One main AC motor driven full duty oil pump
- A DC motor driven emergency oil pump
- 2 × 100% capacity oil coolers and 2 × 100% oil filters
- An oil storage tank, complete with a vent and a vapor extraction fan with a motor
- A filter type oil purifier with a motor, permanently connected to the lubricating oil system (One set for each Unit)
- An AC motor driven filter type oil purifier with a motor, portable for general use around the station (One set for the station)
- All necessary pipework, valves, fittings, temporary flushing strainers
- A dirty oil drain tank, a transfer pump with a motor and pipe work
- A transfer lubricating oil tank, a transfer pump with a motor and pipework

Proximity of the oil system to high temperature steam pipes shall be avoided to minimize the fire hazard. Guards and drip trays shall be provided where necessary to prevent contact between oil escapes and hot piping and other hot parts of the machine.

Appropriate hydrogen detraining section shall be provided to prevent hydrogen from entering the main lubricating oil system.

If required as per OEM's standard, a complete jacking oil system shall be provided including two 100% duty high pressure jacking oil pumps, suction strainers, pressure transmitters, pipings and supports.

The Offshore Supply Contractor shall provide a filter type lube oil purifier for each Unit. The unitized lube oil purifier shall be permanently connected in a bypass to the unit lube oil module such that it can be used whether the turbine generator is running or not. The capacity of the purifier shall be adequate to process at least 10% of the total oil charge in the system per hour. The purifier shall be capable of removing all solid particles down to size meeting NAS class 7 as per OEM's standard.

4.15.2 Lubricating Oil Storage Tank

The lubricating oil storage tank shall be located within the turbine hall as remote as practicable from high temperature steam pipework, and may be situated at basement level or alternatively in an elevated position, so long as adequate gravity drainage of the system to the tank exists, and adequate drainage facilities are provided from the tank to the transfer lubricating oil tank. Supporting steelwork for the tank of adequate strength and rigidity shall be provided.

The tank shall be of welded construction and manufactured from high quality mild steel plate and shall have sufficient capacity to hold all the oil within the system.

In order to avoid welding at site which could damage the tank internal painting all necessary pipe connections, pump seating facilities, pipe supports, steelwork or handrail supports shall consist of pads welded to the tank. Where such facilities are provided around tank penetrations, they shall be welded during manufacture both internally and externally to the penetration. All pads should be tapped for the supply of studs to effect connections. Alternatively, pipe connections to the tank may be made by means of butt-welding and in this case the tank shall be provided with stubs or nozzles extending out from the tank wall a sufficient distance to prevent the heat of welding causing any damage to the internal painting.

The tank base shall be sloped to a common position to facilitate complete draining of the tank and a lockable drain valve shall be provided directly at the flange. Access to this valve shall be provided and in the event of a high level tank being provided, then a second lockable isolating valve shall be provided, located for convenient operation from the basement level.

Following fabrication of the tank, the internal surfaces of the tank shall be painted with a paint that is totally suitable for long term use both in the oil environment and in the humidity to be expected in the oil tank during all working conditions. Appropriate surface preparation shall be carried out prior to internal painting, including shot or grit blasting and solvent washing.

The tank shall be fitted with manholes on the top surface to allow easy access into each internal section. Access rungs shall be provided under each manhole to permit safe and easy access. All openings shall be on the top of the surface of the tank and arranged to prevent ingress of dirt or debris.

The oil tank shall be vented by an AC motor driven vapor extractor. The vapor extractor impeller and casing shall be in materials that will not cause sparking should contact occur. A vent shall be taken outside the building to above roof level and directed at the extremity of the pipe away from the building. Any horizontal vent pipe shall have a natural drainage back to the tank, and suitable drainage of oil build-up shall be provided at the discharge from the extractor and elsewhere in the vent system as necessary.

The oil vapor vent shall be provided with oil strainer/oil trap arrangement to prevent oil spray into the atmosphere.

The oil tank shall be provided with level measurement devices with two alarm settings as per OEM's standard. A visual oil level gauge shall also be provided.

Suitable removable fine mesh strainers shall be fitted to all oil inlet lines to the tank. In addition coarser mesh strainers shall be fitted to all pump suction.

Special attention shall be taken in the tank design for handling hydrogen entrainment in the generator lubricating oil drains.

Sampling points shall be provided at suitable locations on the oil tank.

4.15.3 Oil Coolers

2 × 100% duty oil coolers shall be provided. The oil coolers shall be of the separately mounted self-contained type. Coolers arranged to be integral with the lubricating oil storage tank are not acceptable.

The coolers shall be provided with change-over facilities such, that under all operating conditions up to full load a change-over of cooler can be effected without throttling the oil supply to the unit. The change-over arrangement shall be such, that one cooler cannot be isolated before the second is in service.

The design shall be based upon tubular heat exchangers with closed cooling water passing through the tubes, and the lubricating oil through the shell and over the tubes. The oil coolers of plate type shall be accepted with stainless steel 304 plates. For this type of heat exchanger, sufficient space shall be arranged for assembly / disassembly purpose.

The materials of construction of the cooler tubes shall be of welded type SS304 stainless steel. Tube plate shall be of carbon steel or better. The tubes shall be straight and not of the hair-pin or coil type.

The cooler shells shall be constructed of carbon steel and each shall be fitted with an air vent at the highest point on the shell and a drain valve.

The cooler water boxes shall be of cast iron or fabricated steel lined with neoprene rubber at least 3 mm thick. The highest point of each water box shall be supplied with a connection for priming. An expansion relief valve, and all appropriate drain and air release valves, shall be provided with pipework leading to the nearest convenient drain trench.

The coolers shall be supported such, that the bottom water boxes can be removed without difficulty providing adequate access to the bottom tube plate.

The arrangement of water box connections shall be such, that tubes can be cleaned without breaking any oil or water pipe joints and withdrawal of the tube bundle should be possible without breaking more than the water joints to the water boxes. Cooler top covers shall be fitted with adequate tube bundle withdrawal facilities, and lifting lugs shall also be provided for lifting of the coolers to place.

The oil pressure in the coolers shall be greater than the cooling water pressure under all running conditions. Where this cannot be achieved under all conditions, the oil pressure shall always exceed the cooling water pressure while oil is being pumped through the coolers.

The temperature of the oil leaving the oil coolers shall be locally controlled by closed cooling water flow rate to ensure optimum oil inlet temperature to the bearings.

4.15.4 Lubricating Oil Pumps

The main oil pump shall be shaft driven type and be situated in the front pedestal or at the main oil tank. The pump shall be rated at 100% duty, sufficient to supply all oil requirements for the lubricating oil system as appropriate under all conditions of load and at all turbine shaft speeds between the speed that the auxiliary oil pump either cuts in or cuts out, and the turbine over-speed.

Provision shall be made for automatic priming of the main oil pump and for the maintenance of adequate suction head.

The auxiliary oil pump shall be AC motor driven and located integral with the lubricating oil storage tank. This pump shall be rated at 100% duty, as for the main oil pump, and shall be self-priming. The pump shall be capable of allowing the turbine generator to operate under all conditions of load and speed with the main oil pump out of service, and will be available primarily for starting up of the turbine generator unit and for automatic standby upon rundown of the unit. The auxiliary oil pump stop and start initiation shall be accommodated by the provision of pressure transmitter. The system shall be so designed to ensure that the operation of the auxiliary oil pump, automatic start sequence does not allow the oil pressure to drop to a level that would trip the turbine generator unit.

The emergency oil pump shall be provided, mounted integral with the lubricating oil storage tank, for the supply of sufficient lubricating oil to the bearings, when the unit is running down after a trip, or being barred, or during oil pipes flushing procedures. The emergency oil pump shall be rated to satisfy the turbine shutdown running condition, being DC motor driven, and it shall be arranged so that the pump comes into operation automatically on a fall of lubricating oil pressure. The emergency oil pump shall be arranged to enable the turbine-generator to be brought to rest without damage under mishap conditions such as the total loss of ac supplies.

4.15.5 Lubricating Oil Pipes and Valves

Lubricating oil supply line from oil filter to bearings shall be stainless steel and other pipes shall be carbon steel. No pipes shall run close to high temperature steam pipes, and there shall be no flanged joints except at the terminating flanges on pedestals, pumps, tanks, etc. All welded joints shall be subject to 100% radiographic examination. Alternative NDT method as per OEM's standard, if applicable and Inspection and Test Plan (ITP) shall be submitted for Owner's review. Oil pipes shall be supported to reduce the effects of shock loading and vibration, and particular attention shall be paid to the integrity of areas of high stress concentrations caused by expansion and contraction of the turbine generator unit. All pipe bends shall preferably be of long radius.

Temporary coarse and fine mesh oil filters shall be provided for initial commissioning, in all oil return pipelines. The drain pipes shall be designed such that they cope adequately with all oil flows to be encountered without flooding.

Sight flow inspection boxes shall be provided on all return oil pipes from each bearing and shall be located adjacent to each bearing pedestal in a position that is plainly visible to personnel standing on the operating floor local to the bearing pedestal. The boxes shall be fitted with toughened glass to withstand breakage. They shall be designed such, that oil splash does not occur, and that good vision within the box is maintained at all times.

Plug plates shall be installed by the Onshore Contractor before each bearing and elsewhere as required, for use during oil flushing procedures. The installation shall be such, that strainer basket mesh size can be readily changed to suit the particular flushing operation. Provision shall be made to effect maximum flow disturbances during oil flushing by way of air injection and thermal cycling techniques.

Permanent full flow strainers, having sufficiently fine mesh size to protect the plant from damage, shall be fitted in the lubricating oil supply pipes.

4.15.6 Lubricating Oil Purifier

An off-line centrifuge static type oil purifier, for water and sludge removal from the lubricating oil shall be provided for each unit with an hourly capacity of not less than 10% of the total oil quantity in the lubricating, relay and seal oil systems, and shall be permanently connected in a by-pass to the lubricating oil system in such a manner that it can be operated when the turbine generator is either running or shut down.

If oil heating is required for optimum purifier performance, the purifier shall be provided with an indirect heating system with thermostatic control arranged to heat the incoming oil.

The purifier shall be provided with all necessary tools and equipment to effect maintenance cleaning, which shall be supplied in a portable rubber wheeled trolley. The trolley shall include provision for handling oil waste and cleaning fluid.

One portable oil purifier of AC motor driven filter type and same rating as the permanently installed purifier shall be provided on a suitable rubber wheeled trolley with provision for towing by vehicle, and lifting by crane. It shall be complete with all necessary reinforced flexible hoses and connections to suit each location in which it will be required to operate. Hose end fittings shall be of an approved type allowing connection and shall be made for the portable purifier to operate in several different locations.

4.15.7 Control Oil System

Operation of the turbine governing, stop valves and other regulating valves shall be carried out through the use of control oil system. If lubricating oil is proposed to be used for the turbine control purpose, the details of this arrangement shall be elaborated, in particular the prevention of fire hazards.

Two 100% duty relay oil pumps shall be provided. The configuration of control oil

pump may either be (i) 2 × 100% AC pumps or (ii) 1 × 100% AC pump and 1 × 100% DC pump.

It shall provide an accumulator for stabilizing oil pressure in the system. The cylinder shall be charged with inert gas.

The supply shall be filtered through duplicate 100% duty filters, which will allow on-load cleaning and change over from one filter to the other without interruption to full oil flow.. The filters shall be fitted with differential pressure indication and alarm.

The pipes shall be of all welded system with 100% radiographic examination with a minimum of flanges situated at valve gear, strainers, pedestals, etc. Alternative NDT method as per OEM's standard, if applicable and Inspection and Test Plan (ITP) shall be submitted for Owner's review. Suitable provision shall be made to accommodate oil weeps at such flanges, in particular in the vicinity of or above high temperature components and thermal lagging.

The pipes shall be properly cleaned prior to operation of the governing system, and if necessary, shall be chemically cleaned by the Onshore Contractor at site.

4.15.8 Dirty Oil Drains System

One dirty oil drains collection tank and all associated pipes shall be provided to collect oil by gravity drainage from all oil leakage, venting and draining points such as pump glands, control valves, oil strainers, drip trays, vent and drain and oil purifier. The tank details and the location shall be specified in full to suit these requirements.

The dirty oil drains shall be pumped to a disposal point and to the transfer lubricating oil storage tank. The pump and pipes shall be permanently installed.

The materials and procedure of construction of the tank and system shall comply fully with the requirements of the lubricating oil system. The drains tank shall be suitably protected internally with an oil resistant paint, and shall be located within a bundled area. A tank drain shall be provided, suitably valved and fitted with a blank flange. The drain pipe shall be suitable for use as a puddle pipe where it passes through the bund.

The tank shall be fitted with two level switches which shall have start/stop control on the transfer pump. A high level alarm switch shall also be provided.

4.16 Extraction Steam

The extraction steam system shall supply bled steam for regenerative feedwater heaters and direct contact heater (deaerating heater). HP extraction steam pipes shall be manufactured by a seamless process. Extraction steam system piping shall be routed and supported in such manner as to provide for thermal expansion and to maintain end reaction forces and moments within allowable limits.

At least one reverse flow check valve shall be provided in each extraction line as required by relevant code. The check valves shall protect the turbine from over-speeding due to reverse flow of steam from the extraction piping when the turbine is tripped and is under a vacuum condition. The motor operated shutoff valves, operating in conjunction with the automatic drain valves which are located both upstream and downstream of the valve station containing the reverse flow check valve, shall protect the turbine against water induction. During a turbine trip, controls shall automatically initiate the closing of all the in-line motor operated shut-off valves and the opening of all the drain valves. High water level in a heater shall initiate these operations in the extraction line supply steam to and bypass the feedwater of that heater. All piping drains shall cascade and be routed to the condenser.

4.17 Feedwater System

A feedwater system shall be designed to provide sufficient and reliable feedwater supply to the boilers. It shall include but not be limited to necessary feedwater heaters, deaerators, feedwater pumps, control valves and auxiliaries.

The feedwater system shall operate with varying flows, pressures, and temperatures. Feedwater flow shall be regulated by a feedwater control system measuring feedwater and steam flow. Steam temperature controls shall regulate flow of feedwater into superheater and cold reheat steam attemperators. The feed regulation valves shall be pneumatic or hydraulic operated. The feed regulation isolation valves shall be motorized.

4.17.1 Boiler Feed Pump (BFP) System

4.17.1.1 General

The Offshore Supply Contractor shall provide 3 × 50% BMCR capacity motor driven boiler feed pumps for each Unit. Normal operation shall be with the two feed pumps as duty and one as standby and all heaters in service. Each feed pump set shall be completed with booster pump, drive motor, variable speed drive, speed increasing gear box, pressure stage pump and leak-off valving. Flow variation shall be controlled by the feedwater flow control system automatically regulating the speed of the pressure stage pump with the variable speed drive to respond to steam-feedwater flow.

During periods of low flow, a pump recirculating valve shall permit sufficient flow to protect the pump.

The off line pumps shall be maintained in a standby condition by the provision of a continuously operating warm-up line, if required.

Motor driven boiler feed pumps shall be designed and tested in accordance with the

latest applicable requirements of the relevant international codes and standards.

Type:	Multi-stage barrel type - motor driven pump with variable speed drive and booster pump
Quantity:	Three per Unit.
Design capacity:	Each pump 50% of the BMCR condition plus 10% flow and 21% friction head (system resistances) margin. The pumpset shall be capable of withstanding up to 5 minutes the reverse rotational speed which would result from non-return valve failure under maximum discharge pressure condition. Pump sizing to be finalized during detailed design.
Net positive suction head:	Pump minimum requirements to be met by normal flow conditions and upset flow conditions based on turbine trip and complete loss of turbine extractions.
Desuperheating water source:	
Superheater -	Pump discharge.
Reheater -	Pump intermediate stage.
Minimum flow:	For each pump, suction flow meter shall control individual modulating control valve in recirculation line to deaerator storage tank.
Pump drives:	Electric motor plus variable speed drive
Motor brake horsepower:	To match with total of the pressure stage and booster pump cold water brake horsepower. Two-pumps and single pump operation including runout without using the motor service factor.
Variable speed drive :	The variable speed drive shall be rated for continuous operation at full speed and the maximum motor shaft power to the pressure stage pump. The variable speed drive shall be capable of continuous operation at any speed from 30% to 100% maximum speed.
Leak-off control :	The leak-off control shall be independent of the feed pump sequence control system and shall be rated at 25% of the pump rated flow, with isolation valves upstream and downstream of the

leak-off valves, and connected to the deaerator.

Characteristic curves for pumps shall be submitted to be Owner for review during detailed design stage. The curves shall show total head, NPSH, power, horsepower, and efficiency all plotted against pump capacity in litres per minute.

Transient analysis is required for suction piping for turbine trip condition to ensure adequate NPSH for continued pump operation and submitted to Owner for review during detailed design stage.

4.17.1.2 Operating Conditions

For feed water system there shall be a feed control station comprising one 30% feed control valve and a motor operated valve arranged in parallel. This control valve shall have isolation valves at upstream and downstream sides.

The net positive suction head shall be designed with consideration given to worst case of deaerator level and pressure, and upset flow conditions based on turbine trip.

The leak-off control shall be independent of the feed pump sequence control system and shall be rated at least 25% of the pump rated flow, with isolation valves upstream and downstream of the leak-off valves, and connected to the deaerator.

Characteristic curves for pumps shall be submitted as part of design submission to the Owner for review. The curves shall show total head, NPSH, power, horsepower, and efficiency, all plotted against pump capacity flow.

Single pump operation shall be used for start-up and at low-loads as dictated by overall pumping efficiency consideration. Accordingly the pump sets shall also be capable of operating singly over their full speed range.

The feed pumps shall be capable of withstanding without damage for a period up to 5 minutes the reverse rotational speed, which would result from non-return valve failure under maximum discharge pressure conditions. In addition a reverse rotation alarm shall be provided for each pump.

The pump sets shall be free from harmful vibration under all conditions of load and speed. Any inherent resonant frequencies of the feed pump sets and their mounting arrangement shall be suitably divorced from any operational forcing frequencies. The rotors shall be designed such that no critical speed in water is less than 20% above the rated speed, also such that there shall be no torsional resonant frequency within 10% of the operating speed range. The details of the anticipated critical speeds and resonant frequencies shall be provided.

At turbine loads below approximately 30% output, a back-up steam supply system shall come into operation to prevent the deaerator pressure dropping below

atmospheric pressure so that saturation conditions are maintained.

In abnormal circumstances at start-up or low loads, the deaerator may be placed under vacuum conditions, and the design of the feed pumps shall be such as to accommodate this without problem or risk of damage.

4.17.1.3 Discharge Non-Return Valves

A non-return valve of approved design and manufacture shall be fitted in the discharge line of each main feed pump.

The design shall be such as to avoid chatter at low flows, and shall take into account that the admission of small solid particles into the valve shall not prevent it from functioning correctly.

The valve shall close freely with reduction in flow within the time taken for the pump to stop feeding after unit trip. The closing time of the valve (defined as the free fall time of the valve in water) shall be not more than one second.

Any sliding surfaces in the valve shall either be of corrosion resistant materials or be protected by an adequate thickness of a corrosion resistant layer. The sliding surfaces shall not be distorted by impact with the valve movement. The design shall permit the inspection or servicing of the valve internals without the need to remove the valve from the pipeline.

4.17.1.4 Suction and Discharge Isolating Valves

The suction and discharge isolating valves shall be provided with motorized actuators, and the suction valve shall have locking facilities to prevent inadvertent closure.

Suction and discharge isolating valves shall be fitted with by-pass connections which incorporate motorized by-pass valves. Control of the main and by-pass valve actuators shall be interlocked to ensure the bypass valve opens before the main valve.

A relief valve shall be provided for each pump to protect the suction line, strainer and booster pump from over-pressurization.

4.17.1.5 Suction Strainers

A basket type strainer shall be installed at the booster pump suction of each feed pump. The strainer shall be designed and sized so as to minimize the pressure drop across it and, at the same time, effectively remove from the feedwater any foreign matter that may be harmful to the pump.

The strainer element shall be made of stainless steel and mesh size shall be defined

by supplier's standard.. The strainer elements shall be easily replaceable and their design shall provide for alternative mesh sizes to be used if required.

4.17.1.6 Electric Motors

The motors shall be suitable for continuous operation for long periods without adjustment or attention, for starting up or shutting down daily, and also for immediate starting following long periods at standstill on standby duty. The motors shall be of the squirrel cage induction type suitable for direct on line starting. The motors shall be continuously rated for single pump operation delivering the maximum possible flow or running at maximum speed delivering the rated flow of the coldest water.

A torque/speed curve for the motor super-imposed on the anticipated torque/speed curve for the pump when starting under the most arduous condition (including fluid coupling speed control at maximum setting) shall be provided.

The starting times of the motor shall be as short as possible consistent with the specified starting requirements. The starting current of the motor shall be limited to less than 6 times the full load current.

4.17.2 Deaerating Heater and HP Feed Heaters

4.17.2.1 General

The deaerator and HP feedwater heaters shall be designed and tested in accordance with the latest applicable international codes and standards including Heat Exchange Institute standards, ANSI and ASTM standards. Heaters shall be of horizontal arrangement and shall have integral desuperheating and drain cooler sections to minimize risk of water entering the turbine due to flooding within that section.

Internal baffles shall be arranged within the storage tank to prevent surging of the condensate. Deaerator level indicators and alarms shall be provided in the CCR. The feedwater tank shall store a minimum quantity of feedwater corresponding to 5 minutes of 100% BMCR feedwater flow, assuming that the condensate is initially at the normal working level. The deaerator shall remove all dissolved oxygen in excess of 0.007 mg/litre from 10% to 100% load. Auxiliary steam shall be supplied to heat the deaerator content.

Feedwater heaters shall handle extraction steam flows consistent with the bypass arrangement chosen and the operation with heaters out of service.

The deaerator shell shall be made of carbon steel and the internals made by stainless steel. The HP feedheater tube material shall be made of seamless stainless steel TP304N. The LP feedheater tube material shall be made of stainless steel.

4.17.2.2 High Pressure Feedwater Heaters

High pressure feedwater (HP) heaters shall be designed according to HEI Standards for Closed Feedwater Heaters.

The HP heaters shall incorporate desuperheating and drain cooling sections for reducing the thermal difference between the feed water at the heater inlet and outlet compared to the incoming bled steam saturation temperature and the outgoing drain temperature such that an economic thermodynamic gain is achieved. The desuperheating or subcooling zones can be externalized in a separate heater if it is beneficial for the thermal integration

The feed water velocity in the tubes shall not exceed 2.5 m/s at the flow rate corresponding to the 100% TMCR of the Unit.

The steam leaving the desuperheater section at full duty shall be above the saturation temperature by a sufficient margin to ensure, that no condensation on the tubes within the desuperheating section will occur under normal load conditions and the steam leaving the desuperheating zone will not cause droplet impingement in the condensing section.

Heater shells shall be of fabricated steel suitable for design steam temperatures. The tubing of the heaters shall be of seamless stainless steel U tubes (A213-TP304N) and welded into the tube plate or headers at both ends. The tube plate or headers shall be forged steel and weld overlaid with low carbon steel on the tube joint surface to ensure weldability with the tubes.

Means of inspection of the tube-nests and tube-ends shall be provided. In water box type heaters, arrangements (which do not require the breaking of pipe joints) for inspection and repair of the water box division plate shall be provided. Water boxes shall be made of carbon steel (ASTM A515-65 or equivalent).

4.17.2.3 Low Pressure Feedwater Heaters

Low pressure feedwater (LP) heaters shall be designed according to HEI Standards for Closed Feedwater Heaters. The LP heaters shall be designed on the water side to withstand the maximum discharge pressure of the condensate extraction pump including shut off head.

The tubes shall be of the U type, of stainless steel (ASTM A688-TP304) and suitably expanded into a steel tube plate at both ends.

4.17.2.4 Gland Steam Condenser

The gland steam condenser shall be constructed in accordance with the requirements laid down for the LP heaters except that the tube arrangements may be either hair pin or straight type expanded at both ends into tube plates of carbon steel. The return end

tube plate and covers, if provided, shall be arranged to allow for expansion of the tubes.

4.18 Condensate System and Condenser

4.18.1 General

The surface condenser shall be designed and constructed to HEI Standards, or equivalent, with sufficient margin and spare surface area for the maximum heat rejection duty under both normal operation and turbine bypass operation conditions for the operating regime specified. An automatic on-line condenser tube cleaning system shall be provided.

The condenser shall be of the divided waterbox type arranged to allow waterside cleaning of one half of the condenser with the turbine capable of generating a minimum of 50% TMCR.

Condenser of titanium tubes / plates with temperature rise not more than 6°C, under-slung solid mounted with cathodic protection shall be provided. The design cleanliness factor shall not be larger than 95 %. Condensate outflow from the condenser shall be at a temperature not less than the saturation temperature corresponding to the shell pressure and dissolved oxygen content below 0.015 mg per litre.

On load detection of condenser tube leak location shall be provided. Capacity of condenser hotwell at normal water level shall not be less than 3 minute flow at BMCR. Manhole of not less than 500 mm diameter shall be provided.

The condenser along with the rest of the condensate system shall be designed to ensure freedom from hydraulic shock under all conceivable conditions of operation.

The condenser shall be protected from ingress of debris from the cooling water system by a fine screening system. The fine screening system shall be self-cleaning and operate automatically. Motorized valves shall be provided on the inlet and outlet connections of the water boxes and shall be capable of providing effective isolation to facilitate access into the water boxes.

The condenser shall be constructed with split water boxes in such a way that one half of the tube bank may be taken out of service under loading conditions of the turbine. Under such condition, the turbine shall be capable of carrying at least 50% TMCR load.

4.18.2 Construction Features of Condenser

The condenser shall be rigidly supported with expansion joint between turbine and

condenser, under-slung solid mounted with cathodic protection.

The water boxes or covers shall have at least two access manholes with hinged doors for each compartment. Minimum size of the manhole shall be 500 mm.

The water boxes and covers shall be fitted with lifting lugs. The water boxes shall be provided with vents and drain connections. All bolts and nuts which come in contact with circulating water shall be of stainless steel.

Sufficient numbers of tube support plates shall be fitted inside the shell to avoid resonance vibration and excessive bending of the tubes which may cause fretting and tube to tube rubbing. Tube holes on the tube support plates shall be accurately drilled to have suitable clearances on tube diameter. Edges of the tube holes shall be chamfered and deburred.

The tube nest as a whole shall be slightly inclined to provide self-draining of tubes when the condenser is shut down. The tubes shall be welded titanium tubes with average wall thickness of 0.7 mm for top rows of the tube nests and 0.5 mm for the remainder.

The tube plates shall be of solid titanium or titanium clad carbon steel. The thickness of clad shall be 5 mm and clad by explosion welding or hot rolling and checked by ultrasonic inspection for the integrity of the bond according to a recognized standard.

The tubes shall be fixed to the tube plates by TIG welding after contact rolling.

The tube plates shall be supported by shell flanges by means of collared studs so that it will be possible to dismount the water box without affecting the joint between the shell flanges and the tube plates.

A connection with a diffuser of proven design inside the shell shall be provided to receive steam from LP bypass station without undue noise, vibration and erosion of any parts. Connections for the interconnecting pipes with flash box and connections for steam, drain and vents directly led to the condenser shall be provided with necessary impingement plates and or diffusers on the proper location on the shell.

The condenser shell shall be provided with manholes with hinged covers, one on the exhaust neck and at least one on the hotwell.

Inside the condenser shell, suitable gratings and ladders for the inspection of the tubes and the internals shall be provided so that stepping on the titanium tubes can be avoided.

Condensate outlet connections shall have 75 mm high internal upstands and shall be fitted with hoods arranged to prevent solid objects from passing into the condensate extraction pumps.

The Offshore Supply Contractor shall pay attention to the condenser tube withdrawal space for maintenance work in his equipment location design and shall provide the

rail for a hoist if needed.

At least SS 316L stainless steel shall be used for the material of those equipment contacting with seawater. Waterboxes material shall be defined as carbon steel with soft rubber lining.

4.18.3 Condensate Extraction Pumps

3 × 50% duty AC motor driven condensate extraction pumps with motors shall be provided. The pumps shall be turbine type, vertical, multi stage barrel pumps.

Two pumps shall be capable of transferring the total of the condensate and any drain or make-up water from the condenser at BMCR to the deaerator through polishing plant, the gland steam condenser and the low pressure feedwater heaters.

Limit switches shall be provided on the pump discharge isolation valves with an interlock to prevent pump operation unless the valve is open.

The pumps shall be constructed of materials specially chosen to resist deterioration by erosion or corrosion. All parts subject to wear shall be fitted with renewable liners and all bearings shall be automatically lubricated.

The pumps shall be arranged to enable the pump internals to be removed without disturbing the suction and discharge piping connections.

The pump glands shall be suitably sealed to avoid the ingress of air into the condensate under all load conditions and when the pump is on standby duty. Sealing liquid shall be taken from the same condensate water.

Each condensate extraction pump shall be provided with stop valve for manual operation on the suction side and a non-return valve on the discharge together with a manual isolating valve. The pumps shall be provided with all necessary water sealing valves and any other fittings that may be necessary for the efficient working of the plant.

The barrel shall be welded steel designed for direct grouting into a sub-floor pit and extending to the depth necessary to ensure proper operation under the highest anticipated vacuum in the condenser hotwell. Both the suction and discharge connections shall be flanged. Impeller casings are to be of high quality cast iron and fitted with renewable stainless steel wearing rings. Pump bowl bolting shall be stainless steel.

Rotors shall consist of stainless steel impeller securely keyed and locked to a stainless steel shaft. Condensate lubricated sleeves and cutlass rubber bearings shall be provided at a number and locations sufficient to ensure continuous vibration-free operation under all conditions of operation. Each pump's thrust bearing shall be incorporated in its motor or alternatively in the pump head.

The condensate piping shall be fitted with a recirculation pipe to the condenser, complete with an automatic valve, which would allow circulation of the condensate through the polishing plant and the gland steam condenser during start-up operation of the turbine or during low load operation.

The layout design of the pipework shall be such as to allow any of the pumps to be isolated for maintenance work or held as standby while the other unit is in service.

A suitable permanent strainer shall be provided at pump suction to prevent foreign matter being drawn into the pumps, particularly during the initial commissioning period and subsequently after any maintenance work involving the condenser steam space.

The strainers shall be generously sized so as to keep the pressure drop across them to a minimum. In its permanent configuration each strainer shall be capable of passing flows up to the maximum specified without detriment to the associated condensate extraction pump.

Basket type strainers shall be provided. The strainers shall be fitted with local differential pressure indicators and with differential pressure switches for alarm initiation.

4.18.4 Condenser Tube Ball Cleaning System and Debris Filters

Condenser tube ball cleaning system shall be provided for on-load condenser tube cleaning to maintain optimum cleanliness factor.

The aim of the system is to provide continuous automatic cleaning of condenser tube without shut down or interruption to the heat transfer process. It shall remove the deposit inside the condenser tubes and maintain heat transfer efficiency at optimum level.

Sponge rubber balls shall be injected into the cooling water flow before it enters into the condenser. The diameter of the cleaning balls shall be only slightly larger than the nominal diameter of the condenser tubing. With their elasticity they shall generate a contact pressure on their way through the condenser tubes by which fouling is removed from the inner tube walls. At the condenser outlet a strainer shall be installed in the connecting pipe which separates the balls from the water flow. The balls shall be then drawn off and pumped through a collector to be subsequently discharged into the cooling water inlet to the condenser. This system shall remove fouling and hard deposits from the internal surface of the tubes, which would otherwise lower the heat transfer characteristics of the tubes, with a resulting loss in condenser efficiency.

The ball collection efficiency shall be higher than 95 % when the ball cleaning system is running automatically at designed circulation water flow rate in normal running of the Facility. Local control instruments, on-load automatic ball counting devices at ball

injection and receiving chamber, gauges, motor driven pumps and actuators, local control cabinets, cabling and wiring and other devices as required shall be included in the above plant

Ball recirculating pump shall have sufficient capacity with sealing on pipe connection. Ball injection shall be designed as counter-flow type in order to evenly distribute the balls injecting to condenser tubes.

Electrical change-over valve shall be provided at ball collector to change-over the position of the collector between air and water exhaust. Non return valve shall be provided in the suction of condenser for ensuring no water and ball would flow back. Electrical change over valve shall be capable for balls collection, normal operation and balls replacement.

Debris filter shall be provided at each cooling water inlet pipe having a mesh or hole size of approximately 10 mm diameter and to prevent the loss of balls in case of reverse flow of water in the event of tripping of cooling water pumps. The filters shall provide fully automatic on-load cleaning arrangements by means of suitable debris discharge piping and valves.

The material of components of condenser ball cleaning system contacting with seawater shall be at least SS 316L stainless steel.

4.19 Condenser Air Extraction System

3 × 50% vacuum pumps shall be provided for venting and removal of non-condensable gases during normal operation and to rapidly reduce the condenser pressure from atmospheric during start-up and admission of exhaust and/or bypass steam to the condenser. The holding capacity of two vacuum pumps shall be selected according to the condenser worst air-in leakage rate.

When operating in hogging mode type three vacuum pumps must be able to create vacuum in the condenser of 0.3 bar (a) in less than 30 minutes. Sealing liquid makeup shall be from the demineralization water system. The open-loop auxiliary cooling water is suitable for that purpose regarding the achievable cold water temperature.

The three pumps shall be capable of operating individually or together (operating in parallel).

Water vapour entrained with the extracted air shall be returned to the sealing water recirculation system. Air flow indication shall be provided at the separator vent. Exhaust piping shall be suitably silenced and arranged to avoid creating a hazard to Facility and personnel.

4.20 Condenser Water Box Priming System

If required during design stage, the Offshore Supply Contractor shall provide the condenser water box priming system to evacuate air from cooling water side of the condenser. Automatic start/stop control shall be provided to ensure the cooling water boxes of condenser to be fully primed (i.e. no air locked inside tubings and water box).

4.21 Turbine Bypass System

HP and LP bypass system shall be provided to optimize startup times and load rejection capability. The capacity of the HP (at least 70% BMCR) and LP (at least 40% BMCR + spray) bypass system shall meet the requirements of all start-up during cold, warm, hot, very hot conditions and house load operation. The capacity of turbine bypass system shall be confirmed during execution stage. The turbine bypass systems shall be fast-acting and reliable in operation during start-up, shut-down and transient conditions to protect the steam turbines. The turbine bypass system shall be capable of operating at full load, part load, various start-up, shut-down conditions and house load operation.

A non-return valve shall be provided on the cold reheat piping to safeguard against the back-flow of steam into HP turbine while HP bypass is in operation. Appropriate preventive measures shall be provided to avoid back flow of bypass steam to the turbine cylinders. Desuperheating facilities shall be available to reduce temperature and pressure of the bypass steam. Desuperheating of the LP bypass system flow shall be achieved outside the condenser. Source of desuperheating water shall be feedwater from the boiler feed pump discharge for HP bypass and condensate from condensate extraction pump discharge for the LP bypass.

All HP and LP steam bypass valves, associated control, desuperheating equipment and pipework shall be supplied as fully integrated and designed systems to suit the chosen boiler and turbine designs. The bypass valves shall be superior quality devices of the pressure-reducing and desuperheating type, specifically developed for turbine by-passing duty or other special purpose valves proven for steam turbine by passing duty.

The bypasses shall be arranged to open quickly on direct signaling of an excessive partial or total loss in generator electrical load, turbine trip, and on detection of boiler pressure exceeding a target value below the setting of the boiler safety valves, so that the bypass operation reduces the frequency of safety valve operation in order to reduce safety valve maintenance, water loss and noise emission. In addition to the fast opening of the bypasses on generator sudden load reduction or boiler overpressure, the bypass control shall be arranged to receive signals from the turbine generator start-up and loading control scheme (or from remote manual operation from CCR), to facilitate the matching of steam and turbine metal temperatures, so as to assist the starting or re-starting of the unit.

The control for the LP bypass shall be generally similar to that for the HP bypass, but will include a feature to cause the LP bypass valve to close quickly on excessive condenser backpressure. When one of the cooling water pumps tripped due to failure of the cooling water pump, the control shall, if required, partially close the LP bypass valve to limit the flow capacity of the bypass to prevent thermal overload of the condenser.

Special attention is to be paid to the positioning of the valve, desuperheating and other pressure reducing equipment in order to reduce noise levels to those acceptable within this OTR and if required acoustic hoods shall be provided.

4.22 System Drains and Recovery

Under normal operating conditions the HP drains shall cascade flow through various stages of HP heaters to the deaerator. LP drains shall cascade through LP 1 and 2 heaters to the condenser. LP heaters 1 or 2 drain water shall be incorporated with U tube to hot well. For other heaters, control valves located downstream of each heater shall control the drain flow. All heaters shall be each provided with an alternate drain line to the condenser. Alloy piping shall be used downstream of drain control valves.

Each feedheater, including the deaerator, shall be provided with a venting system for removal of non-condensable gases from the heater shell. The flow of non-condensables from the heaters shall be controlled by an orifice. Gases from the HP heaters shall flow to the deaerator. Gases from the deaerator vent and those from the LP heaters shall flow to the condenser.

Each feedheater shall be provided with safety valves at steam condensate side for protection from over-pressurization. Channel side relief valves shall be provided for the heaters to protect them from over-pressurization due to thermal expansion of the fluid when the heaters are isolated on the channel side only. Channel side vents and drains and shell side drains shall be provided to facilitate filling and draining of the heaters and drain cooler.

Vents and drains from turbine shall go to enclosed drains vessels with suitable desuperheating facilities and finally discharge to the condenser hotwell. Vents and drains from low points of main, reheat and extraction steam system shall go to the turbine hall sewer. Each drains line shall be run separately from its collection point towards the point of discharge with continuous flow. All valves shall be located in accessible positions local to the drains vessels. Vents from drains vessels shall be connected to the condenser.

Each drain shall be separately controlled and run separately to its point of discharge to avoid preferential draining as will tend to occur under conditions of different system resistance. Drains shall have a continuous fall towards the point of discharge, and shall be not smaller than 25 mm nominal bore.

Drainage from the HP and IP turbine cylinders, steam chests, loop pipes and associated bled steam pipes shall be controlled by identical motorized isolating

valves mounted in series arranged to operate sequentially as master/martyr valves. The martyr valve shall carry out the throttling duty and the master valve shall be arranged to provide tight isolation.

All drain valves required to be operated during start up and shut-down shall be power operated and remote controlled from CCR, with features for local operation and maintenance.

4.23 Acoustic Walls

Turbine acoustic wall without ceiling shall be provided in accordance with the Offshore Supply Contractor's standard.

The acoustic walls shall be of pleasing appearance, suitable stiffened and secured to avoid thermal distortion. They shall be resistant to corrosion and mechanical damage, and shall have adequate provisions to allow free access to all areas of the turbine for maintenance and operation. The color of the walls shall be agreed with the Owner.

For the purpose of reducing maintenance outage periods to a practicable minimum, the acoustic walls shall be designed to be readily removable in those sections where maintenance would normally be carried out. Such movable sections shall be strengthened to facilitate removal and replacement without damage and arranged so that when sections are removed, instruments, instrument panels, gauge boards and connections need not be disturbed.

Adequate door access to the inside of the walls shall be provided.

All local instrumentation gauges, oil sight glasses and the like, local to the turbine are to be visible from the outside or inside of the acoustic walls.

Chapter 5 – Generator and Auxiliaries

5.1 System Description

The proposed generator and its auxiliaries shall have a minimum short circuit ratio of not less than 0.50. The generator shall comply with IEC 60034, the grid connection agreement and the local Vietnamese national grid code. Generator and exciter windings shall possess insulation that is non-hygroscopic and of Class F type complying with IEC 60085, but having a temperature rise not exceeding that of class B under any operating condition within the specified output.

Quality of the generators and accessories shall be in accordance with the requirements of ISO 9001, EN 29001 or BS 5750 Part 1 and other similar equivalent international quality standards.

Temperature detectors shall be provided to monitor the maximum operating temperature of the machine.

The generator shall be able to operate continuously at 804 MVA or above at the following operating conditions:-

- | | |
|---|---|
| - Power factor | 0.85 (lag) – 0.90 (lead) |
| - Terminal voltage | 23.5 kV +/- 5% |
| - Frequency | 49 – 51Hz |
| - Hydrogen pressure | 0.45 MPa(g) (to be confirmed during detailed engineering stage) |
| - Generator cold hydrogen gas temperature (Tcg) | 45 °C (to be confirmed during detailed engineering stage) |
| - Generator cold water inlet temperature (Tcw) | 38 °C |

The generators shall each be capable of supplying their rated output of 600 MW_{net} within 49 Hz and 51 Hz and $\pm 5\%$ of nominal rated voltage with the power factor range 0.85 lagging and 0.90 leading.

The Facility shall be capable of operation at frequencies below:

Frequency Range (Hz)	Minimum Sustainable Operation
47.0 to 47.5	Not required
47.5 to 48.0	10 min.
48.0 to 49	30 min.
49.0 to 51.0	Continuous
51.0 to 51.5	30 min.
51.5 to 52.0	1 min
>52.0	Not required
For further information, please see Vietnamese national grid code	

The generator shall be designed and provided with all the following necessary accessories:

- 1) Automatic high speed digital dual channel (auto-manual) automatic voltage regulator (AVR) and power system stabilizer (PSS), be able to maintain steady-state terminal voltage within $\pm 0.5\%$ of the preset value under all operating conditions and be able to smooth and continuous running over the operating band width.
- 2) Excitation system and excitation transformer (if required).
- 3) Surge protection equipment.
- 4) Neutral grounding equipment.
- 5) Generator control and relay panels.
- 6) Generator protection equipment.
- 7) Generator hydrogen cooling system.
- 8) Generator stator water cooling Systems.
- 9) Jacking oil, bearing oil and seal oil systems covered under turbine.
- 10) CO₂ system for purging hydrogen/air.
- 11) Supervisory and control System.
- 12) All the connecting cables.
- 13) Internal wiring, include wiring to the correlative clamp rows of all the supplied boards.
- 14) The terminals of wires connecting to the outgoing feeders by phase isolated busbar.
- 15) Current transformers (CT's) and voltage transformers (VT's) used for metering, tariff meters shall meet the precision level specified by local regulations and IEC 61869 Standard, IEC 62053 standard and other correlating standards.
- 16) Duplex thermocouples or resistance temperature detectors (RTD's) for sensing the generator temperature.
- 17) Sensors to detect vibration at the end of stator windings
- 18) Internal on-load partial discharge monitoring system.

5.2 Design Criteria

The generator and its auxiliaries shall conform to IEC-60034 and other relevant standards. The generator and its excitation system shall have capability of matching the declared VWO condition in all range of Site conditions. It shall be ensured that when the generator is working at this capability and secondary cooling water temperature is at its designed value, no part of the generator shall attain a temperature in excess of the temperature limit specified for class B insulation as per IEC 60034. A generator capability curve shall be provided showing the generator MW/MVAR operating limits at rated hydrogen pressure and temperature.

The combined moment of inertia (GD^2) of the main shaft (high and intermediate pressure (HIP) turbine rotor, 2 × low pressure (LP) turbine rotors, generator rotor) shall be higher than 140 t.m².

- (a) Generators windings shall be double star connected with all phase and neutral terminals brought out. The neutral of the generator shall be earthed through a grounding transformer, the secondary of which is loaded with a resistor. The core design shall permit the flow of earth current of at least 15 amperes for one second without any core damage.
- (b) The generator shall be capable of delivering continuously its rated power at rated power factor with voltage variations within $\pm 5\%$ of the rated value and frequency variation is from $\pm 2\%$ of rated value of 50 Hz with the temperature rise not exceeding class B limits. It shall still be capable of delivering the rated power when the voltage variation does not exceed 5% and the frequency variation is from -5% to +3%.
- (c) The generator shall be capable of operating continuously under unbalance load as per IEC 60034-1. It shall be capable of withstanding during any transient unsymmetrical short circuit condition a negative sequence current (I_2) expressed in per unit of rated current for a duration of t seconds such that the value of $I_2^2 \times t$ shall comply to IEC 60034-1. It shall be capable of withstanding 3-phase short circuit at its terminals when operating at rated MVA and power factor with 5% over voltage.
- (d) For voltage wave form, the telephone interference factor (TIF) shall be within the limit specified in IEC 60034-1.
- (e) The generator shall be capable of withstanding the electrical, mechanical and thermal stresses developed during fast re-closing of high voltage line, transmission line switching, faults, out of step operation, out of phase synchronism and worst-case short circuit fault at generator terminal etc.
- (f) Generator coolers shall be designed such that the maximum temperature rise is not exceeding Class B limits. The coolers shall be so designed that due to outage of one section, the generator can still carry 67% of the rated load continuously without exceeding the temperature limit specified for class F insulation.

5.3 Stator

5.3.1 Casing

The joint and seal design of the casing, seals, valves and pipework shall be such that there is no leakage of air or gas.

Facilities shall be provided at two positions on the generator casing for connecting to the isolated phase busbar (IPB) and station earthing system. Sufficient cross-

section of material to carry safely the maximum unrestricted earth fault current shall be provided.

5.3.2 Core

The stator core shall be made of high permeability low loss material, insulated on both sides. The material specification for the stampings, the insulation between stampings, and the procedure adopted in debarring the plates and checking the finished stampings, shall be such as to eliminate the possibility of interstamping shorts and resulting core faults.

5.3.3 Winding

The windings shall be copper with class "F" insulation system, and temperature rise shall be limited within class "B". The choice of materials, proportioning, clamping, and arrangement of the core in the casing, shall ensure the minimum noise and 100Hz vibration within the core plus the minimum transmission of vibration to the generator foundations, pipes and associated equipment.

The winding insulation system shall be designed and manufactured in such a way that it will meet the specified machine life for the given duty cycle. Insulation thickness and coil support structure shall be amply proportioned to withstand operation at the maximum over-voltage, system switching surge voltages and the fault conditions.

5.3.4 Terminals

The windings shall be in star configuration, and the terminals of each stator phase winding shall be conducted to connection box on racks. All the six connection points for the generator shall be able to be taken apart from the generator and can link with outside connections. Generator connection box shall be installed below the stator. Connection box shall be non-magnetized material type to hinder overheating because of fault currents. Three supporting porcelains shall be installed for the either side of phase line and neutral and concurrently, supporting porcelains of voltage/current transformer required for measuring devices, protection relays and AVR shall be designed.

Adequate provision shall be made in the design of the foundation block for accommodation of the metering, protective and other current transformers required for operating the unit. It may be taken that the current transformers will not be in contact with the terminals and will be supported independently of the generator. Creepage distances shall be equivalent to those specified in IEC 60137 for medium polluted atmospheres as defined in IEC 60815.

5.3.5 Connection

On the line side, provision shall be made for disconnection of the generator from the phase isolated busbars to enable the establishment of safety clearances during maintenance.

On the star point side, provision shall be made for disconnecting the star point from the generator terminals to enable the establishment of safety clearances between the generator and the earthing transformer and to enable testing of individual phase windings.

5.3.6 Others

All fixing and locking components in the ends of the fully assembled generator shall be non-magnetic except nuts for frame rib. This shall apply to the volume enclosed by the core end plate and corresponding frame rib, the casing wrapper plate and the end shield inner covers.

Class F materials shall be used. Materials liable to shrink in the course of time or by heating, and moisture-absorbent materials shall not be used for packing and wedging. The general construction of the stator shall be such as to provide adequate cooling surfaces so that temperatures greater than those permitted for the class of insulation do not occur.

The design shall be such that no retightening is required between major maintenance overhauls. Full instructions shall be given in the maintenance manual for this work. All equipment necessary for this operation shall be provided.

The stator windings shall be tightly bound with glass roving in radial and circumferential direction but support brackets, bearing pads and sliding pins permit the windings to move in axial direction. The design shall prevent the winding overhang under normal operating conditions. It shall eliminate slot discharge and ensure that slot wedge tightness and radial pressure on the coils is maintained to provide satisfactory operation over the life of the plant. The end winding structure shall incorporate provision for re-tightening the end winding clamping after a period of service.

The stator slot wedges shall be able to withstand the force imposed during the most arduous electrical short-circuit between adjacent coil connections of different phases at the turbine end, such that there is an indeed of fault current from the system.

5.4 Rotor, End Ring and Brush Gear

The design of the rotor shall be such that earth and inter-turn faults in the windings will not occur in service. Insulation shall be of Class F temperature classification. The limit of winding insulation temperature rise shall not exceed that nominated in IEC 60034-1 for Class B insulation.

The Offshore Supply Contractor shall incorporate all latest design features to avoid permanent deformation of the winding during service. Packing blocks used in the rotor winding shall be of approved material and entirely suitable for the high temperatures and mechanical forces occurring in large rotors. Particular attention shall be given to the insulating and securing of the rotor winding connection leads to avoid vibration and the possible failure of either the lead or its insulation.

The rotor shall be manufactured from a single forging, however, that separate couplings shall be allowed. No castings shall be used in the rotor assembly, and the number of small parts shall be minimized. The Offshore Supply Contractor shall incorporate suitable features in the rotor design to reduce the variation in rotor stiffness about the direct and quadrature axes, and thus minimize the resultant twice-per-revolution vibration levels. The shaft shall have a reference mark of known relationship to the location of the magnetic poles which can be detected in-service for synchronism of shaft position with portable test equipment. The mark shall be located such that it is always accessible notwithstanding shaft expansion and provision shall be made for mounting of a detecting device.

The centre of the shaft forgings may be bored to permit the inspection and testing of the interior. If, following metallurgical examination, it is necessary either to bore a previously unbored rotor, or to increase the bore diameter, it shall be demonstrated that the effect on the magnetic characteristics shall not limit the performance of the Unit.

The packing blocks used in the rotor winding shall be entirely suitable for the high temperature and mechanical forces existing in generator rotors. The rotor insulation system shall be of sufficient mechanical strength and flexibility to cope with the stress in service without cracking or otherwise becoming damaged. Attention shall be given to the insulation and securing of the coil connections to avoid vibration and possible failure of either the connections or their insulation. An effective damper winding shall be provided in the rotor. It shall be designed to divert negative sequence and 50 Hz current from the rotor teeth, wedges, pole-faces and end rings.

A suitable discontinuity shall be provided on an accessible part of the rotating shaft system for use as a vibration phase reference. Means shall be provided for detecting rotor winding shorted turns in service e.g., air gap search coil or similar device. The rotor shaft shall be earthed.

The satisfactory and consistent dynamic behavior of the generator rotor shall be assured by the design, manufacturing, testing and balancing. The generator rotor shall be capable to withstand an over-speeding process. Asymmetric rotor inertia compensation shall be incorporated. There shall be adequate provision for making in-situ correction to the generator rotor balance without major dismantling.

The rotor end winding retaining rings shall be made of 18/18 Mn/Cr material, which is resistant to stress corrosion cracking in the operational environment.

The brush gear shall operate continuously at full load for at least two years without the need to shutdown for brush gear or slip ring maintenance. The brushes shall be capable of being safely replaced with the unit on load. The slip ring brush gear

assembly shall be totally enclosed to prevent the ingress of oil. Constant brush pressure devices or individual brush tension devices, capable of radial adjustment, shall be provided. The design of slip rings and connections shall be such that carbon dust from the brushes cannot accumulate on insulation surfaces.

Slip rings where necessary shall be mounted on the shaft in a separate enclosure at the outboard end of the generator.

5.5 Stator Water Cooling

The generator stator windings shall be directly cooled by closed loop stator winding cooling water. The stator winding water cooling system supplied shall be complete in all respects and include pumps, water tank, filters, coolers, gauges, level indicator switches and alarms. Any treatment necessary to make the condensate or demineralised water suitable for make-up to the stator winding water-cooling system or for continuous treatment of the water circulating in the winding circuit shall be provided in this system.

For each generator, $2 \times 100\%$ water coolers in parallel shall be provided. Also, two 100% AC motor-driven pumps shall be provided to circulate the stator winding coolant, one acting as standby and the electrical power supplied from a separate source. The two pumps shall be selective to be either active or standby and the standby one shall automatically cut in as required.

The capacity of the coolers shall be adequately to cool the stator water to 48°C when secondary cooling medium is supplied at 38°C .

The water pressure at the inlet to the coolers shall not exceed the minimum permitted pressure for generator operation, but shall provide a pressure margin over saturation pressure corresponding to the highest pressure.

All headers, water boxes, section of pipework, that can be isolated by the operation of valves but through which heat could flow, either by leakage of media or by conduction, shall be fitted with butterfly valves set to relief at a pressure just below the maximum design pressure. Pipework, pumps, strainers, valves, coolers and other ancillary equipment in the stator winding water circuit and in contact with the stator winding water or water vapour shall be made of stainless steel.

The stator winding cooling water manifolds shall be insulated from the stator frame to enable stator winding insulation resistance (IR) test.

A control panel shall be provided complete with pressure gauges, temperature and flow indicators and conductivity recorder, together with alarm indications for the efficient monitoring of the water-cooling system. The water pressure shall be automatically controlled so that the maximum pressure is maintained below the hydrogen pressure. A “liquid in casing” detector shall be provided together with the necessary alarm system.

5.6 Hydrogen Cooling

A hydrogen cooling surface system shall be required at middle of generator shell to maintain the hydrogen temperatures within IEC60034 specified limits. Hydrogen gas shall be used as the cooling medium for the rotor and air gap inside the generator. The hydrogen gas is forcedly circulated through coolers by cooling fans coaxially fitted with shaft of generator. Water for coolers is supplied from cooler inlet water system via automatic controlled valve.

The generator gas shall be circulated through the main components by at least one radial flow fan mounted at end of the rotor. This fan shall deliver cold gas to both end regions of the generator. The gas shall flow through separate paths in stator core, air gap, rotor winding and rotor end winding/air gap of the machine where it shall leave through vents. By virtue of the cold gas flow the outside of the stator casing shall be cooled.

The generator hydrogen pressure shall be maintained automatically at a set value which shall be adjustable.

Control equipment shall be provided to automatically maintain generator coolants at the required temperature by regulation of water flow through the various coolers.

A covered water head tank shall be provided complete with orifice and discharge piping and valves.

Any apparatus necessary for the withdrawal of the hydrogen coolers from the generator casing shall be provided.

Facilities shall be provided for filling the generator with dry air and for passing dry air continuously when the generator is open for maintenance.

An indicating and integrating flow meter shall be provided for measurement of the hydrogen consumption rate of the generator if necessary as per OEM's standard. The range of the meter shall cover the flow rate for a "tight" generator up to the flow rate at which the Unit shall be taken out of service. The maximum rate of hydrogen leakage from the generator shall be stated.

Detraining facilities as mentioned in section 8.3.1 shall be made for the releasing of hydrogen leakage into the cooling water system in service and initiating alarm when measured water level value is lower than set value. Portable/handy type hydrogen leakage detector shall be provided as per OEM's standard.

Equipment shall also be provided for measurement of the gas mixture purity of hydrogen in carbon dioxide and air in carbon dioxide and signals provided for computer monitoring.

All necessary hydrogen vent lines shall be provided and shall discharge at least 2m above the turbine house roof. All hydrogen piping routes shall avoid areas subject to a hazard from mechanical or fire damage.

The generator gas coolers shall provide adequate cooling capacity to support Unit full load. The generator shall maintain at least 67% output should one of the coolers be out of service in accordance with IEC 60034-3.

The capacity of the coolers shall be adequate to cool the hydrogen to 45 °C when secondary cooling medium is supplied at 38 °C.

Wherever an area is subject to the potential leakage of hydrogen, equipment shall be designed and arranged to ensure that the area shall have natural and adequate ventilation in order to avoid the formation of dangerous pockets of gas. These areas shall be classified in accordance with IEC 60079 or equivalent, and all equipment installed in the area shall comply with the requirements of explosion proof or of the nature of intrinsically safe.

Three separate temperature detectors for each side hydrogen coolers shall be provided to initiate the casing hydrogen high temperature alarms and trips.

5.7 Gases Exchange System

Carbon dioxide, for scavenging, and hydrogen shall be supplied to the gas control rack and valve station by bulk storage and bus mains.

Filling with hydrogen or purging with CO₂ shall be a safe and efficient operation and the time taken to purge a generator of hydrogen shall not exceed 6 hours. Heaters shall be supplied if necessary to ensure that the temperature of the CO₂ entering the generator casing during purging is not damaging to the welded structure or other components.

A mechanical interlocking arrangement for the hydrogen and air admission/venting valves shall be provided to ensure the correct order of gas admission. Provisions shall be made for physically disconnecting and isolating the hydrogen supply during maintenance periods.

A gas exchange control rack located in an accessible area remote from where burning oil or other hazardous material can collect in the event of a major disruption and above the maximum flood level shall be supplied to monitor gas supplies and to maintain casing hydrogen purity at not less than 96%.

Pipework and equipment containing hydrogen shall be in one section of the rack and sealed from the electrical sections, and ventilated to atmosphere to eliminate the possible formation of an explosive mixture of hydrogen and air.

If required as per OEM's standard, a silicon gel dryer equipped with auto regeneration device through which hydrogen from the generator casing is continually circulated shall be provided. The desiccant in this dryer shall be automatically reactivated and the water extracted shall be automatically drained. If a hand operated single dryer system is proposed as an alternative, the regeneration shall be done at a frequency of approximately every two weeks and the regeneration cycle shall be completed in 8 hours.

If required as per OEM's standard, the dew point of hydrogen at the gas dryer inlet and outlet shall be continuously monitored and recorded. Alternatively, it is acceptable to install dew point monitor at inlet only. Hydrogen pipes shall be run well clear of cable runs. An integrating flow meter shall be supplied to monitor the consumption of hydrogen during normal operation if necessary as per OEM's standard. An advance gas leakage detection system shall be provided to detect the gas leakage in the system and the generator. Alternatively, gas leakage detectors can be integrated to control system. Excessive leakage shall be alarmed to CCR. Hydrogen purity measurement device shall be provided with alarm signal to CCR.

All pipework and equipment that may be subject to the leakage of hydrogen shall be positioned so that the surrounding area has natural and adequate ventilation in order to avoid the formation of dangerous gas pockets. This applies particularly to the generator connection area under the stator casing and to slip ring housings. In these areas, flanged joints shall not be used unless essential for maintenance purposes. On the whole, all hydrogen pipe line shall be designed above ground level.

5.8 Excitation

5.8.1 General

A continuous fast acting automatic excitation control system (AVR/PSS) of a proven design shall be provided to control the generator voltage without hunting/instability over the entire operating range of the generator.

The excitation system shall be provided with a fast-acting MVAR limiter so as to prevent the generator output falling outside its safe limit. A power system stabilizer shall be incorporated in the excitation system of the generator. Protection features as part of the system shall include over-voltage, over-current, VT fuse failure, over-fluxing, and AVR power supply failure. A field shorting or discharge switch feature shall also be included in the system as protection against overstressing the generator insulation in the event of a fault. The PSS shall provide supplementary control signals to the AVR for damping voltage oscillation in the power system.

Manual excitation control facilities shall be provided as a backup to the automatic control, and shall have an adequate range to allow for control of excitation mainly for commissioning and maintenance purposes. A true null balance shall be provided to allow for smooth excitation transfer between manual and automatic control.

The excitation equipment shall be rated for continuous operation over the full range of generator rated conditions and comply with IEC 60034-3.

It shall be possible to remove the exciter for repair without the need to dismantle the main generator.

5.8.2 Performance

The ceiling level of field voltage shall be at least 2 times the maximum continuous rating of field voltage when generator is at rated terminal voltage and rated current shall be 120% of the normal field current. The excitation system shall be capable of providing the peak power required for operation at ceiling voltage level.

The excitation system shall be of the high initial response type, with a response ratio such that excitation increases from rated load excitation voltage 1.0 pu to a voltage greater than 2 pu in 0.1 secs in response to a step change in generator terminal voltage of 2.5%.

This requirement is based on an operating duty cycle of rated output for a sustained period, then ceiling voltage for 10 seconds, then return to rated output for a sustained period.

Alternatively, ceiling level/rated field voltage, rated field current of different values and different parameters to define high initial response (e.g. response ratio, response time, magnitude of step change, duty cycle) as per OEM's standard can be accepted but details shall be submitted for the Owner's review during contract execution stage.

The Offshore Supply Contractor shall provide, during the design stage, performance details and test curves for the excitation system.

5.8.3 Static Thyristors Excitation System

5.8.3.1 Thyristors

The thyristor bridge shall comply with the requirements of IEC 60146.

A three-phase fully-controlled thyristor bridge shall be provided with sufficient redundancy to permit operation of the generator at full excitation power with 20% of the thyristors failed in any set of parallel paths i.e. arm of the bridge. It shall be possible to either replace faulty thyristors with the generator in service, or the extent of thyristor redundancy shall be sufficient, having regard to their reliability, to permit full excitation power without replacing thyristors more often than once every 8000 hours.

Sufficient redundancy shall be included in the thyristor cooling system to ensure that a single failure will not prevent the achievement of continuous 100% generator rated output.

The Offshore Supply Contractor shall provide, during the design stage a description of mitigation measures to prevent thyristor damage in the following situations and what tests are performed to confirm the effectiveness of these measures.

- A short-circuit at the generator slip rings at excitation ceiling voltage

- The overload occurring with a sudden short-circuit at the generator terminals
- High voltage across the bridge due to reverse excitation current
- Non-uniform current distribution between thyristors

5.8.3.2 Excitation Transformer

A power transformer shall be provided to supply the power requirements of the excitation system under all operating conditions. This transformer shall be a three-phase dry type natural air cooled and manufactured to the appropriate IEC standard for indoor installation.

The design of excitation transformer shall take into account the additional stray and eddy current losses generated by the harmonics supplied to the load. The Offshore Supply Contractor shall calculate the extra losses due to harmonics by reference to ANSI/IEEE C57.110.

The expected demand for excitation power (MVA) at rated generator output shall be stated.

5.8.3.3 Slip Rings and Brush Gear

The slip rings and brush gear shall be of proven design and, in addition to the requirements of IEC 60034, shall comply with the limits of temperature set out in BS 4999: Part 101 Table III, or an equivalent standard.

The Offshore Supply Contractor shall state the slip ring temperature at generator rated output.

The Offshore Supply Contractor shall describe how the generator will be tested to determine that this limit is not exceeded.

The wear rates of the slip rings with the generator operating at continuous maximum rating shall not be less than 8000 hours of operation whereas the wear rate of brushes depends on operation condition, which can be replaced on line

The brush pressure shall not require adjustment between the time of installation and the time of replacement of the brush. The construction shall permit safe and convenient replacement of the brushes with the generator in service. The Offshore Supply Contractor shall provide details of the provisions for safeguarding personnel engaged on "in service" brush replacement. Bolted connections shall be provided that it will permit the polarity to be reversed.

5.8.3.4 Field Suppression

A main field circuit breaker complying with IEC60947 Parts 1 and 2, and a non-inductive resistor shall be provided for generator de-excitation. The circuit breaker shall be capable of breaking the field current under the most onerous operating conditions. On opening the breaker, the field winding shall be short-circuited through the resistor which shall ensure that the rotor winding voltage remains adequately below its test voltage. The field circuit breaker shall have two trip coils which are electrically and mechanically independent.

5.8.4 Supervision

Equipment shall be provided for detection of a generator rotor earth fault and measurement of generator rotor field voltage and current. The field voltage and current measurements shall be monitored by DCS which can be used to compute the field winding temperature.

5.8.5 Excitation Control

The excitation control equipment shall consist of an AVR (with power factor controller) and PSS.

The excitation control shall include the following protective device which remove the faulty equipment from service, trip to manual if necessary, and bring up appropriate alarms and indications:

- a) Under-excitation - operates if the converter output is low in relation to demand.
- b) Over-excitation - operates if sustained high levels of excitation are attained.
- c) Over-current - operates when excessive current draw from the excitation supply.
- d) Over voltage - operates when terminal voltage is high in relation to demand.
- e) VT supply failure - operates if the VT supply is lost.
- f) Over-fluxing - operates if the terminal voltage is high in relation to frequency.
- g) AVR power Supply - Operates if the AVR system power supply fails.

The excitation shall be rapidly suppressed in the event of a major fault or main protection trip.

5.9 Seal Oil System

A seal oil unit shall be provided to contain safely the hydrogen gas within the generator housing to prevent hydrogen from leaking out of generator casing to atmosphere

The seal oil source shall be taken from the main lube oil pump upstream of lube oil control valve or alternatively, a separate seal oil system. The seal oil pressure in the system shall be automatically maintained at a preset point by a seal oil regulator. The seal oil in the air side discharged from sealing packing shall be routed back to the main oil tank through a set of stabilizing and degasifying equipment.

The system shall also be provided with adequate monitoring and protecting equipment including liquid detectors etc.

Shaft oil seals of a proven reliable design shall be provided. Details of the seals shall be provided by the Offshore Supply Contractor. The seal oil to hydrogen differential pressure shall be maintained automatically at all pressures up to the maximum hydrogen pressure, irrespective of the source of seal oil. The system shall be capable of sealing rated gas pressure in the generator under any conditions.

Two 100% seal oil pumps (1 AC motor driven main seal oil pump and 1 DC motor driven emergency seal oil pump), shall be provided to maintain hydrogen purity inside generator frame. Seal oil shall be taken from separate oil system. The loss of all seal oil pumps shall not result in damaging the seals.

Two 100% seal oil coolers shall be provided if required as per OEM's standard. These shall be cooled from the auxiliary cooling water system and shall be parallel on the water side.

Equipment shall be provided for the treatment of seal oil to removed dissolved gas so as to minimize the contamination of hydrogen in the generator. Alternatively, if continuous hydrogen scavenging system as per OEM's standard is applied, seal oil storage tanks and solenoid valve, vacuum treatment system could not be provided.

To minimize fire risk, the tanks (if required), pumps, coolers (if required), valves and pipework shall be assembled as a compact unit supported from the basement floor. All necessary earth bonding shall be provided to eliminate the possibility of a spark occurring in a pipe or tank (if required) which may contain hydrogen. Stainless steel pipes shall be used between generator and the seal oil system.

5.10 Generator Earthing

5.10.1 Neutral Point

The generator stator winding, which will be star connected, shall be suitable for earthing through a single-phase distribution-type double-wound transformer, the secondary winding of which will be shunted by a non-inductive resistance. The Onshore Contractor shall supply details of the capacitance to earth of the stator windings, which is necessary for selection a final resistance value, as part of the commissioning report. The Offshore Supply Contractor shall nominate the expected value of capacitance.

5.10.2 Casing

Earthing terminals or bolts shall be provided at two positions on the stator bedplate for earthing the generator casing. These positions shall be connected at two different locations on the station earthing system.

Major turbine generator components shall be electrically bonded together and connected to the generator earth terminal to provide a low impedance return path for possible Radio Frequency (RF) voltage transients generated by thyristor switching in the generator static excitation circuit.

5.10.3 Turbine/Generator Shaft Voltage/Current Control

The shaft shall be earthed at one position between LP turbine and generator bearings by suitable multiple (at least two) shaft earthing devices.

- (a) The shaft earthing devices shall be protected from the effects of oil, dust, steam and other contaminants.
- (b) Each shaft earthing device shall have 0.01 to 0.1 Ω in-service contact resistance to the shaft at the rated speed and within permissible shaft vibration levels.
- (c) The contact element of each shaft earthing device shall be able to be replaced with the generator in service without safety risk either to a person or to the machine.

Each shaft earthing device shall be connected to the turbine/generator earth bonding conductors or lower bearing case via suitably sized low impedance current measuring shunt.

The generator outboard bearings and preferably HP turbine bearing No. 1 shall be electrically insulated from the generator earth.

Shaft voltage test terminals grouped in one single easily accessible location shall be provided for checking the voltages and insulation integrity for the following:

- (a) Insulated bearings

- (b) Insulated hydrogen seals
- (c) Insulated oil piping
- (d) Other insulated components as necessary

A continuous on-line shaft earthing monitor shall be provided as an integral part of the shaft earthing system.

The monitor shall measure earthing current and shaft voltage, and display these at a local panel adjacent to the generator. It shall also activate at least one "clean" changeover alarm contact.

Alarms shall be generated to DCS for the following conditions:

- (a) "Loss of power supply" to the monitor
- (b) "Shaft earthing current low" (deterioration of shaft earthing device)
- (c) "Shaft insulation low" (loss of insulation integrity)

The Offshore Supply Contractor shall provide details of shaft voltage/current control scheme monitors the bearing insulation, shaft coupling insulation, shaft earthing devices and the like during the design stage.

5.11 Generator Condition Monitoring

The Offshore Supply Contractor shall provide the generator condition monitoring equipment.

A generator core monitor shall be provided and of the type which samples hydrogen gas for the presence of particulates arising from severe overheating of the core and other machine components and raises an alarm.

The rotor shorted-turn detection coil or equivalent system shall be fitted to the generator with monitors provided.

End winding vibration sensors shall be fitted to the stator end windings at both ends of the generator with monitors provided.

High speed electronic fault recorder shall be provided for transient fault recording of electrical parameters.

5.12 Generator Casing Purging System

The system shall permit purging the generator of hydrogen using carbon dioxide, or of carbon dioxide using compressed air, in less than 150 minutes with the generator stationary.

The characteristics of the carbon dioxide supply shall satisfy the design requirement.

A panel shall be provided from which all operations necessary for purging the generator can be carried out. The panel shall accommodate all monitoring equipment associated with the purging system. The panel shall be located on the basement level of the turbine house remote from the turbo-generator foundation block, so that it can be reached safely in the event of a generator fire. Alternatively, valve station with instrument rack can be applied instead of panel as per OEM's standard.

The Offshore Supply Contractor shall provide all necessary heating equipment should the carbon dioxide require warming prior to injection into the generator.

Provision shall be made for a 'visible-break' between the generator and the sources of hydrogen, carbon dioxide and air so as to provide proof of isolation before access is made available for maintenance of the system. The visible breaks may consist, for instance, of flexible hoses with quick disconnect couplings, or removable sections of pipe work.

5.13 Temperature Measurement

Temperature detectors shall be provided as described below and in compliance with IEC 60034. The detectors shall be either copper constant in thermocouples or platinum resistance thermometers, and shall be cabled to and terminated in junction boxes at the generator.

Duplex type (redundant) embedded temperature detectors shall be provided for measurement of generator temperatures as follows:

- (a) A minimum of six detectors per phase in the generator stator slots between the coil sides.
- (b) A minimum of twelve detectors in the stator core

Resistor Temperature Device shall be provided for temperature measurement as follows:

- (a) Cooling air temperature across the generator slip rings
- (b) Hydrogen temperature at the inlet and outlet of hydrogen coolers
- (c) Seal oil temperature at the inlet of each hydrogen seal.
- (d) Cooling water temperature at the inlet and outlet of all heat exchangers

Each generator shall be equipped with one internal on-load discharge or insulation monitoring system and this system shall comprise the following equipment and functions:

- (a) Coupled capacitor, Radio Frequency Current Transformer (RFCT), coils to get signal about the generator condition, radio frequency voltage transformer and RTD module.
- (b) All of the output signals will be converted into 4-20mA standard signal.
- (c) Equipment shall have the ability of not being impacted by noise and ensuring the precision of input signals.
- (d) There shall be an individual channel to reduce noise signal and get rid of noise interference.
- (e) The system shall be capable of displaying and alarming internal discharge/heating situations of generator at 2 levels: abnormal warning and fault alarming (forecasting faults and alarming when faults happen)
- (f) Easily install program through control keys and display monitor.
- (g) Information ports for remote and local communication.
- (h) Local analysis packs for each unit. All of the local analysis packs for each unit shall be common connected together (in ring configuration) having one display monitor and one history printer and it shall be connected to the DCS system of the station to transfer and display all signals at the central control room.

5.14 Not Used

5.15 Sub-synchronous Resonance Protection Relay

Sub-synchronous resonance (SSR) is a condition that can exist on a power system where the network has natural frequencies that fall below the fundamental frequency of the generated voltages and produce sub-synchronous oscillation of the steam turbine and generator shaft.

Turbine-generators can be subject to electrical environments that react with turbine-generator shafts to produce sub-synchronous resonance at shaft natural frequencies.

During design submission stage, the Offshore Services Contractor shall submit user's manual including factory test report reports and standard drawings to the Owner for review. The Offshore Supply Contractor shall provide two SSR protection relays (one for each turbine-generator) for continuous protection of the turbine-generators from SSR. SSR event data capture shall be provided.

Chapter 6 – Electrical Systems

6.1 General

The proposed electrical systems shall comply with the relevant IEC or equivalent. The design shall also meet the Vietnamese national grid code, grid connection agreement and system impact study for connection requirements. The Facility shall operate satisfactorily under the specified normal loading conditions, as well as sudden variations of load and voltage during emergency conditions, including those due to starting loads and short circuit and other fault conditions. All equipment shall be designed for a rating equivalent to a minimum fault level of 50 kA (rms) for 1 second for low voltage (LV) system (below 11 kV), 40 kA (rms) for 1 second for medium voltage (MV) system (11 kV) and 50 kA (rms) for 1 seconds for high voltage (HV) system (500kV).

The design shall include safeguards and provisions for the safety of all personnel during commissioning, operation, and maintenance of the Facility. Where provision is essential to ensure continual operation of the Facility, appropriate redundancy shall be provided so as to permit maintenance with minimum interference.

The design of the electrical systems shall be such that all maintenance and operational activities can be performed safely and with ease. Means shall be provided to prevent unauthorized access to the equipment. In addition, the electrical systems shall be provided with interlocks to prevent accidental contact with live parts. Equipment such as switchgear and control cubicles shall be provided with a main earth bar for the equipotential bonding to earth of all exposed conductive parts, which may become alive under fault conditions.

Equipment enclosure shall provide a degree of protection of not less than class IP32 for indoor installation and IP55 or equivalent international standard for outdoor installation. Degree of protection for DCS control cabinets shall be of not less than IP21. Equipment enclosure shall be provided for hazardous area and dusty areas. Indoor equipment shall be designed for an ambient temperature of 40 °C (except for air conditioned battery rooms) and outdoor equipment for 50 °C. Cable access to outdoor equipment shall be by bottom entry. Cable trays and ducts, where ever possible, shall be laid above ground level for easy maintenance. For outdoor panels, connection boxes or cubicles, it shall be made of stainless steel or hot dip galvanized steel with suitable shelter against rain water ingress.

The electrical systems shall be designed to minimize the generation of harmonics and the effects of electrical interference between power and control/instrumentation circuits, and comply with international requirements on electromagnetic compatibility.

The Offshore Supply Contractor shall provide a complete and comprehensive protection system for the generators/generator step-up transformers/unit auxiliary transformers, transmission lines and the station electrical distribution systems during the detailed engineering stage. The Onshore Contractor shall undertake the installation of the protection relay panels within the control room or unit electrical building, and commissioning tests of the protection schemes. Protection shall be coordinated with the Vietnamese grid requirements.

6.2 Not Used

6.3 Plant Voltage Levels

The voltage levels for the power distribution system including generation, auxiliary power, controls, transmission and start-up power shall be designed as stated below. Nominal system frequency shall be 50 Hz. Power station and system voltages shall be selected from IEC 60038 and shall conform to the existing voltages currently used by the power grid. In addition, all equipment shall be capable of operating over the range $\pm 10\%$ of the nominal voltage.

<u>System</u>	<u>Nominal Voltage</u>
Power Evacuation	500 kV
Generator Voltage	23.5 kV
Station Voltage	11 kV / 400 V
DC Power / Control	220 V / 24 V

Comprehensive fault level study, load flow study, voltage regulation / drop study and protection coordination study shall be submitted in design stage to ensure that the equipment provided is adequately rated for the power supply services. These studies shall cover worst case short circuit conditions. Calculations of these studies shall include voltage drop when starting the largest motor with all other required auxiliaries running. Fault level studies shall take full account of the contribution from motors and the increased fault levels arising from the running of diesel generators parallel with the supply system during on-load testing.

The switching sequence of circuit breakers for the startup of the First Unit and the startup of the Second Unit shall be stated in the commissioning documents for Owner's review during Project execution stage.

The electrical auxiliaries system shall be designed on the unitized principle. Power for the supply of all the auxiliaries associated with the starting and running of each Unit shall be supplied from Unit boards. Equipment which does not have an immediate effect on the running of the Unit shall be supplied from station or common services boards.

Power supply for Unit startup shall be a back-energisation type. No starting transformer is required.

6.4 Electrical Standards

All equipment shall comply with the following requirements or equivalent.

IEEE C37.23	- Standard for Metal-Enclosed Bus
IEC60071 Part 1	- Insulation Co-ordination - Definitions, Principles and Rules
IEC60071 Part 2	- Insulation Co-ordination - Application Guide
BS 159	- Specification for High Voltage Busbars and Busbar Connections
BS 1977	- Specification for High Conductivity Copper Tubes for Electrical Purposes
BS 2898	- Specification for Wrought Aluminum and Aluminum Alloys for Electrical Purposes - Bars, Extruded Round and Sections
IEC60168	- Tests on Indoor and Outdoor Post Insulators of Ceramic Materials or Glass for Systems with Nominal Voltages Greater than 1000 V
IEC60137	- Bushings for Alternating Voltages above 1000V
IEC60076 Part 1	- Power Transformers – General
IEC60076 Part 2	- Power Transformers - Temperature Rise
IEC60076 Part 3	- Power Transformers - Insulation Levels and Dielectric Test
IEC60076 Part 3-1	- Power Transformers - Insulation Levels and Dielectric Tests - External Clearances in Air
IEC60076 Part 5	- Power Transformers - Ability to Withstand Short Circuit
IEC60726	- Dry Type Power Transformers
IEC60129	- Alternating Current Disconnectors (Isolators) and Earthing Switches
IEC60186	- Voltage Transformers
IEC60060 Part 1	- High Voltage Test Techniques - General Definitions and Test Requirements and Test Requirements
IEC60060 Part 2	- High Voltage Test Techniques - Measuring Systems
IEC60185	- Current Transformers
ISO 9606-2	- Approval Testing of Welds - Fusion Welding -Aluminum and Aluminum Alloys
ISO 3777	- Radiograph Inspection of Resistance Spot Welds for Aluminum and its Alloys - Recommended Practice

- ISO 2437 - Recommended Practice for the X-ray Inspection of Fusion Welded Butt Joints for Aluminum and its Alloys and Magnesium and its Alloys 5 to 50 mm thick.

6.5 Isolated Phase Busbar

Isolated phase busbar (IPB) shall be used for connections between the generator and its associated generator circuit breaker, and generator step-up transformer, with taps for high voltage terminals for Unit auxiliary transformer, excitation transformer, and the generator voltage transformers and surge protection system cubicles.

The IPB shall be outdoors and enclosed type with weather-proof dust-proof housing to degree of protection of IP 55. It shall be naturally cooling type. Conductor shall be made of copper or aluminum which possess high conductivity, and supported by porcelain insulators at appropriate intervals. Conductor, insulator, and supporting steel frame shall be so constructed to withstand the mechanical and thermal stress due to momentary current and short circuit conditions. Flexible connections shall be provided at connecting points to adjacent equipment and at intermediate points in order to alleviate the stress which is due to vibration and thermal expansion. At the point of bolted connection, the terminal surface shall be silver plated. The enclosure of isolated phase bus shall be made of aluminum alloy which shall possess high conductivity. Flexible connection of enclosure shall be provided where the vibration-proof connection is needed, and rubber or aluminum bellows shall be adopted. The inspection manhole shall be provided in suitable place of main bus enclosure such as connection part with generator and transformers. Access shall be provided for periodical inspection.

Necessary accessories like capacitive voltage transformer (CVT) shall be installed in an enclosed metallic cubicle. Generator neutral equipment shall also be installed in the isolated metallic cubicle. Removable links shall be fitted in the IPB adjacent to all transformer connections such that any given transformer can be disconnected readily for maintenance and testing. The earthing type of the IPB shall be multiple points earthed in stipulated distance. Earthing points shall be provided on the busbars to enable maintenance to be carried out.

Tee-off connection to Unit auxiliary transformers shall be equipped with disconnection between the transformers and the bus duct for maintenance or reparation.

IPB cubicle shall be of sheet steel construction. The cubicles are designed for the safe isolation and earthing of respective voltage transformers. Indication of transformer service status- isolated or earthed shall be provided. Live parts of a voltage transformer shall be solidly covered to prevent contact with operational personnel upon their access to the voltage transformer panel. For the prevention of the insulators from getting dew, the IPB shall be supplied with the amount of clean and dry air from instrument air system and to ensure that the IPB can work under positive pressure condition. A set of moisture detector to monitor the moisture in the IPB shall be provided.

Generator neutral enclosure including the neutral make-up connections with phase segregated bus bars shall be provided. This shall also include short piece of bus duct connection from the star point to generator neutral grounding cubicle.

Generator neutral grounding cubicle containing neutral grounding transformer and secondary loading resistor shall be provided.

Surge protection and voltage transformer cubicle containing voltage transformers, lightning arresters and surge capacitors shall be provided.

The IPB shall carry the rated current continuously at ambient temperature of 50 °C without exceeding the following temperature limits:

- Main conductor 105 °C
- Bus connections 105 °C
- Busduct enclosures and support structures: 80°C

The effect of solar radiation to the IPB shall be considered.

6.6 Generator Circuit Breaker

Generator circuit breaker (GCB) shall be provided between the generator and the generator step-up transformer. The GCB shall be used to synchronize the generator to the 500kV system. The system configuration shall facilitate the back feed of plant auxiliary loads through the generator step-up transformer and Unit auxiliary transformer. The GCB shall be the three-single pole, sulphur hexafluoride (SF₆) insulated, self-extinguishing interrupting single throw design type, with a pneumatic or hydraulic high-speed operating mechanism and mechanical linkage to open and to close all three poles of the breaker in synchronous operation. The SF₆ gas shall be capable of interrupting all currents between 0 and 100% of rated interrupting capacity. The operating mechanism shall be designed for successful local and remote electrical operation. The operating mechanism shall be provided with two electrically independent trip coils and all necessary circuits, including pressure switches and reset devices to permit tripping the circuit breaker from primary and backup protection relaying.

The GCB shall comply with the following:

- Quantity: 2
- Nominal voltage rating: generator rated voltage
- Maximum generator operating voltage: nominal voltage rating + 5%
- BIL 50 Hz, min: 60kV
- BIL Crest, full wave 1.2 x 50 µs: 125kV
- Continuous current rating: generator rated current

- Short circuit rating at maximum rated kV, Amp, 1 Sec: 100 kA, rms
- Momentary current rating: 270 kA rms
- Rated interrupting time: 5 Cycles

Surge protection and VT cubicle containing VTs, lightning arresters and surge capacitors shall be provided with the GCB assembly.

6.7 Power Transformers

All power transformers shall meet the following requirements, unless otherwise specified in the respective section. Transformers shall comply with IEC 60076.

The transformers shall be ONAN/ ONAF/ ODAF cooling type. The Offshore Supply Contractor shall provide ONAN/ ONAF/ ODAF capacities and the graph of MVA against ambient temperature during detailed engineering stage. The transformers shall be provided with $2 \times 50\%$ cooling units each complete with its radiator banks, fans, pumps and other accessories. The design shall be such that the rated transformer output corresponding to the specified temperature rise can be maintained:

- (a) Continuously with one pump and one fan in any of the cooling bank out of service.
- (b) For at least ten (10) minutes in case of failure of the complete cooling system when the transformer was delivering its rated output.

The transformer shall be capable of continuous operation at specified rating under the following conditions:

- (a) Voltage variation $\pm 10\%$
- (b) Frequency variation $- 5\%$ to $+ 3\%$
- (c) Combined voltage and frequency variation (absolute sum) 10%

The transformer shall be capable of withstanding the short circuit stresses owing to a terminal fault on one winding with full voltage maintained on the other winding for minimum duration of two seconds.

Bushings rated above 38kV shall be oil filled, self-contained, hermetically sealed condenser type with anti - fog shed. Bushings rated 38 kV class and below shall be solid porcelain type. Bushings shall be made with adequate creepage distance for the heavily polluted atmospheric condition.

6.7.1 Generator Step-up Transformer

One three-phase generator step-up transformer shall be provided for each Unit. It shall be placed in the transformer yard on a rail for easy transportation. The

transformer shall include the on-load tap changer of $\pm 12 \times 0.833\%$ on HV side with Automatic Voltage Regulation (“AVR”), and the tap changer shall be able to adopt both manual and automatic operation mode by operator's choice. The cooling system of the transformer shall be ONAN/ONAF/ODAF type.

Two additional years of warranty of the transformer shall be provided.

There shall be $4 \times 50\%$ oil pumps for the generator step-up transformer. Each transformer shall have adequate number of cooling fans. The fans shall be divided into two groups. Each group shall have one standby cooling fan as a backup. If all oil pumps (4 numbers) are out of services, the transformer shall still be able to maintain 80% capacity by ONAF. If all cooling fans are out of services as well, the transformer shall still be able to maintain 60% load by ONAN.

HV windings can be equipped with graded insulation while LV winding shall be uniformly insulated. The impedance of the transformer at normal tap shall be around 15%. The noise level of the transformer shall be restricted not to exceed 85 dB(A) measured at 1 meter.

HV, HVN and LV bushings shall be supplied by ABB (Switzerland).

The generator step-up transformer shall be provided with winding temperature indicator having contacts for temperature high alarm and trip functions.

Each phase of generator step-up transformer bushing connections to external circuits shall be provided with lightning surge arresters .

Generator step-up transformer shall be completed with the following fittings:

- Air bag + silicone gel breather.
- Buchholz relay - double float with alarm and trip contacts.
- Buchholz gas, on load sampling point at ground level.
- Spring operated self-resealing pressure relief device (two for oil quantities exceeding 45,000 liters).
- Tank drain valve.
- Combined filling and filter valve.
- Oil sampling device.
- Approved anti-vibration mountings.
- HV and LV winding temperature indicators having contacts for control of pumps and fans, and alarm and trip contacts for high temperature.
- Thermometer pocket.
- Oil flow indicator with alarm contact for loss of flow.
- Cooler isolating valves.
- Pump inlet and outlet isolating valves.

- Low oil level alarm device.
- Earthing terminal
- Surge arresters mounted on the high voltage terminals shall be connected to the high voltage leads to protect the generator step-up transformer from voltage surges.
- On-line dissolved gas analyzer (DGA) shall be provided.
- A concrete containment basin with sump shall be provided to contain oil in the event of a tank rupture with a provision for draining it out. Fire walls shall be provided on three sides of the transformer.

A transformer condition monitoring system shall be provided. Failure of the monitoring system shall not affect the transformer operation. The system shall provide the following benefits on the transformer operation:

- Early detection of incipient faults;
- Continuous supervision;
- Enabling condition-based maintenance and hence reduction of life cycle cost;
- Knowledge of lifetime consumption.

The transformer condition monitoring system coverage shall also include on-line condition monitoring on the following areas:

- Generator step-up transformer insulation;
- Generator step-up transformer bushing;
- On-line tap changer;
- DGA.

The MVA rating of each generator step-up transformer shall be selected such that they shall be capable of handling the maximum output of the respective generator at turbine valve wide open (VWO) rating minus by the auxiliary electrical power consumption required for the full load operation of the unit, coupled with the electrical power for the supporting balance of plant.

The transformer shall be able to withstand over excitation condition value at the instant of total full load rejection for a period of 5 seconds and the over excitation value after 5 seconds for a period of 1 minute.

Restricted earth fault (REF) protection shall be provided for generator step-up transformer.

Information on the proper installation of bushing shall be provided. The Onshore Contractor shall connect the prepared termination of the cable ends to the transformer bushing and surge arrestor accordingly.

After completion of site acceptance test, full finger print of the transformer shall be provided. The finger print shall include but not be limited to the following:

- Partial discharge distribution;
- Temperature distribution;

- Thermal images;
- Noise level distribution.

The Offshore Supply Contractor shall establish a strict supplier management system, including supplier selection, classification audit, annual process audit, inspection in supplier workshop and incoming inspection. The quality performance of supplier shall be closely followed with a supplier quality database. X ray picture shall be taken for all insulation mould parts, such as angle ring. Scrap shall be implemented whenever particle, bubble or air gap is found. A grade inspection rule & hold point special inspection system shall be implemented to locate and correct the problems during manufacturing process on time. In addition, the cleanliness in the manufacturing workshop shall under continuous improvement. All insulation, winding, other components and material shall be covered.

Master insulation coordination for a transformer shall be defined and the design for the project shall be validated. Simulation and analysis shall be made to evaluate voltage distribution on terminals and in winding using computer program. Maximum over voltage and safety margin shall be taken into account for design. The capability of short circuit withstanding shall also be calculated and validated for the design. Design scheme shall be optimized and short circuit force endured by winding & active part shall be reduced. Reliable structure design shall be developed to strengthen the transformer capability of short circuit withstanding. A number of measures including: strictly control winding dimension, twice constant pressure vacuum drying of winding, pre-assembly process for active part and pre-pressure of short circuit withstanding of active part shall be implemented during manufacturing process to strengthen the transformer capability of short circuit withstanding.

Standardized technology shall be applied, including design rules and tools. Standard design review and quality control program shall be implemented.

SLIM tool shall be used to calculate 3D magnetic field of the transformer and analyze distribution of stray loss and temperature field. Thermo Hydraulic Model & Excool program shall be used for temperature calculation to analyze oil flow distribution and winding temperature distribution. Technical information on all the applied measures to mitigate the transformer insulation breakdown at the bend of HV winding lead shall be provided.

In additional to the standard shop test, the following additional factory tests shall be provided to the transformer:

- Oil flow partial discharge Test;
- Heat Imaging Tests;
 - o Test on iron core right after manufacturing;
 - o Test on generator step-up transformer as part of the factory acceptance test;
- 18 hours heat run test;
- 1.1Ur, 12 hours over exciting test.

Frequency response test shall be performed to check if there are any loosen parts in the transformer.

The Silicon steel of the transformer shall be supplied by Nippon Steel Corporation, Japan. Nynas shall be subcontractor of the transformer insulation oil. Supplier of the breather shall be MR Messko Germany. The insulation material of the transformer shall use imported material from Weidmann, Germany. The Offshore Supply Contractor shall perform sampling checks on insulation materials as part of their quality plan, the percentage of sampling and the procedures shall be reverted back to the Owner. Suppliers and subcontractor list of the major components of the transformer shall also be provided. The manufacturing of the transformers shall be supervised by the Offshore Supply Contractor senior management to ensure high level of excellence in terms of manufacturing. A Quality Control Plan (QCP) shall be provided after detailed design review at contract kick-off with the leadership support of power transformer manufacturer and the Offshore Supply Contractor head quarter Quality and R&D team. The quality control measures specific to the transformer for the Project shall be provided.

The Offshore Supply Contractor shall advance transformer manufacturing in the overall project time schedule to give more float on the project programme to cater for any possible issue arising during the manufacturing, factory testing and commissioning of the transformer.

The Offshore Supply Contractor shall provide skill level distribution of all technical staff take part in the transformer manufacturing. Measures shall be taken to reduce the chance of transformer manufacturing human induced errors, in particular the winding work which is still largely a manual task. The three areas requiring special attention are provision of adequate training, elimination of time pressure/burden, and strengthening of supervision. An experienced supervisor from the Offshore Supply Contractor's headquarters shall provide shop supervision of the transformer manufacturing. Curriculum vitae of the proposed experienced supervisor shall be provided for the Owner review during Project execution stage.

6.7.2 Unit Auxiliary Transformer

The Unit auxiliary transformer (UAT) shall be of the split winding type with a rating not less than 75/40-40MVA. UAT shall be ONAF/ONAN cooled, oil-immersed type with on-load tap changer (OLTC) having range of $\pm 8 \times 1.25\%$ on HV side.

The UAT shall be provided with winding temperature indicator having contacts for temperature high alarm and trip functions.

The main auxiliary power supply system shall be supplied from a UAT with the high-voltage side directly connected to a branch of IPB between the GCB and generator step-up transformer. Each Unit shall have two UATs of three-phase three-winding transformer capable of supplying its associated Unit loads, and the station auxiliary load for both Units. Each 11kV winding shall have its own Unit auxiliary switchboard.

The 11kV Unit auxiliary switchboards of the Facility (two Units) shall be able to interconnect.

If one UAT of any Unit fails, this Unit shall still be able to restart and be operated by the electrical supply from the remaining healthy (three) UATs of both Units. This Unit shall be able to achieve at least 40% rated output power. During the restart of the failed Unit, the healthy Unit is allowed to be ramped down. The Offshore Service Contractor shall submit the startup procedures for Owner's review during design submission stage.

The Unit shall be able to start up by back energization from the grid. Furthermore, UAT shall be ready in advance to produce back energization power through EVN's 500 kV switchyard during commissioning period.

When two Units are generating, they shall be self-sufficient to provide power to all in-house loads of the Facility without the need to import power from the grid. The Offshore Services Contractor shall therefore calculate and add a certain margin to the UAT rating in order to achieve this "self-sufficient" requirement. A load of 1 MW shall be included into the rating of each UAT for the use by Owner's facilities (Owner's ash pond etc.). This 1 MW for each Unit shall be counted in the auxiliary power consumption guaranteed values.

UAT shall be completed with the following fittings:

- Silica-gel breather.
- Buchholz relay - double float with alarm and trip contacts.
- Spring operated self-resealing pressure relief device.
- Tank drain valve.
- Combined filling and filter valve.
- Oil sampling device.
- Oil temperature indicator having alarm and trip contacts.
- Low oil level alarm device.
- Cooler isolating valves (if separate free-standing cooler is provided).
- Radiator inlet and outlet isolating valves for each radiator (if tank mounted radiators are provided).
- Earthing terminal.
- On-line DGA.

6.7.3 LV Transformer

The LV transformers shall generally include 11 kV/ 400 V service transformers and 11kV/ 400 V Dry type lighting transformers.

Bushings shall be solid porcelain type. Bushings shall have adequate creepage distance corresponding to heavily polluted atmosphere.

Dry type transformer shall have a class F isolation: as per IEC 600076-11. Insulation system of 155C and average winding temperature rise limits of 80 K (class B). Dry type transformer shall be enclosed in sheet steel enclosure of at least IP 21 degree of protection, complete with door and interlocking facility.

Power from 11kV switchgears shall be stepped down to 400V voltage levels through 11kV/ 400V Dyn11 connected service transformers.

11kV/400 V transformers shall be dry indoor type, and their maximum capacity and minimum impedance shall not be limited, to restrict the 400V system fault level to 50 kA (rms). The design load of each LV transformer shall not exceed 90% of its maximum capacity.

LV neutral of transformers shall be solidly earthed for fast detection and clearing of earth fault.

Off circuit taps of $\pm 2 \times 2.5\%$ shall be provided on HV windings for controlling the voltage on the LV side.

The LV transformers shall be naturally air cooled type.

Temperature control and indicating system shall be provided for each dry-type transformer. Temperature shall be measured and controlled by the temperature control box and the measuring element in the low-voltage winding. When the temperature is over the set value, the alarm signal and over-temperature trip signal shall be initiated.

6.7.4 Voltage Transformer

The voltage transformer shall be provided at GCB side to measure the generator terminal voltage.

6.8 MV and LV Switchgear

6.8.1 General

All switchgear, control gear and fuse gear shall be of metal enclosed indoor type, with degree of protection class IP40 for all other rooms and IP32 for all air conditioned room and be manufactured and tested in accordance with relevant international standards. Each 11 kV switchboard shall be equipped with one empty panel as spare.

Each 400 V switchboard shall be equipped with reasonable number of contactor module per column for ease of operation and convenient maintenance, where two or more contactors are mounted in the same column they shall be separated by metal barriers. Different rating shall be located along the switchboard as spares.

The main busbars together with the tee-off busbars connecting the main busbars to the isolating/protective devices on all incoming and outgoing circuits shall be contained in a separate compartment(s) within the switchboard. The busbars shall be air insulated except where solid insulation is a design feature. The main busbars shall have the same current rating throughout their length.

11 kV switchgear shall be vacuum or GIS (SF6) circuit-breakers to IEC 60056 or equivalent international standards. Use of fuse contactor at 11kV is not acceptable.

Low voltage switchgear and control gear shall be IEC 60439 Form 3b or equivalent standard with air break circuit-breakers of fault interrupting capacity of 50kA for LV switchboards and 40kA for MV switchboards, contactor units, fuse-switches or moulded case circuit-breakers. All points of isolation shall incorporate a feature for fitting a padlock in the open position.

Switchgears shall be of indoor, metal enclosed, fully draw out type, sheet steel units, assembled to form a rigid, free-standing, floor mounted structure. The busbar shall carry the rated current in compliance with the relevant IEC standards without exceeding the specific temperature.

Integral earthing switches with fault making capacity shall be provided for all incoming and outgoing circuits of 11 kV switchboards. In addition, provision shall be made to earth each 11 kV busbar at two separate points.

Withdrawable units shall be provided with lockable shutters to prevent inadvertent access to live connections when the unit is withdrawn.

All operational and maintenance interlocking and any other feature necessary to comply with IEC or equivalent international standards shall be provided, including a padlocking facility on:

- Busbar and circuit shutters.
- Voltage transformer shutters.
- Truck position selector mechanism (busbar earth/service/circuit earth).
- Circuit-breaker/fused switching device open/close control switch.
- Circuit-breaker/fused switching device local/remote control selector switch.

Operational interlocking shall be provided to ensure the correct sequence of operation. This shall be of the electrical type in compliance with a standard for safety circuits to be agreed with the Owner.

Full maintenance interlocking shall be provided to render the primary and tee-off connections safe for access by personnel from all points of supply. The maintenance interlocking shall be of the mechanical interference (trapped key) type employing a system of unique keys.

All instrument transformers shall meet the requirement of approved metering and protection schemes. Secondary of all current transformers shall be rated for 1A.

Voltage transformers shall be single phase. The voltage transformers for MV switchboards shall be draw-out type and for LV switchboards shall be fixed type.

The current transformers shall be cast-resin, ring type with high stability under short circuit conditions, mounted within the busduct and suitable for continuous operation at an ambient temperature of 50°C. The terminal boxes shall be of IP 54 construction for indoor and IPW55 for outdoor installation. The minimum requirement of accuracy class of the current transformer (CT) and voltage transformer (VT) shall be as follows:

- For other protection - Class 5P20
- For general metering - Class 0.5
- For performance test / tariff metering - Class 0.2S
- For differential protection - Class X

All protective relays, transducers or transmitters shall be comprehensive numerical type capable of measuring all electrical parameters like voltage, current, power, power factor, frequency with selectable output. All protective relays shall be of draw out type or flush mount with built-in testing facilities. Small size auxiliary relays are allowed to be mounted within the cubicle. All relays shall have hand-reset means for visual indication of their operation.

Metering and protection schemes of all incoming, bus coupler and outgoing / transformer / motor feeders shall be subject to Owner's review. All motors feeders above 90kW shall be provided with micro-processor based comprehensive motor protection relays. Differential protection shall be provided for motors rating 1000kW and above.

All breaker controls shall operate on $2 \times 100\%$ station DC supplies with change-over facility. All breaker spring charging motors and lighting & heating of switchgear cubicles shall operate on $2 \times 100\%$ single phase AC supplies with change-over facility.

6.8.2 MV Switchgear

MV switchgear shall generally include, but not limited to, the following:-

- 1) 11 kV switchgears;
- 2) Integral earthing switches with fault making capacity for all incoming and outgoing circuits; and
- 3) Circuit breaker maintenance trolley, one for each type of circuit breakers.

Surge suppressers shall be provided for motor feeders controlled by vacuum circuit breaker.

The circuit breakers shall be of vacuum or SF6 gas insulated type, with Service / Test / Disconnected positions with positive indications for service & test positions. It shall be possible to charge the springs manually, if required. For SF6 breaker, necessary

pressure / density monitoring switches with auxiliary relays shall be provided to monitor SF6 gas pressure.

For each breaker panel, suitable interposing relays shall be provided for remote opening and closing of the breaker.

The design data for MV switchgear shall comply with the following table

Nominal voltage	kV	11 kV $\pm$ 5%
Frequency	Hz	50 Hz - 5% to + 3%
Degree of protection		IP32 for breaker and bus bar compartment IP21 for VT and relay compartment
IAC classification		AFLR (as per IEC62271-200)
Sheet steel thickness of Enclosure	mm	2 mm for panel body / 1.6 mm. for doors / covers
Circuit breaker		Vacuum or SF6
Contactor		Not allowed to use
Breaker duty cycle		3 min-CO-3 min-CO

6.8.3 LV Switchgear and Motor Control Centre

400V switchgears and motor control centre (MCC) shall be of the indoor, and fully draw out type. They shall be sheet metal enclosed, assembled to form a rigid, free-standing floor mounted structure. Vertical units shall be assembled to form a continuous line up of panels. Compartmentalised multi-tier configuration shall be provided.

All switchgears shall be provided with air circuit breakers (ACB's) for all incomers / bus coupler / bus tie / motor feeders and moulded case circuit breakers (MCCB's) for all other outgoing feeders which will not be closed from remote. All MCC's / distribution boards (DB's) switchgears shall be provided with load break isolator / MCCB for all incomers / bus coupler and MCCB for all other outgoing feeders which will not be closed from remote.

All AC motor starters shall be suitable for direct on line (DOL) starting. Motors rated above 90kW shall be breaker controlled and provided with micro-processor based comprehensive motor protection relay. Motors rated for 90kW and below shall be provided with MCCB, contactor and over load relay and the motor starter shall comply with type-2 coordination conforming to IEC 60947-4-1.

Draw out type ACB / MCCB shall have service / test / disconnected positions with positive indications for service & test positions. It shall be possible to charge the springs manually, if required. All vacuum contactor controls, breaker spring charging

motors and lighting & heating of switchgear cubicles shall operate on $2 \times 100\%$ single phase AC supplies with change-over facility.

Above requirement also applies to DC switchgear, if applicable.

The design data for the Facility shall comply with the following table.

1.	Nominal voltage	kV	AC: $\pm 10\%$ DC: $- 15\%$ to $+10\%$
2.	Frequency	Hz	50 Hz - 5% to $+ 3\%$
3.	Degree of protection		IP32 for indoor air conditioned rooms and IP 40 for all other rooms / IP54 for outdoor use
4.	Sheet steel thickness of Enclosure	mm	2 mm for panel body / 1.6 mm. for doors / covers
5.	Circuit breaker		Air
6.	Contactor		Air
7.	Breaker duty cycle		3 min-CO-3 min-CO

ACB shall be of the withdrawable, non-fused, air break, trip-free type. The circuit breaker rating shall be in accordance with IEC 60947-2.

The ACB closing and tripping mechanism and closing, tripping and indication supply voltage(s) shall be as detailed on the Offshore Supply Contractor's design data sheets. An anti-pumping device shall be fitted. Where mechanical closing and tripping facilities are provided there should be adequate safeguards to prevent unintended operation. The condition of the closing spring (charged or free) shall be indicated.

Facilities shall be provided for electrically testing the circuit closing and tripping mechanism when the ACB is in the isolated/test position. The facilities shall be provided for electrically testing the ACB closing and tripping mechanism in the withdrawn position.

Documentation shall be provided to verify that MCCB's have been type tested to demonstrate effective through fault co-ordination with contactor/overload combinations and other downstream devices in accordance with IEC 60947-4-1. Under short circuit conditions, the contactor shall cause no danger to persons or installation and shall be suitable for further use.

Where an MCCB is used as a disconnecter, its performance shall also satisfy the requirements of IEC 60947-3 with respect to isolation, indication and positive separation of contacts.

When used for motor over-load protection, thermal overload devices which are an integral part of the MCCB shall have a performance characteristic in accordance with

IEC 60947-4-1. Thermal overload protective devices for Ex (e) machines shall comply with the requirements of the machine's hazardous area certification.

The closing and tripping devices and the closing, tripping and indication supply voltage(s) shall be 220 VDC for the 400V switchgear and Motor Control Centre circuits, 110VAC for Motor Control Centre contactor closing circuits can be accepted also.

The number of direct driven spare (unused) auxiliary contacts required shall be at a minimum, four (4) normally open and four (4) normally closed. These shall be wired to accessible output terminals and normally-closed contacts are to be maintained with the circuit breaker withdrawn. The Offshore Supply Contractor shall advise Owner of the maximum number of contacts available.

6.9 Motor and Motor Protection

Motors shall comply with the following standards as applicable:

IEC 60034 Rotating Electrical Machines

IEC 60072 Dimensions and Output Ratings for Rotating Electrical Machines

IEC 60085 Classification of Insulation and Materials for Electrical Machinery and Apparatus on the basis of thermal stability in service

All motor designs shall be proven and suitable for tropical application. The efficiency of all motors shall be classified and tested to a level of at least IE2, according to IEC 60034-30:2008

AC motors shall be suitable for a 3 phase 50Hz supply and unless otherwise specified or approved, shall be of the squirrel-cage induction type suitable for direct on-line starting. The rating of each motor shall be adequate to meet the requirements of its driven load. Motor shall be the continuously rated model unless otherwise specified or agreed. The voltage level for AC motor rating shall be as follows:

- Below 0.2 kW --- 230V, single phase
- From 0.2 kW up to including 200 kW --- 400V, 3 phase
- Above 200 kW --- 11kV, 3 phase

(For fire fighting motors for all rating -- 400V, 3 phase)

DC motor shall generally be of shunt / compound wound type rated for 220V DC. All DC motors which are operated from the station batteries shall be rated in accordance with specified DC voltage. All DC operated valve motors shall be rated for operation from +10% to -20% of rated voltage. All other DC motors shall be suitable for continuous full load operation from + 15% to -15% of rated voltage.

All motors provided shall be adequately protected against electrical faults, mechanical faults and overloads. Motors rated at 50kW and above shall be provided with earth protection.

The motor rating shall be at least 10% over the maximum input power requirement of the driven equipment at rated load except MV (11kV) motors that the minimum 10% margin can be reduced but subject to Owner's review. Continuously operating motors shall be of high efficiency of class IE2 according to IEC 60034-30:2008.

Starting current of AC and DC motors shall be in compliance with IEC 60034 with IEC tolerances. DC Motors shall be suitable for operation with fixed resistance starter for maximum reliability.

Motor enclosures shall conform to the degree of protection IP-54 (indoor) and IPW55 (outdoor). For hazardous areas, approved type of flameproof and increased safety enclosure shall be provided. Specific motors shall have local emergency pushbutton. The Offshore Services Contractor shall submit a list of motors with and without local emergency pushbutton for Owner's approval.

The motors shall generally be of self-ventilated type Totally Enclosed Fan Cooled (TEFC). Alternatively for large motors, Closed Air Circuit Air Cooled (CACA) / Closed Air Circuit Water cooled (CACW) / Totally Enclosed Tube Ventilated (TETV) type cooling arrangement shall be adopted.

MV and LV motors shall have class F insulation with temperature rise limited to class B.

Motors of rating 30 kW and above shall be provided with space heaters, rated for 220V, single phase, 50 Hz.

All MV (11kV) motors shall be provided with six simplex type winding temperature detectors, two per phase and two duplex type temperature detectors, each at driving and non-driving end. These temperature detectors shall be resistance type, 3 wire, platinum wound, 100 ohms at 0°C. Local indicator with alarms to DCS shall be provided for the motor bearing temperatures and hot and cold air temperature of the closed air. Sight glass shall be provided for the motor for monitoring cooling water flow and oil flow of forced lubrication bearing, if used.

Current transformers for differential protection for motors rating 1000 kW and above shall be provided with three differential current transformers mounted over the neutral leads within the enclosure.

Motor terminal boxes shall be detachable type. For MV motors, the terminal box shall be phase segregated type and neutral leads shall be brought out in a separate terminal box (not necessarily phase segregated type) with shorting links for star connection. All accessory equipment such as space heater, temperature detector, CT etc., shall be wired and terminated in enclosure, separate from motor (power) terminal box. The degree of protection for accessory terminal box shall be same as that of motor.

Motors which are specially made for specific application, and cannot fully comply with the above requirement shall be specified by the Offshore Supply Contractor for Owner's review e.g Boiler recirculation pump shall conform to the ASME Section I, Power Boiler Code. A wet squirrel cage motor shall be provided, where the pump

casing & motor casing are pressure parts installed in the circulating system of the boiler with a wet stator. The motor bearings shall be lubricated by boiler water.

6.10 DC System

DC system shall feed all station DC loads for main power plant, UPS system, all DC drives, emergency DC lighting etc. Considering the geographical spread, separate DC systems shall be designed for main power block, coal handling plant, and ash handling plant etc.

Two sets of 220V DC systems equipment including batteries, float & boost chargers and distribution boards (DBs) shall be provided for each Unit and common services auxiliaries such as water treatment, coal handling, ash handling and sewage water systems.. These two system sets shall be 100% redundant, i.e. one (1) battery, charger, and distribution board.

In total, there are six sets of 220V DC systems (two for each Unit and common services)

For each of the above DC systems, the float and boost charger shall be paralleled with its battery and will feed to one bus-section of 220V DC board. In case of outage of one DC source, the bus tie breaker / isolator shall be closed and the healthy source will continue to supply 100% load of the system as well as supply boost charging current to one battery simultaneously. Float charger shall be sized to accommodate the following:

- Float charging current of the battery
- Maximum DC running loads of the complete system
- 20% spare capacity over the maximum charging rate

Boost charger unit shall be rated such that the completely discharged battery can be fully charged in about 10 hours.

Charger shall have provision of manual control if the auto mode fails. The load limiting features shall be provided with chargers. The charger shall be provided with automatic voltage regulation in float mode and automatic constant current regulation in boost mode.

The DC voltage regulation shall be within $\pm 1\%$ for a 0 – 100% load variation and 10% combined input AC voltage and frequency variation on the AC side.

Without the charger operating, the batteries for each DC systems shall support the maximum load for a minimum period of 1 hour, considering design margin as 15% and aging factor as 1.25. Critical loads on batteries like UPS system DC lube oil, seal oil, etc. motors back up time of minimum of 1 hour shall be provided.

The batteries shall have at least 10 years' design life.

The mean ambient air temperature surrounding the batteries shall be maintained in the range 20 – 25 °C by means of air conditioning. Adequate ventilation shall be provided for the removal of hydrogen.

The batteries shall be supplied completely with all necessary connectors, racks/cabinets, stand insulators, connector bars and accessories together with local and remote alarms for abnormal conditions.

The design data for the DC system shall comply with the following table:

1.	Type of charger	Float and boost charger, Silicon Controlled Rectifier (SCR) controlled
2.	DC system voltage	220 V DC
3.	System earthing	Unearthed
4.	Maximum time for boost charging the battery	10 hours
5.	Battery support time	1 hour for maximum load
6.	Battery type	Maintenance free vent type Lead Acid
7.	Minimum end cell voltage	1.75 V per cell
8.	Charger and battery location	Air-conditioned room
9.	The ripple content in charger output	Maximum. 1% (rms.) without the battery in circuit.

6.11 Uninterruptible Power Supply System

The uninterruptible power supply (UPS) system shall supply 230V, 50 Hz, single phase to all essential instruments, control, computer, supervisory control and data acquisition (SCADA) and any other AC loads, which cannot tolerate interruption of power. Additionally, the UPS system shall supply loads for the intercommunication system of the power station. Each unit and common services shall have two sets of UPS system. These two system sets shall be 100% redundant, i.e. one (1) rectifier and distribution board.

In total, there are six sets of 230V UPS systems (two for each Unit and common services)

Each UPS system shall consist of the following major equipment:

- a) 1 × 100% inverters with isolation facility
- b) 1 × 100% static switches with isolation facility
- c) 1 × 100% manual bypass switch
- d) AC distribution board

The rated capacity of the each UPS system for each Unit shall be 1×100 kVA.

One DC power shall be fed from the unit ungrounded battery and one AC power that shall serve as the power supply to the inverter.

Inverter units shall be static type consisting of SCR / IGBT. The inverter unit shall be provided with automatic voltage regulator to regulate the output voltage to $\pm 1\%$ for variations of input voltage and load current of 0 to 100%. Provision shall also be made to manually vary the output voltage. The output frequency shall be maintained within 50 Hz $\pm 1\%$ under all conditions of loading. Necessary filters shall be provided to make the output wave shape sinusoidal and the distortion shall not exceed 5% under the most unfavourable condition under full load or no-load and whatever be the power factor.

The static transfer switch shall be solid state type using SCR for automatic / manual transfer of load from "inverter" to "stand-by" source and vice-versa.

Manual bypass switch shall be used to isolate any static transfer switch for maintenance or repair without interruption to the UPS load.

6.12 Earthing and Lightning Protection

Earthing system design shall be carried out as per IEEE-80 and IEEE-142. Suitable corrosion factor shall be considered for a plant life of 40 years. The earthing resistance shall be less than 1 ohm. The Offshore Services Contractor shall design the grounding system based on the soil quality & resistivity and site conditions.

For underground earthing system, copper electrodes and copper rods shall be used. Alternatively, corrosion protected galvanized flat steels and galvanized steel pipes may be proposed for Owner's review during design submission stage. The earthing grid at different areas of the power plant shall be interconnected by minimum two connections. This shall be designed to meet permissible step potential. Treated earth pits, as required, shall be provided. Vertical risers shall be provided at suitable places for connecting to equipment earthing conductors. Equipment earthing conductors shall be galvanised steel flat or hard drawn high conductivity copper conductor suitably sized to withstand the fault current of the system. Test electrodes shall be bonded to the earthing grids through disconnecting links which shall permit periodic maintenance testing.

Where resistance earthing is employed, the protective/earthing conductors shall be designed to withstand the effects of an unrestricted earth fault at the most onerous point on the system.

The main protective/earthing conductors shall be installed as ring circuits with duplicated connections to items such as main switchboards.

All exposed conductive parts of plant control cubicles, valve actuators, cable glands and armours etc., shall be effectively earthed.

All extraneous conductive parts such as structural steelwork, cable support steelwork, steel tanks and pipe work, compound and station perimeter fences, gates etc., shall be effectively bonded to the earthing system. Supplementary equip-potential bonding shall be applied where it is necessary.

A low-noise earthing system shall be provided for earthing of sensitive electronic equipment. This system shall consist of an insulated earth conductor connected to an isolated earthing system junction box. All signal earth and other low level earth connections shall be made on the copper earth bus located in this isolated earth junction box. Insulated cables shall connect this isolated earth junction box with a separate earth grid consisting of interconnected earth electrodes. Earth resistance shall not be more than one ohm.

Control and instrumentation cable screens shall be single point bonded to the instrument earth network to minimize the effects of electrical interference.

The size of the control and instrumentation protective/earthing, conductors and lightning protection scheme shall be in accordance with an internationally recognized standard.

Fences shall be earthed in areas of high personnel traffic, near transmission or distribution lines, and where the fence crosses the earthing grid. All fence openings shall be provided with an earthing jumper.

Lightning protection system shall comprise vertical air terminations, horizontal air terminations, down conductors, test links and earth electrodes. Air terminations, down conductors and test links shall be of galvanised steel conductors and earth connection below ground level shall be of mild steel rod.

Hazardous area shall be protected by a system of aerial earth as per IEEE 142 or equivalent standards.

1) Outdoor Earthing

The final resistance to earth shall be 1 ohm (measured at driest season) or less. Earth rod shall be installed if required to achieve final earthing resistance. The earthing conductor size will be given by calculation in accordance with International Standards.

Air terminal of lightning protection shall be connected to earth grid through down conductor, having size of at least 16 mm².

2) Equipment Earthing

All equipment shall be connected to outdoor earthing using suitable earthing conductor.

Minimum earthing conductor for equipment supplied from 230V and 400V shall be 16 mm² and equipment supplied from medium voltage shall be 35 mm².

Cable tray shall be connected to earthed building steel with suitable galvanized steel bonding jumper at 30 meter intervals or at least two places for shorter runs.

Suitable galvanized steel bonding jumper shall be used at expansion joint or section of cable tray where required to ensure electrically continuity.

Motors frame shall be connected to building steel at the nearest practical location by a copper cable safety earth conductor.

Small motor frames shall be earthed through the conduit to cable tray system.

Electrical equipment room shall be provided with earth bus bar and connected to the main earth grid as well as the nearest building steel.

6.13 Cathodic Protection System

The cathodic protection system shall be designed in accordance with IEC Standard. The impressed current cathodic protection system shall be provided in conjunction with the appropriate plant material selection, for the plant listed below:

- Intake cooling water screen and trash racks
- Pumps
- Condenser water boxes
- Miscellaneous heat exchanger
- Grounding grid
- All below ground pipeworks
- All other corrodible portions of water system, pipework and valves.

The cathodic protection system shall be designed for a minimum of 40 years operation and shall include adequate margin for 20% coating deterioration.

6.14 Power & Control Cables

6.14.1 General

For hazardous areas and directly buried cables, multi core cables shall have galvanised steel armour and single core cables shall be aluminum wire armour.

Cables shall be adequately sized with design margins for the factors of full load current, short circuit current, voltage drops during starting and running condition and derating factors due to ambient temperature and grouping / method of installation. Motor starting conditions shall be as specified in IEC 60034.

Minimum size of power cables shall not be less than 2.5 mm² for copper conductor and 6 mm² for aluminum conductor. Minimum size of control cables shall be not less than 1.5 mm². 20% spare cores shall be provided in all control cables except for CT cables (one spare pair for CT cables).

Cables shall be delivered in non-returnable cable drums having ends of each length of cable properly sealed before dispatch.

All cable runs shall be continuous. Control and instrumentation cables may be interconnected for marshalling purpose in suitable enclosures. All cable laying shall be designed as above ground level for ease of operation and maintenance. If above ground installation is not appropriate, then underground cable laying will be used.

The cables shall include measures to prevent insect damage e.g. cables exposed to direct sunlight shall have ultra-violet resistant over sheaths. Control and instrumentation cables shall be capable of withstanding the voltage transients associated with unsuppressed coils etc.

Cables shall have either copper or aluminium conductors with extruded insulation/sheath in accordance with internationally recognized standards, and be of the following or equivalent types:

- 1) All MV(11kV) cables shall be of stranded annealed copper / aluminum conductor, XLPE insulated, screened, PVC inner sheathed, fire retardant low smoke (FRLS) overall PVC sheathed type with IEC 60502. The cables shall be suitable for earthed system. These cables shall have phase identification colour coding. As to directly buried cable, it shall be wire / strip armoured.
- 2) All LV(400V) power cables shall be of 1000 V grade stranded annealed aluminum / copper conductor, XLPE insulated, PVC sheathed, extruded FRLS PVC outer sheath. The cables shall be suitable for earthed system. These cables shall have phase identification colour coding. As to directly buried cable, it shall be wire / strip armoured.
- 3) Control cables shall be of multi-core 1000-volt, grade 1.5mm² stranded annealed copper conductor, PVC insulated, PVC sheathed, FRLS, and PVC sheathed type. These cables shall have core identification coding. As to directly buried cable, it shall be wire / strip armoured. Instrumentation cables shall be of multi-pairs 500 V, grade 0.5 mm² rigid copper conductor, PVC insulated, , FRLS, and PVC sheathed type. These cables shall have core identification coding. As to direct buried cable, it shall be wire / strip armoured.
- 4) Fire resisting cable tested to the highest test temperature in IEC 60331 shall be used for essential circuits required to remain functional in the event of a fire. This shall include the trip circuits (necessary to safely shut down the plant), emergency communication and audible alarm circuits. Mineral insulated cable shall be used for fire detection and fire fighting equipment.

6.14.2 Cable Segregation

In order to limit economic loss in the event of a fire or similar incident, segregation shall be provided among cables as far as practicable. The following segregation groups shall be applied but not limited to:

- 1) Unit 1, Unit 2 system.
- 2) Main and essential (steam turbine and boiler essential loads) supplies.

Segregation methods shall be designed based on international standards (IEC). And the detailed cable route shall be submitted for the Owner's approval.

Fire barriers and penetration seals through walls, floors and ceilings shall have a minimum fire rating of at least one hour to ISO 834.

Adequate physical separation shall be provided between power cables and control and instrumentation cables to minimize the effects of electrical interference and other disturbances including those associated with unbalanced circuit working.

6.14.3 Cable Support and Fixing System

Adequate provision shall be made in buildings for cable flats and risers, steel support structures, holes through walls/floors, ducts etc., for the accommodation of cable routes and access to plant and equipment. Enclosed cable ways shall incorporate suitable personnel access and emergency escape routes and emergency lighting.

Heavy duty galvanized steel ladder rack or perforated tray shall be used for cable support on major cable routes. For minor routes where the weight is not high, a light duty galvanized steel cable support system may be used.

All cables shall be installed by Onshore Contractor in a neat and orderly manner as far as possible arranged to allow removal of any single cable in a run without undue disturbance to adjacent cables. Adequate support shall be provided with bearing surfaces free from sharp edges. The cables shall be protected against mechanical and other potential hazards e.g., heat sources, at vulnerable positions. Cable minimum bending radii shall not be exceeded.

Single core cable shall be installed by Onshore Contractor in either trefoil or flat formation and secured using cleats. The method of cleat shall ensure that cables are restrained to prevent damage during short-circuit conditions whilst allowing movement due to thermal cycling at bends.

6.14.4 Cable Installation

Adequate precautions shall be taken by Onshore Contractor to prevent cable damage during installation. Winch pulling of cables shall be subject to Owner's review.

Cables ducts shall be free from debris and sharp edges. Following cable installation the ducts shall be sealed at both ends to prevent the ingress of water etc.

Cable installation documents shall be provided to clearly indicate the physical locations, the connection, the termination of the individual core wires, the terminal blocks of the cables at its two ends and intermediate marshalling kiosks (if any).

6.14.5 Cable Ends Identification

The two ends of all cables shall be allocated with unique identification numbers which shall be clearly and securely fastened as soon as possible after installation. All cables shall be terminated using mechanical glands of the metallic or insulated type. Non gland termination shall be to the Owner's review during detailed engineering stage.

MV cable terminations shall be fully insulated where the risk of failure due to condensation is high e.g., outdoors and in plant areas. Air terminations shall have adequate clearance and creep age distances.

Cable termination boxes for high current single core power cables shall have nonmagnetic gland plates etc., to avoid cable heating due to closed magnetic loops.

Conductors shall be terminated using compression fittings. Solid aluminium conductors may be terminated using formed termination techniques. Cores shall be adequately supported and arranged in a neat and orderly manner. All cable cores shall be identified with suitable ferrule labels.

6.15 Emergency Diesel Generator

Emergency diesel generator (DG) sets shall be provided for safe shut down of the Units and their auxiliary equipment, in the event of total power failure. The emergency DG sets shall include, but not limited to, the followings:

- 400 V emergency DG Set (One set per Unit). Each power rating will be calculated in execution stage.
- Auto mains failure (AMF) Panel for each DG set, enabling the DG set to auto start and synchronize to the emergency 400 V board upon detection of AC mains failure. When AC supply is restored, loads on DG set shall be manually changed over to the normal source of power supply
- DG set shall be capable to supply the emergency loads and additional essential loads, such as:
 - Steam turbine and boiler essential load, substation essential loads
 - Plant control system
 - Battery charger
 - Communication systems
 - DC loads
 - Emergency lighting etc.

The capacity of the emergency diesel generator shall meet the above essential loads, provided with sufficient margin capacity for overload facility.

The diesel engine generator set shall be capable in every respect to be operated in parallel with the power plant auxiliary system.

Engine emission shall meet all local government regulation and the sound level requirement shall be less than 85 dB(A).

The diesel generator set shall be complete with all necessary components as required for successful operation, such as: control panel, synchronizing facilities, radiator, fuel oil system, lubrication oil system, facilities for controlling from control room, etc.

Each DG set shall be suitably sized for catering loads for safe shut down of one Unit and 50% of emergency station auxiliary loads in the event of total black out. The starting time period shall be limited to thirty (30) sec. DG sets shall be installed indoor and housed in a dedicated container.

The diesel engine shall be of stationary type and directly coupled to the generator. The engine shall be capable of driving the generator at 10% overload at the rated speed for one hour. The engine shall be air cooled / water cooled with separately mounted radiator fans.

Electrical starting arrangement complete with starting DC motor, starter, batteries and battery charger shall be provided. Alternatively, starting arrangement by compressed air system may also be provided. Starting arrangement shall be suitable for consecutive six (6) starts. No external power shall be used for starting of the engine.

The generator shall be rated for 400 V, 3 phase, and power factor of 0.8. The generator shall be capable to run at 10% overload condition for at least one hour at ambient temperature of 35 °C. It shall be mounted on a common base frame together with the prime mover. The generator shall be of the totally enclosed air-cooled type. Insulation shall be designed to class F but operating temperature shall be limited to class B. In the stator winding, six numbers of resistor temperature detectors shall be provided. Space heaters suitable for operation on 1 phase, 220V AC shall also be provided. The generator shall be provided with brushless type excitation system with automatic voltage regulator (AVR).

Each DG set shall have its own auto mains failure (AMF) panel. This panel shall comprise of AVR, protection & metering equipment, indicating instruments and start / stop interlocking hardware. The panel shall be free standing type, sheet steel enclosed having a degree of protection of IP 40.

All auxiliary devices for control, indication, measurement, protection and alarm shall be mounted on the front door of the panel. Alarm / indications shall also be annunciated in central control room. Remote start /stop operation facility shall be provided from control room.

6.16 Protection, Control and Tariff Metering

All electrical equipment shall be supplied with their protection hardware, complete with remote and local control selection. The switching of GCB and 11 kV switchgear circuits shall be initiated from the DCS operator consoles in the CCR.

The control & synchronizing of generators, generator excitation systems, generator step-up transformers, unit & station auxiliary service transformers, 11 kV/400 V incomers, ties and main outgoing circuit breakers to the auxiliaries shall be also achieved from DCS operator consoles in the CCR.

All necessary isolation points, interlocks, alarms, indications, instrumentation, data logging, condition monitoring/recording equipment, operational metering, test and diagnostic facilities etc., shall be provided for the secure, reliable and safe operation of the electrical systems.

The protection system shall comply with relevant international standards and shall be designed so that in the event of faults occurring, the faulty plant is safely disconnected, whilst continuity of supply consistent with system stability is maintained.

One export/import (bi-directional) metering system i.e. back up #2 metering system shall be provided.

All protective relays shall be comprehensive numerical type capable of measuring all electrical parameters like voltage, current, power, power factor, frequency etc. The relays shall be capable of communicating all analogue and digital signals with DCS / SCADA system, as applicable. All protective relays shall be of draw out type or flush mount with built-in testing facilities. Small auxiliary relays may be of non-draw-out type and mounted within the cubicle. All relays shall have hand-reset means for visual indication of their operation.

The protection system from generator to generator step-up transformer and UAT shall be arranged to provide two independent, high performance and reliable systems with separately monitored DC supplies, separate CT / VT cores, separate cables & trip / hand reset lock out relays to obtain 100% redundancy. Associated trip relays of these two redundant systems shall be separate, having sufficient number of contacts for all the functions. Each protection shall individually energize both trip coils of the circuit breakers to be tripped.

6.16.1 Generator Protection

Following typical protections, but not limited to, for each generator shall be supplied. Additional protection, if any as per manufacturer to meet the system requirements, shall also be provided.

- Generator differential protection (87G)
- Stator earth fault protection covering 95% (64G1) and 100% (64G2) of winding

- Two zones of loss of field protection (40G1 & 40G2), combined with under voltage check feature if required
- Back-up impedance protection (21G)
- Negative phase sequence current protection with alarm and trip element (46G1 & 46G2)
- Duplicated low forward power interlock (37G1 & 37 G2)
- Two stage rotor earth fault protection (64F1 & 64F2) for alarm & trip
- Definite time delayed over voltage protection (59G1 & 59G2) for alarm & trip
- Over fluxing protection (99G & 99T) for generator and generator step-up transformer having inverse time characteristics suitable for over fluxing capability of generator / generator step-up transformer
- Generator reverse power protection (32G)
- Generator under frequency protection (81G1 & 81G2 or more stage) for alarm & stage trippings
- Generator over frequency protection (81G3)
- Generator pole slipping protection (98G)
- Accidental back energisation protection (61G) of generator for accidental
- Breaker failure protection (50BF)
- Voltage restrained over current protection (51V)
- Over current protection (50G) for alarm
- Generator VT fuse failure protection (60G) having alarms and blocking of all protections fed by voltage transformers
- Excitation transformer instantaneous and time delayed over current protection (50/51ET), if applicable

6.16.2 Generator Step-up Transformer Protection

Following typical protections, but not limited to, for each generator step-up transformer shall be supplied. Additional protection, if any as per manufacturer to meet the system requirement, shall also be provided.

- Overall differential protection (87GT)
- Generator step-up transformer differential protection for individual phase (87T)
- Generator step-up transformer HV side backup over current protection (51GT)
- Generator step-up transformer HV side standby earth fault protection (51NGT)
- Breaker failure protection (50BFGT)
- Provision for necessary protection, control & monitoring interface from individual alarm & trip inputs given by the transformer for all operating conditions for the followings :
 - i. Oil temperature protection (26GT)
 - ii. Winding temperature protection (49GT)
 - iii. Buchholz protection (63GT)
 - iv. Oil level (71GT)
 - v. Pressure relief device protection (PRDGT)
 - vi. Power supply failure to marshalling box
 - vii. Cooler bank failure

6.16.3 UAT Protection

Following typical protections, but not limited to, for each UAT shall be supplied. Additional protection, if any as per manufacturer to meet the system requirement, shall also be provided.

- AT differential protection (87AT)
- AT HV side backup directional over current protection (67AT1)
- AT HV side restricted earth fault protection (64R1AT)
- AT HV side directional earth fault protection (67N1AT)
- AT LV side backup directional over current protection (67AT2)
- AT LV side restricted earth fault protection (64R2AT)
- AT LV side directional earth fault protection (67N2AT)
- Breaker failure protection on both sides (50BF1AT & 50BF2AT)
- Over fluxing protection on HV side (99AT)
- Provision for necessary protection, control & monitoring interface from individual alarm & trip inputs given by the transformer for all operating conditions for the followings :
 - i Oil temperature protection (26UT)
 - ii Winding temperature protection (49UT)
 - iii Buchholz protection (63UT)
 - iv Oil level (71UT)
 - v Pressure relief device protection (PRDUT)
 - vi Power supply failure to marshalling box
 - vii Cooler bank failure

6.16.4 Control System

The complete control, indication, annunciation and supervision of the electrical systems for the power plant shall be performed from through plant DCS system with interface to SCADA for communication with local grid despatch centre. Necessary mimics shall be provided with dynamic values and status. Necessary command feedback signals and measurement signals shall be wired up to the plant DCS / SCADA system. It shall be possible to call any electrical systems on DCS screen and control the various switchgears and MCCs. Synchronization of generators shall be carried out from DCS terminal during normal operation. Communication modems and protocol adaptation as required by local dispatch centre shall be provided as required.

The measurements / indications available on DCS screen shall include but not be limited to the following:

- Current, voltage, power, MVAR, MWh, MVARh, power factor and frequency for generator and all transformers.
- Stator core, winding, bearing and cooling system temperature for generator
- Winding / oil temperature and tap position for all transformer
- Current, voltage and power for each 11 kV /400 V incomer
- Current for each MV and LV large motor and motors for important process
- Current for each Circuit Breaker (CB) of large outgoing feeder (more or equal to 90kW) for each 11 kV /400 V switchgears
- Voltage for each 11 kV / 400 V bus
- Voltage for DC and UPS system
- Excitation system parameters
- Emergency DG set parameters

The status of the following shall be monitored as a minimum, through DCS

- Status (ON, OFF and Trip) of all GCB's, 11 kV and main 400 V circuit breakers, isolators & earth switches.
- Status (ON, OFF & Trip) of all motors operated through DCS
- Status (ON, OFF & Trip) of UPS & DC system
- Operation of all protection relays for 11 kV and 400 V systems (up to switchgear bus)
- Operation of protection relays of generators and unit transformers
- Operation of transformer protection mounted on transformers for all HV, MV and LV transformers
- Generator capability curves indicating actual operating point shall also be made available on DCS

6.16.5 Synchronisation Panel

A hardwired synchronizing panel for each generator shall be provided. It shall consist of all synchronizing equipment : double voltmeter, double frequency meter etc., generator selector switch, 'Auto / Manual / Remote' selector switch, auto-synchronizer and other necessary relays / components.

Separate check synchronizing relay, required synchronizing GCB's, shall also be accommodated in each panel.

6.16.6 Tariff Metering System

6.16.6.1 General

An electronic and programmable (with password controls) tariff metering panel shall consist of the following meters:

- Main meter for each generator
- Main meter at HV side of each unit auxiliary transformer

The main meter will be acted as “backup 2 metering system” as required by PPA.

The metering system shall include the meter, current transformer, voltage transformer, the automatic system integrated with the Facility's plant control system, and other equipment described in this section.

6.16.6.2 System Requirements

The metering systems shall

- (a) receive signals from the meters of main meter, check meter and other measuring and metering devices required for safe, reliable and automatic data compilation and logging for determining the appropriate data required to provide accurate measurement of energy import and export from the Facility and support the verification of invoices in conformance with the tariff;
- (b) have the facility for main meter for downloading and remote reading of data by EVN;
- (c) contain devices capable of measuring kWh's, which are both imported into and exported out of the Facility and compiling them into specified demand intervals of 30 minutes to derive half-hourly kWd;
- (d) be capable of interpreting and compiling the kWh and kWd data into readings and performing the required computations and adjustments required to:
 - (i) measure, compile and record the necessary parameters for testing the Unit or the Facility, as the case may be, to determine the net electrical output;
 - (iii) automate the metering data collection at metering points for collection and storage in the computer as basis for determining the energy delivered to EVN and record active and reactive capacity at the metering points with duration for analysis cycle being 30 minutes or 1 hour for energy delivered from 0:00 hours to 24:00 hours of each day; and
 - (iv) ensure that the active and reactive capacity at the metering points are: (1) automatically recorded on a daily basis, (2) stored at the database of DCS, (3) main meter is available for EVN to access and transfer to the data storing and processing

system of EVN, (4) used as the basis for the operation management of metering systems, and (5) utilized for comparing the actual dispatched capacity of the Facility with the capacity as per the Dispatch Instruction by National Load Dispatch Centre.

- (e) be installed at metering points and shall be the basis for determining the delivered energy (sale and purchase) for the purpose of billing and payment to EVN.

6.16.6.3 Technical Requirements of Metering Systems

6.16.6.3.1 Configuration

- a) Meter;
- b) Current transformer (CT);
- c) Voltage transformer (VT);
- d) Electrical circuit and secondary cable;
- e) Equipments serving for collecting data and data transmittal line;
- f) Safety protection equipment, seals; and
- g) Contingency equipment, connection conversion equipment, current isolated equipment serving for testing purpose.

6.16.6.3.2 Technical Requirements

- a) Meter shall have the following features:
 - (i) 3 phases, 4 wires;
 - (ii) Standard electronic meter can be programmed;
 - (iii) Multi tariff meter;
 - (iv) Measuring import/export active and import/export reactive energy separately ;
 - (v) Measuring reactive energy under 4 quarter angles;
 - (vi) Capability to measure maximum load and load profile;
 - (vii) Capability to connect to computer, collecting and reading data at the site and from distance;
 - (viii) Power supply for meters shall be provided from secondary measurement voltage system and shall be ensured to be alive when 1 or 2 phases are failure;
 - (ix) Capability to be protected at multi security password levels;
 - (x) Meters shall have seal locations to ensure no illegal approach can be made to any circuit point or any changes to data programmed can be made without breaking the seals;
 - (xi) All functions and hardware configuration shall be compatible with data collecting and processing system of EVN; and
 - (xii) The accuracy class of meters shall be 0.5s for active energy comply with IEC 62053-22 and shall be 2.0 for reactive energy comply with IEC 62053-23 or equivalent other standards.
- b) Current transformer shall have the following features:

- (i) Secondary measuring windings for meters of main meter and check meter shall be separate and totally independent to each other;
 - (ii) The nominal secondary current is 1A or 5A;
 - (iii) Sealing on the cover of secondary windings of all measuring equipment and meter of main meter and check meter in order to ensure that no illegal intervention can be made to the circuit point without breaking the seal; and
 - (iv) Accuracy limit of 0.5 as per IEC 61869-2.
- c) Voltage transformer shall have the following features:
 - (i) Secondary measuring windings for meters of main meter and check meter shall be separate and totally independent to each other;
 - (ii) The nominal secondary voltage is 100V or 110V;
 - (iii) Sealing on the cover of secondary windings of all measuring equipment and meter of main meter and check meter in order to ensure that no illegal intervention can be made to the circuit point without breaking the seal; and
 - (iv) Accuracy limit of 0.5 as per IEC 61869-3/5.
- d) Metering circuit shall have the following features:
 - (i) In any circumstances, secondary windings of current transformers and voltage transformers and secondary cable connecting to meters of main meter and check meter shall be independent from each other and shall not be used for any other. Metering circuit of metering systems (secondary current circuit, secondary voltage circuit) shall be totally independent of the secondary circuit of protection relay system;
 - (ii) Conductor size of secondary cable shall be suitable for the load of metering circuit, secondary cable of energy metering systems shall go on the shortest possible way with least possible number of connection points through the seal. The load resistance of metering circuit and changes causing impact to CT and VT shall be within the allowable limit as per relevant standard equivalent to capacity and accuracy limit of CT and VT; Capacity load of secondary circuit of CT and VT as well as meter shall not exceed fix load mentioned on the label of CT and VT; and
 - (iii) All test blocks shall be collectively installed for the purpose of inspection of metering systems and shall be satisfactory for sealing.
- e) Meter Data Collecting and Data Reading System
 - (i) The distant data collecting and data reading system (data reading system) shall be compatible with transmittal and protocol of the distant data collecting and data reading software, which will process data at the data centre of EVN (characteristics, protocol and requirements for integration with the software);

- (ii) Data reading system shall have information gate, integrated information transmittal equipment and modem capable of meeting standard RS485 communication protocol via public switch trunk network (PSTN);
- (iii) Data shall be transmitted to EVN via telephone system or leased line protection from illegal interventions;
- (iv) Communication devices shall be equipped with lightning protection device to avoid the lightning pulse transmitted via communication network causing damage to the meter; and
- (v) Subject to metering data collection mode at the terminal point agreed by EVN and the Owner, the Onshore Contractor shall arrange station computer for reading and archiving data on-site and at centralised data centre. Data collected by the station computer of the Facility shall be transmitted to the server located at the EVN's office; and
- (vi) The devices installed inside the panel shall comply with applicable Vietnam safety requirements and facilitation for management work.

f) Technical Requirements for Seal and Security of Metering Systems.

Metering systems shall meet the following requirements on sealing and security:

- (i) Sealing method and form shall be agreed by EVN;
- (ii) Whole measuring and metering system including components such as: wire connection box for CT, VT, meter, seals, connecting components, secondary circuit, supporting equipment, logical circuit, meter cabinet, information lines and network shall be sealed to avoid illegal interventions;
- (iii) Software of meter shall have password protected with multi accessing levels in order to decentralize to different users with different passwords;
- (iv) The Onshore Contractor shall ensure that the readings in the meters shall be protected against illegal onsite access or remote access by passwords and relevant isolating devices; and
- (v) The Onshore Contractor shall supervise the following hierarchy of passwords in at least three levels:
 - (I) Password for “read only” (level 1) for reading and monitoring the readings of capacity and energy;
 - (II) Password for “resetting the meter clock” (level 2) with ability to synchronise the time of the meter for remote periodic synchronisation of the meter's time with the reference time source (e.g. GPS Clock installed by the Onshore Contractor at the Facility);

and

(III) Password for resetting the password (level 3) which was already set for the meter.

6.16.6.4 Data Logging

The metering systems shall each day compile kWh and kWd data from the meters of main meter and check meter in half an hour intervals, starting from 0:00 hours and ending at 24:00 hours. Subject to the agreement with EVN, twenty four (24) or forty eight (48) records in each hour or half an hour as the case may be will be prepared each day for each of the following parameters:

- (a) Total exported electric energy provided to EVN: as summation of exported kWhs.
- (b) Net energy output as summation of exported kWhs during periods of operation.
- (c) Exported electric demand provided to EVN: as a record of exported kWd in demand in half an hour intervals.
- (d) Total imported electric energy from EVN: as summation of imported kWhs.
- (e) Imported electric demand from EVN: as a record of imported kWd demand in half an hour intervals.

6.16.7 Ambient Conditions Measuring Systems

- (a) This system shall include instrument, conversion, collection and automatic recording equipment for recording every one minute the ambient air temperature at the point determined during design stage, and cooling sea water temperature at condenser inlet, to provide data for correcting the output and heat rate as described in Schedule 5 (Determination of Tariff) of the PPA;

This system shall also comprise a computerized data management system complete with monitor, data retrieving facility, off line data storage facility and printer.

- (b) Technical design of this system and its detailed explanation shall be provided by the Offshore Services Contractor through the Owner to EVN and agreed by EVN;
- (c) Accuracy limit, tests and sealing method, processing, storing and managing the ambient air temperature and cooling sea water temperature recording system shall be agreed in detail in the technical design and detailed explanation for this system.

Measured Parameter	Maximum Error
Ambient air temperature	± 0.2 °C
Cooling sea water temperature	± 0.2 °C

- (d) It shall be sealed independently from the operational metering system. It shall be independent from the tariff metering system. Independent measurement channels shall be provided for each unit and collected in a single computerized data management system. This system shall be located in main central control room.

6.17 Neutral Earthing Resistor

The neutral earthing resistor (NER) shall be used for resistive earthing of 11 kV system. Each NER shall be connected between neutral point of secondary side winding of each of unit transformers, station transformers and all 11 kV service transformers and earth. Rated current for NER shall be 300 A minimum for 10 seconds.

The NER unit shall be air-cooled type suitable for installation at outdoor locations.

The resistor element shall be made of non-aging stainless steel or equivalent corrosion resistant material having high electrical resistivity and low temperature coefficient of resistance. All element connections shall be bolted type to ensure stable resistance value throughout the working life of the unit.

Each NER shall have structural steel work enclosed on all sides and also on top by sheet steel. Suitable ventilating louvers shall be provided on sides to ensure proper ventilation. The louvers shall be provided with fire wire mesh to make it vermin proof.

Each enclosure shall be self-supporting, weather proof type suitable for outdoor mounting having a protection class of minimum IPW-55 or equivalent standard. Removable bolted disconnect link shall be provided for the purpose of isolation. Bushings shall be solid porcelain type. Bushings shall have adequate creepage distance corresponding to heavily polluted atmosphere.

The current transformer shall be cast resin type suitably mounted inside the NGR cubicle in such a way as to ensure easy removal and replacement. The minimum requirement of accuracy class of the current transformers shall be Class X for restricted earth fault protection and Class 5P20 or equivalent standard for other protections. VA burden shall be selected to suit the system requirement. CT secondary shall be rated for 1 A.

6.18 MV Busducts

If busducts are included in the design, the design has to follow the hereunder requirements.

The continuous rating of the each busduct shall be such that the rated current can be delivered continuously at site ambient conditions without exceeding the following temperatures considering an ambient of 50 °C duly considering the effect of solar radiation:

- Main conductor : 105 °C
- Bus connections : 105 °C
- Busduct enclosures and support structures : 80 °C

MV busducts shall be metal enclosed and natural air-cooled. The enclosure shall be made of sheet steel. The enclosure shall have a degree of protection of minimum IP

54 for indoor and IP55 for outdoor portion. Suitable rain hood shall be provided for outdoor run and also at enclosure joint.

Bus conductor shall be of aluminum alloy, adequately sized for continuous rated current and short circuit current for duration of minimum 1 second.

Flexible joints shall be provided in the busduct run and at transformer / switchgear end to accommodate thermal variations. Seal off bushings shall be provided at wall crossings between outdoor and indoor. Wall frame assembly shall be provided at wall crossings.

The busducts shall be provided with space heaters to control moisture. The bus conductor and inside surface shall be treated with matt black paint for efficient heat dissipation. All supporting steel structures shall be hot dip galvanised.

6.19 Electric Actuators

Actuators shall have totally enclosed weather & dust proof construction with IP65 enclosure and shall be suitable for outdoor application without the necessity of a canopy. The actuator shall be suitable for mounting directly on the valve. Each actuator shall have a hand wheel fitted on it for emergency operation. It shall be provided with required no. of position limit switches / torque switches of adjustable type.

Each actuator shall have a space heater in the limit switch compartment suitable for 220 V, 50 Hz, single-phase supply.

All motorised actuator shall be provided with integral starter with MCB, Contactor, overload relay, Local/ Remote selector switch (lockable), ON-OFF-Trip Indication lamps etc. The motor starter shall comply with Type-2 coordination conforming to IEC 60947-4-1.

The drive motor shall be three phase, squirrel cage, induction type with class B insulation and IPW-55 enclosure, designed for high torque and reversing service. The motor shall be designed for full voltage direct on-line start, with starting current limited to 6 times full-load current.

Each actuator shall have one built-in local position indicator for 0 -100% travel and one transducer for remote position indicator for 0 -100% travel.

6.20 Plant Illumination System

Illumination systems shall be provided consisting of general area lighting, street lighting, lighting for the plant and equipment, emergency lighting, chimney aviation warning lights, etc. The systems shall generally include, but not limited to various types of lighting fixtures with lamps and accessories, Lighting distribution boards and lighting panels, Lighting poles / towers, required quantity of conduits, wires, lighting cables, switch boards, junction boxes, supports and other accessories.

For administration building and central control room, energy saving fluorescent lighting (T5 bulb) shall be provided.

The plant illumination system shall provide illumination to all plant areas and ash water recirculation area. It shall also provide illumination to selected areas during plant emergency conditions. The basic illumination system shall include the following:

- Normal AC lighting system
- Emergency AC lighting system
- Emergency DC lighting system
- Aviation lighting
- Fence & security lighting
- Area lighting
- Escape lighting & emergency exit signs

Normal AC lighting shall be provided by AC lighting fixtures distributed throughout the plant and shall be ON as long as the normal AC power supply is available.

Emergency AC lighting shall be provided by AC lighting fixtures in critical areas of the plant and shall be ON as long as the normal AC power supply as well as emergency AC power supply, fed by Emergency DG Set, is available.

Emergency DC lighting shall be provided by DC lighting fixtures for operator at Central Control Room (CCR) and Emergency DG set area. DC lighting fixtures shall be fed by station batteries through DC Distribution Board (DCDB) and DC Emergency Lighting Panels (DCELP). These fixtures shall be automatically switched in the event of failure of both normal and emergency AC power supplies and be switched off automatically with a time delay after resumption of AC power supply.

Self-sustained batteries pack emergency lightings shall be provided at strategic locations : entrances, staircases, landings, control rooms except CCR.

Illumination levels and design for various areas are subject to Owner's review. The minimum illumination level at any point shall not be less than 75% of the average levels on the respective areas. All lamps shall be of energy efficient type.

Fluorescent fixtures with electronic ballast shall be generally provided for a room height up to 5m. For rooms / areas of more than 5m high, medium / high bay fixtures with HPSV lamps shall be used. Fluorescent fixtures shall be industrial trough type with vitreous enamelled reflector or decorative type to suit the aesthetics. For corrosive areas, corrosion proof lighting fixture shall be provided. For hazardous areas, flame proof type equipment shall be used for lighting fixtures and associated equipment. For false ceiling areas, recessed mounted decorative fixtures with mirror optic louvres shall be used. For areas like boiler platforms, well glass type HPSV fixtures shall be used. All indoor lighting fixtures shall be provided with selective switch control or group control as per layout requirements. For all outdoor lighting fixtures, control shall be automatic through photo cells / timers.

Lighting poles shall be swaged type tubular of approved height as per relevant standards complete with weatherproof junction box and all other accessories. Lighting towers shall be of hot dip galvanised steel lattice structure with platforms or of raising / lowering type steel tubular structure. A weather proof marshalling box with water tight hinged door and suitable locking arrangement shall be provided at the bottom of the each flood lighting pole / tower. This box shall contain MCBs, neutral links, terminal blocks, ground studs etc.

Chapter 7 - Control and Instrumentation

7.1 Design Principles

The proposed control and instrumentation (C&I) systems shall apply to all equipment including, but not be limited to, all mounted measuring devices and related ancillaries, conditioning devices, transmitters and related circuits, microprocessor based and computer based devices (such as programmable logic controllers (PLC), and distributed control system (DCS)), control drives and actuators, control pipework and control circuits, marshalling racks and panels, cabling and wiring, and the control side of all switchgear.

The design for security and integrity of the control, interlock and protection systems throughout the Facility shall conform to the grid connection agreement and to the following principles:

- The system design shall adopt “Fail safe” philosophy to ensure safety of equipment in case of failure or malfunction of related instrumentation.
- The system shall be designed to ensure redundancy at controller level, I/O level, bus level and power supply level.
- No single component failure will cause the loss of the Unit protection.
- No single component failure will cause an immediate risk to the Unit trip. The Offshore Services Contractor shall provide, as part of design submission, details of Unit protection and tripping scheme for review by the Owner during detailed engineering stage.
- No single component failure will cause the failure of the DCS which renders the Unit inoperable from CCR.
- All important C&I equipment which affect Unit operation shall be supplied by a direct current (DC) or uninterruptible power supply (UPS) system.
- Essential transmitters and sensors, which directly or indirectly trip the Unit(s), shall be triplicated and arranged in two-out-of-three voting logic for binary signals or median selection logic for analogue signals. Details shall be submitted for Owner's review if this principle cannot be applied in specific cases.
- All protection, tripping or interlock instruments/circuits which affect Unit operation shall be redundant.
- The startup, normal running and shutdown operation shall be carried out in CCR with no attendant required to attend various plant outside the CCR. The design shall allow only one operator to operate one Unit during normal operation. Assistance from another operator is allowable during Unit start up and shut down.
- All main Plant equipment including boiler and boiler auxiliaries, turbine-generator and auxiliaries shall be controlled by the DCS. The turbine control shall interface with the DCS and shall comply with all requirements as if they are part of the DCS. As a minimum, the turbine control shall send back to the DCS all protection and control signals for the operation of the power plant with time tags accurate to their

acquisition scanning rates in the DCS. All trip signals shall be hardwired to the DCS to give accurate time stamp. The interface shall enable the operator to control the power plant as if this turbine control system is part of the DCS. Trip signals, major control signals such as turbine load control & group start/stop from DCS shall be connected by hardwire. Other signal interface between DCS shall be connected by communication link (OPC or TCP/IP). And the detailed signal interface shall be designed and reviewed during contract execution stage.

- All major balance of plant (BOP) controls and electrical switching (i.e. closing and opening of switchgear) of GCBs, station MV/LV electrical supplies and tap changing of generator step-up transformer and Unit auxiliary transformer shall be achieved by operator commands via the DCS in CCR.
- A panel of minimum hardwired equipment shall be provided in CCR, for emergency shutdown in case of total failure of DCS. The panel shall be operated independent of the DCS.
- All motorized actuators shall be provided with torque switch and position switch.
- All outdoor cubicles shall be of stainless steel or hot dip galvanized steel type.

The Offshore Services Contractor shall provide Unit protection and tripping schemes during detailed engineering stage.

The Onshore Contractor shall cooperate with Owner for implementation of the Plant Information (PI) system during DCS commissioning/testing time.

7.2 Programmable Logic Controller

Programmable logic/process controllers can be used in common services as well as standalone control systems. The memory of the PLC at controller level shall be non-volatile e.g. EEPROM or Flash memory. PLCs shall be arranged in redundant configuration. Single configuration PLC can be only accepted provided that the system can be operated in manual by operator if the PLC fails. The controller shall be 100% hot standbys with bumpless transfer facility. Each controller shall have 40% spare functional capacity to implement additional function blocks, over and above implemented logic/loops under worst load conditions.

PLC shall be designed to a popular communication protocol that shall be able to communicate to the DCS. Only global controls (group orders, group signals, group alarms) shall be called up in the DCS in CCR.

The Offshore Supply Contractor shall standardize their proposed PLC supply by limiting the choice of suppliers, where the equipment supplier's standard allows. The Offshore Supply Contractor shall use the PLC from suppliers in the Permitted Subcontractor list, unless approved by the Owner.

Personal computer type programming tools with colour monitors and printers shall be provided. The programming tools shall display and print the control algorithm in either ladder circuit diagram form, boolean logic form, or program statement form. Each PLC system shall be provided with its own programming tool and shall not share the

programming tool of other systems. Training for PLC hardware installation and application programming shall be provided.

PLC for BOP systems shall be controlled and monitored by DCS in CCR. BOP PLC will form its own network or integrate with the main DCS. PLC such as coal handling system, and water treatment system shall have their own operator station in dedicated local electrical building and will interface with the main DCS via Modbus.

7.3 Distributed Control System

The Offshore Supply Contractor shall provide a microprocessor based DCS for the overall control of the Facility. The DCS shall accommodate the data acquisition, modulating and sequence controls of the Unit to a high degree of automation. The CCR control console shall enable centralized and fully automatic operation of the Facility with minimal staffing level in CCR and outside Plant. The Offshore Supply Contractor shall provide the system architecture of DCS in detailed engineering stage.

The DCS shall consist of data acquisition system (DAS), modulating control system (MCS), sequence control system (SCS), furnace safeguard supervisory system (FSSS), digital electro-hydraulic control system (DEH) and so on. Emergency tripping system (ETS), steam turbine safety inspection system (TSI) are independent systems (which may be supplied by turbine manufacturer) shall interface to DCS by hardware and/or software communication. The turbine control systems (DEH, ETS, TSI, etc.) shall be operable through DCS as if they are part of the DCS as per section 7.1.

7.3.1 Data Acquisition System

Data acquisition system (DAS) shall acquire, manipulate, store and retrieve different kinds of parameters (analogue variable, switching value and pulsed quantity); and shall provide to the operators with the following functions: picture display, beyond-limit alarm, scheduling and print, performances calculation, trending, event sequence record, history data storage, operational instruction etc.

The system shall also have provisions for generation of following logs as minimum:

- Equipment running hours
- Daily log
- Performance log
- Analogue trend log
- Boiler log
- Turbine log, etc.

7.3.2 Modulating Control System

Modulating control system (MCS) shall control the boiler, steam turbine and generator as an integrated controlled object to meet the requirement of unit start-up and shutdown, either in fixed pressure and sliding pressure mode. MCS shall control the whole process from the start of boiler lighting, steam turbine rolling up, synchronization, load ramp up, up to full load.

The Plant control mode shall be designed with the following four modes by coordinated control system:

- 1) Coordinated control mode
- 2) Steam turbine follow mode
- 3) Boiler follow mode

4) Manual operating mode

For any modes of the MCS, the following plant functions shall be controlled in a coordinated manner.

- 1) Fuel flow control
- 2) Fan group air flow control
- 3) Furnace negative pressure control
- 4) Feedwater control
- 5) Superheated steam temperature control
- 6) Reheat steam temperature control
- 7) Exit temperature and air flow control of coal mill
- 8) Primary air/Secondary air pressure control
- 9) Fuel oil pressure control
- 10) Boiler drain control
- 11) Steam pressure of sootblower system control
- 12) Deaerator water level control
- 13) Deaerator pressure control
- 14) Steam condenser water level control
- 15) Steam turbine shaft seal pressure control
- 16) Steam turbine shaft seal temperature control
- 17) Feed pump sealing water differential pressure control
- 18) High pressure heater water level control
- 19) Low pressure heater water level control
- 20) Generator cooling water temperature control
- 21) Generator sealing oil pressure control
- 22) Generator cold gas temperature control
- 23) Aux steam pressure and temperature control
- 24) HP/LP bypass control

7.3.3 Sequence Control System

Sequence Control System (SCS) shall be designed to be capable of operating various Plant equipment (such as pumps, fans, motor operated valves, dampers, etc.) in a sequential manner with pre-start check conditions, either automatically through the DCS or manually by the operator. Remote operation from the CCR shall be available

sequentially and also local manual on-off operation (if required for some specific equipment) shall be available.

- 1) The automatic SCS shall ensure safety oriented sequential startup and shutdown of Plant auxiliaries/equipment.
- 2) Interlock and protection shall be provided between the various drives under different conditions of operations.

SCS shall include the following sub-group systems, but not limited to:

- 1) Air pre-heater sub-group control
- 2) Induced draft fan sub-group control
- 3) Forced draft fan sub-group control
- 4) Primary air fan sub-group control
- 5) Boiler attemperator sub-group control
- 6) Superheater PCV sub-group control
- 7) Boiler drain sub-group control
- 8) Boiler sootblower system sub-group control
- 9) Condensate extraction pump sub-group control
- 10) Steam extraction system sub-group control
- 11) Low pressure heater drainage & steam release sub-group control
- 12) High pressure heater drainage & steam release sub-group control
- 13) Condenser vacuum pump sub-group control
- 14) Water recycled pump sub-group control
- 15) Auxiliary steam system sub-group control
- 16) Turbine-generator bearing oil sub-group control
- 17) Turbine EH oil sub-group control
- 18) Auxiliary steam system sub-group control
- 19) Motor-driven boiler feed pump sub-group control
- 20) Turbine rolling (ATRS) sub-group control

7.3.4 Furnace Safeguard Supervisory System

Furnace Safeguard Supervisory System (FSSS) include burner control system (BCS) and fuel safety system (FSS).

Main functions of BCS shall include:

- 1) Ignition oil gun start-up and shutdown sequential control
- 2) Oil burner automatic management
- 3) Coal burner automatic management

Main functions of FSS shall include:

- 1) Furnace purge
- 2) Fuel oil system leak test
- 3) Flame detection
- 4) Master fuel trip (MFT)
- 5) Oil fuel trip (OFT)
- 6) Damper control

Conditions requiring a master fuel trip shall include the following but not be limited to:

- 1) Emergency Unit trip push button pushed.
- 2) Air flow less than 30%.
- 3) Loss of all flame.
- 4) High furnace pressure.
- 5) Low furnace pressure.
- 6) Loss of both forced draft fans.
- 7) Loss of both induced draft fans.
- 8) Loss of both primary air fans.
- 9) Loss of all control power.
- 10) Loss of all fuel.
- 11) Turbine trip (unless bypass is in operation or on automatic).
- 12) OFT relay operated.

7.3.5 Electrical Control System

Electrical equipment control and switching shall be realized by DCS in CCR and no local routine operation is required. However, a minimum of hard-wired operating equipment for ensuring Unit shutdown safely and their associated signals shall be provided.

Electrical Control System (ECS) shall control the following electrical equipment. The scope of supply shall include, but not limited, to the following:

- 1) Generator
- 2) Generator step-up transformer
- 3) Excitation regulating system
- 4) Unit auxiliary transformer
- 5) Low voltage auxiliary transformer
- 6) Emergency power supply and diesel generator

- 7) Switching of circuit breakers at GCB's, 11 kV and 690 V levels
- 8) Direct current power supply system
- 9) UPS device

7.3.6 Reliability of DCS

The DCS shall include on line self-surveillance monitoring and diagnostic facility so that a failure or malfunction can be diagnosed automatically down to the level of individual input / output channels or hardware modules with display and print out. The reliability of DCS shall at least fulfill the following requirement:

- 1) System availability: $\geq 99.9\%$ (test time is 30 days)
- 2) Data acquisition system MTBF > 8600 h (for design criteria)
- 3) Control system MTBF > 16000 h (for design criteria)

7.3.7 DCS Capability Margin

The DCS shall be designed with sufficient hardware so that the loading can be distributed evenly among CPU processors, memory and I/O channels. The following minimum requirement shall be met:

- 1) Controller CPU load rate at the busiest time: $< 80\%$
- 2) Operator station CPU load rate at the busiest time: $< 70\%$
- 3) Occupancy capacity of internal memory: $< 70\%$
- 4) Occupancy capacity of external memory: $< 70\%$
- 5) Communication bus load rate: $< 50\%$

7.3.8 Compatibility and Redundancy

The DCS system shall be flexible for future modifications and extensions. The system (hardware and software) shall constitute a pedigree of products long been used in the market and has demonstrated its upgrade compatibility. The system shall utilize a common programming language, and as far as possible common hardware units. Future modifications or improvements in the product line shall be designed to be compatible with the present products in this respect. The man machine interface in CCR shall be compatible to commercially available operating systems (such as a Windows environment).

All components in the DCS including the network cable, communication controllers, local controllers, group controllers, power supplies and operator stations shall be provided in redundancy and arranged to operate in parallel-running mode. If parallel-running mode is not possible, duty/standby mode is acceptable. Changeover between the duty and standby modules shall be automatic upon detection of failure of the duty

module or manually by the operator. The changeover shall be bumpless and shall not cause any disruption to control functions or loss of data. However, the operator shall be alerted of this changeover by some alarms.

Triplicate or redundant signals from plant shall not be connected to the same input/output card to avoid common mode failure.

Each DCS cubicle shall be fed by two different power supplies and redundant power feed to every module rack.

In the event of complete failure of the DCS, a minimum hardwired back-up controls in the form of emergency trip pushbutton shall be provided in CCR to allow the safe shutdown of the Unit.

The system shall be designed to ensure redundancy at controller level, I/O level, bus level and power supply level so that failure of any of them does not affect the normal functioning of the system. The battery backup for the system in case of normal power failure shall be designed to ensure availability of the system for at least 1 hour.

7.3.9 System Capacity

The DCS of each Unit shall be capable of receiving at least 2500 analogue and 5000 digital signals. Scan rates for digital signals shall range from 1 ms to 100 ms and for analogue signals shall range from 0.1 s to 1.0 s. If separate sequence of event (SOE) logger is used, the capability of the SOE logger shall not less than 250 input points per Unit. Operator station monitor shall be of flat LCD type and screen size of 559 mm minimum. A minimum of 6 operator stations for the operator of each Unit, another 4 operator stations for common services and another 2 operator stations for the CCR supervisor shall be provided. In addition, another two large screen LCD displays for each Unit of minimum diagonal size of 1500 mm shall also be provided and hanged as mimic walls. Any DCS screen mimics or graphics shall be callable onto the large screen displays. At least one engineer workstation for each Unit shall be provided.

Notwithstanding the above requirement, the DCS shall provide at the I/O rack with 15% spare slots, fully equipped with input / output cards and wiring terminals. A space shall be reserved for installing a new I/O cubicle in the computer room.

7.3.10 Operator Functions

The DCS shall provide diagrammatic mimics with real time data representing the various plant equipment and processes. The production of mimics shall be arranged with the Owner to give comments at the preparation stage. The color coding, mimic symbol, and convention shall be consistent throughout all mimics. The DCS shall also be capable to produce the various logs, on request or at predetermined times.

7.3.11 Performance

All control algorithms for processing digital or pulse inputs/outputs shall be executed every 1 - 100 ms depending on the operational requirement. All control algorithms for processing analogue inputs/outputs shall be executed every 0.1 - 1.0 s depending on the operational requirement.

All displays, graphics, curves (except historian curve), texts and control functions shall appear and be complete on the LCD screen with real equipment signal values within 1.0 second on completion of the operator command (i.e. after pressing the pushbutton or keys). Operator command or action shall reach the final drives within 1 second. Alarms from field shall appear on the LCD screen with status within 1 second after the field contact is initiated.

The data transfer speed in any part of the network shall not be slower than 10 Mbits/s. The busy time (i.e. network is occupied during transfer of data) shall not be more than 50% in any circumstances, as recorded by network analyzer.

The loading (processing time) on each CPU (or microprocessor) of each local controller, group controller, operator console and interface computer of the control systems shall not be larger than 80% in any circumstances. The memory usage shall not be more than 70%.

All modules of DCS shall be time synchronized to a master clock signal with an accuracy not exceeding 1 second per day. Time synchronization shall be carried out at least once per day.

On line performance diagnosis shall be provided in the DCS to indicate the various performance figures against ref. figures.

7.3.12 Cyber Security

Appropriate measures shall be taken to ensure the cyber risks are negated. These measures shall include but not limited to:

- 1) All servers and PC's for DCS will have security provisions to prevent access by unauthorized personnel. All portable media shall be disable (Floppy, CD, DVD and USB sticks). Only authorized media to be accessed by workstation by exception.
- 2) System security features which address system management, electronic access control, user management and software patch management.
- 3) Anti-virus function shall be supplied according to DCS manufacturer standard to prevent, detect and remove malware.

7.3.13 Testing

DCS shall be tested in the following three stages.

- (a) Factory acceptance test (FAT), which shall be carried out during and after the completion of the manufacturing of the DCS. The Offshore Supply Contractor shall submit the FAT procedure of DCS for review by the Owner during detailed engineering stage. The Owner shall be given the opportunity to witness the test

and the DCS shall not be shipped until the Owner is satisfied that there has been a successful completion of FAT. The FAT shall include the hardware testing of assembled panels, and software testing of control algorithms

- (b) Site test, which shall form part of the installation and commissioning of the DCS, shall be witnessed by the Owner. The Site test shall be performed by the Onshore Contractor after the completion of the assembly of the whole DCS system
- (c) Availability tests, which shall be included within the overall reliability test of the Unit, shall achieve an availability of 99.9% or better.

The Offshore Supply Contractor shall provide the testing details and procedures for above tests for Owner's review.

7.4 Monitoring System

The DCS system shall provide the following packages. If the DCS system cannot provide the packages due to its own limitation, special dedicated systems shall be provided and connected to DCS for display.

Some of the packages may require special instrumentation mounted onto the equipment of the Facility to provide the necessary measurement signals for inputting to the DCS. The Offshore Supply Contractor shall ensure that all such special additional instrumentation, as well as sufficient signals as may be required from instrumentation that has been provided for other duties, shall be supplied for the effective implementation and operation of each of these packages.

The details of calculation algorithm shall be released to the Owner at the start of commissioning of DCS and the Owner shall be given the right to modify the calculation.

The DCS shall allow Owner's management information system (MIS) to obtain plant parameters from DCS and display them in remote sites. Sufficient spaces along the cable routes shall be reserved for MIS cables.

7.4.1 Efficiency Monitoring System

The efficiency monitoring system shall serve to monitor and provide information on the following:

- Boiler performance and efficiency;
- Steam turbines performance and efficiency;
- Unit performance and efficiency.

Sampling and calculations shall be performed on-line, so that the operator is capable to adjust the operation mode (if necessary) to optimize the economic generation of the Unit. However, off-line evaluation shall also be provided for tracking of performance.

7.4.2 Condition Monitoring System

All major pieces of rotating and reciprocating equipment shall be provided with permanent vibration monitoring equipment and integration of electronic hardware and software, which is so called condition monitoring system.

The vibration monitoring equipment supplied shall be suited to the design and operating conditions of each individual piece of mechanical equipment. The vibration monitors shall provide indication and control signals to facilitate alarms and equipment shutdown as well as data for preventive maintenance diagnostic and operating efficiency.

A microprocessor based monitoring unit shall be permanently wired to the field sensors and signal conditioners. The monitoring unit shall be mounted in a cabinet located in the central equipment room in control building.

The monitoring unit (computer) shall collect, store and display the dynamic values from the field sensors and signal conditioners. The monitoring unit shall also provide alarm status at signal levels above (operator) entered set points. External outputs for alarm/control functions shall be provided.

The vibration monitoring system shall be capable to trend information and to facilitate analysis of troubles of the turbines and generators.

The system shall provide sufficient sensors data to the Unit control system for on-line operator information and alarm on excess levels of vibration.

The machinery management system shall include the following equipment and functions:

- (a) Transducers.
- (b) Monitoring Systems.
- (c) Data Acquisition and Display Systems.
- (d) Machinery Analysis System.
- (e) Machinery and Plant Performance Monitoring System.

Continuous on-line machinery protection and management shall be implemented for equipment items identified as having a large impact on production and for equipment items which constitute major investments or which may require expensive or long duration repairs in the event of catastrophic failure, such as:

- Turbine.
- Generator
- Boiler feed pumps
- Forced draft fans,
- Induced draft fans,
- Primary air fans,
- Cooling water pumps,
- Condensate extraction pumps

The monitoring points of steam turbine shall include but not limited to the following:

- (a) Shaft speed (RPM) using a keyphasor
- (b) Shaft vibration and radial position at bearings using orthogonally mounted X and Y proximity sensors for radial vibration.
- (c) Axial position within thrust bearing using two proximity sensors
- (d) Critical process parameters
 - Metal temperature at radial and thrust bearings
 - Steam flow rate
 - Lubricating oil pressure
- (e) Performance process parameters
 - Inlet steam pressure and temperature
 - Discharge steam pressure and temperature
 - Extraction temperature, press, and flows (if applicable)
 - Shaft HP or kW
- (f) Non-critical process parameters
 - Inlet valve position
 - Lubricating oil temperature

The monitoring points of generator shall include but not limited to the following:

- (a) Shaft speed (RPM) using a keyphasor (if not already available on driver shaft)
- (b) Shaft vibration and radial position at bearings using orthogonally mounted X and Y proximity sensors for radial vibration.
- (c) Critical process parameters
 - Metal temperature at radial bearings
 - Winding temperature
 - Megawatts - MW
 - Lubricating oil pressure
- (d) Performance process parameters
 - Mega Volts Amps React. - MVARs
 - Power factor
- (e) Non-critical process parameters
 - Lubricating oil temperature
 - Stator coolant pressure and temperature
 - Field current
 - Field coolant pressure and temperature

The monitoring points of boiler feedwater pumps shall include but not limited to the following:

- (a) Shaft speed and phase using a keyphasor transducer

- (b) Shaft vibration and position on pump and motor bearing housing using X and Y orthogonally mounted proximity sensors for fluid film bearings (seismic for roller bearings)
- (c) Axial position within thrust bearing using two proximity sensors
- (d) Critical process parameters
 - Pump bearing metal temperature
 - Feedwater tank pressure and temperature
 - Flow rate
 - Lubricating oil pressure
 - lubricating oil temperature
- (e) Performance process parameters
 - Discharge pressure and temperature
 - Motor power, voltage and current from MV board

The monitoring points of cooling water pumps and condensate extraction pumps shall include but not limited to the following:

- (a) Vibration on pump and motor bearing housing using X and Y orthogonally mounted accelerometers for bearings.
- (b) Critical process parameters
 - Pump thrust bearing metal temperature or lubricating oil temperature
 - Flow rate (for condensate extraction pump)
- (c) Performance process parameters
 - Discharge pressure and temperature
 - Motor power, voltage and current from MV board

The monitoring points of forced draft fans, induced draft fans and primary air fans shall include but not limited to the following:

- (a) Shaft speed and phase using a keyphasor transducer
- (b) Shaft vibration and position on fan and motor bearing housing using X and Y orthogonally mounted seismic sensors for the bearings.
- (c) Axial position within thrust bearing using two proximity sensors
- (d) Process parameters
 - Flow rates at forced draft duct and primary air duct
 - Flue gas velocity at stack for induced draft fan
 - Lubricating oil pressure
 - lubricating oil temperature
- (e) Performance process parameters
 - Discharge pressure and temperature
 - Motor power, voltage and current from MV board

7.4.3 Electrical Metering Package

With signals from tariff meters, the metering package shall calculate and provide the following minimum information on a real time and regular basis:

- Generator MWh exported;
- Generator MVARh exported;
- Unit auxiliary transformer MWh for in-house use by import power
- Diesel generator MWh for in-house use.

7.5 Master Clock

The DCS system shall have master clock which shall be provided with self-battery or UPS or DC system. The master clock shall operate for a minimum 10 hours in the event of loss of AC power and with a long term error of less than 30 seconds per year. All processors and clocks within a particular sub-system including the sequence of events (SOE) tags shall be synchronized to this master clock. The clock shall have a resolution of less than 1 ms.

The DCS, plant information/application processor and SOE hardware and software shall support time schemes such that updating the clock shall occur instantly and seamlessly across all systems and without loss of data.

Station clock system consists of one (1) master clock and twenty (20) slave clocks. The master clock shall have a capacity to connect maximum 50 pieces of slave clock. Time correction signal to DCS shall also be provided

Time correction of slave clock shall be automatically controlled by master clock. Twenty (20) slave clocks shall be assigned as follows.

- One (1) in central control room
- One (1) in GIS control building
- Six (5) in administration building
- Two (2) in power house
- Five (5) in control building
- One (1) in workshop
- One (1) in ESP electrical house
- One (1) in ash electrical house
- One (1) in jetty electrical house
- One (1) in coal handling electrical house
- One (1) in SWFGD electrical house

7.6 Continuous Emission Monitoring System

7.6.1 Stack

7.6.1.1 General

A continuous emission monitoring system (CEMS) shall be provided for continuous monitoring of emission gas condition at the stack and boiler combustion control of each Unit. The CEMS shall measure flow, NO_x, SO₂, CO, CO₂, O₂, particulate, velocity, moisture, pressure and temperature of flue gas at the stack. Measurements of NO_x, SO₂, CO, CO₂, O₂ and particulate shall be corrected to dry, 0°C, 101.325 kPa(a) and 6% O₂. Both raw and corrected data shall be able to display locally and transmit to DCS.

The output of the CEMS shall be connected to DCS by I/O interface for monitoring purposes, including displays on DCS, logging, reporting, alarm display, data storage media. The CEMS shall also provide interface to transmit data directly to local Environmental Protection Department of Vietnam Government.

Different sensors and sample probes shall be directly installed in the Unit 1 and 2 flue gas chimney for above CEMS measurement. Chimney analyzer enclosure with air conditioner for each Unit shall be provided at the same working platform of chimney near to the sensors and sample probes to minimize the length of heat tracing/sampling line from sampling probe to the analyzers.

The flue gas shall be extracted through a heated probe and transferred along heated sampling line to the conditioning units. There the flue gas shall be cooled down and separated from the condensed water. The dried and cooled flue gas shall be routed to the different measuring devices, where the specified parameters shall be measured by the following techniques:

- | | |
|--|---------------------------------|
| • SO ₂ , CO, CO ₂ , moisture | Non-Dispersive Infra-red (NDIR) |
| • NO _x | NDIR or Chemi-luminescence |
| • O ₂ | Paramagnetic |
| • Standard | EPA protocol standard |

The CEMS shall measure flue gas flow and velocity by a common measuring instrument. Also, the CEMS shall measure particulate, pressure and temperature by separate sample probes. The measurement methods shall be as follows:

- | | |
|---------------------|---|
| • Particulate | In-stack opacity monitor |
| • Temperature | Thermocouple |
| • Pressure | Pressure transmitter |
| • Velocity and flow | Common Pitot tube or thermal flow or ultrasonic |

For calibration of analyzers, calibration gas cylinders shall be provided near to each chimney analyzer enclosure.

The Offshore Supply Contractor shall provide a CEMS container at ground level near the chimney. The electrical power distribution panel, UPS, communication panel, PC for data acquisition system, printer, PLC, working desk, etc. shall be provided in the CEMS container. Necessary power supply shall be provided in the CEMS container.

The CEMS container shall provide UPS and non-UPS power supply for the chimney analyzer enclosures. The enclosures shall send necessary signal and contacts signals to the CEMS container.

Analyzer enclosure at the platform of chimney shall send the signal from fire detector, sample pump stop command, calibration start command, etc. to CEMS container at ground floor. Then the CEMS container shall provide all these signals and reporting to DCS for data logging, reporting, alarm and display.

The CEMS container shall provide necessary interface to connect through modem or public telephone line for remote monitoring.

7.6.1.2 Design

The gas analyzer equipment shall include but not limited to the following:

1. Gas analyzers
2. Extraction equipment, heated tracing
3. Gas cooler and collecting tank for condensate
4. Pumps
5. Filters
6. Complete auxiliaries
7. Cabling
8. Automatic calibration equipment
9. Temperature control for extraction heater

The following requirements shall be complied:

- (a) All gas analyzers shall be digital, microprocessor based systems. Each analyzer shall be equipped with heater and temperature control plus high temperature alarm.
- (b) The analyzer shall be designed and arranged to facilitate periodic testing, calibration maintenance without removing them from the duct. Test jacks or terminals shall be provided.
- (c) All gas analyzers shall be able to be auto-calibrated every 24 hours.
- (d) SO₂ analyzer
 - i. Non-linearity less than or equal to 1% of the measuring span
 - ii. Zero drift (week) less than or equal to 2% of span
 - iii. Span drift (week) less than or equal to 2.5% of span
 - iv. Response time less than or equal to 180 second
- (e) NO_x analyzer

It shall meet the same technical features as the above specified SO₂ analyzer.
- (f) CO analyzer

It shall meet the same technical features as the above specified SO₂ analyzer.

(g) Moisture

It shall meet the same technical features as the above specified SO₂ analyzer.

(h) CO₂ analyzer

- i. Measurement range: 0 ~ 20%
- ii. Accuracy less than or equal to 1.5% of span
- iii. Non-linearity less than or equal to 0.5% of span
- iv. Response time less than or equal 30 seconds

(i) O₂ analyzer

- i. Measurement range: 0 ~ 25%
- ii. Accuracy less than or equal to 1.5% of span
- iii. Non-linearity less than or equal to 0.5% of span
- iv. Response time less than or equal 30 seconds

(j) Particulate

- i. Zero drift (week) less than or equal to 2% of span
- ii. Span drift (week) less than or equal to 5% of span
- iii. The particulate monitor shall include a calibration system for simulating a zero in-stack opacity and an upscale in-stack opacity value for the purpose of performing periodic checks of the monitor calibration while on an operating stack. This calibration system shall provide, as a minimum, a system check of the monitor internal optics and all electronics circuitry including the lamp and photo-detector assembly.
- iv. Instrument air shall be provided to clean the optical surface exposed to the effluent stream automatically.
- v. Each monitor shall provide a mean of access to the optical surfaces exposed to the effluent stream in order to permit the surfaces to be cleaned without requiring removal of the unit from the source mounting or without requiring optical realignment of the unit.
- vi. If the system has a feature that provides automatic zero compensation for dust accumulation on exposed optical surfaces, the system shall also provide some means of indicating when a compensation of 4% in-stack opacity has been exceeded.
- vii. The monitor shall provide a design that accommodates the use of an external calibration filter to assess monitor operation.
- viii. Each monitor shall provide some method for visually determining that the instrument is optically aligned.

(k) Temperature

- i. Uncertainty less than or equal to 3 °C

(l) Velocity

- i. Measuring range: 0 to 40 m/s
- ii. Accuracy less than or equal to 3% of span
- iii. Uncertainty less than or equal to 0.1 m/s
- iv. Response time less than or equal to 60 seconds

(m) Data Acquisition System

- i. The CEMS shall include a data acquisition system that accepts the output of the analyzers and flow monitors and converts these to emission rates of the gaseous pollutants in appropriate units. These systems shall be capable of interpreting and converting the individual output signals from each monitor to produce a continuous readout in appropriate units. The system shall maintain a permanent record of all parameters in a format acceptable to the Owner and agreed during detailed engineering stage. The system shall also record and compute daily zero and span drifts. It shall read and record the full range of pollutant concentrations from zero through to span, and provide a continuous, permanent record.

7.6.2 Cooling Water Outlet

The continuous monitoring instruments shall be provided to measure flow, pH, residual chlorine and temperature at the cooling water discharge of each Unit.

The output of the instruments shall be connected to DCS by I/O interface for monitoring purposes, including displays on DCS, logging and data storage.

The continuous monitoring instruments shall be inspected by the Owner, and also provide an interface for the transmission of data directly to local Environmental Protection Department of Vietnam Government. The instrument shall be calibrated in accordance with the O&M Manuals.

The type of instruments for pH, Chlorine and temperature measurement shall be provided as follows:

The type of instruments

- | | |
|---------------|-------------------|
| • pH | In-situ method |
| • Temperature | In-situ method |
| • Chlorine | Extractive method |

The measurement method and reference list shall be provided for Owner's review during detailed design stage.

7.6.3 Seawater Discharge of SWFGD

A monitoring point for monitoring the quality and various parameters of seawater discharged from each SWFGD shall be installed, before the seawater flow to the discharge canal.

The monitoring point shall monitor the seawater quality continuously, and shall monitor, but not limited to, the following parameters:

- Flow
- Temperature
- pH value

- Dissolved Oxygen (DO)
- Total Suspended Solids (TSS)

The output of the instruments shall be connected to DCS by I/O interface or serial link for monitoring purposes, including displays on DCS, logging and data storage.

The continuous monitoring instruments shall be inspected by the Owner, and also provide an interface for the transmission of data directly to local Environmental Protection Department of Vietnam Government. The instrument shall be calibrated in accordance with the O&M Manuals.

The Offshore Service Contractor shall propose a technical solution to perform continuous monitoring of total sulphite (HSO_3^- and SO_3^{2-}) for Owner's review during design submission stage, to satisfy the EIA requirement.

7.6.4 Waste Water Treatment Plant Outlet

The continuous monitoring instruments shall be provided to measure inlet and outlet flows, pH, temperature, COD, TSS, turbidity and oil-in-water at the waste water treatment plant outlet.

The continuous monitoring instruments shall include sealed automatic samplers and cameras connected to the internet to observe the waste water treatment plant outlet and store images taken within the last 3 months.

The output of the instruments shall be connected to DCS by I/O interface or serial link for monitoring purposes, including displays on DCS, logging and data storage.

The continuous monitoring instruments shall be inspected by the Owner, and also provide an interface for the transmission of data directly to local Environmental Protection Department of Vietnam Government. The instrument shall be calibrated in accordance with the O&M Manuals.

7.7 Interface with Grid Dispatch Centre

Supervision/control, monitoring and information systems shall comply with the system control concepts and procedures and will be subject to the approval of local grid company.

Automatic and manual synchronizing facilities, with such check facilities as may be required shall be provided as a minimum for all circuits associated with generators.

The Facility shall be provided with a CCR so that operators can control the generators and perform switching and load dispatch duties. A DCS shall be provided to coordinate the control and supervision of the generator control, Facility auxiliary systems, and generator step-up transformer breakers.

The necessary data and information to be exchanged between the Facility and the dispatch centre via the EVN's 500kV switchyard and SCADA/EMS systems shall include but not limited to the following list:

	From Vung Ang II Power Plant	NLDC	RLDC
1	Status of GCB's and two generator step-up transformers	X	X
2	Status of 500kV earthing switches, GCB's and two generator step-up transformers	X	X
3	Status of operated relay protection system of two generator step-up transformers	X	X
4	Measured signals of voltage, current, power and reactive power of all 500 kV lines	X	X
5	Measured signals of voltage, current, power and reactive power and tap changer positions of transformers	X	X
6	Measured signals of voltage, current, power and reactive power from the Units	X	X
7	Measured signals of voltage of the bus bars	X	X
8.	Measured signal of frequency	X	X
9	Raise/lower signals for controlling the Unit outputs by either set point or pulses	X	X
10	Operation orders (open/close/trip) circuit breakers, disconnectors, and earthing switches	X	X

An interface to the power plant's telecommunication system shall be provided in order to receive signals from the dispatch center. Details of control modes and interfaces shall be designed in the project stage with the agreement from the dispatch centre. All the necessary interface hardware and software for this purpose shall be provided.

The Onshore Contractor shall determine the signals to be exchanged between the dispatch centre and Facility during Project execution stage. If necessary, hardwiring of not more than 100 signals and redundant links with protocol compatible (IEC 60870-5-104) with the dispatch centre shall be provided to the interface marshalling box where the signals will be transmitted to dispatch centre. The Contractor shall provide the interface marshalling box located within the Site.

The Facility shall respond to the load demand under different control modes from the dispatch center. The actual calculation algorithm for the control modes shall be programmed to suit the requirement of the dispatch centre. It is expected that the control modes shall at least include the following:

(1) Automatic Generation Control (AGC):

Unit output shall closely follow the load demand signal from dispatch centre or the local station.

(2) Automatic Frequency Control (AFC):

Unit output shall respond to power system frequency deviation and shall be adjusted accordingly. Another mode of AFC shall enable the output to respond not only to frequency deviation but also the difference between the actual output and the setpoint.

(3) Emergency Generation Control (EGC):

In case power system frequency deviates beyond the normal range, Unit output shall respond at its maximum instantaneous allowable rate in order to pull back the frequency to the normal range.

7.8 Communication Systems

The communication system shall be furnished complete with required accessories, and devices for the following system:

7.8.1 Telephones and PC LAN Cables

A 120 extension 25 incoming line digital private automatic branch exchange (EPABX) shall be provided. Adjacent to each telephone socket in the workplace, a PC LAN socket shall also be installed. A PC LAN cable shall be also provided together with the telephone cables to all the telephone / PC sockets.

7.8.2 Inter Communication System

A solid state public address system with page and party channels and a number of walkie-talkie or radio pagers shall be used throughout the plant.

The EPABX shall be fully compatible with public telephone system. The EPABX shall be connected to public telephone system. The extension set in the noisy plant environment shall be provided with handset station, which consists of hand set and acoustic hood. The hand set shall be of telephone type with receiver and low impedance, noise canceling transmitter designed for talking in close proximity.

All telephone sockets shall be provided with dual sockets (one for telephone and one for computer jacks).

7.8.3 Telecommunication between the Facility, 500kV Switchyard, EVN National Load Dispatch Centre, EVN Regional Load Dispatch Centre and Viettel

Figures 7.1 and 7.2 are the schematic diagrams for reference only. They may change due to actual site condition and legal regulation during Project execution stage.

The Onshore Contractor shall coordinate and work closely with Viettel and EVN to provide and install necessary communication equipment, communication cable and connection, and LV power supply system (if not available) of installed equipment INSIDE Viettel and EVN units for the communication interface between

- i. the Facility and the EVN 500 kV substation to serve for the Facility,
- ii. the Facility and National Load Dispatching Centre to serve for the Facility,
- iii. the Facility and Regional Load Dispatching Centre to serve for the Facility, and
- iv. the Facility and Viettel to serve for the Facility.

The communication cables between EVN 500kV substation and EVN units & Viettel along the transmission lines are excluded.

The definition of such communication equipment shall be agreed with EVN during Project execution stage.

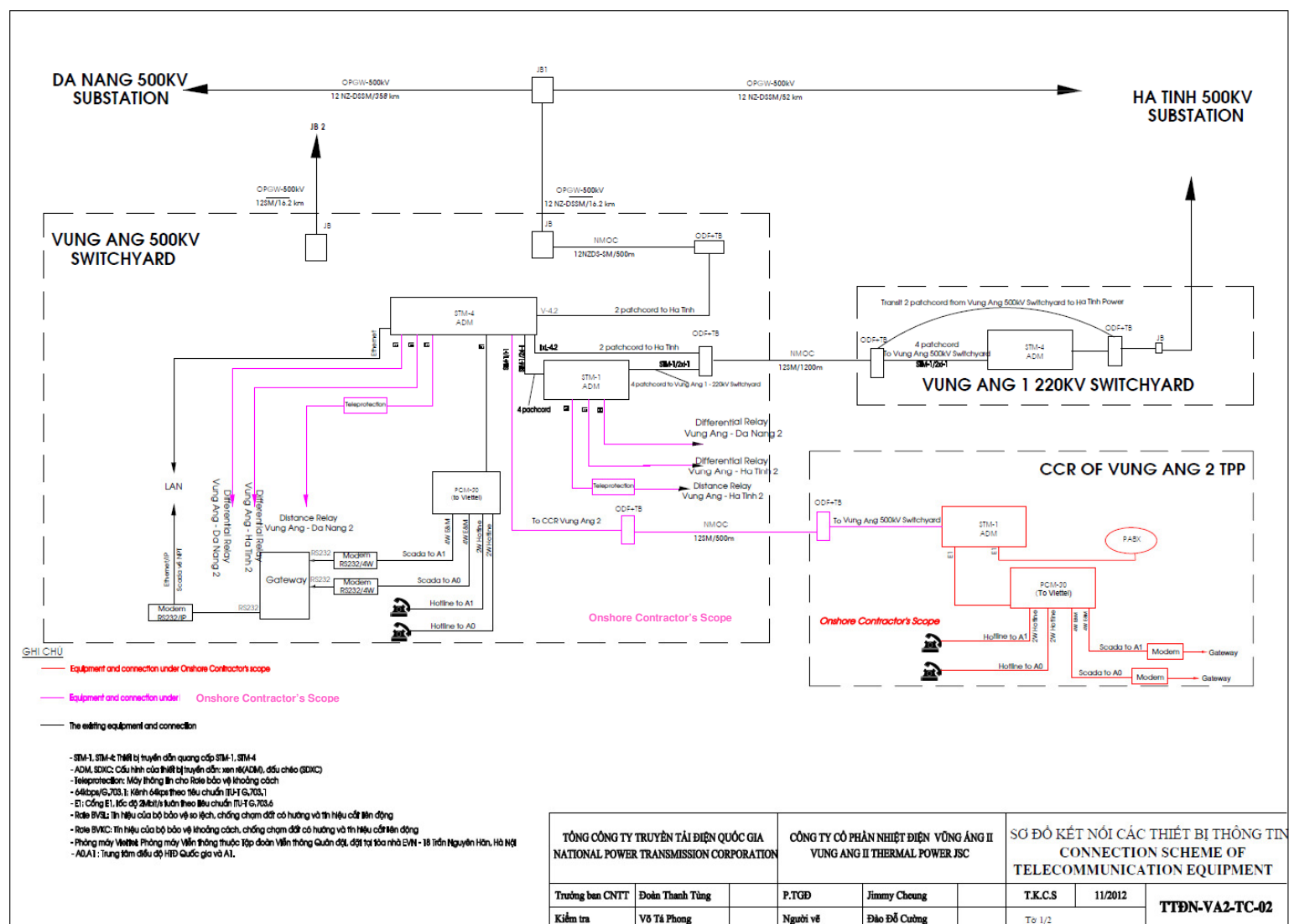


Figure 7.1 TTDN-VA2-TC-02 Telecommunication Interface (1 of 2)

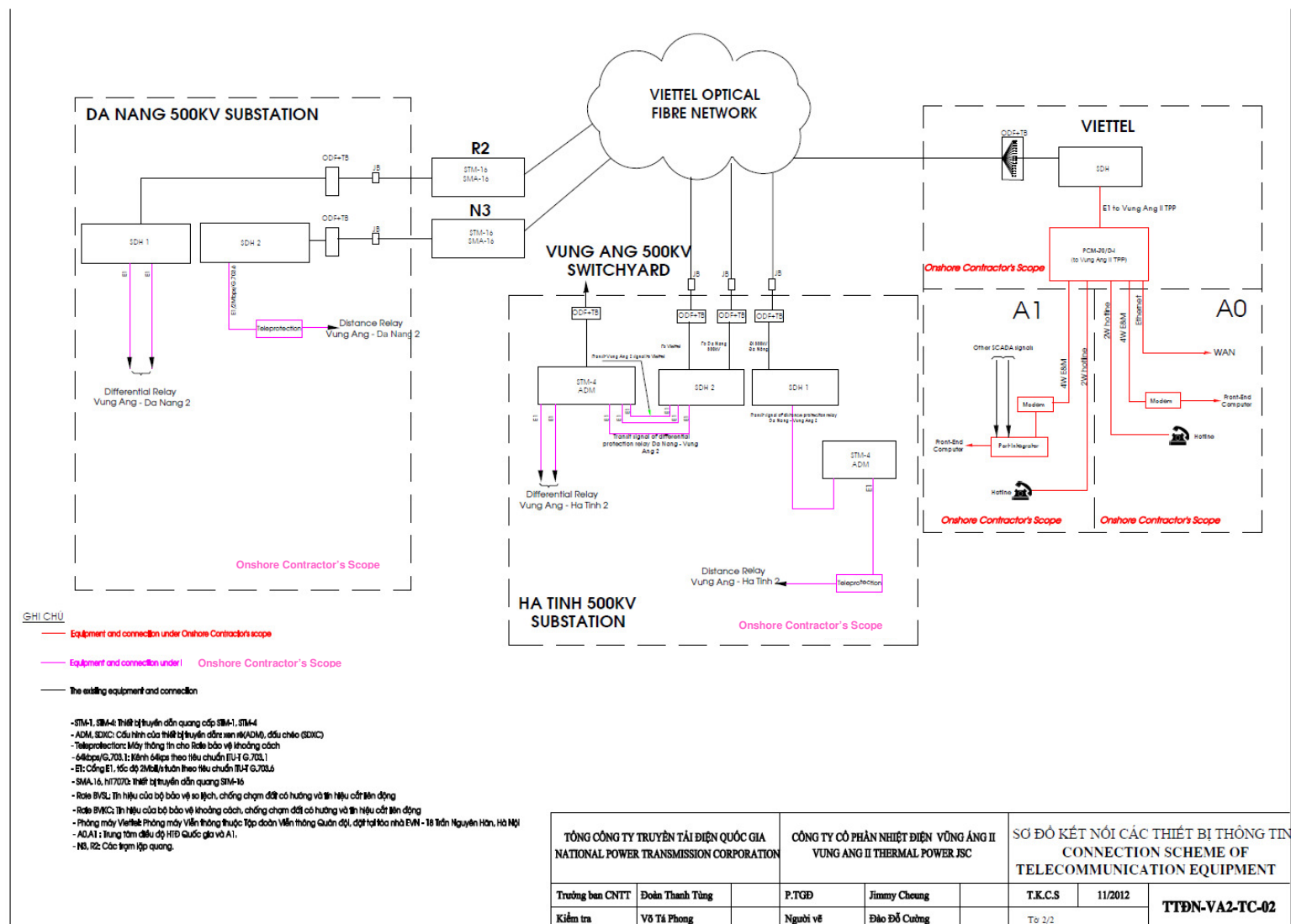


Figure 7.2 TTDN-VA2-TC-02 Telecommunication Interface (2 of 2)

7.9 Closed Circuit Television System

Closed circuit television system (CCTV) shall be provided for security purpose and operational purpose.

The security-purpose CCTV system shall monitor the access road, entry gate area, boundary line (fence) and other yards considered for security observing purposes. This set of CCTV shall be installed in the guard house at the main entrance of the Facility.

The operational-purpose CCTV shall monitor the operational area such as burner area, turbine hall, coal unloading jetty, stacking of coal in coal yard, coal transfer tower, sea water FGD, water treatment plant, transformers, switchyard and other machineries and equipment. This set of CCTV shall be installed in the CCR.

The CCTV shall be located at strategic points to monitor the movements of personnel for vigilance. Cameras with remote pan, tilt and zooming facilities shall be furnished. These cameras video images shall be monitored in TFT-LCD in the guard house.

Fibre optic cable shall be provided to link between the cameras and CCTV monitoring system in the guard house. The monitors of system shall be mounted on the auxiliary back up panel in the guard house.

Each monitor CCTV shall be capable to be divided into minimum 4 (four) sections. A control console shall be provided in the gate house, to enable an operator to select and control individual camera. High resolution color monitors shall be provided and time-lapse data optical recorders which include a time/date generator for recording the images from the selected camera shall be provided. Recorders up to 7 days recording capability shall be provided.

7.10 Steam Turbine Digital Electro-hydraulic Control System

Steam turbine digital electro-hydraulic control system (DEH) shall interface with the DCS in such a way that control of the steam turbine is realized in DCS as if the DEH is a part of the DCS as per section 7.1. DEH shall execute the following functions:

- 1) Rotational speed control of steam turbine
- 2) Load control function
- 3) Main steam pressure control.
- 4) Valve management
- 5) Run back

7.10.1 Rotating Speed Control Function

DEH shall ensure the steam turbine adopts the maximum run-up rate conforming with the turbine thermal conditions, admission steam conditions and permitted steam turbine life consumption. It shall automatically control the steam turbine turning gear from rotational speed to rated speed. The design of automatic run-up system shall coordinate with the design of steam turbine and its bypass system and be adaptable to the operation of steam turbine with and without bypass.

The system shall switchover automatically in each run-up stage among the three valving modes of high pressure main steam valve, high pressure regulating valve and medium pressure regulating valve according to the start-up requirements for various thermal conditions.

The runup rate in steam turbine shall either be selected automatically by DEH or by hand according to the thermal state of steam turbine. Critical speed zones shall be rush-passed automatically.

DEH shall have an interface with automatic synchronizer to realize automatic synchronization of generator.

7.10.2 Load Control Function

DEH shall realize automatic control of the turbine generator from initial load to full load after the generator is connected to the grid. DEH shall take part in the tasks of primary frequency modulation and secondary frequency modulation according to the power grid requirement.

The target load and load variation rate of the system shall be given by operators or received the instruction from DCS.

The system shall have adjustable maximum and minimum load limit values and rate limit value. When the operating conditions appear abnormal, the system shall perform the following limit controls:

- 1) Power feedback limit
- 2) Variable load rate limit
- 3) Maximum and minimum load limit
- 4) Acceleration limit
- 5) Main steam pressure limit

In order to improve the dynamic stability of the power system or restrain the over speed of steam turbine generator units, DEH shall have load shedding control function. When the Unit sheds off load by a big margin, DEH shall automatically control the rotating speed of steam turbine, prevent the Unit from tripping because of over speed, and control the Unit at the rated speed.

7.10.3 Over Speed Protection

Over speed protection is a control function restraining over speed. Over speed protection shall regulate the high and intermediate pressure till normal rotating speed is resumed.

7.10.4 Over Speed Trip Protection and On-line Test

When the speed of steam turbine reaches a pre-determined speed under 110% of rated speed, the system shall operate high and medium pressure regulating valves. The over speed (110%) trip protection function of steam turbine shall be provided by the emergency trip system (ETS).

DEH system shall provide the instruments for performing the pre-determined speed under 110% and 110% over speed tests by operators during normal operation so the operator can confirm the correct functions of over speed control and trip system.

7.10.5 Run Back

In order to prevent complete Unit shut down when the load is more than 60%, the Unit shall be possible to reduce the load to a safe operating level.

The Unit load runback shall be initiated if any one of the following equipment fails:

- 1) One induced draft fan tripped
- 2) One forced draft fan tripped
- 3) One primary air fan tripped
- 4) One air pre-heater tripped
- 5) One boiler feed pump tripped
- 6) Condenser vacuum decrease

Any load runback condition shall initiate operation of the steam generator controllers and the turbine control system in order to reduce the steam generator load to the possible allowable steam generator performance. All information and/or conditions shall be provided to the operator or load-dispatching center in a clear and understandable display on the operator monitoring stations to limit the load demand in runback operation.

All functions of the load runback shall be provided and implemented into the hardware and software of the Unit load demand controller.

7.11 Emergency Trip System of Steam Turbine

Emergency trip system of steam turbine (ETS) shall monitor the turbine generator parameters and states affecting safe operation. When the parameters are beyond the limit or the state is abnormal, ETS shall send out the trip signal of steam turbine and shut off the inlet valve of steam turbine. ETS shall have on-line test function.

When any one of the following and other conditions as per OEM's standard happens, ETS shall send out a signal to shut off main steam valve:

- 1) Steam condenser vacuum too low
- 2) Steam turbine is 110% over speed
- 3) Lubricating oil pressure is too low
- 4) Axial displacement not within limits
- 5) Shaft vibration not within limits
- 6) Failure of generator step-up transformer
- 7) EH oil pressure too low
- 8) Boiler MFT
- 9) Important control system mains fail (e.g. ETS, DEH loss control power)
- 10) Manual shut down

7.12 Steam Turbine Safety Inspection System

Steam turbine safety inspection system (TSI) shall monitor all important operating parameters of steam turbine body continuously. These parameters reflect the operation state of steam turbine and are related to safe operation of steam turbine closely. When the parameters are beyond the normal values, alarm shall be sent out and record. When the parameters are beyond the limit values, tripping signal shall be sent to ETS.

TSI monitors the following parameters:

- 1) Speed or key phase
- 2) Differential expansion
- 3) Casing extension
- 4) Eccentricity
- 5) Shaft vibration
- 6) Bearing vibration
- 7) Thrust bearing position

7.13 BOP System

The BOP system shall not require any operation attention, but engineering station for each local control room, complete equipment with all components and accessories shall be supplied for BOP for the following system:

- 1) Coal handling equipment control system
- 2) Ash handling equipment control system
- 3) Water treatment control system
- 4) Wastewater treatment control system

7.14 Power Supply

7.14.1 AC and DC Power Supply for Control and Monitoring System

The design of the power supply for the control and monitoring systems shall meet the following requirements:

- 1) Power supply for important C&I equipment affecting Unit operation shall be fed from UPS.
- 2) High degree of reliability and availability to ensure that under no condition does the supply to the control system fail.
- 3) Ability to use the normal power supply available in the Facility under maintenance conditions.

7.14.2 DC Power Supply for Control Portion of the System

The redundant 24V DC supply for I/O cubicles shall be derived from the UPS which shall be secured by the diesel generator.

Chapter 8 – Balance of Plant Systems

8.1 Cooling Water System

8.1.1 General

The cooling water (CW) system shall consist of the Plant required to supply CW to the condensers, the Unit auxiliary cooling water (ACW) system, the closed cycle cooling system, the electro-chlorination Plant, the seawater desalination Plant and the sea water flue gas desulphurization system (SWFGD).

The Offshore Services Contractor shall be responsible for the design of the Vung Ang I Outfall Extension Works and the Onshore Contractor shall be responsible for the procurement and construction of the Vung Ang I Outfall Extension Works from existing near shore surface discharge by 1.8 km into the sea. This is to satisfy the recommendations from the warm cooling water recirculation study on the combined operation of the Vung Ang I and Vung Ang II Projects. The structure of Vung Ang I Outfall Extension Works shall be totally and physically separated from that of Vung Ang II. To facilitate the above Works, the Owner will provide updates to the contractor of the Vung Ang I CW system design (with any changes to the design or dimensions of the pipeline forming part of the Vung Ang I Outfall Extension Works after the date of the Contract to be treated as a variation). The Offshore Services Contractor shall design the Vung Ang I Outfall Extension Works based on the information provided by the Owner. Vung Ang I will provide a diversion chamber with removable stop logs to enable initial near shore discharge when Vung Ang I becomes operational, and then subsequent diversion to off shore discharge after the said extension is complete. The Vung Ang I Outfall Extension Works forms part of Section 2 of the Works and shall be completed prior to operation of the First Unit. The Offshore Services Contractor shall include and provide a scheme to resolve the interface with Vung Ang II CW system and jetty.

The Onshore Contractor shall implement quality control measures according to Best Industry Practice and international standards on the construction of the Vung Ang I Outfall Extension Works to ensure the integrity of the extension and achievement of leak-free construction.

The quality control measures shall include but shall not be limited to the following:

- Cement strength tests;
- Site sample quality checks and tests; and
- Inspection and testing of the on-shore / off-shore pipe and culvert seals and connections and record the inspection results.

The Onshore Contractor shall provide the following:-

- (a) CW intake system including CW intake culvert/pipework, forebay, pump station, access ladders and platforms in the CW pump station;
- (b) Civil construction and structure;
- (c) In-situ interconnecting pipework from CW pump station to the Facility;

- (d) All services including fresh water, compressed air, fire-fighting and electric power etc. for a manned arrangement of the CW pump station.
- (e) CW discharge and outfall culvert;
- (f) Interconnecting pipework from treated contaminated storm water from the Site to the CW discharge;
- (g) Aeration pond; and
- (h) Vung Ang I Outfall Extension Works.

The Offshore Supply Contractor shall provide the following:

- (a) CW intake system including CW pumps, valves, trash racks, trash-removal screens, crane for handling pump, screen washing pump;
- (b) Interconnecting pipeline from pump station to the Facility;
- (c) Electro-chlorination Plant;
- (d) Cathodic protection system for the CW equipment including the associated CW pipework; and
- (e) Discharge from SWFGD in conjunction with the CW system.

The Offshore Services Contractor shall design to have the CW supplied to the Facility via two pressure CW pipes. The two pipes at the pump discharges shall be inter-connected by sectional valves. Any CW pumps shall be designed to supply seawater to any Units.

The intake chamber shall be adequately sized to give satisfactory flow condition at the pump bellmouth. The CW pumps shall be of the wet pit or concrete volute type, direct driven type or high speed motor with speed reducer (gear box) and soft starter, complete with electric motor. There shall be 2 × 50% CW pumps supplying water to each Unit.

The CW system shall ensure that there is no unacceptable surge developed in the system following a pump trip or under system transient conditions. The CW pumps shall be protected from excessive reverse rotation when the butterfly valve is closing and any damage resulting from reverse rotation under these conditions.

Stop gates shall be provided upstream of the bar screen to enable each pump intake channel to be isolated and dewatered for maintenance. Storage racks with hoists of same number as stop gates supplied shall be provided.

Similarly, provisions shall be made to allow the isolation of the SWFGD aeration basins so that they can be isolated for maintenance and inspection.

Submersible type dewatering pumps shall be provided for dewatering of screen chamber and intake chamber. Local power supplies shall be supplied for each pumping out areas for the dewatering pumps.

Appropriate numbers of access manholes shall be provided along the intake and outfall culverts for inspection purposes.

Impressed current type cathodic protection shall be provided to protect all wetted parts against corrosion. Impressed current type cathodic protection shall also be applied to the protection of the CW pipework as well as the submerged intake, the steel pipework between the CW pump discharge and Vung Ang I crossing area, and outfall structures. In addition, these submerged structures shall be protected from fouling by suitable chlorination dosing. The design of the CW system shall be designed to avoid the settlement of silt. The design shall allow for measures to achieve the removal of silt by appropriate settlement basin such as at the inlet to the CW pump station forebay.

CW shall be taken off from the CW system at a point upstream of the condenser for use for the ACW system. The cooling water of ACW system after passing the Unit auxiliaries shall be returned to the CW system via a connection downstream of the condenser. Isolation valves shall be provided upstream and downstream of ACW system. Whenever a direct seawater cooling system is not considered appropriate for the cooling of essential auxiliaries, eg due to the risks of contamination due to leakages, a second cooling medium with demineralized water shall be used.

Seawater supply to the SWFGD shall be taken downstream of the condenser and shall not increase the flow requirement of the seawater discharged by the CW pumps.

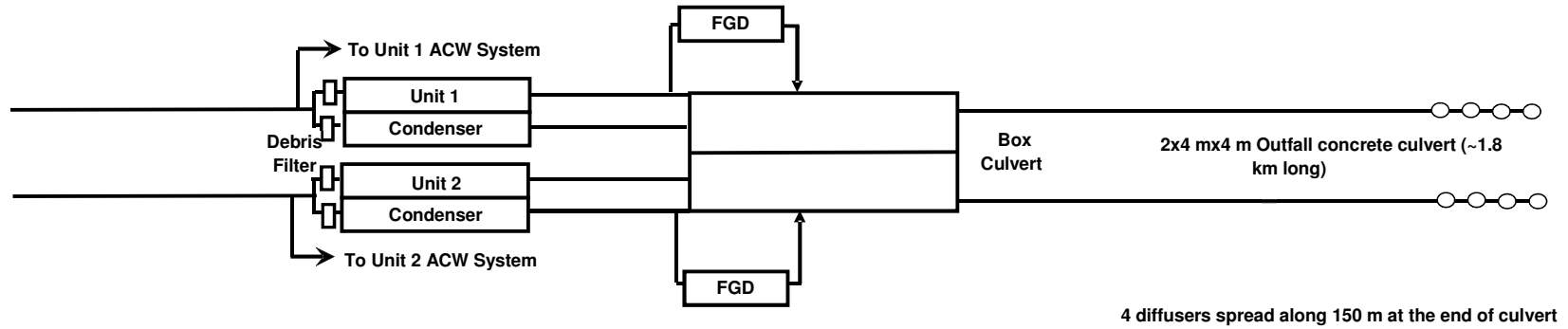
Sea water for the desalination plant shall be taken upstream of the condenser. Sodium bisulfite injection or other proven measures shall be adopted for de-chlorination of sea water before feeding it to the desalination plant as the sea water has been dosed with sodium hypochlorite.

The scope of supply for the CW system shall include, but not be limited, to the following:

- (1) Four (4) sets of bar screens and accessories
- (2) Four (4) sets of traveling screens and accessories
- (3) 2 x 100% of screen wash pumps with motors, duplex filters, valves and water jets
- (4) Stop-gate enabling the isolation of two Units;
- (5) Four (4) 50% capacity CW pumps with motors, and accessories
- (6) Four set (4) of portable submersible pumps with motor drive for evacuation of sea water from intake chamber
- (7) All CW pipework including:
 - (a) All necessary valves, expansion joints, fittings, flanges
 - (b) Manholes with fittings on the circulating water pipe
 - (c) Branches and blanks for anti-surge valves
- (8) Power operation of all valves and other items which are necessary for the remote starting-up and shutting-down of the CW system
- (9) Supports, access steelwork and platforms, including necessary walkways, ladders and handrails required for convenient access for operation, inspection and maintenance of the Plant

- (10) All fixed and roller supports, foundation steel work, embedded steelwork holding bolts and grouting materials
- (11) Painting and surface protection for components during and after manufacture and at site after erection and testing
- (12) Local instruments, gauges and necessary test connections
- (13) Electrical equipment
- (14) Controls and instrumentation
- (15) Anti-surge valves
- (16) CW chlorination plant to arrest biological growth and its associated chlorine dosing and distribution systems
- (17) Impressed current type cathodic protection system
- (18) Hydraulic oil system
- (19) CW pumps station based on outdoor design with the associated cranes and lifting facilities
- (20) CW intake with navigation aid facility at the protrusions such at inspection manhole, intake head, etc.
- (21) CW outfall with navigation aid facility at the protrusions such at diffusers, etc.

The general definition on the scope of supply for the CW system is as indicated in the following figures.



8.1.2 Design Conditions for CW System

The CW system shall be of once through type and is envisaged as follows:

Bar screens, traveling screens with motors and screen wash pumps with motors installed outdoor shall be of weather proof structure.

Heated CW from condenser shall be routed to the SWFGD system for absorbing SO₂ in the gas and then discharged into sea.

Sea water level at Site varies significantly depending on the seasons and is also affected by the tides. The CW system shall be designed to be suitable for 600 MW net at CW system at average fouling condition, LLWL at CW intake, inlet cooling water temperature at 33 °C and HWL at CW outfall. The Offshore Services Contractor shall provide design submission on CW system during detailed engineering stage.

The system shall be designed to recover the maximum possible siphonic head, but the CW pumps shall be capable of priming the system without resort to vacuum or priming devices. The Offshore Services Contractor shall select an optimum difference in level between the top of the condenser water box and the top of the weir at CW seal pit during the detailed engineering stage.

The design pressure rating of the CW system shall be the larger of :

- 1) 6 barg, or
- 2) Maximum pressure anticipated under surge condition, or
- 3) The maximum pressure at pump discharge (considering shut off head and high sea water level).

Hydrostatic pressure shall be 50% excess of the design pressure.

The CW pumps shall be identical and shall be suitable for parallel operation. Components of the CW pumps shall be interchangeable. The head versus capacity and shaft power versus capacity characteristics should match to ensure even load sharing and trouble free operation throughout the full load range. The pumps shall operate satisfactorily without cavitation or deleterious effects or under vibration or noise at all water levels from minimum to maximum.

A minimum margin of 10% on friction head shall be provided for the CW pumps. In case the Offshore Services Contractor want to apply in house design margins for the sizing of the main cooling water pumps, details shall be submitted for Owner's review during the detailed engineering stage.

The CW pumps shall be directly driven by a constant speed squirrel cage induction motor. Continuous motor rating (at 40 °C ambient) for CW pump shall be at least ten percent (10%) above the maximum load demand of the driven equipment in the operating range (including run out condition of pump). The CW pump motors shall be designed to be protected from overloading by interlocking CW pump motor operation with CW pump discharge valve open/close operation, CW pump motor winding temperature and motor overloading time limit.

The thickness and the construction of the pipes / ducts including joints shall be able to withstand the maximum pressure/ vacuum encountered during the transient conditions.

Pump design with single thrust bearing at motor top or separate thrust bearings at pump and motor is acceptable. All thrust bearings shall be oil lubricated, centrally pivoted tilting pad type capable of absorbing axial thrust in both direction of rotation.

The CW pumps shall be of the vertical suspended or concrete volute, single stage type. Inlet to the pumps shall be formed in the pump station foundation. The pumps shall be designed so that entire pumps (internal metallic parts of concrete volute pump) can be pulled out as a unit by disconnecting the pump discharge pipes. The Offshore Services Contractor shall design the CW pumps for easy maintenance e.g. component shall be anti-rusting, sufficient access shall be available for maintenance activities. Any special tools for facilitating easy maintenance shall be provided if necessary. Duplex stainless steel shall be applied to high velocity areas which are subject to erosion and corrosion.

Each CW pump and motor drive shall be capable of starting against a closed discharge valve and shall be capable of operating between zero flow and run out flow without overheating the motor drive. The pumps shall be capable of running under throttled discharge condition for the purpose of filling and priming the circulating water system and the condenser. Necessary speed switches to detect reverse rotation shall be provided to prevent motor running. The indication and alarm in DCS shall be provided for reverse rotation.

Electro hydraulically operated butterfly valves with its own actuator shall be provided at the discharge of each CW pump.

8.1.3 Construction of CW Pumps and Accessories

For vertical suspended pump, the head piece shall be of fabricated steel construction. The pump casing, bellmouth, rising main and flanged discharge bend shall be constructed from duplex stainless steel. Replaceable wear rings of stainless steel shall be provided in the casing. The wear rings help to maintain correct clearance between the pump casing and the impeller. If the clearance exceed design limit, only the wear rings are need to be replaced, not the whole casing. This is to ensure right clearance between casing and impeller and protect the casing without replacing the whole casing. For concrete volute pump, the water suction is ensured by a concrete elbow guiding water to the impeller. At the impeller outlet, the concrete volute guides the fluid and allows it to flow regularly to the discharge. Above the impeller, a casing cover is installed to close the upper hole of the volute, keeping the shaft and bearings in a dry environment.

The pump shaft shall be manufactured from stainless steel, with mixed flow impeller securely keyed and locked. Journal sleeves shall be of stainless steel. Bearings shall be of cutlass rubber and/or ceramics, and lubricated by pumped water and, if necessary, also by service water. Wearing rings shall be of stainless steel.

The drive motor shall be supported by the head piece made of fabricated steel or supported on the concrete floor with a sufficient access to the coupling and upper pump seal for inspection and maintenance.

An intake chamber level indicator shall be provided. The level indicator of suitable corrosion resistant materials shall be graduated in centimetres and shall have easily readable division numbers engraved every ten centimetres.

A float type level switch or an ultrasonic type level sensor with transmitter to be installed near the level indicator and all cables and accessories for monitoring the minimum water level compatible with the safe operation of CW pump shall also be furnished.

8.1.4 CW Piping, Expansion Joints and Valves

The piping shall include valves, manholes, expansion joints, air vents, foundation bolts, supports, hold-down straps and other accessories needed for the complete functioning of the system.

Bolted manholes shall be provided on the pipe at intervals of not more than 100 m. Covers shall be designed and manufactured to seal and not to increase head losses. Suitable reinforcing plates shall be provided around the manholes.

One double hole pre-cast concrete culvert with two circular intake heads shall be used for the off shore part of the CW intake system. One corrosion resistance inspection manhole shall be provided at the intake head. Corrosion resistance inspection manholes shall be installed along the off shore intake culvert. Distance between the manholes shall not be more than 300 m. On shore part shall be provided with steel pipe from CW pump station to the crossing area at Vung Ang I with appropriate corrosion protection on both inner and outer surfaces of the pipework and then PCCP or RCCP shall be provided up to the turbine hall in CW system. Detailed information and reference (including in Vietnamese projects) of GRP piping material shall be submitted for the Owner's acceptance in detailed design stage. Two (2) single concrete culverts, one per Unit, shall be used from the turbine hall CW discharge to the SWFGD aeration basin. Pre-cast concrete culvert shall be used from the SWFGD aeration basin to CW outfall. Pipe support and stopper shall be provided for CW system pipe. Settlement monitoring poles shall be provided for those underground CW pipework. Carbon steel piping with rubber lining and appropriate outer surfaces corrosion protection shall be adopted for CW piping inside the turbine building for connection to the condensers.

PCCP or RCCP pipe and CW culvert shall be in full compliance with relevant international standards such as the American Water Works Association (AWWA) Standard, ASME or equivalent. The pipe shall provide the necessary structural strength and rigidity to withstand heavy external loads as well as a smooth interior wall for optimum water flow. The supports and foundations of the pipework shall be designed to prevent pipe damaging caused by differential settlement. The pipes shall be capable of withstanding high internal water pressure and large external loads. The Offshore Services Contractor shall demonstrate to the Owner that the CW pipes and

culvert can withstand the maximum pressure experienced under system surges. Detailed calculations shall be submitted to the Owner for review.

The bifurcations and branches shall be fabricated of suitably shaped plates which shall be stiffened and ribbed, as necessary.

For all types of steel CW pipes, the exterior of the pipe shall be applied with adequate protective paint/coating e.g. one with interposed layers of fiberglass for protection. The steel pipework shall have suitable allowance for corrosion design for its service life.

A discharge valve and expansion joint shall be furnished at each CW pump discharge.

8.1.5 CW Pump Discharge Valves

The Offshore Supply Contractor shall provide for each CW pump one rubber seated butterfly valve with hydraulic actuator. Valve operating speed shall be selected to prevent the occurrence of water hammer and to avoid damage to the system. The type of discharge valves and actuators as well as their characteristics shall be confirmed as per hydraulic analysis results during the detailed design stage.

The details of the valves and actuators shall be submitted for Owner's review during the detailed design stage.

The valves shall be rated to withstand the design pressure of the pipework and under the maximum surge conditions with adequate margins.

The CW pump discharge valves shall be capable of modulation in order to limit surge effects during opening and closing and attention shall be paid to the flow and modulation characteristics. The local hand-wheels shall automatically disengage, when the valves are operating in the automatic mode. The valves and their actuating equipment shall be capable of operating satisfactorily under all possible conditions of CW pressure and flow including operation against the CW pump maximum shut-off pressure in the operating range.

All necessary contacts and interlocks shall be prepared together with each motor drive unit for remote electrical control and for automatic stopping of the valve disc at the limits of its travel in both opening and closing operations. Provision for protection of the valve disc from damage due to clogging with foreign materials on the valve seats shall be provided. Motor drive unit shall be suitable for installation in any position without lubricant leakage or other operating difficulties. Conduit boxes shall be waterproof and corrosion-resistant.

Each valve shall be fitted with arrangements which provide for the valve to be securely locked into either the fully open or fully shut positions by means of a manually inserted locking pin. With the valve locked in position by means of this pin, it shall be possible, while the pump is operating to remove the actuator drive unit for maintenance purposes.

The CW pump discharge valves shall be arranged to open on remote manual control as well as remote sequential control from local.

The shutdown of the CW pump shall be interlocked with the closing of its associated discharge valve. When a normal shutdown of the pump is initiated from the Plant control room, the pump motor shall not be isolated immediately from its power supply and shall continue to run under full power. Initiation of the shutdown will cause the discharge valve to start closing and when the valve reaches a predetermined percentage of closure a limit switch, fitted to the valve, shall then initiate the isolation of the pump motor power supply. Each pump discharge valve shall be closed at two rates: for the initial 85% of closure the valve shall close rapidly and very slowly for the remaining 15%. If the Offshore Services Contractor proposes to apply their CW pump shutdown sequence, it shall be submitted to the Owner for review during detailed engineering stage.

The rates of closure for the valves shall be determined by the surge study and shall be selected to minimize hydraulic surge and prevent possible damage to the pump resulting from reverse rotation.

Emergency shutdown of the pump may occur due to any of the automatic protective trip devices or due to interruption of the electrical supply to the pump motor. Under such conditions the relevant discharge valve shall be arranged to close immediately to minimize hydraulic surge.

8.1.6 CW Screens

8.1.6.1 General

One set of traveling water screen shall be supplied for each CW pump, complete with washing system capable of automatic operation based on preset level differential and all required accessories to obtain a proper installation and successful operation of the equipment. The screens shall be installed side by side in the intake channel. The screen shall be of the single or double flow, horizontal spray type suitable for vertical installation in the intake civil structure. The screen shall be furnished with electric motor, reduction gear drive, wash pumps and nozzles, guides for casting into the concrete intake walls, structural frame and enclosed housing. The selection of the screen materials and protection system shall take into account the corrosive nature of the CW. All wetted parts shall be protected by the application of corrosive resistant materials, protective coating and cathodic protection.

8.1.6.2 Traveling Screens and Accessory Features

The traveling screens and accessories shall be of the outdoor type and completely protected from wind, dust and storm. The main frame shall be constructed of stainless steel plates and shapes to provide a rigid structure. Chain strands shall be confined and guided in tracks on both the ascending and descending sides. Jacking screws and

sole plates shall be provided to free the frame from fixed side guides in maintenance operations.

The main frame, sprocket shaft assemblies, shaft bearings, and boot section shall be designed to resist all forces without damage resulting from the full breakdown torque of the motor while the screen is jammed at any point in its travel including jamming in the boot section.

For traveling band screen, they shall have baskets mounted on two strands of chains, running at the speed specified and shall carry trash above the operating floor to be washed off the baskets into a concrete trough cast into the operating floor. The screen baskets shall have frames of stainless steel construction provided with curved end stainless steel plates or shrouds to provide sealing between adjacent baskets and the traveling screen boot and frame. The crack width at the seal shall not exceed 3 mm.

The screening material shall be attached to the screen basket frames in a manner which provides easy replacement of screen with minimum parts to be removed. All screen clamp bars and bolts to be used shall be, at least, of type 316L stainless steel.

Screen chains shall be of the offset side bar, roller type with a pitch equal to the screen basket spacing. The chains shall be especially designed for traveling water screen service and shall operate satisfactorily without lubrication.

The drive chain from the reduction gear unit to the head sprocket wheel shall be of the roller type. The chain shall be constructed entirely of stainless steel for screen designs, which have the drive chain operating within the main housing. For screen designs having the drive chain outside the main housing, the chain may be either constructed of carbon steel or stainless steel.

Each screen unit shall be provided with a drive unit assembly mounted directly in the traveling screen unit. The drive unit shall have an electric motor directly connected through an all metal flexible couplings to a reduction gear unit. The reduction gear unit shall drive the head sprockets through a chain. All chain drive sprockets shall be constructed from carbon steel.

The reduction gear unit shall be of the helical or herringbone gear type operating in an oil bath and completely enclosed in a heavy weatherproof oil tight housing.

All couplings, chain drives, and exposed shafts shall be provided with suitable steel guards.

A complete spray cleaning system shall be provided for each screen unit. The system shall be designed for front cleaning to wash the screen baskets near the top of the unit.

The spray nozzles shall be of the non-clogging type constructed from stainless steel. The nozzles shall have bores of adequate size to pass any object passed by the screen wash water strainers. The spray nozzles header and all feed piping between the spray nozzle header and the spray water valve shall be constructed of stainless steel pipes. Flanged joints shall be provided in the piping for easy removal of the spray nozzle header.

The screen unit shall be provided with stainless steel and fiber glass housing to enclose the portions of the screens and head sprocket assembly projecting above the operating floor.

The front and rear splash housing shall be molded fiber glass with stainless steel hardware. They shall be easily removable for maintenance.

All surfaces of the main frame, boot section, head and foot sprocket assemblies, screen frames and other carbon steel parts subject to wetting shall be thoroughly sandblasted and given one coat of bitumastics or an approved equal protective coating before shipment.

All bearing surfaces and surfaces subjected to rubbing shall not be painted but shall be given an easily removable protective coat to prevent rust.

8.1.6.3 Wash Water Pumps and Traveling Screen Controls

2 x 100% wash water pumps with motors shall be furnished with automatic self-cleaning duplex strainers to supply the spray water requirement of four traveling screens. Each pump shall be capable of supply band screen cleaning water to all four traveling band screens. The screen washing shall be done automatically. Each pump shall be capable to remain as a standby to the other pump. A suitable selector switch shall be provided for this purpose.

Duplex stainless steel shall be adopted for the wash water pump.

If filtered cooling or lubricating water is required for the glands, the Offshore Supply Contractor shall include the required piping and valves in his supply.

One gate valve and one check valve of material resistant to the action of sea water and all required fittings and gauges shall be provided.

8.1.6.4 Bar Screens

One set of bar screen for each pump shall be installed across the inlet to each traveling water screen with the following specifications:

- (a) Inclination 90 °
- (b) Bar pitch <100 mm
- (c) Bar Size (To be determined during detail design stage)
- (d) Material 316L Stainless steel

The supply shall include all welding pads and embedded items required for the installation of the bar screens. Each bar screen shall be fitted with lifting lugs for easy installation and removal.

8.1.7 Anti-surge Valves

Flanged anti-surge valves shall be provided and fitted after the CW system hydrostatic test at points on the pipework to allow air ingress to minimize surge effects. The locations of the valves shall be finalized during detailed engineering stage.

The Offshore Services Contractor shall provide a complete hydraulic analysis of the CW system from the water intake to the water discharge canal to establish the system surge characteristics under the most onerous operating conditions during detailed engineering stage.

8.1.8 CW Discharge and Outfall System

The Offshore Supply Contractor shall provide a CW discharge and outfall system starting from outlet of stilling basin and the diffuser system. The outfall system shall be laid under the sea bed, in parallel with the coal receiving jetty. At the point, where outfall pipelines and the coal dock for the Project is about 300m, the submerged pipelines turn away from the dock at 45 degrees and terminating in the diffusers lines to avoid negative effect to the coal collier due to discharge flow. The diffuser at the end of the outfall system shall be laid at a depth of approximately 12 metres. For each diffuser line, the total length shall be at least 150 m consisting of 4 diffuser elbows, each with a single diffuser port with diameter of 1.5 m and 1 m above the sea bed. The diffuser ports discharge shall be directed horizontally and in perpendicular to the shoreline. Water shall be blown at the velocity at the discharge port of 5 m/s. The outlet discharge shall be designed to achieve the maximum dispersion of the CW to achieve the design temperature rise of the surrounding water. At the outlet point, the outlet CW temperature rise shall comply with the requirement of the local environmental clearance limits.

Means of access to the sea water discharge and outfall system for internal inspection shall be provided.

The Onshore Contractor shall conduct dredging for the construction of the CW discharge culverts and diffusers. Due to the depth of the dredging and soil of the sea bed is sandy, sheet piles shall be used to maintain stability and prevent sand deposit. The design of sheet pile types and driving depth shall be carried out and reviewed by the Owner during detailed engineering stage. Sheet piles shall be driven by vibrators. Due to big length of dredging, it shall be divided into sections. After completing installation of the piping system, the sheet piles shall be removed and filling-back shall be provided along the piping bed.

The design of all marine Works shall allow for the protection of the submerged structures, pipework as well as achieving an objective to minimize the marine hazards to the navigation of the vessels in the adjacent areas. Navigation aid facility shall be provided at the diffuser of the CW discharge culvert.

8.1.9 Environmental Monitoring

Sodium hypochlorite dosing is carried out at the CW intakes and forebay areas to minimize the extent of bio-fouling growth within the system.

Chlorine monitors shall be provided for the Facility at the following locations:

- i. CW pump intake channel
- ii. Inlet of condenser water box
- iii. The end of discharge basin

In addition to chlorine monitoring equipment the seal pits shall also be provided with pH monitors and temperature elements to monitor the discharge seawater quality.

8.1.10 CW Discharge Temperature

The CW system shall be designed to be suitable for 600 MW net at CW system at average fouling condition, LLWL at CW intake, inlet CW temperature at 33 °C and HWL at CW outfall. The CW discharge temperature to the sea shall be less than 40 °C. The Offshore Services Contractor shall optimize the CW piping and condenser design to meet such a requirement.

8.1.11 CW Intake

8.1.11.1 General

The CW intake shall include the civil Works and structures of an offshore CW intake up to the pump station forebay.

The scope shall include the following:-

- Two intake heads;
- Two submerged intake culvert of about 1100 m long;
- Sodium hypochlorite (NaClO) dosing lines from main Plant to intake head;

8.1.11.2 CW Intake

The intake water system consists of CW intake heads and conveyance Works to the pump station.

The intake culvert shall allow sea water to flow by gravity from sea to forebay located at the CW pump station at LLWL.

The intake shall consist of two concrete culverts with intake heads located in approximately 5 m of water. The Onshore Contractor shall construct the water intake heads 1100 m far from the shore line and the submerged pipeline from intake heads to the forebay of the pump station.

Means of access to the sea water intakes structure and intakes pipes for internal inspection shall be provided.

The design of the CW intake shall allow for the protection of the submerged structures, including the pipework and shall minimize the marine hazards to the navigation of the vessels in the adjacent areas.

The CW intake shall satisfy, but not be limited to, the following:

- a. The minimum water depth the inlet heads shall be located 5 m below the LLWL.
- b. Navigation aids and protection structure shall be provided to avoid damage of the intake head by the pass-by vessels.
- c. The CW Intake and outfall culvert shall be semi-buried in the seabed.

8.1.11.3 Sodium Hypochlorite Dosing

The Offshore Supply Contractor shall provide the NaClO dosing lines starting from forebay of CW pump station to the submerged CW intake culvert. NaClO shall be dosed at the CW intake system to control biological growth and destroy creature that adhere to inner side of culvert. The dosing pipe shall be installed along the culvert with suitable protection from the damage of anchor. Materials of the dosing line and dosing nozzle shall be resistance to the attack of sodium hypochlorite. The system shall be designed to flush with seawater, if required and the flow rate shall be designed to prevent accumulation of deposit inside the dosing pipe.

8.1.11.4 Civil Measures

The Onshore Contractor shall conduct dredging for CW intake culvert. Due to the depth of the dredging and soil of the sea bed is sandy, sheet piles shall be considered, if necessary to facilitate formation of the foundation and to prevent sand deposit. Detailed designs of sheet pile types and driving depth shall be carried out and reviewed by the Owner during detailed design stage. Sheet piles shall be driven by vibrators. Due to big length of dredging, it shall be divided into sections. After completing installation of the culvert, the sheet piles shall be removed and filling-back shall be provided along the culvert bed.

8.1.12 Vung Ang I Outfall Extension Works

The scope shall cover design by the Offshore Services contractor, procurement by the Offshore Supply Contractor and construction by the Onshore Contractor of the Vung Ang I Outfall Extension Works to enable off shore discharge. This is to satisfy the recommendations from the warm cooling water recirculation study on the combined operation of the Vung Ang I and Vung Ang II Projects. The structure of Vung Ang I outfall extension shall be totally and physically separated from that of Vung Ang II.

The Vung Ang I Outfall Extension Works shall comprise two 1.8 km long concrete culverts, with its near shore end connected to the Vung Ang I CW diversion chamber and the off shore end terminated at a set of diffuser elbows.

Vung Ang I will provide the diversion chamber with removable stop logs to enable initially near-shore discharge when Vung Ang I becomes operational, and then subsequently diversion to off shore discharge after the said Vung Ang I extension is complete. The Vung Ang I Outfall Extension Works forms part of Section 2 of the Works and shall be completed prior to operation of the First Unit.

The on shore part of the Vung Ang I CW concrete culvert after the diversion chamber will have the inner dimensions of $2 \times 3.0 \text{ m} \times 7.0 \text{ m}$. On-shore section of the culvert shall be made of reinforced concrete structure cast-in-situ, while the off-shore section shall be made from pre-fabricated reinforced concrete structure of $2 \times 4 \text{ m} \times 4 \text{ m}$.

The outfall culverts shall be laid under the sea bed in between the Vung Ang II coal jetty and the future port breakwater. It shall run first in parallel with the above coal jetty, then turned at around 45 degrees away from the jetty toward the future port breakwater for a distance of 390 m before they turn again at some appropriate angle to enable effective CW dispersion.

Each culvert shall have a set of 4 diffuser elbows some 150 m long terminated at 1 m above the seabed. The diffuser ports shall be directed horizontally and perpendicular to the shoreline. The design shall consider minimal effect to the berthing and harbour breakwater and maximize the effect of thermal dispersion.

The Onshore Contractor shall implement all necessary quality control measures according to Best Industry Practice and international standards on the construction of the Vung Ang I Outfall Extension Works to ensure the integrity of the structure and leak free operation. The diversion box provided by Vung Ang I will have the stop logs which need to be watertight to enable construction of the Vung Ang I extension and the stop logs will allow for removal and replacement during operation of Vung Ang I.

8.2 Electro-chlorination Plant

An electro-chlorination plant shall be provided for onsite production of sodium hypochlorite from seawater. The plant shall be based on electrolysis process of seawater to produce and distribute sodium hypochlorite solution for the chlorination of the CW and ACW systems to control biological growth and fouling of heat exchanger. The injection points shall be at the CW intakes heads and just upstream of the stop gates of the CW pump station.

The Offshore Supply Contractor shall provide $2 \times 50\%$ electro-chlorination units rated for total residual chlorine concentration in CW of 0.2 mg/l continuous dosing with all CW pumps in operation. The design of the dosing regime shall be allow both shock and continuous dosing be implemented. All electro-chlorination units shall be able to operate at maximum production rates at the same time.

CW pumps shall supply seawater to feed the electro-chlorination plant and sea water

pre-treatment plant. The supplied seawater for electro-chlorination plant from CW pumps shall pass through 2 x 100% automatic self-cleaning strainers which protect the electrolyzers from oversize material above 400 microns.

A bypass shall be provided around the cell modules, dis-entrainment tanks and dosing pumps, to the intake dosing lines. This shall maintain flow velocities in the intake dosing lines when a continuous dosing regime is in operation with a limited number of CW pumps running.

Water from the self-cleaning strainers shall be delivered direct to the cell modules. The cell modules shall be provided with passive flow controllers at the inlet to each cell so that a constant flow of seawater shall be maintained.

The electro-chlorination plant shall be interlocked with the operation of the CW pumps for NaClO injection as well as to prevent NaClO injection when the pump is shutdown. NaClO solution shall be fed as required for control of biological fouling with manual selection of either shock or continuous chlorination.

Initiation and termination of continuous chlorination shall be manual. A timer control system shall be provided to control the duration and frequency of shock chlorination.

The electro-chlorination plant shall eliminate entrained hydrogen, if produced, so that no fire or explosion risk exist at any time in the Facility or in any building or part of the CW system. A hydrogen detection system shall be provided in the cell room. Each cell module shall be provided with individual sensing unit. Alarm panels shall be located at each entrance to the cell room. The alarm signal shall also be sent to central control room (CCR).

Two hydrogen dis-entrainment tanks/sodium hypochlorite storage tanks each of at least 70 m³ shall be provided. An occasional foam formation, caused by phytoplankton blooms, may be produced on the surface of the solution within the dis-entrainment tank. Spray water shall be provided to take seawater from the seawater strainer outlet pipework to dissipate the foam via spray bars located across the top of the tank.

The electro-chlorination plant shall be capable of operation normally unattended and shall incorporate all necessary safety features to ensure its safe operation when unattended. One bank of cell shall be easily taken out of service for inspection and maintenance whilst maintaining full dosing rate with the remaining cells. An allowance shall be made for deterioration in cell performance.

An automatic acid washing system, which is started by operator locally, shall be provided. Acid washing of the cells shall serve to remove any calcareous deposits and prevent anode degeneration.

A spent acid neutralization system shall be provided to utilize caustic soda to correct the pH of the solution in a neutralizing tank. Once neutralized, the solution shall be discharged to the waste water treatment system or seawater intake chambers.

To prevent the build-up of manganese on the cell anodes, which reduces chlorine

generation efficiency, system shall be provided to reverse the cell operating polarity for a short period at irregular intervals firing operation. A system, for measuring the voltage drop across each individual cell of each module shall be provided.

Two 100% duty gland flushing pumps, if necessary, shall be provided to supply service water for seal flushing of the seawater booster pumps and dosing pumps as well as chemical dilution water and fume scrubber water. The gland seal water shall be pumped around a ring main fed by the gland flushing water break tank. The dosing pump gland seal water shall pass directly to drain to prevent possible hypochlorite contamination.

Chlorine analysers shall be used to monitor the residual chlorine level and to assess the need for supplementary dosing or otherwise. Sampling equipment including sampling pumps shall be provided at the monitoring points where required. The chlorine analysers shall provide feedback signals for control of hypochlorite injection rate.

Conductivity meters located upstream of the cell modules shall be provided to give an average signal to the transformer rectifier to enable adjustment of the applied current when the salinity level falls below the design threshold.

The power supply system for each hypochlorite generating unit shall be separately isolatable for maintenance. The electro-chlorination plant shall be fully automatic with control from 10% to 100% output. The electro-chlorination plant shall be designed for, in addition to local controls, remote operation.

8.3 Closed Cycle Cooling Water System

8.3.1 General

Unitized closed cycle cooling water (CCCW) system shall be provided for removing heat from the components of various Plants and rejecting heat to the auxiliary cooling water (ACW) system. The CCCW pumps shall circulate the water through the closed system, and the heat exchangers shall provide the means for dissipating heat to the ACW system.

The Offshore Services Contractor shall use 33°C as ambient seawater temperature for sizing of equipment.

It is envisaged that the CCCW system shall at least serve the following:-

- Coal mill coolers
- Air heater oil coolers
- Induced draught fan coolers
- Furnace doors
- SWFGD Plant equipment coolers
- Stator water coolers

- Exciter air coolers
- Generator hydrogen coolers
- Boiler feed pump oil coolers and motor coolers
- Turbine lube oil coolers
- Other auxiliary coolers

The CCCW system shall consist of, but not be limited to, the following components for each Unit:-

- Two 100% duty ACW pumps if necessary
- Two 100% duty auxiliary cooling water strainers if necessary
- Two 100% duty closed circuit cooling water pumps
- Two 100% duty ACW titanium shell and tube type or three 50% duty plate type heat exchangers
- One CCCW elevated head tank
- One chemical dosing tank
- Associated piping, valves, instrumentation and controls.

The system shall continuously circulate demineralized water as a cooling medium for equipment in the closed loop cooling system. Make-up to the system shall be obtained from the demineralized water system.

The system shall be sized to supply adequate CCCW to the closed loop system when the exchangers are supplied with highest CCCW system temperature. Strainers shall be provided at inlet to the CCCW pumps. The system shall permit shutdown and maintenance of the individual items of equipment without interruption of the cooling function of the rest of the system.

An elevated head tank shall be provided for surge capability, system make-up, emergency cooling of emergency diesel generator running at full load, venting and adequate net positive suction head for the CCCW pumps. The size of the CCCW head tank shall be adequate to provide necessary emergency cooling for the whole Unit to shut down safely in case of black out situation. A chemical dosing tank complete with associated dosing facilities shall be provided to add corrosion inhibitor to the system. Sampling connections shall be provided in the pipework to enable checks to be made of the quality of water within the system. Detraining facilities shall be provided to allow any hydrogen that may have leaked into the CCCW to be released to atmosphere via vent pipe.

8.3.2 ACW Coolers

2 × 100% duty demineralized water/ACW coolers shall be supplied. The coolers shall be titanium shell & tube type heat exchangers. Alternatively, the heat exchanger can be 3 x 50% duty plate type which must have titanium plates and a debris filter shall be provided at the cooling water inlet to the plate type exchanger.

For tubular heat exchangers, the ACW shall pass through the tubes and the demineralized water shall pass through the shell and over the tubes. The tubes shall be welded titanium tubes with solid titanium or titanium clad carbon steel tube plates.

The entry of the sea water to the heat exchanger shall be from the bottom. The maximum velocity of circulating water through the heat exchanger tubes shall be 2 m/s. For the purpose of calculating the overall heat transfer coefficient of the coolers, a fouling factor shall be included. The cooler shells shall have provision for differential expansion between the cooler shell and the tube bundle. A floating head type design is not acceptable. The cooler shells shall be constructed of carbon steel and each shall be fitted with an air vent at the highest point on the shell and a drain valve. The cooler water boxes shall be of cast iron or fabricated steel in both cases coated with equivalent to coal tar epoxy paint of min. 0.25 mm thickness. The highest point of each water box shall be supplied with a connection for priming and another fitted with an isolating valve for use with the siphonic system vents. If different tube and tube plate material from the preferred ones, cathodic protection on water box side shall be included depending on selected materials. The arrangement of water box connections shall be such, that tubes can be cleaned without breaking any pipe joints. The water box covers shall be fitted with lifting lugs and inspection doors (manholes if possible).

8.3.3 ACW Pumps (if necessary)

2 × 100% ACW pumps or other means shall be provided to supply seawater from a point upstream of the condenser for the ACW system. ACW pump discharge pressure remote monitoring shall be provided in the CCR. The ACW system water shall be returned to the main CW system via a connection downstream of the condenser. Isolation valves shall be upstream and downstream of ACW. Whenever a direct seawater cooling system is not to be used for essential auxiliaries, due to the risks of contamination, a second cooling medium with decalcified water shall be used.

The pump impeller shall be made of stainless steel, and keyed to a shaft made from stainless steel. The bearings shall be of ball or roller pattern.

The pump glands shall be fitted with mechanical seals but so designed that reversion to packed type glands can be achieved conveniently if desired. A spare gland flange to facilitate this reversion shall be provided for each pump. Removable stainless steel sleeves are to be provided where the shaft passes through the gland.

The pumps shall be supplied complete with driving motors of the squirrel cage induction type wound for operation. Air release cocks shall be provided on each pump casing.

The necessary inlet and outlet isolating valves and non-return valve shall be supplied together with a pressure gauge complete with isolating valve on the suction and discharge branches from each pump.

Water discharged from air cocks and glands shall be piped via tundishes to the nearest convenient drain trench.

8.3.4 ACW Strainer (if necessary)

Two full duty strainers of automatic self-cleaning type shall be provided at the discharge of auxiliary cooling water pumps. These strainers shall be arranged in parallel with automatic change-over provision, so that when the duty strainer is in self-cleaning mode then the standby one comes into service automatically.

The mesh size of the strainer element shall not be greater than 15% of the inside sectional area of a tube utilized in the auxiliary cooling water coolers, and total path area of the apertures or holes of the strainer element shall be greater than three times of the total bore area of the tubes. The supplier's recommended mesh size can also be adopted for the strainers provided full details and relevant project reference are submitted for the Owner's approval.

The strainers shall be fitted with locally mounted differential pressure indicators and alarm initiation differential pressure switches, as well as pressure gauges arranged to reach both inlet and outlet pressures. Timer control and local mounted selector switch having Auto/Manual/Stop selection shall also be provided.

The strainers shall be of cast iron or fabricated mild steel coated with equivalent to coal tar epoxy paint. The baskets and other internal components shall be of stainless steel.

All necessary vents and drains, and equalizing connections, complete with valves shall be provided, and shall include appropriate connections to the siphonic vacuum venting bus main.

8.3.5 CCCW Head Tank

A stainless steel CCCW head tank shall be provided, being supplied with demineralized water from the outlet header of the condensate extraction pump.

Level control valve shall be provided on the make-up water supply line. An overflow pipe and valve drain shall be led to the rain water down pipe and a vent led outside the building.

8.3.6 Sampling and Dosing

Provisions shall be made for sampling the demineralized water at convenient points in the system.

The chemical dosing system shall be supplied to suit materials of coolers included in the demineralized cooling water system, nature of water and operating condition. The Offshore Services Contractor shall state their recommended dosing procedure and equipment.

8.4 Water Treatment System

Seawater shall be treated by filtration and a reverse osmosis (RO) process, as a source for the service water required. The RO system shall be common to both units and shall be designed to pre-treat and desalinate seawater for use as boiler feedwater makeup after treatment by the demineralization plant, as potable water after treatment by potable water system and service water for other plant general services. The water treatment system shall be controlled automatically by a microprocessor-based (PLC) system with required instrumentation for monitoring water quality and interlock function to accept/discard outlet quality and initiate regeneration. A DCS/PLC link shall be provided to allow the CCR operator to monitor and control the water treatment plant.

8.4.1 Seawater Pre-treatment System

CW pumps shall supply seawater to the seawater pre-treatment plant and electro-chlorination plant. The seawater pre-treatment system shall be designed as $3 \times 50\%$ process streams. The final process scheme for seawater pre-treatment shall be submitted to the Owner's for review during the detailed design stage. All backwash pumps, flushing pumps, effluent pumps, sludge pumps, dosing pumps and air blowers within the seawater pre-treatment system shall be of $2 \times 100\%$ configuration (1 working, 1 standby). NaClO, coagulant, pH correction, and polymer chemical shall be added to the clarifier to aid in filtration. Flash mixing of the chemicals with raw water and previously formed precipitates, flocculation, and final sedimentation shall take place within the clarifier. The clarifier shall also feature a separate sludge scraper and drive with provision for sludge blow-off and backflush. Treated clarifier effluent shall be distributed equally between dual filters by means of individual filter pressure drop based on sequential backwashing. Filtered water shall be piped to a filter well. To avoid damage on RO system from residual chlorine in seawater by NaClO dosing, chlorine shall be removed in seawater before pumping to RO system. Filtered water shall then be pressurized by high-pressure pumps and passed through the RO system module skids.

The seawater pre-treatment system shall include following major equipment:

- $1 \times 100\%$ capacity (sized for both Units), circular, conical base, solids contact type clarifier, if necessary.
- $3 \times 50\%$ capacity (sized for both Units), gravity type filters, if necessary.
- $1 \times 100\%$ capacity (sized for both Units), chlorinator, if necessary.
- One common (50% solution, sized for both Units) coagulant bulk storage tank of 7 days storage capacity and $2 \times 100\%$ feed Pumps
- One common (50% solution, sized for both Units) sodium hydroxide bulk storage tank of 7 days storage capacity and $2 \times 100\%$ feed pumps
- One common (1% solution, sized for both units) polymer storage tank of 7 days capacity and $2 \times 100\%$ feed pump
- $2 \times 100\%$ capacity (sized for both Units), sludge sump transfer pumps
- $2 \times 100\%$ capacity (sized for both Units), backwash sump transfer pumps
- One (sized for both Units) filtered water clearwell and $2 \times 100\%$ transfer pump.
- De-chlorination treatment system.

8.4.2 Seawater Desalination Plant

Booster pumps in the RO system shall pump the pre-treated seawater from the filter water clear well through a set of primary and secondary media filters or ultrafiltration with self-cleaning strainers. The filtered water shall then be treated with scale inhibitor and acid and then passed through cartridge filters. Residual chlorine shall be removed by adding sodium bisulphate. The filtered water shall then be pressurized by variable frequency drive (VFD) high-pressure pumps and passed through the RO system module skids. Permeate shall be forwarded to the common product water tank. All backwash pumps, flushing pumps, effluent pumps, dosing pumps and air blowers within the seawater desalination plant shall be of $2 \times 100\%$ configuration (1 working, 1 standby).

The RO system shall be designed to operate on a continuous basis. Multiple units and pumps shall be provided to allow for on line maintenance, membrane replacement, and cleaning operations. The more concentrated reject brine water generated by the RO process shall be stored in brackish water storage tank and then to SWFGD system for further treatment before discharge back to sea. Minerals such as sodium bicarbonate/calcium chloride shall be added at the outlet of the RO system to meet the required pH value of service water quality.

The seawater desalination system shall supply treated water to meet the following requirement for service water quality.

pH	: 6.5 – 8.5
Suspended solid	: < 10 mg/l
Turbidity	: < 2 NTU
Organic matter	: None
Silica content	: < 10 ppm
Total dissolved solids	: < 350 mg/l

A membrane design cleaning skid shall be provided to periodically clean the RO units. For system utilizing media filter, filter backwash/brine application tank shall be provided for periodic cleaning and sanitizing of the filter units. All cleaning wastes shall be treated and recovered as much as possible.

The system configuration of the RO plant shall consist of $3 \times 50\%$ capacity processing trains. The RO system shall consist of the following major equipment:

- Booster pumps
- $3 \times 50\%$ primary and secondary media filters or ultrafiltration with self-cleaning strainers
- Seawater reverse osmosis : $3 \times 50\%$
- Backwash pumps
- Backwash tank
- Air scour blowers
- Chemical systems
- Cartridge filters
- VFD high-pressure pumps

- Seawater RO module skids
- Permeate transfer pump.
- Brackish water storage tank
- Filter backwash equipment if applicable
- RO membrane cleaning and flushing system
- Interconnecting piping, valves, instruments and controls, electrical and PLC control system.

The desalination plant shall be controlled automatically by a microprocessor-based (PLC) system from the control panel located in the water treatment plant area and it can also be operated manually from the local control panel. A DCS/PLC link shall be provided to allow the CCR operator to monitor and control the desalination plant.

It is recognized that the Onshore Contractor will have used the desalination plant during construction for an extended period of time. The Onshore Contractor shall refurbish the desalination plant including replacement of RO membrane elements if necessary prior to Taking Over of Section 2 of the Works.

8.5 Demineralization Plant

8.5.1 General

The demineralization plant shall treat the service water supplied from the desalination system and provide high quality demineralized water for make-up to the steam cycle, bearing cooling water system and for regeneration of the ion exchange resins. The plant shall consist of RO, cation exchanger, anion exchanger and mixed bed demineralizers. Reverse osmosis shall reduce the 350 mg/l dissolved solids concentration in the service water to approximately 35 mg/l. The mixed bed demineralizers shall further reduce the dissolved solids concentration to 0.050 mg/l.

The water from the demineralization system outlet shall meet the following quality:

- Conductivity : < 0.15 μ S/cm at 25 oC
- Sodium : < 0.015 ppm as Na
- Silica : < 0.02 ppm as SiO₂
- Total dissolved solids : < 0.05 mg/l

The demineralization system shall consist of 2 $\times$ 100% trains with one train in operation while the other train in regenerating or standby. Each train shall be sized for a net demineralized water capacity equivalent to minimum of 1% of the total boiler flow (both Units) at BMCR.

Each train shall consist of a pressure type activated carbon filter, forced draft degasifier, cation exchanger, anion exchanger and mixed bed polisher. Filters, reverse osmosis modules plus mixed bed polishers as the demineralized process is also acceptable. The regeneration system and neutralization system shall be commonly used for both trains.

Each train of the demineralization plant shall be designed to operate independently from the other train. Start-up of the system shall be manual and the system shall be automatically shut down on receiving a high level signal from the demineralized water storage tank.

In each train, the water shall pass through in the following sequence, at first water enters in the pressure type activated carbon filter, cation exchanger, degasifier, anion exchanger, mixed bed polisher and then discharge to the demineralized water storage tank.

Each train shall be designed for not more than one regeneration per 24 hours operation at rated output. Regeneration shall be carried out for less than 4 hours duration per week for mixed bed polisher and 1 hour per day for cation and anion exchanger. Under emergency conditions, both trains shall be capable of being operated at the same time.

Demineralized water tanks shall be provided at the outlet of demineralized water plant, which will cater the demineralized water requirement for plant operation at BMCR, even during the interruption of demineralized water plant operation for 5 days.

8.5.2 Controls

The demineralization system shall be controlled automatically by a microprocessor-based (PLC) system from the control panel located in the water treatment plant area and it can also be operated manually from the local control panel. A DCS/PLC link shall be provided to allow the CCR operator to monitor and control the demineralization plant

The normal mode of operation of the control system shall be automatic except for the initiation of a regeneration sequence. In this auto mode, all motors and control valves shall be actuated automatically. Demineralized water quality shall be monitored continuously on the local control panel.

Regeneration sequences shall be manually initiated and monitored while regeneration is in progress to ensure that each function is performed correctly. The signals from CCR to the local control panel shall be able to initiate and stop the operation of demineralized water treatment plant and to change the operating stream to the stand-by one as a function of manual override. Sequential logic shall be carried out using reliable industrial type programmable controllers, mounted on the demineralized water treatment plant control panel.

The automatic regeneration cycle shall be initiated manually from the demineralized water treatment plant control panel.

Volumetric flow, integrated flow, tank levels and treated water conductivities shall be indicated in the control panel along with the diagrammatic representation of the plant.

8.5.3 Major Components

The demineralization plant shall include the following major components for $2 \times 100\%$ trains:

- Micron pre-filters
- 1st pass RO module skid
- 2nd pass RO module skid
- RO high-pressure feed pumps (for 1st and 2nd pass RO module skid)
- Chemical cleaning system
- Two (2) activated carbon filters
- Two (2) strong acid cation exchangers, if necessary
- Two (2) forced draft degasifiers (including blowers), if necessary
- Two (2) degasified water pumps, if necessary
- Two (2) strong base anion exchangers, if necessary.
- Two (2) mixed bed polishers
- One (1) acid storage tank
- One (1) acid measuring tank for cation exchanger
- One (1) acid measuring tank for mixed bed polisher
- Two (2) acid transfer pumps
- One (1) caustic soda storage tank
- One (1) caustic soda measuring tank for anion exchanger
- One (1) caustic soda measuring tank: for mixed bed polisher
- Two (2) caustic soda transfer pumps
- Two (2) regeneration pumps
- Two (2) air blowers
- One (1) neutralization basin
- Two (2) neutralized effluent pumps
- Interconnecting piping, valves, instrumentation and controls and PLC control system.

8.6 Service Water System

The system shall take service water from the desalination water and store in 2×8000 m³ service water tanks and then supply to various plant buildings via general service water pumps and a ring main distribution system. Each of the service points shall be provided with one fixed hose connection and one flexible hose connection with a 30 m length flexible hose reel.

The service water system shall supply water for cleaning/washing requirements of floors/galleries/equipment in the plant area. In addition, service water will be used for the requirements of dry ash handling, coal handling plant dust suppression and extraction systems, gardening, etc.

Service water pumps shall discharge directly in to distribution piping network covering entire plant area. Service water pumps shall get water supply from service water storage tank.

The minimum terminal pressure at any point of water connection shall not be less than 0.15 MPa. Service water shall be made available close to all installation in required quantity. Pressure throttle valves with locking provision shall be provided at tapping points from main lines to distribute water flow as per requirement.

Separate service water lines shall be routed all along the coal conveyors. User connections shall be provided in the entire conveyer system, at all floor levels of junction towers and crusher house, bunker floors, tunnels, coal unloading point, etc.

All piping, tubing, fittings; other wetted parts, pumps and tanks shall be of suitable material as per Vietnamese standards. Service water pipe shall be HDPE pipe for buried pipe and ASTM A106B pipe for above ground pipe.

8.7 Potable Water System

The potable water system shall take service water from service water tanks to provide potable water to various consuming points in the Plant for potable water usage. The system shall consist of potable water tanks and pumps, and chlorine dosing facilities with ring main supply pipes to various buildings. The system shall satisfy the potable water requirement of at least 200 m³/day. 200 m³ potable water tank shall be provided. Potable water shall be used for washing, cooking and sanitary purposes. Toilets (including of washbasins, cleaning sink, shower and hand spray(health faucet)) shall also use potable water.

Water from clarified water pumps discharge is routed through pressure filters, sterilized with ozone/chlorine and stored in overhead storage tanks. Further the potable water from storage tank is routed to various consuming points by 2 × 100% potable water distribution pumps with the associated pressurization units.

Potable water system comprises of pressure filters, ozone/chlorine generators, ozone/chlorine feeding/mixing equipment, overhead storage tank, piping, valves and fittings, etc. Other potable water treatment process which can produce potable water with a quality meeting World Health Organisation (WHO) and Vietnamese standards can also be accepted.

The turbidity level in the potable water generated shall be less than 2 NTU. Potable water generated shall be free from harmful bacteria for human consumption and excess ozone/chlorine in the water shall not exceed permissible limits given in Vietnamese regulation and the WHO.

The filtering media shall be sand of suitable size, shape and quality. The generation period between two back wash operations shall be less than 20 hours. 2 × 100% air scouring blowers shall be provided. Alternately, auto back washing based on differential pressure across the filter shall be provided.

The back wash water effluent from pressure filters shall be routed to sludge water treatment plant or clarifiers located in pre-treatment plant area.

Ozone shall be generated from atmospheric air in the required concentration. The ozone generators and ozone mixing equipment shall be from reputed manufacturers. The entire equipment shall be located indoors in pre-treatment plant area.

All piping, tubing, fittings; other wetted parts, pumps and tanks shall be of suitable material as per relevant international codes and Vietnamese standards. Carbon steel is not acceptable.

8.8 Drainage System

Plant drainage system shall be provided to collect the chemical drain, rain water, fire water, equipment drains, floor drains, and equipment washing water drains in the respective drain pits. The collected drains shall be transferred to the waste water treatment plant, the oil separator, or the sea according to the quality of the drain.

8.8.1 Design Basis

The collected drain generated from the Plant shall be transferred and treated according to following concept:

(a) Chemical drain

The chemical drain generated from the laboratory shall be transferred to the waste water treatment plant.

(b) Oily drain

The oily drain such as rain water, fire water from diesel oil storage tank yard, transformer yard and turbine oil storage tank yard shall be transferred to the oil separator to separate oil and water. Then separated water from the oil separator shall be transferred to the sea through the same drainage system which discharge uncontaminated rain water to the sea.

(c) Non-oily drain

Non-oily but contaminated drain such as boiler startup flash tank, equipment and floor drain from various buildings and rain water from coal yard sedimentation pond, etc., shall be transferred to the waste water treatment plant.

(d) Seawater drain

The seawater drain from seawater blow pit, CW pump intake pit or CW pump outlet valve area shall be discharged back into the sea without treatment in the waste water treatment plant if they haven't been contaminated.

(e) Surface drain

Surface drains from the buildings, roads, etc., generated due to rain, which are not contaminated, shall be routed to the sea directly.

Portable sump pumps shall be used for elevator pit, cable tunnel, work shop drain pit, fuel oil pump house drain pit and chlorination house drain pit, etc.

8.8.2 System Requirements

Equipment drains and floor drains containing oil, chemical, solid, etc., generated in the various area of the Plant shall be routed to the respective drain pits. The drain pits containing non-oily waste water or chemical drain shall be directly routed to the waste water treatment plant for treatment and disposal. The oily drains shall be transferred to the oil separators, where oil and water are separated. The separated oil collected in the oil separators, will be disposed by the Owner. The separated water from the oil separator shall be pumped to the sea through the same drainage system which discharge uncontaminated rain water to the sea. The sea water drain shall be discharged into the sea directly.

The drains from the drain pit shall be transferred by submerged type pumps, which are operated by a signal from the drain pit level switches.

The drainage piping material shall be typically carbon steel, and if required, lined carbon steel, PVC, etc. shall be provided.

The rainy water around the jetty shall be collected into the jetty drain pits and the drain shall be discharged into the coal yard pond by the drain pump.

8.9 Compressed Air System

8.9.1 Duty

The compressed air systems shall supply and distribute instrument air quality compressed air for the instrument air system for control and instrumentation purposes and a separate general service air system for pneumatic maintenance tools and general usages.

Both the maximum transient and steady state flow rates shall be taken into account when sizing the air compressors. All air compressors shall be rated at the same size. A minimum of four compressors which are used for both instrument air system and general service air system shall be supplied in total. When two Units are in normal operation, at least two compressors shall be in standby. When one Unit is in normal operation and one Unit is in start-up operation, at least one compressor shall be in standby. The Offshore Services Contractor shall submit detailed air compressor sizing calculations for the Owner's review during detailed engineering stage.

The duty of the instrument air and general service air systems shall be determined by

the Offshore Services Contractor to meet the requirements of the Plant and to meet the additional requirements of the Project as defined herein:

Instrument air:

- Oil-free.
- Dust particles in the compressed air shall be smaller than 1 micron in diameter.
- Dust particles in the compressed air shall be smaller than 0.01 mg/m³ in concentration.
- Outlet air shall have an exit pressure dew point less than -40 °C.

General services air:

- Oil-free.
- Dust particles in the compressed air shall be smaller than 1 micron in diameter.
- Outlet air shall have an exit pressure dew point less than 3 °C.

The compressed air systems shall supply dry, oil-free compressed air at the required pressure and capacity for the whole power station. Two air distribution ring main systems shall be provided.

The compressed air systems shall be capable in all respects of meeting the varying demands for air that could arise without leading to unduly onerous operating conditions for the air compressors. In particular, the frequency of starting and stopping of compressors shall be maintained, at all times, within limits acceptable to the compressors and compressor drives.

The instrument air system shall be of high integrity and, in particular, shall be arranged such that faults on the distribution system to anyone Unit shall not lead to a loss of instrument air on any other Unit or on common plant. Similarly, the effects of any faults on any parts of the air supply system that are common to more than one Unit shall be minimized so that a single such fault shall not lead to a loss of instrument air to more than one Unit or to common plant.

Sufficient storage of air, irrespective of the general service air back up, shall be available in the instrument air system to permit safe shutdown of all power station Plant in the event of loss of electrical supplies to the instrument air compressors. Without prejudice to this requirement, a minimum air storage capacity equivalent to thirty (30) minutes of full flow demand at above the minimum acceptable operating pressure shall be provided. The Offshore Services Contractor shall submit detailed receiver sizing calculations for the Owner's review during detailed engineering stage.

Centralized compressed air system shall be provided for the supply of general service air system and instrument air system. Sufficient storage of air shall be provided on the general service air system to maintain pressure fluctuations during normal operation within reasonable limits.

Permanent general service air supplies shall be provided for the following plant systems:

- Turbine house
- Boiler areas

- Workshops (mechanical, electrical and C&I)
- Warehouse
- Laboratory
- Waste water treatment plant
- Water treatment plant
- CW pump station
- Coal handling building
- Chemical building
- Ash handling building
- Fuel oil pump house
- Garage
- Fire fighting pump house
- Fire station
- Compressed air Plant house
- Hydrogen plant
- FGD
- Other auxiliary Plant house and amenities buildings

8.9.2 Extent of Supply

The extent of supply of the compressed air systems shall include, but not be limited to, the following:

- All air compressors, drives and associated equipment such as couplings, guards, bedplates, holding down bolts, air filters, coolers, oil and moisture separators, silencers, driers, automatic governor and unloading devices, lubrication equipment and sound attenuated enclosures
- Air receivers
- All air distribution pipework and valves as required by the station operation requirements
- On-plant control and instrumentation equipment and local panels
- Air compressor cooling system including all associated equipment and control
- All necessary fittings and devices for the safe operation of the compressors

8.9.3 Design

8.9.3.1 Overall Design

For interchange-ability purpose, identical equipment shall be used for both air systems wherever possible.

In the event of low air system pressure, the non-essential air requirements shall be automatically isolated from the general service air system and all available air shall supply to essential station and control air demands.

The compressed air systems shall be arranged as a minimum of four (4) air

compressors and air dryers streams. Output from the system shall be delivered to the two (2) instrument air receivers and two (2) service air receivers at the compressor plant house. Two (2) additional instrument air receivers shall be provided next to the turbine hall (one per each Unit basis) and one (1) buffer instrument air receiver shall be provided at the water treatment plant building. Except for the 10m³ instrument air receiver at the water treatment plant, all the air receivers supplied shall be of a minimum size of 20 m³.

The air compressors shall be sized to cater for the regeneration purge air required by the air dryers in addition to the actual system demand. A dedicated compressor cooling system shall be provided for each of the air compressors.

The system shall be designed to ensure that no single equipment or system failure, mechanical or electrical, will cause the loss of the plant compressed air needs.

8.9.3.2 Air Compressors

The compressors shall be of oil free rotary type, furnished with all accessories as a complete package unit to provide oil free compressed air. Each unit shall be free-standing, with all equipment completely supported on a common baseplate.

The compressors shall be of the water cooled type or direct air cooled type as an alternative. For air cooled compressors, the cooling medium shall be derived from the intake air to the compressor without the need of an additional cooler outside the compressor baseplate.

A lubricating oil system shall be provided for lubrication of the compressor bearings and gearbox, if any. The oil system shall include all pumping equipment, pipework, strainer, pressure relief device and oil cooler as required.

Each compressor shall be complete with all necessary drain valves, drain pipes, relief valves, inlet and outlet air valves, sight feed lubricators, pressure gauges, thermometer pockets and other necessary fittings.

The vibration of the air compressors and the effect on adjacent plant shall be kept to an absolute minimum.

Lifting lugs shall be provided on each compressor unit to facilitate handling.

8.9.3.3 Compressor Coolers

Each compressor shall be completed with water cooled (or air cooled) air coolers and oil coolers for reducing the temperature of the discharge air and oil temperature. The coolers shall be a compact assembly with the coolant supplied from a closed recirculating service water system.

The cooling water pipework system to be supplied shall include a cooling water inlet manifold, an outlet manifold and pipes connecting to and from the coolers arranged on a Unit basis.

The pipework system shall be complete with all necessary drains, vents, water and expansion relief at appropriate locations.

Thermometer pockets shall be provided on both air side and water side, inlets and outlets of each cooler.

An automatic drain trap shall be provided for drainage of condensate from the cooler.

8.9.3.4 Inlet Air Filter-Silencers

Each compressor shall be furnished with a full capacity, dry type, intake air filter-silencer.

Each filter shall be properly sized for the individual application in accordance with the recommendation of the filter manufacturer. Each filter shall be capable of removing all particles one micrometres and larger. Filters capable of removing smaller particles shall be provided if required by the compressor manufacturer.

8.9.3.5 Oil and Moisture Separators

Means shall be provided for separating oil and moisture from the compressed air.

8.9.3.6 Air Dryers and Filters

Efficient air dryers shall be installed downstream of the air compressors. The dryers shall be of twin column absorbent type with automatic heatless regeneration facility.

The air dryers shall be complete with local colour change moisture indicator, dew point indicator and alarm, and an exhaust silencer if required.

Filters shall be provided upstream and downstream the air dryers together with bypass pipe and isolating valves across each dryer-filter assembly to facilitate cleaning of the filters. The filters shall be designed to intercept any break down of dryer desiccants without causing excessive pressure drop.

8.9.3.7 Sound Attenuated Enclosures

Enclosure shall be constructed of sheet steel and reinforced as required to assure a suitably rigid structure.

Quick release panels and hinged doors shall be provided as required for ease of access to the equipment for inspection and maintenance.

8.9.3.8 Control and Instrumentation

Compressors shall be provided with the following control functions:

- (i) loading and unloading of the running compressors
- (ii) automatic cut-in of the standby compressors

Pressure tappings for function (i) shall be located immediately downstream of each compressor discharge non-return valve. The automatic air governors shall be of a proven type.

Pressure tappings for function (ii) shall be located at the compressor discharge manifold. The size of impulse lines shall be such as to eliminate the possibility of blockage and also minimize pressure transmission time lag.

Selector switches shall be provided for each of the compressors. Operating modes of the compressors shall be arranged as - Start, Stop, Standby 1, Standby 2 etc.

The selector switches shall be grouped together and displayed on a local control panel. Remote operation shall also be provided in the CCR.

Flow indicators fitted with alarm contacts shall be provided in the cooling water outlet of each compressor. In addition each compressor shall be arranged to trip in the event of a high cooling water temperature being attained.

8.9.3.9 Air Receivers

Air receivers of suitable sizes shall be provided for the instrument air and general service air systems, capacity and full details shall be submitted for review by the Owner during detailed engineering stage. Material of construction shall be carbon steel plates joined together by the fusion welding process. The receivers shall be designed to withstand an internal working pressure of not less than the safety valve setting pressure.

Air receivers shall be provided with the following documents during construction stage:

- Mill certificate of the receiver materials
- Maker's certificate and structural drawings
- Third party certificate by a recognized inspection body in the country in which the receivers are constructed.

Each receiver shall be provided with the following fittings to BS 1123 or equivalent:

- A stop valve on discharge pipe.
- A direct spring loaded safety valve.
- A fusible plug fitted on top of receiver.
- A well designed inlet port for maximum separation of moisture.
- A pressure gauge having a dial of not less than 150 mm diameter, with provision

- for fitting of a check gauge.
- A thermometer pocket complete with a dial thermometer.
- A suitable manhole.
- An automatic water trap and a drain lock at the bottom of the receiver.
- A metallic plate displaying all the relevant data of the receiver.

The air receivers shall be supported on legs to give a bottom clearance of at least 150 mm. The interior surface of the air receivers shall be sand blasted, cleaned and coated with moisture-resistant material.

8.9.3.10 Air Compressor Cooling Water System

The Offshore Supply Contractor shall provide a dedicated cooling water system for each of the air compressors. The air compressor shall be cooled by a closed loop service water cooling system, which is in turn cooled by a separate heat exchanger using air cooled fin fan radiators.

Initial filling and make up to the compressor cooling water system from the service water system and the cooling water shall be circulated by pumps to the compressor coolers.

Cooling capacity of the fin fan radiators shall be sufficient to meet the requirements of the compressor at maximum design temperature. The fouling factor shall be not less than 20% at peak rating under site conditions.

All necessary drains, air releases, water expansion relief as well as control and instrumentation shall be provided by the Offshore Supply Contractor.

8.10 Carbon Dioxide and Nitrogen

The carbon dioxide system shall be sized to provide storage for the minimum quantity for initial air purge plus an emergency hydrogen purge for both generators. The capacity for initial air purge shall be calculated as a minimum 1.25 times the generator volume and that for the hydrogen purge as 3 times the generator volume.

Carbon dioxide shall be stored by bottle liquefied CO₂ storage systems and CO₂ bottles replenishment shall be handled effectively and quickly. The system shall be equipped with 2 × 100% vaporizers (one duty and one standby). A low temperature cut off device shall be fitted downstream of each vaporizer.

The station compressed air shall be available for purging CO₂ system.

A centralized nitrogen blanketing system shall be provided for the feed heating systems and the boilers. Nitrogen shall be used to pipe into the vapour spaces of the feed system to minimize deterioration of any feed water left in the system and to inhibit corrosion of the vessels when the feed system is out of service for extended periods. Nitrogen shall also be used for sealing to avoid oxygen intrusion to the boiler waterwalls, superheaters, and economizer, when the Unit is shutdown for a short

period of time.

The capacity of the nitrogen system shall be designed to be sufficient for wet lay up of two boilers. The system shall consist of cylinder racks, manifolds, distribution pipework and all necessary valves. The system shall be designed based on racks of cylinders and for replacement by exchange of complete racks. Changeover between the racks shall be a manual operation. Flexible hoses shall be used to connect the rack of cylinders to the manifolds. The racks shall be equipped with high pressure relief valves, low pressure gauges and manifold changeover valve assemblies.

8.11 Hydrogen Generation Plant

A supply of dry and pure hydrogen gas shall be provided for the generators of the steam turbines. The hydrogen generation plant together with storage tanks shall be provided to generate > 99.9% purity of hydrogen required for the scavenging and filling of one generator and the full make-up requirements for both generators for a period of at least ten days. The maximum working pressure of the hydrogen generation plant shall not exceed 3.2 MPa.

The storage area shall be suitably segregated with fire resistance wall and shall be designed complying with relevant statutory requirements and international recognized standards.

The Offshore Supply Contractor shall provide the hydrogen generation plant, hydrogen manifold station, pressure regulators, leak detection system, and associated distribution piping system for this purpose. The design details and the generation process shall be submitted for Owner's review during detailed engineering stage.

The Offshore Services Contractor shall design the system taking the following into considerations:

The manifold system shall be provided with pressure reducing valves, safety valves, flash back arrestors at upstream of each branch connection.

All the pipelines shall be designed with a maximum allowable velocity of 10 m/s. If pipelines are made of stainless steel, they shall be designed with a maximum allowable velocity of 20 m/s.

Hydrogen quality shall be maintained at above 99.9% purity and less than 0.05 g/Nm³ in moisture.

Lockable double block valves with a blank or plug shall be provided for all points of systems which can discharge hydrogen to atmosphere, vent or other system. This does not apply to pressure relief valves.

The Onshore Contractor shall obtain the approval from statutory authorities for the design and installation of the hydrogen generation Plant.

The storage area shall be suitably segregated with fire resistance wall and shall be

designed complying with relevant statutory requirements and international recognized standards. Adequate double isolating valves and relief valves shall be provided.

Hydrogen shall be piped from the manifold to the inlet connection of the generator gas control cubicles. The manifold shall be installed with automatic and manual shut-off valves. The distribution pipework shall be run in open air (above ground) and the routing shall be in such a way to minimize mechanical damage. All hydrogen pipework shall be copper to BS 2871 or tungum to TCC 100B or equivalent material. All fittings shall be naval brass to BS 2874 CZ 112 (soldered) and brass to BS 2874 CZ 121 (non-soldered). All pipework joints shall be welded. All valves shall be of copper alloy and have provision for padlocking. A bypass provision shall be made in the control valves for maintenance use.

Facilities for sampling shall be provided so that the hydrogen or CO₂ purity can be tested in each section by portable meter. Hydrogen gas detectors shall be provided and arranged to initiate a visual flashing and audible alarm outside each entrance to the area monitored. Remote alarm signals shall also be provided in the CCR. Infra-red or other suitable fire detection system, manual call points, water spray system, and portable extinguishers shall be provided in the storage area to comply with local requirements. All electrical equipment shall be designed to NFPA 50 and to be suitable for hazardous area applications in accordance with BS 5345 or equivalent.

8.12 Chemical Monitoring System

Unitized chemical monitoring system shall be provided to monitor the chemistry of the steam and water cycle of each Unit to ensure that a consistently high quality is maintained. The chemical monitoring system shall consist of sampling probes, the sampling conditioning units, the chemical analytical equipment, and associated pipework, controls, fittings and valves.

Condensate, feedwater, boiler water and steam samples shall be collected by sampling probes at specified sampling points, which shall be conditioned via the sampling conditioning units prior to the chemical analytical equipment for analysis.

All the sample lines from each Unit shall be run to a chemical monitoring room where the sample conditioning units and the analytical equipment room will be positioned.

The monitoring rooms shall be air-conditioned, and shall be equipped with bulkhead for incoming connection from the Plant, bulkhead for drain connections, furnishings and facilities in enabling on-plant analytical test to be conducted.

Chemical analytical equipment and its corresponding sampling probes and sampling conditioning units shall be provided at the following sampling points. Extra manual sampling ports shall also be provided at each of these locations.

Sampling Point

1. Extraction pump discharge

Monitor

Dissolved oxygen
Cation conductivity

	pH
2. Deaerator outlet	Dissolved oxygen
3. Economizer inlet	Dissolved oxygen Cation conductivity Hydrazine Specific conductivity Silica Sodium pH
4. Boiler water (separator)	(Grab Facility only)
5. Superheated steam (after superheater)	Cation conductivity Sodium Silica

The chemical monitoring system shall be designed for continuous unattended operation except for hand sampling, routine inspection and maintenance. The sampling probes shall be designed to suit the conditions at the sampling points and shall be made from a material with high corrosive resistance. All sampling lines and valves shall be of 316 stainless steel material. Isolating valves shall be provided in double isolation valve configuration.

To enable operators to take a manual sample, a manual sample grab facility shall be incorporated into all continuous sample locations inside the monitoring room. In addition, manual grab facility shall also be provided for sampling of HP heater drains, LP heater drains, cold reheat steam and steam separator drains.

On-line continuous specific conductivity measurement equipment shall be provided in the condenser hotwells of both units to detect any leakage of cooling water into the condensate due to condenser tube failure.

All monitoring parameters shall be equipped with alarm facilities to provide local alarms in the chemical monitoring rooms, as well as alarm initiation in CCR via the DCS. All signals from the analysers shall also input into the DCS where continuous monitoring and historical trending of each parameter shall be carried out.

A high pressure (HP) and low pressure (LP) chemical dosing system shall be provided as skid mounted units consisting of a solution tank, metering pumps and accessories, level gauges, level switches, pump isolation valves and pressure gauges. All chemical feed solution tanks and accessories are of welded stainless steel construction. The dosing systems shall be provided with all necessary instrumentation and control systems. Emergency shower and eye wash station shall be provided at the HP and LP chemical dosing equipment compound.

The chemical dosing system shall include all the equipment required for operator to control cycle steam and feedwater quality in oxygenated treatment (OT) water chemistry mode. In addition, an independent hydrazine dosing system shall be provided for the operator to control dissolved oxygen level.

8.13 Waste Water Treatment Plant

8.13.1 General

Waste water treatment plant (WWTP) shall treat the Plant effluents such as water from boiler startup flash tank, coal handling plant area drains, ash handling plants drains, treatment of chemically contaminated effluents. The treated effluent shall be recovered from internal station usages as much as possible. The discharge effluent shall be in compliance with the relevant Vietnamese codes and standards.

Waste water treatment plants shall treat the following contaminated waste water generated in the Plant;

- pH adjusted waste water from the neutralization basin of demineralization plant
- Boiler startup flash tank water
- Boiler and turbine area equipment and floor drains
- Contaminated rainwater collected at coal store
- Air pre-heater washing water
- Stack washing water
- Electrostatic precipitator (ESP) washing water
- SWFGD wasted water
- Sampling rack blow down water
- Water from coal stores
- Chemical laboratory drain
- Workshop drain

Rain water from oil tank yard, transformer yard and turbine lube oil area shall be treated by oil/water separator. Rain water from boiler and ESP area shall be treated and reused or discharged after treatment. Other rain water from buildings, roads, etc., are not contaminated, which are within the effluent limits can be discharged in to the sea without treatment. The waste water shall be treated by flocculation, clarification and neutralization process and treated effluent shall be fed into the treated waste water basin. The treated effluent from the treated waste water basin shall be discharged into the CW discharge channel after satisfying the waste water discharge limit of Vietnamese standards.

8.13.2 System Requirements

The WWTP shall be designed to treat the waste water in order to achieve the effluent quality specified in OTR Clause 2.3.2 and the Vietnamese Code QCVN40:2011/BTNMT.

One (1) 100% capacity WWTP shall be provided. One (1) 100% waste water stream is capable to treat steady waste water from the Units, unsteady waste water such as air pre-heater washing water . The WWTP shall have 140% capacity of steady waste

water quantity as a common facility for two (2) Units. The capacity of WWTP shall be at least 200 m³/hr and comply with EIA and ESHIA.

The WWTP shall be controlled by a microprocessor-based (PLC) system from the control panel located in the WWTP area and it can also be operated manually from the local control panel.

Waste water generated from the Facility is classified as follows:

- Regular waste water
- Irregular waste water
- Oily waste water

Regular waste water from the Facility, such as demineralization Plant regeneration waste water, boiler startup flash tank blowdown, sampling rack drain, submerged scraper conveyor overflow, chemical laboratory drain, floor drains, etc., shall be fed into the primary holding sump which has minimum 1600 m³ capacity. Irregular waste water from the Facility, such as air heater washing water, rainwater from air heater and ash silo area, etc., shall be fed into the irregular waste water storage basin which has minimum 1500 m³ capacity. Regular and irregular waste water shall be mixed in the primary holding sump before treatment in the WWTP. The Offshore Services Contractor shall provide detail calculation of waste water flow rates to justify the selected size of the regular holding sump and irregular waste water storage basin and satisfy the requirement of the incident tank of defined capacity required in Decree 40/2019/ND-CP in detail design stage.

In the pH adjustment tank, sulphuric acid and caustic soda shall be added to adjust the pH of the water. Sodium hypochlorite and coagulant shall be dosed in the pH adjustment tank to disinfect and coagulate suspended solids in the waste water. The waste water from the pH adjustment tank is overflowed to flocculation tank, where coagulant aid shall be injected to coagulate the suspended solids present in the waste water. Then the waste water shall be transferred to the clarifier, where sludge and water are separated by the clarification process. The clarified water from the clarifier shall be fed into the final pH adjustment tank, where sulphuric acid and caustic soda shall be added to finally adjust pH of the water. Treated waste water from the sewage treatment shall be fed into the final pH adjustment tank. Treated waste water shall be stored in the treated waste water basin and finally discharged into the CW discharge channel by treated water pumps.

Sludge generated in the clarifier shall be transferred to sludge thickener by clarifier sludge pump. Sludge from the raw water system and the sewage treatment plant shall also be collected in this sludge thickener. In the sludge thickener, water shall be separated from the sludge and collected in the drain pit and transferred to regular waste water storage basin. Sludge from the sludge thickener shall be transferred to dehydrator by sludge feed pump to dehydrate the sludge. Polymer shall be mixed with the sludge at the dehydrator inlet to enhance the dehydration process in the dehydrator. Dehydrated sludge shall be transferred by cake conveyor to cake hopper, from there sludge is disposed using the trucks by the Owner. The storage capacity of the cake hopper shall be 3 days of maximum sludge flow rate. Sludge thickening and dewatering of the sludge from the water pre-treatment Plant, the WWTP and the

sewage treatment Plant may be separate, depending on the Site configuration and supplier's technologies.

Oily waste water from the Facility, such as fuel oil storage yard drain, transformer yard drain, turbine lube oil area drain, shall be diverted to the oily waste water sump of the oil separator. From the oily waste water sump, oily water is pumped or overflowed to the oil separator, where oil and water shall be separated. The separated oil shall be collected in the oil pit and it will be disposed by the Owner periodically. Treated oily water will not be mixed to the primary holding pond. It will be discharged to the sea together with non-polluted rain water.

Oil separator shall be capable of treating the contaminated rain water when it rains with the design rainfall value. In the rare case, the amount of rainfall in a few days exceeds the design rainfall value, at that time the contaminated water in the coal yard and tank yard shall be stored within each bund and treated subsequently in the wastewater treatment system.

Effluent resulting from acid clean of electro-chlorination plant equipment shall be treated and achieve the effluent quality specified in the Vietnamese Code QCVN40:2011 column B requirement before discharge into the sea.

Effluent collection sumps and effluent sump pumps shall be provided for each chemical storage and battery rooms to collect any effluent generated in these areas for offsite treatment.

8.13.3 Control and Monitoring Requirements

The normal mode of operation of the WWTP shall be of automatic type with a facility for manual operation. The WWTP shall be primarily operated/monitored from local control room. In this automatic mode, all motors and control valves shall be actuated automatically. The following treated water quality shall be monitored continuously on the local control panel;

- a) Inlet and outlet flows
- b) pH
- c) Temperature
- d) COD
- e) TSS
- f) Turbidity
- g) Oil-in-water

Sequential logic shall be carried out using reliable, industrial type programmable controllers, mounted on the local control panel. Sample grab facility shall be provided.

8.14 Sewage Treatment Plant

8.14.1 Function

The function of sewage treatment Plant is to treat the sanitary wastewater generated in the Facility. Effluent shall be treated in activated sludge, sedimentation, and sterilization process, and treated water shall be transferred to the final pH adjustment tank in the WWTP.

8.14.2 Design Basis

Sewage and sanitary waste water generated in the buildings around the Facility shall be first collected in lift station and shall be transferred to the sewage collection cum equalization tank by sewage pumps. The sewage and sanitary waste water transfer pipe material shall be PVC and the pipe shall be buried under the ground. The sewage treatment plant shall be designed to reduce biological oxygen demand (BOD) and suspended solids (SS). After mixing the treated sewage effluent with the treated wastewater in the wastewater treatment plant, the water shall meet the requirement of the effluent quality standard values as required by Vietnamese standards before discharge outside station.

The sewage treatment plant shall have net capacity of at least 150 m³/d, which shall meet 300 permanent personnel on daily basis and additional 280 maintenance personnel during overhaul period. The train shall consist of sewage collection cum equalization tank, activated sludge aeration tank, sedimentation tank, treated effluent collection, disinfection tank and sewage effluent sump.

The sewage treatment plant shall be controlled automatically by a microprocessor-based (PLC) system from the control panel located in the WWTP and it shall also be controlled manually from the local control panel.

8.14.3 Major Components

The sewage treatment plant shall consist of the following major components:

- One (1) sewage collection cum equalization tank
- One (1) combined biological device + sedimentation tank
- activated sludge aeration tank, if necessary
- One (1) sedimentation tank, if necessary
- One (1) treated effluent collection and disinfection tank
- One (1) sewage effluent sump
- Two (2) treated effluent disposal pumps
- Two (2) aeration blowers
- One (1) sodium hypochlorite dosing system
- Two (2) sludge recirculation and disposal pumps
- Interconnecting piping, valves, instrumentation and controls, and PLC control system
- Sewage pumps

8.14.4 System Requirements

The sewage treatment Plant process, major components, control and instruments shall be as below.

8.14.4.1 Process Description

Sewage water generated in the Facility shall be collected in a lift station and pumped to the sewage collection cum equalization tank, and then it shall be sent to the activated sludge aeration tank where BOD and SS in the waste water are decomposed with help of activated sludge in the tank. Treated water from the activated sludge aeration tank shall be transferred to the sedimentation tank. In the sedimentation tank, the activated sludge in the treated water is precipitated on the tank bottom and collected into the centre of the tank. The precipitated matters shall be returned to the activated sludge aeration tank, or transferred to the thickener in the WWTP, which depends on the concentration of the activated sludge in the tank. The clear water shall be transferred to the treated effluent collection and disinfection tank and treated with help of sodium hypochlorite. Treated water shall be collected in the sewage effluent pump sump, from there treated water shall be routed to the final pH adjustment tank in the WWTP using the treated effluent disposal pumps.

8.14.4.2 Major Component Description

- a. Sewage collection cum equalization tank
One (1) 100% capacity tank shall be provided, which will collect sewage wastewater. The tank shall be made of concrete.
- b. Activated Sludge Aeration Tank
One (1) 100% capacity activated sludge aeration tank shall be provided. The activated sludge aeration tank shall be made of concrete. This tank is installed to decompose BOD and SS contained in the sewage wastewater. Two (2) 100% capacity aeration blowers shall be provided for supplying air to the activated sludge aeration tank.
- c. Sedimentation Tank
One (1) 100% capacity sedimentation tank shall be provided. The sedimentation tank shall be made of concrete. This tank is installed for sedimentation and to remove activated sludge from treated water. In this tank, solids and liquids are separated and the clear water is collected into the treated effluent collection and disinfection tank by gravity. The sedimentated matters shall be collected into the centre of the tank bottom by gravity and shall be returned to the activated sludge aeration tank or transferred to the thickener in the wastewater treatment plant, which depends on the concentration of the activated sludge.
- d. Treated effluent collection and disinfection tank

One (1) 100% capacity tank shall be provided to treat the over flow water from the sedimentation tank with sodium hypochlorite. The tank shall be made of concrete.

e. Sewage Effluent Sump

One (1) 100% capacity sewage effluent sump shall be provided. The sewage effluent sump shall be made of concrete.

f. Treated effluent disposal pump

Two (2) 100% capacity pumps shall be provided to transfer the treated wastewater. The pumps shall be of centrifugal type. The selected pump material will be suitable for handling the purified wastewater.

g. Sodium hypochlorite Feed System

One (1) 100% capacity sodium hypochlorite feed system shall be provided to inject the sodium hypochlorite into the treated effluent collection and disinfection tank. The sodium hypochlorite feed system shall be provided with storage tank and dosing pump. The selected equipment material shall be suitable for handling the sodium hypochlorite.

h. Sewage water pump

Two (2) 100% capacity sewage water pumps shall be provided to transfer the sewage and sanitary waste water to the sewage collection cum equalization tank. The selected pump type and material shall be suitable for handling the sewage and sanitary waste water.

8.14.4.3 Control and Instruments

The normal operation mode of the Plant shall be basically automatic and operated/monitored from local control room. In this auto mode, all motors shall be actuated automatically. Treated water quality shall be monitored continuously at the outlet of the sewage treatment plant.

The system shall be capable of operation under remote manual control from the water treatment system control panel.

Sequential logic shall be maintained by using reliable, industrial type programmable controllers, mounted on the WWTP control panel.

8.15 Fire Protection and Detection System

8.15.1 System Description

A suitable fire protection and detection system shall be provided which shall consist, but not be limited to, the following:

- Outdoor hydrant system
- Hose box
- Wet riser system
- Indoor hydrant
- Automatic water sprinkler / deluge system
- Fixed water spray system
- Air foam system
- Automatic CO2/FM200 system
- Portable fire extinguishers

The fire fighting water shall be supplied from fire fighting water storage tank which can be shared with service water tanks. 1 × 100% capacity electric motor driven fire pump, 1 × 100% diesel engine driven fire pump and 2 × 100% capacity electrically driven jockey pumps shall be installed for supply of prescribed water to hydrant/sprinkler mains.

Motor driven fire pump shall be controlled by pressure switch to be automatically started or stopped when pressure is at pre-set level. At the time of fire accident, the fire fighting system shall be operated resulting in pressure drop in water main to permitted level. Then, motor driven fire pump will be automatically started. In emergency conditions such as power source failure or motor failure, the diesel engine driven fire pump will start automatically as a result of the fire main pressure dropping below the set pressure of the pressure switch. Two electric motor driven jockey pumps along with a hydro pneumatic tank shall fill fire mains in the event of minor leaks. The jockey pump stops automatically via a pressure switch when the pressure in the fire main returns to a set level. The pressure of the pneumatic tank shall be maintained by electrical motor driven jockey pumps with the help of level switches.

The entire built up area of the proposed unit shall be covered by pressurized water supply pipe network for supply of firewater. The system consists of a supply from the pressurized ring main system through automatically operated deluge valve stations, a water distribution pipe network and a set of high velocity water spray nozzles to cover the equipment and its accessories.

8.15.2 Design Criteria

The fire protection and fighting system shall be designed and built in accordance to the Annex C Fire Protection Report (to be followed only for the scope of supply) in this OTR and NFPA Standards. It shall be acceptable by the local regulation.

If no relevant Vietnamese standards are applicable to equipment design, related international standards shall be applied i.e. NFPA (National Fire Protection Association) or IEC 60079-10 (International Electromechanical Commission). The overall fire detection and protection system including its design and materials etc. shall be to the approval of the Vietnamese fire protection bureau.

The capacity of each of the fire water pumps shall be sufficient to meet the demand of the largest single risk on the station, i.e. all automatic and manually operated high velocity water spray zones for the turbine generator lubricating oil systems operating simultaneously.

In the event of a fire, and the operation of an automatic or manual fire protection system or operation of a hydrant, the electric pump is automatically started via a pressure switch when the pressure in the fire main falls below a set value.

If one electric pump fails to start or the pressure in the fire main continues to fall, the second electric pump shall start. In case of failure of electric pump to start the diesel pump shall be automatically started. Once started, the pumps shall continue to run until they are manually shutdown.

Pipe work for the trunk mains and hydrant mains shall be arranged on a ring main basis with a number of isolating valves to allow maintenance and servicing without the necessity to close down the whole system.

The operation of fire ring wall system for built up area system shall be controlled by an automatic deluge valve in conjunction with a thermo-pneumatic detection system containing quartz bulb or other suitable heat detector, which will open deluge valve on a sensing of fire.

The Facility shall be covered by suitable number of hydrants and hose reels at strategic locations such that water can be sprayed to any equipment from at least two hydrant/hose reels.

Fire detectors of appropriate type shall be installed at suitable locations to detect and alarm in case of fire. Centralized fire alarm panel shall be located in the CCR. Repeater alarm panels shall be located in the fire water pump house, existing fire station and at the main gate.

The Onshore Contractor shall obtain approval of concerned agency for the entire fire detection and protection system design and installation, to obtain maximum benefit in insurance premium.

8.15.3 Fixed Water Sprinkler Spray System

Fixed water spray systems shall be provided for the protection of following items of the Facility:

- Transformers (generator transformer, unit transformers, station transformer and other transformers larger than 3 MVA)
- Boiler burner areas
- Lubricating oil tank and seal oil supply unit
- Bunker area
- Coal conveyor tower and connection between the power house and cable spreading roomH₂ storage house
- Steam turbine above operating floor (for lubricating oil lines)

- Boiler feed pump (for hydraulic and lubricating oil hazard)
- Fire fighting pump house
- Dangerous goods store. The Onshore Contractor shall install alternative system should spray water system is not applicable for the storage substances.
- Offices (Administration Building and central control building)

The fixed water spray system for the transformers, boiler burner areas, and lubricating oil tank and seal oil supply unit shall be of automatic actuation type with hydraulic release system consisting of quartz bulb sprinkler heads (wet pilot sprinkler).

The sprinkler heads shall be pressurized by the main water and the spray system shall be actuated hydraulically when pressure loss occurs in the detection line.

The fixed water spray system for the cable spreading room shall be of automatic actuation type consisting of quartz bulb water spray units initiated automatically by smoke detectors installed in the immediate vicinity of the cables in the cable spreading room.

The fixed water spray system provided to protect the tank roof and shell surface of the fuel oil storage tanks from external heat radiation shall be of manual actuation type and heat detection system. The deluge valve shall be actuated manually at local and remotely at the fire alarm panel in the CCR.

The fire risks in the Facility shall be identified and suitable water spray systems (high velocity sprays, medium velocity sprays, and sprinkler) shall be provided to various locations. High velocity spray systems shall be used to protect against all risks involving oil.

Water application rate shall meet the requirement of NFPA 15 or equivalent standard.

The spray nozzle shall be of the directional solid-cone open head type and of non-clogging type. The deluge valve shall be of hydraulic actuation type.

System configuration

The automatic spray system shall consist of the following major items:

- (a) Deluge valve
- (b) Pressure switch
- (c) Spray nozzles
- (d) Sprinkler heads (wet pilot sprinkler)
- (e) Smoke detectors
- (f) Heat detectors
- (g) Solenoid valve
- (h) Fire fighting local panel (FFLP)

8.15.4 Outdoor Hydrants & Hose cabinets

Outdoor Hydrants

The coal storage area shall be provided with fire protection and fighting system using outdoor hydrants. Outdoor hydrants shall be connected with the ring main pipe laid underground around the plant. The hose connection and valve shut-off device on the hydrant shall be exposed above the surface.

Hose boxes including two fire hoses and two light alloy jet/spray water nozzles shall be located near each hydrant.

Indoor Hydrant

Indoor hydrants consisting of angle valve, rack, synthetic fiber hose and water nozzle shall be installed at each floor of the buildings. Moreover, a manual call point, an alarm bell and a location lamp shall be provided on each indoor hydrant cabinet.

The system design for the indoor hydrants, including pitch, etc., shall be in accordance with NFPA 14 or equivalent standard.

Wet riser system

The wet riser system shall be installed in the boiler galleries and the power house. It shall be provided near the base of stairways and landing valve shall be provided at each floor.

Hydrants system pipe shall be HDPE pipe for buried pipe and ASTM A106B pipe for above ground dry pipe and A53B galvanized for wet pipe.

8.15.5 Air Foam System

The fuel oil storage tanks, day tanks, tank farm bund, fuel oil pump house, and station oil unloading area shall be provided with fixed foam installations in accordance with the requirement of NFPA 11 or equivalent. A subsurface foam injection system complete with necessary pumping equipment or a surface foam injection complete with proportioning equipment shall be used for protection of fuel oil storage tanks.

The foam system shall be designed for both local and remote manual operation and control, and shall have facilities for coupling to the hydrant system to provide a manually actuated foam spray. Foam concentrate filling pump for filling of the foam tanks shall be provided.

Fuel oil storage tanks shall be provided with an air foam system using 3% AFFF concentrate in accordance with NFPA 11 and the major components of the system shall consist of a proportioner, a foam concentrate tank, foam hydrants and hose cabinets, air foam chambers, a fire fighting local panel and foam distribution piping.

Manual operation of the starting push button on the fire fighting local panel shall open the motor valve provided to the water supply pipe.

8.15.6 Portable Fire Fighting Equipment

The appropriate numbers of the required portable extinguishers shall be specified to meet the local regulations. The Onshore Contractor shall be responsible for formulating the Site protection strategy, and the disposition of portable equipment around the Facility.

Portable fire fighting equipment shall include portable extinguisher, portable and trolley foam container that are equipped for protection requirement of each area. TCVN 4878: 89 (ISO 3941) – Foam and gas fire fighting system shall be based to select the type of extinguisher.

The portable fire fighting equipment shall be provided for all buildings next to door.

The extinguisher shall be sealed, and manufacturer trade stamp shall clearly mark shell weight and gas weight. When installing, extinguisher top shall not be 1.5 m higher than floor.

Selecting fire fighting substance and extinguisher weight for each area shall ensure protection capability for construction structure of building and tool/device at this area. The mobile extinguisher with trolley shall be designed and installed for large protection area with high requirement on substance quantity and concentration.

8.15.7 Gas Protection System

Manual and / or automatic FM200 gas suppression system in conformance with BS:ISO14520 or equivalent shall be provided for fire protection purposes of the following areas.

- CCR
- Equipment rooms
- DCS equipment rooms

Local control and indication panels shall be provided in areas covered by gas protection system. Continuously manned areas shall also be protected by portable fire extinguishers for use on small fires. A dedicated detection and alarm system shall be provided to allow automatic initiation of the discharge and to allow personnel to evacuate the risk area.

In addition, CO₂ fire fighting system shall be applied for rooms with electronic equipment, tunnels and cable trench, central control house and system control house. CO₂ fire fighting system shall also be installed for MV cable trenches in steam turbine house. Consideration will be given to use sprinkler system instead, subject to agreement by local authority.

Emergency diesel generator shall be protected by fixed CO₂ system, fire detectors shall provide signs to activate CO₂ automatic fire fighting system in addition to indoor fire hydrants protection.

Low pressure CO₂ fire fighting system shall be applied for the coal bunkers.

Warning signs, time delay and alarm shall be provided in gas protection system as NFPA standards on carbon dioxide and FM200. Cares shall be taken to dissipate hazardous atmospheres, not merely move them to another location, using forced ventilation system or mobile ventilation system depending on accessibility, for verification by trained people that it is safe.

8.15.8 Fire Alarm and Detection System

Manual call points shall be used at strategic points throughout the Facility with alarm and indications to the fire alarm panels. A sounder system shall be installed covering all areas of the Facility. A separate sounder/beacon alarm system shall be provided for CO₂ protection areas, with sounder alarms differentiated from other type of fire alarm.

Fire alarm panels shall be provided in gate house and security building. A main fire fighting monitoring panel shall be provided in the CCR with mimic maps of all area of the Facility. The system shall be designed as a microprocessor-based information system, powered by UPS with back up from DC.

For the purpose of fire detection, detectors of smoke, heat, linear heat or flame shall be used and a selected type most effective to a given area shall be provided. A manual call point, an alarm bell and a location lamp shall be provided on each indoor hydrant cabinet.

Because of strict features and requirements for control cubicles at the CCR, such location shall be provided with very early smoke detection aspiration (VESDA) system.

For detecting temperature increase due to spontaneous combustion in the coal bunkers, metal thermocouples shall be installed at exterior surfaces of the coal bunker shells (3 measuring points per bunker). Their measuring temperature data shall be sent to the DCS and their alarms shall also be indicated on the fire alarm panel.

A fire alarm panel and all local panels shall be provided with a backup battery ensuring at least 24 hours' continuous operation of equipment and at least 3 hours' continuous operation during fire in considering an electricity supply failure respectively.

The fire alarm and detection system shall comprise of the followings:

(a) Fire Alarm Panel

Fire alarm panel shall be located in the CCR, fire station building and gate house.

(b) Fire Alarm and Detection Local Panel

Fire alarm and detection local panel shall be installed in all buildings and outdoor areas as per Vietnamese standards.

(c) Fire Fighting Local Panel

Fire fighting local panel shall be installed near the water spray systems, the automatic water sprinkler system and the air foam system.

(d) Manual Call Point

The manual call points for buildings and outdoor areas shall be provided as per Vietnamese standards.

(e) Smoke / Heat detector

Smoke detectors and heat detectors (rate-of-rise and fixed temperature type) shall be provided as per Vietnamese standards.

(f) Combination box

Combination boxes consisting of a manual call point, an alarm bell and a location lamp shall be provided as per Vietnamese standards.

8.15.9 Fire Station and Fire Engine

The Onshore Contractor shall provide a fire station building for firemen on duty, storing of essential fire equipment for instance fire hose, nozzle etc. In addition, a fully equipped fire engine meeting the relevant Vietnamese standards shall also be provided together with a covered car park at the ground floor of the fire station.

8.16 Coal Handling System

8.16.1 General

The coal handling system shall be based on a well proven design and shall be capable of operation on a 24 hours continuous basis with minimum down time. Coal handling Plant shall be controlled by DCS (if this is not possible, by PLC) and the local and remote control in CCR shall be provided. In case the coal handling plant is controlled by PLC, local control and remote control shall be provided in the coal handling control room in the coal plant control building. Communication interface shall be provided between the coal handling control room and the CCR. Coal handling plant status shall be transmitted to the Plant DCS for remote monitoring in the CCR. Remote CCTV monitoring system shall be installed for coal handling plant surveillance. The layout shall be as simple as practicable to minimize the number of conveyors, conveyor transfer points, diversion chutes and changes in direction of the coal flow.

Starting from the jetty, the Offshore Supply Contractor shall provide all coal handling system including two grab type ship unloaders each rated capacity of 1500 t/hr and maximum capacity of 1800 t/hr and single jetty conveyor with capacity of 3000 t/hr. The coal can be conveyed directly to coal stores or boiler bunker.

During the detailed engineering stage, the Offshore Services Contractor shall provide as part of design submission calculations for the respective conveyor sizing. Means shall be provided to prevent the upstream conveyors from overloading the bunker conveyors on occasions when coal is transferred directly from ship to bunker.

In order to protect the coal from rainfall events and to minimize fugitive dust, the coal storage area shall be covered and civil foundation of the cover shall be provided. The dry coal shed shall be a large-span steel grid supported by reinforced concrete columns. Dimensions of the coal shed shall satisfy the operation of coal stacking and reclaiming equipment used. The steel structure of the coal shed shall be painted according to coating system and meet relevant Standards' requirements as per the Contract. Material specification shall be submitted to Owner for review during detailed engineering stage and colour coding is to be agreed by Owner.

Covered coal store based on the arrangement of two storage areas shall be provided with a total capacity not less than 555,000 tonnes, which is the 45-day BMCR operation of the Facility at 85% capacity factor burning performance coal.

The coal storage yard shall be full covered (Dry Coal Shed) with open end at seaside only. In this case, the wind barrier will not be provided. A 12,000 m³ settlement pond for coal yard run-off water shall be provided with local treatment system which discharges the treated run-off water to the CW system. There shall also be a provision for piping the coal yard run-off water to the WWTP. All treated effluents shall comply with OTR Clause 2.3.2. In addition, emergency overflow to avoid flooding of the site shall be provided to allow discharge directly to the CW system when the settlement pond is full with the pumps for emptying the pond running in full capacity. The sizing of 12,000 m³ rainwater settlement pond shall be reviewed and optimized

to achieve the coal yard capacity and wastewater treatment scheme during design submission stage.

The coal stores shall be equipped with fire detection and protection systems as well as a remote CCTV monitoring system to enable surveillance.

The conveying system shall discharge coal from conveyors system from jetty onto a stackout conveying system and then the coal shall be stacked in designated areas. Provisions shall be made to bypass the coal stores and discharge coal directly onto the reclaim conveying system.

Coal from the storage piles shall be reclaimed by stacker-reclaimers with variable capacity inside the coal stores and conveyed to the coal screen house via the reclaim conveying system. The coal shall then be conveyed to coal bunkers inside the boiler building via tripper conveyor. Interlock protections shall be provided for stacker/reclaimers to avoid over feeding.

The receiving/stackout and reclaim conveying systems shall be furnished with the following auxiliary equipment and/or systems:

- Self-cleaning, magnetic separators to remove ferrous material and metal detectors to detect non-ferrous material
- Belt scales for inventory purposes and to control the conveying rate of the belt feeders
- “As-Received” and “As-Fired” sampling systems
- Conveyor hood covers for all outside conveyors
- Enclosed transfer towers
- Dust suppression and dust control systems
- Wash down systems for the enclosed transfer towers
- Sumps and sump pumps for the enclosed transfer towers
- Fire detection and protection system
- Ventilation systems
- Service air

Mobile equipment shall be supplied. Magnetic separators, belt weighers, traveling belt trippers, diverter gates, vibratory feeders, belt scales, roller screens, ferrous and non-ferrous metal detectors, samplers and belt speed sensors shall be installed in appropriate locations to protect and monitor the Facility.

8.16.2 Grab Type Ship Unloaders

The grab type ship unloaders shall be a shore based, electrically powered, rail mounted, gantry ship unloaders with, but not limited to, the following features

- Boom be capable of being raised and lowered to provide clearance of ship super structures
- Grab be capable of being opened and closed, raised and lowered, travelling forward/backward through mechanical differential mechanism.
- Gantry be cable of travelling on rail in parallel with the berth side of the jetty.

- Operation and emergency braking system for boom motion, grab motion and gantry motion.
- Electrical power be supplied by electrical facilities on jetty through cable reeling system on board ship unloaders.
- Spray water dust suppression system at all coal transfer points on the ship unloaders and at interface between the ship unloaders and jetty conveyors
- Water be supplied from jetty through water reeling system on board ship unloaders.
- Coal feeding system receiving the coal unloaded by grab and transferring the coal to the jetty conveyor in stable manner.
- Firefighting and fire detection system
- Lift for transporting O&M personnel to all working levels of the ship unloaders
- Access way, staircase, platform satisfying all O&M activities need
- Unloader motion control system including
 - semi-automatic grab/trolley motion control
 - equipment interlocking function
 - Grab/trolley/gantry motion protection system including overtravel protection
 - Unloader/unloader anti-collision system
- Anchoring system to secure the ship unloaders in stowage position on the jetty when the unloaders are not in operation.
- Electrical protection system
- Land/ship unloader data communication and telecommunication system
- CCTV allowing unloader operator to monitor positions critical to unloader operation.
- Alarm and boardcasting system
- HVAC system for operator's cab and equipment room.
- Overhead crane in the equipment room for O&M activities

Coal will be transported to the jetty by coal collier. The coal delivered by coal collier with maximum 100,000 DWT shall be unloaded by two (2) grab type ship unloaders, each with rated capacity of 1,500 t/hr and maximum capacity of 1800 t/hr with coal density at 0.84 t/m³, traveling on common rails. Normally, two ship unloaders will be in operation at the same time. Suitable protection shall be provided to avoid over-loading of the conveyor.

The size of a typical 100,000DWT coal collier is as follows:

- Overall length 250 m
- Molded breadth 43 m
- Molded depth 20.3 m
- Loaded draft 14.5 m

The normal coal unloading operation is with the boom in level position. The handling operation shall be carried out from the operator's cab. Grab open/close motion, grab up/down travel and trolley travel shall be capable of simultaneous operation.

The ship unloaders shall be capable of operating at any feed rate up to the design rate, without spillage.

Provisions shall be made to prevent coal spillage from falling into sea.

Dust suppression sprays shall be provided on the unloaders, using re-used water.

The ship unloaders receiving hoppers shall discharge coal on to the jetty conveyor mounted on top of the jetty deck for the full length of the jetty.

Bag filters shall be provided in the dust collector to collect air and dust from the transfer chute during unloading. The bag filters shall be evacuated by time control and return the dust back to the coal flow in the conveying system. The bag filters shall be cleaned automatically by compressed air system after each evacuation.

Unloader/unloader anti-collision system provided shall stop the unloaders from colliding into each other by alarming the operator, braking the gantry to stop and cutting the gantry motors power according to limits of distance between adjacent unloaders preset in the unloader motion control system. A permanent communication with signals of distances of nearest part of other ship unloaders shall be installed in each of the ship unloaders.

Emergency stops shall be provided at several places of the ship unloaders.

The ship unloaders shall be installed with wind sensors and alarm shall be initiated to alert operator when the wind speed exceeds preset limit for safe operation. The ship unloaders shall be equipped with anchoring system to secure the unloader in stowage position during heavy wind condition. With the anchoring system, the gantry shall be able to be manually locked with special storm lock at the sea side. The crane shall be operable under the wind velocity up to operating design wind. If the wind exceeds the operating limit which is measured at the anemometer furnished on the crane, the operation of the crane will be stopped and secured in the stowed mode. The crane shall be secured in the following stowed mode:

- The boom shall be fully raised and securely latched.
- The crane shall be anchored manually to the foundation fittings which are provided at the stowage position on the berth.
- The trolley shall be positioned at the stowage position.

The operator's cab shall be of an independent drive type. It shall be of steel construction, insulated and provided with wide windows and an entrance door at rear wall. The floor shall be of double bottom construction to allow cable wiring inside. The operator's seat shall be uphoisted with head and arm rests, fore and aft adjustment, height adjustment. Air conditioner shall be provided for the design maximum atmospheric temperature.

Ready to run signal shall be provided from the shore when the jetty conveyors are running. Then, the ship unloaders can be able to start. If this signal disappears, the grab unloaders shall stop unloading immediately.

The ship unloaders shall be capable of lifting front end loader of the following specification in or out of ship holds.

- Weight: up to 22 tonnes
- Dimension: up to 9 x 3 x 3.5 m (L x W x H)

The hoist shall be located on the horizontal arm.

One set of front end loader of ~190 hp shall be provided for assisting in cleaning coal up in the ship holds.

Automatic grease system shall be provided for the ship unloaders to grease the majority of the lubrication points. A grease barrel with a grease pump shall be provided on the gantry. From this point, the grease is distributed via this system of pipes and hoses to lubricators located on different places on the ship unloaders for major unloader operation motions and the wire rope. Each lubricator shall be equipped with closed containers with level guards. The electrical control system shall automatically fill up each respective lubricator when indication is made by the low level guard. The lubricator shall distribute grease to each lubrication point with correct quantity at pre-set intervals. An alarm shall notify the operator when the main grease barrel has to be replaced and when the grease system is not functioning correctly. The volume of the main barrel shall be at least 200 litres.

An overhead crane with minimum capacity of 10t shall be provided in the machinery house to facilitate maintenance and replacement of the components in the machinery house. The lift hoisting capacity shall be enough to lower the components to the dock level. The hoisting, traveling shall be electrically operated by pendant switch.

The grab bucket shall be of the clamshell type. The grabs shall be dead-end type operated by holding ropes and closing ropes. Shell of the grab shall be welded steel construction and the teeth edge shall be reinforced by welding of wear resistant electrode or used wear-resistant steel. The grab bucket shall be provided with lugs for the attachment of lifting slings for lifting a front end loader or other clean-up equipment in and out of the coal holds of coal colliers.

8.16.3 Conveyor Systems

All conveyors shall be of dust tight hood cover design with 1000 mm wide walkways (on both sides), fire detection equipment, fire protection equipment, wash down equipment, lighting and power outlets for maintenance along all coal plant conveyors.

To minimize environmental impact, all transfer towers shall be of the full enclosure design and slanted roof with internal walkways, fire detection and protection system. Dust suppression and full enclosure shall be allowed at the transfer towers together with the provision of primary sampler cutter facilities, and sample transfer to a sampling station flop gates. The enclosure shall be weather proof and water tight and able to withstand the adverse weather conditions at site. Automatic fire detection and sprinkler system shall be provided along the whole length of the conveyor.

Weighers with totalizer counters shall be installed on each line to measure the quantity of incoming coal from first junction at the site boundary, the coal conveying

to boiler bunkers, the coal handled by each stacker reclaimer and the hot coal transported out of the coal stores to the emergency dump area..

Each conveyor shall be fitted with a non-contact type speed detecting switch for use in the sequential starting and tripping interlock scheme and for ensuring that the driving motor can be shut down automatically in the event of the belt slipping or breaking.

Protection system shall be installed for safe operation of the conveyor, including but not limited to pull chords, belt sway switch, zero speed switch etc.

The driving gear of each conveyor shall be by electric motor and reduction gear. All gears shall be totally enclosed and oil immersed and capable of operating on a 24 hours continuous basis.

Chutes shall be of the shuttling or fixed type. All surfaces of the chutes liable to come into contact with the coal shall have sufficient lining to assist free flow of coal and to minimize wear. To prevent blockage of chutes by high moisture coal, suitable vibrators shall be installed for the chutes. The detailed design scheme shall be provided at the design stage.

(a) Concrete Clean Up Slabs, Spillage and Clean Up

Concrete clean up slabs shall be located under all conveyor transfers. The slabs shall be kerbed and sloped to clean up sumps to suit clean up with a 1500 mm wide bucket. Sump floor shall be sloped to suit. Embedded steel shall be used to protect concrete edges.

(b) Belt Weighers

Belt weighers shall have a minimum of four (4) carry idler sets located on the weighing frame. Idler rolls shall be fully machined and balanced. The belt scale system shall have a floating frame mounted on four hermetically sealed load cells. The system shall have an overall accuracy of +0.5% at 20% to 100% of total belt capacity and comprise an integrator, load cell digitizer and digital speed sensor.

(c) Pulley Bearings Temperature Sensor

All pulley bearings shall have temperature sensors.

(d) Trailing Cables

Cable trays shall be provided to support all trailing cables to machines.

(e) Conveyor Chutes

Chutes shall be enclosed and provided with adjustable deflector plates to centre material flow and abrasion resistant liners at areas subject to wear. Areas not subject to wear but subject to build up of fine sticky coal shall be lined with HDPE plastic. Chute angles shall be designed to promote free material flow. Chute liners shall be 6 mm stainless steel.

(f) Vulcanizing Press

Vulcanizing press shall be supplied and used for the connection and repair of conveyor belting.

(g) Belt Reeling Equipment

Suitable belt reeling equipment shall be provided for reeling up and off of all belts for the first installation and future maintenance. Two winding devices, with each set shall be provided; one with drive, the other without drive.

(h) Sampling Installation

Sampling operations shall be per ISO 1988 or equivalent and shall be automatic. The primary sample shall extract a full cross section cut from the moving stream of coal on the conveyor. The unit shall be installed above the belt. Stainless steel shall be used in the fabrication of the moving parts. The total system comprising the primary sampler and out bound sampling station shall include, but not be limited to, belt elevator, sample feeder belt, primary and secondary crushing, secondary and tertiary sampling and sample collection.

(i) Magnetic Separator

The electromagnet shall have a wear resistant discharge belt running around the unit and be complete with take up arrangements.

(j) Drive Pulley Surface Friction Factor

The friction factors used for design shall not exceed 0.3 under all operating conditions, unless otherwise approved.

(k) Conveyor Take Ups

Conveyor take ups shall be provided on all belts to cope with any elongation of the belting plus splicing requirements and to prevent belt slip. Mechanical devices shall be provided for tensioning. Sufficient clearance for machine clear up of spillage shall be allowed for in the design. Safety protection screens shall be provided.

(l) Snub Pulleys

Where the angle of wrap is less than 180 degrees and snub pulleys not provided, heavy duty rubber covered flat idlers shall be used.

(m) Conveyor Pulleys and Idlers

Conveyor pulley fabrications shall be stress relieved. Carbon steel shafting with compression type hubs for non-drive pulley shall be used. The lagging of non-drive pulleys shall be plain with a minimum thickness of 10 mm. The lagging on drive pulleys shall be herring bone grooved with a minimum thickness of 12 mm. Troughing idlers are to be of steel construction, three roll 35 degrees' troughing wide base type. Return idlers may be V-type with flat rolls and/or rubber disc type.

(n) Head Pulley Belt Scrapers

Head pulley chutes shall be provided with one double belt scraper with elastic mounts and HDPE skirts.

(o) Conveyor Belting

All conveyor belting shall be of fire resistant rubber. Belting may be of fabric synthetic or steel cord construction spliced endless during construction. Belt design grades shall be designed for up to 16°.

(p) Conveyor Drives

Conveyor drives shall consist of gear reducers directly coupled to or shaft mounted on pulley assemblies. Motors shall have torque control characteristics or fitted with torque control (fluid) couplings. Drives may either be of parallel shaft or right angle orientation. For standardization, drive units shall be of like horse power and orientation, where possible.

8.16.4 Coal Store

Two separate covered dry coal stores shall be provided to store a total combined storage capacity of 45 days BMCR operation of the Facility at 85% capacity factor at performance coal.

Below ground level is allowed to be used for coal storage. However, the coal at the below ground level shall be accessible by stacker/reclaimer and mobile plant for operation and maintenance of coal yard. Owner will put about 0.5m height of bedding coal (coal reject and ash) at the bottom of the coal storage area. This amount of storage shall not be counted as part of the capacity of coal storage. For example, if the coal stock pile is designed with 1m underground, bottom level of 0.5m for bedding coal will not be counted for coal capacity and upper level of 0.5m will be counted for coal capacity.

Common structure for two coal sheds shall be provided to maximize the capacity of coal storage. Maintenance accesses to the coal sheds and its structural steel shall be provided.

The store foundation shall be designed with rail supports for the stacker/reclaimer located inside the coal store. The rails provide support and allow movement of the traveling stacker/reclaimer.

At least 6 m width mobile plant access shall be provided around each side of coal pile, except at the rail side of stacker/reclaimer and at the front and rear sides of coal piles, within each coal shed for operation and maintenance purpose.

The coal yard base shall consist of layers placed on a compacted sub-base which shall be compacted satisfactorily to a compaction factor $k \geq 0.95$:-

- Layer 1: Geotextile (impermeable layer, Single layer), 1.5mm thick
- Layer 2: Excavated soil (screened), 200mm thick;
- Layer 3: High-density Polyethylene (HDPE) composite geomembrane (impermeable layer), 1.5mm thick;

- Layer 4: Protection excavated soil (screened), 300mm thick.

If underground coal storage is used, water leakage protection measure shall be applied to the wall of the underground coal storage to avoid contaminated water leakage to subsoil and reviewed during design submission stage.

The design of the coal handling system shall ensure that coal unloading at 3000 t/hr from the jetty and coal reclaiming at 1500 t/hr to the coal bunkers can be carried out at the same time.

In order to facilitate blending of coal reclaimed from the two coal stores and unloaded from the jetty in correct ratio and feeding the blended coal to the boiler, the reclaiming capacity of the two stacker reclaimers in the two coal stores shall be able to maintain precisely in any loading set by the operator between 0 to 1500 TPH. The rotation of the bucket wheels and the traveling of the two stack reclaimers shall be able to operate at variable speed to enable the stacker reclaimers running at variable reclaiming capacity.

The stacker/reclaimer inside both coal yard shall be provided with a stationary reclaim hopper at one end of the store. If any one of the stacker/reclaimer is out of service, mobile plant equipment shall be provided to reclaim the coal, generally located at the end also, and feed it to the stationary reclaim hopper.

Any one type of coal shall be able to be reclaimed from each coal store by the stacker/reclaimer. There shall be flexibility that a third type of coal can also be blended with the two primary coals. In this case the third type of coal need to be reclaimed manually by plant mobile equipment and dumped in any one of the reclaim hoppers.

The store foundation shall be designed with a 1.0 m high retaining wall inside the cladding to prevent coal spillage from covering either the store sheeting or the rail. The design of retaining wall within coal yard shall be reviewed during design submission stage.

Mobile type emergency reclaim hoppers shall be provided on the reclaim belt conveyors. If reclaimers are mal-function, coal shall be reclaimed by wheel loaders or bulldozers and dump on the emergency reclaim hoppers.

The coal shed shall be provided with all required lighting, ventilation equipment fire-fighting (Protection and Detection) system and dust suppression system, which shall satisfy relevant Standards and the Contract requirements. Safe access and platform for maintenance of these auxiliary installations shall be provided.

Sufficient lighting shall be provided for safe operation to be carried out any time in a 24-hour period. Light fittings shall be specified for heavy coal dust environment in accordance with relevant standards.

Adequate ventilation of the coal shed shall be provided to ensure safe working environment and to facilitate the drying process of the coal and to remove gases of

the high volatile matter content coal to minimize the risk of spontaneous combustion and to prevent the build-up of toxic gases underneath the shed.

The fire protection and detection system shall satisfy Vietnamese standards and the relevant requirements of the Contract. Fire protection / detection of the coal store shall include the following:

- Fixed temperature alarms
- CCTV's with self-cleaning lenses
- Wet type automatic sprinkler systems and fire hydrants

Fire-fighting equipment shall include water cannons mounted around the coal store, giving total coverage of the coal area.

Fire escape shall be located such that distance from any positions inside the coal store to the fire escapes shall satisfy relevant Contract requirements.

Dust suppression spray system shall be provided to meet ESHIA and local regulatory requirements avoiding dust nuisance in the coal store and to the Plant and surroundings.

These units shall be monitored from CCR and coal handling control room where audible and visible alarms shall be provided.

The reclaim conveyors from both stores shall be equipped with heat resistance belts to enable hot coal to be transported out of the stores. The hot coals shall be removed. The stacker shall be capable of stocking hot coal to the emergency dump area inside the store.

If mobile plants are proposed to be used to remove hot coal from the coal store to the emergency dump area, all necessary mobile plants shall be provided.

Internal drainage shall be by an internal perimeter collection drain. Water collected here shall be drained to the coal run-off settlement pond and from there the water shall be discharged to the waste water treatment plant for treatment prior to reuse or discharge off site.

8.16.5 Not Used

8.16.6 Coal Store Auxiliary Facilities

Coal store auxiliary facilities shall include the bulldozer garage/store, car park, petrol station, car washing station, dust-suppression spray system, and fire fighting system. Coal store administration and auxiliary facilities shall be provided.

(a) Bulldozer Garage/Store

The bulldozer garage/store shall be reinforced steel and concrete in structure. A portion of the mechanical plant workshop shall be open-spaced. Overhead cranes shall cover all the working areas. Trenches shall be provided to facilitate the

under chassis maintenance work on mobile machines. The workshop shall also be equipped with an office, store room, welding bay, changing and washing rooms. There shall be trenches surrounding the whole building to collect the drainage from the shop. All such drainage shall be treated as contaminated water and shall be treated before dumping or reuse.

Station garage shall include diesel storage and filling facilities and covered compound for mobile plant and station vehicle maintenance. It shall have similar requirements as the car washing station regarding the location and layout. In addition, its design shall follow all applicable Vietnam laws, codes, standards and regulations.

(b) Car Washing Station

The construction of car washing station shall match the configuration of different types of vehicles. The location and layout shall allow easy access, entry and exit of large size coal handling mobile machines.

(c) Coal Contaminated Water Sump

A coal contaminated water sump shall be provided. The settled water from the sump shall be treated by the waste water treatment plant and then re-circulated to the spray water tank for dust-suppression use.

(d) Dust-suppression Spray Water System

Spray water tanks and dust suppression system by water sprays shall be incorporated into areas as specified but not limited to the following:

- Jetty
- Transfer towers and transfer locations
- Reclaimer wheels
- Stacker discharge locations
- Stockpile faces

Each transfer point in the conveying system shall also be provided with a hose connection for wash down purposes of transfer chutes and surrounding areas. Spray nozzles and power operated valves shall be installed along the perimeter of the coal stockpile.

(e) Fire Fighting System

Fire fighting facilities shall be installed for the store. The fire fighting system shall meet the NFPA requirements, fire protection and fighting report in Annex C of OTR and local fire codes and standards. Adequate ventilation shall be provided for all enclosed areas such as transfer towers, bunker houses and coal stores.

(g) Screening Equipment

Roller screens shall be provided to protect the coal handling Plant from damage by any extraneous material delivered in the coal including, but not limited to, rock, large metal items and other tramp material and to ensure that the correctly sized coal is fed to the bunkers and the coal mill. $2 \times 100\%$ roller screens shall be provided in the flow from coal stores to bunkers. A cross-over shall be

provided at the roller screen outlets. The screen out materials from the coal roller screens shall go to the ground level for removal by trucks.

(h) Mobile Plant

The following mobile plant shall be provided for coal handling including but not limited to:

- 1 set of bowl scraper (if the coal yard is designed for stacker reclaimer only, coal scraper is not needed.)
- 2 sets of bulldozers of 305 hp (net power) and 328 hp (gross power) (Caterpillar D8R), trimming blade, enclosed cab with air conditioner & dust and sound suppression.
- 4 sets of front-end loaders of 197 hp (net power) and 217 hp (gross power) (Caterpillar 950H) with at least 3m³ bucket and equipped with air-conditioner. Size shall be in line with the lifting capacity of the CSU. The front-end loaders shall be equipped with suitable auxiliaries for vessel hold cleaning applications, such as straight bucket cutting blade and lifting eyes or auxiliary lifting appliances.
- 1 set of high level (so as to reach the area of the highest point at the conveyor system, except bunkering conveyors, for replacing parts) cherry picker or other suitable hydraulic mobile access equipment
- 1 set of water bowser of 6 × 4 chassis with 10 m³ water tank and chemical additives. The water bowser shall be equipped with water jet, spray heads in the front or rear side of the vehicle for dust suppression on coal yard and water pump for spray nozzles and water jet.
- 1 set of supersucker with at least 3 m³ tank. A supersucker shall have sufficient power to assist cleaning work in precipitator / ash silo with a long hose and high vacuum pump for clearing blockage of ash pipes and cleaning silo, etc., with vacuum power of -380 mmHg to -760 mmHg.

All the mobile plant shall be purchased in Vietnam to the extent available by Onshore Contractor. Authorized service providers with dedicated service centres and the capability to provide after-sale service support and spare parts shall be available in Vietnam.

Mobile plant shall be compliant with Vietnam public road license requirements where required.

8.16.7 Coal Plant Control Building

A coal plant control building shall be provided. The coal plant control room shall house the equipment for the operation and administration of the whole coal system.

Computerized control with multi monitor display shall be provided inside the control room. It shall include electrical equipment room, air conditioning plant room, office, conference room and control room. The building shall have water tank and air-conditioning.

8.16.8 Station Vacuum Cleaning System

A station vacuum cleaning system shall be provided to clean the following areas for each Unit.

- Coal feeder floor
- Bunker floor

The station vacuum cleaning system for each Unit shall consist of the following

- Fixed permanent pipework with an air inlet port and air outlet ports.
- The air inlet port shall be equipped with a secure and air tight connection fitting to facilitate connecting and disconnecting the port and the station mobile supersucker.
- Outlet of each air outlet port shall be equipped with quick release fittings to facilitate connecting and disconnecting the port and flexible hoses. The fittings shall automatically air tight shut the ports when the flexible hoses are disconnected from the ports.
- Flexible hoses to be connected to the air outlet ports during cleaning operation to facilitate cleaning at different locations of the covered areas near the engaged air outlet port.
- Cleaning tools to be connected to the other end of the flexible hose during cleaning operation to facilitate sucking up debris into the flexible hoses and the fixed permanent pipework.

The vacuum power of station vacuum cleaning system is provided by the mobile supersucker connected to it. The capacity of the station vacuum cleaning system, pipework arrangement, locations of the air outlet ports, the number and the sizes of the flexible hoses shall be designed for effective cleaning operation at all locations of the abovementioned areas.

Pipework arrangement, pipe sizes, pipe material and fittings shall be designed for blockage free and minimum air inleakage cleaning operation.

The complete station vacuum cleaning system shall also be designed to comply with relevant international standards for safe operation including preventing coal dust explosion.

8.17 Ash Handling Plant

Bottom ash shall be collected and disposed in wet form. Fly ash shall be collected in dry form to promote utilization. Fly ash shall be collected and transported to ash pond or sale by truck. Each Unit shall have independent bottom ash and fly ash collection systems.

The fly ash from ash collection hoppers of air-heaters, economizers and ESPs shall be extracted and transferred pneumatically to the fly ash silos. The fly ash transporting capacity shall be designed assuming 90% of the total ash to be fly ash.

A design capacity margin of 15% shall be provided for the fly ash transporting capacity. Separate fly ash conveying pipes shall be provided for different fly ash sources. At least three fly ash conveying pipes shall be provided for each unit for transporting fly ash to fly ash silos.

The dry fly ash system shall be conceived such that coarse and fine fly ash can be collected, stored and transported to the main silos separately.

Ash handling plant shall be controlled by PLC based control system. Separate PLCs shall be provided for bottom ash system and fly ash system of each unit. The control panels and MCC for both Units shall be located in a common room near the ESP's.

8.17.1 Bottom Ash Handling

The bottom ash handling system shall be provided to remove bottom ash from the bottom of the boiler and to transport the bottom ash to the bottom ash discharge bin by submerged scraper conveyor (SSC). The SSC shall be designed for fresh water quenching. Alternatively, dry bottom ash system with clinker grinder shall be provided.

Two bottom ash silos or one bottom ash silo with two unloading outlets shall be provided for each boiler. The total capacity of the bottom ash silo(s) shall be sized to store bottom ash for 48 hours operation under BMCR output firing Performance Coal with ash content of 14% (ADB). To prevent the ash becomes stagnant, suitable vibrator or air cannon shall be provided for the silo. From the silo, the ash shall be dumped into ash disposal area by using mobile truck.

The quenching water for bottom ash handling shall be fully recycled and reused. The temperature of quenching water overflows from the submerged chain scrapers shall be 60 °C. All the equipment in the ash conveying system shall be provided with at least 10% margin on the actual flow requirement of the system. The design head of the equipment shall be selected based on the 15% margin on the pressure drop.

Pump shall have a continuously rising and stable head capacity characteristic rising towards shut-off. The head capacity characteristics combined with its drives shall ensure equal load division in parallel operation at all times.

Storage silos shall be of cylindrical RCC or steel construction fitted with access doors, and local and remote level indication. The silos shall be supported such that there is head room below for truck access. Each silo shall be complete with covered roof, manhole with ladder for the access inside the silo, pressure relief doors, handrails and access staircase with intermediate platforms.

Inclined drag chain conveyors shall be provided to convey the wet bottom ash to the bottom ash silos. Alternatively, submerged scraper conveyor (SSC) with clinker grinder shall be provided to convey the wet bottom ash to the bottom ash silos.

8.17.2 Fly Ash Handling

8.17.2.1 General

The ESP and fly ash handling system components shall comply with Vietnamese standards and equivalent International standard as applicable.

The fly ash handling system shall be provided to remove the fly ash from ESP hoppers, economizer hoppers and air-heater hoppers. The equipment shall be suitable to handle the dust burden from the specified fuel under all loading conditions. Each hopper shall be furnished with the ash transmitters and use positive pneumatic fly ash handling system to convey the ash into fly ash silo via ash transfer piping system. Three fly ash silos with storage of 48 hours operation under BMCR output firing Performance Coal with ash content of 14% (ADB). Two fly ash silos are provided for coarse ash and one fly ash silo shall be provided for fine ash. Each silo shall be equipped with vent bag filter, fluidizing air admission ports, heating elements, level monitors, etc. The ash silos shall be designed for ash unloading in dry type and wet type to truck.

The piping arrangement of fly ash from each Unit shall be able to be delivered to each of three fly ash silos.

Silo dust removal, either dry or conditioned, shall be automatic following initiation by the truck driver. The system shall be designed to function with minimum operator intervention.

The hopper emptying system shall operate continuously whilst the boiler is on load and for a sufficient time after shutdown to ensure that all hoppers are clear of dust.

The dust handling plant shall be designed to cope with possible moisture present in the dust collection plant during cold starts on the boiler.

The hoppers shall be emptied on a continuous cyclic basis so that dust level building up in the hoppers does not occur.

All major components shall have inspection and cleaning equipment in the event of system blockage. It shall be possible to isolate each section of the system for maintenance whilst retaining the remainder of the system in service.

All ash conveying pipelines shall be of carbon steel.

All bends on the ash pipelines shall be long radius and ceramic lined or other approved material for wear resistant. Wear resistant linings shall be used where it is required. An increase in pipe internal diameter approaching the silo is desirable on long pipe runs in order to reduce the material velocity.

The system components shall be adequately supported and due allowance made for thermal expansion.

Components subject to high temperatures shall be insulated to protect personnel. Other components shall be insulated as required. Due allowance shall be made for the expected ambient conditions local to the Plant.

The dry fly ash evacuation from the ESP hoppers, economizer hoppers and air-heater hoppers to the ash silos shall be continuous (cyclic operation and in stages as per fly ash generation in each zone). An individual ash transporter shall be provided below each of the ash hopper which is connected with ash pipeline for transport of fly ash to silos. RCC construction silo shall be provided with ash unloading provision, in dry or moist form, into truck for the final disposal. Provision shall be made for ash discharge from any unit to any of the RCC silos. Each silo shall be provided with two nos. of gates with rotary feeders/ dust conditioners units and telescopic chutes for these purposes. The minimum ash unloading capacity of the dry fly ash unloaders and wet fly ash unloaders at each fly ash silo shall be 150 T/hr.

All the equipment in the ash conveying system shall be provided with at least 10% margin on the actual flow requirement of the system. The design head of the equipment shall be selected based on the 15% margin on the pressure drop.

Local control panels shall be provided near the storage silos for carrying out the ash unloading operations.

The fly ash handling system shall be principally monitored and operated from DCS level with controls and interlocks through a local control panel. In case PLCs are used for the control of the fly ash handling system, the PLCs shall be connected to the DCS for remote monitoring and operation in the CCR.

Storage silos shall be of cylindrical RCC fitted with access doors, pressure relief vents and local and remote level indication. The silos shall be supported such that there is head room below for truck access. Each silo shall be complete with RCC covered roof, manhole with ladder for the access inside the silo, pressure relief doors, handrails, access staircase with intermediate platforms. Silo vent filters, silo fluidizing equipment complete with air slides, complete piping and valve for the entire air supply system shall be supplied with silos for complete separation of ash from air.

Pneumatic changeover valves shall be provided to direct the fly ash flow to a particular fly ash silo and the changeover valves shall be designed for remote operation from the CCR.

8.17.2.2 Fly Ash Pneumatic Conveying System

The fly ash pneumatic conveying system shall be designed to remove fly ash by a positive pressure pneumatic conveying system (similar to a dense phase system) from the two Units while firing Coal B with ash content of 28% (ADB) under BMCR output condition.

The extent of supply of the fly ash pneumatic conveying system for each Unit includes, but not be limited to, the following:

- (a) 2 x 100% (1 working + 1 stand by) transport air compressors. Each compressor shall be equipped with its own standard accessories including silencer, filter, drive motor, after-cooler, air compressor cooling system, acoustic enclosure, oil and moisture separator. Any other necessary accessories, supporting structures and platforms shall also be provided if necessary for operation and maintenance.
- (b) 2 x 100% (1 working + 1 stand by) refrigerant type air dryers for the two compressors along with other necessary accessories.
- (c) A control system and associated instrumentation equipment to be interfaced and interlocked with the control system of the fly ash handling system of the main plant with PLC, coupled with the local panels for the compressors and air dryers. The control system shall be designed to ensure that no single equipment or component failure will result in loss of transport compressed air for the fly ash handling system.
- (d) One air receiver along with safety relief valve, automatic drain trap and other accessories as required.
- (e) 2 x 100% (1 working + 1 stand by) fluidizing blowers. Each fluidizing blower shall be equipped with dedicated silencer and air heater. Any other necessary accessories, supporting structures and platforms shall also be provided if necessary for operation and maintenance.

The compressor shall be screw type and along with air dryer shall meet the air quality requirements as defined herein to ensure suitability for normal and trouble free fly ash conveying from fly ash hoppers to ash silos:

- Oil carry over in the transport compressed air shall be less than 2ppm
- Outlet air after the refrigerant dryer shall have an exit pressure dew point less than 3°C.

Ambient temperature of 40 °C and RH of 100% shall be considered in the design of compressor capacity. The compressor capacity shall be at least 110% of the maximum flow requirement. The sizing of the compressor shall also consider the fly ash conveying distance, topography of the area, routing of pipelines etc. The Offshore Services Contractor shall submit detailed air compressor design basis for Owner's review during the detailed design stage.

8.17.3 Wet Ash Slurry System to Ash Pond

8.17.3.1 General

In addition to trucking, unutilized ash will be disposed to a remote ash pond. Ash shall be conveyed by using plant waste water and with minimal raw water from local raw water supply. Excess water from the ash pond shall be returned to ash slurry system. The wet ash slurry system shall be designed to remove all the ash from the

two Units while firing Performance Coal with ash content of 14% (ADB) under BMCR output condition.

Ash pond will be built by Owner. The pipelines route distance from the power plant to the ash pond is about 6 km. In addition to the ash truck loading system, the Onshore Contractor shall provide and construct wet ash slurry system to transport ash from power plant to the ash pond. The system shall be independent of the fly ash and bottom ash truck loading systems. 2 x 100% ash slurry pipes and 1x 100% ash water return pipe shall be provided with associated pump sets and auxiliaries.

Air venting system shall be provided for the ash slurry pipes and ash water return pipe.

When the slurry reaches the ash pond end of pipelines, distribution pipelines shall be provided to discharge the slurry into the ash pond.

A pigging system shall be fitted to the ash slurry pipelines and return pipeline to allow the pipework to be cleaned.

8.17.3.2 Overland Pipework and Pigging System

Two overland ash slurry pipes, one duty and one standby laid side by side, shall be provided to transport the ash to ash pond. One return overland pipe shall be provided to pump water from ash pond back to the power station for re-using at ash slurry system to minimize raw water usage.

The pigging equipment shall be manually controlled locally and requires operatives to be present at both the launching and receiving stations during the pigging operation. Pigs shall be transported to the launching stations on a suitable vehicle.

Three pig alerts shall be fitted in the system. These consist of flags raised when the pig passes that position. They are situated in the following locations:

- (a) at the launching station to check that the pig has started its journey;
- (b) just prior to the receiving station to indicate imminent arrival of the pig; and
- (c) at the receiving station to show that pig has arrived.

An electronic pig detection system shall be provided to locate the pig inside the line. To facilitate pig insertion and removal, the stations comprise an oversize barrel connected to a concentric reducer. Its neck is of such a size as to ensure that the pig makes an effective seal before entering the line.

A pig launching and receiving station shall be fitted at the beginning and at the end of each overland line. The launching station shall consist of a single pig launcher fitted with an air vent, by-pass water line, pressure gauge and water drain valve. At the rear is a door for the insertion of the pig. A drain sump shall be fitted below the door. The receiving station shall be similar to the launching station but arranged to allow for the reception and removal of two pigs. The doors of all stations shall be fitted with quick seal closures.

Chambers shall be provided at intervals along the pipelines to give access to air and drain valves. Internal access to the pipelines shall be obtained at these chambers by removing a short section of pipe. Additional chambers shall be provided solely for access. Air valves shall be provided with isolating valves fitted at the pipeline high points. Any slurry and water drained from the overland pipework shall be able to be pumped into suitable vehicles for subsequent disposal. Low lying chambers shall be pumped dry using portable pumping equipment before access of personnel is permitted. Drain cocks shall be fitted at low points in the pipework.

The pipework shall be designed with additional thickness of pipe wall and coating to allow for wear due to the passage of ash slurry for 25 years. Design shall be based on the assumption that two slurry lines will be used equally and alternately.

Ash disposal pipelines shall be routed on concrete pedestals. Facilities for easy rotation and replacement of the disposal pipelines shall be ensured to overcome the problem of wear and erosion.

8.17.3.3 Bottom Ash Disposal

The bottom ash generated shall be disposed of by slurry system through piping to the ash pond. The bottom ash from bottom ash silos shall be conveyed to ash slurry system. Bottom ash shall be crushed and collected in a common slurry sump. Then, it shall be disposed of through ash disposal pump sets of adequate capacity to disposal area.

8.17.3.4 Fly Ash Disposal

The ash from fly ash silos shall be disposed of by slurry disposal system to the ash pond. The fly ash for disposal shall be fed by rotary feeders into the slurry mixing tank for conditioning. The conditioned dry fly ash shall be mixed with water in the mixing tank and the entire ash shall be blended to achieve homogeneous mixture by the mixer.

8.17.3.5 Ash Slurry Pumping System

Water flow control valve shall be modulated to maintain constant level in the slurry mixing tank. Rotary feeder speeds shall be variable to maintain density set point. The tank shall be designed to operate at constant level and shall be located at an elevation sufficient to provide NPSH required for pump. The ash slurry so formed in the mixing tank shall be gravity feed to the slurry pump. The pump shall be provided with variable speed drive so that pumping rate can be automatically and infinitely varied from 10% to 100% of full flow. The ash slurry pumps shall be designed for continuous, heavy duty and shall be suitable for disposing of ash to dump area.

The ash mixing and pumping process shall be capable to operate in cyclic mode and automatically dispose of ash to accommodate changes in ash production caused by fluctuating boiler loads or other variables.

On the other hand, the fiber-optic cables for control and status indications of return water pumps, which is connected between the central control room of the Facility and ash pond, shall be supplied inside the area of Ash Pond, and shall be supplied and installed outside the area of ash pond. Status indication and control of equipment in ash pond shall be provided in the central control room of the Facility. The fiber-optic cable shall be buried underground or installed above-ground along the ash slurry pipe and wire/strip armoured. If the fiber-optic cable is installed above ground, suitable protection shall be reviewed by the Owner during Project execution stage. The Owner will install the control cables inside the ash pond area.

8.17.3.6 Ash Disposal Water System

The water required for ash disposal shall be tapped from plant waste water system, raw water supply and recycled water from ash pond routed to the ash water sump. Raw water of 1500m³/day will be provided at Site boundary by the Owner.

The pumps required for ash disposal viz. water pumps, seal water pump etc. shall be provided. At least 50% capacity of pumps shall be standby.

8.17.3.7 Electrical Connection

The Owner will provide an electrical connection of 400 VAC at 165kW for the power supplies of two return water pumps at ash pond. The Onshore Contractor shall pick up from the defined terminal point and provide the necessary switchgears etc. to receive and distribute the power for return water pumps.

8.17.3.8 Design and Supply of Systems and Equipment Inside Ash Pond Area

The Onshore Contractor shall design and supply the following various systems and equipment inside the ash pond area:

- Return water pipe;
- Ash slurry distribution ring and associated pipes;
- Pigging system, including the pigging receiver and pigging station within the ash pond area (if applicable);
- Power and control cables for the return water pumps;
- Return water pumps;
- Local control panels;
- Anchor plates and
- Support.

Above systems and equipment shall be provided with design document to the Owner no later than NTP + 12 months.

The Onshore Contractor shall provide the design information of civil and structural works to the Owner no later than NTP+8 months.

The Owner will be responsible for the construction of all works inside ash pond area with technical support from Onshore Contractor.

Upon completion of installation above systems, Onshore Contractor shall conduct the inspection together with the Owner.

8.17.4 Provision of Space for Future Dry Ash Processing Plant

The Offshore Service Contractor shall make provision of space for a dry ash processing plant which comprises of; ash classifier, 3x ash silos of 3,000m³ each, ash conveying system, ash unloading system at jetty, electrical feed and control room in the current layout and 1 ha brick making facilities in the current laydown area. The Offshore Service Contractor shall submit plot plan drawing which identifies the provision of space for these ash handling items during design submission stage.

8.18 Fuel Oil Storage and Handling System

8.18.1 System Requirements

Diesel Oil in accordance with TCVN 5689 shall be designed to be used for start-up and low load operations of the boilers. Diesel oil shall be received by the road tankers.

2 × 2000 m³ diesel oil tanks shall be provided. Diesel oil transferring system shall include 2 × 100% transferring pumps, transferring header, hose pipes, valves and fittings, etc. Two numbers of hoses with quick coupling valves connecting to unloading header shall be provided. Fuel oil metering system shall be provided for fuel oil transferring system to the diesel oil delivered from road tankers. The diesel oil tanks shall be cross-connected.

Fuel oil transferring pumps and forward pumps shall be located in a common pump house near the transferring area. A local control panel shall be provided for start / stop of the pumps, indication and alarm of the status of the pumps, tank levels, fuel oil metering system etc. in the fuel oil pump house. Fuel monitoring system shall be provided for pressure monitoring and cutting in of standby pump when oil pressure drops.

Bund shall be provided for IDO Tanks. In case of leakage of tanks as per NFPA requirement the bund shall be able to contain the amount of oil leaked from the larger tank which is 2000 m³. An air/foam system shall be provided for fire fighting system on IDO tanks as per NFPA 11 standard.

8.18.2 Design Criteria

Diesel oil system shall allow simultaneous oil firing of both Units at 30% BMCR capacity at the same time.

Two fuel oil strainers shall be provided. Each strainer shall be sized to supply diesel oil to meet the 30% BMCR capacity of two Units without coal firing on continuous basis plus the minimum recirculation flow.

Two fuel oil forwarding pumps shall be provided. Each pump shall be sized to supply diesel oil to meet the 30% BMCR capacity of one Unit without coal firing on continuous basis plus the minimum recirculation flow.

In no case the capacity of the forward system is less than the flow requirement of diesel oil by two Units with 10% flow margin.

All motors in fuel oil system shall be totally enclosed/totally enclosed fan cooled with IP-55 degree of protection and shall be of flame-proof type. All valve actuators shall conform to IP-67 protection.

8.18.3 Design and Construction

The storage tanks and diesel generator day tank shall be provided with all necessary nozzles, appurtenances, drain and vent connections, instrument tapings, access stairs, platforms, man holes, etc. Necessary fire detection and protection system shall be installed for the storage tanks and day tanks. Diesel oil tanks shall be provided with flame arrestor with the vents.

All fuel oil pipe line shall be seamless conform to ASTM 106. All the piping system shall be capable of withstanding the maximum pressure in the lines at the relevant temperature. All the pipe work shall be routed over ground. Where underground routing is unavoidable, the pipe line shall be in built in trench or on trestles. Fuel oil piping shall be provided with vents at high points and drains at low points. Suitable slope shall be provided in oil pipelines towards the drain point. Suitable support, anchor, hanger, saddles, clamps etc. shall be provided for the piping system.

8.19 Workshop and Stores

Equipment and tools shall be provided for the workshop and store building, as appropriate, to meet the Facility maintenance requirements. As a minimum, the extent of supply of the equipment for the workshop and store shall include the following.

- All equipment and tools required for the proper maintenance of the mechanical, electrical, instrumentation and milling equipment including electrical and pneumatic driven machine tools, hand tools, benches, welding sets, heating cupboards, precision and calibration equipment etc. complete with guards and

- lubricating equipment.
- All bins, shelves, containers and steelwork structures for the stocking of equipment, tools, spare parts, pipes etc.
- All portable partitions and protective devices required in the workshops and stores.
- Mobile lifting vehicles for transportation of medium weight objects;
- All portable blowers, vacuum cleaners, and dust extractors etc.

The following vehicles for common use and stores equipment shall be provided:

- Water spray truck
- Pool car - 2 off around 1500cc
- Five ton truck with hiab
- Fork lift trucks - 5 ton and 10 ton
- Pickup truck
- Hand pallet truck-3 ton
- Elevated platform (scissors type)
- Electric operated mini bus 20-seater.
- Flatbed trailer (30 ton).

All the vehicles shall be purchased in Vietnam to extent available. Authorized service providers with dedicated service centres and the capability to provide after-sale service support and spare parts shall be available in Vietnam.

Vehicles shall be compliant with Vietnam public road license requirements where required.

Electric mini bus shall be supplied with battery charging station in the Facility.

The store building shall be large enough to house the extra Strategic Spares whether or not the Owner decides to do so. Store shall be fitted with heavy duty and light duty racks to accommodate all Spares.

The stores shall be equipped with suitable plant handling system (updated requirement: electric overhead crane of 15 ton capacity on rails covering full width of the building for the loading /unloading area) and mobile plant.

A comprehensive review of station maintenance and storage requirements shall be conducted to recommend further equipment considered necessary in addition to the above minimum provisions.

The Onshore Contractor shall provide a combination of secure open storage areas and enclosed buildings as appropriate to maintain the spares in usable condition.

The workshop shall be divided into areas for mechanical workshop, electrical workshop, and instrumentation workshop to facilitate maintenance of different equipment. The store shall be divided into different functional areas such as light store, heavy store, tool store, paint store etc. to enable safe and proper storage of spares/tools. Open store area of about 300 m² with concrete load bearing floor shall be provided. The Offshore Services Contractor shall submit general arrangement

drawing of workshop and store with equipment layout as part of the design submission for Owner's review.

Essential mechanical, electrical, instrumentation workshop equipment, chemical laboratory equipment, mobile plant workshop equipment, stores tools and equipment shall be provided.

Mechanical workshop shall be provided with centre lathe, milling machine, shaper, bench type drilling machine, grinder, pipe bending machine, cutting saw, two working benches (2 m × 1 m each) fitted with four vices, TIG welding machine, electrical welding machine and gas cutting set.

C&I workshop shall have one small lathe.

Coal plant tools and equipment shall also be included.

8.20 Laboratory Equipment

Necessary laboratory equipment shall be provided in the following areas.

- Chemical, water and thermal laboratory
- Coal sampling and analysis
- Metallurgical laboratory
- Electrical laboratory
- C&I laboratory
- General laboratory
- Other equipment accommodating the Contractor's requirements

A comprehensive review of the station requirements shall be carried out, and the list of equipment shall be provided for the Owner's review.

8.21 Cranes and Lifting Equipment

8.21.1 System Requirements

All overhead traveling cranes, hoists, runway beams, etc. necessary for the safe and efficient operation and maintenance of the Facility shall be provided.

All parts of the Facility shall be provided with a convenient arrangement for slinging or handling during erection and overhaul. Any item of Plant normally stripped or lifted during periods of maintenance that weighs 55 kg or more shall be permanently marked with its weight.

Where permanent cranes, hoists, etc. are used for the construction of the Facility, these shall be refurbished to an "as new" condition before handing over to the Owner.

Two overhead traveling cranes shall be provided for the two turbine generators in the turbine building. The crane shall be a double girder electric overhead traveling crane with hoist, cross travel and long travel motion. It shall be equipped with a main hoist and an auxiliary hoist. Control of the crane shall be either through a push-button pendant or from an open shell control cabinet. A lift beam is used to lift the turbine HP and IP cylinder modules.

Adequate number of cranes and hoists shall be provided with the proper capacity and lift to facilitate assembly, repair and maintenance of

- CW pump station (pump hoist);
- CW pump station (bar screens and stop gates);
- CW wash water trash basket;
- Workshop (suitable to lift and turn the heaviest motor, at least of 15 Ton overhead on rail lifting crane shall be provided.);
- Store;
The hoist in the stores building for ground floor shall be of 5 ton. Electric overhead crane of 15 ton capacity on rails covering full width of the building for loading/unloading area.
- Feed water heater;
- Boiler feed pumps;
- Air pre-heater;
- Coal mill;
- Electric-static precipitator;
- Auxiliary buildings hoists;
- FD, ID, PA fans and condensate extraction pump.

The following criteria shall be adopted for selection of type and capacity of cranes and hoist:

- a) Suitable lifting arrangements shall be provided for the equipment required for handling during maintenance of heaviest part / component of weight equal to or more than 500 kg covered.
- b) For the equipment part / component to be handled having weight more than 500 kg but less than 2 tons, electrically operated hoists of adequate capacity shall be provided.
- c) Crane of suitable capacity shall be provided for lifting the weight 3.5 ton or more than 3.5 ton of weight unless there is a space constraint in which case an alternative solution can be proposed subject to agreement by the Owner.
- d) Hoists of capacity up to 1.5 tones and lift of 3 metres shall be manual hoists. Hoists of capacity above 1.5 tones or lift of more than 3 metres shall be electric hoists.
- e) The location and number of hoists shall be finalised during detailed engineering.
- f) Capacity of each hoist shall be 1.2 times of the anticipated maximum equipment load in the area that the hoist serves.

Cranes and lifting equipment shall be designed based on International standards and compliant with Vietnamese regulations and code of practices.

Further lifting and crane requirements shall be finalized during detailed engineering stage.

8.21.2 Design Requirements

Operation of hoisting, cross travel and longitudinal travel of specified speeds shall be electrically controlled. The creep speed shall be 10% of main speed. The main speeds for various movements shall be as follows as minimum:

- Hoisting : 3 m/min.
- Long Travel : 6 m/min.
- Cross Travel : 6 m/min.

In the design of components on the basis of strength, factor of safety shall not be less than five (5) based on ultimate strength. For compression members, the slenderness ratio shall not exceed 120. In case of other load carrying members and subsidiary members the slenderness ratio shall not exceed 180. For girders, the following values of maximum span to depth ratio shall be governing

- Plate girders : Span/depth = 18
- Lattice girders : Span/depth = 12

The crane shall be capable of raising, lowering, and holding in position and transporting loads equal to the rated load over the entire load and speed range without damage or excessive deflection to any crane parts.

The crane shall be controlled and operated from ground level or from elevated walkway by means of pendant control box or control cabin. Any crane walkway shall be accessible by ladders, which shall be placed at each end.

The crane girder shall be adequately designed for hoist load plus all applicable factors as per Vietnam regulation. The maximum vertical deflection of the girder at any location and under worst operating conditions shall not exceed 1/900 th of the girder span.

An electro-mechanical brake of fail safe type, shall be provided for hoisting motion. Hoisting shall be designed to hold 1.5 times of the full load torque while cross travel and long travel brakes shall be designed to hold 1.25 times the full load torque.

Motor ratings shall be calculated keeping margin of at least 25% over the maximum power requirement. Further, the hoist motors shall be rated to lift 120% of the design load on the hook at the rated speed.

Conventional rotor resistance control shall be used in both hoisting and lowering directions, with a minimum of five (5) speed steps in each direction.

Hoist control shall be designed to achieve "Inching Speed" of 10% the rated speed in both hoisting and lowering directions with loads (no-load to full-load) on hook.

The supply of electrical crane shall include but not limited to following:

- Crane structure complete with bridge girder, end carriage, trolley, crane runway and trolley rails with fixing clamp, wheels, end stops for bridge and trolley, walkway platforms and access ladders etc.
- Main hook complete with load blocks, sheaves and fittings.
- Drum and wire ropes.
- Drive unit for bridge, trolley and hoist complete with drive shafts, gear reducer, brakes, shaft bearings, limit switches and necessary controls.
- All electrical system such as illumination, controllers for hoist, bridge, trolley travel, resistors, contactor & crane interlock, grounding system etc.

The crane shall have single girder or double girder as required. The bridge girder shall be box section type or braced I beam type as per standard design of the manufacturer. The exterior surface shall be smooth and as free from projections etc. as possible to minimise dust collection on it.

The highest position reached by the lifting hooks shall be such that during operation, the minimum vertical critical clearance between bottom of the equipment being handled and the top of any permanent structure or equipment in the operating area shall be at least one metre.

Hooks shall be solid, forged, heat treated carbon steel in rugged construction. Swiveling hooks shall be mounted on thrust bearings and protective skirt shall be provided to enclose the bearing. The hook shall swivel freely through 360 degree Celsius. rotation with full load. The hoist hook shall be provided with safety locking arrangement to provide accidental unhooking.

Provision shall be made for lubrication of all moving parts and bearings while the hoist is in service except where life lubricated bearings are used. All exposed bearings shall be suitably sealed or shielded.

Bridge girder for the crane shall be of plate welded box type construction fabricated from tested steel plates. Box section shall be adequately reinforced by internal diaphragms and ribs. Box girder shall be provided with end plates for sealing. Diaphragms shall be provided to transmit the trolley wheel loads to the web plates.

Trolley / crab runway rails shall be standard rail section or equivalent. The rails shall be accurately aligned, leveled and properly fixed to bridge girder. End stops and curved wheel stoppers on crab/ trolley runway rails on both end of each bridge girder shall be provided in approved construction. The rail section shall be sufficient to cover the whole runway length and shall be adequate for maximum permissible trolley travel.

The trolley frame shall be built up from heavy steel sections welded together to form a rigid piece frame adequately braced to resist vertical, lateral and torsional strain and support all loads without undue deflection. Trolley frame shall be provided with equaliser sheaves and spring bumper at each side for safe operation of the crane assembly.

Manually operated hoists shall be of spur gear chain pulley block type. It shall be suspended from the trolley by a hook. The load hook shall be swiveling type fitted with a locking device. All manually operated hoists, unless stated otherwise, shall be trolley suspended type. The trolley of hoists shall be manually operated.

Electric hoist shall be electric wire rope trolley suspended type. Minimum speed for hoisting shall be 3 m/min. and that of for trolley motion shall be 15 m/min.

Further lifting and crane requirements shall be finalized during detailed engineering stage.

8.22 Miscellaneous Pumps, Portable Compressors, Tools etc.

All the necessary miscellaneous pumps, portable compressors and tools for the effective operation of the Facility shall be provided.

Further requirements shall be finalized during detailed engineering stage.

Chapter 9 – Sea Water Flue Gas Desulphurisation

9.1 General

The duty of the seawater flue gas desulphurisation (SWFGD) system shall remove the sulphur dioxide (SO₂) from the flue gas produced by the Facility, with a sulphur dioxide emission concentration of less than 200 mg/Nm³ for the Performance Coal with sulphur content of 0.85% (ADB). The SWFGD system design shall be based on such that the design sulphur dioxide emission concentration is less than 200 mg/Nm³ and the design SO₂ removal efficiency is at least 92% at 100% TMCR load for the Performance Coal with maximum sulphur content of 1.1% (ADB).

Performance Coal with sulphur = 0.85% (ADB) shall be used for the testing of SWFGD. Furthermore, the SWFGD shall be designed in such a way that the sulphur dioxide emission is less than 200mg/Nm³ when firing Performance Coal with sulphur content of 1.1 % (ADB). The SWFGD system shall form an integral part of the overall station systems, controlled via the main distributed control system (DCS).

9.2 Scope

The scope of the SWFGD system starts from the induced air (ID) fan outlet to stack inlet, and the SWFGD seawater treatment plant which receive seawater from condenser and discharge treated seawater to the cooling water (CW) discharge system.

Each Unit shall consist, but not be limited to, the following:

- 100% absorber;
- 100% flue gas bypass duct with bypass damper;
- SWFGD inlet and outlet dampers;
- Gas-gas heater;
- 3 × 50% absorber spray pumps with intake pit;
- 3 × 50% or 6 × 25% aeration air fans
- 1 × 100% aeration air filter for each aeration fan;
- Absorber spray pump pits with strainers for absorber spray pumps' suction;
- One seawater treatment system;
- 1 × 100% portable aeration basin dewatering sump pump (common for Units 1 & 2);
- 100% aeration basin with distribution and diffuser pipes;
- Stop logs for insertion at the aeration basin inlet, if required;
- Interconnecting ducts and dampers.

9.3 System Requirements

9.3.1 General

The Offshore Supply Contractor shall supply a SWFGD system based on a once-through, wet, seawater scrubbing process. The capability of the SWFGD and its auxiliaries shall be designed according to the flue gas flow at 100% boiler maximum continuous rating (BMCR) plus 10% flow margin. The scope of supply shall include all necessary treatment stages, ductwork, fans, dampers, pumps, pipework, and treatment facilities for all solids, liquids and gaseous wastes generated in the SWFGD system. Within the scope of supply, it shall also include all the provisions required to render the seawater used for scrubbing fit for discharge to the sea.

The scope of supply shall include a 100% SWFGD bypass for flue gas with associated isolation and bypass dampers. The dampers shall allow 100% flue gas bypass as well as suitable for isolation purposes.

9.3.2 Operating Principle

The SWFGD process shall utilize a portion of the circulating seawater downstream of the condenser as the absorbent to absorb the SO₂ in the flue gas. The Offshore Supply Contractor shall check if the alkalinity of seawater is sufficient to neutralize the SO₂ in the flue gas.

The design of absorption system shall allow sufficient seawater flow in counter current arrangement with the flue gas for effective scrubbing of SO₂ and the formation of bisulphite, which may also convert to sulphite. Then, the absorber effluent shall mix with the condenser outlet circulating seawater in a dedicated mixing zone, and then flows to the seawater treatment system where it is air-sparged to complete the oxidation of absorbed SO₂ (as bisulphite and sulphite) into sulphate. The alkalinity in seawater, present as carbonates and bicarbonates, neutralizes the hydrogen ions to increase the seawater pH.

The design of the SWFGD shall ensure that there is no SO₂ release from the seawater returned from the SO₂ absorber around the seawater treatment system. The pH of the seawater discharged from the seawater treatment plant shall be within the range of 6 to 9.

9.3.3 Absorption system

The SWFGD system shall be installed downstream of the electrostatic precipitators (ESPs). 100% bypass shall be provided with pneumatically actuated isolation dampers to enable bypassing 100% flue gas flow, as would be required when the SWFGD system is shutdown. The SWFGD system inlet and outlet dampers shall be double-blade tandem louver shutoff type, while the bypass damper shall prevent any

backflow of flue gas from the SWFGD system outlet and consequent unstable operation.

The booster fans, when required, shall be single-stage axial fans with hydraulic pitch control of the fan blades. The absorber shall be an open spray tower of vertical cylinder type vessel made of steel plate with steel structure, and it shall be designed to facilitate close contact between the absorbent (seawater) and the flue gas in the counter-current direction. Seawater flows shall be vertically downward, while flue gas shall rise upwards through the absorbing section of the vessel. Along the height of the absorber, the vessel has subsections performing certain specific functions. The lowest section of the absorber shall collect the seawater rich in sulphates and sulphites formed during the absorption process. The flue gas duct connected to the absorber shall be above the seawater level in the absorber. The duct arrangement shall be inclined downwards as it enters the absorber, which reduces absorbent spray-back into the inlet duct. Seawater flowing down from the absorption section, and being distributed over the entire vessel cross-section, quenches the hot flue gas in order to protect the absorber internals from exposure to high temperatures. The flue gas, thus cooled, enters the absorption section of the absorber.

The absorption section of the absorber shall be open spray tower type with seawater distribution system of headers and nozzles installed. The fluid flow across shall be countercurrent, which provides sufficient interfacial area for enhanced mass transfer.

The absorber spray pumps of 3 x 50% shall be designed with 3 layers of spray bank for absorber. Each pump has a duty of spray at each spray layer. 2 x 50% absorber spray pumps for 2 x 50% spray layers shall be able to supply seawater for the treatment of flue gas from the boiler at 110% BMCR. 1 x 50% absorber spray pump and 1 x 50% spray layer shall be standby.

Treated flue gas shall leave the absorber and enter the final section in the absorber comprising of a mist eliminator. This device shall trap entrained seawater in the flue gas leaving the absorbing section. The limit drop size of the mist eliminator shall be 30 microns or smaller. Flue gas treated leaves the absorber through the isolation damper in the SWFGD system discharge line.

Fire fighting connection (blank off without nozzles) shall be provided for future retrofitting of emergency cooling system by Owner.

The flue gas duct leading to the absorber and the bypass duct shall be insulated for personnel protection. Expansion joints shall be provided at suitable locations to accommodate thermal expansion.

9.3.4 Seawater Supply System

One seawater treatment Plant (SWTP) shall be provided for each boiler. The discharge canal upstream of the SWTP shall be separated and the canal downstream of the SWTP shall be combined. 3 x 50% absorber spray pumps for each boiler shall supply seawater from the downstream of condenser to the spray layers of the absorber for use as absorbent and as a quenching medium. Independent intake pits shall be

inclined to the normal direction of flow in the seawater canal in order to ensure smooth flow of seawater with minimal pressure drop to the pump suction.

Each intake pit shall be a concrete structure equipped with a strainer. The strainer protects the pump from any entering debris that could possibly fall in the open seawater canal. The absorber spray pumps shall be of vertical wet pit pump design which can be extracted out for maintenance. Permanent lifting equipment shall be provided for maintenance of the absorber spray pumps.

The absorber spray pump shall be equipped with motorized or hydraulic operated discharge valve to enable individual pump isolation. Fiberglass reinforced plastic (FRP) material pipes shall be provided to supply the seawater from the absorber spray pumps to the absorber.

The arrangement and layout of seawater distribution system in the absorber shall optimize seawater consumption and ensure maximum coverage. Other proven design to ensure optimized seawater utilization for flue gas quenching and SO₂ removal can also be accepted subject to Owner's review.

Aeration fans shall be installed.

9.3.5 Absorber Aeration and Sea Water Treatment System

Seawater collected in the absorber shall firstly be drained into a relay pit located adjacent to the absorber vessel. The relay pit shall have a weir to prevent the absorber from completely draining out in order to keep the minimum water level in the absorber, which is essential to maintain the water-seal. The relay pit shall be covered with the concrete slab and installed with openings to avoid any pressure fluctuations. Beside adopting a relay pit design, other proven design for maintaining the required water level in the absorber can also be considered subject to Owner's review during detailed design stage.

The absorber effluent seawater flows under gravity from the absorber or the relay pit to the mixing zone of the aeration basin where it mixes with the seawater released by the condenser. FRP lining or concrete with acid resistance with relevant project reference shall be provided to protect the mixing zone from exposure to the low pH absorber effluent. The mixing pit shall be sized adequately for complete mixing before aeration to eliminate any fugitive sulphur dioxide in the seawater, thus preventing any foul odour in the vicinity.

Necessary provisions (Fe²⁺ rectifier) shall be made to maintain active iron (Fe²⁺) level in the aeration basin in order to optimize the aeration process.

The seawater treatment system shall comprise of an aeration basin arranged along the length of the discharge canal. Sufficient aeration nozzles or dispersion pipes shall be distributed along the aeration basin to generate bubbles to the effluent, reducing its chemical oxygen demand. The width and the cross-section area of the aeration basin shall be increased to reduce the seawater velocity for effective treatment and also increase the contact time between air and the seawater.

During a major overhaul of the seawater treatment system, it would be essential to empty the discharge canal for the Unit taken under shutdown. Accordingly, stop log guides shall be provided on the intake of the discharge canal if they are required for the effective isolation of the discharge canal. A portable outfall dewatering sump pump shall also be provided to facilitate the emptying of the seawater treatment plant.

9.3.6 Gas-Gas Heaters (GGH)

A 100% GGH shall be provided for each Unit. The hot side of each GGH shall see flue gas from the outlet duct of a booster fan or the ID fan. The cold side of each GGH shall see saturated flue gas from the outlet duct from the SO₂ absorber. Leakage of flue gas from untreated side to treated side of GGH shall be minimized. The GGH shall be designed to limit the leakage rate at full load to be not larger than 1%. Suitable purge and scavenger seal system shall be provided if necessary to meet the allowed leakage limit.

The GGH shall be the rotary regenerative type with bisector arrangement and vertical axis of rotation.

A reheated flue gas outlet temperature of 80 degree Celsius minimum shall be designed from the GGH at full load for dispersion with minimum formation of white plume. The GGH shall be fully equipped with sootblowers and water hoses and drainage for the cleaning. Low pressure water wash system shall be provided so as to permit off load water washing for removal of bonded deposits and adequate arrangement for disposal of effluents from water washing to nearest drain shall be provided. High pressure water wash system shall be provided for on load water washing. A bottom hopper along with a drain line shall be provided for this purpose. Alternatively, the washing water shall be drained to absorber by gravity.

Galleries and platform surrounding the GGH shall be of sufficient height and load carrying capacity to store the elements from at least one of the largest sections.

A dedicated fire hydrant in the absorber area shall be provided for fire-fighting purpose on the GGH.

The GGH heating element shall be enamelled with coating resistant to corrosion.

9.3.7 Mist Eliminators

Absorber mist eliminators shall be provided to remove entrained water and contained solids. The limit drop size of the mist eliminator shall be 30 microns or smaller. These mist eliminators shall be oriented such that the flue gas passes through the mist eliminators in the vertical direction.

Access to each layer of mist eliminator shall be provided for maintenance and operation activities.

9.4 Operational Requirements

The SWFGD system shall be designed to accommodate flexible operation of the Plant from BMCR down to the maximum turndown capability of the SWFGD system. The SWFGD system shall have a rundown capability compatible with existing technology.

The SWFGD system shall operate in a safe and efficient manner, taking account of the complete boiler operating regime, the full range of fuel specification and ambient conditions including low load operation, boiler and SWFGD system start-up, shut down and transient operation.

The SWFGD system shall operate automatically through DCS in central control room (CCR) with limited operator intervention. The proposed system shall include manual override / intervention facilities. A local control room shall be provided, if manual operation is required during abnormal conditions.

Details of the procedures to be used during the start-up of the SWFGD system and the main boilers shall be provided. The Offshore Services Contractor shall specify the minimum boiler load at which the flue gas can be introduced into the SWFGD system. In addition, the Offshore Services Contractor shall also state the sequences and timing for the opening of all dampers and the start-up of individual main components.

When the flue gas bypasses the SWFGD system, or individual flows are taken out of service, for routine or breakdown maintenance, with the unit on load, the SWFGD system as a whole or (as appropriate) the inoperative streams shall be capable of being safely isolated. Sufficient mechanical and electrical isolation shall be provided to accommodate this.

9.5 Plant Performance Requirements

The SWFGD system shall be capable of removing SO₂ from the boiler flue gases, to the performance guarantee level specified and to the required standards of the Vietnamese authorities, when the SWFGD system is operating at BMCR load. It shall also be capable of achieving this over the normal ranges of the Plant operation.

The Onshore Contractor shall perform a series of guarantee tests to demonstrate the compliance of emissions as described in the Schedule of Performance Guarantees to demonstrate the performance of the SWFGD system.

9.6 Plant Design and Fabrication Requirements

At the project implementation stage, the Offshore Supply Contractor shall supply complete set of design and fabrication information, except the proprietary information of the Offshore Supply Contractor, for the Owner's review prior to commencement of the SWFGD system manufacture and construction.

Ductwork shall be sized for a flue gas velocity of not higher than 18m/s. Bypass duct shall be free of dead legs and has adequate drains to avoid condensation. Separate one booster fan, in addition to ID fans, shall be provided if necessary.

The SWFGD system shall be designed to minimize the consumption of power and chemicals, subject to its meeting all other performance criteria

Appropriate materials and lining for various parts of SWFGD system shall be reviewed and agreed by the Owner. Brick and mortar, if used, shall be acid resistant.

9.7 Materials

Materials of construction are a significant design consideration for the SWFGD system. Seawater released into the absorber quickly turns acidic, and becomes very corrosive in nature. In addition, flue gas at temperatures below the acid dew point is also very corrosive. Accordingly, the Offshore Supply Contractor shall incorporate the following corrosion prevention measures.

9.7.1 Gas Inlet Duct

The inlet duct from the GGH to the absorber shall be lined with corrosion resistant lining. The portion of the flue gas inlet duct connected to the absorber, with an increased cross-sectional area at the absorber inlet, requires protection from corrosion caused by exposure to seawater spray back along with the dry hot flue gas. The corrosive condition at this location is therefore the most formidable with acidic environment at high temperature. Wet-dry zone at absorber inlet shall be protected with proven corrosion resistant method which shall be reviewed during design stage. Relevant project reference shall be provided to demonstrate its corrosion resistant performance in the same applications. Before wet-dry zone, the inlet duct shall be lined with organic flake lining.

9.7.2 Absorber Vessel

The absorption process in the absorber exposes its internals to very severe conditions. The absorber inlet and quenching zones are subjected to a wide change in operating temperatures. The absorption process decreases the pH of the sorbent seawater. The seawater spray nozzles are located in the quenching zone and above the absorber

plates. For packed tower design, the seawater distribution system is located above absorber packing. The design shall position the spray nozzles or the seawater distribution system to minimize spray impingement force, without sacrificing spray coverage. High temperature and acid resistant glass-flake lining shall be provided for withstanding the incoming hot flue gas, very acidic environment and spray impingement. Accordingly, the entire shell of absorber vessel shall be internally lined using glass flake lining.

In addition to the glass flake lining, the absorber bottom shall also be rubber lined of at least 3 mm thick to protect the glass flake lining from mechanical damage caused by outage inspection and maintenance activities. Glass flake lining for corrosion and mechanical protection of the absorber bottom without rubber lining can also be considered when relevant project reference is provided to demonstrate its performance in the same applications and subject to Owner's review during detailed design stage.

Corrosive attack by sulphurous acid caused by the presence of moisture and residual sulphur dioxide is the main corrosion problem. The bypass duct (downstream of the bypass damper), and the flue gas duct up to the chimney shall be lined with glass flake lining for effective protection against sulphurous acid attack. The absorber outlet duct up to the GGH shall be acid resistant glass reinforced plastic (GRP) duct or glass flaked lined steel duct.

The mist eliminator shall be fabricated from polypropylene, which has effective acid resistance.

Other absorber internals, such as the seawater distribution pipes, plates or packing, aeration pipes, supports, etc. shall be manufactured in FRP, polypropylene (PP) or polyethylene (PE).

9.7.3 Absorber Spray Pumps, Pipes and Pits

The main absorber spray pump discharge piping shall be manufactured in FRP. The smaller size lines which are in carbon steel shall be with PE lining. If the piping downstream of butterfly valves is made of steel, it shall be rubber-lined to a length of 2.5 pipe diameters. The seawater return pipe shall be lined with acid resistant glass flake lining or made of acid resistant FRP.

When provided, the relay pit (Being in concrete) shall be lined with 1.5 mm thick acid resistant FRP lining.

The mixing zone, being in concrete shall be constructed with concrete with acid resistant capability with relevant project reference.

9.7.4 (Not Used)

9.7.5 GGH

The heating surface of the GGH shall be liberally sized so as to achieve maximum possible temperature of treated gas at the exit of SWFGD plant. Suitable measures shall be taken to prevent corrosion of heating elements from acid attack. Full details of the measures proposed for mitigating corrosion of heating elements along with anticipated life of heating elements shall be incorporated, which shall in design case not be less than 60,000 operating hrs during detailed engineering stage. The arrangement provided shall be a proven arrangement satisfactorily working for units of similar capacities and operating conditions.

9.8 SWFGD Process

At the Project implementation stage, the basis of the SWFGD process shall be stated including the chemistry, and flow sheets provided and mass balances completed for all parts of the Plant. This shall include as a minimum:

- Flue gas flow, composition and conditions at all stages through the Plant
- Water flows, compositions and conditioning at all stages through the Plant
- Oxidising air flow rate

The Offshore Supply Contractor shall provide a clear and detailed explanation of how they intends to prevent the build-up of scale on the absorber internals, surfaces and packing and any other vulnerable part of the SWFGD system, including details of the precautions to be taken during the Plant commissioning, start-up, shutdown and during periods of operation at reduced throughput.

The Offshore Supply Contractor shall indicate the precautions to be taken to ensure that any scale present which detaches can be removed from the system without causing damage or blockage to any part of the SWFGD system.

The Offshore Supply Contractor shall provide all necessary facilities and equipment, if required, to counteract foaming in the absorber and shall indicate what, if any, anti-foaming agents will be suitable.

The Offshore Supply Contractor shall include in their scope of supply all reasonable facilities and equipment required to prevent and/or minimise corrosion of any parts of the SWFGD system, on both the gas and water sides, including that caused by conditions during start-up, shutdown or operation at part load. Where corrosion is inevitable, details of the approximate corrosion rate, lifetime of components, and a justification for the selection of materials used shall be provided.

9.9 Electrical and Control Systems

All medium voltage (MV)/low voltage (LV) electrical switchgears, motors and actuators shall use the same model and same suppliers as those used for the main Plant. The MV/LV voltage levels shall follow the levels identical to those of the main Plant. DCS for the SWFGD shall be of the same model and same supplier as the one used for the main plant. Whilst the local control room for SWFGD may not be manned, the local control equipment shall be provided to enable local and manual control by the operator during commissioning and abnormal operating conditions. During normal operation, the control and monitoring of SWFGD shall be carried out in the central control room of the Facility. There shall not be necessary to have operator at local. Earthing systems for electrical and control equipment shall be of the same design as the main earthing systems of the Facility.

The SWFGD DCS shall control the start-up, shutdown and flow capacity of the SWFGD system. If total flue gas flow measurement for SWFGD control is necessary, at least two redundant flow transmitters shall be provided to measure the flue gas flow through the SWFGD duct, which enable the computation of the total gas flow as average of these two flow signals.

An interlock shall fully open the bypass damper, if any of the following conditions occurs:

- ID fan tripped; or
- All absorber spray pumps tripped.

Chapter 10 – Civil Structural and Architectural Works

10.1 General Requirements

This chapter establishes the minimum basic requirements for all civil, structural and architectural Works for the Project.

Each contractor shall fully satisfy itself about the Site conditions, nature of soil, ground water, contour levels etc. Otherwise, the Onshore Contractor shall conduct investigations to ascertain the correctness of the data furnished.

All civil and structural works shall be designed to give a minimum working life of forty (40) years before any significant repair or replacement. This requirement shall include the proposed working life of the main structural elements and secondary structural elements.

Civil works shall be designed, furnished and installed, complete as specified in this Owner's Technical Requirements (OTR) herein.

The scope of the civil and structural works by Onshore Contractor shall include, but not limited, to the following:

- Site investigation and clearance, excavation, filling, embankment, piling, foundations, landscaping;
- Site level shall be at +8.0 m, the same as existing Vung Ang I's site level, except for the coal stockyard level which may be lower subject to review during design stage. Historical tidal and wave data shall be considered as well as the risk of flooding in determining an appropriate site level for Vung Ang II main Plant and lay down area with a proper drainage system. Cooling water pumping station can be at +7.0 m;
- Temporary construction facilities including concrete mixing plant, water supply facilities, electricity supply facilities from the defined terminal points and site camps;
- All embedment required for anchorages, supports, trench and duct or pit curb angles, manholes, etc.;
- Buildings and structures to contain or house the specified Plant;
- Cable trenches or pits;
- Road, drainage, fences and bridges across the cooling water intake/discharge channel;
- Cladding, flashing, gutters, downpipes, doors, windows, glazing, ventilation louvers, and other fittings;
- Brickworks, rendering, plastering, tiling and suspended ceilings;
- Painting, varnishing and surface treatment;
- Plumbing including septic tanks and provision of fixtures

- Toilets and washing facilities (including in the turbine hall and all office environments);
- Ventilation and air conditioning system for enclosed building and structures;
- Earthing and lightning protection;
- Miscellaneous power and lighting systems;
- Permanent boundary wall/fencing; and
- Temporary fencing.

The Offshore Services Contractor shall perform engineering studies, calculations and analysis to ensure good civil and structural design and submit such information to the Owner for review during the detailed engineering stage. Such studies shall include calculation/analysis required by the individual civil specification or requested by the Owner which in his opinion is necessary to obtain a good civil and structural engineering design and approvals from the local authorities.

The minimum studies/calculation/analysis for the civil design that shall be performed by Offshore Services Contractor shall include, but not be limited, to the following:

- 1) Site level design and analysis;
- 2) Storm drainage system analysis and calculations;
- 3) All foundation design analysis and calculation and test;
- 4) All structural design analysis and calculation including stack, power building, turbine generator pedestal, cranes etc.;
- 5) Architecture conceptual design;
- 6) Environmental protection;
- 7) Report on chimney design; and
- 8) Analysis of water table level and special provision for the protection of underground structures, plant components, pipework and cabling.

The minimum studies/calculation/analysis for the civil design that shall be performed by Onshore Contractor shall include, but not be limited, to the following:

- 1) Geotechnical investigation report;
- 2) Piling tests report; and
- 3) Analysis of potential land sliding.

The above design of the civil works shall be carried out in accordance with international standards. Chinese GB standards can be accepted if they are proved to be more stringent. However, the design shall meet the local Vietnamese regulations in terms of seeking applicable regulatory approvals for the Facility. The Offshore Services Contractor shall submit reports for the above studies/calculation/analysis to the Owner for review.

The Onshore Contractor shall submit the instructions/installation procedures of the

civil works during the detailed engineering stage. If required, the Offshore Services Contractor shall also submit the relevant civil and structural calculations of the Plant buildings for the review by the Owner.

10.2 (Not Used)

10.3 Site Investigations

The Onshore Contractor shall carry out its own detailed topographical, hydro-geological and soil investigations, bathymetric survey, and other investigations, as deemed necessary.

The Onshore Contractor shall carry out the topographical survey in all areas identified for locating the Facility before he commences detailed design and Site levelling. Based on field observations, the Onshore Contractor shall prepare and submit for the Owner's review the survey maps of the Site on a suitable scale. The survey map shall show the spot levels and contour with grid lines. Established methods of surveying like triangulation, traversing, levelling etc. shall be adopted for the survey work. Spot levels shall be taken at a maximum of 20 m intervals and at closed intervals where pits, undulation etc. are met with. These levels shall be taken in two orthogonal directions. Contours shall be plotted at 0.5 m intervals.

The Onshore Contractor shall carry out his own geotechnical investigation in all work areas for establishing the subsoil conditions and to decide type of foundation for the structures envisaged, construction methods, any special requirements / remedial measures / treatment, called for sub soil / foundations before the Offshore Services Contractor commence detailed design and working drawings;. The Onshore Contractor shall submit the field and laboratory testing scheme proposed by them for the Owner's review before undertaking the geotechnical investigation work.

The field tests shall include, but not be limited to, the boring, trial pits, standard penetration test, collection of disturbed and undisturbed soil samples, in situ permeability test, plate load test, field vibration test, dynamic cone penetration test, static cone penetration test, field vane shear test, seismic refraction test, electrical resistivity test, chemical analysis of underground water samples / soil samples.

Field investigations shall be done in sufficient numbers as necessary up to the appropriate depth for complete determination of subsoil conditions. The laboratory investigations shall be conducted on soil & rock samples collected during field investigations in sufficient numbers. Geotechnical investigations shall be carried out in accordance with the relevant Vietnamese standards and established practices.

On completion of all field & laboratory work, the Onshore Contractor shall submit the geotechnical investigation report for the Owner's review. Such reports shall contain the geological information of the region, procedure adopted for investigation, field and laboratory observations / data / records and recommendations on type of foundation for different type of structures envisaged.

Tests shall be conducted at the laboratory of the soil investigation agency and shall be witnessed by Owner, if required.

10.4 Site Formation Requirement

10.4.1 Unexploded Explosive Ordinance (UXO) Clearance

The Owner will carry out UXO clearance as per Vietnamese Government requirements, in the following areas

- Main Plant
- Onshore cooling water system
- Ash pond
- Laydown areas
- Jetty and trestle
- Onshore dredging material disposal area
- Offshore cooling water pipelines (intakes, outfalls and VA1 extension)
- Access road
- Route of wet ash slurry pipelines

The final UXO clearance reports other than onshore dredging material disposal area will be provided by the Owner prior to the issue of the Notice to Proceed (NTP). The UXO clearance report of the onshore dredging disposal area will be provided by the Owner no later than 10 months after NTP or schedule agreed with Onshore Contractor.

10.4.2 Site Catchment Drainage System

The Project Site is located at foothill of nearby mountains in the south, and partly at cultivated field. The levelling areas comprise of approximately 38 ha within the Plant boundary. As the Site topography gently slopes towards the sea at the port side, drainage trench system shall be constructed of concrete storm water drainage channel design, with the adequate slope along the Facility boundary and discharged directly to the sea. The catchment drainage system shall be designed to ensure the drainage of surface water for the area of Facility as well as storm water received from surrounding mountains, offshore, and 500 kV switchyard area with a flooding frequency of one in 100 years. The Onshore Contractor shall coordinate its stormwater drainage design with Others.

The drainage trench at the north collects water and routes to the sea. It shall be designed by the Offshore Services Contractor and constructed by the Onshore Contractor to avoid i) excessive flow velocity in the channel for erosion control, ii) damage from erosion arising from direct sea waves and stormwater and iii) adverse effects on the coastal environment.

10.4.3 Site Levelling

Unless otherwise specified, the principal datum of the site is suggested at EL +8.00

m, same as VA1's site level. Cooling water pumping station can be at EL +7.0 m.

As the Project Site is partly located on cultivated fields, during levelling process, the loose top soil layer and slurry shall be dredged during the levelling process prior to embank top soil layers. Soil layer of thickness of 0.3 m shall be removed during average levelling. Soil base shall be compacted to design thickness to ensure base stability.

The surface of the cutting/filling up areas after reaching final level shall be dressed to the required levels and slopes.

The Site levelling of 500 kV substation is not in the scope of any contractor.

The Onshore Contractor is required to source the filling materials for Site levelling.

10.5 Foundation Design

10.5.1 General

The entire foundation system shall be designed such that it shall ensure the proper and due performance of the Facility and afford adequate protection against the effects of any long term settlement, against seismic activity, and against periodic flooding of the area.

Depending on the structures to be supported, foundations shall either be piled foundations or spread foundations. Design shall be able to accommodate total and differential settlements between structures supported on different types of foundation. Due consideration shall be given to minimize the chance of liquefaction of the underlying strata.

The turbine pedestal and foundation shall be designed to resist the static and dynamic forces imposed by machine operation as well as seismic loading. Deflections shall be maintained within limit defined by the turbine manufacturer and natural frequencies shall be such as to preclude resonant conditions during operations. The pedestal foundation design report shall be submitted to the Owner for review during detailed engineering stage.

The Offshore Services Contractor shall, subject to the restrictions herein, select the type of foundation which he considers to be most appropriate for the location and ground conditions on the basis of technical suitability, economy, availability of materials and equipment, ease of installation, rate of progress of construction and his experience in the performance of similar works to the Works.

At least two weeks prior to the commencement of piling, if applicable, the Onshore Contractor shall submit for Owner's review the detailed design calculations and drawings received from the Offshore Services Contractor together with a complete piling schedule on individual civil package showing the type, location, design bearing capacity and proposed piling schedules to be incorporated in the master

programme.

The Onshore Contractor shall submit for comment full details of the proposed method of installation, the equipment to be used and the sources of supply of any pre-cast concrete or prefabricated steel pipe material.

Subject to satisfactory pile loading tests, the Onshore Contractor shall provide information for the Offshore Services Contractor to revise and update the drawings and pile schedule, as and when required by the Owner throughout the duration of the piling Works.

The minimum spacing in plan between centres of adjacent piles shall be one metre or three pile diameters, whichever is the greater. Full allowance shall be made in the design for the action of closely spaced piles as a group and the design capacity of individual piles in the group shall be reduced accordingly.

The Onshore Contractor shall maintain accurate records of all pile installation, copies of which shall be submitted to the Owner daily.

10.5.2 Loading Tests

Based on the geotechnical investigation report, the Offshore Services Contractor shall propose the capacity, length and diameter of pile and the scheme of initial load test in vertical, lateral and uplift modes and submit the same to the Owner for review along with supporting design calculations. The Onshore Contractor shall carry out piling for initial load tests to confirm the proposed pile capacities. Loading test shall be done in compliance with Vietnamese standards or equivalent.

Initial pile load test

The number of initial pile load tests shall consider the importance of the project, the complexity of the ground conditions, the design experience, the type of piles used and the quality of pile construction in the field and it shall comply with the applicable design standards. In no case less than 2 piles shall be tested.

The test specification shall be prepared by the Contractor and reviewed by the Owner before starting the test. Pile load tests shall be witnessed by Owner. A report on the pile load test shall be furnished to the Owner for his review.

Test pile shall be loaded to at least twice the design load capacity of the pile. Deflection readings shall be taken immediately before and after the addition of each increment of load and at hourly intervals thereafter. The maximum test load shall be maintained for at least 24 hours and further readings shall be taken during, and at the end of this period.

Measurements shall be taken during the unloading of the test pile when the remaining load is 75%, 50%, 25% and 0% of the test load and the interval of time between each decrement of load shall be not less than half an hour. Further measurements shall be taken 24 hours after the test load has been removed.

In the event of failure of a test pile, the Onshore Contractor shall test additional piles and submit the remedy proposal for review by the Owner.

The loading test shall be compliance with TCVN 9393:2012 or ASTM D1143-81

Routine pile load tests

The test specification shall be prepared by the Contractor and reviewed by the Owner before starting the test. The Onshore Contractor shall statically test for vertical mode a minimum of 0.5% of total number of piles but not less than 2 piles. .

For the routine piles, the Dynamic loading test (Piling Driving Analyzer – PDA system) can be carry out as alternative to the static loading above mentioned. The onshore Contractor shall test dynamically a minimum of 1% of each capacity of pile or the minimum percentage required by Vietnamese code (TCVN 11321:2016), whatever is more stringent.

The locations of test piles shall be usually at places with typical soil conditions. In the case of complex soil conditions or in areas with large loads, the test pile shall be chosen at the most unfavourable location. When selecting testing pile, it is necessary to pay more attention to the actual quality of construction piles.

10.5.3 Driven Piles

Pre-cast reinforced concrete pile, pre-cast pre-stressed reinforced concrete pile, or prefabricated steel pipe driven piles shall comply with the requirements of this OTR for concrete and structural steelwork. They shall be in one full length wherever practicable and the Onshore Contractor shall provide equipment of sufficient capacity to handle and install them without cutting or splicing.

The Onshore Contractor shall protect all pre-cast or prefabricated piles from damage due to excessive bending stresses, impact, abrasion or other causes during delivery, storage, handling and driving and any pile so damaged shall be repaired or replaced. Any pile which is lifted during the driving of adjacent piles shall be re-driven.

10.5.4 Bored Cast-in-Place Piles

The Onshore Contractor shall install bored cast-in-place concrete piles by drilling with the use of bentonite mud, placing of reinforcement steel bar cage, depuration of the bentonite and placing of concrete by tremie pipe.

The bentonite mud shall be produced by thoroughly mixing water and bentonite. The concentration of bentonite shall range from a minimum of 4% to a maximum of 10% by weight, according to the soil conditions encountered.

The depuration plant shall be capable of controlling and keeping, to a minimum, the percentage of suspended materials in the mud during the placing of concrete. It shall also be capable of ensuring that, at the time of placing of concrete, the specific weight of the mud to a depth of one metre from the bottom of the hole shall be 1.3 ton/m³

with a sand content of 6% by volume.

The reinforcement steel bar cage, comprising vertical reinforcement with spirals or hoops, shall be placed in the hole so that it is at least 100 mm from the bottom of the hole, with spacers at the sides of the cage to maintain the specified concrete cover between the outside of the cage and the sides of the hole. The PVC pipe ultrasonic test shall be installed in 0.5% of the total number of piles.

10.5.5 Piles Near Existing Installations

Special precautions shall be taken to avoid damage to existing installations due to piling operations. No piles shall be driven within 10 m of cast-in-place concrete which has been in place for less than 7 days.

10.5.6 Tolerances

All permanent piles shall be installed within 150 mm of their designated positions in plan and the underside of the pile cap, as shown on the reviewed construction drawings. The maximum inclination of any pile from its design centre line shall be 1 in 75 and the top of each pile shall be within 50 mm of its design level.

10.5.7 Inner Drilling

The Onshore Contractor shall perform drilling inside steel pipe piles to required depths in accordance with the construction drawings. During drilling, the optimum method and equipment type for the inner drilling work for bearing rock stratum shall be selected upon thoroughly studying data such as boring. A works plan shall be submitted to Owner beforehand for comment.

The Works shall be executed with greatest care not to damage the steel pipe, and pollution control measures shall be taken in order that matter such as slime shall not be directly disposed of to the sea. During drilling, the elevations of soil and rock surface inside steel pipe piles shall be measured, and the data submitted to Owner.

10.5.8 Sheet Piling

Driven piles shall comply with the requirements of this specification for structural steelwork. The Onshore Contractor shall provide equipment of sufficient capacity to handle.

The Onshore Contractor shall protect all piles from damage due to excessive bending stresses, impact, abrasion or other causes during delivery, storage, handling and driving and any pile so damaged shall be repaired or replaced.

The interlocks shall be thoroughly cleaned before assembly and sheet piles shall be driven to the required line and level. Guide walling and/or other forms of control shall be used.

Any piles found to have deviated from the correct alignment or which are out of plumb by more than the permissible deviation either longitudinally or transversely,

out of clutch with adjacent piles or insufficiently driven, will be liable to rejection.

Walling shall be installed on the level with designed height and tie-rod shall be installed at right angles to sheet piles.

Prior to laying tie-rods or tie-cables under the ground, the Contractor shall carry out thoroughly tension of tie-rods or tie-cables and tie up of tune buckles, bolts and nuts.

The Onshore Contractor shall notify the Owner when all preparation, as specified above, have been completed and are ready for inspection. No backfill shall be carried out without the prior inspection of Owner. Backfill of the tie-rod shall be taken greatest care of it, and the Onshore Contractor shall protect tie-rods from any damage for the backfilling, if necessary.

10.6 Chimney

10.6.1 General

The chimney wind-shield shall be designed in accordance with applicable Vietnamese laws, codes, standards and regulations. A dynamic analysis and design for wind vortex excitation, period and resonant response shall be performed and a dynamic analysis of the chimney for seismic excitation shall be submitted for Owner's review.

Concrete chimney shall comply with ACI 307 Cast-in-place concrete chimneys or equivalent.

The design shall take account of any special dynamic loading or movement effects, such as vortex shedding and wake buffeting, arising from the proximity of other chimneys or nearby tall structures.

Individual steel flue stack shall be provided inside the chimney wind-shield for each Unit. The chimney (wind-shield and the two steel flue stacks) shall include thermal and corrosion protection to ensure that no maintenance is required over the operational life of the Facility. The flue gas emission point shall be 210 metres above the plant grade level.

Lightning protection shall be provided appropriate to the configuration and location of the Facility. Lighting for aircraft warning shall also be provided.

10.6.2 Design Requirements

The flue gas velocity at the exit of chimney shall be higher or equal to 24.1 m/s in accordance with the environmental impact assessment (EIA). The flue gas velocity inside the chimney shall be acceptable to the supplier of the Pennguard or Ecocera lining, such that the 10-year product warranty is not affected. Detailed information and reference of lining material shall be submitted for the Owner's approval in detailed design stage.

Both the wind-shield and the steel flue stacks shall have a design life of not less than 40 years performing their duties under all working environment.

Platforms and provisions for sampling and monitoring flue gases shall be provided. Other provisions shall include access staircases to various platform levels, aircraft warning lights, aviation safety painting, lightning protection, reference monitoring the long-term inclination changes and settlements. Provision of power and lighting services shall be provided.

The structural steel transition inlet ducting shall be bottom supported. This transition ducting shall be suitably profiled from a rectangular shape at the chimney inlet to a circular shape up inside the chimney where it shall be connected to the suspended circular steel liners through suitable (non-metallic) fluoro elastomeric fabric expansion joints. These joints shall be a single component or multi-layered construction capable of withstanding the worst flue gas conditions. One such spare joint shall also be supplied. Transition ducting shall also be thermally insulated. Platform shall be of structural steel construction.

The chimney roof shall however, comprise of a reinforced concrete (RC) slab supported over a grid of structural steel beams.

An internal structural steel staircase, supported from the shell wall, shall be provided for full height of the stack. Suitable embedment shall be retained in the shell wall for this purpose.

The flooring panels of the platforms and risers of the staircase shall be of chequered plate construction. Handrails of platforms and staircase shall be of tubular construction. Welding on handrails shall be minimized. Where this is necessary, they shall be smoothed to avoid personnel injury.

The top external portion of the wind shield shall be coated with acid and heat resisting paint in alternate bands of colours red and white to meet the aviation safety requirements. The mini-shell and the top few metres of the internal surface of the windshield shall be painted for acid and heat protection with bituminous paint.

Galvanized mild steel discrete strakes shall be provided, at the top (usually 1/3rd height) if found necessary from design requirements.

Normal and emergency alternating current (AC) power supply for chimney shall be derived from main plant power supply system. Emergency supply shall feed 20% of platform lighting, 50% of staircase lighting, aviation obstruction lighting and elevator load. All other loads shall be connected on normal power supply.

Ambient temperature for design of all equipment shall be considered as 60 °C which is likely to be encountered inside the chimney.

Aviation obstruction lighting system shall conform to the requirements of the latest applicable rules of International Civil Aviation Organisation (ICAO) and Vietnamese regulations. The type of obstruction lighting system shall be of high intensity aviation obstruction lights. The lights shall be of automatic rotary type; each set of lights consists of at least 6 lamp bulbs. Upon failure of one lamp bulb, the

second lamp bulb shall rotate to the position and take over the duty of the failed lamp bulb. During construction stage, temporary lighting shall be provided.

Suitable lightning protection system shall be all along chimney height and connected to the main plant earthing system. Above ground level, earthing and lightning protection system shall comprise of galvanised steel strips. These materials provided at top 12 metres shall have additional coating of 2 mm thick seamless lead cover or it shall be of stainless steel to take care of corrosion.

Communication system comprising of telephone socket at every internal platform level and at grade level, necessary wiring installation, a telephone hand set, junction boxes, etc. shall also be provided.

10.6.3 Construction Requirements

Only ordinary portland cement (OPC) shall be used for all concrete work associated with the chimney.

The Onshore Contractor shall carry out concreting of the foundation only after the steel reinforcement for the entire foundation has been placed and tied in position. Only vertical construction joints with joggles shall be permitted between any two segments. Horizontal construction joints shall not be permitted.

The chimney shell shall be constructed using shuttering and the norms, precautionary and remedial measures with respect to the rates of sliding, delay in sliding, discontinuous (or stop start) sliding, curing, protection, finishing, providing for opening and embedment and maintaining true vertically shall be strictly adhered to.

Special attention shall be given to proper curing of the chimney shell during construction so that the shell wall does not remain dry for long periods. Concrete curing shall be carried out either by moist curing or by membrane curing to be decided based on the site conditions.

The formwork system selected by the Onshore Contractor shall be of a proven make and shall ensure that adequate control can be maintained on setting out, concrete quality finish and fixing of all pockets, chases and blockouts and providing of corbels, etc.

A water level system together with plumb-bobs shall be established. Hanging walkways, safety netting and adequate access ladders shall be provided below the level of the shutter to allow for inspection, curing and finishing of the shell. The slipform shuttering shall have independent controls to counteract any tendency of the structure to travel out of plumb or to slew, rock or float.

The Onshore Contractor shall measure at six hourly intervals the following dimensions and positions and shall record these immediately in tabular form and on graphs:

- Progress height
- Centre line in relation to the centres at the base.

- Internal wall faces in relation to the concrete at the base.
- Wall thickness
- Twist

The Onshore Contractor shall have supervisors in attendance at all times whilst sliding is in progress. Such supervisors shall be experienced in the control of slides.

Forms shall be designed to produce hardened concrete having the shape, lines and dimensions indicated on the drawings. Forms shall be constructed and maintained in proper position and accurate alignment. Accurate alignment shall include maintaining uniform taper, round cross section and plumbness of concrete shell. Forms shall maintain vertical alignment of form seam marks.

Forms for the exterior of the shell, shall produce surface having a smooth uniform appearance. Curved forms shall be used for the shell construction to ensure true circularity.

Tolerances for chimney construction shall be in accordance with the following:

Deviation of shell horizontal centre from plumb at any section	1 horizontal to 2500 th of the height of the above ground level plus 12 mm
Deviation of horizontal centre, of any shell section in any 15.0 m height from plumb	12 mm
Variation of shell from true circular cross-section and the specified diameter	(+/- 25 mm)
Shell thickness	Minus 5 mm to plus 25 mm
Opening in chimney shell	
Top elevation	0 mm to plus 15mm
Bottom elevation	Minus 15 mm to 0 mm
Sides	Plus or minus 25 mm, but width of opening not less than that indicated on the drawings
Deviations from plumb of walls and other vertical surfaces (beam, columns, slabs etc.)	1 horizontal to 500 vertical
Variation of cross-sectional dimensions (beams, columns, slabs etc.)	Minus 5 mm to plus 5 mm

All structural steel work shall be fabricated using materials conforming to the requirements of ASTM or equivalent standards.

Visual inspection and dye penetration test shall be carried out on 100% of the welds. Radiographic testing shall be carried out for any portion of the weld length at random for not less than 10% of the shop butt welds and 15% of the site butt welds. Where radiography is not possible ultrasonic testing shall be carried out.

Anchor bolts/fasteners to be embedded in concrete shall be galvanised.

The complete fabrication and erection procedure of all structural steelwork and steel flue giving the construction sequence, details of all equipment, hoisting arrangements, temporary supports, restraints etc. to be used, shall be submitted for Owner's review prior to commencing of erection. The Onshore Contractor's report on the erection methodology shall include the method statement for construction of individual steel flue stacks and for the assembly of the complete liner system.

10.6.4 Chimney Lining

The chimney lining system shall be suitable for continuous operation of the Units with and without the seawater flue gas desulphurisation (SWFGD) system for indefinite periods. The lining inside the steel flues shall be of Pennguard supply of foam glass bricks. The lining system will be installed in the presence of the relevant Permitted Subcontractor and the Owner shall be named as a beneficiary of any extended warranty provided by the Permitted Subcontractor for the chimney lining system. The design life of the chimney lining system shall be at least 40 years.

The chimney lining system shall consist of three layers, namely, ground painting, adhesive and foamed glass bricks. The density of the foamed glass brick shall be more than 200 kg/m³. The ground painting shall have good bonding and coverage. Adhesive shall be high acid resistance and long-term flexibility. The foamed glass brick shall be good for anti-corrosion and insulation. Inspection schedule and repair procedure shall be provided during the detailed engineering stage.

10.7 Building and Structures

The Onshore Contractor shall construct the following:

- Boiler house (with one lift per boiler);
- Turbine house, weatherproofed and ventilated, suitable for location of the steam turbine generator and auxiliary equipment with suitable areas for plant lay down during periods of major plant maintenance;
- Central control building;
- Passenger and goods lifts for central control building (1000 kg passenger lift) and Administration Building (1000 kg passenger lift); power house (3000 kg passenger cum goods lift); chimney (rack and pinion type, 400 kg passenger lift); The lifts shall be provided with variable frequency drive (VFD);
- Gatehouse/security building with electronic access control system. The main gate house shall have a safety briefing room for 50 persons and a waiting room. Alternatively, the safety room can be constructed as a separated building nearby a gatehouse with conditions of same structural material and architectural finish and other facilities;
- Station weighbridge adjacent to the main gatehouse;
- Administration Building with appropriate office areas, conference rooms;

- Visitor centre and lecture theatre which shall be a separate building near entrance of power station for the seating of 100 people;
- Production services building;
- Canteen (with catering utensils and capacity for two sittings, each for 120 persons);
- Clinic (with emergency medical equipment and a fully equipped ambulance);
- Three workshops (mechanical, electrical and C&I);
- Stores;
- Water treatment building with chemical storage annex;
- Hydrogen generation Plant;
- Compressor house (can be integrated within other building);
- Fire fighting pump house and tanks;
- Garage for housing and maintenance of mobile plant and other vehicles, plus petrol / diesel fuel filling station;
- Process steam & water sampling room (can be integrated within turbine house);
- Coal unloading control building;
- Coal handling control building;
- Ash handling building & ash slurry pit;
- Cooling water pumping station based on outdoor design;
- Chlorination plant house ;
- Cooling water culverts;
- Dangerous goods store;
- Laboratory (can be integrated within central control building);
- Battery rooms;
- Coal sample preparation room;
- Electrostatic precipitator (ESP) electrical and control building;
- SWFGD electrical and control building;
- Coal plant transfer towers;
- Coal store;
- Raw water tanks and demineralised water tanks;
- Auxiliary boiler house;
- Oil transfer pump house;
- Chimney with elevator;
- Waste water treatment plant and sewage treatment facilities;
- Car and motorcycle shed with cover;

- Temporary offices for owner staff during construction;
- Permanent boundary wall/fencing;
- Temporary fencing; and
- Owner's Site office (as more fully described in section 10.12.1).

The Offshore Supply Contractor shall supply the following:

- Station weighbridge adjacent to the main gatehouse; and
- Hydrogen generation Plant.

The Offshore Services Contractor shall provide the following details of each building during the detailed engineering stage:

- General schedule
- Type of building chosen;
- Materials to be used. For structural steelwork, the type of surface treatment shall be specified;
- Design drawings for construction
- Method of construction;
- Foundation preparation;
- Layout plan with area shown;
- Services requirements;
- Finishes schedules;
- For each manned building, number of staff allowed.

10.8 Elevators

10.8.1 General Elevator

Elevators shall be designed based on following criteria:

Service	Passenger Elevator	Passenger cum goods Elevator
Quantity	Administration Building Central control building Turbine hall	One no. for each boiler
Location	One no. each for Administration Building,	Front corner of each boiler

Service	Passenger Elevator	Passenger cum goods Elevator
	central control building and turbine hall.	
Design/construction	International standards or equivalent standards compliance to Vietnamese regulations and code of practices	
Load carrying capacity	1000 kg	3000 kg
Rated speed	0.55 m/s	0.55 m/s
Drive/motor	Worm geared traction machine driven by suitable motor or two speed motor with proven microprocessor based controls	
Power Supply	400 V, 3 phase, 50 cycles, AC supply	
Method of control	Single speed Servo-drive control type with automatic level adjustment	
Type of doors	Vertically split, Imperforated collapsible door of hollow metal construction from 16 gauge thick steel sheet.	
Machine room	Machine room will be located at top of Elevator. Machine shall be pressurised dust proof. Sound reducing material below machines in machine room shall be provided	
Operation of Elevator	a) Selective duplex collective automatic operation with or without attendant b) Provision for locking control in auto or attendant position	

The Onshore Contractor shall obtain operation licenses for all elevators.

The supply of all elevators shall take into account the requirement for regular maintenance and services for the equipment. The proposed vendors shall therefore be chosen from those ones who can provide the service from approved service agency in Vietnam for the maintenance of the equipment. The proposed vendors shall be reviewed and to the agreement of the Owner.

All proposed elevators shall be of the industrial type of rugged design and duties, suitable for the dusty power plant environment. The drives and control shall have the appropriate degree of protection.

Metallic wire mesh/screen in the area of lift travel shall be provided where car and the counter weight cross each other. Elevator shaft shall be enclosed with 3.15 mm

thick galvanized mild steel (MS) cladding sheet from top of elevator pit up to machine room level. Appropriate sealing shall be provided for the cladding sheet to prevent dust/ water entry in the shaft.

Construction of the elevators shall have the following features:

Flooring of Cabin	Chequered plate flooring with heavy timber underlay over steel sheeting and replaceable felt spreading, each 25 mm thick.
Design Construction	Car inside enclosure shall be scratch proof stainless steel (SS) sheet.
Car entrance and landing doors	Type 'F' doors of British Standard (BS) BS:5655 (Part-I) or equivalent fire resistant doors defined and tested as per Appendix 'F' of BS:5655 (Part-I) or equivalent. Landing doors shall have fire resistance of at least one hour. These doors shall also be smoke tight as far as possible.
Car Door Construction	Imperforated collapsible door of hollow metal construction from 16 gauge thick steel sheet.
Signals	Car position indicator with directional arrows both audio and visual, hall position indicator at all floors, tell lights at all floors, battery operated alarm bell and emergency light with suitable battery, charger & controls, fan and telephone
Type of indicators	Standard indicators or alternatively luminous hall buttons and luminous button in car operating panel and digital hall position indicator on all floors. (All fixtures in stainless steel face plates)
Emergency indicator	In case of emergency hold-ups in between the floors, automatic flashers shall indicate its location of hold-up both audio & visually as well.

Further, elevators shall have provisions to meet following operational requirements.

- (i) Selective collective, automatic operation with or without operator through illuminated push button station located inside the lift car;
- (ii) Power operated with automatic opening / closing car and landing doors;
- (iii) Two push buttons, one for upward movement and the other for downward movement at each intermediate landing, and one push button at each terminal landing shall be provided in order to call the car;
- (iv) Push buttons shall be fixed in the car for holding the doors open for any length of the time required;
- (v) All other safety / protection / operation interlocks as required by BS:5655 (Part-I) or equivalent; and

- (vi) After sales services shall be provided for the spare support, maintenance and routine locally.

Indication for position and call indication inside the car, and up/down travel direction indication in car and at each landing.

10.8.2 Chimney Elevator

A rack & pinion elevator shall be provided inside the windshield to access all internal platforms. The specification of the rack and pinion elevator is:

- a. Carrying capacity : 400 kg
- b. Operational speed at 50Hz : (approx.) 40 m/min.
- c. Cab floor size (inside) : 1000 mm x 1100 mm
- d. Landing Levels (including ground) : Same as the number of platforms
- e. Electrical power supply system : 400V, 3 Phase, 50 Hz.
- f. Type of loading : Passenger
- g. Motors : One
- h. Automatic stop equipment shall be provided in the elevator cab.
- i. All necessary safety, equipment and devices shall be installed.
- j. Phase failure relay shall be provided for the protection of the electrical equipment in the elevator system.
- k. The elevator shall be provided with front opening and single call push button controlled.
- l. The elevator shall have local maintenance and service agent for routine maintenance and inspection.
- m. The elevator shall be provided with VFD.

10.9 Building Design and Superstructure

Each building shall be designed by the Offshore Services Contractor and constructed by the Onshore Contractor with a minimum life expectancy of not less than 40 years. The local building regulations shall be met for all aspects to include but not be limited to the floor and superimposed loadings, safety evacuation, emergency exits, fire protection and adequate stairs and access.

The turbine hall shall be steel framed structure having an external cladding system. Elevated floors of the turbine hall shall be made of concrete slab with welded steel grating in certain areas. The wall and roof shall not cause leakage of rainwater, which shall also be structurally strong to withstand the wind loading caused by the maximum wind speed. The walls and roof shall be of profiled metallic cladding to be reviewed and agreed by the Owner. Interconnection walk-ways between the turbine building levels and the adjoining boiler house shall be provided.

The Offshore Service Contractor shall take considerations on the following factors and reviewed by Owner during design stage of the turbine hall:

- Loading of each steel beam in turbine hall structure;
- Spacing concern on storage of refurbish components such as turbine rotor, generator rotor, bearings, cylinders, etc.;
- Generator rotor preservation concern, temperature and humidity control, access control, inspection tools, equipment storage and working areas;
- Transport of critical components in/out of turbine hall, loading and unloading area and transport methods;
- Huge size of components shall not be stored in the frequent lifting path as concern of safety;
- Identification and demarcation of walking / emergency exit path shall be provided by planning of material storage;
- Resources and auxiliary requirement for inspection/refurbish/repair such as power supply, water supply and drainage, pneumatic air, environment issues, noise pollution, NDT requirement, etc.; and
- The accessibility and working loading of the overhead crane at individual and combined operation

The structural steel buildings shall consist of steel columns, lattice trusses or girt & purlin members. It shall be used with hot rolled sections or built-up sections for columns, girder and braces (trusses). To resist walls and roof of structure, it shall be considered with hot rolled section, cold formed sections or built-up sections for girt and purlins. The Offshore Service Contractor shall state the minimum yield strength of the steel members for Owner's review during design submission stage. All structural steelwork shall be protected against corrosion by use of appropriate surface finish as required to maintain the integrity of the structure over the life of the plant. The steelwork shall be shot blasted to minimum Swedish Standard SA 2.5 in view of the proximity of the plant to the harsh marine environment.

The Offshore Service Contractor shall propose the corrosion protection system for Owner's review during design submission stage.

The scope of supply for the Administration Building shall include provision and distribution of all services (electrical/telecom, local area network (LAN) conduits and ports, fire/potable/service water, accommodation and facilities for operations and canteen services and dining areas, first aid room and other sanitary & medical facilities, lifts, parking spaces for vehicles, roads and walkways, sewage treatment, general drains, foul/storm drainage) within the Facility. The Administration Building shall have UPS provision for computers, printers, servers etc. The Owner will provide the furniture.

The civil buildings and their associated services shall be designed to meet with the following manning requirements:

- Total permanent operation and maintenance (O&M) work force of 300;
- A space for containerized office for additional 280 temporary maintenance staff during major overhaul periods.

The buildings shall be designed by the Offshore Services Contractor and constructed by the Onshore Contractor so that there are safe means of personnel escape during emergencies (e.g. in the event of fire, access to a place of safety at all times). Equipment location and arrangement shall ensure that not less than two escape routes are available for personnel in the event of fire or other hazard during operation and maintenance.

The maximum travel distance for personnel where there is a normal fire risk shall be less than 25 m where travel is possible in one direction only and 45 m otherwise. The maximum travel distance where there is a high fire risk shall be 12 m where travel is possible in one direction only and 25 m otherwise.

All floors, walls or other structural elements which provide a fire compartment or barrier function shall have an appropriate design fire resistance. The Onshore Contractor shall ensure that all services ducts, cables etc. which are required to penetrate such fire protection walls shall be sealed with appropriate fire barriers, dampers etc. after installation as to give an equivalent fire rating not less than that of the wall.

The road access within the Facility shall be designed to allow unobstructed access by emergency vehicle at all times, including during an emergency situation.

For enclosed buildings, wall and roof cladding and building service systems shall satisfy the following criteria throughout the design life of the Facility:

- Provide a completely watertight envelope around all Plant for equipment and personnel;
- Create an acceptable, controllable working environment for all building users;
- Be sufficiently flexible to allow necessary change and expansion with minimal disruption to Facility operation;
- Have a clear minimum maintenance programme based on design life cost appraisal;
- Be readily replaceable if damaged during the design life;
- Be sufficiently robust to resist all anticipated wind, thermal, loading and other movements during the design life;
- Be of an acceptable consistent appearance;
- Satisfy all appropriate standards, regulations and quality requirements;
- Be capable of resisting any environmental or chemical effects produced by the Facility or external environment; and
- Allow penetrations in the building enclosure to be formed and sealed easily.

10.10 Design Loadings

The design of the civil works shall take account of all loads applied including dead, imposed, wind, thermal, dynamic, settlement, movement, seismic and other loading conditions where appropriate. Temporary loads during maintenance and erection shall also be considered. The standards used and loading assumptions made shall be stated in the design basis and calculations.

Foundations shall be designed to support the anticipated loadings without any distress or excessive settlement. A factor of safety against exceeding allowable bearing capacities shall be selected which shall not result in excessive settlement. The maximum permissible absolute and differential movement of the foundations shall be stated in the design basis for each building and structure.

All structural components shall be designed for the following minimum loads;

1) Dead Loads

The dead loads consist of structure and equipment of a permanent and semi-permanent nature. The load includes the weight of framing, roofs, floors, walls, partitions, platforms, and all permanent equipment and materials. The vertical and lateral pressures of liquids shall also be treated as dead loads.

2) Live Loads

Live loads consist of assumed unit loading deemed sufficient to provide for movable or transitory load such as people, portable equipment, fixtures and parts lay down.

3) Floor and Roof Live Loads

The following imposed loadings are the minimums that shall be used:-

Roof	150 kg/m ²	Turbine House
	500 kg/m ²	Mechanical Annexe
	75 kg/m ²	Non Accessible Roofs
	300 kg/m ²	Catwalk for maintenance of roof fans
RC Floors	2,500 kg/m ²	Turbine House Ground Floor
	1,000 kg/m ²	Turbine House Mezzanine Floor
	1,500 kg/m ²	Turbine House Operating Floor
	2,500 kg/m ²	Mechanical Annexe Ground Floor
	1,000 kg/m ²	Mechanical Annexe Mezzanine Floor
	1,500 kg/m ²	Mechanical Annexe Operating Floor
	1,000 kg/m ²	Mechanical Annexe Deaerator Floor
		Warehouse bottom floor with 5t forklift truck operation
Boiler Area	500 kg/m ²	Platform

	1000 kg/m ²	Concrete coal feeder floor and tripper floor.
ESP / SWFGD Bottom Floor Area	5,000 kg/m ²	
Walkways	500 kg/m ²	
Jetty deck	2000 kg/m ²	
Jetty trestle	1000 kg/m ²	

4) Pipe Hanger Loads

Piping loads for main steam, boiler feed, condensate, and extraction lines shall be specifically determined and located. Piping expansion loads and dynamic load shall be considered on an individual design basis. Loads imposed on perimeter beams around pipe chase areas shall also be specifically determined. Pipe dead loads for other area shall be determined as uniform loads per square metre of floor area and shall be carried to columns and foundations as pipe dead loads.

5) Impact Loads

The following minimum impact loads shall be used in addition to other loads for components supporting reciprocating or rotation machines, hoist, cranes, or other equipment creating dynamic forces:

- For supports of elevators 100%
- For cab-operated traveling crane support girder and connections 25%
- For pendant-operated traveling crane support girder and connections 10%
- For supports of light machinery, shaft or motor drive 20%
- For supports of reciprocating machinery or power driven units 50%
- For hangers supporting floors and balconies 33%

6) Construction Loads

All structural components shall be designed for anticipated loading imposed by construction. Appropriate areas of the ground floor slabs of the power plant building and adjacent underground utilities shall be designed to support the operating loads of large construction cranes. The steel framing system shall be designed for stability at all stages of erection.

7) Wind Load

Wind loads shall be calculated in accordance with the Vietnam Ministry of Construction Codes or other approved internationally recognized standards.

8) Seismic Load

All the project construction items shall be designed for seismic conditions. According to “Vietnam Construction Norm”, volume III, appendix 2.8, published in 1997 by Ministry of Construction, and earthquake zoning map of Vietnam territory of 1:1.000.000 scale established by Institute of Geophysics – National Technology and Natural Science Centre, Vung Ang II thermal power project is located in the maximum seismic area with $I_{\max} = 7$ (MSK-64). Calculation of earthquake load shall take into account the work dynamics, which are fluctuation cycle, deaden specific fluctuation and work specific fluctuation. The essential items such as stack, boiler, and turbine base are designed for anti-seismic at higher seismic level than the maximum level according to the zoning map of Institute of Geophysics.

9) Crane Load

Crane lateral forces shall be determined in accordance with the requirement of the crane.

10) Abnormal Load

These loads shall include loads by postulated accident condition such as emergency torque of generator, and other loads as given by equipment manufactures.

11) Road Loading

The design of the roads shall be satisfactory for the maximum load anticipated during both the construction and operational period of the Project.

10.11 Cladding and Roofing

Materials shall be durable and fit for purpose. Materials for external surfaces shall be selected for aesthetic qualities and proven durability to withstand all climatic conditions likely to be experienced, ease of maintenance and economy of cost. Every endeavour shall be made to create as pleasant a working environment as is practical for the staff working in the Facility's buildings.

The selection of cladding materials shall be compatible with the constraints of the buildings and structures in accordance with Best Industry Practice. Details of materials proposed for building exteriors and their respective finishing colours shall be submitted for Owner's review when the design starts.

For the material of pipeline cladding, aluminium cladding shall be used.

Equipment located on flat roofs shall be avoided wherever possible. If location of equipment on flat roofs is unavoidable then the waterproof and surfacing system shall take account of maintenance and other access required.

10.12 Office Facilities

10.12.1 Provision for Temporary Offices for Owner's Staff

Temporary offices and car parks shall be provided by the Onshore Contractor for Owner's staff to accommodate 50 construction management staff, and 100 operations and maintenance staff to be recruited gradually from 26 months after NTP until 35 months after NTP. The office shall be air-conditioned with all necessary electrical conduits, wiring, power supplies, LAN connections and telephone connections for each desk as well as furniture, office equipment (excluding computers) and other utilities (including pantry, toilet and canteen services). The Owner's Site Office shall be based in a separate building from the Onshore Contractor's office. The Owner will pay for public utility service (e.g. electricity, water, internet and telephone) bills for the use of their own services. It is noted that, subject to handover of Administration Building as part of Section 1 of the Works, providing office space for O&M staff, the requirement for temporary offices for O&M staff during construction period will reduce depending upon the staff build-up schedule.

The Onshore Contractor shall make provision for, and pay all charges, in connection with high quality Site office accommodation for the Owner's project management staff within the Onshore Contractor's central administration area.

This accommodation shall be based on the following:-

- An office for the Owner's Project manager;
- A conference room;
- Offices for engineers;
- Accommodation for clerical staff with filing and reprographics facilities;
- Store room;
- Kitchen;
- Day rest room;
- Office furnishings;
- Photocopying and facsimile equipment;
- Suitable WC facilities;
- A training room for 30 trainees;
- Server room of adequate size; and
- 70 no. car parks (including 30 covered car park lots) shall be provided.

The accommodation provided shall be new and of modern construction complete with ventilation and air conditioning, and be of insulated design.

This accommodation shall be available for occupation by the Owner at NTP+6 months after the Site access date and up to the time of departure from Site of the Onshore Contractor or following handover of all of the Facility within his scope of works. From NTP+1 (4 weeks) till NTP+6 months, suitable, fit for purpose and meeting all functional requirements as described for the Site Office Accommodation above - temporary portable Site office shall be provided to the Owner's project management staff.

Provision of furniture include but not limited to:

- Office tables/chairs/filing cabinets/cupboards/waste bins
- Telephones
- Fax machine
- Photocopier
- Cooker/microwave oven
- Refrigerator
- Water heater and kettle
- First aid box - fully equipped and serviced.

All facilities and equipment provided for use to the Owner by the Onshore Contractor shall remain the property of the Onshore Contractor and be retained by the Onshore Contractor upon completion of the Works.

Owner Telecom Facility

The Onshore Contractor shall provide telecom or a two-line satellite facility that will permit communications to all major cities in the country and access to the international telecommunications system. The corresponding communication lines will be provided by the Owner.

The facility shall be maintained until completion of the Works

The Onshore Contractor shall provide an electronic telephone exchange unit (multiple incoming and outgoing lines) for the Owner's accommodation connected to the various accommodating rooms provided as above.

The Owner will pay for the public utility service (e.g. electricity, water and telephone) bills for the use of his own services. The rest of the cost of the above accommodation including furniture and facilities shall be included and such facilities are to be to the Owner's satisfaction.

The Onshore Contractor shall provide a gate house with electronic access control system. Gate house shall also include a site induction room facility and a waiting room.

10.12.2 **Provision for Permanent Offices**

Permanent office facilities at various locations such as control room, workshop, laboratory, stores etc. shall be consistent with suggested number of staff required to operate and maintain the Facility. Each office facility shall be provided with electrical, telephone, LAN ports and conduits. List of all buildings showing designed number of occupants, telephones, LAN points, etc. shall be submitted. As a guide, approximate area is specified below:

- **The Administration Building** shall be three stories. It shall be centrally air-conditioned. Adequate number of toilet bowls shall be provided for 50 persons

as per building codes. (A toilet for the disabled shall be provided at ground floor.) The Administration Building shall be ready for use by the Owner and handed over as part of Section 1 of the Works.

- **Canteen building** shall be including kitchen, store, lockers and change room for O&M staff.
- **Lecture theatre and visitor centre** shall be two separate compartments which may be combined with the Administration Building, or near or adjacent to the Administration Building. It shall contain a reception area, washroom facilities and function room which can be used for tea/coffee break. The lecture theatre shall accommodate at least 100 people. The visitor centre shall be located near the entrance or has its own entrance near the Administration Building. The Onshore Contractor shall provide model, poster, display board, photos, etc. for the visitor centre.
- **Stores** shall be of two levels with offices for 4 staff. Office, toilet and shower room shall be located on ground floor of the store. It shall have a folder gate opening fit for the access of 5 ton forklift and truck. The loading/unloading bay of the store shall be designed to facilitate movement of store items.
- **Workshops** shall have separate ventilated painting and welding rooms. It shall have a folder gate opening fit for the access of 5 ton forklift and truck. The workshops shall have toilet and shower rooms. The wall of the workshop shall be fitted with louvers for ventilation.
- **Chemical laboratory** preferably next to demineralization plant, shall have area including office accommodation for at least three staff.
- **Central Control Building** shall have rooms and offices for at least 7 staff (staff other than shift). Space shall be provided for permit to work office, document room, C&I inspection room, meeting room, pantry and store room.

The floor area requirement of the structures, facilities and buildings are shown in the following table. During detailed engineering stage, the actual floor plan and area shall be optimized by referring to the no. of people working there, the actual equipment dimension, maintenance activities and tools used through design review, and a detailed review shall be conducted with the Owner prior to finalizing the building size.

Table: Building Schedule

	Building	Item Nos.	Structure	Total Floor Area (m²)
1.	Power house			
	Turbine hall		Steel	22500
	Central control room		RC	500

	Building	Item Nos.	Structure	Total Floor Area (m²)
	Boiler house		Steel	n/a
2.	Administration Building		RC	4500
3.	Warehouse		RC	3300
4.	Workshop (Mech / Elect. / C&I)		RC	2500
5.	Production services room (office for operators)		RC	500
6.	Canteen		RC	850
7.	Clinic		RC	300
8.	Main gate house / security building		RC	200
9.	Gate house – rear		RC	100
10.	Ash handling building		RC	500
11.	ESP, boiler electrical building & SWFGD electrical building		RC	1200
12.	Demineralization plant building		RC	3000
13.	Waste water treatment plant		RC	3000
14.	Coal handling control building		RC	1000
15.	Compressor plant room		RC	800
16.	Dangerous good store		RC	500
17.	Fuel oil pump house		RC	400
18.	Fire pump house		RC	200
19.	Heavy equipment & coal mobiles garage		RC	900
20.	Fire station (with covered car park for fire engine, duty room and fire fighting equipment store room)		RC	1000
21	Visitor centre and lecture theatre (Separate building near entrance of power station)		RC	350
22	Chemical laboratory		RC	600

Floor finishes in industrial areas shall be designed to withstand vehicular and pedestrian traffic, wash down and/or chemical exposure without flaking, dusting or excessive cracking. Finishes will be specified as below.

(a) Administration Building and central control building

Exterior Wall: Masonry and Mosaic Tile

Interior Finishes:

- Office etc.
 - Floor : vinyl Tile
 - Wall : emulsion paint
 - Ceiling : suspended false ceiling panels
 - Control rooms
 - Floor : false raised floor of 300mm high to facilitate cable laying beneath
 - Wall : emulsion paint
 - Ceiling : suspended sound-proof ceiling panels
 - Lobby, stair and corridor (interior)
 - Floor : terrazzo tile
 - Wall : emulsion paint
 - Ceiling : emulsion paint
 - Storage
 - Floor : cement towelled finish
 - Wall : emulsion paint
 - Ceiling : emulsion paint
 - Changing rooms, showers & toilets (ladies & gentlemen)
 - 1st Floor: unglazed ceramic tile
 - 2nd Floor and above water proofing on shower and toilet then unglazed ceramic tile
 - Wall : glazed ceramic tile
 - Ceiling : suspended plastic stripe shaped panel
 - Locker
 - Floor : vinyl Tile
 - Wall : emulsion paint
 - Ceiling : emulsion paint
 - Kitchen
 - Floor : unglazed ceramic tile
 - Wall : glazed ceramic tile
 - Ceiling : suspended plastic stripe shaped panel
 - Canteen
 - Floor : unglazed ceramic tile
 - Wall : glazed ceramic tile
 - Ceiling : emulsion paint
 - Stair (interior)
 - Floor : terrazzo tile
 - Wall : emulsion paint
 - Ceiling : emulsion paint
- (b) Gate house
- Exterior Wall: Masonry and mosaic tile with Owner's logo and name
- Interior Finishes:
- Office
 - Floor : vinyl Tile

Wall : emulsion paint
Ceiling : suspended false ceiling panels

- Toilet

Floor : unglazed ceramic tile
Wall : glazed ceramic tile
Ceiling : suspended plastic stripe shaped panel

(c) Electrical building

Exterior Wall: Masonry and paint

Interior Finishes:

- Cable Spread room

Floor : chipping Paving
Wall : chain link Fence
Ceiling : Injection type insulation w/protective coating

- Medium voltage (MV)/low voltage (LV) switchgear room

Floor : epoxy paint
Wall : emulsion paint
Ceiling : emulsion paint

- Transformer-rectifier room

Floor : hardener
Wall : emulsion paint
Ceiling : emulsion paint

- Stair(exterior)

Floor : hardener
Ceiling : Acrylic paint

- Stair(interior)

Floor : terrazzo tile
Wall : emulsion paint
Ceiling : emulsion paint

- Corridor(interior)

Floor : terrazzo tile
Wall : emulsion paint
Ceiling : emulsion paint

- Safe alternating current (AC) & direct current (DC) system room

Floor : epoxy paint
Wall : emulsion paint
Ceiling : emulsion paint

- Protection relay room

Floor : epoxy paint
Wall : emulsion paint
Ceiling : emulsion paint

- heating ventilation and air conditioning (HVAC) room

Floor : hardener
Wall : emulsion paint

Ceiling : emulsion paint

- Battery room

Floor : chemical resistant paint

Wall : chemical resistant paint

Ceiling : emulsion paint

- Toilet

Floor : unglazed ceramic tile

Wall : glazed ceramic tile

Ceiling : suspended plastic stripe shaped panel

- Office

Floor : vinyl Tile

Wall : emulsion paint

Ceiling : emulsion paint

(d) Coal yard

Storage floor:

The coal yard base shall consist of layers placed on a compacted sub-base which shall be compacted satisfactorily to a compaction factor $k \geq 0.95$:-

- Layer 1: Geotextile (impermeable layer, Single layer), 1.5mm thick
- Layer 2: Excavated soil (screened), 200mm thick;
- Layer 3: High-density Polyethylene (HDPE) composite geomembrane (impermeable layer), 1.5mm thick;
- Layer 4: Protection excavated soil (screened), 300mm thick.

(e) Steam turbine

Exterior Wall: metal cladding

Interior Finishes:

- Turbine hall

Ground Floor: hardener for loading bay and access area

13m Floor: hardener

Wall : metal cladding

Ceiling : metal cladding with insulation

(f) Workshop & warehouse

Exterior Wall: masonry and emulsion paint

Interior Finishes:

Workshop

Floor : hardener

Wall : emulsion paint

Ceiling : emulsion paint

10.13 Doors and Windows

The location of pedestrian and vehicular door accesses and windows shall be coordinated with the requirements for the Facility and service penetrations through the building elevations. Door locations shall be determined from access requirements identified during the layout planning study. All electrical room doors shall be equipped with a master key system.

Windows shall be provided for office type environments or where daylight is a specific benefit to movement about buildings of personnel or of carrying out maintenance functions. Opening shall be specified where appropriate for ventilation or cleaning purposes. The location of windows shall take into account building exposure orientation and control of solar gain. They shall be visually harmonious with the building elevations.

Where surface openings in the high level cladding areas cannot be avoided they shall be carefully coordinated and their appearance shall be such that they relate to the building visually and have an uncluttered appearance.

10.14 Access Ladders and Handrails

Pipes for all handrails and ladders shall be of 32 mm nominal bore MS material (medium class). It shall then be galvanized to a minimum of 610 g/m² for operating floor or outdoor application.

Adequate access shall be provided to allow satisfactory and safe operation and maintenance of all buildings and equipment. The requirements for equipment and crane access, access to manholes, pits and trenches, maintenance access and staircase access to the upper floors of buildings shall be considered during the layout planning study. Safety barriers and handrails shall be provided wherever necessary to meet the requirements of the adopted standards.

In any case the design superimposed loading for handrails or access ways shall not be less than 2.5 kN per linear metre. Surface finish to access ladders, stairs and handrails shall be compatible with the design environment and shall give satisfactory performance without deterioration during the design life and shall have a non-slip surface on walking areas.

10.15 Site Drainage

The Site shall have segregated process, storm and foul drainage systems. Site areas likely to be exposed to spilt oil or oily water shall be discretely drained into appropriate oil interceptors prior to discharge into the Site effluent drainage system. Facilities for periodic removal of sludge and debris from interceptors and containment areas shall be provided. A waste water pond of 5000 m³ shall be provided. The size of this waste water pond can be optimized with the regular holding sump and irregular waste water storage basin in OTR 8.13.2 and satisfy the requirement of the incident waste water storage of defined capacity required in

Decree 40/2019/ND-CP during design submission stage. All Site drains, except non-contaminated rainwater drains, shall be routed through this pond and, after treatment, discharge into the CW system.

Foul water drainage shall run in a separate system from surface water drainage and be designed in accordance with Vietnamese standards. All foul drainage shall comply with the minimum technical requirements of the local authority.

The basis of design shall include full drainage calculations and design details for permit application when required. The fouling drainage system shall take into consideration build-up of staff during plant outages.

The foul drainage system design shall be based on the maximum flow conditions occurring including domestic discharge from the on-site sewage treatment facility in accordance with relevant standards and all plant effluent including boiler blowdown and water treatment effluent.

Equipment shall be provided to allow access for regular sampling and monitoring of total discharge flow.

The surface water drainage system needs to ensure the drainage of surface water for the area of facilities as well as storm water received from surrounding mountains, offshore, and 500 kV switchyard area with a flooding frequency of one in 100 years.

10.16 Roads and Hardstandings

The materials selected for road construction shall not be deleterious and shall be of adequate strength and durability to meet the design requirements for the whole of the design life.

Provided with minimal maintenance, the road pavements shall be designed for lifespan of 40 years with necessary maintenance and shall provide a level of service which shall be free from significant cracking or rutting. Roads and access ways shall be designed to carry a minimum level of traffic of 150 of the heaviest laden commercial vehicles per day and shall also be capable of carrying all construction traffic, operational traffic and heavy maintenance traffic.

Road alignment, shall allow for any construction, equipment delivery and abnormal loads or heavy maintenance requirements and shall be designed to acceptable standards as stated in the basis of design. At the point that road crosses the trench, the trench shall be designed in consideration of the maximum vehicle load using the road.

All areas which may be used for chemical, oil or other unloading shall have a pavement material capable of withstanding the likely exposure to chemicals or oil. The design life of these areas shall be 40 years.

Car and vehicle parking areas shall be laid to falls and be kept well drained. The design life of car parks shall be 40 years with necessary maintenance.

Depending on the ease of access and equipment transportation, roads shall be designed to three sizes of 8 m, 6 m, and 4 m in width. Road construction shall be as follows,

- Sand bed compacted,
- 20 to 40 mm grade aggregate compacted, $k \geq 98$. Thickness 350 mm,
- 10 to 20 mm grade aggregate compacted, $k \geq 98$. Thickness 300 mm,
- Primer coat,
- Bituminous binder course 60 mm thick,
- Bituminous wearing course 40 mm thick.

Pavement of 1.0 m width on both sides of main roads and single side of secondary roads shall be provided. Storm water collection sumps and associated piping shall be provided along the roads for drainage services. The designed load shall not be less than H30, XB80 load level of Vietnamese standard, with the frequency of not less than 150 times/full day.

The roads within the Facility shall connect and reach to every functional area. In addition, there are ring roads arranged surrounding the station boundary, the main power house, the coal storage yard, the oil tank area etc.

There shall be a large hardening field arranged in front of both the production services building and the central control building which can act as temporary parking lots. The coefficient between road and field is 15%.

10.17 Fencing to Electrical Equipment

Fences to transformer compounds and other electrical areas shall be galvanised steel palisade type of minimum height 2.4 metres.

Fire walls as per NFPA850 or equivalent standard shall be provided for generator transformer.

10.18 Scaffolding

All scaffolding, temporary working and access platforms used throughout the construction shall be metal type construction. Support frames shall be of steel or aluminium tube construction, with all couplers, spigots, base plate accessories and fittings of steel or aluminium as appropriate. Scaffolding boards and ladders may be of timber construction. Scaffolding shall be erected, altered and dismantled by qualified scaffolders provided by the Onshore Contractor working under the supervision of experienced supervisors and shall comply with the relevant local

health and safety regulations. All scaffolds shall be registered and be inspected by competent persons weekly for defects (or as dictated by the appropriate regulation).

10.19 Landscaping

Sufficient space in the Site shall be reserved for landscaping and the Onshore Contractor shall perform and complete the landscaping. Landscape plan shall be submitted to the Owner for comment during detail design stage.

Minimum maintenance approach shall be adopted. The concept shall be to create an environment within the Facility that is easily sustainable with a minimum of maintenance (i.e. requiring minimum application of fertilizers, watering etc.) Landscaping (grasses, shrubs, plants and trees) areas shall be maximized within the Site.

The Onshore Contractor shall provide suitable hydrant systems for watering.

10.20 Heating, Ventilation and Air Conditioning (HVAC)

The Facility shall be provided with all necessary HVAC equipment including Plant, ductwork, pipework, controls, instrumentation, interlocking and cabling systems designed to maintain appropriate environmental conditions for equipment/personnel. Environmental friendly refrigerants shall be used. The HVAC systems shall be designed with the following conditions.

	Summer			Winter		
	Temp (oC)	RH (%)	Air Change Rate (V/hr)	Temp (°C)	RH (%)	Air Change Rate (V/hr)
Turbine hall and all ventilated areas			5			5
All air- conditioned areas	25.5±1	≤ 60		20±1	≤ 60	

Air conditioning system shall be, as a minimum, provided for the following areas:

- Gatehouse and security building;
- Administration Building;
- MV electrical rooms, battery rooms and motor control centre rooms;

- All laboratories and coal samples preparation room;

Central type air conditioning system shall be, as a minimum, provided for the following areas:

- Administration Building; and
- Central control building;

All other areas of the Facility shall be ventilated. 100% standby capacity shall be provided for all major equipment including the air conditioning systems serving the central control room (CCR), distributed control system (DCS) equipment room, electronic and computer rooms. The duty/standby units shall be arranged so as to changeover automatically in the event of failure. The ventilation system shall be provided adequate ventilation to ensure each room is kept at an acceptable temperature level for the equipment, facilities, etc. and not exceeding ambient temperature +5 °C.

The design philosophy of the power plant shall make use of natural ventilation system, especially at spacious locations. The forced ventilation system shall be used at specific areas which are designed with particular requirements such as operating rooms to ensure the internal temperature and air suitably with human's biological demand. In addition, for a number of locations, combination between natural and forced ventilation systems can be made. Manned locations shall be provided with air condition. Places where demand on air condition is high shall be equipped with central conditioning such as central control building and Administration Building.

10.21 Lighting and Small Power

Miscellaneous power and lighting required for the safe and reliable operation of this power station shall be provided. This includes the normal lighting, emergency lighting, exit warning sign, distribution boards, etc.

An exit and an emergency lighting system (derived from the DC) shall be designed and implemented for the safe passage of personnel and for the plant to be operated safely in the event of loss of normal AC supplies. This system shall be designed as a maintained system being supplied normally from AC and switching over to the DC when normal power is lost.

Lighting in hazardous locations shall be specifically designed for use in those areas. Lighting at hazardous areas shall be in accordance with the relevant international and Vietnamese standards for hazardous areas. Plant/area which may be classified as hazardous based on industry practice and experience include, but not limited to the followings:

- (a) Fuel oil storage area

- (b) Fuel oil pump station (house)
- (c) Turbine lubrication oil tank area
- (d) Generator sealing oil tank area
- (e) Diesel generator oil storage area
- (f) Fire fighting pump (diesel driven) oil tank area
- (g) Chemical storage area
- (h) Hydrogen plant & bottle house
- (i) Station bunker room
- (j) Conveyor transfer towers
- (k) Coal feeders and coal mill area
- (l) Burner operation floors

The emergency lighting shall have a minimum luminance as follows:

Emergency access/personnel safety lighting	20 lux
Other operational Facility areas under emergency lighting	50 lux
Normal office and places for precision works	550 lux

Energy efficient road galvanized iron (GI) lighting poles shall be provided for road lighting.

10.22 Medical Facilities

The Onshore Contractor shall provide a medical facility and health care during construction for the Owner's Site staff and those of the Onshore Contractor and his Site sub-contractors. This temporary facility can be located anywhere within the Site perimeter. A permanent medical facility is also to be provided, located within the Administration Building, where it will service the operational station. Alternatively, a permanent medical facility can be combined with a clinic house as a separated building nearby the administration building with conditions of same structural material and architectural finish and other facilities.

The medical facility (both permanent and temporary) shall comply with all local statutory regulations and have adequate communications to central / state medical facilities.

The Onshore Contractor shall equip the medical facility (both permanent and temporary) with emergency medical equipment and a fully equipped ambulance. The mobile (portable) and fixed medical equipment shall be provided.

Within the Site perimeter a suitable hard standing helicopter pad shall be constructed. Suitable markings and other accoutrements (wind-sock and the like) shall be

provided and the immediate area cleared of any obstructions that may impede the landing or take off of a helicopter.

10.23 Coal Jetty

10.23.1 Load by Coal Collier

A coal unloading jetty shall be designed by the Offshore Services Contractor and constructed by the Onshore Contractor to accommodate one coal collier from 60,000 up to 100,000 deadweight tonnage (DWT).

The main dimensions of a typical 100,000 DWT coal collier are as follows:

- Overall length L 250 m
- Molded Breadth B 43 m
- Molded Depth H 20.3 m
- Loaded Draft T 14.5 m.

Loads acting on the berth by the collier consist of the mooring load and berthing load. The load shall be calculated in accordance with Vietnam Standard 22 TCN 222-95. The mooring load acting on the bollard shall be determined in consideration of the wind force and the current force acting on the vessel. The design current speed and wind speed for operation of the jetty shall be at least 0.5 m/s and 14 m/s respectively.

The berthing load shall be consideration of the berthing velocity of 0.15 m/s and berthing angle to be $0^{\circ} \sim 6^{\circ}$.

10.23.2 Marine Structures

To accommodate the marine traffic and unloading the ships that deliver coal to the VA II plant, a coal jetty is required for the berthing of 60,000 to 100,000 DWT coal collier. The berthing of vessel shall be at the eastern side of the jetty. A trestle from Site boundary shall extend to the jetty.

The required berth depth at low water level shall be calculated by adding the maximum draft of the design coal collier to the needed under keel clearance. The minimum under keel clearance shall be at least 15% of the maximum design ship's draft. Consideration shall be given to potential silting and the cost of dredging versus moving the unloading pier further into deeper water.

The berthing location at the jetty and turn around basin shall be deep enough to allow free manoeuvre of a fully loaded collier.

The top concrete elevation of the marine structures shall be determined by the need to avoid water breaking over or inundating the deck. Historical data of wave heights and high water levels shall be used to establish the top of concrete elevation. When

the waves reach or flood the deck, the very high lateral and vertical loads shall not damage the mechanical equipment on the deck. Suitable nominal air gap shall be provided between the crest of the design wave and the underside of the deck.

The trestle shall consist of a concrete deck, supported on piles. Navigation aids shall include flashing yellow lights, day beacons, and transit lights.

The precast concrete decking shall be provided to minimize the use of cast-in-place concrete because of the great length of the access trestle and the difficulty of transporting and pouring concrete in deep water away from the shore. There shall be vertical piles that support the weight of the deck and equipment and batter (inclined) piles that will resist lateral and horizontal loads.

All concrete exposed to marine environment shall be in compliance with TCXDVN327:2004.

If steel pile design is adopted, the Offshore Services Contractor shall incorporate the following corrosion protection measures in the design:

- A corrosion allowance in the selection of steel pipe thickness;
- Anti-corrosion coating;
- Cathodic /impressed current protection; and
- Anti-corrosion sealing tape and physical protection at the tidal zone.

The berthing structures shall be designed to absorb the wind impingement forces of a ship anchored and unloading during a design defined current and wind speed. It shall also be designed to resist the impact of a fully loaded coal collier moving in compliance with BS6349 requirements. The structure shall be equipped with high efficiency dissipating rubber fender units with rubbing panels consisting of a steel frame and low friction facing material. Each fender shall be reinforced with a backing frame, and restrained with tension and shear chains or cables. The mooring points at the two far ends of the jetty shall be equipped with mooring bollards capable to accommodate up to three ship's hawsers.

The principal functions of the deck platform shall meet the following requirements:

- Provide lateral berthing support for the ship and marine laser docking system;
- Support the coal receiving mechanical equipment;
- Provide vehicle access to the coal receiving equipment for operation, maintenance and repair; and
- Allow the transfer using the unloading crane, of a small front-end loader into the ship's holds, to bring the coal from corners and sides toward the crane's bucket, during the final part of the unloading cycle.

The jetty shall be connected to the shore by an access trestle. The access trestle as well as the jetty shall be designed such as not to impede small boat navigation (special openings shall be provided between rows of piles) and allow marine life free travel access across and along the trestle.

The jetty approach trestle shall be designed with the following:

- A single lane vehicular access for jetty operations with 2 passing zones; and

- The structural supports for running the coal conveyor from the jetty to the shore.

The deck shall have areas for equipment component laydown during repairs and maintenance, and for parking a few personnel cars and maintenance equipment.

For berthing and de-berthing tugboats will be provided by third party.

The jetty and access trestle shall be provided with ladders to low water level at regular intervals along the access trestle and at the jetty.

The jetty shall satisfy the following conditions:-

- One coal collier berths and is unloaded at the jetty with two ship unloaders are operating; and
- One of the ship unloaders is in maintenance. Coal unloading can be carried out by the other ship unloader without shifting the coal collier.
- The length of the jetty shall be at least 310m.

The internationally recognized British Standard BS6349 or Vietnamese standard on maritime structure, whichever is more stringent, shall be adopted in the design of the jetty and associated structures. Other standards that shall also be complied with are listed in section 10.23.4 of the OTR.

The jetty shall be designed to give berthing based on average hourly wind speed of 14 m/s. The wind force, the current force and the wave force on the vessel shall be taken into account in the calculation of the berthing load and mooring load on the bollard. The selected wind, current and wave forces shall be based on return of 100 years. The jetty design shall also allow for the load of ship unloaders acting on the jetty under the normal operation and extreme storm condition without unloading operation on the jetty, as well as that of one conveyor, supported on the jetty. The vertical (settlement) and lateral displacement (horizontal movement) of the jetty and trestle shall be limited to avoid damage to or interruption of operation of facilities on jetty deck.

The jetty, whilst it shall be designed for accommodating coal collier of 100,000 DWT, it shall also cater for berthing of coal colliers down to 60,000 DWT, with a suitable choice of fenders. The maximum hull pressure on vessel shall be adopted in the jetty fender design.

The berthing speed shall be selected according to the standards with a choice of the environmental condition and berthing direction. The berthing structure and the fender shall be designed to resist the impact of a fully loaded coal collier moving at a maximum angle of 6° to the berth with an allowance of minimum safety factor of 1.50.

The historical wave height and the high water levels shall be used to determine the jetty top and soffit level to ensure they are clear of crest of wave under worst sea condition. The wave height based on a return of 100 years shall be adopted.

The jetty, trestle and dredged area shall be completed with navigation aids to the local port authority requirement. The jetty shall also be completed with office for a small number of staff, toilet, equipment room, small workshop and store, waste water collection and treatment system, potable water, service water, fire detection and fire fighting, lighting and electrical supply etc. as appropriate.

The materials to be used, tests and inspection for the jetty construction work and the acceptance criteria shall comply with internationally recognized standards.

The Onshore Contractor shall conduct an initial survey for establishing the base level of dredged area, post dredging and final acceptance surveys for jetty berthing box, approach channel and turning basin.

10.23.3 Construction and Dredging Works

The Onshore Contractor shall carry out but not limited to the following works:

- Performing bathymetric and soil survey;
- Preparing foundation and structure;
- Casting concrete of deck slab;
- Installing steel walkway;
- Installing bollards and fenders; and
- Dredging works required for the approaching channel, turning basin and coal unloading jetty.

The Onshore Contractor shall formulate a dredging and dumping plan, as well as mitigation measures, for the dredging work of marine structures for the Project in accordance with the environmental impact assessment (EIA) report for the Owner's review during design submission stage. The Onshore Contractor shall dump surplus dredged materials to an identified area as shown in the Figure – Identified Dumping Area, subject to Owner's confirmation of location prior to start of dumping.



Figure – Identified Dumping Area

The bottom level of approaching channel, turning basin and berthing box shall be at least -17 m Chart Datum (CD), -17 m CD and -18 m CD respectively. Bathymetric survey shall be conducted on the dredged area at the Taking Over of the Facility as confirmation of the required depth.

10.23.4 Codes and Standards for Marine Structure Design

The Offshore Services Contractor shall comply with relevant international standards and codes of practice as listed below in the design and construction of the marine works. Any additional requirements such as the local codes and standards, regulations, approval by local Government Bodies, including but not limited to the local environmental requirements, to be complied with shall also be specified in the Exhibit 4 of the relevant contract.

- 22 TCN 222-95 Impact of Waves and the Ship when Designing New or Improved Design of Waterway Traffic Works in the River and the Sea.
- API-RP-2A-WSD: Recommended Practice for Planning, Design and Construction Fixed Offshore Platforms (20th Edition)
- API-Spec-2B: Specification for Fabricated Structural Steel Pipe
- API-Spec-5L: Specification for Line Pipe

- API-Spec-2H Specification for Carbon Manganese Steel Pipe for Offshore Platform Tubular Joints
- AWS D1.1 Structural Welding Code
- AISC (ASD) Specification for the Design, Fabrication and Erection of Structural Steel
- AISC A58.1 American National Standard Minimum Design Loads or Buildings and Other Structures
- OCIMF Guidelines and Recommendations for the Safe Mooring of Large Ships at Piers and Sea Islands
- ASTM Application Materials Specified in Civil / Structural Specifications
- UBC Uniform Building Code
- BS 6349 Maritime Structures
- BS 5950 Structural Use of Steelwork in Building
- BS5400 Steel, Concrete and Composite Bridges
- ACI – 318 Building Code Requirements for Structural Concrete
- AASHTO LRFD Bridge Design Specifications
- PIANC Fender Design

10.24 Dumping of Dredging Materials

10.24.1 Dredging Areas

Dredging shall be conducted in the ship approaching channel, turning basin, coal unloading jetty and the excavation of the foundation of the CW culvert of CW intake and outfall in the sea. The Onshore Contractor shall be responsible for the technical design and the permit for the dredging and dumping works. The Onshore Contractor shall implement those environmental monitoring system and migration measures as per EIA as well as EIA supplement for dredging and dumping.

Depending on the property of the dredging materials, it is the responsibility of the Onshore Contractor to use the dredging materials for site levelling and/or onshore dumping depending on evaluation of cost and schedule impact by using dredging materials for site levelling and/or import soil for site levelling.

10.24.2 Dredging Materials for Site Levelling

The area to be filled shall be cleared of vegetation and the top soil shall be stripped and stockpiled. All soft yielding material shall be removed and replaced with granular selected material, and it shall be disposed on a designated onshore disposal area which is allocated by the Owner. Filling to the formation level shall be brought up from the bottom in uniform compacted layers. Excavated material obtained from the Site could be used for filling, as well as part of dredging material.

The Site filling, levelling and compaction on the Site shall be carried out into four zones: power block, K-buildings area & construction area, BOP area, coal yard area.

The foundation backfilling shall be carried out in layers not exceeding 400 mm thickness each one. The site filling layer thickness shall be agreed with the Owner after evaluation of the soil report. All necessary quality control works including in-situ soil density tests, moisture content and other laboratory testing shall be carried out to ensure that all materials used in the filling elsewhere are compacted in accordance with the specified requirements which shall be reviewed by the Owner during design submission stage.

Prefabricated Vertical Drain (PVD) and /or vibro-compaction shall be carried-out to ensure proper consolidation of the soft substratum. Dynamic compaction may be adopted for the work units not supported by piles. The sequence, the area and the extent of the dynamic compaction shall be determined and submitted to the Owner for review during design submission stage.

The final soil compaction and improvement method statement shall be confirmed as per site soil tests and shall be submitted to the Owner for review during design submission stage.

10.24.3 Onshore Dumping of Dredging Materials

The onshore dumping area will be allocated by the Owner agreed with local government and approved in EIA. The dumping area will be about 50 to 70 ha. The elevation of the backfill shall be not higher than the road elevation next to the dumping area.

The laydown areas can be backfilled with the excess of dredged material subject to the approval by the local authorities and EIA. In addition, the location of backfilled onshore areas may change during Project execution stage.

The dumping area shall be enclosed and surrounded with bagged sand cofferdams. The sand bags shall be made of anti-aging polypropylene split filament woven fabric, in specific length, width, height and layers, subjected to the requirements by the Owner with relevant Government Bodies and EIA. The boundary of the cofferdam shall be the same as the dumping area in terms of shape and size.

The dredging water shall be treated by sedimentation. The sedimentation tank shall be constructed underground, using brick for the tank wall and applying waterproof mortar to the interior of the sedimentation tank. The dredging water shall be treated and discharged to the sea as per local regulations and EIA.

Chapter 11 - Operation and Maintenance

11.1 General Operational Requirements

The Facility shall be operated from a single central control room (CCR) permitting safe and efficient operation to be achieved with minimum manpower requirements. Fully automatic control systems over the full load range shall be provided where practicable, incorporating high reliability, load following capabilities with fail safe features.

The Facility shall be designed such that all routine operations, e.g. start-up, shutdown and load changing, can be initiated and controlled from a single CCR. The Facility shall also be designed to minimize through life costs.

Provision shall be made for automatic turbine run-up and automatic grid synchronization.

Facilities adjacent to, but separate from the CCR, shall be provided for data amendment, design of software and formats, control, storage and printing of associated documentation, loading into the appropriate system and monitoring of the network and attached equipment. At least one engineering work-station shall be provided for the station, connected to distributed control system (DCS) and provide commercially available gateway for connection to other computer systems.

11.2 Plant Operating Patterns and Staffing

Operational staff shift patterns will be 24 hours per day 7 days per week for main Plant operation such as boiler and turbine operation with the possibility of reduced shift cover for other Plant areas dependent upon the provided Plant systems.

11.3 Commissioning Personnel

The Onshore Contractor shall provide sufficient numbers of competent personnel, who are fully conversant with the commissioning and operation of the Facility, to place the Facility in full commercial operation and also train, instruct and supervise the Owner's operating staff to a standard of safe and efficient operation of the Facility. The Owner will be fully involved in the commissioning process with his staff working to the instructions of the Onshore Contractor as part of the training regime. To keep the O&M staff costs at optimal level, it is critical that O&M staff is recruited close to the time when they are required to support commissioning (and to attend classroom training). Therefore, the schedule of availability of each type (e.g. coal handling system, water treatment system, etc.) of Owner's operators shall be agreed during the construction phase to align with the Contractor's commissioning schedule for the Facility.

11.4 Maintenance

The Offshore Services Contractor shall provide the original suppliers' recommended maintenance duration (the proposed working/shift pattern in order to meet the proposed planned outage duration) and frequency of any minor and major maintenance outages required to maintain and achieve the required plant availability.

The Facility shall be designed such that where maintenance work is required the appropriate Plant or part thereof as applicable can be isolated electrically and if necessary mechanically, drained, vented or purged as required to ensure safety from the system and the isolation made secure by the use of lockable devices.

Adequate access shall be provided for the safe operation and maintenance of all equipment requiring regular access and where local operation of the Plant is required in case the remote control function is inoperative and reverted to manual operation. Final access requirement shall be determined during the detail design stage.

The Offshore Services Contractor shall provide a planned maintenance programme to enable assessment of the preventative maintenance. The programme shall indicate how the planned maintenance will be phased for the whole Plant life in order to maximize the overall Plant availability.

The Offshore Services Contractor shall provide the planned maintenance schedule indicating the duration and frequency of the maintenance outage for each major Plant item i.e. boiler and turbine island. The basis for determining the interval between the planned maintenance outages shall be stated. If equivalent operating hours are used the definition and formula shall be given with appropriate values for the constants for starts, shutdowns and trips. Consideration shall be given to the requirement of outages permitted in Power Purchase Agreement, if any.

The Offshore Services Contractor shall provide the maintenance programme one year prior to the programmed date for commencement of the Tests on Completion for the First Unit.

11.5 Operation and Maintenance Training

The Onshore Contractor shall provide comprehensive training courses at Site to all operation and maintenance (O&M) staff covering all aspects of the Facility, and operations and maintenance. The Offshore Services Contractor shall provide training on control system, Information Management System and Automation Trouble Shooting on MFC cells outside Vietnam. All adequate training resources, experienced tutor, information and material to train the O&M staff of the Owner shall be provided. The necessities and sufficiency of the training scope and depth shall be reviewed and agreed with the Owner. If the training results for some topics are not to the Owner's satisfaction, a repeat training session may be necessary. Training shall be delivered in English, with all training documentation to be provided in English. Training shall specifically include appropriate periods for the Owner's control and

instrument maintenance personnel to be in the manufacturer's works, or training facility for DCS system and familiarisation and maintenance training on Site training during installation of DCS, and training during commissioning. Operators shall be trained on simulator or an operating station that is similar to the supplied DCS.

The Owner will provide operations personnel to be involved in the commissioning process, working to the instructions of the Onshore Contractor as part of the training regime.

The Offshore Services Contractor shall submit a training programme for the O&M staff within one year of the issue of the Notice to Proceed. All types of training shall be completed no later than 8 months prior to the commencement of the Tests on Completion of the First Unit. The number of trainees and duration of each training programme shall be reviewed and agreed with the Owner, as and when required. The Offshore Services Contractor's programme for training and commissioning shall be agreed by the Owner twelve months in advance of the commencement of commissioning of the First Unit or earlier if possible. Training shall commence sufficiently in advance of commissioning to enable staff to become familiar with the Works and intended operational techniques. Training sessions shall be staggered to support Contractor's commissioning schedule and Owner's recruitment schedule (For example, a schedule which is staggered for water system, coal handling system and main plant classroom training courses.). The last session of plant overview training shall be scheduled for the last batch of staff recruited at about 9 months prior to Unit 1 Commercial Operation Date.

The Onshore Contractor shall ensure that all of his commissioning staff is fully conversant with the commissioning, performance testing and operation of the Facility.

The trainees will have appropriate education and experience in their field of activity. The number of trainees will be agreed between the Owner and the Onshore Contractor during construction stage.

To provide training to the batches of operators recruited subsequent to the classroom training, video recording of the classroom training courses shall be allowed and conducted by the Owner. The purpose of video recording is purely for training and shall not be used for any other purpose other than the training of Vung Ang 2 O&M staff. Owner will provide one copy of the video recordings to the Onshore Contractor.

The test period can be used as part of the training of operations personnel.

A complete hardware training course shall be provided for Owner's Personnel. It shall be highly practical, incorporating considerable contact with the equipment and ensuring familiarity with it. The Offshore Services Contractor shall take particular care that equipment included in the system, but manufactured by others, are adequately covered in the training course.

Similarly, a comprehensive software training course shall be provided for the Owner's Personnel. Detailed instructions shall be provided on all aspects of operating system software, system software, and application software. A detailed course of instruction on the writing of application algorithms (database system, control logics,

sequence, graphic mimics, historian and other relevant programs) in the language appropriate to the proposed computers shall be provided.

Training modules shall be designed for maintenance and operation function separately so that requirement of both the functions is covered in detail.

The training shall include plant/system/control/protection design philosophy (concept) in the overview of individual modules.

Simulator

General

computer-based, emulated DCS simulator including, but not limited to, the performance, testing, training, documentation,.

The simulator models shall be developed so that they meet the performance criteria detailed in “Simulator Performance Criteria” of this specification during steady state and transient operations.

A Generic version of the simulator shall be provided in advance for training purpose. This Generic version shall be similar to Vung Ang 2 project and it will be further updated with the modifications and tunings done during the commissioning and tests of the power plant.

Following should be provided:

- Provide one set of Virtual DCS Simulation software and hardware (and one set of Virtual Simulation software and hardware, if applicable).
- The virtual simulator should directly read/convert the files from DCS and manufacturer's standard steam turbine control systems with zero error.
- The simulator shall replicate all graphics and control logics etc. for the Vung Ang 2 DCS and manufacturer's standard steam turbine control systems.
- Plant simulation model for Vung Ang 2 generation unit(s) and associated relevant Balance Of Plant/System/Equipment.
- The simulator response and behaviour shall be similar to Vung Ang 2 and associated auxiliary systems/plants continuously in real time over the entire range of normal, abnormal and emergency operations.
- At least 30% spare computation time and spare memory capacities for background tasks and any expansion.

Purpose

The Simulator shall be designed based on Vung Ang II's plant and control configuration. The Simulator will be used by Owner as a training device supporting a systematic program for training operators in the safe and efficient operation of Vung Ang II. The Simulator shall be adequate fidelity to serve as a training device for normal and emergency operations of the plant. .

The Simulator shall be capable of simulating:

- (a) Complete cold, warm and hot start-ups
- (b) Shutdowns
- (c) Load changes
- (d) Normal operating conditions
- (e) Pre defined Abnormal operating conditions
- (f) Pre defined Emergency operating conditions

The Simulator shall have the capability and fidelity to allow operators and I&C personnel to evaluate control, process and equipment dynamics changes.

The simulator computer values for the principal mass and energy balance will be satisfied and will not detract from training.

Scope

The equipment will be completed for a computer-based simulator installation. .

Simulator Hardware

The simulator shall consist of operator workstations and processors that will run the software model for this Project. The simulator hardware shall consist of:

- Computer system completes with all peripherals, control modulus, and associated support equipment.
- Four operator computer stations with necessary peripheral device(s), e.g. printer, LCD, mouse, keyboard, etc.
- At least one disk to store instructor data, calculations, and programs associated with the system. Storage should be sufficient to save software system, past records and training record
- Instructor station with one computer, one colour CRT or LCD monitor, one keyboard and mouse and one printer in monitoring trainee performance.
- I/O interface: Emulated I/O routing, conversion and processing of signals between the simulator computer, operator interface and instructor's stations.
- Hardware design of the simulator should be available for upgrade/modification..

A drawing showing the proposed configuration and a recommended spare parts list shall be provided. The drawing shall detail how the Simulator and instructor station, their simulation computer, emulation computer, and instructor station will interface.

Simulator Software

The major software functional areas of simulation operation shall include operating system software, applications programs required to simulate the plant systems programs and diagnostics and maintenance support programs. The simulator software shall include:

- Operating system software of latest version Microsoft Windows that can be easily updated or replaced with future recommended operating systems.
- Application programs such that the simulation response is the same as the response of the reference plant.
- Fault injection capability from instructor station including concatenated faults
- An emulated man machine interface (MMI) such that the operator cannot discern any difference between the operation of the simulation and the actual plant DCS.
- Diagnostic software with special peripheral devices to assist in the detection and repair of the hardware system problems

The simulator software shall be capable of evaluating the performance of trainee and generate performance reports

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The Offshore Supply Contractor shall provide a license for the simulator software, on the same terms as the DCS software license, limited to 4 computer stations. License to modify the simulator applications shall be included.

Simulator Performance Criteria

The capacity criteria shall include:

- A list of critical parameters (~30 per unit), the steady state value of each parameter at different load points, and the allowable variation of the parameter during the steady state operation should be provided to the supplier.
- The acceptance tolerance of the algorithm of different components should be around 5-8%, even at different load in steady state.

The Simulator will present to the operator trainees quantitative values of plant parameters within the tolerance specified for those conditions that the Simulator is designed to simulate.

The Simulator will be based on data that pertain to the performance, operation, physical properties, and other characteristics of the reference plant. The Contractor is responsible to collect “base line” performance data on specified reference unit.

Steady State Operation

The Simulator accuracies will be related to full power values and interim power levels for which design plant data is available. Tests will be conducted at several points over the operating range including:

- 100% rated megawatt output
- 75% rated megawatt output
- 50% rated megawatt output
- Minimum megawatt output

The following tolerances, accuracies, and limits will also apply:

Transient Operation

Simulation performance under transient conditions shall meet the following conditions:

- The observable change in parameters shall correspond in time response and direction to those expected from a best estimate for the simulated transient, and will not violate the physical laws of nature
- The Simulator shall cause an alarm or automatic action similar to main plant
- The Simulator shall demonstrate realistic response to operator action
- The shape of any plant performance curve shall be similar to that of the operational power plant. The transient responses of the displayed variables shall have slopes and trends that are characteristic of the actual power plant and shall have the proper time-dependent relationship to the related parameter.

Malfunctions and transients not tested in accordance with the above stated criteria will be tested and compared to best estimate or other available information.

The rate of simulation software processing shall be sufficiently fast to assure that the response of the display meters and recorders shall be smooth and that control delays shall not be noticeable to the operator. Instrument noise that simulates typical variations in steady-state signals from plant processes shall be provided. Instrument noise shall either be random or instructor selectable. Random Noise - The noise frequency shall be random in nature and there shall be no detectable noise correlations between any two displays in close physical proximity to each other. Instructor Selectable Noise - The instructor shall be able to define the amplitude of the noise.

Simulator Operating Limits

The requirements of real-time simulation may necessitate simplification in mathematical modeling that results in limited ability to simulate certain events. In addition, it may be possible to create events on a simulator which progress beyond plant design limits. To avoid negative training which could result from simulator operation during such events, detail design specification of the simulator will allow the instructor to realize when certain parameters approach values indicative of events beyond the implemented model or known plant behavior

Training

The Contractor shall include in its proposal at least two training sessions for the Owner instructors who will operate the Simulator and two sessions for personnel who will maintain and modify the Simulator. The Contractor shall provide training manuals.

Training sessions for instructors who will operate the simulator and personnel who will modify and maintain the simulator shall be provided with descriptions of all courses available.

Instructor training shall include the following:

- Overview of hardware and software, simulation model and modifying simulator models
- Review of instructor functions
- Hand-on practices using the system and building malfunctions
- Training on tuning of the models to match the plant behaviour (training on change to models shall also be included)
- “Trainee Performance Report” function: The simulator instructors shall also be trained on how to define the performance criteria.

Engineer training shall include the following:

- Details of hardware, software, simulation models including process and controls
- Review of instructor functions
- Hands-on practice using the system, building models and troubleshooting
- Details of diagnostic software
- Removal and reinstallation of specific components

The system modelling shall include, but not be limited to, boiler/furnace system, boiler air & flue gas system, fuel burning and boiler control system, steam turbine system, condense system, feed water system, generator system, station electrical system, circulating water system, closed circuit cooling water system, auxiliary steam system, distributed control system.

The generic simulator shall be ready and handed over to the Owner as a pre-condition to the issue of the Taking Over Certificate for Section 1 of the Works, with the modified Project simulator to be handed over to the Owner as a pre-condition to the issue of the Taking Over Certificate for Section 3 of the Works.

If the training for the instructor is to be conducted by the Onshore Contractor at the Site, then the training shall commence upon handover of generic simulator and shall be completed within a month of handover of generic simulator.

Cybersecurity (if applicable) shall include:

- Cyber security of the simulator platform should be considered when developing a simulator even if the simulator is with or without network (solely having its own environment) connectivity with other ICS (Industrial Control System) or external systems

11.6 Spares

Spares for Erection and Commissioning

The Offshore Supply Contractor and the Onshore Contractor shall provide as part of their base scope all of the spares required to be used for erection and commissioning of the Facility.

Spares for the First 2 Years Operation

Both the Offshore Supply Contractor and the Onshore Contractor shall provide a detailed, itemized and priced list of all consumables, breakdown spares and routine spares expected to be used for the first two years' operation and maintenance of the Facility. The Owner will notify the Offshore Supply Contractor and/or the Onshore Contractor as to which items should be supplied, against a Provisional Sum to be set aside for this purpose.

A soft copy of the complete spares list, in a format suitable for input to the particular computerized maintenance management system to be selected by the Owner, shall also be provided.

Spares for Overhauls

Both the Offshore Supply Contractor and the Onshore Contractor shall provide an itemized price list for those O&M spares associated with planned overhauls to ensure plant conditions meet availability or efficiency targets.

Strategic Spares

The Offshore Supply Contractor shall provide an itemized price list for Strategic spares, ie. those non-recurring items or assemblies, required for major overhauls and insurance purpose, which are generally high in capital cost and/or long in lead-time (greater than 6 months) ordering, the deployment of which will either be unique in the life of the Facility or required infrequently.

This category of spares shall only consist of the high cost or long lead items such as generator transformers, steam LP turbine last row blades, BFP cartridge and high volt motors etc.

Spares – General

All spares shall be directly interchangeable with the corresponding parts installed in the Facility and shall meet the requirements of the original Equipment specification.

11.7 Plant Description Manuals, Operating and Maintenance Instructions

The Offshore Services Contractor shall provide 5 complete sets of manuals containing a detailed description of the whole Facility and all the operating and maintenance instructions necessary for the operation and maintenance of the plant throughout its life.

All documents including manuals, instructions, Subcontractor literatures shall be given in English.

The manuals shall contain information suitable for personnel who have received a basic training and/or have a knowledge and experience of similar plant to the Plant.

The manuals shall alert the Owner to any hazard inherent in the Plant or likely to arise in the implementation of operating or maintenance procedures.

Where manufacturer's standard brochures are available in a suitable form and relating specifically to the plant supplied, these may be used providing they are edited to remove irrelevant material and integrated into the overall manual.

The nomenclature, terminology and abbreviations used throughout the manuals shall be consistent and conform to internationally recognized engineering terms.

The manuals shall use ISO A4 format and shall be punched to fit standard four-hole fixings. The use of fold-out pages shall be kept to a minimum. The manuals shall be produced by Xerographic, Lithographic or similar processes. The use of coated papers and Dyeline prints for drawings or scripts is unacceptable.

The content of the manuals shall conform to the table of contents and shall be complete and specific to the Plant supplied under the Contract. The manuals shall cover all the auxiliary systems and equipment supplied under the contract. Material which does not contribute to the understanding of the design, operation and maintenance of the plant shall be excluded from the manuals. Use shall be made of drawings, diagrams, pictures or photographs when they add to the understanding of the text.

The Offshore Services Contractor shall provide details of the standard of the manuals they propose to supply together with a synopsis of the content during detailed engineering stage. The following information shall be included:

- Soft copies of all the manuals and instructions;
- Computer aided design (CAD) format and Adobe Acrobat portable document format (PDF) format of as-built drawings;
- Performance test plans, procedures and results in detail;
- List of plant items with item codes shall conform to Kraftwerk Kennzeichnungs system (KKS) coding system. (soft copy required as an input to computerised maintenance management system);
- Draft manuals shall be submitted for review by Owner; and
- Design basis reports where applicable shall be submitted. (system design criteria)

A plant system handbook shall be provided along with O&M manuals at completion.

11.8 Safety

All plant systems shall be designed for 'fail safe' operation with appropriate audible and visual warning of any plant fault in the control room.

All maintenance operations which in the opinion of the Offshore Services Contractor, have a potential for safety impact on personnel, shall be highlighted as such and a risk assessment relating to these operations shall be provided.

Hazardous Materials

Asbestos shall not be used in this Project. Materials which could be hazardous to health in normal or fault condition, shall not be used in the construction of Plant, Facility or in components used therein. Where such use cannot be avoided, then the prior approval of the Owner shall be obtained, i.e. assessment shall be required and the materials shall be clearly and unambiguously indicated on the equipment, together with any special instructions. Hazardous shall be taken as meaning of explosive, toxic, carcinogenic, mutagenic or biologically active in any way.

The instruction manual and the operating instructions shall detail the extent to which any hazardous materials are used in the Plant, Facility or both. Instructions shall be given regarding the disposal of such materials or components containing or incorporating such materials.

Information relating to hazardous materials need not refer to the use of lead in solder but shall refer to the use of beryllium, mercury, selenium, and other toxic elements or compounds which might constitute a hazard in use or in the event of damage to or overheating of a component or Plant.

Materials used in the Plant shall as far as possible be either non-combustible or shall be resistant to flame propagation.

Vents and safety relief valves shall have their outlets directed away from areas where their discharge could endanger personnel or other plant items.

Plant Isolation for Maintenance

The Offshore Services Contractor shall ensure that each item of Plant comprising the Facility that may require isolation for maintenance purposes shall be capable of being locked off with a standard pattern of padlock. Sufficient suitable padlocks shall be provided for retention by the Owner, to satisfy all isolation requirements during commissioning, normal operation and maintenance periods.

Where the isolation involves high temperature, high pressure or corrosive/hazardous to health working mediums, double isolation facilities shall be provided.

11.9 Special Tools

Before start of manufacture of any equipment package, the Offshore Supply Contractor and/or the Onshore Contractor (if applicable) shall confirm to the Owner whether or not any special tools are required for the operations and maintenance of that equipment. If any special tools are so required, they shall be delivered to the Site together with the equipment concerned. All special tools supplied by equipment suppliers for the operations and maintenance of their equipment are considered to be part of the equipment and are required to be a) listed (consolidated list) upon completion of procurement of equipment, and b) handed over to the Owner at or before Taking Over of Section 2.

Further requirements should be finalized during detail engineering stage.

11.10 Safety Gear

For General Personal Protective Equipment:

General Personal Protective Equipment (e.g. safety shoes, helmet, ear plugs and safety glasses) will be purchased by the Owner for all Owner's staff.

For High Voltage (HV) Earthing and Isolation:

2 sets of the following safety gear for HV earthing and isolation shall be provided:

- HV rubber gloves;
- HV electronic live tester;
- Earthing poles;
- Earthing bar; and
- General purpose portable earthing leads

For Other Safety Gear:

2 sets of the following safety gear shall be provided:

- Gas detector
- Breathing apparatus (BA sets)
- Chemical handling protective gear
- Fire retardant protective wear

Chapter 12 – Commissioning and Testing at the Site

12.1 General

All tests shall be conducted by the Onshore Contractor, and the Owner shall be given the opportunity to witness these tests.

The whole of the Works shall be tested in accordance with the standards adopted for the Project and the Onshore Contractor's quality plans and testing procedures which are reviewed and agreed by the Owner.

All measurements shall be taken with appropriately calibrated test instruments as per the requirements of the relevant Vietnamese standards and international test codes.

12.2 Pre-Commissioning Checks

An erection checklist prepared and maintained by the Onshore Contractor shall be countersigned by the Owner after the satisfactory completion of each part of the Facility. Copies of Site erection inspection document files for items of Facility or systems shall be distributed prior to pre-commissioning.

The correct alignment of equipment which is to be coupled on Site installation with other rotating parts (e.g. motors, generators, reduction gears etc.) shall be demonstrated by the Onshore Contractor to the satisfaction of the Owner. The alignment results shall be entered into an appropriate alignment report sheet which shall be countersigned by the Owner and the Onshore Contractor.

After completion of erection, non-destructive and pressure testing as appropriate, further tests including pre-commissioning function trials shall be made on individual items of Plant and their alarm and shut-down systems to satisfy the Owner that the Plant is ready to be safely commissioned.

The Onshore Contractor shall submit a comprehensive description of each test. This shall include the type and classification of all test equipment and shall be submitted for review at least 90 days before the proposed date on which each test is to be carried out.

Certificates shall be issued for each test made. Group certificates are acceptable for the tests of cabling and similar items.

Pre-commissioning checks on mechanical Plant shall include:

- Individual pre-commissioning runs (test runs) of all rotating equipment such as pumps, dampers, compressors, fans, dosing equipment etc.; and

- Functional tests of the mechanical equipment, alarm and tripping systems.

The Onshore Contractor shall perform testing for all Plant and shall not be limited to the following items of Plant and tests mentioned.

(a) Boiler and Auxiliaries

Boiler pressure parts shall be hydrostatically tested following ASME procedures.

The Onshore Contractor shall visually inspect 100% of site welding of furnace membranes after fit-up and completion of the butt-welding of furnace wall tubes. These inspections shall be readily performed from scaffolding during erection of the furnace walls. When the ducts are fabricated at site, additional dye penetrant/visual inspection of duct seal welds shall be conducted. This sampling welding inspection during site fabrication of ducts shall be readily performed. Inspection of section to section joint welds shall be performed by dye penetrant visually as the sections are erected at elevation in the boiler structure. The Onshore Contractor shall perform thorough visual inspection of all welds forming parts of the gas seal on the furnace enclosure to ensure that air tightness is achieved.

The Onshore Contractor shall conduct air leakage test in cold condition to ensure that the flow path is clear and there is no significant leakage into the system.

Individual pre-commissioning runs of all rotating auxiliary equipment such as fuel oil pumps, fans shall be performed following erection. Boiler pressure parts shall be chemically cleaned, except superheater and reheater which shall be cleaned by steam line blowing.

Boiler pressure parts and materials shall be tested in accordance with ASME. In addition, the tests on the completed boiler shall be acceptable by relevant Vietnamese authorities. The Onshore Contractor shall be responsible for the application for the approval of the boiler installation under the Vietnamese requirements.

(b) Steam Turbine Generator and Auxiliaries

Mechanical and electrical function tests shall be performed on all parts of the turbine-governing and associated systems, and of the generators.

(c) Electrical Equipment

The following checks and test measurements shall be made during erection and pre-commissioning (as appropriate) in accordance with agreed standards and procedures with the Owner:

- Screwed connections for correct assembly

- Terminals and terminal connections for correct assembly
- Checking of earthing connections and testing of earthing resistance
- Measurement of insulation values
- Fire proof partitioning
- Marking, inscription, provision of designation plates
- Voltage checks
- Polarity checks in the case of D.C. voltages
- Fuses, over-current trips, short-circuit trips, time settings, relay settings
- Transformer oil levels
- Safety signs and warning signs
- Setting indicators, feedback signals to the central control room etc.
- Checks on wiring and cabling for conformity with the constructional circuit-drawings and plans
- Dielectric tests
- Trip tests
- Functional test on Plant.

Tests on the transformers

- During installation: checking of complete delivery, checking of core earthing and insulation of active part from tank, HV sample tests of transformer oil in accordance with IEC, or equivalent Standards.
- Checking for perfect electrical and mechanical correctness and for the airtightness of individual main connections system.
- The Onshore Contractor prior to energisation of respective auxiliary equipment or systems as pre-commissioning check shall be reviewed with Owner.
- The Onshore Contractor shall submit schedule of site commissioning test details to Owner for review.

Tests on the earthing and lightning protection system

- Acceptance tests and measurements of the earthing installations in accordance with IEC Standards.

Tests on the lighting system

- Proof of the average and minimum new value of lighting densities, checking of correct operation both electrically and mechanically.

(d) Control and Monitoring Equipment

During the erection period and before the associated plant item is put into operation all control and monitoring equipment shall be tested.

- i) These tests shall include calibration and function tests in addition to on line commissioning for the following:

- Calibration tests of all analog measuring loops including all remote indications and recorders and the input signals used for closed loop control.
 - Testing of all binary transmitters by actuating the plant mounted transmitters. Those which cannot be actuated at the primary element shall be simulated as close as possible to their sensors.
 - Wiring test of all control cabling in the field, relay room and control rooms combined with the function tests.
 - Testing of all control modules in the control room
- ii) Functional testing of remote control of drives, circuit breakers, solenoid valves, actuators etc.
- Testing of open loop devices especially all sequence logic equipment using simulated inputs.
- iii) Testing of all interlocks to ensure safe operation.
- iv) Testing of the alarm annunciation and event recorder system in connection with all field and control room devices using simulated inputs.
- v) Testing of all closed loop controls.
- vi) Testing of insulation between cable screen and ground.

12.3 Commissioning Period

After erection and pre-commissioning checks of individual Plant have been completed, the Onshore Contractor shall submit the test report of pre-commissioning checks to the Owner for review. This is the period in which the Plant shall be prepared, operated and adjusted to obtain specified performance so as to ensure the subsequent initial operation and performing the Tests on Completion.

During the startup and commissioning period, the Onshore Contractor shall maintain documentation of test procedures and data records.

The Onshore Contractor shall provide qualified and experienced mechanical, electrical, instrumentation, operations, testing, and supervisory personnel, as required, to support all phases of the start-up, commissioning, and plant testing.

The Onshore Contractor shall furnish standard and special test instruments, tools, equipment and normal startup consumables (excluding coal, oil and back energization power as per Contract), spare parts, and materials required to perform

testing, startup, and commissioning. The Onshore Contractor shall provide skilled construction craftsmen and labour support as required.

The Onshore Contractor shall arrange and schedule adequate sub-contractor's service engineer for installation, checkout, testing, start-up, and initial operation of Plant, components, and systems, as required.

The Onshore Contractor shall give sufficient detail in its commissioning program to satisfy the Owner that,

- The sequence and duration of the proposed activities are logical, realistic and in accordance with safety and permit regulations in force on the site.
- Owner's operating personnel shall be allowed to work with the Onshore Contractor to gain practical experience during commissioning under the supervision of the Onshore Contractor.

The commissioning shall be performed in accordance with the procedures contained in the commissioning program agreed by the Owner. The commissioning activities shall demonstrate the proper functioning of individual systems and overall Facility in accordance with the Contract's requirements.

The commissioning activities shall cover start-up, no-load and load operation, functioning of safety and control equipment, shut down and down time operation of the entire Facility or Units of the Facility.

During the conduct of the commissioning, a report comprising of observations and recordings of various parameters measured in respect of the above commissioning tests shall be prepared by the Onshore Contractor. The reports, besides recording the details of the various observations during the tests shall also include the start and finish dates and shall be signed by the Owner and the Onshore Contractor. The reports shall have sheets recording all the details of interruptions occurred, adjustments made, modifications and any repairs undertaken or necessary to be made subsequently.

The Onshore Contractor shall ensure the Plant is able to be operated safely prior to initial operation.

Prior to energisation of the Plant, the Onshore Contractor shall complete quality assurance (QA) and quality control (QC) procedures which is agreed by Owner.

The Onshore Contractor shall be responsible for any defective component replacement of the equipment and systems during commissioning period.

The Onshore Contractor shall submit the list of equipment and system which will be energized with construction power supply.

12.4 Initial Operation and Trial Run

The Onshore Contractor shall provide commissioning and start up specialists identified for the period commencing with start up and extending through all tests. During these periods, the Onshore Contractor shall make available all test equipment required to prepare for and conduct the tests. The Owner will make available their operating personnel and necessary supporting staff. The Onshore Contractor's commissioning, and start up supervisory engineering personnel shall train the Owner's Personnel prior to and during this period so that they will be able to operate and maintain the Facility satisfactorily after acceptance by the Owner.

After completion of erection and/or installation and/or before being put into operation, the Units and all appurtenances shall be thoroughly cleaned and inspected, for correctness and completeness of installation. All piping system where applicable shall be flushed, chemically cleaned, steam blown, air blown as required and cleanliness demonstrated using acceptable industry standards. Procedures to accomplish this work shall be subject to Owner's review.

The check outs during the pre-commissioning period shall be programmed to follow the construction completion schedule. Each system, as it is completed by construction and turned over to the commissioning engineers, shall be checked out and cleaned. The checking and inspection of individual systems shall then follow a prescribed schedule. On completion of inspection, checking and after the pre-commissioning tests are satisfactorily over, the Unit itself shall be placed on initial operation during which period the Unit shall be operated integrally with all sub systems, controls and supporting equipment.

When the Unit is operating properly, its characteristics shall be recorded on the start up report sheets. Copies of start up report shall be given to the Owner. Start up reports for all equipment shall be completed before the start of the trial operation period.

12.5 PPA Tests

The following tests are extracted from PPA. Some of them are overlapped with the testing requirement in this chapter. The Onshore Contractor shall perform the test to comply with PPA.

12.5.1 Functional Tests

Functional tests shall be undertaken to demonstrate that key features of the design operate satisfactorily, in particular those associated with plant safety. The Onshore Contractor shall undertake the following functional tests for the First Unit and the Second Unit:

a. Test and start-up of auxiliaries

All auxiliaries shall be tested to verify that they can be operated safely, that their performance is up to the design specifications, and that all protective devices, mechanical as well as electrical, are functioning effectively and at their correct settings. Interlocks which prevent start up under dangerous conditions, the operation of pressure relief devices, over temperature devices, and over current devices are especially important. Automatic start-up of stand-by auxiliaries upon the loss of running auxiliaries shall also be tested.

b. Control System

Automatic control systems shall be tested for correct functioning, although it is anticipated that the final trimming of the controls needs to be done at a later stage. Operation and control of air flow, flue gas flow, feeding coal into furnace, mills, thermal boiler load and turbine load shall be in mode Auto.

c. Test trip of Unit at 100% load, no malfunction or endangering of the equipment shall take place and all controls shall operate safely.

d. Testing the interlocking protection system.

e. Testing the generator and transformer protection system.

f. Testing the batteries, charges and Uninterrupted Power system (UPS).

g. Synchronising Checks

Before the machines are permitted to operate in parallel with other machines, tests shall be performed to ensure that it is safe to do so. These tests can be witnessed by Owner and EVN, and Owner and EVN shall be satisfied that all the instruments associated with the synchronising operation are functioning correctly.

h. Electrical Protective Devices

All electrical protective systems, circuits, devices, and instruments shall be tested on site to prove operation and stability, as well as the compliance of the actual relays and current transformers with the manufacturer's published information. Owner and EVN shall be entitled to receive from the Onshore Contractor full documentation of these tests before accepting any system as operational. This documentation for electrical protection devices shall be used as a basis for the system acceptance by Owner and EVN.

i. Mechanical Protective Devices

Tests on over-speed trip and other mechanical protective devices for turbine(s) shall be conducted to prove the effectiveness of their operation. The speed

governor shall be capable of operating over its range with the droop being adjustable between four and six percent (4% and 6%).

j. Stability

The automatic voltage regulator ("AVR") of the generator(s) shall be checked for proper and stable operation over zero to maximum load at the specified power factor. The tests shall demonstrate that each Unit is able to operate at rated voltage, frequency, and specified range of power factors. Each Unit shall be capable of operation in a stable condition at capacity load of 240,000 kW(net) without supplemental firing.

k. Load Rejection Test

Each Unit shall be able to sustain a 100% load rejection by opening the Generator Circuit Breaker (GCB).

l. Any other functional tests agreed by the Parties where required to demonstrate compliance with the Grid System and Grid Code requirements.

12.5.2 Other Tests

The Onshore Contractor shall perform other tests to demonstrate the suitability of each Unit and the Facility. These tests shall comprise:

- a. Environmental compliance tests;
- b. Noise level test;
- c. Reactive capability (only per Unit);
- d. Minimum sustainable load capability (only per Unit);
- e. Plant automatic startup and loading times and rates from various initial conditions (only per Unit);
- f. Step load change tests (only per Unit);
- g. Ramp rate tests (only per Unit);
- h. Runback tests (only per Unit);
- i. Safety system tests: test all safety valves of boiler; and
- j. House load tests following 100% load cut-off (only per Unit).

Such tests shall meet the requirements of the environmental permits and regulations, technical limits and contracted operating characteristics.

12.6 Record of Initial Condition

The Onshore Contractor shall record the initial condition of the major Plant as listed below. These tests shall not be repeated if the tests have been conducted properly and the result shall be provided as information only and not guarantee.

The Onshore Contractor shall submit the procedures to record the initial condition of the major equipment as listed below for review and agree with the Owner prior to the test.

12.6.1 Boiler

a) **Boiler Maximum Continuous Rating (BMCR)Capacity**

The Onshore Contractor shall record the actual steam flow rate at rated steam parameters at superheater outlet with one standby mill and the coal being fired within range specified, corresponding to 100% BMCR at rated CW temperature and maximum aux. steam consumption with firing Performance Coal.

b) **Part Load Performance**

The Onshore Contractor shall record the actual boiler efficiencies corresponding to 85%, 70% and 40% TMCR with firing Performance Coal.

c) **Mill capacity at rated fineness.**

12.6.2 Turbine Generator

a) **Part Load Performance**

The Onshore Contractor shall record the actual Unit net electrical output and gross turbine heat rate at Unit part load of 85%, 70% and 40% of TMCR.

b) **Turbine Generator Set Capability**

The Onshore Contractor shall record the Unit net electrical output at VWO conditions. .

c) **Condenser**

The performance of the condenser, i.e., the back pressure achieved at design CW flow and actual inlet temperature corresponding to the sizing of condenser and cleanliness factors at VWO condition shall be recorded. The Onshore Contractor shall provide the performance curve of condenser for predicting the condenser back pressure against heat load to the condenser at various CW temperatures.

12.6.3 Control and Instrumentation

- a) The Onshore Contractor shall record the actual performance of control and instrumentation system specified elsewhere in the OTR. The record shall include the parameters within specified limits under various transient and steady state load conditions as per the requirement of C & I in OTR.
- b) The Onshore Contractor shall record the actual performance of the C&I system of the boiler and its auxiliaries and turbine generator and its auxiliaries when integrated with balance of plant C&I system.

12.6.4 Balance of Plant

The following systems of the Plant shall record the actual functions, capacities and performances if applicable:

- Cooling water system
- Fire protection and fire fighting system
- Air compressor system
- Hydrogen generation plant
- Water treatment plant
- Steam purity and water chemistry
- Electrostatic precipitator
- Waste water treatment plant
- Coal handling system
- Mill throughput
- Ash handling system
- Desalination plant
- Service water system
- Potable water system
- Sewage treatment system
- EDG
- FGD SO₂ Removal Efficiency

The measurement of SO₂ of the Unit shall be taken at 100% TMCR load condition at the stack and inlet of FGD. All SO₂ shall be corrected to the basis of 6% excess O₂, 0°C and 1.013kPa dry conditions and expressed in mg/Nm³. The FGD SO₂ removal efficiency is calculated as dividing “corrected SO₂

emission at the inlet of FGD minus corrected SO₂ emission at the stack” by “corrected SO₂ emission at the inlet of FGD”.

- SCR Tests:
 - a. Ammonia slip of SCR
 - b. Ammonia consumption rate
 - c. SCR removal efficiency

12.7 Tests on Completion

12.7.1 Test Sequence

The Onshore Contractor shall perform the tests on Completion as soon as practicable after successful completion of above tests, to demonstrate that the Units (and the Facility, as applicable) meet the Performance Guarantees.

The Tests on Completion shall consist of the following tests:

- Emissions Guarantee Tests
- Performance Guarantee Tests
- Reliability Guarantee Test
- Initial Dependable Capacity Test (IDCT)

12.7.2 Measuring Equipment

All necessary test grade instruments for Performance Guarantee Tests of various equipment and systems shall be brought and installed by the Onshore Contractor. All test and measuring instruments shall be calibrated and test certificates from authorized institution to be submitted to the Owner 30 days in advance of the test.

The calibration certificates shall be valid for a period of 6 months from the date of their calibration. Measurement uncertainties shall be calculated in accordance with ASME PTC codes and standards to verify the quality of the Test.

No tolerances can be applied to the measured and corrected test results.

12.7.3 Test Codes

A detailed test procedure document, including all correction curves, shall be prepared by the Onshore Contractor and shall be agreed with the Owner in advance of the start of the Completion Tests. The procedure shall refer, where appropriate, to the latest edition of Vietnam standards, international test codes and standards (current as at the time of Contract signing) such as:

- Performance test code on fired steam generators
- - ASME PTC 4 (Boiler fuel efficiency based on HHV shall be determined using the energy balance method.)
- Performance test code on steam turbine
 - ASME PTC 6 (Full-Scale)
 - ASME PTC 6 (Alternative Tests - Condensate flow measurement basis, uncertainty @ 1/3%)
- Performance test code on Overall Plant Performance
 - ASME PTC 46
- Steam Tables
 - IAPWS-IF97 for the Thermodynamic Properties of Water and Steam
- Test uncertainty
 - ASME PTC 19.1
- Method of measuring total NO_x concentration in flue gas
 - US EPA Method 7E
- Method of measuring total SO₂ concentration in flue gas
 - US EPA Method 6C
- Methods of measuring particulate in flue gas
 - US EPA Method 5B, 17 and 201A
- Method of measuring flue gas flow rate
 - US EPA Method 2
- Method of measuring O₂ concentration in flue gas
 - US EPA Method 3A
- Method of measuring CO₂ concentration in flue gas
 - US EPA Method 3A
- Method of measuring CO concentration in flue gas
 - US EPA Method 10
- Method of sample and velocity traverses for stationary sources
 - US EPA Method 1
- Set up of Measurement System
 - US EPA Method 20
- Noise far field
 - ISO 6190 or ANSI B 133.8
- Noise near field
 - ISO 3746 or ANSI S 12.36

The Onshore Contractor's test procedure documents shall set out the scope of tests and a programme, the Onshore Contractor's responsibilities and Owner's responsibilities, as well as details on the measurements to be made, methods of measurement, test duration, supply of staff and testing apparatus, methods of calculation, test correction curves, calibrations and reporting etc.

12.7.4 Emission Guarantee Tests

The performance of the Unit when operating in accordance with the Site Performance Conditions shall be measured as follows:-

A. NOx Emission Test

The measurement of NOx emission of the Unit shall be taken at 40% to 100% TMCR load condition at the outlet of the stack. The measurement shall be corrected to the basis of 6% excess O₂, 0 °C and 101.3 kPa dry conditions and expressed as mg/Nm³.

B. Other Emissions Tests

Other emissions guarantees to be tested include SO₂ and particulate. The measurements shall be taken at 100% TMCR load condition at the outlet of the stack. The measurement shall be corrected to the basis of 6% excess O₂, 0°C and 101.3kPa dry conditions and expressed as mg/Nm³.

C. Noise Emission Test

The measurement of noise shall be taken at any load conditions for central control room, Equipment inside the building, Equipment outside the building and at Site boundary according to the procedures and methods specified in Annexure 15 to the Contract - Schedule of Performance Guarantees.

D. Effluent Discharge Emission Test

The measurements of the effluent discharge shall be taken at any load condition at the discharge point at Site boundary specified in the OTR.

12.7.5 Performance Guarantees Tests

The Onshore Contractor shall conduct Performance Guarantees tests to verify the following Performance Guarantees:-

Performance Guarantees at Site performance conditions at 100% TMCR condition as more fully described in Schedule of Performance Guarantees (Annexure 12 of the Offshore Supply Contract, Annexure 11 of Offshore Services Contract and Annexure 14 of Onshore Contract):-

- Net electrical output guarantee at site performance conditions;
- Net heat rate guarantee at site performance conditions; and

Turbine heat rate data shall be furnished by the Onshore Contractor indicating basis and procedures. In case the Onshore Contractor anticipates the use of spray for superheater and reheater during the performance guarantee test, the corresponding effects shall be taken into consideration and the related heat balances shall be furnished with the proposal. The Onshore Contractor shall furnish correction curves applicable for heat rate calculations taking into account effect of ambient air temperature, barometric pressure, relative humidity, power factor, grid frequency, cooling water temperature and sea water level. Corrections curves shall be provided for the corrections to Site performance conditions.

12.7.6 Reliability Guarantee Test

Without prejudice to the more detailed requirements as given elsewhere in the Contract, the Onshore Contractor shall demonstrate that while the Unit is operating in accordance with normal operating procedures (except as necessarily modified for testing purposes) and within the limits of output specified, either continuously or intermittently as may be more convenient for the Owner and in compliance with the specified environmental requirements, all applicable laws and statutory approvals.

The Unit (including all unitized and station common auxiliaries which are deemed by the Owner to be essential for the safe, efficient and reliable operation of the Facility) shall be available for a period of not less than 168 consecutive hours. During this time, the Unit shall operate continuously at full load for at least 72 hours, and for the rest of the period at any load as required by the Owner per what is nominated by grid company without any forced outage caused by the Facility itself.

Should any failure occur in any portion of the Unit, due to or arising from faulty design, material, workmanship, or operation, which requires or causes the Unit available capacity to be reduced, a new Reliability Guarantee Test shall be undertaken after rectification has been completed. If during the carrying out of the Reliability Guarantee Test of the Unit any interruption in operation or reduction in loading occurs as a result of fault at grid connection and the same is not due to a cause attributable to the Onshore Contractor, then for the purpose of determining whether or not the Unit has passed its Reliability Guarantee Test, such interruption in operation or reduction in loading shall be deemed not to have occurred; and during the occurrence thereof, the Unit shall be deemed to have been operating continuously.

The Reliability Guarantee Test procedure shall be agreed between the Owner and the Onshore Contractor prior to the Tests on Completion being commenced.

When Reliability Guarantee Test starts, the generating Unit being tested shall have been in a stable condition at the loading specified by the Owner and the grid company.

12.7.7 Initial Dependable Capacity Tests (IDCT)

IDCT is the last of the Tests on Completion and will be conducted on the First Unit and subsequently on the Facility. It forms the basis for demonstrating the initial dependable capacity of the First Unit at a minimum of 600MW net and the Facility at a minimum of 1200MW net.

Prior to conducting an IDCT, the Onshore Contractor shall ensure that the First Unit or the Facility, as the case may be, is ready in terms of mechanical and electrical stability and can be operated safely. The IDCT shall be undertaken after the relevant Unit's Performance Guarantee Tests and 7 days of the Reliability Guarantee Test have been completed. To establish that the First Unit or the Facility, as the case may be, is ready for the IDCT.

The following shall be in place prior to the IDCT:

- (a) EVN and Owner agree to conduct the test;
- (b) The metering system (including the Main Metering System and the Back-up Metering Systems as defined in the PPA) shall have been installed and functionally tested;
- (c) Instruments shall have been set up to monitor the ambient air temperature and cooling water temperature;
- (d) Means for timing the tests shall have been set up. The test timing devices shall be either:
 - (i) master clock signal from the plant control system (DCS); or
 - (ii) synchronised watches or clocks, as agreed by the Owner.
- (e) All other Tests on Completion and PPA tests as referred to in Section 12.5 of OTR shall have been completed.
- (f) For IDCT of the First Unit, the First Unit shall be ready for dispatch at a minimum of 600MWnet; For IDCT of the Facility, both the First Unit and the Second Unit shall each be ready for dispatch at a minimum of 600MWnet.

The IDCT, performed at relatively constant ambient conditions, shall last for eight (8) hours. The Onshore Contractor, Owner, EVN and independent engineer shall record the readings of the tariff metering system and Site conditions at the beginning of the test and thereafter every hour for tariff metering and every minute for Site conditions until test completion.

During the IDCT, the Onshore Contractor shall provide the overall leadership for the execution of the test and be responsible for the result of the test. Prior to IDCT of the Facility, the First Unit and Common Systems shall be verified in an acceptable condition according to a mutually agreed work programme on Site. In case the First Unit and Common Systems are not found in an acceptable condition, technical reasons and responsibility shall be investigated.

The initial dependable capacity shall be the sum of the kWh recorded in each hour by the tariff metering system corrected to the PPA Site Reference Conditions for the relevant hour during the IDCT, divided by the total duration in hours of the test. The resulting kW shall be deemed the initial dependable capacity of the First Unit or the Facility, as the case may be.

PPA Site Reference Conditions	Ambient air temperature	24.3	°C
	Relative air humidity	84	%
	Ambient air pressure	100.5	kPa abs
	Sea water temperature at inlet	25	°C
	Power factor (at gen terminals)	0.85 lagging – 0.9 leading	-

	<i>Grid frequency</i>	<i>50+/-1.5</i>	<i>Hz</i>
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12.8 Test Report Format

All test reports submitted to the Owner by the Onshore Contractor shall include, but not limited to, the following information

- List of testing equipment
- Test procedures
- Exceptions observed during the test
- Calculated result
- Calculation assumption
- Result discussion
- Conclusion
- Detail calculation steps
- Basis of calculation method
- Equipment calibration certificates
- Raw data log sheets
- Equations/graphs of performance/correction curves applied

In addition, raw data and result calculation in electronic spreadsheet format shall be provided along with the reports by the Onshore Contractor for the Owner to understand the test report.

Chapter 13 – General Technical Requirements

13.1 Overall Requirements

The Units shall be designed, installed, constructed and commissioned to ensure that the Units are suitable to be run continuously at high availability and reliability.

All Plant shall be fit for purpose.

Plant proposed to be used by the Offshore Supply Contractor and/or the Onshore Contractor shall be new and unused and shall be manufactured by manufacturers with sufficient experience of the particular equipment proposed for the Project.

Areas with potential flammable atmosphere shall be classified in accordance with the provisions of the internationally recognised standard. Suitably certified equipment shall be used in the designated hazardous areas. To the maximum extent practicable, equipment requiring operator attention and/or electrical equipment shall not be installed in hazardous areas.

For design of all platforms and footpaths, the minimum clearance height is 2.2 m and all the lifting platforms and walkways shall be provided with stop bars.

Dangerous areas for equipment shall be designed in conformity to the appropriate safety regulations by striving to reduce the hazard to the minimum.

Motorised lifting devices, i.e. hoists, chain pulleys, jacks, etc. shall be provided for handling and carrying out maintenance of any equipment and / or part having weight in excess of 500 kg. Suitable beams, hooks etc. for this purpose shall be provided in the buildings.

All Plant shall be protected from ingress and damage by dust and humid atmosphere and be designed to prevent the entry of vermin, insects or animals which could result in short circuits or damage. Outdoor equipment shall be weatherproof and designed so that water cannot collect at any point. The undersides of all tanks shall be ventilated or suitably treated in an approved manner to prevent corrosion.

Rotating or any other moving part of the machinery, hot parts or any part that may cause accident to the staff shall be adequately protected, in accordance with the safest method known.

A uniform system of padlocks, keys and key cabinets shall be supplied for all the Facility items such as valves, switchgear isolators etc., which have to be locked off for operation safety and/or prior to the issue of "Permit to Work" for the maintenance. All high voltage (HV) room doors should be lockable with a master key. All access ways, platforms stairs, handrails etc. shall be in accordance with the relevant international and Vietnamese standards, whichever is more stringent. The minimum width of access ways shall be 900 mm and the minimum headroom shall be 2200 mm. Where the travel distance to the nearest exit from a platform exceeds 23 m, a secondary means of escape from the platform shall be provided.

Safety showers and eye washes shall be provided in all areas where chemical or hazardous materials are stored or areas that have higher fugitive dust than normal.

13.2 Plant Identification and Labelling

13.2.1 Plant Identification System

International SI system of units shall be used for design, drawings, diagrams, instruments, etc.

The Offshore Services Contractor shall establish a single identification numbering system for items throughout the Facility. A Kraftwerk Kennzeichnungs system (KKS) shall be used to identify all equipment and components of the Facility. The coding of an item shall consist of the following fields:

UU/SSS/SS/EE/CCC/NN

where UU = Unit

SSS = System

SS = Sub-system

EE = Equipment

CCC = Component

NN = Sub-component or other use, if applicable

KKS system shall be applicable to all the electrical (including cable) & mechanical Plant & equipment and civil structures/buildings.

The identification numbers shall be shown on the manufacturer's drawings.

The Offshore Services Contractor shall prepare and submit comprehensive Plant identification schedules to Owner for review showing the name and identification code of each item of the Facility. The codes shall be shown in the respective drawings of the items.

The proposed identification system shall be integrated into the distributed control system (DCS) and other computer management systems to facilitate the searching of items, order replacement and maintenance planning.

The Onshore Contractor shall provide detailed training sessions for the use of KKS system.

13.2.2 Numbering System, Nameplates and Labels

All labels, nameplates, tags, instruction and warning plates necessary for the identification and safe operation of the Facility shall be provided. All nameplates on equipment shall be written in Vietnamese and English. The format shall be submitted to Owner for review during detailed engineering stage.

All labels, nameplates, instruction and warning plates shall be securely fixed to items of Plant with stainless steel rivets, plated self-tapping screws or other approved means. The use of adhesives will not be permitted. Marks shall be undeletable and clearly visible. Marks made only with paint will not be accepted. Nameplates and labels shall not hinder the normal operation of the Plant.

The Offshore Supply Contractor and the Onshore Contractor shall submit for Owner's review a list showing the inscription (in the form which it will appear on the label), type, size and fixing method proposed to be used for each label. The inscription shall include the Plant identification number, Plant system and equipment description and function.

Nameplates for Plant identification and record purposes shall be manufactured from stainless steel with a matt or satin finish and engraved with black lettering of a size which is legible from the working level.

Warning plates shall be manufactured from stainless steel with a matt or satin finish, engraved with red lettering and sited in a position which affords maximum personnel safety.

All equipment within panels and desks shall be individually identified by satin or matt finish stainless steel labels, laminated plastic labels or manufacturer's standard labels subject to the review of Owner.

Each circuit breaker panel, electrical control panel, relay panel, etc. shall have a circuit designation label engraved with black lettering in accordance with the circuit designation system. Corridor type panels shall additionally have circuit designation labels within the panels.

Each valve shall be fitted with a stainless steel nameplate indicating the valve service and reference number.

Valve nameplates shall be fixed where necessary to a 3 mm thick steel backing plate with steel bracketing and painting agreed by Owner. The mounting shall be such that the label is absolutely clearly associates with the respective valve, and fixed in such a way that the valve may be maintained or removed and replaced without disturbing the label. Groups of small instrument valves may be labelled by a single label plate which shows the valve arrangement and otherwise meets the requirements as listed above.

The pipe identification shall include the flow direction arrow in addition to what is specified.

All unfired pressure vessels and lifting tackle shall have permanently attached to it or stamped on it, in a conspicuous position, a rating plate giving such details as the design code, design parameter, test parameter, date of test, etc., to the review of Owner.

Labels shall be provided on front and rear access doors of all cubicles. Labels shall also be provided inside cubicles to assist the identification of apparatus and terminals.

The description and number of each item of Plant shall be consistently applied throughout all drawings, manuals, documents and labels.

Labels/signage for safety, hazardous and dangerous goods including flammable and toxic gases, acids and alkalis shall be in accordance with local works safety requirements and relevant Vietnamese standards.

All interconnecting pipes including those in trenches and all pipes carrying potentially hazardous substances shall be colour coded and have pipe markings on the outer surfaces complying fully with relevant Vietnamese standards.

13.3 Hazards & Operability (HAZOP) Study

A qualified facilitator shall be appointed to organize and run a comprehensive HAZOP study for the whole Facility once the preliminary design for all systems are complete. With the advice of the facilitator, the Offshore Services Contractor shall organize various HAZOP teams consist of personnel of design, operation and maintenance expertise to review the hazards and operability of various systems. The Offshore Services Contractor shall track and report on progress associated with resolving actions raised during the study and present the Owner with a complete dossier showing the original minutes and the actions taken to resolve issues raised. The design shall then be modified to incorporate the recommendations from the HAZOP study.

Wherever possible, the HAZOP study shall be carried out before the finalisation of the design to allow the incorporation of the modifications made

13.3.1 Hazardous Area Classification

The Offshore Services Contractor shall carry out a thorough, comprehensive plant-wide assessment of all areas that could be deemed hazardous in accordance with IEC 60079/10 and shall design the electrical equipment to meet all the requirements of the relevant codes and standards.

The Offshore Services Contractor shall submit a list of potentially hazardous areas as part of the HAZOP Study.

The Owner has identified that the following Plant areas or systems may be classified as hazardous based on industry practice and experience. Such areas include, but not limited to:

- Fuel oil storage area
- Fuel oil pump house
- Turbine lubrication oil system
- Generator sealing oil system
- Generator hydrogen system
- Diesel generator oil storage Area
- Fire fighting pump (diesel driven) oil storage area
- Chemical storage area
- Hydrogen generation Plant and hydrogen pipe routes
- Boiler coal bunker area
- Coal conveyors and transfer towers
- Cover coal yard
- Coal feeders and pulverizer mill system
- Boiler burner operating floors
- Chlorination Plant
- Absorbing tower of seawater flue gas desulphurisation (SWFGD)

13.4 Piping, Valves & Fittings

All the piping systems be designed to operate without replacement and with normal maintenance for a Plant service life of 40 years, and shall withstand the operating parameter fluctuations and cycling which can be normally expected during this period.

This section of specification covers all low pressure piping handling fluids like raw water, clarified water, cooling water, demineralised water (DM) water, diesel oil (IDO), condensate, compressed air, hydrogen etc.

13.4.1 Codes and Standards

The design, manufacture, shop testing, erection, testing and commissioning of piping and valves shall conform to the latest revisions of the following codes and standards and others which are acceptable to Government Bodies of Vietnam.

Examples are as follows:

Table A - Material for construction for low pressure pipes

BS: 5142	Cl globe valve.
ANSI:B 16.5	Steel pipe flanges and flanged fittings.
ANSI:B 31.1	Power Piping code.
ASTM A 106, Gr. C	Schedule 80 - Seamless carbon steel pipe
ASTM A 312 Grade TP-316	Schedule 40s - Stainless Steel pipes
ASTM D 3517 / 4024	Glass reinforced plastic (GRP) pipes

Table B - Material of construction for pipes carrying various fluids

S.No	Service	Recommended Material
1.	Raw water	Carbon Steel /GRP
2.	Clarified water/Filtered water	High density polyethylene (HDPE)/GRP
3.	Potable water	Buried pipe: HDPE Above ground pipe: Stainless steel pipe (ASTM A312-TP304)..
4.	Fire fighting water	Buried pipe: HDPE Above ground wet pipe: Carbon steel A106 B Above ground dry pipe: Carbon steel A53 B galvanised
5.	Acidic Water	Rubber lined Steel/stainless steel (SS)
6.	Demineralised Water	Mild steel (MS) Rubber lined /stainless steel (SS)
7.	Acid (hydrochloric)	Mild steel (MS) Rubber lined /stainless steel (SS)
8.	Alkali (Sodium Hydroxide)	
a)	Strong (5% & above)	Stainless Steel
b)	Dilute	Rubber-lined/ Stainless Steel
9.	Alum Solution	Rubber-lined Steel
10.	Lime Solution/Suspensions	Galvanised Steel
11.	Coagulant aid Solution	Rubber-lined Steel
12.	Liquid & Gas Chlorine (Under Pressure)	Seamless carbon steel schedule 80 (Heavy Duty)
13.	Chlorinated Water	Rubber-lined Steel
14.	Wet Chlorine gas (Under Vacuum)	Polypropylene
15.	Sludge	Cast Iron/ Ductile Iron
16.	Instrument Air	Galvanised Steel
17.	Waste effluent from DM	HDPE/ Mild steel (MS) Rubber lined

13.4.2 Valves

All valves, shall be suitable for service conditions i.e. flow, temperature and pressure under which they are required. The valves shall be of standard pressure rating as per the relevant code/ standard. All valves shall be listed, and major control valves and actuated valves specified on specific data sheets to be returned for Owner's review.

Sluice valve/knife edge type slide valves shall be used in the sludge and drain pipe line.

Valves for acid & alkali services shall be proven patented diaphragm type. Diaphragm shall be of reinforced teflon or equivalent for acid services and reinforced neoprene/hypalon or equivalent for alkali services.

13.4.3 Piping Runs

Piping runs shall be arranged aboveground as far as practicable. When the Site conditions dictate other arrangement, the following criteria shall be observed.

Underground : Living, fire fighting, storm drains and waste water pipes;
Overhead : Hydrogen, fuel gas, fuel oil and heating pipes;
Overhead or inside trench: Compressed air, oxygen, acid, alkaline and demineralised water;

Racks shall be used for cabling and piping which crosses roads. Where piping, valves and fittings are installed below ground (either in trenches, pits or buried in ground), they shall be made of suitable material to withstand corrosion due to accidental flooding or chemical attack. Where piping are installed in trenches or underground, the design shall allow for free and ready access to reach all parts of the pipe run and, in particular, valves for repair and maintenance. The underground pipe except plastic materials shall be wrapped by appropriate wrapping materials for corrosion protection. In addition, there shall be sufficient clearance between adjacent pipe runs for maintenance.

No pipework shall be run in trenches carrying electrical cables. Centralized piping support racks, and piping trenches shall be used as far as practicable.

The piping layout and arrangement shall be clearly shown on flow diagrams, pipe routing and erection drawings. Pipe dimensions, materials, operating conditions, insulation thickness and piping and equipment identification numbers shall be shown on the relevant drawings.

The Offshore Services Contractor shall design all piping systems with the piping arranged in orderly pipe runs giving due consideration to pipe expansion, valve locations, practical pipe supporting and access to all piping accessories for operation and maintenance purposes.

All saturated steam and compressed air pipeworks shall be arranged with a slope of not less than 1:100.

Piping arrangements shall be provided for the removal and/or maintenance of equipment and equipment internals without removing the piping. A minimum of 2.2 m headroom shall be maintained to the bottom of all piping components or insulation in walking areas.

A minimum passageway of 750 mm shall be provided for access between and around all equipment and a 50 mm minimum clearance shall be provided between piping, including insulation, and any point of adjacent equipment or piping.

Additional clearance shall be provided, where required, for pipe movement due to thermal expansion.

Control valves, strainers, orifices, valve motor drives and all other piping accessories shall be provided with adequate space for instrument piping manifold and/or removal of parts for maintenance. Valve hand-wheels, bracket supports and other protruding attachments which constitute a hazard to operating personnel shall be elevated to a minimum of 2.2 m from the operator's access elevation. All horizontal runs of outdoor piping shall be supported by pipe racks with a maximum distance between supports of 6.0 m. Pipe accessory handles, stems, linkages, instruments, brackets, etc., shall not project into passageways. All instruments shall be positioned to allow easy observation.

A 230 mm minimum clear space shall be provided from the bottom surface of pipe to trench bottom or finished grade.

All piping shall be grouped where practicable and shall be routed to present a neat appearance while making maximum use of the Plant, steel and concrete structures for hanging and support.

13.5 Electrical, Control and Instrument Panels / Cubicles

All the external wiring/cabling/PVC conducts shall be UV rays resistance.

13.5.1 Panel Lighting and Heaters

Panels containing apparatus that requires regular routine testing and maintenance shall be fitted with a lamp (or lamps) to adequately illuminate all wires and apparatus as evenly as possible without glare.

Electrical equipment including switchgear, panels and cubicles shall be provided with totally enclosed, metal clad type anti-condensation heaters with thermostat control. The surface temperature of the heater with which accidental contact may be made shall not exceed 60 °C.

13.5.2 Wire and Terminal

Internal wiring shall be of copper conductor. All wiring shall be protected from abrasion and the location and routing shall minimize the risk of damage from heat, steam, oil and hydraulic fluids.

To prevent accidental contact, shrouds or covers shall be provided over terminals that are energized above 50 V. Low voltage 3 phase alternating current (AC) circuits shall be separately bunched and segregated with appropriate warning labels. They shall not be run in common wiring troughs or through flexible trunking with circuits at other voltages unless the higher voltage wiring has double insulation.

Adequate segregation of circuits susceptible to alternating current “pick-up” from cables carrying AC shall be provided.

Terminal block for outgoing connections shall provide 20% spare ways, in excess of those required for active cores. Not more than two connections may be made to a terminal on one side of a terminal block. Internal and external connections of a cubicle etc. shall not be on the same side of a terminal.

Terminals shall be numbered systematically and all wires shall have an identification marker at each end numbered. Terminal numbers and wire marker shall be shown and matched in relevant drawing.

13.5.3 Electromagnetic Interference

Equipment shall not generate electrical or electromagnetic interference at a level which could be detrimental to the performance of other equipment or which could cause annoyance or discomfort to personnel.

Particular attention shall be paid to minimize the effects of radio frequency interference which could emanate from a 3 Watt effective radiated power (ERP) mobile radio telephone operating at VHF or UHF. In this connection, undue reliance shall not be placed upon the fact that the equipment is earthed, as at radio frequencies the earth impedance may be significant. As a minimum, the operation of 3 Watt ERP radio (walkie-talkie) shall pose no detrimental effect to the control and distributed control systems, in particular, the keyboard, mouse and trackball of these systems.

13.5.4 Panel / Cubicle Classification

Equipment housings shall normally be made of appropriately coated steel sheet. Adequate framework shall be constructed internally to support the housing and the equipment mounted on it. Panels / cubicles placed outdoor shall be made by stainless steel or hot dip galvanized steel.

The appropriate class shall be determined from the following table for the enclosure and its enclosed equipment for the intended location.

Class	Typical Environment	Ambient Temperature	Maximum Relative Humidity	Degree of ingress protection (IP) rating Protection
1	Air-Conditioning Area	0°C - 40°C	90%	IP32
2	Outdoors or Equipment-mounted Area	7°C - 50°C	100%	IP55
3	Seafront, salty or corrosive area	7°C - 50°C	100%	Stainless Steel IP55

13.6 Local Instruments

Instrumentation shall perform sensing, indicating, transmitting functions as required for safe, efficient, reliable operation of the Facility.

Where required, shock absorbing mountings shall be supplied for instrumentation equipment. Instrumentation shall be protected from the weather and also accidental tampering.

Field instruments shall be grouped in rack with rain shelter.

13.6.1 Transmitters

The transmitter shall be of the 2-wire design or of smart type and its output shall be isolated from earth and input. The transmitter output shall be clamped below 125% of the maximum rated output (e.g. 25 mA for 4-20 mA transmitter) during input overload, input fault, output short circuit or other fault conditions. Transmitters shall be capable of withstanding open or short circuits of the output without damage.

The transmitters except head or rail mounted temperature transmitter shall be fitted with an integral indicator and shall be provided with terminal strip with screw terminals for ring lug connections to which all outgoing and test connection are made. Pin-type terminal lug according to product specification can be accepted if ring lug connection is not practicable.

13.6.2 Temperature Measurement

Temperature measurements and controls shall have linearizing facility for non-linear temperature sensors such as thermocouple/resistivity temperature detector (RTD)/thermistor such that the output from the circuit is linearly proportional to the temperature measured.

RTDs shall be industrial heavy duty type and shall be contained in an electrical insulating material and a sheath of sleeve of stainless steel. Resistance thermometers shall comply with and shall have been type tested in accordance with Standard BS 1904 or equivalent. Type Pt100 resistance thermometers shall be used unless otherwise specified.

SS316 stainless steel sheathed thermocouples with pockets shall be used unless otherwise specified. Only K-type and E-type thermocouples shall be used throughout the whole Facility.

Thermocouples shall be manufactured from a continuous length of cable from the hot junction to the first termination box or cold junction cabinet whichever is appropriate in the design concerned. Compensation cable shall be used for further extensions.

Where measurement of a small temperature differential at high absolute temperature is required, thermocouples may be used providing that they are manufactured from the same batch of material. The differential signal shall preferably be obtained by connecting the thermocouples 'back to back' at the first terminating box from the hot junctions. The terminal arrangement shall be designed to minimize the risk of picking up stray EMFs. The transmitter or the measuring device shall be able to detect the positive or negative differential signal.

For high accuracy requirements, 3 wire bridge measuring system shall be used. It shall comply with ANSI MC96.1 and ASTM E-230-87.

13.6.3 Pressure Measurement

Pressure and differential pressure sensing elements shall be stainless steel type, unless otherwise specified on the instrument data sheet. Pressure, including absolute pressure, transmitters shall be designed for an over range pressure greater than 100 psig or 150% of the maximum sensor range without affecting the calibrating of the instrument. All pressure measurements on steam & water service over 70 °C and the length of signal pipelines less than 3 m shall be adopted with siphons. Pulsating dampers shall be furnished.

All pressure transmitters and switches shall be located as far as practicable near to the equipment to minimize the pipe length to avoid the pressure drop and high pressure lines. Filters and snubbers shall be added to pressure measurement pipework as required to reduce undesirable effects that cause fluctuations and other imperfections on the measured signal.

Only one pressure measuring device for control or interlock or protection purpose is allowed to be connected to one tapping point. Not more than two pressure measuring devices for indication purpose shall be connected to one tapping point.

13.6.4 Flow Measurement

The flow of liquid, steam and gases shall be measured by the differential pressure unless otherwise specified. The sensing element shall be of venture nozzle, insertion multi-throat type or orifice with stainless steel material.

Positive displacement meter and turbine meter shall be used for the following conditions:

- They shall be used where a higher degree of accuracy and reliability in liquid flow integration is required than that which can be obtained by orifice.
- For flow metering of oil.

The Offshore Services Contractor shall furnish the flow elements with certification reports and in accordance with the requirements of ASME "Fluid meters, their Theory & Application".

13.6.5 Level Measurement

Temperature correction shall be provided whenever necessary (e.g. at oil tank, separator, etc.). Level transmitters analogue output signal shall be directly proportional to the level of the measured process medium unless otherwise specified. Local level indicator shall be provided.

Level transmitter for vacuum service shall be of the displacement type and shall be provided with 50 mm socket-welded connections.

Local level indicators of tanks shall be of the magnetic type and shall be equipped with a scale of sufficient length according to the tanks. For local measurement at pressured vessels the bi-colour water indicator type shall be used, for clear readability, electric illumination shall to be installed.

Indicators on tanks under vacuum shall have special sealing system. All indicators shall be provided with fast closing valves and protective sleeves.

13.6.6 Actuator

Motorized actuators shall have motor protection, a hand wheel for manual operation, position and torque limit switches, and remote position signal. Fault signal (such as power supply, air supply and signals in case) shall be provided. Preferably, all actuators shall be supplied from one manufacturer.

For modulating duty either pneumatically or electrically powered devices shall be used. Each modulating actuator shall be equipped with some forms of failure detection equipment which shall trigger any necessary control action to be taken for Plant safety in the event of actuator failure detection. Actuators with hydraulic (in place of pneumatic) or electric is preferred.

Failure detection shall include loss of output signals, loss of feedback signals as well as loss of motive power to the actuator.

All actuators used for isolating duty shall be provided with local open/close controls mounted on the actuator but accessible only via a lockable local/remote selector switch.

Suitable isolation shall be provided between C&I and electrical cable in the actuator terminal box to avoid possibility of shorting. The selection of pneumatic actuator in various areas shall follow “fail safe” philosophy.

13.6.7 Hardwired Indicators

Electrical indicating instruments shall comply with IEC 60051 Direct Acting Indicating Electrical Measuring Instruments and their Accessories.

Digital indicating instruments shall have an accuracy of $\pm 0.1\%$.

Scales shall be printed in black figures and divisions on a white background. The quantity measured shall be marked on the instrument dial in black capital letters in abbreviated form. Initials only of the manufacturer together with instrument serial numbers shall be printed in the lower right hand side of the dial. The alarm, trip and normal operation regions of the measurements shall be emphasized by providing colour bands throughout the scale. Colours shall generally be red for trip, amber for alarm and white for normal operating regions. It is envisaged that the above colour coding shall be done beside rather than superimposed on the scale divisions and numbering.

Motor current indicators located in the central control room (CCR) shall use current transducers as the source of measurement.

Motor current indicators shall be graduated 0 to 125% with motor full load current corresponding to 100%; the indicators shall be able to handle motor starting current of up to nine times full load current without damage.

Instruments shall be capable of operating with an electrically offset zero where they share signals with the control system.

Instruments on panels shall be flush-mounted and provided with a narrow bezels having a uniform high grade matt finish. Low reflection glass shall be used on all instrument faces.

Instrument cases shall be dustproof.

Instruments shall be adequately shielded to prevent stray fields influencing the reading.

Instruments with circular scales shall have a minimum pointer deflection of 240 degrees.

Normal working indication shall be at a point corresponding to approximately 75% of full scale deflection.

Instruments shall have black bar-type pointers, except instruments having more than one point, in which case only one of the pointers shall be black.

Suspensions for instrument movements shall be of the taut band type of the spring-loaded jewel and pivot type. Instruments shall be adequately damped. End stop devices shall be provided.

Digital indicators shall have a minimum character height of 20 mm. Characters shall be of the high luminosity type on a dark background. Where indicators are installed under high level lighting, digital indicators shall be provided with shades to improve the intensity of the characters.

Devices for routine zero adjustment shall be accessible from the front of the instrument. Devices for routine re-calibration shall be easily accessible.

If alarm initiating devices are provided integrally with instruments, they shall not restrict the movement of the point. The alarm contacts shall be adjustable.

Multiple range instruments shall be provided with a range selector mounted on the front of the instrument.

Terminals of electrical instruments shall be in the form of studs projecting not less than 40 mm from the rear of the instrument case. The studs shall be completely threaded and provided complete with nuts and washers. Instruments with metallic cases shall be provided with similar 20 mm long studs for connection to the station earth grid.

13.6.8 Indicating Lights

Indicating lights shall be of the miniature type using LED clusters which shall be replaceable from the front of the panel without the use of tools.

Lenses shall be of translucent glass. Plastic lenses will be accepted if fitted with a robust anti-shrink ring. Painted lenses, coloured glass lamps or coloured lamp covers are not acceptable. The colours of the lenses shall be permanent.

Colours for indicating lights, illuminated pushbutton domes and screen based graphical Plant symbols shall be used consistently and in accordance with the colour standards approved by Owner during the detailed engineering stage.

13.6.9 Operating Hours Counter for Major Equipment

Counters for actual record of operating hours shall be provided for at least but not limited to the following equipment:

- Generator Transformer
- Unit Transformer
- Mill
- Stacker-reclaimer
- Electric boiler feed pump
- Fire fighting pump
- Fuel oil pump
- Diesel Generator
- Air compressor
- Cooling water (CW) Pump
- Sodium hypochlorite (NaClO) generator
- Resin in DM Plant
- RO system
- Condenser ball cleaning system

13.7 Prohibited Materials

The Facility shall not incorporate:

- Asbestos or asbestos-containing materials
- Cast iron for any oil service;
- PCBs
- Askerals
- CFC's or other ozone depleting substances
- Carcinogenic materials, or
- Any other materials generally known in the construction industry at the time of use to be harmful to health.

13.8 Corrosion & Paint

Due attention shall be paid to the environmental conditions prevailing at Site subject to seawater corrosion. All painting systems including the proposed surface preparation, application of primer and topcoat materials shall be reviewed and agreed by the Owner during the execution stage.

All metallic surfaces except stainless steel and galvanized steel of the Facility that are subject to corrosion shall be properly cleaned (e.g. sand-blasted) and prepared prior to the application of suitable anti-corrosion primers and top coats in the factory shops or on Site. All paints shall be products of renowned paint suppliers and painting standards shall be provided.

Where dissimilar metals are in contact or close proximity and corrosion may occur through electrolytic action or differences in electric potential, protection shall be afforded by electroplating, suitable gaskets, cathodic protection or other means.

All surfaces shall be adequately protected in transit and any damage shall be renovated immediately on off-loading and on completion of erection.

Materials subject to corrosion, including all exterior equipment surfaces and structural steelwork, shall be protected with suitable coatings systems in relevant International or Vietnamese Standards.

Surfaces that will be inaccessible after shop assembly shall be finish coated/painted for lifetime protection. All access ladders, security fencing and hand railing etc. shall be hot-dip galvanized. Galvanized coating shall also be applied to steel fencing, handrail and platform, where deemed necessary.

Colour strip indication system shall be used for pipes. These strips shall be painted on the joint of pipes, entrance, valves of pipe. This pipe without outside protection layer shall be marked by some colour in whole length. RAL code for piping shall be submitted for Owner's review during detailed engineering stage.

The principal colour of field equipment is outlined as follows:

	Equipment	Colour	RAL Color Code
Major building			
1	Sandwiched corrugated metal sheet wallboard	Main colour: navy blue	5002
		Auxiliary colour: grey white	9002
2	Sandwiched corrugated metal sheet wallboard	Light grey	7035
3	Architectural external wall painting	Main colour: grey white	9002
		Auxiliary colour: Navy blue	5002
4	Indoor steel handrail	Light grey	7035
5	Outdoor steel handrail and stair	Light grey	7035
6	Steel door	Light grey	7035
7	Wood door	Light grey	7035

	Equipment	Colour	RAL Color Code
8	Steel window	Light grey	7035
9	Steel door	Light grey	7035
10	Crane beam	Navy blue	5002
11	Steel column and beam	Agate grey	7038
12	Steel coal bunker	Light grey	7035
13	Crane	Signal yellow	1003
14	Boiler coping	Navy blue	5002
Boiler			
1	Boiler top	Navy blue	5002
2	Boiler coating	Agate Grey	7038
3	Flue gas duct from SWFGD	Silver grey	7001
4	Boiler steel supporting		7000
5	Air heater	White aluminum	9006
6	FD Fan	Light grey	7035
7	PA Fan	Light grey	7035
8	Induced air (ID) Fan	Light grey	7035
9	Coal mill	Light grey	7035
10	Coal feeder	Light grey	7035
11	Seal air fan	Light grey	7035
12	Core air fan	Light grey	7035
13	Air compressor	Sky blue	5015
14	Equipment in oil system	Traffic yellow	1023
15	Ash handling system	Light grey	7035
16	Coal handling system	Light grey	7035
17	Stacker cum Reclaimer	Navy blue	5002
18	Coal crusher	Navy blue	5002
Turbine			
1	Overhead cranes	Signal yellow	1003
2	Main steam turbine	Light blue	5012
3	Gland steam condenser	Pale green	6021
4	Gland steam condenser fans	Pigeon blue	5014
5	Main oil pump	Sulphur yellow	1016
6	AC lub oil pump	Sulphur yellow	1016
7	Direct current (DC) lub oil pump	Sulphur yellow	1016
8	Jackling oil pump	Sulphur yellow	1016
9	Oil tank fans	Sulphur yellow	1016
10	Oil cooler	Sulphur yellow	1016
11	Main oil tank	Sulphur yellow	1016
12	Generator	Light blue	5012
13	Hydrogen cooler	Light blue	5012
14	DM Water cooler	Pale green	6021
15	Hydrogen water cooling pump	Pale green	6021
16	DM water cooling tank	Pale green	6021

	Equipment	Colour	RAL Color Code
17	Condenser	Silver grey	7001
18	High pressure (HP) heater	White aluminum	9006
19	LP heater	White aluminum	9006
20	Deaerator	White aluminum	9006
21	Deaerator tank	Pale green	6021
22	Vacuum pump	Pigeon blue	5014
23	Condensate pump	Pale green	6021
24	Boiler feed pump	White aluminum	9006
25	Dirty oil storage tank	Clay brown	8003
26	Oil storage tank	Sulphur yellow	1016
Turbine pipework			
1	Main steam and reheat steam pipes		
2	Extraction steam pipes		
3	Other steam pipes		
4	Condensate pipes (insulated)		
5	Condensate pipes (non-insulated)	Pale green	6021
6	Feed water pipes		
7	DM water and Chemical dosing pipes	Pale green	6021
8	Drain pipes		
9	CW water and Industrial Water pipes	Signal black	9004
10	Oil pipes	Sulphur yellow	1016
11	CO ₂ and N ₂ pipes	Pearl light grey	9022
12	Air pipes	Distant blue	5023
13	Underground pipes	Signal black	9004
14	Support	Silver grey	7001
15	N ₂ pipes	Silver grey	7001
Electrical			
1	Generator excitation panel	Gray	7004
2	11kV panel	Grass green	6019
3	415V panel	Gray	7004
4	Generator Transformer	Olive grey	7002
5	Unit Auxiliary Transformer & Station Transformer	Olive grey	7002
6	22kV generator isolated phase busbar (IPB)	Pastel Turquoise	indoor 6034 outdoor 7002
7	11kV busbar	Grass green	indoor 6034 outdoor 7003
8	Generator excitator casing	Olive grey	7002
9	Electric panel in control room	White ivory	1015
10	415V AC low voltage (LV) motor	Gray	7002
11	11kV AC HV motor	Grass green	6019
12	220V DC motor	Orange	2007

	Equipment	Colour	RAL Color Code
13	Electric local control panel	Grey	7004
C&I			
1	DCS panel	Light ivory	1015
2	SWFGD DCS panel	Light ivory	1015
3	Balance of plant (BOP) programmable logic controller (PLC) control panel & console	Light ivory	1015
4	CCR panel	Light ivory	1015
5	CCR console	Light ivory	1015
6	Electronic room panel	Light ivory	1015
7	Air preheater local panel	Agate grey	7038
8	Local panel, cubicle junction box and termination box	Agate grey	7038
Chemical			
1	Acid storage trough	Red orange	2001
2	Alkali storage trough	Signal yellow	1003
3	Acid pump	Red orange	2001
4	Alkali pump	Signal yellow	1003
5	Acid spray	Red orange	2001
6	Alkali spray	Signal yellow	1003
7	Anion exchangers	Leaf green	6002
8	Cation exchangers	Leaf green	6002
9	Mixed beds	Leaf green	6002
10	Resin trap	Grass greet	6010
11	DM water pump	Pale green	6021
12	DM water tank	Pale green	6021
13	Washing pump	Grass greet	6010
14	Back-washing pump	Leaf green	6002
15	Resin transport Trolley	Grass greet	6010
16	Water pump	Grass greet	6010
17	Acid metering tank	Red orange	2001
18	Alkali metering tank	Signal yellow	1003
19	Acid metering pump	Red orange	2001
20	Alkali metering pump	Signal yellow	1003
Water supply			
1	Cooling water pump	Pigeon blue	5014
2	Washing pump	Pigeon blue	5014
3	Rotating type screen	Pigeon blue	5014
4	Service water/Potable water/Wastewater system	Pigeon blue	5014
5	Potable water pump	Pigeon blue	5014
6	Service water pump	Pigeon blue	5014
7	Mechanical filtration equipment	Grass greet	6010
8	Sanitary sewage wastewater equipment	Grass greet	6010

13.9 Thermal Insulation

All pipeline insulation materials, cable materials and wall sealing materials shall be of the non-combustible type.

Any hot surface ($>60\text{ }^{\circ}\text{C}$) shall be insulated with an approved material covered by aluminium cladding so that the temperature of the outside of the cladding or protection does not exceed $60\text{ }^{\circ}\text{C}$ when the ambient temperature is $25\text{ }^{\circ}\text{C}$ and assuming a wind speed of no more than 4 m/s for outdoor and 1 m/s for indoor and the surface in question is not in direct sunlight.

Where regular inspection of the equipment is required, the insulation and/or protection shall be readily removable. Where access across pipework or insulation is required, the insulation shall be protected by checker plating or other suitable protection.

Access doors to the boiler casings and the exhaust stacks shall be insulated to a similar degree as the casings. Insulation of doors shall be removable and replaceable so that access can be gained without causing damage to nearby insulation.

Cladding shall be made of aluminium and not less than 0.7 mm. (22 SWG) in thickness and shall be adequately supported to prevent sagging or excessive deflection. All aluminium cladding shall be capable for preventing surface oxidation. In the case of exterior surfaces, it shall be designed to resist wind loadings. Due allowance shall be made in the design and fixing of the cladding for thermal expansion of the lagged surface. Where necessary, expansion joints shall be provided in insulating materials. All joints in cladding shall be waterproof.

Thermal insulation of valves, strainers and fittings shall be of modular design structure to allow efficient dismantling and installation and “cladding and insulation” together. Ceramic insulation or equivalent blankets shall be used to insulate turbine cylinders, steam chests and loop pipes.

Cladding and insulation over all flanges and valves shall be easily removable and replaceable without disturbing the surrounding insulation. The separate portion of the cladding and insulation shall extend sufficiently far along the pipe to allow the joint bolts and nuts to be withdrawn without disturbing the cladding and insulation on the pipe. The casing over flanges shall have a drain hole to indicate the presence of a leak.

13.10 High Pressure (HP) Valves

For the pressure above 4 MPa in thermal system, all valves shall be welded excepting valves integral with the main Plant and instrument valves. Flange connection valves are not allowed unless they are specifically agreed by the Owner.

Parallel slide or globe valves shall be used for all high pressure piping including but not limiting to steam and HP feedwater.

Manually operated HP valves shall be provided with reduction gear unit for valves of size 250 nominal bore (NB) and above. Valve provided with motorised or pneumatic actuator shall be provided with a hand wheel for manual operation.

All the HP valves shall be fitted with a position indicator, which clearly shows the open and shut positions. Facilities shall be provided for isolation locking of the valves in the open and closed positions using a padlock where required for equipment isolation or personnel safety.

Chapter 14 Selective Catalytic Reduction System (SCR)

14.1 General

An SCR system shall be designed, procured, supplied, installed, commissioned and tested for each boiler unit to reduce the nitrogen oxides of the flue gases entering the chimney. The performance of the SCR shall be guaranteed for all loads from 40% TMCR to 100% BMCR, not greater than the limit specified below.

$$\text{NOx} < 84 \text{ mg/Nm}^3$$

Above figure is measured at hourly average, volumetric dry, at 273K, 1 atmosphere and 6% O₂, when firing the range of coal in accordance with Chapter 2, paragraph 2.2.2 of the Owner's Technical Requirements.

The anhydrous ammonia consumption rate per Unit, while still meeting NOx of 84mg/Nm³ on a one (1) hour average from load range 40% TMCR to 100% TMCR, when firing the range of coal in accordance with Chapter 2, paragraph 2.2.2 of the Owner's Technical Requirements shall be not more than 280 kg/hr.

The Contractor shall provide the auxiliary power and ammonia consumption for NOx Emission levels (420 and 300 and 84 mg/Nm³).

The SCR shall be designed to handle flows from 0 to 110% of the design flue gas at full load. The SCR shall be designed to accommodate all reasonably foreseeable operation scenarios, including but not limited to, long term outage, operation of the SCR after a long term outage, short term outage, and all other possible modes of start-up, ramp-rate, load following, operation, and shut-down. The SCR shall be designed for an expected plant services life of 35 years, except the catalyst lifetime which shall be not less than 24,000 operating hours.

The SCR shall utilize anhydrous ammonia as reactant. The SCR shall operate in conjunction with the designed anhydrous ammonia handling system, ammonia unloading system, ammonia delivery system, ammonia supply, electrical or steam heated ammonia evaporators and ammonia storage system that shall provide an ammonia stream to the SCR. Storage for ammonia shall be sized for 30 days consumption, based on two units operating at full load. Design of handling and storage facilities shall comply with the local Vietnamese Standard on Hazardous Materials.

The Offshore Services Contractor shall design the SCR with a spare catalyst layer for the future installation of an additional layer of catalyst as part of a strategy to optimize catalyst exhaustion.

The ammonia consumption rate shall be designed for both as new condition and at the end of the 24 months life of catalyst, when the Unit is firing coal within the design

range at TMCR condition and the corresponding NO_x emission level not to exceed the above NO_x performance guarantee. The design rates shall be given for the design coal. For other coals within the design range, the design figures shall be those corrected for the actual coal particulars in accordance with the sets of correction curves to be provided by the Offshore Supply Contractor after the Commencement Date.

14.2 Scope of Supply

The scope of SCR shall include but not limited to the following:

- A reactor system including reactor housing, catalyst, flue gas flow mixing devices, dilution air system, ammonia distribution and ammonia injection system;
- Ammonia unloading system;
- Ammonia evaporators;
- Ammonia storage and handling system;
- Ammonia dilution facility complete with effluent sump and effluent pump;
- Dilution air fans;
- Nozzles and distribution pipework;
- Catalyst cleaning system;
- Catalyst hoist;
- Gas path, complete with ductwork, dampers, SCR bypass
- Safety related systems including fire protection system, eyewash shower facilities etc.;
- Ammonia analyzer to monitor ammonia slip measured from SCR outlet;
- NO_x analyzer at inlet and outlet of SCR;
- Electrical and control equipment;
- Sonic horns;
- Steam Sootblowers;
- Compressed air system; and
- Operation and maintenance access platforms and staircases;
- Spare DCS I/O; and
- QRA and HAZOP Study on SCR, ammonia plant and ammonia pipes;
- Instrumentation and Control System and interface with DCS.

14.3 Design Principles and Operating Capability

As a minimum, Offshore Services Contractor shall comply the following design principles:

- The SCR shall be designed for an expected plant service life of 35 years.
- Model flow test shall be carried out to verify the design of flow mixing and distribution devices to given homogenous and uniform distribution of gases over the catalyst.
- Removal efficiency of SCR shall be designed to meet the above emission limit;

- Flue gas flow rate shall be designed for 110% BMCR;
- SCR shall be controlled by DCS;
- The design life of catalytic elements shall not be less than 24,000 operating hours;
- The ammonia vaporizing process shall be designed by electrical or steam heated evaporators to provide a stable ammonia stream to the SCR reactor. The ammonia vaporizing system shall be sized to output 105% of the SCR ammonia flow requirements of two units operating at full load.
- SCR shall be designed to operate for the full loading range (40% TMCR – 100% BMCR) of the boiler ; and
- Ammonia slip shall be designed to be not more than 5 ppmvd (dry, 6% O₂).

14.4 SCR Reactor System

The reactor system shall be designed to include the reactor access doors, catalyst modules of plate type, catalyst seals, catalyst replacement system (with carts and hoists), catalyst ash cleaning system, grid and screen, and other components to form a complete SCR. The flue gas shall pass through the SCR once the minimum reactor temperature is achieved. Space of one layer shall be designed for additional catalyst layer to be installed in the future. A gas side temperature regulation system for maintaining the minimum reactor temperature shall be designed. The SCR shall be designed with permanent thermocouples used to measure the flue gas temperature at the economizer outlet and shall also include analyzers for measuring NO_x concentration upstream and downstream of the catalyst.

The catalyst composition and geometry shall be designed to minimize plugging of catalyst open area by fly ash, erosion of leading edge of catalyst surfaces due to abrasive nature of the ash, and poisoning from trace elements. Catalyst sealing shall be designed to prevent flue gas bypassing the catalyst and the sealing shall be at least of the same lifetime of the catalyst elements. Sample catalyst coupons shall be designed to be located in each layer to allow for catalyst activity analysis. The catalyst shall be supplied by any of the Permitted Subcontractors of the Contract.

All materials in contact with the product gas from the ammonia conversion system shall be compatible with ammonia, carbon dioxide, moisture, trace compounds, and other reaction products from ammonia conversion system.

The entire SCR shall be designed to meet the safety codes applicable to ammonia. Due consideration of design shall be given to hazard classifications, labeling, fire and explosion, exposure to human, spill and leaks, water and soil pollution, protection and emergency treatment procedures. The areas housing the SCR and its auxiliaries shall be classified according to the hazardous area classification requirements. All electrical equipment located in the hazardous areas shall be explosion proof design and certified in accordance with the applicable international standards. HAZOP study on entire SCR shall be conducted.

The Onshore Contractor shall construct access platforms, stairs, and access doors as required for maintenance, testing, and inspection of the entire system, and removal/installation of the catalyst layers. Personal access doors shall be of a minimum opening of 600 mm x 600 mm. Additional platforms and access shall also

be provided for equipment, instrumentation, and flue gas sample ports and/or test locations, as required. Trolley beams and electrically actuated hoists shall be included for removal and replacement of the catalyst to and from grade elevation.

14.5 Catalyst

Sufficient catalyst shall be provided to treat the flue gas to meet the NO_x emission requirements without requiring off-line cleaning, rejuvenation, or replacement for the specified guaranteed life span. The catalyst “performance guaranteed life” shall be designed with at least 24,000 hours during which the NO_x emission limit and ammonia slip levels shall be complied with. Catalyst operation hour is defined as an hour of time the catalyst is exposed to flue gas.

The catalyst composition and geometry shall be designed to minimize:

- plugging of catalyst open area
- erosion of the leading edge or catalytic surfaces, due to abrasion
- pressure drop across each layer
- poisoning from trace elements

Catalyst materials supplied should not to promote or support combustion.

14.6 Gas Distribution System

The Offshore Supply Contractor shall provide all gas mixing, gas distribution devices, and gas turning vanes that are required to meet the above performance guarantees.

The gas mixing and distribution devices shall be designed to break up local gas phase concentrations, velocity profile, and temperature striations and to produce overall mixing for uniform concentrations, uniform velocity, before the SCR catalyst. The gas mixing and distribution devices shall be designed for low-pressure drop while achieving the required mixing performance. The gas mixing and distribution devices shall be designed not to plug with ammonium compounds.

14.7 Ammonia Injection System

The Offshore Supply Contractor shall be responsible for the design and supply of the equipment for the ammonia injection system in full accordance with all applicable local regulations and code requirements. Operation and maintenance procedures established for this equipment shall also be developed in full accordance with all applicable local regulations and code requirements. The equipment supplied shall be a complete, functional, and safe system.

The ammonia injection, upstream of the reactor, shall be designed and spaced for thorough mixing of the ammonia into the flue gas stream. The Offshore Services Contractor's design for the ammonia injection grid at the inlet to the SCR reactor shall ensure a uniform concentration of reagent is delivered and mixed with flue gas so that a homogeneous distribution is delivered across the catalyst bed. The injection

nozzle system shall be designed with the ability and flexibility to balance at site the NH₃/NO_x distribution during startup. Each ammonia injection zone shall be easily adjustable outside of the duct. All piping handling ammonia shall be stainless steel of grade 316L.

14.8 Ammonia Vaporizing System

The Offshore Supply Contractor shall provide all equipment and safety-related items required for the ammonia vaporizing system with steam heated evaporators. The system shall provide a stable ammonia stream to the SCR reactor. The ammonia vaporizing system shall be sized to output 105% of the SCR ammonia flow requirements of two units operating at full load.

14.9 Ammonia Storage and Handling System

The Offshore Supply Contractor shall provide complete on-site ammonia storage and handling system. The system shall have the redundant capacity to store ammonia for 30 days of operation of two Units at full load. The storage of the ammonia shall be divided in three storage tanks. The system shall be designed to be capable of unloading anhydrous ammonia from standard container, or by ammonia tanker. The On shore Contractor shall provide the Ammonia in the provision of full storage tanks (100%) (first fill). Owner will provide the Ammonia for the commissioning and tests as established in the Annexure 3 of the Contract (Owner Supplied Materials and Utilities).

14.10 Control and Instrumentation

The SCR control and instrumentation system shall function automatically over the entire operating range. It shall include all equipment including local instruments, final actuators, I/O, cable, enclosures, and control logic required for control of all Contractor furnished equipment and systems. The system shall be designed and controlled to follow all changes in NO_x mass flow without exceeding NO_x or residual ammonia emission guarantees, and all other Contractor guarantees.

Sufficient alarms shall be provided to ensure that the operators are alerted to any abnormal condition or equipment failure. Interlocks shall be provided to ensure an orderly shutdown of the SCR in the event of the operation of a protective device or an equipment failure.

Instrumentation shall include, but not be limited to, continuous monitoring of system temperatures, pressures, flowrates and inlet and outlet NO_x and O₂ concentrations. 2 of 3 voting is required for any parameter that directly or indirectly causes a runback or trip of the SCR.

The SCR shall include permanent control thermocouples for measuring the flue gas temperature at the SCR inlet. The SCR shall include temporary thermocouple grids

used to measure the flue gas temperature at each SCR upstream of the catalyst layer, and downstream of the catalyst layer.

During startup, injection of ammonia shall not be permitted to initiate until the SCR catalyst is at proper operating conditions. The permissive for ammonia injection shall be contingent upon adequate temperature being available both downstream of the last catalyst layer on cold startup and upstream of the ammonia injection grid on shutdown. The Offshore Services Contractor shall design the minimum allowable continuous operating temperature for ammonia injection.

An ammonia leak detection system shall be provided in the ammonia storage and injection plant areas. The system shall monitor, alarm and shutdown the ammonia source if dangerous levels of ammonia are detected. Detectors shall be located near any SCR equipment that may potentially leak ammonia. The ammonia leak detection system shall monitor ammonia concentration in plant areas, provide local alarms in each area and provide input to the SCR control system. Any additional fire protection systems, such as water spray or ventilation that may be required to meet the statutory requirements, shall be provided.

Chapter 16 – Switchyard

16.1 General

The Phase 2 Vung Ang 500 kV Switchyard (Phase 2 Switchyard) shall be an extension to the existing Phase 1 Vung Ang 500 kV Switchyard (Phase 1 Switchyard). The Phase 2 Switchyard shall be designed to match with the design and equipment supply of Phase 1 Switchyard. A survey of Phase 1 Switchyard shall be carried out by the Onshore Contractor.

The Onshore Contractor shall be responsible for the design, engineering, procurement, construction and commissioning of the Phase 2 Switchyard.

The Phase 2 Switchyard shall provide the following as an extension to the existing Phase 1 Switchyard.

- Two (2) bays for transmission lines 500kV Ha Tinh no. 2 and Dang Na no. 2,
- Two (2) generator transformer bays. The choice between underground cable feeders or overhead lines shall be determined during design stage.

The proposed configuration is set out on the following drawings:

- Drawing no. TTDN-VA2-EL-10.02: MAIN SINGLE LINE DIAGRAM (Figure 16.3);
- Drawing no. TTDN-VA2-EL-10.03: GENERAL LAYOUT ARRANGEMENT (Figure 16.4); and
- Drawings no. TTDN-VA2-EL-10.04 & 10.05: RESPONSIBILITY OF INVESTMENT – SECTION (Figure 16.5 & 16.6).

16.2 Scope of Work

16.2.1 Common Equipment and Services

The following supplies and services shall be included:-

16.2.1.1 General

- a) All tests and inspections mandated by legislation;
- b) Engineering design of the complete equipment supply including interface coordination;
- c) All as-built documents;
- d) Quality control plan and safety plan;
- e) Operating manual;
- f) Detailed operating and maintenance instructions;
- g) Maintenance program;

16.2.1.2 Electrical, instrumentation and control

- a) All local measurements and field control loops (thermometers, pressure gauges, local regulating devices, etc.) as well as all testing and checking instruments.
- b) All necessary electrical drives;
 - i. complete installation material including wiring, cabling and piping material, all fastenings, conduits, brackets, cable trays and other supports;
 - ii. all required junction boxes and cubicles;
 - iii. all field control boxes.
- c) All instruments mounted on instrumentation racks;
- d) Outdoor lighting;
- e) Lightning protection;
- f) Electrical earthing of the equipment;
- g) All logic interconnections including sequencing, interlocking, protection, safeguarding of the equipment operation/start-up/shut down.

16.2.1.3 Civil

- a) Design and construction of all buildings and foundations;
- b) Detail design, calculations, supply, fabrication, installation and finishing works for all steel structures;
- c) Earthing;
- d) Fire protection;
- e) Painting/varnishing;
- f) Air conditioning systems where required, e.g. control cubicle rooms, etc.

16.2.1.4 Packaging and transportation

- a) Suitable packaging and transportation of the equipment supply;
- b) Transport insurance;
- c) Disposal of packing and transportation material;
- d) Customs clearance;
- e) Crane or hoisting facilities at seaport and site;
- f) Transportation to site;
- g) Unloading at site
- h) On-site temporary storage and transportation.

16.2.2 Detailed Supply Scope

The Onshore Contractor shall provide all the necessary equipment, materials, workforce, etc., to complete the project in every respect. The detailed scope supply is set out in the following sections.

16.2.2.1 Electrical System

The supply scope shall cover the design, engineering, procurement, erection, inspection and testing of all equipment/materials necessary to complete the Phase 2 Switchyard, including the following systems:

- a) 550kV Circuit breakers;
- b) 550kV Voltage transformers;
- c) 550kV Current transformers;
- d) 550kV Disconnectors and earth switches;
- e) Mechanical interlocks between isolators and earth switches;
- f) Surge Arresters;
- g) Post insulators and string insulators;
- h) All towers, gantry towers and equipment supports;
- i) Voltage (>1kV) connection within the switchyard up to the interface points;
- j) LV Switchgears and Distribution Boards;
- k) 220/48V DC unit power supply systems with battery chargers and batteries;
- l) UPS systems;
- m) All voltage (>1kV) conductors and/or underground cables, LV cables, FO cables and accessories;
- n) Cable trays and supports, conduits and fittings for cables within the Phase 2 Switchyard and up to interface point with other facilities;
- o) Protection and measuring systems for:
 - i. Generator transformers circuits outgoing OHLs and the Da Nang and Ha Tinh substations;
 - ii. LV switchgears and Distribution Boards as necessary;
- p) The main and back-up no.1 tariff metering systems and meter data collect systems for Vung Ang II Thermal Power Plant;
- q) Grounding/earthing;
 - i. The Onshore Contractor shall provide the design, supply and installation of the earthing grid with earthing conductors, earthing rods, etc. for Phase 2 Switchyard up to interface with the existing Phase 1 Switchyard earthing grid,
 - ii. Design, supply and installation of grounding/earthing of all electrical systems and equipment within Phase 2 Switchyard to the main Switchyard earthing grid;
- r) Lightning protection for structures and equipment;
- s) Indoor and outdoor lighting and socket outlet systems;

- t) Network Communications and SCADA/EMS - Synchronous Digital Hierarchy (SDH) Network and Access Equipment Plesiochronous Digital Hierarchy (PDH) as necessary for SCADA and tele protection purpose;
- u) CCTV system;
- v) Collecting grid data, performing electrical system studies (including the sub-synchronous resonance study), relay protection calculation and setting;
- w) Sufficient spare parts and special tools for 2 years operation;
- x) Protection system, telecommunication system and accessories at Ha Tinh and Da Nang substations, Viettel room (18 Tran Nguyen Han, Hanoi), National Load Dispatch Center, Northern Load Dispatch Center.

16.2.2.2 Instrumentation and Control

The Phase 2 switchyard control system shall be an extension to the existing HV Switchyard Control and Monitoring System of Phase 1 Switchyard. The Phase 2 Switchyard systems shall be fully integrated with the existing HV Switchyard Control and Monitoring System of Phase 1 Switchyard. The Onshore Contractor shall design, manufacture, test, inspect, supply, deliver, erect, install, test and commission the computerised HV Switchyard Control and Monitoring System of Phase 1 Switchyard.

- a) Extended Control System with bay control units, network devices and cables, servers, workstation, engineering station, printer including desk and chair as necessary;
- b) Interface panel for connection to Vung Ang II Thermal Power Plant DCS;
- c) All required input data for DCS implementation;
- d) Sufficient spare parts and special tools for 2 years operation.

16.2.2.3 Civil Works

The Scope of Civil Works will include but not be limited to the following:-

- a) Design and construction of all buildings, equipment, substructures, and superstructures;
- b) Detailed design, supply, fabrication, installation, construction and finishing works for all steel structure and access, platforms, cable supports including all necessary walkways, ladders and handrails required to provide convenient access for their operation, inspection and maintenance.

16.2.2.4 Miscellaneous

The Onshore Contractor shall also provide the following in this scope of supply:-

- a) All Cranes, stationary hoists, lifting tackle and other lifting devices required to access and maintain the equipment;
- b) Cleaning, paint and protective coatings of all equipment and surface protection requirements during and after manufacture, at the site, and after erection and testing;

- c) Anti-condensation heater equipment and anti-fungoid preparation for outdoor electrical plant and I&C equipment. The Onshore Contractor shall provide preventative measures to eliminate/minimize the congregation of insects at night;
- d) Tools and maintenance equipment;
- e) Fire protection during construction;
- f) Power, water and telecommunication for construction and commissioning;
- g) Consumables, etc. required for construction, erection and commissioning;
- h) Special tools and fixtures as required;
- i) Supply of welding rods, filler wire oxygen, acetylene, propane, argon, nitrogen gases and the like for all welding, heating, burning and purging operations on Site;
- j) All equipment for connections and attachments of plant items to concrete or steelwork;
- k) All necessary temporary works support including cranes, vehicles, trailers, scaffolding, dampers, ladders, temporary accommodation, offices, toilets, washing facilities, consumable materials, accommodation for the Onshore Contractor's expatriate and local staff, canteen facilities, tools, machine tools, welding machines, stores, workshops, welding rods stores, temporary electrical equipment, temporary roadways required by the Onshore Contractor for his own use, quality control facilities and equipment and temporary compressed air supply;
- l) All necessary additional tapping points, complete with isolation for commissioning tests and performance tests of all plant supplied under this Contract;
- m) All other necessary equipment for the satisfactory, safe and efficient pre-commissioning, commissioning, testing and operation of the plant;
- n) Site offices with all necessary working facilities and accommodation for the Onshore Contractor's staff;
- o) All outdoor panels shall be stainless steel.

16.2.2.5 Services

Onshore Contractor's scope of work shall include the complete design, manufacture and delivery of equipment to Site; the construction, erection, testing and commissioning of the equipment including the provision of:-

- a) A first aid facility until the Taking Over Certificate is issued. The Onshore Contractor shall provide the necessary qualified personnel to support this facility. The first aid facility will revert to the Owner upon completion of the Works;
- b) All necessary information and co-ordination during the Contract;
- c) Design drawings necessary to operate and troubleshoot the Onshore Contractor supplied equipment;
- d) Periodic design and scheduling meetings as required;
- e) Factory tests on material and equipment for the Works;

- f) All necessary measuring and recording precision instruments necessary to carry out the required commissioning and performance tests on all plant and materials supplied under this Contract;
- g) Photographs and video recordings of important events, critical erection and commissioning of the Contract ;
- h) The Onshore Contractor shall clean the Site, his own accommodation and office facilities and generally keep the Site in an orderly and safe state;

16.2.3 Interfaces and Limit of Supply

The Onshore Contractor shall actively manage interfaces with the Owner, authorities and external personnel participating in the Project.

The Phase 2 Switchyard interfaces shall be as defined below:-

SYSTEM	CONNECTION POINT
1. Interface to existing Vung Ang 500kV Phase 1 Switchyard	- Bus-bar connection: connecting two bays to the existing Vung Ang 500kV switchyard bus bars.
	- Switchyard control system: Connecting network devices/bay control devices to existing switchyard control system. The Onshore Contractor shall provide any equipment/material/manpower necessary to connect and upgrade the switchyard control system hardware and software.
	- SDH network: connecting to the existing system.
	- LV power supply: at the Vung Ang 500kV Switchyard LV Switchgear.
	- Earthing/grounding: connecting to the existing Vung Ang 500kV Switchyard earthing system.
	- Lightning: connecting to the existing Vung Ang 500kV Switchyard lightning system.
	- CCTV: connecting to the existing Vung Ang 500kV Switchyard CCTV system
	- Fire protection : connecting to the existing Vung Ang 500kV Switchyard fire protection and fighting system
2. Interface to the Grid	
2.1 At Vung Ang 500kV Switchyard	- Outgoing OHL connection: The Onshore Contractor is responsible for the connection at the gantry tower of the switchyard.
	- SCADA/EMS: Onshore Contractor to coordinate with A0/A1 to design and implement the signals exchange list with LDC into the SCADA system.
	- Series capacitor and shunt reactor: AC/DC supply source and connecting cable, integrating their control signal list into the switchyard control system.

2.2 At Da Nang 500kV Substation	Connecting to and extending the existing teleprotection system. Connecting new 48VDC power supply for the installed teleprotection equipment to the existing LV power supply system if EVN agrees.
2.3 At Ha Tinh 500kV Substation	Connecting to and extending the existing teleprotection and telecommunication systems. Connecting new 48VDC power supply for the installed teleprotection equipment to the existing LV power supply system if EVN agrees.
2.4 At A0	Connecting to the installed telecommunication system at Viettel Telecom. Connecting new 48VDC power supply (if any) for the installed telecommunication equipment to the existing LV power supply system if EVN agrees.
2.5 At A1	Connecting to the installed telecommunication system at Viettel Telecom. Connecting new 48VDC power supply (if any) for the installed telecommunication equipment to the existing LV power supply system if EVN agrees.
2.6 With Viettel Telecom	Connecting to and extending the existing telecommunication system. Connecting new 48VDC power supply (if any) for the installed telecommunication equipment to the existing LV power supply system if EVN agrees

16.3 Co-ordination with Statutory Bodies and Agencies

The Onshore Contractor shall carry out all the necessary co-ordination & liaison work required by the Electrical Inspector, Factory Inspector or other statutory bodies. The Onshore Contractor shall submit the inspection application including copies of drawings and certificates before requesting the Electrical Inspector & other statutory bodies to inspect the Works. The Inspection application must be made well ahead of time so that the actual commissioning of equipment is not delayed. The Onshore Contractor shall arrange for the actual inspection work by the Electrical Inspector. Any official fees paid to the Electrical Inspectors/statutory bodies, etc. shall be borne by the Onshore Contractor.

16.4 Codes, Standards and Regulations

The entire scope of work shall be carried out in accordance with established engineering practice, this OTR and the relevant Vietnamese specifications and codes.

16.4.1 Design Codes, Standards and Regulations

All equipment and materials shall be designed, manufactured and tested in accordance with the latest applicable Vietnamese Standards and IEC except where modified and/or supplemented by the OTR. The Works shall also satisfy the following minimum requirements:

- a) Circular no. 25/2016/TT-BCT dated 30 November 2016 by Ministry of Industry and Trade (MOIT) regarding regulation of the power transmission system;
- b) Circular no. 39/2015/TT-BCT dated 18 November 2015 by MOIT regarding regulation of the power distribution system;
- c) Decision no. 19/2006/QĐ-BCN dated 11 July 2006 by Ministry of Industry (now MOIT) promulgating the codes of electricity equipment;
- d) Circular no. 40/2009/TT-BCT dated 31 December 2009 by MOIT regarding the national electrical technical regulation;
- e) Circular no. 04/2011/TT-BCT dated 16 January 2011 by MOIT regarding the national electrical technical regulation;
- f) The national technical regulation on electricity safety QCVN01:2008/BCT;
- g) Circular 44/2014/TT-BCT dated 28 November 2014 by MOIT regarding the operation procedure of the national power system;
- h) IEEE-Std 80-2000: Guide for safety in AC Grounding Systems;
- i) Structures such as towers, beams, supports and their foundations etc. are to be designed in accordance with the following standards:-
 - i. TCXD 51-1984: Drainage – Outdoor network and facility – design standard,
 - ii. TCVN 9379:2012: Building structures and foundations - basic calculation rules,
 - iii. TCVN 2737-1995: Load and impact – design standard,
 - iv. TCVN 9346-2012: Concrete and reinforced concrete structures – requirement of protection from corrosion in a marine environment,
 - v. TCVN 5575-2012: Steel structures – design standard,
 - vi. TCVN 5574-2012: Concrete and reinforced concrete structures - design standard.
 - vii. QCVN QTD-5: 2009/BCT: National Technical Codes for Testing.

Major standards covering the manufacture and testing of equipment are listed below.

16.4.2 Standards covering the selection of primary equipment

The latest IEC standards shall apply to equipment selection.

IEC 62271-100	High voltage AC circuit breakers;
IEC 62271-102	AC disconnectors;

IEC 62271-200	Air insulated switchgear
IEC 61869-2	Current transformers;
IEC 61869-5	Capacitor voltage transformers;
IEC 60099- 4	Surge arresters;
IEC 60273	Insulators;
IEC 60305; IEC 60120	String insulators;
IEC 60502	Power cables;
IEC 61089	Round wire concentric lay overhead electrical stranded conductors.

16.4.3 Codes, standards and regulations covering the control and protection requirements

- 16.4.3.1 Circular no. 40/2014/TT-BCT dated 05 November 2014 by MOIT stipulating the procedure for dispatching national power;
- 16.4.3.2 Circular no. 42/2015/TT-BCT dated 01 December 2015 by MOIT regarding the regulation of power metering;
- 16.4.3.3 Decision no. 176/QĐ-EVN dated 4 March 2016 by Vietnam Electricity stipulating the regulations for control systems to 500kV, 200kV, 110kV substations;
- 16.4.3.4 Decision no. 55/QĐ-ĐTĐL dated 22 August 2017 by Electricity Regulatory Authority of Vietnam regulating the technical requirements, management and operation of the SCADA system;

16.4.4 Codes, standards and regulations covering telecommunication

- 16.4.4.1 Fibre optic cable: ITU-T G.655, ITU-T G.652, TCN:68-160, IEC-60793, 60794
- 16.4.4.2 Transmission equipment: ITU-T G.704, ITU-T G.707, ITU-T G.708, ITU-T G.709, ITU-T G.957, ITU-T G.691, TCN 68-173:1998, TCN 68-178: 1999, ITU-T G.703, ITU-T G.707, ITU-T G.7041, G.7042 and IEEE 802.x, ITU-T G.841, ITU-T G.842 and ETSI, IEC-60384-1(10/1999); IEC-60870-2; IEC 61000-4-2 , ITU-T G703
- 16.4.4.3 PCM equipment: ITU-T G.703, ITU-T G.714, Q552, ITU-T V.24, RS232.
- 16.4.4.4 Modem: EIA RS-232-C/CCITT V.24/V.28, EIA RS-422/RS-485/CCITT V11, CCITT V23. 2- or 4-wire.

16.5 Electrical Work Safety Requirements

16.5.1 Running Out or Lowering on Overhead Line Conductor which is under or over Other Live Overhead Lines

16.5.1.1 Introduction

Under normal condition, overhead line circuits should not run underneath or across another overhead line. However, when it is deemed necessary to run out or lower overhead line conductors in the vicinity of other live overhead lines special precautions shall be taken to ensure the safety of personnel and equipment. Under some situations, outage of the live overhead line may be required in order to provide a safe working environment.

16.5.1.2 Scope

This section sets out the necessary precautions and requirements to be taken when running out or lowering overhead line conductors under or over other live overhead lines from low voltage to 500kV.

16.5.1.3 Electrical safety, codes, standards and regulations

- a) The National Technical Regulation on Electrical Safety: QCVN01:2008/BCT;
- b) Circular no. 31/2014/TT-BCT date 02 October 2014 by Ministry of Industry and Trade on stipulating certain details of electrical safety;
- c) Warning label shall be bilingual (English & Vietnamese).

16.5.1.4 Working near overhead line conductors

- a) Running out / lowering of overhead line conductor over / under Live High Voltage conductors shall require:-
 - i. A Competent person to conduct a pre-work risk assessment on site to determine the risks to the Safety Distance if there is live conductor located near the work site;
 - ii. An Authorized Person to define the limits of safe access in section 16.6.1.4.b) below before work commences.
 - iii. An Authorized Person, with the consent of a Control Engineer, to cut out the live conductors auto-reclose circuit(s) facility if necessary.
- b) An Authorized Person to define the limit of safe access by assessing the Safety Distance from a Live Voltage >1kV conductor taking into consideration:-
 - i. Insulator and conductor movement due to wind conditions;
 - ii. Tools and equipment to be used;
 - iii. Work nature, and

iv. Points of work.

16.5.1.5 Running out or Lowering of Overhead Line Conductors above Live Voltage (>1kV) Overhead Lines

- a) Where a possibility exists that the Safe Distance from a live high voltage conductor cannot be maintained when running out or lowering on overhead line conductor . Then:-
 - i. An Authorized Person shall assess the work method to ensure that adequate guards are provided to prevent any infringement of the Safety Distance, or
 - ii. The Live Voltage (>1kV) overhead line underneath shall be isolated and earthed. A Permit to Work shall be issued for the work. The Permit to Work shall make specific reference to the overhead line. (see Figure 16.1).

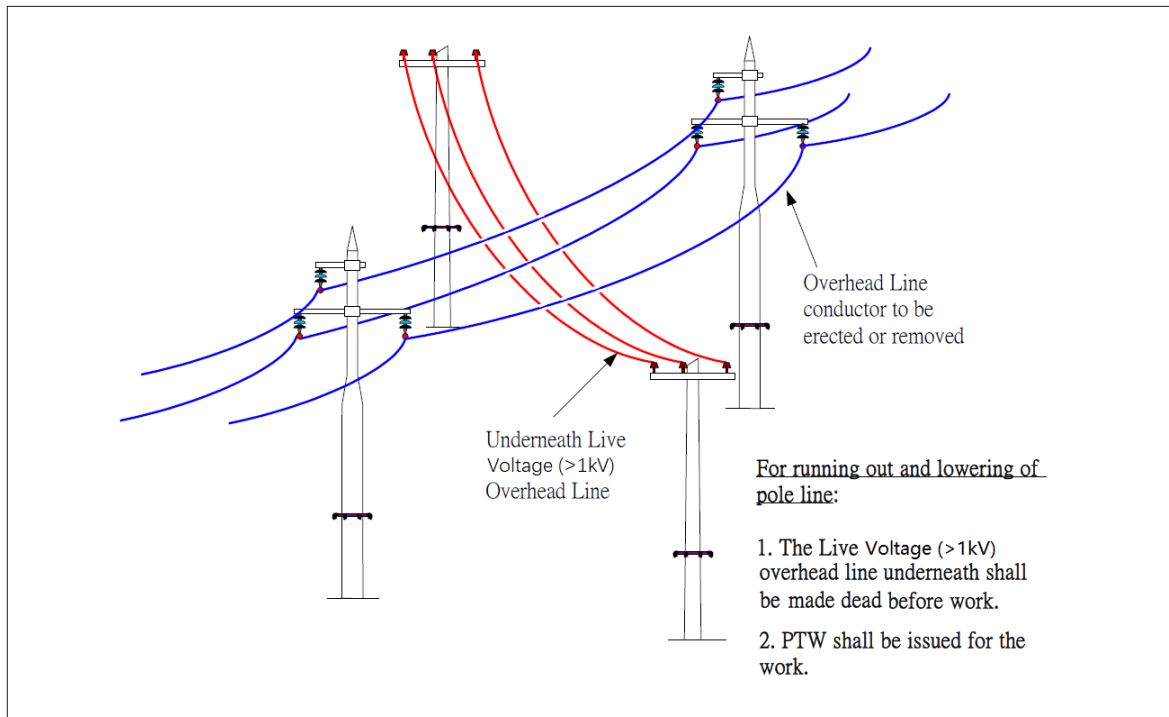


Figure 16.1: Running out or Lowering of pole line conductor ABOVE a Live Voltage (>1kV) overhead line for the case 6.1.5.a)ii).

16.5.1.6 Running out or Lowering of an overhead line conductor UNDERNEATH a Live Voltage (>1kV) overhead line

When, after a pre-work risk assessment by a Competent Person, there is a risk of infringement of the required Safety Distance the safe access limits shall be defined by an Authorized Person before work commences. Adequate precautions shall be

taken by establishing a clear demarcation of the safe working area to prevent infringement of the Safety Distance. If the risk of infringing the Safety Distance cannot be eliminated, the upper overhead line shall be isolated and earthed. A Permit to Work shall be issued for the work making reference to the upper High Voltage overhead line. The required Safety Distance depends on the voltage level of the upper overhead line.

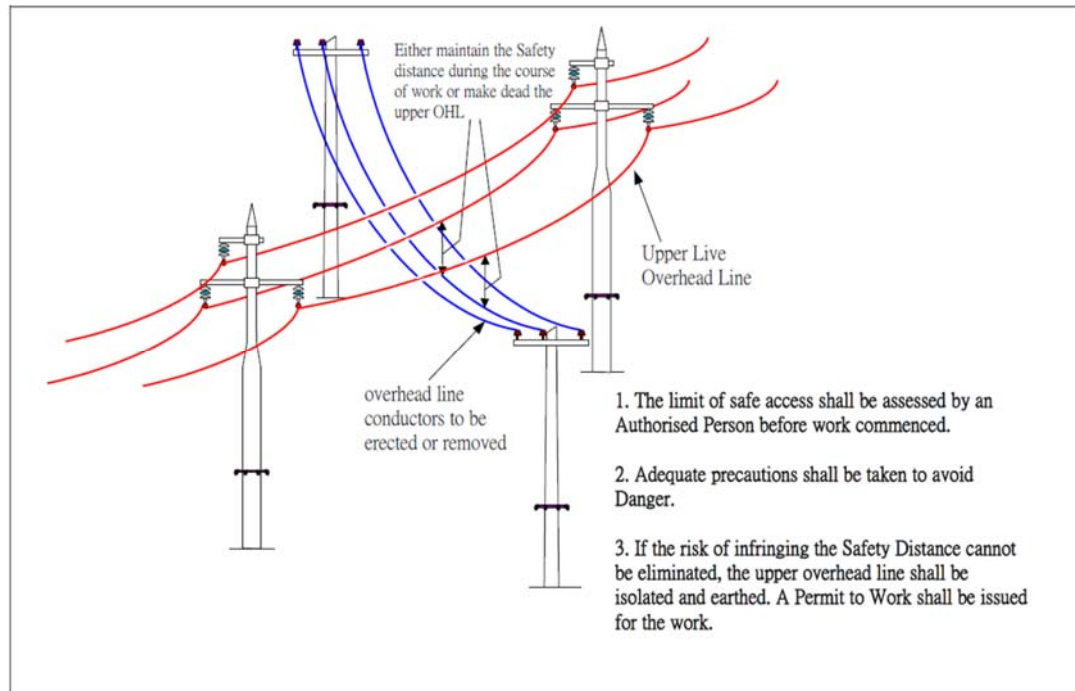


Figure 16.2: Running out or Lowering of overhead line conductor UNDERNEATH a Live Voltage (>1kV) overhead line. (16.6.1.6)

16.5.1.7 Running out or Lowering of overhead line conductor above a Live Low Voltage overhead line

If, when running out or lowering a live low voltage overhead line conductor, there is a risk of touching a bare live low voltage conductor, then:-

- a) A Competent Person shall assess the work method to ensure that adequate safe guarding is provided to prevent any infringement of the Safety Distance, or
- b) The live low voltage overhead line underneath shall be made Dead before work commences.

16.5.1.8 Running out or Lowering of overhead line conductor UNDERNEATH a Live low voltage overhead line

The work shall be supervised by a Competent Person who shall ensure that either:-

- a) Adequate precautions have been taken to avoid Danger by establishing a clear demarcation of the safe working area at all crossing or overlapping points or,
- b) The Live low voltage overhead line shall be made dead.

16.5.2 Work in Enclosures Containing Exposed Live Voltage (>1kV) Conductor

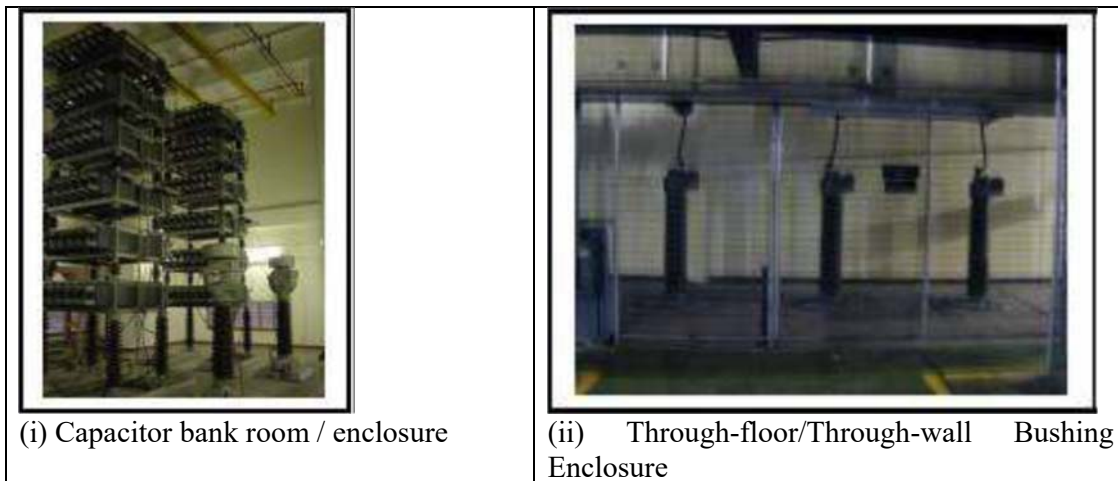
16.5.2.1 Introduction

Exposed live voltage (>1kV) conductors are used for connecting purposes in some substations. These connections are normally located inside a designated room or fenced off by an enclosure. This section stipulates the safety working requirements in these areas, enclosures or rooms to prevent any infringement of the Safety Distance.

16.5.2.2 Areas Containing Exposed Live Voltage (>1kV) Conductors

The areas containing exposed conductors are classified into two categories where:-

- a) Safety Distance **cannot** be maintained when accessing these enclosures; for example



- b) Safety Distance **can** be maintained if adequate precautions are taken.
- c) Examples of area where Safety Distance **can** be maintained are:-



(i) Substation with exposed conductors and open-type switchgear



(ii) Reactor / Transformer with open-type bushing



(iii) GCB enclosure for capacitor bank/reactor



(iv) Enclosure for transformer Neutral Earthing Resistor



(v) OHL Landing Platform / Enclosure





(vi) Exposed live apparatus mountings on substation walls.

16.5.2.3 Working in the Area where a Safe Distance CANNOT be maintained in the enclosure requires that:-

- a) Access to such areas shall be locked or;
- b) Access to such areas shall be allowed only when the apparatus located in the area is made dead, isolated and the circuit main earths have been applied;
- c) Additional earthing shall be applied to the exposed conductors (if required);
- d) All work or tests in these areas shall be conducted under the terms of the Permit to Work and Test Sanction;
- e) The person gaining access into these areas shall be the named recipient of a Permit to Work, and Test Sanction or under the on-site supervision of the named recipient.

16.5.2.4 Working in the Area where a Safe Distance CAN be maintained if adequate Precautions are taken

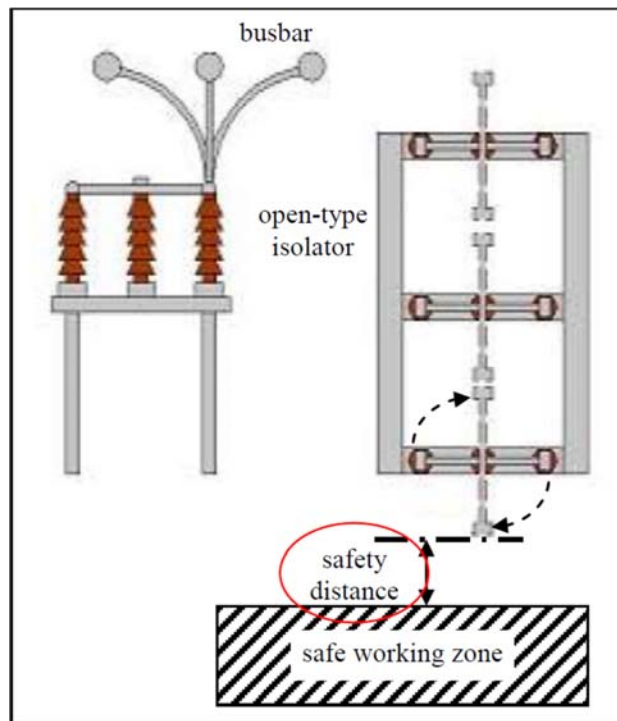
- a) Assessing the Safety Distance Requirement:-

When working in an area containing exposed live voltage (>1kV) conductors. Senior Authorized Person (SAP) shall:-

- i. Assess and define the safe working zone and,
- ii. Give clear instructions to an Authorized Person (AP) who will supervise the work within the defined safe working zone.

- b) When defining a safe working zone, the following factors shall be taken into consideration:-

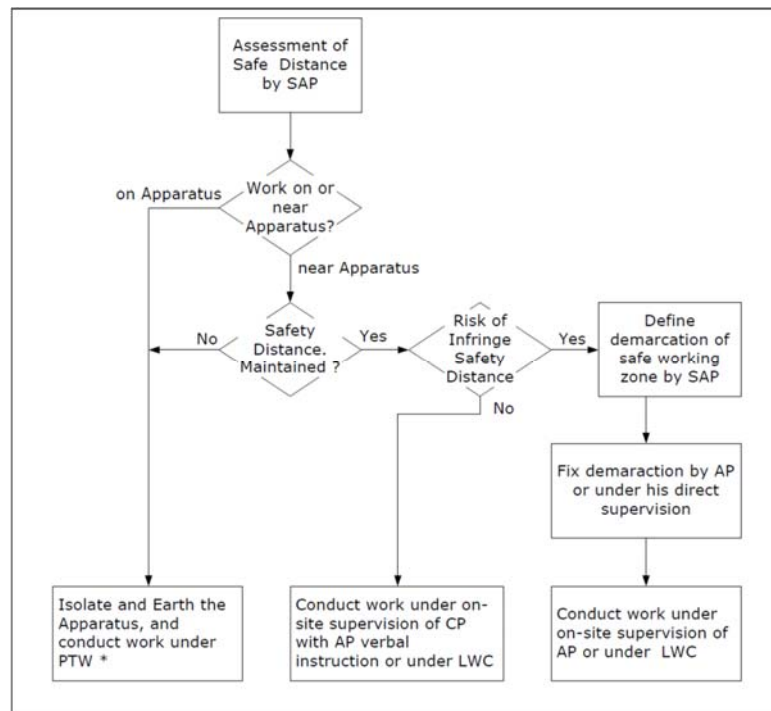
- i. Distance and height of the working position from the live conductors;
- ii. Work nature;
- iii. Physical size of the tools and equipment to be used;
- iv. Length and material of the portable ladders and objects to be used;
- v. The movement path of tools and equipment during work;
- vi. Movement of the exposed live parts.



- c) In accordance with the assessment under clause 6.2.4a) above, precautions as shown in table below shall be enforced:-

	Assessment results	Precautions taken
1	Safety distance can be maintained and no risk of infringing the safety distance exists.	Refer to Section 6.2.5.
2	Safety distance can be maintained but there is a risk of infringing safety distance.	Follow Section 6.2.4, make dead the apparatus and exposed conductors, or Refer to Section 6.2.6.
3	Safety distance cannot be maintained.	Follow Section 6.2.3, make dead the apparatus and exposed conductors.

Refer to the below flow chart.



* Note : If any personnel work is located in an area containing other exposed live voltage (>1kV) conductors, the requirements in Section 16.6.2.5 or 16.6.2.6 shall be followed under a Permit-To-Work (PTW).

16.5.2.5 Working inside an area containing exposed live voltage (>1kV) conductors without a risk of infringing the Safety Distance

- a) The AP shall give a Competent Person (CP) clear verbal instructions limiting the work and/or work area; or
- b) When verbal instructions are considered insufficient the AP shall issue a Limited Work Certificate (LWC) to the CP.

16.5.2.6 Working inside an area containing exposed live voltage (>1kV) conductors when there is risk of infringing the Safety Distance

- a) The SAP, who defines the safe working zone, shall determine the work to be conducted under:-
 - i. On-site supervision of an AP, or
 - ii. Limited Work Certificate or Permit to Work where appropriate.
- b) When the work is to be conducted under an AP's supervision the SAP shall give clear instructions to the AP and the working team demarcating the work zone and limiting the safe working zone.
- c) All demarcations and the Danger Live Notice demarcation of the safe working zone shall be fixed or moved only by or under the direct supervision of the AP.

- d) The working personnel shall be provided with sufficient illumination to clearly see the safe working zone.
- e) If there is more than one entrance to the enclosure containing exposed live voltage (>1kV) conductors, the SAP shall identify the gate or door to access or egress the enclosure.
- f) Only SAP approved portable ladders, long objects, cranes, scaffold or mobile access equipment can be brought into and used in these areas.
- g) Movement or erection of portable ladders, long objects, cranes, scaffolds and movable equipment, shall only be carried out by the Competent Person (CP) under the direct supervision of the AP.

16.6 Tests on Completion

All tests shall be conducted by the Onshore Contractor, and the Owner shall be given the opportunity to witness these tests.

The whole of the Phase 2 Switchyard Works shall be tested in accordance with the standards adopted for the Project and the Onshore Contractor's quality plans and testing procedures which are reviewed and agreed by the Owner.

All measurements shall be taken with appropriately calibrated test instruments as per the requirements of the relevant Vietnamese standards and international test codes.

16.6.1 General Test Requirements

All test methods together with acceptance criteria shall be subject to approval of the Owner before commencement of tests.

At least 30 days' notice shall be given to the Owner that the plant is ready for progress inspection or for witnessing of tests. Five copies of the records of all tests shall be furnished to the Owner.

KEMA, CESI or accredited Short-circuit Test Laboratory (STL)'s type test certificate and reports for the offered design shall be submitted.

16.6.2 Factory Acceptance Tests

The following tests shall be carried out at the Onshore Contractor's Works

16.6.2.1 Circuit Breakers

In accordance with the requirements of IEC 62271-100

The Owner may require in addition to the following tests to be carried out, the details of which will be agreed between the Owner and the Onshore Contractor

- ☐ Electrical endurance tests
- ☐ Power-frequency voltage test at reduced gas pressure

16.6.2.2 Disconnectors and Earthing Switches

In accordance with the requirements of IEC 62271-102

The Owner may require in addition to the following tests to be carried out, the details of which will be agreed between the Owner and the Onshore Contractor

- ☐ Electrical endurance tests
- ☐ Power-frequency voltage test at reduced gas pressure

The following type tests shall be included for earthing switches with fault making capacity

- ☐ Inductive current breaking test
- ☐ Capacitive current breaking test

16.6.2.3 Metal – Enclosed Switchgear

The requirements of this section are intended to cover complete switchgear assemblies. Component parts such as circuit-breakers, disconnectors, current and voltage transformers, instruments shall be submitted to separate tests as specified in the relevant IEC Recommendation.

In accordance with the requirements of IEC 62271-203.

The Owner may require in addition to the following tests to be carried out, the details of which will be agreed between the Owner and the Onshore Contractor

- ☐ Operation tests at limit temperatures as specified in the Site Operation Conditions
- ☐ Internal arc type test shall be carried out on the most critical gas compartment as mutually agreed between the Purchaser and the Onshore Contractor or waived as agreed by the Owner.

16.6.2.4 Compressed Air System

All compressors, air receivers, pipework, valves and ancillary apparatus shall have applied to them the tests called for in the appropriate specifications and standards. Pressure tests shall be carried out at 1.5 times the pressure at which the associated safety valves operate. The test pressure shall be maintained for a time sufficient to ascertain whether or not leakage occurs.

16.6.2.5 Gas Insulated Pressure Containers

The tests shall be carried out at the pressures given below for a period of 15 minutes and components shall thereafter be marked in an agreed manner.

- (a) For containers subject to static gas pressure loads only, and not subject to mechanical shock in service, the test shall be made at 1.5 times maximum working pressure.
- (b) For containers subject to rapid changes of gas pressure, or to gas pressure plus mechanical shock, the test shall be made at three times maximum working pressures.

16.6.2.6 Current Transformers

In accordance with the requirements of IEC61869-1 and IEC61869-2

16.6.2.7 Voltage Transformers

In accordance with the requirements of IEC61869-1 and IEC61869-3

16.6.2.8 Surge Arrester

In accordance with the requirement of IEC60099-4.

16.6.2.9 Control and Indicating Panels, Marshalling Kiosks, Instruments and Secondary Wiring

Routine Tests

All relays, instruments, transducers and other equipment shall be subjected to routine tests in accordance with relevant IEC recommendations prior to assembly in the complete equipment.

Tests shall be carried out to prove the correct functioning and wiring of the complete equipment.

All secondary wiring including panel wiring and control circuits and all apparatus connected directly thereto shall withstand a high voltage test of 2 kV to earth unless subject to other requirements such as detailed under protection equipment.

All apparatus directly connected to pilots which pass outside the switching station shall withstand for one minute a high voltage test of 5kV to earth, or 15kV to earth when specified. During the test all other wirings, connections or contacts shall be earthed.

16.6.2.10 Motors

Type Tests

One motor of each type shall be submitted to the following tests.

- (a) Measurement of winding resistances
- (b) Verification of direction of rotation relative to the phase sequence of the supply.
- (c) No load loss measurement at nominal voltage with check for balance with the machine unbolted from the bedplate.
- (d) Overvoltage test at 125% nominal voltage, no load, for 15 minutes.
- (e) Temperature rise test.
- (f) Verification of load characteristics and momentary excess torque capability.
- (g) Verification of capability to produce rated output at minimum transient voltage, with measurement of slip.
- (h) Measurement of breakaway, pull-up and pull-out torques.
- (i) Measurement of breakaway starting current.
- (j) Overspeed test at 120% nominal speed for 5 minutes.
- (k) Dielectric tests including measurement of insulation resistance.

Routine Tests

All other motors shall be submitted to test (a), (b), (c), (d) and (k) of the above.

16.6.2.11 Materials

Type Tests

Samples selected by the Engineer from metals used in the Contract Works shall be tested to prove compliance with the Specification, including the guarantees stated.

16.6.2.12 Galvanising

Sample Tests

Representative samples, selected by the Engineer of all galvanised material shall be submitted to galvanising tests. Galvanised fittings associated with insulators shall be tested in line with the relevant IEC Recommendation. All other fittings, fabrications, hardware and fixings shall be inspected and tests in accordance with BS EN ISO 1460 and 1461 standard.

16.6.2.13 Sulphur Hexafluoride Gas

Sample Tests

Samples of gas from each consignment shall be tested before dispatch and shall comply with the requirements of IEC 60376.

16.6.2.14 Handling Devices and Lifting Tackle

Routine Tests

All handling devices and lifting tackle supplied for maintenance purposes under the contract shall, unless they are built into and form part of the equipment, be tested, marked and certificates of test provided.

Lifting tackle built into and forming part of the equipment shall be operated with the maximum working load to the satisfaction of the Engineer.

16.6.2.15 Auxiliary Cables

Auxiliary cables shall be submitted to type and routine tests in accordance with the appropriate IEC or other approved standard.

16.6.2.16 Gas-Insulated Transmission Line / Gas-Insulated Bus

In accordance with the requirements of IEC 61640.

The Engineer may require in addition to the following tests to be carried out, the details of which will be agreed between the Engineer and the Onshore Contractor :-

- Power-frequency voltage test at reduced gas pressure
- Test to prove the strength of partitions (refer to 6.104 of IEC 62271-203)

All switchgear components such as circuit breaker, disconnector, earthing earth and busbar etc. shall be assembled together as practicable as possible when carries out the factory acceptance test.

16.6.3 Completion Tests

The Onshore Contractor shall perform the Completion Tests as soon as practicable after successful completion of installation.

16.6.3.1 Site and Commissioning Tests

Site and commission tests shall be carried out by the Onshore Contractor as specified in the contract agreement. The Onshore Contractor shall provide experienced test personnel or supervising engineer when specified. The Onshore Contractor shall provide a schedule of recommended tests and test procedures to be agreed with the Owner.

For gas-insulated switchgear, site tests shall be carried out including but not limit to the following:

- a) Voltage test on main circuit
- b) Dielectric tests on auxiliary circuits
- c) Local air components associated with pneumatic operation, including air compressor
- d) SF6 monitoring devices including pressure gauges, pressure switches and temperature compensating density switches
- e) Measurement of contact resistance and main circuit resistance
- f) Operational tests
- g) Gas quality, its air content, its dryness and its dielectric strength

- h) Timing tests
- i) Current Transformer tests including magnetisation characteristic, insulation resistance, secondary winding resistance, Primary injection test to check ratio, polarity and spill current
- j) Voltage transformer tests including insulation test, polarity check, phasing check, and ratio check
- k) The correct tensions, sags and clearance of flexible busbars and connections
- l) Contact resistance tests for all connections and joints
- m) Earth continuity test
- n) Instruments and instrument transformer circuits
- o) High voltage surge arrester tests including secondary injection tests, Megger test and resistive leakage current test
- p) Interlock test
- q) Coupling Equipment test in accordance with relevant standards and Owner's requirements
- r) Partial Discharge test according to IEC 62271-203
- s) General and visual check
- t) On-line partial discharge monitoring system test and calibration

For gas insulated transmission line / Gas-Insulated Bus, site tests shall be carried out including but not limit to the following:

- a) Voltage tests on the main circuits
- b) Dielectric tests on auxiliary circuits
- c) Measurement of the resistance of the main circuit
- d) Gas tightness tests
- e) Checks and verifications, including:-
 - conformity of the assembly with the supplier's drawings and instructions
 - sealing of all junctions and tightness of all connections
 - conformity of the wiring with the diagrams
 - proper functioning of the monitoring and regulating equipment
 - correct connection of the bonding system
- f) Measurement of gas condition
- g) Tests on enclosures welded on site
- h) Earth continuity test

The Works shall also comply but not limited to the following Vietnamese standards and international test codes with all relevant Acceptance Record to obtain the Taking-over Certificate:

- QCVN QTĐ-5: 2009/BCT: National Technical Codes for Testing.
- Decree 46/2015/ND-CP dated 12 May 2015: Government on Quality Control and Maintenance of Construction Works
- Circular 26/2016/TT-BXD dated 26 October 2016: MOC on Construction Quality Management and Maintenance.
- IEC 62271-100: High voltage AC circuit breakers
- IEC 62271-102: AC disconnectors
- IEC 62271-200: Air insulated switchgear
- IEC 61869-2: Current transformers;
- IEC 61869-5: Capacitor voltage transformers;

- IEC 60099-4: Surge arresters;
- IEC 60273: Insulators;
- IEC 60305, IEC 60120: String insulators;
- IEC 60502: Power cables;
- IEC 61089: Round wire concentric lay overhead electrical stranded conductors.

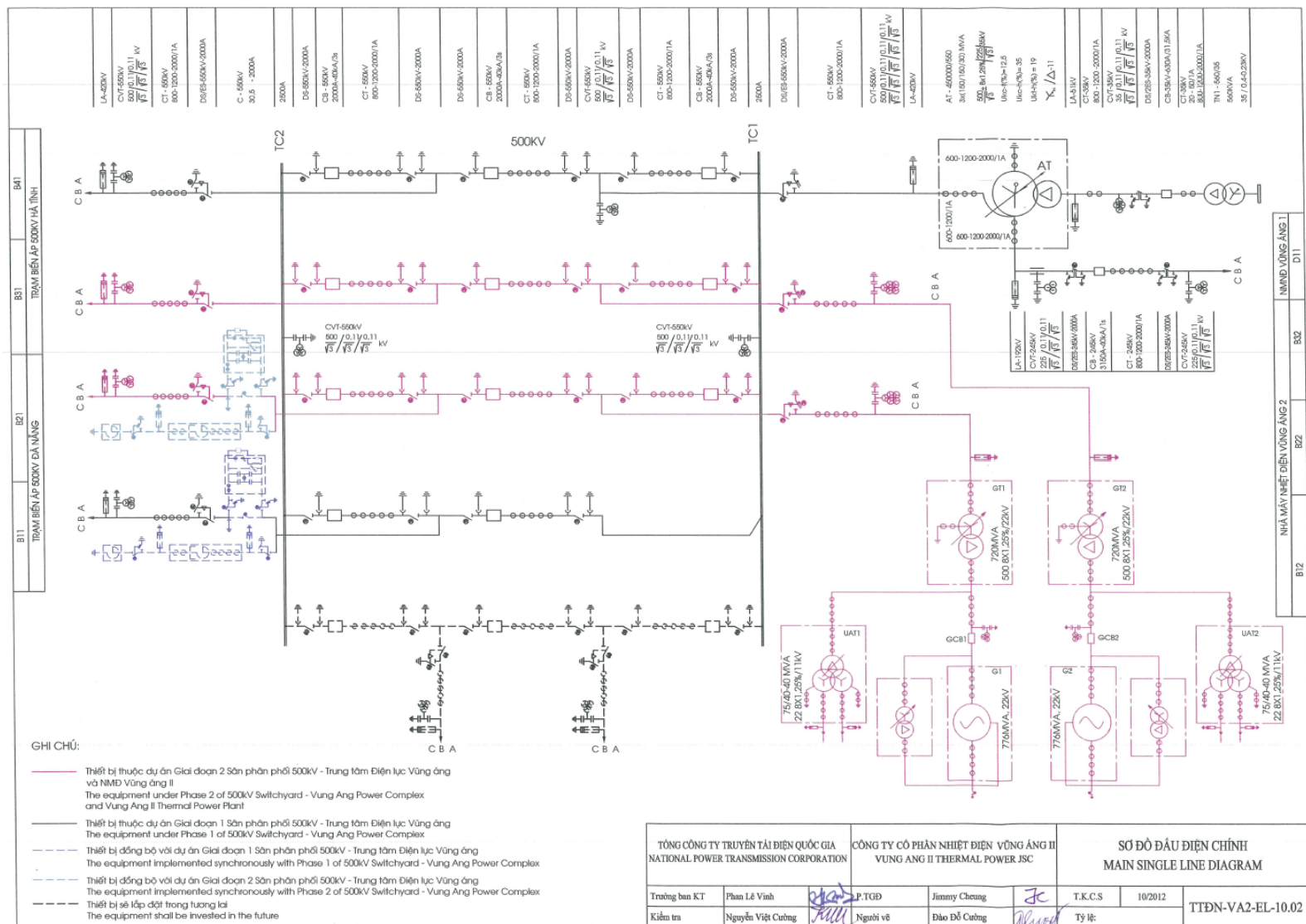


Figure 16.3 Drawing no. TTDN-VA2-EL-10.02: Main Single Line Diagram

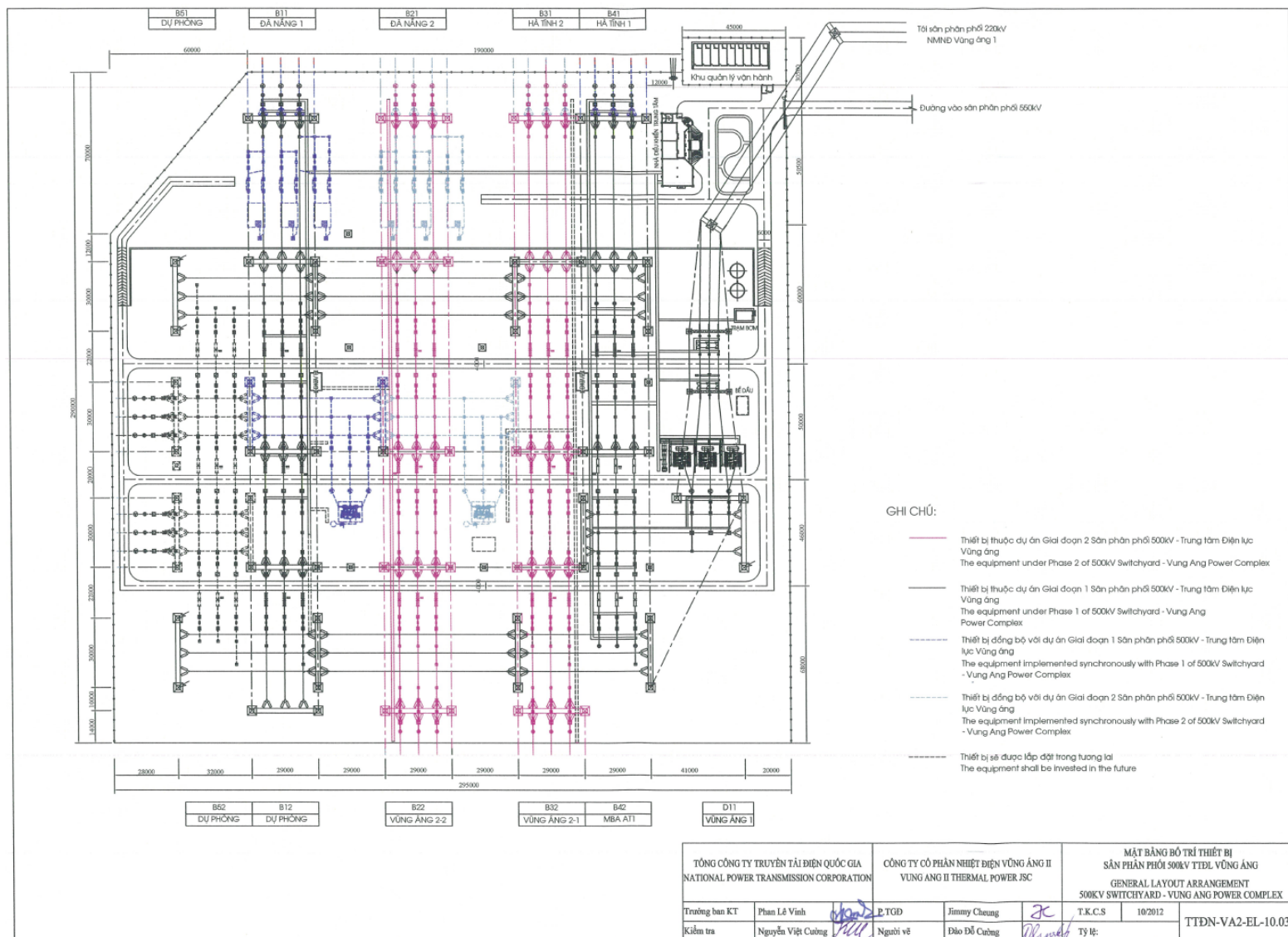


Figure 16.4 Drawing no. TTDN-VA2-EL-10.03: General Layout Arrangement

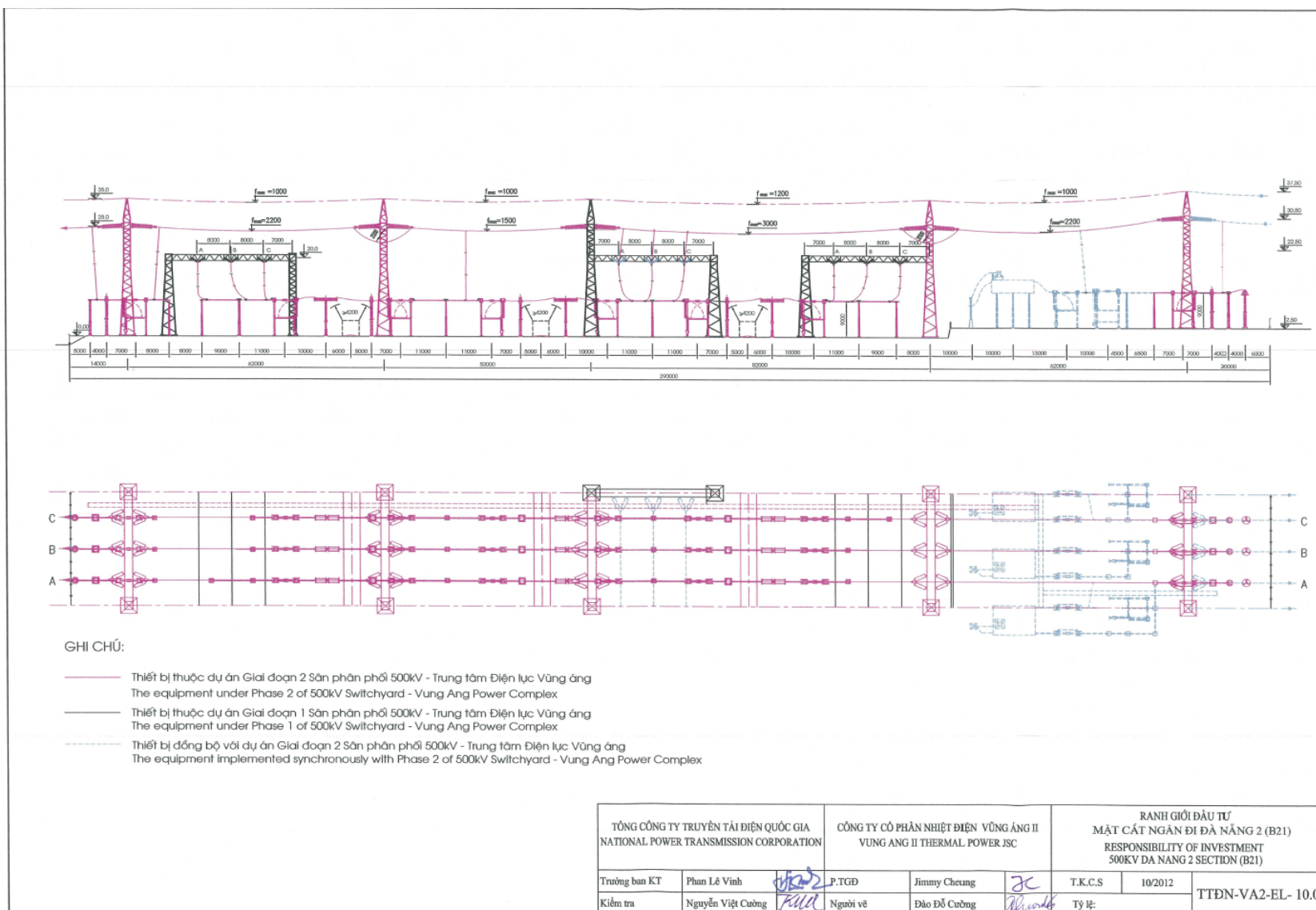


Figure 16.5 Drawings no. TTDN-VA2-EL-10.04: Responsibility of Investment – Section

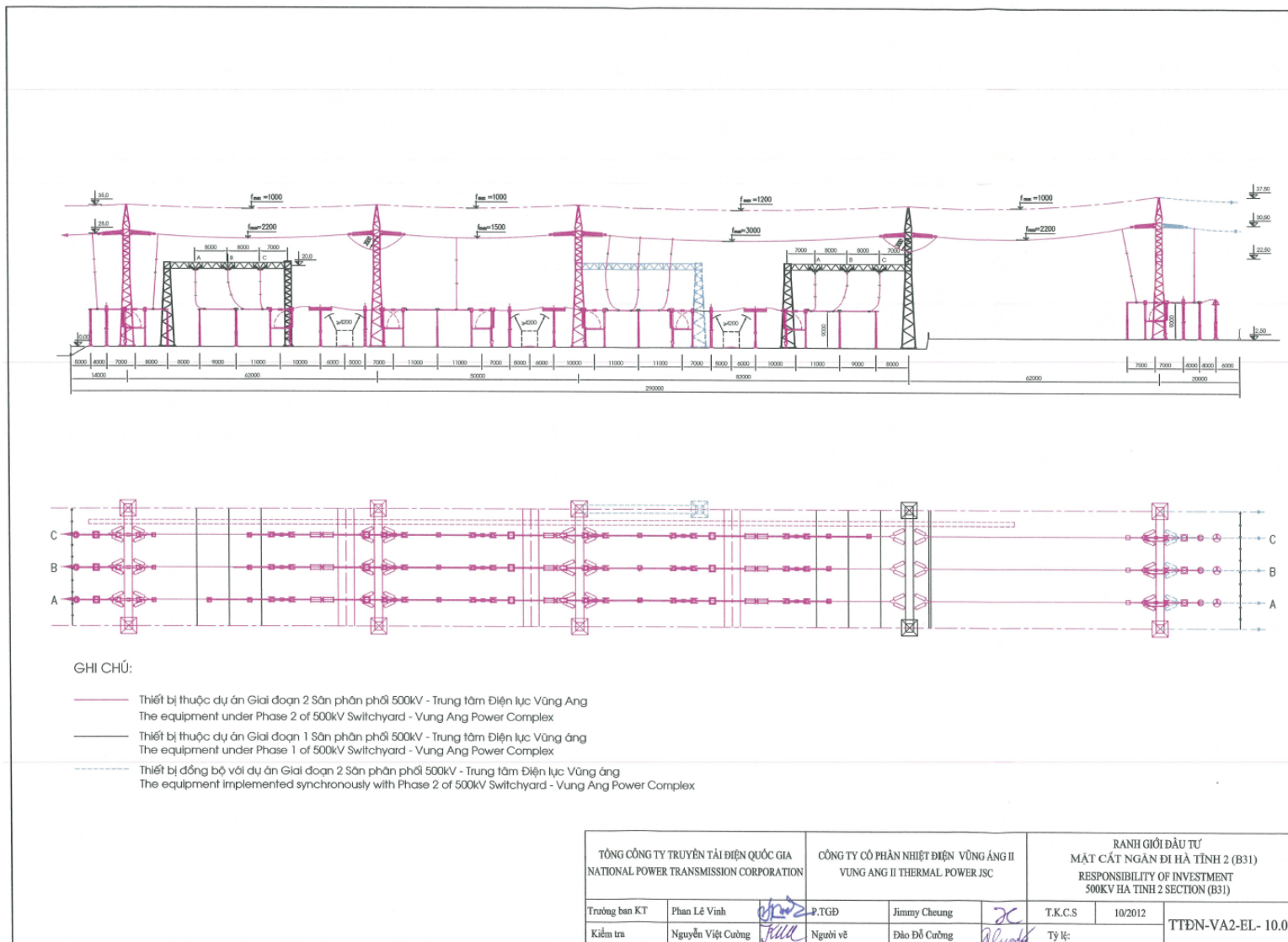


Figure 16.6 Drawings no. TTDN-VA2-EL-10.05: Responsibility of Investment – Section

Annex A – Schedule of Project Management

Schedule of Project Management

1 Construction Programme

1.1 General Requirements

There will be an overall Project Programme containing inputs from the Offshore Supply Contractor, Offshore Services Contractor and the Onshore Contractor which shall fully inter-relate the various activities on design, manufacture, erection and commissioning until the Taking Over of the Works. In addition, information on resources for the construction work shall also be provided. The resources and labour at Site shall be segregated into occupation and trade categories.

The Onshore Contractor shall provide a “base line schedule” in electronic file form incorporating inputs from the Offshore Service Contractor, the Offshore Supply Contractor and the Onshore Contractor. The Onshore Contractor shall submit all primavera programmes up to level 3 (in native XER and PDF formats), versioned uniquely. Any extension of time claim shall be substantiated minimum with a level 3 electronic (soft) programme copy of primavera programme.

1.2 Planning and Progress

A planning representative shall be designated by the Onshore Contractor who shall be readily available at all reasonable times to the Owner and who is of sufficient seniority to have free access to each of the Offshore Supply Contractor's, Offshore Services Contractor's and the Onshore Contractor's management staff and to any Subcontractors' workshops (including those of the Offshore Supply Contractor, Offshore Services Contractor and Onshore Contractor), Site works etc..

All Programmes shall be submitted in the form of Gantt bar charts.

1.3 Form of Programme Reports

Gantt bar charts shall have a date scale at the top and equipment/areas of work identified at the left hand side. Where logical connections exist between activities they shall be indicated.

In all Programmes where time units are identified calendar dates shall be used. For dating purposes, the week shall start on Monday. A list of all standard abbreviations used on each network or Gantt chart shall be displayed on each individual sheet of the schedule.

1.4 Submission of Project Programme

Indicative information on resources proposed for the construction work shall be submitted at Notice to Proceed (NTP). Estimation of Site labour resources shall be segregated into occupation and trade categories. This information shall be updated in

monthly progress report. The Onshore Contractor shall submit the level 3 primavera programme in native XER and PDF formats within 28 days of the date of issue of NTP. That Programme shall include detailed manufacture and Site construction programmes. The Onshore Contractor shall provide the schedule of major construction equipment and labour histograms. Critical path(s) shall be indicated. Master SHE (Safety, Health & Environment) work plan shall be with sufficient resources loaded and committed that shall match the requirements of the construction programme.

With respect to all programmes submitted or to be submitted by the Onshore Contractor, the Owner may reasonably request more detailed information and the Onshore Contractor shall provide such information promptly and at no additional cost.

1.5 Revisions to the Overall Contract Programme

If during the execution of a relevant contract, a relevant contractor modifies for any reason the agreed base line schedule (programme), then a new revision of the base line schedule shall be issued to the Onshore Contractor who will submit this to the Owner in accordance with the Onshore Contract. In addition, the Onshore Contractor shall submit the working electronic native XER and PDF files quarterly or when there is a major update or as otherwise directed by the Owner, in order for the Owner to monitor the Project progress.

If, at any time during the performance of a relevant contract, any critical activity in respect of that contract as identified in the overall programme is delayed for 15 days or more, the Owner may direct the Onshore Contractor to prepare a report on actions to take in order to mitigate the deviation presented on the base line schedule.

1.6 Detailed Setting to Work and Commissioning Schedule

The Onshore Contractor shall develop a detailed commissioning programme which shall be broken down into the systems of Plant and equipment. Electronic native XER and PDF copy of all programmes shall be provided.

2 Project Management

The Onshore Contractor shall provide to the Owner within 30 days after NTP a comprehensive Project Management Plan. The contents of the Plan shall have been developed by the Onshore Contractor (with assistance from the Offshore Supply Contractor and Offshore Services Contractor) to suit the Project for review by the Owner. The Project Management Plan will include the Onshore Contractor's procedures and Project specific plans for effective project execution, communications, document control, and reporting. Specific plans are required for

- a) Quality assurance;
- b) Programme;
- c) Engineering and design management;
- d) Interface management;
- e) Permits and consents;

- f) Procurement;
- g) Materials management (logistics);
- h) Construction and constructability;
- i) Risk assessment; and
- j) Management of design changes.

A single Project management system shall be adopted by the Offshore Supply Contractor, Offshore Services Contractor and Onshore Contractor throughout the Project.

In particular, each contractor shall explain his proposed management organisation, provide the names, qualifications and experience of the proposed managerial and senior supervisory staff to be employed for carrying out the scope. Subject to the terms of the relevant contract, each contractor shall not make any changes to his senior staffing (Project Director and his direct reportees) arrangements without prior approval by the Owner which shall not be unreasonably withheld. The Onshore Contractor shall designate a Project Manager plus a separate Site Manager who shall be stationed at Site and readily available to interact with the Owner. Both of them shall be suitably experienced and qualified and of sufficient seniority to direct design and Site staff and mobilise the qualified staff to exercise the work in order to complete the activities of the Programme.

During the execution of the Project, each contractor shall provide to the Owner access to platform for document exchange.

During the execution of the Project, the Onshore Contractor shall provide both English and Vietnamese versions of certain Project management documents in order to ensure effective communication with all local contractor workers and sub-contractors effectively. The applicable Project management documents are as follows:-

- Project management policies;
- Public notices;
- Signage on Site;
- Induction and training on Site;
- Emergency procedures, especially related to SHE aspects and at working instructions level;
- All relevant safety & health notices advising of potential hazards and precautions to be taken;
- Any other document agreed between the Onshore Contractor and the Owner.

During the execution of the Project, monthly progress reports in PDF format (a Microsoft Word document shall be provided additionally (whereas objects other than texts are images)) for review. In addition, progress review meetings shall also be held at regular intervals between the Owner and the relevant contractors throughout the design, manufacture, construction and commissioning and testing stages, during which information to be provided shall contain the information enabling the Owner to understand the progress and the proposed design and construction processes. If necessary, Sub-contractors may also attend the progress review meetings.

The monthly progress report for the Project shall be sent to the Owner no later than ten (10) days after the end of each calendar month. The monthly progress reports shall cover a wide range of issues including, but not be limited to, the following:

- Progress on SHE plan, statistics and graphical trends on leading and lagging SHE indicators and key items from OHSE requiring management attention;
- Review Site activities, including safety and environmental issues with statistics;
- Records of meetings and events;
- Progress to the end of that month;
- Expected progress to the end of the next month;
- Overall progress, with S-curve showing percentage of completed (values to be given in an Excel table additionally), outstanding and late works;
- Progress summary schedule using dropline;
- Critical path in the overall schedule;
- Progress Level 3 schedule;
- Status of critical outstanding items or issues;
- Review design submissions;
- Schedule of documents and drawings received / reviewed;
- Schedule of major equipment/service being/to be ordered, under manufacturing, received at Site and installed in position;
- Shipping schedule for arrival of equipment with date of FOB where applicable and expected arrival date at Site;
- Review of inspection and testing plans;
- Review quality plans and their implementation;
- Quality issue during manufacturing and on-site;
- Contractor organisation charts;
- Work force at Site, labour relations and community relations;
- Agree commissioning procedures and practices and monitor their implementation;
- Agree test procedures and practices and monitor their implementation;
- Payment;
- Variations;
- Permit and Approvals;
- Losses and claims, if any;
- Other matters, as appropriate.

The format and details required for the submission of the monthly progress report shall be agreed with the Owner.

3 Occupational Health, Safety and Environmental Requirements (OHSE)

The Onshore Contractor shall comply with the Health, Safety, Security and Environmental Requirements set out in Annex B of the OTR, unless otherwise specified in this document.

4 Design and Engineering

4.1 General

Each contractor is fully and exclusively liable and responsible for the design of its scope of works (including design criteria and calculations) and for all documents and information (including the correctness, accuracy and completeness of such documents and information) used, produced or required for or in connection with its scope of works or referred to in the relevant contract.

The Offshore Services Contractor is fully and exclusively liable and responsible for the basic and detailed design of the Facility and all engineering work to fulfil its obligations under the Offshore Services Contract, or where not specified, in accordance with all applicable Vietnamese Laws and with due consideration for reliability, ease of operation and maintenance, safety and the environment.

4.2 Review of Documents and Participation in Meetings by the Owner

The Offshore Services Contractor shall provide design documentation submission for all areas of the works as part of the Offshore Services Contractor's Documents to enable the Owner to fully understand the proposed system design. These shall include a system description, system layout, process and instrumentation diagrams, control and operation descriptions, schedules of major characteristics of equipment items, and any other information necessary for the understanding of the systems.

The Offshore Services Contractor shall prepare and submit all Offshore Services Contractor's Documents to enable the Owner to verify compliance with the Offshore Services Contract. The Offshore Services Contractor shall provide to the Owner a list of design documents. The list shall indicate the design documents which will be submitted to the Owner for review and those which will be submitted for information. The Owner will review the list with the Offshore Services Contractor and agree a list of "for review" and "for information" documents. This list is named as Master Document List.

For documents that will be submitted to the Owner for review, two hardcopies and one electronic version (with no restrictions on further copying) of the documents shall be submitted. Electronic versions of all as-built drawings shall be submitted in AutoCAD format

The Offshore Services Contractor shall allow the Owner to review the documents in advance and invite the Owner to attend the design review meetings.

A RAM (Reliability Availability and Maintainability) study shall be conducted by the Offshore Services Contractor and will be reviewed with the Owner.

The Offshore Services Contractor shall prepare and submit to the Owner the basic designs, master drawing designs and the detailed designs of the Facility and the Works, the documents specified in the Owner's Technical Requirements for the Offshore Services Contract and the Offshore Services Contract for review to enable the Owner to verify compliance with the Offshore Services Contract.

System design manual shall be provided for each major system defined in the Master Document List for Owner's review. The system design manual shall include a system

description stating the function, required equipment, design basis, process data, operating principles and control logic, analysis of failures and interface requirements with other systems. The equipment design manual, which are the specification of the equipment as per Key Components List in Annexure 13 of the relevant Construction Contracts for enquiry purposes, shall be submitted for Owner's review. The Offshore Services Contractor shall agree with the Owner for the schedule for the submissions of these two sets of documents for review.

Within 30 days after receipt by the Owner of any document, the Owner shall notify the Offshore Services Contractor of the results of its review of the document and the modifications, if any, which the Owner suggests. The Offshore Services Contractor shall submit the document in a logical sequence, within a reasonable timeframe and at reasonable and sensible intervals. Consideration will be given by the Owner to shortening of the review period to 15 days for some design critical to the programme.

The Offshore Services Contractor shall at its own cost modify the document to take the Owner's comments into account and re-submit it to the Owner for review. The Offshore Services Contractor shall not depart from any reviewed document. Each part of the works shall be performed in accordance with the documents which have been reviewed and rectified to include all comments of the Owner. If the Offshore Services Contractor wishes to modify any design or document which has previously been submitted for review, the Offshore Services Contractor shall immediately notify the Owner in writing and shall promptly submit revised documents to the Owner for review.

The Offshore Services Contractor shall give not less than 14 days' notice to the Owner in relation to any design review meetings, design liaison meetings and associated special expert meetings in relation to the design of the Facility or the Works. The Owner or its representative may attend such meeting and give comments on the relevant design or Equipment and to the extent relevant, the Offshore Services Contractor shall take such comments into account in the bid assessment process and/or incorporate such comments in the design or documents before submitting the same to the Owner for review.

The other design information which shall be submitted by the Offshore Services Contractor shall include, but not be limited, to the following:

- Civil information,
- General arrangement drawings,
- General layout drawings,
- Equipment arrangement and layouts,
- System/equipment design manual,
- Mechanical flow diagrams,
- Piping and instrument diagrams (P&ID),
- Site services piping / cabling coordination drawings,
- Performance curves and heat balance diagrams,
- Pump and fan characteristic curves,
- Facility start-up and shut-down curves,
- Single line diagrams, relay metering diagrams, protection settings etc.,
- Quality manuals and plans,
- Results of various design calculations, analysis and test reports,
- Complete and comprehensive schedules for summary of Equipment such as

- valves, piping, motor, actuator, painting, lubrication ... etc.,
- Instrument schedules,
- Control philosophy/logic/sequence diagrams,
- Panel wiring diagrams,
- C&I piping installation diagrams,
- Electrical fault level study, and
- Other information as reasonably requested by the Owner, except proprietary design information.

4.3 Discrepancies, Errors or Omissions

Subject to the terms of the relevant contract, the each contractor is responsible for any discrepancies, errors or omissions in the documents prepared by it, irrespective of whether those documents have been reviewed by the Owner. The Owner's approval or review of, comment on or rejection of, any documents does not entitle a contractor to make any claim or in any way limit or change the relevant contractor's obligations or liabilities under the relevant contract.

Subject to the terms of the relevant contract, each contractor's attention is drawn to the fact that all Owner Supplied Information is based on the best information available at this stage, and hence, is given to each contractor for its reference only. Each contractor shall therefore verify all the Owner Supplied Information, including making all the additional measurements, monitoring and investigations, as deemed necessary, at his own costs and under his sole responsibilities, to determine the required design of the Facility so as to ensure that the proposed design of the Facility is compatible with the Site conditions.

5 Key Components

Prior to entering into any Permitted Subcontract of Key Components for which the aggregate order value with any one supplier is expected to be greater than USD 1 million, the Offshore Supply Contractor and the Onshore Contractor (as applicable) respectively, shall provide the following information of their supplier:

1. Company profile
2. Relevant reference list with similar application
3. Brief description of the components

6 Documentation

The Onshore Contractor shall provide to the Owner the Erection Instructions and Commissioning Instructions for all equipment/systems 30 days before respective site work begins.

All drawings submitted during the design stage shall be in PDF format and, if applicable, read only AutoCAD format. All manuals and word document shall be submitted with Microsoft word and excel formats.

The Offshore Services Contractor shall supply adequate documentation for all computer systems embedded in the scope to enable the Owner to obtain a complete

understanding of the operation of the system and of the individual programs, and to enable modifications to be carried out to the programs. The document shall include, but not be limited, to the following.

- Input/output schedule
- Overall system description and network
- Computer architecture
- Detailed programme descriptions with flow charts and block schematics, showing information flow within and between the programs
- Power supply and earthing arrangement
- Programming handbooks
- Standard software manuals
- Application software manuals
- Diagnostic software manuals
- Computer manufacturer manuals
- Peripheral manufacturer manuals

All documentation requirements under the relevant contract apply to Vung Ang I Outfall Extension Works. Documentation submissions for the Vung Ang I Outfall Extension Works shall be separated and separately labelled from that of the VA2 Project. All finalized documentation for the Vung Ang I Outfall Extension Works (including all as-built drawings) shall be handed over by the Offshore Services Contractor to Owner when the Vung Ang I Outfall Extension Works are handed over to Owner. The completion of the Vung Ang I Outfall Extension Works and submission of as-built drawings for the Vung Ang I Outfall Extension Works is a pre-condition to the issue of the Taking-Over Certificate for Section 2 of the Works.

The documentation shall be submitted according to the following table, or in accordance with another time table agreed by the Owner.

Document	Date required:
Health, Safety, Security and Environmental Plan	14 days following NTP
Project Programme	28 days following NTP
Project Management Plan	30 days following NTP
Environmental Management Plan, incorporating management plans for storm water, dust, traffic and noise.	At NTP
Construction Certificates	As required to support the Works
Design Concepts/Philosophies	30 days before detailed design commences
Preliminary Quality Plan, Design Submission Plan	30 days after NTP
Final Quality Plan	60 days after NTP
Inspection and Test Plans	28 days before commencement of respective manufacturing of equipment other than Key Components
Inspection and Test Plans for manufacturing of Key Components, construction, testing and commissioning	2 months prior to commencement of respective work
Material Traceability Procedure for high pressure and temperature metals (e.g. T/P 91 and T/P 92)	60 days before commencement of respective manufacture

Material Identification Procedure for high pressure and temperature metals (e.g. T/P 91 and T/P 92)	60 days before commencement of respective manufacture
Site Welding Control Program for high pressure and high temperature materials	90 days before commencement of welding
List of Common Systems with drawings	7 months after NTP
Finalized list of Common Systems	9 months after NTP
Sample O&M Manual showing structure, format and content list of the future final O&M manual	12 months prior to the Time for Completion of the First Unit.
Preliminary (draft) O&M Manuals	180 days prior to the Time for Completion of the First Unit.
Final O&M Manuals	90 days after the date of Completion of the First Unit and common systems as stated on the Taking Over Certificate for Section 2 of the Works.
Commissioning Plan	90 days prior to the first fire of the First Unit boiler
Maintenance Programme	365 days prior to the programme date for commencement of the Tests on Completion on the First Unit
Training Schedule and Programme	365 days prior to the programme date for commencement of the Tests on Completion on the First Unit
Performance Testing Plan and Procedures	180 days before the programmed date for the commencement of the Tests on Completion on the relevant Unit
As-built “Red Correx”	Prior to commencement of Tests on Completion for the relevant Unit
As-built drawings	For critical drawings which affect operation of the First Unit, they shall be submitted within 90 days after the date of completion of the First Unit and Common Systems as stated on the Taking Over Certificate for Section 2 of the Works. For other drawings, they shall be submitted within 120 days after the date of completion of the Second Unit.
List of special tools and lifting gears for O&M	90 days after NTP
Detailed list of 2-year O&M spares	Provided progressively in accordance with Clause 7.7 of the Contract.
List of other spares (such as overhaul spares, strategic spares) for O&M use after 2 years	Provided progressively in accordance with Clause 7.7 of the Contract.
List of hazardous materials	Within 30 days from NTP

7 Quality Assurance

7.1 General

Each contractor and its suppliers and Subcontractors shall have a quality management system like ISO 9001 or equivalent.

Each contractor shall submit its proposed Quality Assurance Plan for assessment by the Owner, not later than 30 days after the date of issue of the NTP.

The Quality Assurance Plan shall address how the OTR requirements are implemented by the relevant contractor and its Subcontractors. A list of quality control plans shall be included.

After agreement by the Owner of the Quality Assurance Plan, then the plan shall form an integral part of the relevant contract. No changes shall be made without the written agreement of the Owner.

The Quality Assurance Plan shall define the systems and acceptance procedures which will be used to ensure that the applicable scope complies with the relevant contract.

The Quality Assurance Plan shall include a list of the quality plans.

Each contractor shall conform to the Owner's quality requirements.

For the Key Components agreed between the Owner and the Offshore Supply Contractor and the Onshore Contractor in Annexure 13 of the Offshore Supply Contract and Annexure 13 of the Onshore Contract, the Owner will attend to or review the following:

1. Kick off Meeting.
2. Increase in number of inspection (witness and hold) points, as Identified in the Quality Assurance Plan and/or Quality Control Plan (QCP or ITP).
3. Factory Acceptance Test (FAT), as applicable.
4. Pre-Shipment Inspection (PSI)

7.2 Quality Plans

Quality plan shall be prepared, identifying manufacturing and construction, inspections, tests and records specific to the items covered by those quality plans. Each contractor shall be responsible for identifying in a suitable manner, any inspections and tests required by the relevant contractor for statutory purposes.

Each quality plan shall be submitted to the Owner for review at least 30 days prior to manufacturing, construction, commissioning and testing. The Owner will confirm his requirements on such documents as drawings, component specifications and procedures to be included under the quality plan before giving his review.

Each contractor shall adhere to the quality plans which the Owner has reviewed. Any difficulties, deficiencies or other conditions adversely affecting quality shall be reported to the Owner.

7.3 Monitoring of the Quality Assurance Arrangements

The Owner will, after the issuance of the NTP, monitor the implementation of the quality assurance arrangements. Monitoring will be carried out by means of surveillance of activities at the work location and/or by formal audits/reviews of the adherence of the relevant contractor to the systems and procedures, which constitute their quality assurance arrangements. A corrective action programme shall be agreed and implemented with respect to the deficiencies revealed by such monitoring.

Each contractor shall provide reasonable facilities, including access, which may be required by the Owner for monitoring activities.

The contractors shall list the parts of their respective scopes and Plant (as applicable) for which quality plans will be submitted at the same time as submission of quality assurance arrangements. The contractors shall also state the level of detail of the quality plan proposed for each major subcontract.

The Offshore Supply Contractor and the Onshore Contractor respectively, shall perform regular reviews of sub-contractors quality assurance arrangements for their equipment. The review documentation related to Key Components shall be made available to the Owner. The Owner reserves the right to participate on an as needed basis in the Offshore Supply Contractor's and Onshore Contractor's monitoring of Subcontractors' quality assurance arrangements. Quality documentation of other equipment shall be made available to the Owner, on request.

On completion of the manufacture or work and completion of inspections and tests the Offshore Supply Contractor and the Onshore Contractor (as applicable) respectively, shall provide to the Owner a Certificate of Conformity/Compliance or a Quality Release Note for their manufacture or work and completion of inspections and tests. The Offshore Supply Contractor and the Onshore Contractor (as applicable) respectively, shall submit "End of Manufacture Report" for all their equipment and the Onshore Contractor shall submit the "End of Erection and Commissioning and Testing Status Report" for all systems.

Instrument test and calibration records, nonconformance reports and all relevant factory test records of Key Components shall be included in End of Manufacture Reports.

Each contractor shall submit a valid certification to ISO 9001 or ISO14000 or equivalent for the parts of their organisation involved in the execution of the Project, as evidence of their ability to satisfy the above quality assurance requirements. Submission of proposed quality management system is still required for those sections of the organisation and major Subcontractors, which are not covered by such certification. The possession of valid certification will not affect the Owner right to monitor each contractor's quality management system. In the event of certification being withdrawn, cancelled or lapsing, the relevant contractor shall advise the Owner immediately together with a proposal to rectify.

The Offshore Supply Contractor shall allow and arrange the Owner to visit the

manufacturing factories of the equipment, review the quality, observe the manufacturing process and witness the testing.

7.4 Welding Quality Control

The Offshore Services Contractor shall identify the applicable welding standards and implement the quality control procedures for all welding works, which shall as a minimum, be submitted to the Owner for review before the appropriate manufacturing activities commence. The Onshore Contractor shall also implement same for the welding work on Site.

- Welding Procedures and Qualification Records
- Welder Qualification Records
- NDT Procedures
- Heat Treatment Procedures
- Defect Acceptance Criteria

For high pressure or high temperature metals, the Offshore Supply Contractor shall obtain mill certificates and material certificates from suppliers and submit to Owner for review.

7.5 Non-conformance Control

The Offshore Supply Contractor shall develop / implement its own control procedures to address deficiencies / nonconformances during manufacturing while communicating to the Owner on status and actions taken. The Onshore Contractor shall also implement same to address deficiencies / nonconformances identified during fabrication, construction and commissioning. All nonconformances shall be resolved and tracked until their closure. Technical deviations will be dealt with under Design Change Notice management. Non Conformance Reports (NCR) shall be categorized as follows:

a. Type A: NCR's for Owner's approval

are defined as non-conformances/discrepancies to the relevant contract whereby a contractor is requesting "Accept As-Is". This request, if approved, could be considered as a concession to the relevant contract.

b. Type B1: NCR's for Owner's review and comment

are defined as non-conformances/discrepancies on Key Components (as per Annexure 13 of the relevant contract) which shall be restored to original requirement but could affect the performance, reliability and safety of the plant and equipment.

c. Type B2: NCR's for information

are defined as non-conformances/discrepancies on Key Components (as per Annexure 13 of the relevant contract) which have been restored to original requirement with no loss of performance or adverse effect on interchangeability requirement.

Above types of NCR's will be included in the relevant End of Manufacturing Reports. The contractors are required to maintain a NCR log for monthly review by the Owner. However, it is required that

1. All the significant NCR's need to be closed before release for shipment. If NCR's are not closed before shipment, the relevant contractor is responsible to follow up until they are satisfactorily closed. The Owner will be kept informed of the status of NCR's till they are closed.
2. The relevant contractor to submit NCR type B1 as they occur for Owner's review and comment.
3. Owner will provide technical disposition for type A and B1 NCR's within 5 working days subject to availability of all required information.

8 Inspection and Testing Plans

The Offshore Supply Contractor and the Onshore Contractor respectively, shall each provide inspection and testing plans (ITP's), as determined and identified both in the contract documentation, OTR's and as and when required by the Owner, for their procured equipment and the Works for the Owner's agreement within the time specified in the relevant contract stating what inspections / tests will be conducted for what equipment and at what time.

8.1 General

The relevant contractors shall facilitate and coordinate all inspections and tests required under the relevant contracts and as described above.

All special measuring equipment or special apparatus required for carrying out inspections and tests shall, unless otherwise agreed by the Owner, be provided by the relevant contractor without extra cost to the Owner.

8.2 Inspection and Test Certificates

The following documents shall be provided to the Owner: -

1. Two hard copies and one electronic copy of inspection and test certificates on equipment or materials subject to quality assurance in accordance with the Owner's Technical Requirements and quality plan.
2. Two originals of inspection and test certificates for all equipment requiring local regulatory approval.

8.3 Inspection and Testing Plan (ITP)

The ITPs shall be provided for manufacturing, construction, testing and commissioning, as applicable. For commissioning on Site, cross reference shall be made to Chapter 12 of the Owner's Technical Requirements. The Owner has the right to add inspection points in the relevant ITP where appropriate to ensure the quality of product and/or work will meet with the Project requirements.

Each ITP shall include hold points, witness points and review points for equipment manufactured in suppliers' factories as agreed during the review of the submitted ITP's. The Owner shall be notified in advance and given the opportunity to review, hold or witness the inspection and testing before the Equipment is released for delivery. Control procedure, reference codes and standards and acceptance criteria (for acceptance criteria, the relevant contractor will reference the relevant Code/Standard provision.) shall be included in ITP. Dimension tolerances if not shown on the drawings shall also be indicated in the ITP.

Hold Points: The relevant contractor shall not proceed beyond a hold point (Hold) without the presence of the Owner or his representative, or written approval from the Owner to continue the work beyond the hold point. The Owner shall advise within ten (10) working days after notification of the hold point of their intention to attend or to waive. The Hold notification shall be issued by the relevant contractor 20 working days prior to the event taking place. If the Owner or his representative does not advise or is not present, then the relevant contractor will proceed and the Hold will be considered as waived. Activities performed without proper notification may be repeated for the Owner's observation at the expense of the relevant contractor.

Witness Points: The relevant contractor may proceed beyond a witness point (Witness) without the presence of the Owner or his representative, if directed by the Owner. Activities performed without proper notification may be repeated for the Owner's observation at the expense of the relevant contractor. If the Owner or his representative does not advise or is not present, then the relevant contractor will proceed and the Witness will be considered as waived. Witness inspection notification shall be as follow:

- I. For Shop Inspections: issued 14 working days prior to the event taking place and, in case of any change, factory notifies 7 days prior to the event taking place. In the case that the Owner dispatches a resident inspector at the Contractor's or its supplier's factory, the relevant witness notifications shall be replaced by factory monthly/weekly manufacturing and inspection plan and factory confirmation one working day prior to the event taking place.
- II. For Construction Inspections: shall be provided by estimated site weekly installation and inspection plan and final written notification 24 hours before event taking place.
- III. For Commissioning Inspections before the 1st Fire of the Power Plant: shall be provided by estimated site weekly commissioning and inspection plan and final written notification 24 hours before event taking place.
- IV. For Commissioning Inspections after the 1st Fire of the Power Plant: shall be provided by estimated site weekly commissioning and inspection plan and final written notification 12 hours before event taking place.

Review Points: The relevant contractor shall make available the document for review or information by the Owner at the time of the inspection.

The relevant contractor shall ensure that there is valid certification of all equipment subject to statutory inspection available at the time of issue of the Taking-Over Certificate for each respective Unit.

8.4 Non-Destructive Testing

All non-destructive testing procedures and operator personnel certification shall be developed according to established standards and be submitted to the Owner for information. NDT's shall be governed by the ASME code and Boiler Pressure Vessel Code where applicable based on design code. All NDT results shall be kept and transmitted to the owner in a data pack containing hard copies and electronic records of all NDT's performed with associated inspection record sheets and films for X-Ray (conducted on Site).

8.5 Hydraulic and Mechanical Testing

Hydraulic tests on systems or components and mechanical tests on associated materials shall be carried out at the maker's works in accordance with the appropriate standard. Shall be governed by the ASME code and Boiler Pressure Vessel Code (where applicable based on design code).

8.6 Mill Certificates

Mill certificate of chemical composition, mechanical properties and thermal mechanical heat treatment shall be provided as defined in the relevant ITP. Those not specifically referenced in the ITP shall, as far as is reasonably practicable, be submitted together with the QA/QC records and data packages during submission or transmission to the owner.

8.7 Works Testing and Factory Acceptance Testing

Factory acceptance tests of an item of equipment, where practical, shall be carried out at the manufacturer's works and the method and record of test shall be agreed between the relevant contractor and the Owner.

Works tests and commissioning tests on control and instrumentation equipment shall be detailed in relevant testing procedures. Where actual testing is not performed as per prior, recorded agreements with the Owner, a type test report could be accepted on a case by case basis, subject to review and acceptance by the Owner.

Works tests shall be detailed in testing procedures, which shall be submitted to the Owner for review and acceptance. Relevant testing Codes and Standards according to the Contract shall be adhered to. .

8.8 Quality Records

Each Contractor shall document the inspection and test process through all applicable elements of the relevant contract. Such quality records shall be produced commensurate with the design, manufacture, construction and commissioning and be available for review by the Owner throughout the execution of the relevant scope.

- The manufacture quality records shall be delivered to the Owner at Site, unless otherwise advised, within one (1) month of the completion of the relevant scope, but no later than at the time of the receipt of equipment at Site. .
- The construction and commissioning test records shall be delivered to the Owner on completion of the Works within one (1) month of completion.

9 Manufacturing Control

Each contractor shall exercise close quality control and frequent visits to his Subcontractors to ensure the equipment is manufactured to meet the standard stated in the relevant contract. Each contractor shall ensure that Subcontractors follow the requirements of the OTR and have adequate controls over all aspects covered by the Owner's Technical Requirements. Owner and Owner' representative shall have the right to witness the manufacturing processes and testing of the equipment at the suppliers' works, subject to factory rules, including the requirements on confidentiality.

Each contractor shall ensure that any product that does not conform to the Quality requirements is identified to prevent its un-intended use or delivery.

All Manufacturing related Non Conformance Reports (NCR's) shall be managed according to section 7.5 above. These will include the deviations from accepted design as well as deviations to the OTR.

9.1 Manufacturer's Standards

Where the Owner has not specified any particular Standard as included in the Contract, the Offshore Services Contractor shall state at the design stage the Standard to which the Offshore Supply Contractor's and/or the Onshore Contractor's proposed equipment is offered and will request for the Owner's review and acceptance.

10 Technical Support from Equipment Suppliers

Technical support from equipment suppliers shall be arranged by the relevant contractors to cover at least the boiler, turbine, generator, ESP, FGD, desalination plant, DCS, elevators and fire protection and detection system. The scope of technical support shall include the provision of technical advisors on Site giving accurate instruction for erection / commissioning and prompt solution to any problem encountered. Further expert advice from suppliers' headquarters shall also be provided if necessary.

Training programmes shall also be provided to Owner's staff either at Site or at suppliers' place to cover the operation and maintenance aspects of above mentioned Plant and equipment forming part of the Works.

11 Site Management

The Onshore Contractor shall provide, in the area(s) allocated by the Owner, any necessary temporary structures and all sanitary and other accommodation for the personnel employed by the Onshore Contractor and his Subcontractors. The buildings and associated services shall be maintained in a clean and sanitary condition. No residential accommodation will be allowed on the Site.

The Onshore Contractor shall be required to maintain a suitably staffed records office to maintain updated records of all relevant documentation and test certificates and to see that these are available to meet obligations to all statutory bodies and to any commissioning committees, working parties or test teams which are established to meet the needs of the Onshore Contract. These records shall be maintained and retained until handover on completion of all Works on the Site and shall be subject to review by the Owner.

Throughout the period of construction of the Works, the Onshore Contractor shall maintain the whole area of his operations in a clean, tidy and safe condition and shall arrange his materials in an orderly manner, all to the satisfaction of the Owner. The Onshore Contractor shall be responsible for waste collection and transfer to a suitable disposal site. The Onshore Contractor shall clearly state, in the Project Management Plan the provision proposed for disposal of waste materials both on and off Site.

All waste from initial site clearance and site levelling (e.g. trees, rocks, soil) shall be disposed by the Onshore Contractor. Upon NTP, the Owner will request the relevant local Government Body to identify an area which can be used to dispose of the Onshore Contractor's waste from site clearance and levelling activities. If the Onshore Contractor elects to use any other area, such disposal area shall comply with the terms of the ESHIA and the EIA.

General construction waste: the Onshore Contractor shall employ companies authorized by relevant Government Bodies to handle such waste disposal.

Chemical wastes: the Onshore Contractor shall formulate a plan and handle the disposal in a way complying with the Laws of Vietnam

For the dredging material of marine work, please refer to OTR (chapter 10).

The Onshore Contractor and his employees shall not trespass beyond the boundary of the Site on to any adjoining land and the Onshore Contractor shall take any necessary action to prevent trespass occurring. An access control system shall be enforced at Site to identify all personnel working at Site.

The delivery dates of fire engine, ambulance and first-aid clinic shall be agreed with the Owner.

The Onshore Contractor shall install site surveillance cameras (with pan, tilt and zoom

capability) to enable the Owner to monitor site progress during construction, with off-site access by the Owner through internet.

The Onshore Contractor shall provide Construction Management Information System for records keeping and information exchange between the Contractor and the Owner.

12 Efficient and Economical Use of Energy

Vietnamese Laws "Law 50/2010/QH12" and "Direction 34/CT-TTg" state that mandatory measures shall be implemented by any party carrying out construction activities. The Onshore Contractor shall comply with these regulations in undertaking the Works for the Facility.

OWNER'S TECHNICAL REQUIREMENTS

Annex B - Health, Safety, Security and Environmental Requirements

OWNER'S TECHNICAL REQUIREMENTS

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OWNER'S TECHNICAL REQUIREMENTS

Health, Safety, Security and Environmental Requirements

1 Overview

1.1 Purpose

This document entitled 'HSSE requirements' (**HSSE Requirements**) outlines the health, safety, security and environment protection requirements to be observed by the Contractor, including the Contractor's Personnel while working on the Site.

In addition to the HSSE Requirements, the Owner may in accordance with the Contract also identify and from time to time request other specific HSSE Requirements (e.g., project specific HSSE requirements for particular high risk activities), IFC's Performance Standards on Environmental and Social Sustainability and Equator Principles for the Contractor to comply with under the Contract and the Contractor shall ensure compliance with such provisions.

1.2 Application

Capitalised terms in these HSSE requirements are defined as:

- "HSSE" means "Health, Safety, Security and Environment";
- "IFC" means "International Finance Corporation";
- "Regulatory Requirements" include, but are not limited to those applicable laws, and regulations as described in the Contract, as modified, amended or supplemented from time to time.
- "Designated Position" shall mean:
A position identified by the Contractor as being critical to safety at work. It is either: A position or the job holder:
 - o is subjected to high exposure to catastrophic incident that could have negative impact on employee, public or environmental, or
 - o plays a key and direct role in the operating process where failure could cause a catastrophic incident.

Unless otherwise agreed in writing to the contrary, the Contractor shall be responsible for all costs that may be required in order to satisfy these HSSE Requirements.

1.3 Reference Documents

This HSSE Requirements form part of the 'Owner's Technical Requirements' and are to be read in conjunction with any other specific HSSE requirements specified in Contract.

In the event of any ambiguity or discrepancy between the HSSE Requirements and the Owner's Technical Requirements, the Contractor shall proceed to comply with the highest quality or standard specified or perform the more onerous requirement.

2 HSSE Leadership

2.1 HSSE Policies

The Contractor shall observe all relevant provisions as stated in:

- The CLP Value Framework;
- The CLP Group Occupational Safety and Health Policy;
- The CLP Group Environmental Policy;
- The CLP Responsible Procurement Policy Statement;
- The CLP Group Safety Principles,
- The CLP Group No Alcohol and Drug Abuse Policy,
- The IFC Performance Standards, and
- Equator Principles.

The abovementioned documents will be made available to the Contractor upon request. The Contractor shall develop HSSE policies, which are:

- documented and visibly endorsed by top management;
- of a standard comparable to Owner's HSSE policies and principals;
- communicate to all of the Contractor's Personnel;
- implemented and maintained at all levels in the Contractor's organization, and
- periodically reviewed and updated.

2.2 Target and Objective Setting

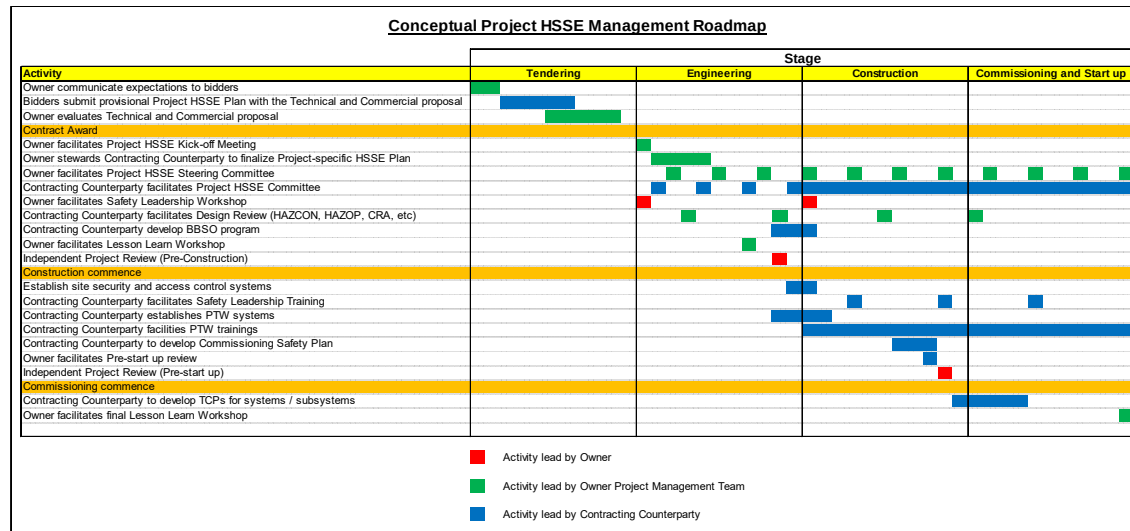
The Contractor's HSSE goal shall be able to reflect Contractor's endeavor to ensure to an incident free workplace. Targets and objectives shall also be focused on leading indicators for measuring proactive efforts to reduce the likelihood and severity of incidents.

2.3 Project HSSE Plan

2.3.1 Project-specific HSSE Plan

The Contractor shall submit a project-specific HSSE Plan which describe key HSSE requirements and a roadmap for the development of project-specific HSSE Management System to ensure proper plans and procedures would be in place prior to the commencement of different stages of the project proposed by the Contractor, e.g. Pre-NTP work, substructure work, superstructure work, mechanical and electrical installation, commissioning, etc.

The Contractor shall agree the Project HSSE management roadmap based on the following conceptual Project HSSE roadmap which summarized the key HSSE processes and deliverables.



The project-specific Project HSSE plan shall be made available and approved by the Owner prior to the commencement of any construction or servicing activity involving Site work or at most, within thirty (30) days of the date of the Notice to Proceed. The project-specific HSSE Plan shall be implemented with periodic monitoring on its compliance status and effectiveness by the Owner. Section 2 to Section 5 of this document identifies the Owner's expectations on the Project-specific HSSE Plan.

The Contractor shall strictly comply with the requirements as specified in the Environmental Impact Assessment (EIA), Environmental, Social and Health Impact Assessment (ESHIA) report, IFC Performance Standards and Equator Principles as well as requirements / conditions specified in the associated permit or approval letters.

2.3.2 Revision of HSSE Plan

The Contractor shall review the Project-specific HSSE Plan:

- At least on a yearly basis,
- Prior to any phase (Engineering, Construction and Commissioning) or part of the work where new HSSE hazards are identified and corrected, and
- Any proposed revision of the Project-specific HSSE Plan shall be submitted to the Owner for review and comment prior to implementation.

2.4 Project HSSE Committees

2.4.1 Project HSSE Kickoff Meeting

The Owner will facilitate a Project HSSE Kickoff Meeting to further explain the Owner's expectations on HSSE management and align with the Contractor and the Contractor's Personnel regarding the HSSE performance to be delivered.

2.4.2 Project HSSE Steering Committee

The Contractor shall actively participate in a Project HSSE steering committee which will be held at least on a quarterly basis, the committee shall comprise of senior project management and senior project HSSE personnel from the Owner, the Contractor and selected critical Subcontractors. The Contractor shall work together with Owner to develop a charter for the committee. The Project HSSE steering committee meetings will serve as a forum to further define, implement, and steward the HSSE management of the Project.

2.4.3 HSSE Committee

The Contractor shall establish a HSSE committee as a forum to demonstrate leadership and commitment to HSSE. A charter shall be established to define the targets and objectives, and the composition of the committee. The committee shall:

- set objectives and target on HSSE aspects;
- resolve HSSE issues by initiating improvement plans;
- establish and monitor performance indicators (both leading and trailing indicators);
- formulate and implement incentive and disciplinary action procedures;
- compose of line management and HSSE personnel from Contractor and Subcontractors;
- invite workers' representative(s) as member and actively seek feedback from the workforce;
- invite the Owner to act as observer of the meeting;
- meet shall be held at least
 - o quarterly during engineering or design phase;
 - o monthly during construction and commissioning phase, and
 - o weekly during the period specify by the Owner agreed by the parties, such as period with peak workforce on Site.
- conduct a Site walkthrough by the committee members before the meeting.

2.5 Safety Leadership and Behaviour Safety

2.5.1 Safety Leadership Workshop

The safety leadership workshop aims to provide management with the appropriate knowledge, skills and techniques to actively & visibly support safety by example (i.e. management walkthroughs, meetings, personal discussions, etc).

The Owner will facilitate a safety leadership workshop after Notice to Proceed and prior to commencement of major civil works, aimed at senior and middle management of the Owner, the Contractor and critical Subcontractors. After the initial leadership workshop, the Contractor shall facilitate the same training to the Project Site management including supervisor level based on project requirements.

2.5.2 Management Walkthrough

The Contractor shall identify line managers and Subcontractor representatives to perform safety walkthroughs on a weekly basis. The objectives of such walkthroughs are to demonstrate commitment and leadership to HSSE by management in a highly visible manner, solicit feedback from workforce, monitor HSSE performance, and most important, reinforce HSSE expectations and safe behaviours.

2.5.3 Behaviour Based Safety Observation (BBSO)

The Contractor shall implement a BBSO program as performed by trained observers to positively reinforce safe behaviours and to change at-risk behaviours through constructive feedback.

Examples of BBSO programmes include:

- B-Safe
- STOP
- Loss Prevention Observations
- customized inspections which utilize a structured mechanism for providing worker feedback and capturing data to evaluate performance

The BBSO program shall include both observation and intervention processes according to an established guideline which describes the methodology on the approaches and techniques to achieve effective observation and intervention. All at-risk behaviours being observed shall be resolved as soon as practicable; all actions shall be tracked for closure.

By analyzing incident data, interviews with personnel and other relevant sources, safety critical behaviours shall be identified as the key focus of the program.

Checklist including the safety critical behaviours shall be used to guide the observers to carry out BBSO systematically. The checklist could be in the form of card or portable hand held device, the data collected shall be used for trend analysis.

The BBSO program shall target to achieve 1 trained observer for every 15 workers to conduct 1 observation per week.

3 Plan

3.1 Hazard Identification, Risk Assessment and Control

The Contractor shall develop a risk management process to identify, evaluate and control health, safety and security hazards and significant environmental and social aspects. The Contractor shall ensure that:

- the appropriate type of risk assessment is used according to the risk and complexity of the Works;
- results of risk assessment are communicated to the relevant workforce prior to commencement of the work tasks to which the risk assessment refers to;
- implementation of control measures are monitored, and

- risk assessments are reviewed regularly.

CLP Group HSSE Management Guideline – Hazard Identification, Risk Assessment and Control (HSSE-GUL-06-001), IFC Performance Standards and Equator Principles will be made available to Contractor for information upon request.

3.1.1 High Level Risk Assessment

The Contractor shall undertake a high level risk assessment for high risk and/or complex tasks by evaluating inherent risks and residual risks utilizing the CLP Group Risk Matrix [Attachment 1]. High level risk assessment team normally involves multi-disciplinary personnel and subject matter experts.

Additional risk assessment such as Specialist Risk Assessment or Task Based Hazard Analysis may be required of the Contractor by the Owner to further study a particular risk upon completion of High Level Risk Assessment.

3.1.2 Task Based Hazard Analysis

The Contractor shall utilize task based hazard analysis as a pre-task hazard identification process for the front line supervisors and work groups to identify potential hazards associated with each step and the control measures that eliminate or mitigate such hazards.

Examples of task based hazard analysis include:

- Job Safety and Environment Analysis (JSEA);
- Job Hazard Analysis (JHA);
- Hazard Identification Risk Assessment (HIRA);
- Task Hazard Analysis (THA);
- Task Risk Assessment (TRA), and
- Total Safety Task Instruction (TSTI).

The task based risk assessment shall be:

- Developed by frontline supervisors with participation from the workgroup and supported by HSSE personnel and other subject matter experts;
- Developed specifically for all high risk and/or non-routine tasks;
- Developed generally for all low risk and/or routine tasks;
- Reviewed with the work team prior to commencement of work tasks to which the risk assessment refers to;
- Written in English and preferably with also the working language of the work team, and reviewed verbally if literacy is a concern, and
- Monitored by the responsible front line supervisor to ensure implementation of control measures.

3.1.3 Pre Work Risk Assessment (PWRA) and Briefing

Each work group of the Contractor shall conduct PWRA and briefing prior to work in each shift, OR after change in methodology, work environment and/or team members. The Contractor shall record the completed PWRA in a prescribed form of the Owner.

The aim of the pre-shift briefing is to conduct pre-task planning including review of permit requirements (if any), toolbox talk and sharing / capturing incidents learning and best practices.

The aim of PWRA is to identify hazards of daily activities (6 major sources of risk: Human, machinery, material, method, work environment and environmental protection) and to record the control measures to be adopted. PWRA shall be conducted by the responsible frontline supervisor at the work location and communicated to all personnel of the work team, visitors and other adjoining parties to ensure the safety of those personnel.

3.1.4 Personal Risk Assessment

The Contractor shall implement a non-written personal risk assessment process, which facilitates involvement of individuals in hazard identification and control prior to performing a specific task or step of a task or after work completion. Tools such as cards and posters of appropriate languages should be provided to assist the workforce to implement.

Examples of Personal Risk Assessment include:

- Stop and Think
- Last Minute Risk Assessment (LMRA)
- Safe Performance Self-Assessment (SPSA)
- Step Back 2X2 or Step Back 5X5

The basis process may utilize the following steps:

- STOP – Step back and observe
- THINK – through the task to be carried out
- IDENTIFY – potential hazards
- ASSESS – controls required to manage the risks
- CONTINUALLY – monitor risks

3.2 Environmental Impact Assessment

Subject to the terms and conditions in the Contract, the Contractor shall comply with the requirements as specified in the Environmental Impact Assessment (EIA), Environmental, Social and Health Impact Assessment (ESHIA), IFC Performance Standards, Equator Principles and the associated environmental permit / approval letters, if any. The Contractor shall formulate and implement an Environmental Monitoring and Management Plan, and also an assessment and management of Environmental and Social Risks and Impacts throughout the construction phase.

3.3 Legal and Owner's HSSE Requirements

Owner's goal is zero injuries for all Works. The Contractor shall comply fully with all applicable Regulatory Requirements and the requirements of any Government Bodies when executing any Works.

The Contractor shall bear the responsibility at all times to identify and exercise all necessary precautions for the health, safety and security of all persons including the Contractor's Personnel, the Owner's Personnel and other persons who may be affected by the performance of Contractor's Works as well as protecting the environment as specified in the Contract.

The Contractor shall observe all relevant clauses as stated in Owner's policy in safety, health and environment as set out in Section 2.1. Copies of these documents and the update will be made available to the Contractor upon request.

In addition to Regulatory Requirements, the Contractor shall also comply, whenever applicable, with all relevant HSSE Requirements as defined by the Owner, in particular, the following:

- Owner / Operator's Energy Isolation Rules;
- CLP Group HSSE Management System Standard;
- CLP's Critical Risk Standards
 - o Vehicle and Mobile Equipment (HSSE-STD-02-001)
 - o Safe Switching Operation (HSSE-STD-02-002)
 - o Working at Height (HSSE-STD-02-003)
 - o Protection against Hazardous Energy (Work Permits) (HSSE-STD-02-004)
 - o Lifting Operation (HSSE-STD-02-005)
 - o Confined Space (HSSE-STD-02-006)
 - o Hand and Powered Tools (HSSE-STD-02-007)
 - o Working around Water (HSSE-STD-02-008)
 - o Excavation (HSSE-STD-02-009)
 - o Hazardous Substances (HSSE-STD-02-010)
- CLP Group HSSE Management Standard – No Alcohol and Drug Abuse (HSSE-STD-02-011);
- CLP Group HSSE Management Standard – Incident Management (HSSE-STD-13-001);
- CLP Group HSSE Management Standard – Performance Monitoring and Reporting (HSSE-STD-14-001);
- CLP Group HSSE Management Standard – Environmental Impact Assessment (HSSE-STD-05-001),
- IFC Performance Standards,
- Equator Principles, and

- any other additional HSSE requirements, instructions, procedures, guidelines etc. as defined in the Contract.

The Contractor shall make itself aware of and comply with the content of the above listed HSSE documents issued by Owner, which will be made available to Contractor upon request.

Gambling is not allowed anywhere on the Site. Smoking is only allowed at the designated smoking areas approved by the Owner.

4 Do

4.1 HSSE Organization

The Contractor's HSSE organization shall provide fit-for purpose HSSE organization onsite and offsite support to plan, execute and monitor the Works.

4.1.1 Provision of HSSE Personnel

The Contractor shall provide one qualified, full-time, and dedicated Project safety and health Manager to provide oversight for safety throughout the planning and execution phase of the Project. The proposed Project safety and health manager shall be nominated to the Owner immediately following the issue of the NTP. The nominated Project safety and health manager shall have a minimum of 10 years of experience in safety and health management, of which at least 3 years must be at managerial level.

The Contractor may extend the responsibility of the safety and health manager to cover security aspects, provided that the nominated candidate has the required competence and experience on managing these aspects, and specialized staff(s) should be staffed to provide technical support on respective disciplines.

The Contractor shall also provide one qualified, full-time, and dedicated Project environmental manager to provide oversight the development and implementation of Environmental Monitoring and Management Plan throughout the planning and execution phase of the Project. The Project environmental manager shall have experience on EIA and ESIA, and associated environmental monitoring and audit process. The proposed environmental manager shall be nominated to the Owner immediately following the issue of the NTP.

The Contractor shall provide qualified, full-time and dedicated safety and health officer, who possessed with a national or international recognized registration, certification or membership, and a minimum of 3 years post-qualification experience.

The Contractor shall provide qualified, full-time and dedicated safety and health supervisor, who possesses a national or international recognized qualification and a minimum of 2 years of experience.

The following table summarizes the minimum number of HSSE personnel to be employed relative to the number of Contractor (including subcontractors) employees on Site

Number of Employees	Number of Safety and Health Officer(s)	Number of Safety and Health Supervisor(s)
1 – 10	0	1

11 – 200	1	1 for every 50 employees
201-400	2	
401-600	3	
601-800	4	
>800	4 + 1 for every 300 employees	

Unless the nominated safety and health personnel are adequately knowledgeable and experienced in Security and Environmental Management, the Contractor shall provide the adequate number of qualified staffs to support the implementation of such systems.

If any HSSE personnel are unable to perform their duties for any reason, the position shall be replaced within fourteen (14) days. The performance of the HSSE personnel shall be continuously monitored by the Owner. The replacement personnel will need to comply with the same qualification requirements specified above

4.2 Personnel Training and Competence

4.2.1 General Requirements

The Contractor shall ensure that all employees employed in the Works are legally entitled to be so employed and that all necessary visas and permits have been obtained.

All Contractor's employees shall be able to communicate and understand the work procedures as provided by their supervisors, be physically fit, possess sufficient competence and skills for the Works .

The Contractor shall report to the Owner the attendance record of their working personnel on Site everyday.

4.2.2 HSSE Training

The Contractor shall ensure that all its personnel have attended relevant HSSE training (including training provided by Owner) as well as specific HSSE competency training in order to acquire necessary competency for fulfilling their job requirements.

The Contractor shall prepare a training matrix to identify training requirements for management, supervisors, general workers, and workers requiring specialist skills, etc. based on:

- Specific job assignments (such as plant operators, divers, HSSE management training, supervisor induction training, etc.)
- Potential exposure hazards and work environment (such as induction training, hazard awareness, hot work, confined spaces, working at height, confined spaces, marine work, permit to work, etc.)
- EIA, ESIA, IFC Performance Standards and Equator Principles requirements and the associated significant environmental aspects identified in Section 3.1.
- Regulatory requirements

The preliminary training matrix shall be submitted to Owner within thirty (30) days after the date of issue of the Notice to Proceed for review. Monitoring of the agreed training matrix shall become part of the Contractor's overall monitoring process.

The Contractor shall develop an Induction Training Program for all Contractor's Personnel which provides a basic familiarization of the Project HSSE expectations and requirements, which should:

- Provide an overview of the project and Site facilities;
- Provide an overview of key HSSE documents, such as policies, plans, procedures and rules;
- Communicate HSSE roles and responsibilities of respective positions (e.g. managers, supervisors, HSSE personnel and workers);
- Identify the types of hazard and significant environmental and social aspects that may be encountered on Site and respective control measures;
- Provide an overview on Emergency Preparedness and Response procedures and Incident Reporting procedures;
- Require training time of at least three (3) hours, and
- Require post training test to ensure understanding of content and refresher training each year.

The Contractor shall establish mechanisms for controlling personnel access into the Site, including validating identity record for annual safety induction training. All personnel must complete induction training before entering the Site.

The Contractor shall utilize knowledgeable internal or external training personnel with adequate experience and qualification on respective subject matters. Visual and audio aids shall be used to optimize understanding of training content, especially where workforce literacy and language barriers are a concern. In general, training provided to the employees of the Contractor shall be conducted in the language which the attendees understand unless specified in the OTR.

Post training assessment (e.g. test, examination and practical assessment) shall be conducted to evaluate the understanding of attendees, especially for competence training.

The Contractor shall provide a process to verify training for all personnel is current and all training records are ready for inspection by the Owner for auditing purpose. Evaluation on the training programs shall be conducted regularly to identify improvements to enhance the training effectiveness.

CLP Group HSSE Management Guideline – Personnel Training and Competence (HSSE-GUL-08-001) will be make available to Contractor for information upon request.

4.3 Subcontractor Management

The Contractor shall ensure that Subcontractors comply with the HSSE Requirements and are accountable for their HSSE performance.

The Contractor shall develop a Subcontractor management process to select, evaluate, monitor and control subcontractors utilized to support execution of work. The Subcontractor management process includes:

- A selection process to select potential bidders who would be capable of satisfying Contractor and Owner's HSSE requirements and expectations;
- An evaluation process of bidders which HSSE performance is an evaluation criteria, Owner reserves the right to participate in the evaluation of HSSE capability of any Subcontractor;
- A control process which formally communicates HSSE requirements to Subcontractors, and
- A monitoring process which includes verification of compliance in the relevant subcontract and HSSE performance indicators.

4.4 Personal Protective Equipment

The Contractor shall equip each of his personnel with the appropriate and approved personal protective equipment ("PPE") for that particular job activity and ensure that all of the Contractor's Personnel are adequately trained and understand the proper use of the PPE.

The Contractor must provide and enforce proper use of the following PPE in all Site areas unless an area is designated as "Safe Zone":

- safety helmet (standard EN397 or equivalent);
- safety shoes with steel toe cap and midsole (standard EN345 / EN ISO 20345 or equivalent);
- reflective vests (with 3M/Non 3M 2"/3" width reflective strips), and
- safety glasses for impact protection (standards EN 166 / EN 172 / ANSI Z87 or equivalent).

The Contractor must provide and enforce proper use of safety harness ("5-points parachute type" in accordance with standards EN 361 / ANSI / CSA) with dual lanyard by workers working at height (where there is a hazard of potential fall exceeding 2m).

The Contractor shall develop a PPE program to:

- identify needs of PPE to protect personnel from hazards to be encountered;
- identify areas where specialized PPE (e.g. fall protection systems, special face / eye protection, gloves, respiratory protection, hearing protection, personal floatation devices) is required and provide signage at prominent location (e.g. high noise, bulk chemical storage and handling and etc);
- select appropriate PPE comply with approved / recognized standard;
- inspect, store and maintain in alignment with manufacturer's requirement;
- provide training on the selection, use and maintenance of PPE, with particular concern on the limitation of PPE;
- monitoring process to verify the compliance of PPE requirements, and
- maintain records of inspection and issue of PPEs in form of register.

4.5 No Alcohol and Drug Abuse Policy and Medical Examination

The Contractor's Personnel shall not use, possess, distribute or sell alcoholic beverages, illicit or un-prescribed controlled drugs, drug paraphernalia, or misuse legitimate prescription drugs while on the Site.

Owner requires Contractor's personnel undertaking Designated Positions to undergo pre-appointment, pre-assignment, and yearly **medical examinations** as well as **random A&D tests**.

Contractor shall comply with CLP Group HSSE Management Standard – No Alcohol and Drug Abuse (HSSE-STD-02-011), and will be made available to Contractor upon request.

Cost of medical exam and A&D test shall be borne by the Onshore Contractor. All Designated position shall be subjected to A&D random test which is conducted, in general, 25% per quarter. Classification shall at least include operator of 50 ton crane, persons issue energy isolation permit, person responsible for HV switching and coach driver. All mobile plant operator shall be included in A&D tests.

Examples of substances covered in the Alcohol & Drug Test include:

List of Substances Under Urine Test	Designated Position	Standard Testing Threshold Levels	
		Screening Level*	Confirmatory Level#
Alcohol	√	60 mg/100 ml [22 µg/100 ml of exhale air for breathalyzer test]	60 mg/100 ml
Amphetamines	√	1000 ng/ml	500 ng/ml
Barbiturates	√	300 ng/ml	200 ng/ml
Benzodiazepines	√	300 ng/ml	200 ng/ml
Cocaine	√	300 ng/ml	150 ng/ml
Cannabinoides (marijuana)	√	50 ng/ml	15 ng/ml
Opiates	√	300 ng/ml	300 ng/ml
Methadone	√	300 ng/ml	200 ng/ml
Phencyclidine	√	25 ng/ml	25 ng/ml
Propoxyphene	√	300 ng/ml	200 ng/ml

Remarks: * Drug test cup or dip card with FDA approval shall be used for screening test

Specimen exceed the screening level shall be sent to approved laboratory for confirmation test

Example of medical examination coverage:

Medical History	• Current complaints • Review of systems • Medication • Dietary history • Social history
Physical Examination	• Full complete physical examination
Associated Test	• Hearing test (conversational voice at 2 meters) • Eye examination and colour vision • Prostate specific antigen • Chest X-ray • ECG • Urine analysis for Glucose and Protein

Blood Test & Analysis	• MCV • Hematology: Hb, WBC
Diabetes Screening	• Biochemical screening profile: glucose
Kidney Function	• Blood urea • Creatinine
Heart	• SGOT
Liver Function	• SGPT • Gamma GT • Bilirubin • Total protein and albumin • Globulin
Gout	• Uric acid
Lipid panel	• Cholesterol • Triglycerides • HDL cholesterol • LDL cholesterol • Chol/HDL chol ratio

4.6 Emergency Preparedness and Response

The Contractor shall establish Site-specific emergency preparedness and response procedure to minimize the impacts to people, environment and/or assets. The procedure shall include:

- An emergency organization which describe the roles and responsibilities of both internal and external parties, and means of communication;
- Identification of training needs for both responders and non-responders;
- Identification of credible emergency scenarios which might encounter during the execution of work, which might include but not limited to the following:
 - o Personal injury and illness;
 - o Spillage or release of hazardous substances;
 - o Fire, explosion, or both;
 - o Adverse weather conditions (e.g. typhoon, thunderstorm, flooding and etc);
 - o Contagious diseases (e.g. SARS, avian flu, pandemic flu, malaria, and etc);
 - o Confined space incidents;
 - o Falls from height;
 - o Drowning (if applicable), and/or
 - o Security incidents such as bomb threat, assault and etc.
- Identification of emergency response and rescue equipment according to the credible emergency scenarios, which include but not limited to:
 - o Stretcher
 - o Spine board
 - o Extrication collars
 - o Rescue Rope
 - o Tripod
 - o Oxygen cylinder

- o Resuscitator
- o Automated external defibrillator (AED)
- o Chemical Suit
- o Breathing apparatus
- o Spill response kit / absorbent
- o Rescue kit for falls from height
- Emergency response procedures to address the emergency scenarios, which define the required training, emergency equipment and frequency of drill, and
- Evacuation procedures and procedure for suspension of operation and resuming normal operation.

The Contractor shall nominate the medical services provider which will be responsible for providing and operating the medical facilities on Site. However, Owner reserves the right to appoint an independent medical service provider if the medical service provided by Contractor is not satisfactory.

The services and facilities to be provided by the medical services provider shall include but not limited to:

- Site clinic provided with proper equipment and medical supplies which enable the medical staffs to handle:
 - o Potential injuries such as bone fracture, thermal/chemical burn, deep cut, amputation and etc.
 - o Medical emergencies such as heart attack, loss of pulse, stop breathing, unconsciousness and etc.
 - o Common diseases
- Examples of critical medical equipment of the Site clinic

Critical First Aid Equipment for a Construction Site Clinic / First Aid Room		
Purpose	Description	More details
Clinic Facility	Two-section couch	with fire-retardant vinyl leatherette
	Examine light	Pantophos (Ultra-high intensity, Color-corrected, Cool Ray)
	Patient's chair / Examine chair	
	Foot rest	
	Autoclave	
	Infection control set	
Responder bag and crash trolley at clinic	Diagnostic set	Stethoscope, sphygmomanometer, etc.
	Airway set	

	Intravenous (IV) set	
	Syringe and needle set	
	Dressing set	
	Burn and eye wash set	
	Emergency drug and consumable set	
Ventilator	Ventilator	provided on ambulance
Resuscitation	Site ambulance	
	Silicone resuscitator (Ambu-Bag)	with case & airways
	Automatic External Defibrillator (AED)	with electrocardiogram function, memory card and spare battery
	Medical oxygen	Full set with oxygen cylinder, regulator, tubing, Masks & airways
	Cardio monitor	With SpO2 and BP monitoring
	Pulse oximeter	1 in medical responder bag and 1 in clinic
	Suction unit	1 manual suction and 1 power-driven
	Laryngoscope with blades	Fiber optic single use
Immobilization	Basket stretcher	with restraints and lifting straps
	Scoop stretcher	with pairs straps and buckles
	Universal head immobilizer	with Head Belt & Fasten Belts
	Spine board	with pins & strap
	Traction splints	
	Emergency air splints	
	Stiff neck extrication collar	
	Padded board splint	
	SAM splint	
Transportation	Stretcher trolley / folding stretcher	
	Vacuum mattress	(for long distance transport)
	Patient wheel chair	
	Emergency carrying chair	

- Suitable number of medical doctors and paramedic staffs to ensure coverage to the Site working hours;
- Site ambulance / Site vehicle that can be equipped for conveyance of injured / ill persons from accident scene to hospital;
- Train up suitable number of persons in basic first aid;

- Provide health and hygiene related training on Site with particular concern on contagious diseases and sexually transmitted diseases (STD), and
- Disposal of medical waste shall be treated according to Regulatory Requirements and all sharps shall be stored in sharp box for disposal.

CLP Group HSSE Management Guideline – Emergency Preparedness and Response (HSSE-GUL-12-001) will be made available to Contractor upon request

5 Check

5.1 Performance Measurement and Monitoring

The Contractor is strongly encouraged to abide by applicable industrial standards for safety and health and environmental performance that meet or exceed Regulatory Requirements.

The Contractor shall provide evidence of its on-going performance of HSSE management. The performance indicators shall report on a monthly basis and breakout by period (e.g. monthly, year-to-date, project-to-date, etc.). The performance indicators shall include but not limited to:

- Lagging Indicators:
 - o Number of injuries – Fatalities, Lost Time Injuries, Disabling Injuries, Restricted Work Injuries, Medical Treatment Injuries and First Aid Cases
 - o Number of incidents (fire, property damage, security, spill, release, regulatory compliance, legal proceeding, dangerous occurrence and etc) and near miss
 - o Incident rates per 200,000 man-hour – Fatalities, Lost Time Injuries, Disabling Injuries, Total Recordable Injuries (Fatalities and Injuries, except First Aid Cases)
 - o Percentage of environmental compliance (e.g. Percentage of environmental monitoring results compliance), number of environmental incidents (definition and categorisation to be agreed and aligned with Owner), (e.g. malpractices on waste handling, illegal dumping, etc.) and environmental complaints (justified).
- Leading Indicators
 - o System Integrity – percentage of scheduled activities (e.g. Management Walkthroughs, Audits, Inspections, Drills, JSA, etc) against planned, and/or activities per 200,000 man-hours
 - o Action Follow-up – percentage completion of actions plans (e.g. Audit actions, incident actions, hazards identified, etc)
 - o Training and Competency – percentage of training / tool-box-talk completed against planned

The Contractor shall liaise with the Owner when the Project commences, to confirm the leading and lagging indicators to be reported, leading indicator shall focus on the key HSSE programs of the Contractor and may change as the Project progresses. The following are the key leading indicators to be reported:

- PWRA/JSA per 200,000 manhours

- BBSO/Inspection per 200,000 manhours
- Management walkthrough per 200,000 manhours

The Contractor shall provide the following information in no later than seven (7) calendar days after the calendar close of the previous months:

- Leading and Lagging Indicators (Contractor and Subcontractors).
- man-hours and Lost Time Injury free man-hours (Contractor and Subcontractors), man-hours may be estimated if it is necessary to achieve the Owner reporting deadline.
- implementation status and outcome of any corrective and preventive actions highlighted in inspections by relevant authorities and/or the Owner, incident investigations, and HSSE audits and reviews in the level of detail required by the Owner.

The Contractor shall provide reports to the Owner on minor incidents (including near-misses, first-aid cases), HSSE inspections, incident investigations, audits, reviews and risk assessments as stipulated in the Contract or upon request by the Owner.

Contractor shall comply with the CLP Group HSSE Management Standard – Performance Monitoring and Reporting (HSSE-STD-14-001), a copy of which will be made available to Contractor upon request.

5.2 Mitigation of Non-Compliance

In the event of any non-compliance with the HSSE Requirements, the Contractor shall take prompt and reasonable action to minimise the harmful outcome, to restore a safe condition, and to seek emergency services if needed. The Contractor shall interrupt the Works and/or Services as necessary to prevent impending or further harm of this nature.

In the event that the Owner has identified any area where the Contractor is not complying fully with the HSSE requirements and has instructed the Contractor to rectify such non-compliance, the Contractor shall promptly take remedial action.

5.3 Incentive Scheme and Disciplinary Actions

The Contractor shall utilize both incentive scheme and disciplinary actions to reinforce desirable behaviors.

In general, incentive scheme shall be based on merit, participation to HSSE programs, and achievement of performance indicator targets. It could be recognition letters, certificates, events, distribution of nominal items and etc.

Disciplinary actions for undesirable behavior shall be defined clearly and communicated to the workforce. In general, disciplinary actions include re-training, suspension from work with / without pay and dismissal, monetary penalties shall be agreed between the Owner and the Contractor before start of work. The Contractor shall also develop a documented procedure for implementation which includes an appeals mechanism.

5.4 Incident Management

The Contractor shall whenever possible, immediately notify the Owner of the following incidents / irregularities:

- any incident resulted in personal injury, occupational illness, environmental incidents, exceedance on environmental monitoring, environmental complaints, dangerous occurrence (if applicable), theft incidents, regulatory noncompliance, complaint or warning from the public or relevant authority, significant near miss etc., during the execution of the Works and/or Services under the Contract.

The Contractor shall verbally report the incident to the Owner immediately within the shift of occurrence, and shall within three (3) working days provide a written report on the incident. The report shall include a summary of what was discovered during the investigation and any actions taken subsequently.

At the Owner's discretion, any incident may be further investigated by the Owner with the Contractor's co-operation. Such investigation is intended to determine the cause of the incident and identify measures that could prevent a recurrence. The Contractor shall provide a monthly summary on the occurrence of the incident and the status of corrective actions.

The Contractor shall establish an Incident Management System which includes:

- A case management program to ensure the injured / ill persons is provided with prompt and quality medical diagnosis and treatment;
- A rehabilitation program to allow injured to resume duty or restricted duty as soon as possible without compromising the recovery process;
- Requirements on classification, notification and investigation of incidents in alignment with:
 - o CLP Group HSSE Management Standard – Incident Management (HSSE-STD-13-001);
 - o CLP Group HSSE Management Standard – Standing Instruction for Accident Investigation Panels (HSSE-STD-13-002),
 - o CLP Group HSSE Management Standard – Performance Monitoring and Reporting (HSSE-STD-14-001).
- Criteria for selection and composition of the incident investigation team and the depth of investigation;
- Defining the root cause analysis technique for conducting incident investigation of recordable incidents and significant near misses;
- Incident sharing with the Owner to communicate lessons learned, and
- Incident analysis every six (6) months to identify trends of incident and formulate action plans for improvement.

5.5 Near-miss and Hazard Reporting

The Contractor shall develop a near miss and hazard reporting mechanism to allow workers to report near misses and hazards (as well as at-risk behaviours and/or condition)conditions. The mechanism shall:

- Allow personnel to document or verbally communicate a near miss or hazard, such as a simple report form, hotline and etc;
- Provide feedback to the person reported in a timely manner, and

- Provide data for different trend analysis, such incident analysis and inspection analysis.

6 Act

6.1 HSSE Inspections and Audits

The Contractor shall define an inspection and audit program for systematically sending people at the Site for the purpose of verifying the implementation of HSSE management plan and compliance with these HSSE Requirements.

Inspection shall be carried out by various levels of Contractor's organization to identify, resolve and trending HSSE issues at Site. The Contractor shall ensure all designated supervisors and HSSE personnel will perform HSSE inspections at least on a weekly basis. The scope and frequency of inspection shall be defined clearly and all inspection shall be recorded on an inspection checklist to enhance the effectiveness of inspection. At risk observations shall be prioritized and followed up by the Contractor, analysis on at risk observations shall be conducted at least quarterly to identify trends and associate actions for improvement.

The Contractor shall conduct internal audit in every six (6) months to:

- evaluate the effectiveness of current HSSE management plan;
- verify the compliance of implementation of HSSE plan;
- identify opportunities for improvement, and
- develop action plans to drive improvement

At Owner's discretion, in-house or third party audit will be conducted at least every six (6) months to evaluate the effectiveness of Contractor's HSSE plan. The Contractor will be notified at least fourteen (14) days in advance of the audit.

The Contractor shall:

- provide all necessary access, assistance and facilities to enable the Owner to inspect and verify that work is being undertaken in accordance with the Contract, to verify compliance with the Regulatory Requirements, approved HSSE Plan, Owner's Technical Requirements and to observe safety practices; and
- provide suitable qualified staff to accompany the Owner and provide all necessary access, assistance and facilities to enable such audit is conducted.

7 Safety Requirements

7.1 Design Safety and Constructability

The Contractor shall develop and implement an integrated design safety and constructability program which is an integral part of detailed design. The program shall:

- Advance the design safety and constructability concepts developed during engineering;

- Engage subject matter experts from the design, construction, operation, and safety disciplines;
- Integrate major construction subcontractors into the construction related program;
- Specific requirements – design safety

The design safety program shall provide for early and continuous input into the design by key construction and operation stakeholders and other disciplines to drive specific focus on Construction and Operation related safety issued during design stage that:

- o Implements a risk-based approach to integrate feedback from subject matter experts into detailed design and early job planning to enhance safety of work during construction
- o Increases focus during the detailed design phase on construction worker safety
- o Increases focus during the detailed design phase on ergonomics and other HSSE related issues for operation, such as:
 - Display and control knobs on control console
 - Illumination and acoustic design for control rooms and simulators
 - Design on layout, size, operating height, operating forces on switches and circuit breakers
 - Access to, height of operating, operating forces of lever or hand wheels
 - Access to and from switches, valves, gauges, sampling ports, etc.
 - Access to equipment at high-bay ceiling, such as ceiling lights in turbine hall, maintenance on services
 - Labeling on equipment and controls (language, font size, colour, positions, etc)
 - Signage
- o Colour coding on pipework and ducting
- o Provides a disciplined approach for identifying, evaluating and eliminating or mitigating hazards which may be encountered in the construction phase
- Specific requirements – constructability

The constructability program shall provide for early and continuous input into the design by key construction stakeholders and other disciplines to establish guidance and direction to the design as it relates to constructability, including:

- o Execution steps and sequence
- o Productivity enhancements
- o Measures to reduce construction costs and schedule
- o Pre-assembly, pre-fabrication, modularization, and standardization
- o Accessibility of laydown and staging areas during construction

- o Personnel accessibility during construction phase as well as on-going operations and maintenance purpose
- The Contractor shall provide training to design engineers, construction engineers and safety personnel regarding the concepts of Design Safety and Constructability in the early stage of detail design.
- Schedule and facilitate design reviews (hazard identification processes such as HAZCON / HAZOP, construction risk assessment and etc) at specific design milestones attended by appropriate design, construction, operation, and safety subject matter experts from the Contractor, major subcontractor and Owner.
- The Contractor shall prepare a design safety and constructability register to prioritize the identified improvement opportunities, assign responsibilities, and update monthly to track closure of action items.

7.2 Permit to Work

The Contractor shall establish a permit to work (PTW) system to systematically plan, execute and control critical tasks at the Site (which is solely controlled by the Contractor). This system shall include administration procedures, training and appointment of qualified personnel for issuing and receiving permit to work, compliance audits, record keeping, lockout/tagout procedures, signage and labeling standards, etc

Only one PTW system will be utilized at each area, the Contractor shall liaise with the Owner to define clear system boundary if needed. Regardless which PTW is implemented, the Contractor is responsible to ensure all permit requirements are complied with while executing the work.

In general, the following types of permit should be utilized:

- Hot work permit
- Permit for protection against hazardous energy (Lockout tagout on electrical, mechanical, pneumatic, etc.)
- Confined space entry permit
- Excavation permit
- Critical lift permit
- Radiography permit
- Diving work permit
- Work in hazardous zone of hazardous atmosphere
- Removal of fall protection (guardrail and grating/temporary cover) permit

Contractor's PTW system shall:

- describe the roles and responsibilities, and authorization of personnel involved in the PTW process;
- define boundaries of application;
- describe the PTW process including:

- o permit request/application
 - o risk assessments which are integrated with the requirements in Section 2.5 of this document
 - o permit issuance defining additional safety precautions and permit validity
 - o specific requirements, such as gas test, isolation and earthing standards
 - o permit acceptance / recipient
 - o working party acknowledgement
 - o work execution and work supervision
 - o shift handover requirements
 - o permit extension and permit suspension
 - o complete work and clearance of permit
 - o working party declaration and communication
 - o permit cancellation
 - o internal effectiveness assessment
- describe the requirements for performing daily coordination meetings in order to evaluate and resolve potential conflicts among operations and tasks to be performed; and

Prior to commencement of commissioning phase, the Contractor shall liaise with the Owner to prepare a structured transition plan to ensure smooth transition from the PTW to the Owner's PTW.

7.3 System Completion

System completion spans the entire life cycle of a project, from engineering to operations. It is a complex activity with multiple interfaces between different discipline, team and organization which require careful management.

It is important to define responsibilities of, and interfaces between the Owner and Contractor. Effective planning, communication and coordination must commence at the early. In general, system completion will be executed in the following stages.

1. Pre-commissioning checks
2. Commissioning
3. Initial operation
4. PPA Tests
5. Tests on Completion (including trial run)

The Contractor shall:

- Comply with the testing requirements as described in the OTR
- Define clear system boundaries

- Identify safety requirements including, vendor cautions, commissioning procedure, hazard awareness, safety system bypasses and simultaneous operations
- Prepare a commissioning safety plan for controlling work permitting during different stages of work, which include:
 - o Construction / installation and commissioning activities which are occurring simultaneously
 - o Interface with Owner / Operator's permitting procedure
- The Contractor and Owner will jointly conduct a pre-start up safety review to confirm that:
 - o the requirements of the OTR have been met
 - o adequate safety, operating, maintenance and emergency procedures are in place
 - o operating procedures, records, etc are verified as installed drawings
 - o defects have been identified and corrective actions are taken
 - o training of affected personnel including contractors has been completed

7.4 Housekeeping

The Contractor is required to keep all work areas reasonably clean at all times. Good housekeeping shall be maintained throughout the day and properly clear the Site following the completion of Works.

Though individual workgroups are responsible for their own housekeeping, but in order to support effective housekeeping, the Contractor shall ensure resources such as trash bin, dump trucks and dedicated housekeeping crews are available.

Housekeeping performance shall be monitored during HSSE inspection and management walkthrough

7.5 Fire Prevention

The Contractor shall utilize risk assessments to identify the relevant fire hazards (e.g. flammable materials and ignition sources) and the control measures to eliminate or mitigate such hazards. Appropriate firefighting equipment (e.g. fire water tank, site water browser, temporary or permanent fire protection systems, and portable fire extinguishers) shall be provided and inspected regularly. Training needs for different level of fire-fighting skills shall be identified to ensure adequate personnel will be available for the foreseeable fire hazards.

The Contractor shall also identify designated smoking area and relevant control measures for the Owner's review, smoking is strictly prohibited unless at approved smoking area.

Hot work shall follow hot work permit procedure to ensure that such work is executed in a controlled manner. All tools, plant and equipment which pose ignition hazards shall be inspected regularly. All hot surfaces shall be properly guarded, insulated and/or signed.

All flammable / explosive substances shall be stored, protected, labeled and handle properly, any spillage shall be clean up immediately. All spontaneous ignition materials (e.g. oily waste) shall be stored in appropriate containers in cool, dry and well ventilated areas.

Fueling procedure shall be established to identify the control measures, with particular concern on bonding and grounding while handling highly flammable substances (e.g. gasoline).

7.6 Pressure

The Contractor shall develop procedures for all pressure tests and leak tests, ensuring that such tests shall:

- be controlled by PTW and JSA;
- be physically isolated (work area) to minimize exposure of non-essential personnel to any type of potential pressure testing failure, leak testing shall have a greater distance for restricted access by general workforce as higher energy release will be resulted in case of any failure occur;
- comply with Work Permit System to ensure the components being testing are isolated accordingly;
- be verified to ensure all facilities and materials used to support pressure testing are designed to withstand the testing pressure;
- be verified to ensure all pressure testing equipment are certified with calibrated gauges to ensure pressure remain with the specified limits;
- utilize water as the preferred pressure testing medium unless otherwise authorized by the Owner, and
- utilize inert gases as the preferred leak testing medium unless otherwise authorized by the Owner, hazards related to Oxygen deficiency due to potential leak of inert gases in confined spaces / areas shall be considered.

8 CLP's Critical Risks (Safety) Requirements

8.1 Vehicle and Mobile Equipment

The Contractor shall ensure all drivers of vehicles and mobile equipment (self-propelled equipment other than vehicle, such as crane, excavator, roller, loader, forklift, etc) are adequately trained and have written authorization from their employer to use the vehicles and mobile equipment provided by the employer. Any mobile equipment that will be used within building or near any temporary structure shall be installed with both visual warning and audible warning devices otherwise will not be allowed to use in such confined area. All heavy vehicle and mobile equipment shall install with reverse audio alarm to alert workers and pedestrian while reversing.

The Contractor shall ensure all vehicles and mobile equipment are under regular maintenance program with proper insurance coverage. All persons riding on a vehicle shall fasten their seat belts (if available), driver / operator using mobile phone is strictly prohibited while operating a vehicle / mobile equipment. Vehicle and mobile equipment shall not be used to convey passengers unless a proper seat is provided, unauthorized personnel shall not be allowed to enter the cabin of mobile equipment.

Parking facilities and cycle sheds shall be provided by the Contractor in a compound within the Site boundary but outside the security fence of the construction area and as near to the Site access as possible. The Contractor's employees' private vehicles will not be permitted

to enter the construction area. Only authorised vehicles shall enter the construction area and these must display the vehicle pass issued by the Contractor.

Designated parking spaces shall be allocated local to the main Site office for the Owner's project manager, Site engineer and other senior staff.

The Contractor shall plan proper Site accesses (both pedestrian and vehicular) within the Site boundary. Designated names shall be provided to Site access roads for easy reference. Pedestrian walkways with proper barricade shall be provided to segregate pedestrian from the heavy vehicle and mobile equipment. Speed limit shall be established and speed bump shall be installed along main haul roads to avoid speeding. Proper traffic sign and lightings shall be provided at all road junctions.

The Contractor shall be responsible for devising his own delivery logistics for this Contract.

The Contractor shall be responsible for any specific and attributable road improvement required for construction traffic access, and for the provision of traffic control where the Site access joins the public road. Public traffic shall not be impeded during the construction period, and access must be maintained for emergency vehicles.

The Contractor shall at all times ensure the public roads, giving access to Site, are kept clean of any materials deposited by traffic from the Contractor or his Subcontractors. The Contractor shall for this purpose provide wheel washing facilities at the Site entrance which will be used as required.

The Contractor shall make good damage to roads, footways, paved areas, boundary walls and fences etc. and clean all fouled roads and pavings etc. whether public or existing Site, caused by or in any way attributable to the handling/transport of Equipment or materials by the Contractor or any Subcontractor and indemnify the Owner against loss, damage or claims by the local authority or others for damage to or fouling of roads pavings etc., by reason of unusual traffic or other causes.

CLP's Critical Risk Standard – Vehicle and Mobile Equipment (HSSE-STD-02-008) is available to provide Contractor more information and will be available upon request.

8.2 Safe Switching Operation (Arc Flash) and Electricity

The Contractor shall identify foreseeable electrical hazards to be encountered while working with various types of permanent and temporary electrical plant and equipment, including specific requirements when electrical plant and equipment is to be utilized in classified areas.

The Contractor shall define the competence required to qualify a person for various types of electrical powered plant and equipment, only authorized person are allowed to perform any types of repair or maintenance to electrical plant and equipment. Signs, tags, and labeling shall be used to identify hazardous areas or hazardous electrical plant and equipment. Use of specialized PPE for different electrical level shall be defined, such as:

- Eye / face protection from arcing;
- Insulated gloves;
- Non-conductive helmet and footwear;
- Insulated tools, and
- Insulated mats and blankets.

Working on live equipment is prohibited unless approved by the Owner, safety working distance shall be established while working in the proximity of live electrical plant and equipment (e.g. overhead line).

All electrical plant and equipment shall be:

- effectively grounded;
- guarded or shielded to ensure all live and exposed components are segregated from inadvertent contact;
- rated appropriately and prohibit overload,
- industrial type power plugs, sockets, and cable leads are to be used. Water-proof equipment shall be used for outdoor environment or near wet areas, and
- inspected to verify integrity of the safety controls (e.g. defective parts, faulty insulation, grounding, loose connections, faults, exposed live parts, etc)

All conduit, cable, and wiring systems shall be protected from damages by various types of mobile equipment, and exposed or unprotected wiring are prohibited.

The Contractor shall develop requirements and guidelines for:

- the use of residual current device (RDC) for all electric machineries and portable tools / equipment;
- the sensitivity shall be maintained below 30mA and tested on a monthly basis with record;
- prevention of inadvertent “backfeeding” from other power sources (such as emergency power generator), and
- working on electrical plant and equipment in adverse conditions (e.g. weather, flooding, etc).

CLP’s Critical Risk Standard – Safe Switching Operation (Arc Flash) (HSSE-STD-02-002) is available to provide Contractor more information and will be available upon request.

8.3 Working at Height

The Contractor shall evaluate the risks associated with working at heights during the design and execution phase to eliminate or minimize exposure whereas reasonably practicable and identify sufficient safeguards in order to mitigate such risks.

The Contractor shall identify various types of fall scenarios and associated control measures, such as fall protection systems and related work procedures and work practices.

The fall protection procedure where there is a potential to fall 2 meters or more shall detail the use, maintain and inspection for fall protection systems including:

- Passive fall protection systems such as scaffolding, elevated working platforms, portable mobile platform, man cage, safety net and etc.
- Active fall protection such as full body harness, dual shock absorbing lanyards, anchorage point, positioning / travel restraint systems, self-retractable / vertical / horizontal lifeline systems and etc.

Personnel exposed to falling hazards and authorized to perform work at height with the use of active fall protection devices, shall be trained to:

- recognize potential falling hazards;
- control measures to prevent falls;
- selection of appropriate fall protection system,
- use, maintain and inspection of fall protection systems; and
- rescue personnel from high levels.

In general, scaffolds shall be erected to provide a safe work platform where any work is carried out over 2 meters above the ground or erect Temporary Works and dismantle it after relevant work has been completed.

Use of full body harness with dual shock absorbing lanyards associated with anchorage system (Single person anchor points shall be capable of withstanding 15kN) shall only be used as a last resort for fall protection where it is impracticable to erect any working platform properly or while personnel are involved in the erection of working platform or staging.

Use of man cages must be strictly controlled and adhered to applicable Regulatory Requirements. Risk assessments shall be undertaken and all involved equipment shall be adequately rated and examined.

All Contractor's Personnel shall assess his/her own physical capability prior to performing work at height. Contractor's Personnel with any visual deficiency shall wear appropriate visual corrective glasses especially when working at height. Personnel that wear reading glasses or eyeglasses consisting bifocal or multifocal lenses are prohibited for working at height or walking at high levels except when working on a safe working platform without risk of fall from height.

When work is carrying on top of grating, the vicinity must be protected to prevent falling of small components and tools to lower level through the opening.

Proper measures shall be provided to safeguard workforce from potential falling objects while performing working at height, such as:

- establish drop zone and restrict access of general workforce;
- utilize tool lanyards and containers;
- scaffold net to enclose elevated working location;
- catch fan, and/or
- covered walkway.

The Contractor shall actively monitor the implementation of working at height activities to ensure compliance to fall protection requirements. A trained rescue team with knowledge in "Suspension Trauma" outfitted with fit for purpose rescue equipment shall be provided at the Site.

All ladders must be uniquely identified, listed in a register and be subject to formal inspections. Self-made ladders are not permitted. Working at height on ladders and rope ladders are prohibited.

CLP's Critical Risk Standard – Working at Height (HSSE-STD-02-003) is available to provide Contractor more information and will be available upon request.

8.3.1 Scaffolding

The Contractor shall appoint a scaffolding superintendent with adequate qualification and experience to ensure compliance of scaffolding standard on the Site.

The Contractor shall provide metal scaffolding as temporary structure or work platform, use other form of scaffold (e.g. bamboo, wood and etc) must be authorized by the Owner. Large (>30m tall) and complex scaffolding, suspended scaffold and falsework shall be engineered to ensure it is designed properly.

In general, scaffold shall be designed to accommodate a minimum of four (4) times the intended working load (including personnel, equipment and material as well as any environmental factors as wind, rain, snow, or ice). Use of scaffolding for hanging or support material which it is not designed, or using scaffold as a structure to pull against in order to move material are strictly prohibited. Mechanical means shall be used to place or remove tools and materials which cannot safely be transported by hand or by tool belts. Housekeeping shall be performed each shift while using scaffolds.

All scaffolds shall be erected, maintained, modified, dismantled, and inspected by competent persons at least on a weekly basis, after any modification, and after any event which may have caused damage to the scaffold.

Scaffold tag in highly visible colour shall be used to identify the condition of scaffold:

- GREEN tag – scaffold is 100% completed and safe for access;
- YELLOW tag – scaffold is in modification but can be used safely with additional controls, and
- RED tag with a diagonal cross – scaffold is incomplete and is not safe to use.
- while erecting, modifying, or dismantling scaffolds:
 - o fall protection must be provided for scaffolders;
 - o hazardous areas (drop zone) must be barricaded (with warning signs) to restrict access;
 - o throwing or dropping of scaffold components is strictly prohibited;
 - o scaffold components must be inspected before use;
- scaffold must be erected on stable and even surfaces using based plates and / or sills when necessary;
- scaffold components shall be square, level and plumb, with all connections properly fitted / tightened;
- use defect free scaffold planks which is adequately rated and secured on the structure, and has an adequate overlap to the end support (between 150mm-300mm if not specified under Regulatory Requirements);
- provide standard guardrail (900mm-1150mm) and toe board (>200mm) for scaffold with adequate strength with work surfaces over 2 meters;

- provide safe access and egress for personnel movement, such as ladders, stairs, gangway, working platform and etc; and
- ties and bracing (e.g. ledger bracing, façade bracing, cross bracing, etc) shall be installed as per design or international standard.

8.4 Protection against Hazardous Energy

8.4.1 Work Permit System / Log-out Tag-out (LOTO)

The Contractor shall develop a Work Permit System / LOTO Procedure to ensure stringent controls will be utilities to prevent unexpected energizing, start-up, or release of stored energy while working with plant and equipment which have an energy source. The Work Permit System / LOTO Procedure shall include:

- Definition on the boundary of application;
- Authorization (include roles and responsibilities), competency and training requirements for different roles (e.g. Issuer, Recipient, affected workers, etc.);
- Description of the process control, which shall include:
 - o Requesting an application for a Work Permit to carry out Work;
 - o Preparation of Work Permit and Isolation of Equipment;
 - o Work Permit issue and acceptance;
 - o Work Permit transfers and
 - o Work Permit Clearance and Cancellation
 - o Restoration of Equipment / returning equipment to operation
- Identification of potential energy sources for work activities (e.g. electrical, pneumatic, mechanical, thermal, hydraulic, chemical, gravity and etc);
- Definition of the isolation and earthing standards, isolation to be performed by authorized persons only;
- Standard for the locking and tagging devices (i.e. colour, sharp and size);
- Locking devices shall be singularly identified with serial number and a unique key, and prohibit their use for other purpose;
- Requirements on the use of personal locking devices or similar control measure such as “sign in” and ‘sign off” by the work party members on the permit ;
- Requirements on the use of a lockbox or multiple locks on a multi-lock hasp in case of a group scenario;
- Requirements on use of tag along with the locking devices with a statement such as “Do Not Start” or “Do Not Operate”, the tag shall also identify the party that applied the lock, the date the lock was applied and the purpose of lockout;
- Shift or personnel change procedure to ensure continuity of protection;

- Procedures for management of keys for locking devices including removal of locking devices due to emergency or loss of key(s), and
- Requirements for testing in which the apparatus will be required to be energized from time to time.

CLP's Critical Risk Standard – Protection against Hazardous Energy (Work Permits) (HSSE-STD-02-004) is available to provide Contractor more information and will be available upon request.

8.4.2 Hot Work

The Contractor shall implement hot work permit for hot work involving the use of, or which will create naked flame, sparks or molten slag. The hot work permit requirements shall include:

- the requirements for periodic gas testing for hot work being performed in the vicinity of flammable substances. Gas testing shall be conducted by trained personnel using calibrated gas detector and results are recorded on the Hot Work Permit;
- identification of designated hot work area where hot work can be performed in a safe and controlled manner (e.g., welding workshop), which hot work permit might be exempted;
- verification of all personnel performing hot work are properly trained and/or certified to the type of hot work to be conducted;
- the use of non-flammable substances to isolate and contain ignition sources (e.g. welding slag or sparks);
- wetting down of area when falling slag may accumulate;
- pre-use check of hot work tools and equipment by a qualified person; additional precautions for vessels / pipe work that previously contained flammable / hazardous substances;
- provision of suitable firefighting equipment in the immediate vicinity of the hot work, and
- provision of trained fire watch to continuously monitor during execution and for a minimum of thirty (30) minutes after the hot work is completed.

For welding equipment used for hot work, the Contractor shall ensure:

- Gas cylinders are labeled and colour coded; not taken to confined space; stored in well ventilated areas in vertically; away from sources of heat; use trolley or cradle for transportation to minimize manual handling;
- All hoses, valves and regulators are in good condition;
- Flashback arrestors are provided to gas welding / flame cutting equipment;
- Friction lighters or stationary pilot flames are used for ignition of flame, and
- When inert gas shielding is used for the welding process, Contractor Counterparty shall ensure that there is adequate ventilation to remove the risk of Oxygen deficiency, follow confined space entry procedure if necessary.

CLP's Critical Risk Standard – Protection against Hazardous Energy (Work Permits) (HSSE-STD-02-004) is available to provide Contractor more information and will be available upon request.

8.5 Lifting Operations

The Contractor shall develop a lifting operation procedure to control:

- Lifting appliances (e.g. cranes, winches, chain blocks, pulley blocks, gin wheels, etc);
- Lifting gear (e.g. slings, chains, shackles, eye-bolts, lifting clamps, lifting beams, etc), and
- Critical and non-critical lifts.

The Contractor shall identify:

- the training requirements for operators and rigging personnel (in-house and / or as per Regulatory Requirements);
- roles and responsibilities of persons involved in the planning and execution of lifting operation (i.e. responsible person for the job, rigging supervisor, riggers, plant operator, signaler, etc);
- safety devices for each type of lifting appliances (e.g. safe load indicators, limit switches, anti-two block devices, anti-freefall devices, reverse alarm, brakes, horns, seat belts, guards, etc) and these safety devices shall be inspected by a competent person at least on a weekly basis;
- the lifting capabilities of all lifting appliances and lifting gears, and marked clearly in appropriate languages;
- Inspection and examination procedure in alignment with manufacturer's and Regulatory Requirements and recommendations, including:
 - o qualifications of inspection and examination personnel;
 - o frequency of inspection and testing for various type of lifting appliances and lifting gears, lifting appliances and lifting gears shall be inspected by a competent person at least every week and every quarter respectively;
 - o pre-use visual inspection criteria;
 - o definition of acceptance and rejection criteria;
 - o calibration requirements;
 - o tagging system to provide visual status (e.g. colour code system), and
 - o documentation.
- process for lifting to the weight and center of gravity for objects being lifted;
- critical lifts which require a more comprehensive planning, control and supervision, including the use of lift plan, pre-lift meeting, permit to lift, and JSA, critical lifts shall include but not be limited to:
 - o loads being lifted over or near a building, operating equipment or overhead power lines;

- o Tandem Lift which lift a Load using more than one crane simultaneously. In no circumstance should the Load to be lifted exceed the SWL of any one Crane in the tandem lift (except overhead gantry Cranes that are designed for coupled operations);
- o non-standard crane configuration being adopted;
- o load is greater than forty five (45) metric tonnes [1,000 Kg];
- o load is greater than ninety percent (90%) of the manufacturer's rated capacity at the working radius;
- the method of communications used to support lifting operations(e.g. radios, hand signals);
- conditions that suspension of operation is required due to adverse weather conditions (e.g. thunderstorm, high wind, rough sea, low visibility, etc), and
- safe work practices and restrictions for different types of lifting appliance and lift gear.

CLP's Critical Risk Standard – Lifting Operation (HSSE-STD-02-005) is available to provide Contractor more information and will be available upon request.

8.6 Confined Space Entry

A confined space is an area that is not intended for normal occupancy and may exhibit the following state of conditions:

- has limited opening for entry and exit;
- may contain or produce harmful / toxic air contaminants;
- may contain a flammable, explosive or oxygen enriched atmosphere;
- may have a high concentration of an inert gas;
- an oxygen-deficient atmosphere may occur;
- has a risk of excessive heat, and/or
- has a risk of an in-rush of free flowing solid or liquid.

The Contractor shall develop a confined space entry procedure to ensure safe entry and work execution in a confined space, the procedure shall:

- identify all confined spaces on Site and must display with proper warning signs;
- define the confined space entry certificate approval process to control entries into confined space;
- require to conduct a confined space risk assessment to identify the potential hazards and control measures required to safely enter confined space;
- define the acceptable entry criteria (at least include oxygen, lower explosion limit, carbon monoxide and hydrogen sulfide) for safe entry;
- require formal training for all persons involved in the confined space entry;

- have control measures to prevent unauthorized entry to confined space, such as barricade with signs, attendants / standby person and etc;
- define the requirements for periodic gas testing. Gas testing shall be conducted by trained personnel using calibrated gas detectors and results are recorded on the confined space entry certificate;
- address the requirements for cleaning and purging of confined spaces required to achieve acceptable entry criteria;
- require the use of forced ventilation when natural ventilation is inadequate to maintain a safe atmosphere for work execution;
- require the use of artificial lighting when natural lighting is inadequate for work execution;
- require gas cylinder to be fed by gas hose with proper maintenance if gaseous product is used inside the confined space (such as oxy-acetylene flame cutting);
- define the communication method to be utilized;
- identify the need for the use of specialized tools and equipment:
 - o low voltage
 - o intrinsically safe
 - o explosion proof
 - o air supplied respirator (air-line hose or self-contained breathing apparatus - SCBA)
- require a trained rescue team with fit for purpose emergency extraction equipment located in the immediate vicinity of the confined space.

CLP's Critical Risk Standard – Confined Space (HSSE-STD-02-006) is available to provide Contractor more information and will be available upon request.

8.7 Hand / Powered Tools and Equipment

Throughout the duration of the Works, the Contractor shall ensure that all tools, equipment, temporary facilities and other items used are in a safe, sound and good condition and are capable of properly performing the functions for which they are intended.

All power activated tools shall be operated by trained / competent personnel. Pneumatic power tools shall be secured to the hose or whip by positive means. Safety pins shall be used in connections where appropriate.

All electric powered tools shall be protected by earth-leakage circuit breakers or meet the double insulation standard. Weatherproof electrical plugs and sockets shall be used for outdoor work and indoor work where a wet or damp atmosphere is envisaged.

The Contractor shall identify inspection and testing procedure in alignment manufacturer's requirements and recommendations, including:

- qualifications of inspection and testing personnel;
- frequency of inspection and testing for various type of tools and equipment, in general:

- o power activated tools and equipment shall be inspected and test at least in a quarterly basis, and
 - o hand tools shall be inspected and tested at least in a monthly basis.
- pre-use visual inspection criteria;
- definition of acceptance and rejection criteria;
- calibration requirements;
- tagging system to provide visual status (e.g. colour code system), and
- documentation.

Use of “shop built” or “shop modified” tools and equipment shall be strictly prohibited, powered hand tools should equipped with dead man switch which will automatically turn off when the worker release the switch.

Removal, bypass or defeating safety devices and guards are strictly prohibited. Tools and equipment to be used in hazardous or classified areas shall be explosion proof or intrinsically safe unless additional controls are identified on the hot work permit.

CLP’s Critical Risk Standard – Hand and Powered Tools (HSSE-STD-02-007) is available to provide Contractor more information and will be available upon request.

8.8 Working around Water

The Contractor shall identify the type of work that drowning hazard exists and the respective control measures, such as:

- Marine operations
- Land-based construction on, adjacent to or over water
- Industrial diving

Such control measures shall include but shall not be limited to:

- Fall prevention such as working platform, guardrail and kerb (shall be at least 150mm height and at a distance of 500mm from the edge near water if mobile plant or vehicle is operated near the water.), safety net, lifeline and safety harness and etc;
- Proper access and egress, ladder, and access to and from vessels;
- Provide proper PPE such as lifejacket and buoyancy aid complied with relevant standards. Example of United States requirements include:
 - o Type I lifejacket for offshore work
 - o Type II buoyant vest for near-shore work
 - o Type III flotation aid for calm/inland water
- Competency of specific positions, such as vessel master, crane operators, industrial divers and etc;
- Training to person engaged in these activities to understand the hazard, control measures and emergency procedures; and

- Rescue and emergency procedures to cover the resources required in case of emergency, with particular interest in the provision of rescue boat.

The Contractor shall comply with the following requirements for any industrial diving work to be carried out on Site:

- All divers must hold a valid annual certificate of medical fitness to dive
- Ensure all divers are competent to perform the assigned task in a safe manner
- A team of at least 3 persons must be presented at each diving operation which comprise:
 - o A diving supervisor;
 - o Diver(s) engaged in the underwater work, and
 - o A standby diver at the diving location (equipped and ready to dive).
- Breathing air supplied by compressor or self-contained breathing apparatus must meet Grade D minimum requirements of Compressed Gas Association (CGA) of United States with a minimum flow rate of 200 liters per minute, or other recognized standard approved by the company.
 - o Oxygen 19.5%-23.5%
 - o Carbon Dioxide < 1000 ppm
 - o Carbon Monoxide < 10 ppm
 - o Condensed hydrocarbons < 5mg/m³
- Properly secured and tended lifeline shall be provided for each diver, except where restriction on mobility would create a hazard.
- Effective communication shall be maintained between all diving team members, such as using the lifeline as a signaling system or a reliable audio system.
- All plant and equipment shall be periodically examined and tested by competent persons in accordance with manufacturer's instruction and recommendations
- Decompression procedure shall be developed according recognized decompression tables to prevent decompression sickness
- Medical emergency procedure shall be developed for decompression sickness
- Diving logs shall be prepared to capture key information of the diving operation

CLP's Critical Risk Standard – Working around Water (HSSE-STD-02-008) is available to provide Contractor more information and will be available upon request.

8.9 Excavation

The Contractor shall notify respective utilities companies including the Owner (brown field work) prior to commencing any excavation works. Drawings of underground utilities and structure shall be collected and these utilities and structure shall be exposed before commencement of any mass excavation. Unless specified in the Regulatory Requirements, the Contractor shall define safety clearance distance for mechanical excavation near existing underground utilities to avoid damage.

Soil stability should be evaluated according to established criteria to design adequate support system such as battering, benching or shoring wherever the depth excavation is greater than 1.2m. Adjacent structure which the stability might be affected by the excavation shall also be considered during designing of support system.

The Contractor shall arrange competent person with adequate qualification and experience to conduct inspection to the excavation at least on a weekly basis and after any adverse weather conditions. Inspection criteria shall be established to ensure the excavation is safe for entry. The excavation shall be suspended if inspections indicate that safety controls are inadequate for entry.

Safe means of access and egress shall be provided for all excavations, rigid barriers and toeboards should be provided if personnel access is required near the edge of excavation when there is a risk of fall exceeding 2m. Temporary barriers for excavation less than 2m in depth need to be located at least 1.5 meters away from the edges of the excavation.

Barrier blocks, cones or signage shall be provided along the edge of excavation when an excavation is located near vehicular access. As a minimum requirement, material including excavated material and backfill must not be stored within 1.5m of the edge of excavations. The actual clearance necessary related to the particular excavation and must be calculated at site with particular concern if there is a need for heavy mobile equipment access or position of crane near the edge of the excavation (minimum 2 meter clearance must be maintained).

Deep excavation may be classified as confined space which requires Confined Space Entry Permit.

CLP's Critical Risk Standard – Excavation (HSSE-STD-02-009) is available to provide Contractor more information and will be available upon request.

8.10 Hazardous Substances

Hazardous substances are those materials that by virtue of chemical, physical or toxicological properties, either singly or in combination, constitute a hazard. They may be explosive, flammable, oxidizing, toxic, harmful, irritant, corrosive, radioactive, carcinogenic or with persistent harmful impacts to the environment, etc.

The Contractor shall ensure that adequate hazardous chemical substances control measures are implemented and maintained in transportation, storage, use and disposal of hazardous chemical substances so that the associated HSSE risks are minimized.

No prohibited hazardous substance (such as asbestos, banned ozone depleting substances, polychlorinated biphenyls (PCBs), substances listed in Stockholm and Rotterdam Conventions and other substances as determined in the Contract) and regulated or banned volatile organic compounds (VOC) containing products shall be brought or introduced on the Site unless previously approved by the Owner and deemed necessary as part of the Works. The Owner would request the Contractor to prove that the material used for the work is -free from the prohibited material when required. The Owner reserves the right to conduct test and examination on material suspected to contain such prohibited material, the Contractor will be responsible for the cost on demolish and replace such material being identified, unless such material does not contain any prohibited hazardous substance.

The Contractor shall:

- prepare an inventory of all hazardous chemical substances on Site and update at least on monthly basis;

- provide additional control measures if the substances is classified as dangerous goods / substances, if necessary, apply license for use / transport / storage from local authority;
- ensure all containers and receptacles of hazardous chemical substances are provided with chemical label in appropriate language(s);
- ensure all hazardous chemical substances are provided with material safety data sheet (MSDS) in appropriate language(s) and readily available on Site;
- ensure the handling, storage, disposal and emergency response (first aid, spillage, release, etc) comply with the requirements stipulated on the MSDS;
- provision must be made for spill containment in each hazardous substances storage, that are in liquid state or substances that would become fluid upon the changes in temperature.
- observe compatibilities for storage of hazardous chemical substances;
- provide proper emergency response equipment according to MSDS (e.g. eye wash, emergency shower, fire-fighting equipment, spill kit, etc);
- provide training to personnel use or expose to the hazards; and
- provide Permit to Work system for the handling of radioactive substance (e.g. Non-destructive testing)
- ensure all Hazardous Substances shall be disposed of an approved manner that is in accordance with local regulations and international codes or standards.

CLP's Critical Risk Standard – Hazardous Substance (HSSE-STD-02-010) is available to provide Contractor more information and will be available upon request.

9 Occupational Health Requirements

The Contractor shall establish the following programs and practices to ensure health related hazards are being identified and mitigated by proper control measures:

Occupational Health:

- Hearing conservation
- Heat exposure
- Respiratory protection
- Fibrous material, both asbestos (might exist for refurbishment project) and non-asbestos (e.g. fiberglass, rock wool, etc)
- Abrasive blasting
- Radiography
- Handling hazardous substances
- Alcohol and drug program
- Fitness for duty – medical surveillance for designated position and as per Regulatory Requirements

General Health and Hygiene:

- General sanitation
- Pest control program (rodent, insects)
- Pandemic flu preparedness
- Food safety
- Provision of potable water
- Provision of hygiene and washing facilities

Potable water purification devices (or point-of-use water treatment system) shall be provided if there is not treated water source to provide potable water for the workforce. All vessels and pipe-works use for containing and supplying potable water shall be cleaned regularly.

A food facility shall be provided on the Site by the Contractor for the reasonable use of both his and his Subcontractor's employees, the Owner and his staff and employees. A central Site canteen may be provided to a suitable standard. There shall be an appropriate canteen for professional managerial staff.

The Contractor shall submit the proposals for the Site clinic and canteen facilities, including detail drawings and operational procedures.

The Contractor shall allow persons authorized to enter the Site to use the first aid and hygiene and washing facilities.

Where relevant, the Contractor shall include in his supply of temporary works, maintenance and operation of clean conditions changing facilities including changing areas, ablutions and amenities. The Owner shall have free access as required.

The Contractor shall conduct regular inspection (at least monthly) to the Site clinic, food production and storage areas, changing and other hygiene facilities to ensure they are in good condition. The Contractor shall also conduct regular inspection (at least fortnightly) at the worker camp-site to ensure reasonable hygiene standards can be maintained.

9.1 Manual Handling

The Contractor's Personnel shall provide training for all persons involved in manual handling as including lifting, carrying, pulling, pushing, stacking, or laying down of loads without any mechanical assistance. In general, the training shall explain the hazards and control measures for manual handling; the following is a typical guidance of carrying out a manual handling:

- Evaluate the estimated weight, center of gravity, and dimensions of the object to be lifted before performing the lift;
- Verify that the route is clear and free from obstacles before performing the lift;
- Verify a clear view of the route before performing the lift;
- Position close to the load, keeping feet firmly on the floor;
- Squat down with back straight, bends knees, and maintain tight stomach muscles;

- Verify the weight of the object is manageable for the individual and a firm grip is in place before handling the load;
- Stand up slowly, using leg strength, holding the load close to the body;
- Do not twist while carrying the load; lead movements with feet;
- Avoid standing for extended periods of time while holding heavy loads;
- Do not change grip while carrying heavy loads; if necessary, rest the load on firm support before changing grip, and
- Lower the load slowly, using leg strength rather than back muscles.

Risk Assessment (i.e. JSA or other risk assessment specialized for manual handling) shall be conducted for:

- repetitive tasks which have record of causing injuries and / or musculoskeletal disorders;
- loads that weight over sixteen (16) kg, and
- loads that require several workers to handle.

Typical control measures to reduce the risk of lifting operation include:

- additional assistance (manual or mechanical);
- appoint an additional person solely responsible for coordination;
- use of gloves when the task might injure hands or a better grip is required, and etc

10 Security Requirements

The Contractor shall provide security facilities for the Site and temporary storage areas, including but not limited to, the following:

- A uniformed security guard force, adequately manned, equipped and trained for both static and mobile duties.
- Road access gates with traffic boom gates manned during construction by uniformed security staff, and Site patrols when necessary. Normal policy should not allow vehicle access into the Site unless the vehicle is official or is integral to the project.
- A secure perimeter with full access controls is to be erected before construction begins. Consisting of:
 - o Security fencing and CCTV at strategic points. The location to be agreed with the Owner (additional temporary security fencing where required over the duration of the construction phase shall be as specified in the Contract).
 - o Chain-link fence around Site boundary shall be provided with lower anchorage and barbed wire on the top.
 - o Flood lights are to be installed along the fencing perimeter at regular intervals and at strategic locations (such as blind spots, entrances, corners).
 - o Perimeter Intrusion Detection System (PIDS) where appropriate.
 - o Security fence shall be regularly inspected and maintained to ensure its integrity.

- Before the security perimeter is complete, a manned, secure building/stores/compound for receipt/temporary storage of equipment is to be made available.

The Contractor shall submit at the Contract execution stage details of his proposed Site security policy and procedures, which shall include, but not be limited to the following:

- Physical layout to include physical protection systems, points for personnel and vehicular access, location of offices, warehouses and sensitive / restricted areas for which higher levels of security are required;
- Procedures for deployment and management of guard forces;
- Access control procedures which shall include use of photo identity badges, vehicle entry and parking permit, turnstiles for personal access control, boom gate for vehicular access control, exit procedures for staff (bag checks / vehicle checks) etc;
- Linkage to Emergency Response and Preparedness systems;
- Linkage to the Alcohol and Drug-free Policy and Procedures;
- Information security;
- Security communications systems;
- Security surveillance systems;
- Threat levels and threat escalation;
- Threat-based baggage searches, and
- Visitor processing and escorts.

The Contractor shall have full responsibility for the security of the Site.

The Contractor and his employees shall not trespass beyond the Site boundary into any adjoining land and the Contractor shall take any necessary action to prevent trespassing.

11 Environmental and Social Requirements

This section is the minimum requirements for all locations. Where the EIA/ESHIA/IFC Performance Standards/Equator Principles has a more stringent requirement than this document, then the EIA/ESHIA/IFC Performance Standards/Equator Principles shall apply.

11.1 Effluent Discharge Management

Site drainage, the locations of any Site water discharge points must be identified before any work is undertaken. This will involve: i) Mapping of the wastewater and storm water networks on Site, including the sources of all discharges and the discharge points, ii) Identification of discharge points for these networks. Drains must be identified as to whether they are storm (surface) water drains or foul (wastewater) drains to aid identification of where a spill will leave the Site.

The permanent and the temporary surface and foul water drainage systems shall all be separate. The Contractor shall make his own arrangements for foul drainage from the Site up to the time of taking over section IV. A sewage treatment equipment of the requisite

capacity to meet the permanent requirements for the Site, shall be installed by the Contractor.

The provision and maintenance of all temporary and permanent foul drainage on Site shall, during the construction period, be the responsibility of the Contractor, who shall ensure that all national and local requirements for the effluent discharge are complied with

The Contractor shall not cause pollution of the Site drainage system or any water course by either dumping rubbish, fill materials or waste products from any source whatsoever.

Proper wastewater treatment facilities and control must be provided to treat wastewater generated from site facilities such as domestic sewage from toilet and catering facilities, drainage serving an open refueling point, vehicle and plant maintenance areas and etc. Surface runoff from Works on the Site must be treated by designed silt removal facilities before discharging to the approved discharge point as per local regulatory requirement.

During construction phase, all discharges to adjacent water bodies or storm drainage, it must be ensured that:

- All discharge points are identified under the EIA and ESHIA Report. If not, agree with the local authority all the discharge points;
- Monitoring and measurement should be conducted according to the EIA and ESHIA Report or the Permit / Approval Letter's requirements, and
- If conditions are breached or exceedance detected, it must be mitigated immediately; reported to the regulatory authority and internally using the Accident / Incident reporting procedure.

Storm water that is essentially "clean" water can be allowed to enter storm drainage or water courses without the need for consent / licence.

In addition to its obligations as stated in the Contract, the Contractor shall make himself familiar with all requirements relating to the discharge of effluent from the Contractor's Equipment and shall take all steps to ensure that these are not contravened by the disposal of effluent from the Site

To prepare for spill and leakage of chemicals to water bodies, an emergency response plan must be established and implemented. This plan will be simulated at least annually to ensure its effectiveness. Emergency arrangements must be periodically tested either by simulations (where practical) or other means such as audits, inspections or desktop review.

11.2 Air Emissions Control

The basis for air monitoring may be set by Regulatory authorities in the form of Permits or Consents or by other requirements.

When planning emissions monitoring, consideration must be given to the following, where applicable:

- During construction phase: Mainly fugitive dust generated from construction activities. Details should be referred to the significant aspects registered mentioned in Section 3.1
- During operational Phase : monitoring and reporting requirements should be referred to the EIA and ESHIA Report requirements, relevant licence/permit issued by or in agreement with local authority, if applicable

- Technologies and limitations of the emission monitoring equipment in particular continuous emissions monitoring system and the associated quality assurance and quality control requirements. International standards should be referenced to;
- Locations of sampling points and relevant standards for setting up monitoring equipment
- Reporting requirements, e.g. frequency (periodic or continuous) and timing (i.e. time of day) of tests /measurements, real time display, etc.
- Emissions limits of Group's emission standard

Unless otherwise specified by regulatory authorities or customer specifications, samples must be taken:

- Before work starts, in order to establish background baseline
- On commencement of the work
- On each substantive change in activity
- At any other time during the work at the reasonable request of the Regulatory Authority
- As part of an investigation into a community related complaint
- If it is suspected that controls are no longer effective

Only suitably trained or experienced personnel will carry out sampling and surveys.

Any equipment used for measuring / monitoring purposes must be calibrated to an approved standard or tested prior to use, as appropriate.

11.3 Dust control

Control / minimise dusts emission from dusty activities / operations by means of water spray, wheel damping basin, dust suppression on aggregate silo, batching plants, etc.

11.4 Soil and Groundwater Protection

In addition to the identified scenarios of emergency situations caused by spillage and leakage under the EIA and ESHIA Report, all potential land and groundwater contamination activities during construction phase under normal, abnormal and emergency conditions should be identified and evaluated in Section 3.1. Based on the EIA and ESHIA results and identified significant environmental aspects, set out associated operational control procedure; prevent through design or other relevant mitigation measures and conduct relevant drill on a regular basis.

For material handling:

- Pre-Delivery of Bulk Substances

Prior to delivery, a suitable risk assessment must be performed, ensuring that appropriate controls have been identified and that they are implemented on delivery. Details of the risk assessment must be recorded.

An advance copy of the Material Safety Data Sheet (MSDS) must be obtained to assist with preparation of the risk assessment.

- Delivery

All deliveries must be supervised by a Responsible / Competent person who has familiarized themselves with the risks associated with the substance and the requisite control measures.

Information required, an MSDS must be provided, no matter how the substance is supplied - in bulk or in packages, no later than at the time of delivery. All MSDS must be retained in a format suitable to allow easy retrieval by any interested parties. All containers must be clearly labelled as to their content.

- Above Ground Storage

All fuels, paints or other chemicals must be stored within a bunded area suitably positioned, or other steps taken, such as the provision of drum trays with integral bunds, so as to minimize any risk of damage by impact (so far as reasonably practicable) the walls and base of which must be impermeable to the substance stored and to water. Bund Drainage, under normal operating conditions all bund drain valves must be locked in the closed position.

- Refueling Activities

Where practical, refueling of mobile plant will be conducted in a designated area, preferably on an impermeable surface, away from drains or watercourses. The operation will not be left unattended or valves jammed open. Diesel pumps and similar equipment will be placed on drip trays to collect minor spillages and must be checked regularly and any accumulated oil removed for disposal.

11.5 Environmental Noise Management

Contractor shall ensure that arrangements are in place for the monitoring and reporting of the Site environmental noise, in accordance with this procedure and local authority and / or customer requirements.

Noise surveys will be conducted at periods stipulated within, as applicable:

- Project-specific HSSE Plan
- Contract
- EIA and ESHIA and relevant Permits/Consents requirements

The locations for noise survey sample positions will be identified on a scale map of the designated area. Noise samples are to be taken using recording equipment calibrated in terms of the Site Quality Management System. Samples will be taken at two intervals:

- During the process/ operation period
- Outside the process/ operation period

The exact time and date of the recording, wind speed and direction and the noise level are recorded at the time of each run of samples.

11.6 Waste Management

The Contractor shall follow the waste management hierarchy and develop a waste management plan for the Owner's review:

- Avoid;
- Reduce;
- Recycle / Reuse, and
- Treatment and disposal

The Contractor shall be responsible for waste collection and transfer to a suitable disposal site. The Contractor shall clearly state, in the appropriate schedule, what provision is being made for disposal of waste materials both on and off-Site.

All rubbish, etc. shall be systematically cleared from all working areas and, if not removed directly from the Site, shall be deposited at general collecting points provided by the Contractor and agreed with the Owner, pending removal from the Site.

Where practicable, screen shall be provided to prevent the ingress of dust and dirt to any part of the Works on Site

The Contractor shall at all times ensure the public roads, giving access to Site, are kept clean of any materials deposited by traffic from the Contractor or his Subcontractors. The Contractor shall for this purpose provide wheel washing facilities at the Site entrance which will be used as required.

In the event of an incident whereby the pollution is attributable to any act or omission on the part of the Contractor, then the Contractor shall clear his pollution as soon as possible and make early arrangements to remedy the cause. The Contractor shall prohibit the committing of nuisance by his employees on the Site or upon either the Owner's land or others, and any employee found violating this provision shall be immediately discharged and not be employed again on this Contract.

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Dangerous or noxious waste products, chemicals or materials shall be disposed of in accordance with HSSE Requirements and Regulatory Requirements.

The Contractor shall separate hazardous chemical waste from non-hazardous waste described above; identify all hazardous waste, including waste oil, arising from all activities and take all necessary measures to ensure the safe/proper labelling, packaging, storage, handling, collection and disposal of waste oils / chemical wastes pertaining to the Works on Site which he has responsibility and to ensure the prohibition of:

- Any discharge of waste oils / hazardous wastes into surface waters, ground waters or drainage systems of any kind;
- Any deposit and/or discharge of waste oils / hazardous wastes on to the ground or floors of buildings and any uncontrolled discharge of residues resulting from the processing of waste oils or the cleaning of oil contaminated surfaces with solvents.
- In order to ensure compliance with this requirement, the Contractor shall provide suitable vessels for the collection of all waste oils. Where large quantities of waste oils need to be stored prior to being disposed of in bulk, suitable storage tanks shall be provided by the Contractor for this purpose, and
- Any processing of waste oils causing air pollution which exceeds the level prescribed by existing Regulatory Requirements. The burning of all waste oils should only be carried out following written permission being granted by the Owner.

Attachment 1 CLP Group Risk Matrix

CLP Group Risk Matrix (Issue 4)

CLP 中電

Version: Issue 4

Issued on: 18 October 2016

Issued by: Group Risk Management

							Likelihood						
							Rare	Unlikely	Possible	Likely	Almost Certain		
							Is expected to occur only in exceptional or extreme circumstances	Could occur at some time but unlikely in the foreseeable future	Not unusual and might occur in the foreseeable future	May probably occur in most circumstances	Already happen or is expected to occur in most circumstances	Qualitative measure	
							Less than once in 10 years	Once in 10 years up to once in 3 years	Once in 3 years up to once per year	Once per year up to once per month	Once per month or more	Frequency	
							Less than 10% chance of occurrence	10% up to 33% chance of occurrence	33% up to 66% chance of occurrence	66% up to 90% chance of occurrence	90% or greater chance of occurrence	Probability	
Consequence Category													
	People, Safety & Health	Environment	Regulatory & Governance	Reputation	Financial	Operations & Systems							
Consequence	Critical	Multiple fatalities	Persistent severe environmental impact affecting extensive areas & requiring serious mitigation response for an extended time	Material breach of external regulations liable of loss of critical operational / business licence	International negative media attention Long-term impact on company brand name	Material financial loss impacting financial viability and strategy execution of the Group (Operating earnings or cashflow impact of US\$125m or more within next 3 years)	Business disruption at Group level Power/system disruption: Severe impact on large community requiring full scale response for an extended time	Medium	High	High	Extreme	Extreme	
	Major	Fatality	Severe environmental impact affecting extensive areas & requiring serious mitigation response	Material breach of external regulations liable of substantial fines imposed	National-wide negative media attention or public dissatisfaction Short term impact on company brand name at national level	Major financial loss impacting the Group business plans and financial results (Operating earnings or cashflow impact of US\$45m up to US\$125m within next 3 years)	Business disruption at BU level i.e. CLP India, Power HK, EA etc. Power/system disruption: Material impact on large community requiring full scale response	Medium	Medium	High	High	Extreme	
	Moderate	Multiple lost time injuries	Moderate environmental impact affecting the site and/or surrounding areas & requiring moderate/serious mitigation response	Non-compliance noted by regulatory body with prosecution instituted	Territory-wide / state-wide negative media attention or public dissatisfaction Short term impact on company brand name at territory / state level	Moderate financial impact to the Group (Operating earnings or cashflow impact of US\$15m up to US\$45m within next 3 years)	Business disruption at Dept Level under a BU Power/system disruption: Significant impact on small community	Low	Medium	Medium	High	High	
	Minor	One lost time injury	Minor environmental impact affecting site and/or surrounding areas & requiring little or no mitigation response	Non-compliance noted by regulatory body with issuance of warning	Local media attention or public concern	Minor financial impact to the Group (Operating earnings or cashflow impact of US\$5m up to US\$15m within next 3 years)	Business disruption limited to specific plant or process Power/system disruption: Minor impact on small community	Low	Low	Low	Medium	Medium	
	Insignificant	Injury without lost time e.g. first aid	Nil	Minor non-compliance to regulatory requirement, internal rules or procedures	Nil	Immaterial financial impact to the Group (Operating earnings or cashflow impact of less than US\$5m within next 3 years)	Power/system disruption: Minimal impact on small community	Low	Low	Low	Low	Low	

DOSSIER CONSTRUCTION

The Investment Project Report for Vugn Ang II - 2x660MW Thermal Power Plant is structured as follows:

PART I	: MAIN REPORT
PART II	: BASIC DESIGN
Volume 1	: Power Plant
Book 1	: Technical Descriptions
Book 2	: Appendices
Book 3	: Drawings
Volume 2	: Cooling Water System
Volume 3	: Coal Jetty
PART III	: SPECIALIST REPORTS
Book 1	: Meteo-Hydrological Conditions Report
Book 2	: Freshwater Supply Study Report
Book 3	: Plant-to-Grid Connection Study Report
Book 4	: Fire Protection Report
Book 5	: Topographic Survey Report
Book 6	: Geological Survey Report
Book 7	: Seismic Hazard Study Report
Book 8	: Coal Supply Study Report
Book 9	: Environmental Impact Assessment Report

This is PART III, Book 4



Vung Ang II Thermal Power Joint-Stock Company (VAPCO)

Investment Project Report

For

Vung Ang II - 2x660MW Thermal Power Plant

PART III: SPECIALIST REPORTS

Book 4: Fire Protection Report

Prepared by: Power Engineering and Consulting
Joint-Stock Company (PECC1)



January, 2010

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Drawings attachment:

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1	Sơ đồ nguyên lý hệ thống PCCC/Fire fighting system - flow diagram	60.09-VA2-ME-25
2	Sơ đồ nguyên lý hệ thống báo cháy/Fire detections alarm system flow diagram	60.09-VA2-ME-26
3	Tổng mặt bằng/General layout	60.09-VA2-GA-02

1. INTRODUCTION

Vung Ang 2 thermal power plant with capacity of 2x660MW is one of two coal-fired thermal power plants belonging to Vung Ang power with General Planning which has been approved by Ministry of Commerce and Trade in the Decision No. 2582/QĐ-BCN dated 29/09/2006. The plant location (and power center) is at Hai Phong village, Ky Loi commune, Ky Anh district, Ha Tinh province. This location is located in Vung Ang economic zone which is one of key economic area in the North Central which has been approved by Vietnamese government in the general planning. The other power plant in the center is Vung Ang 1 TPP which is owned by PVN and has been constructed on date of 27/08/2009.

Vung Ang 2 TPP will be invested under BOT mode. VAPCO, a private joint stock company with founding shareholders of Lilama, OneEnergy and REE, has been appointed by Vietnamese government as the project investor in the Circular No.1266/VPCP-KYN dated 02/03/2009 by Governmental office. At present, VAPCO is preparing the investment project report in complying with stipulations in Decree No.12/2009NĐ-CP dated 12/2/2009 by the Government on the project construction investment management.

This document is one part of the investment project report for Vung Ang 2 TPP. The purpose of this document is to obtain initial approval from the fire protection and fighting authority for fire fighting design and equipment to be applied for the power plant (official agreement for design and fire fighting equipment will be done in the project detail design and construction stages).

The report content will comprise of basic points for design and structure of the plant in relating to fire fighting safety, and basic design for the fire protection and fighting for the power plant.

2. DESCRIPTION OF VUNG ANG 2 TTP

The plant will be located on a 42-hectare site adjacent to Vung Ang 1 TPP. The proposed site is located on a low-lying coastal strip of land that includes marshland, farmland, and a few wooded areas. The terrain slopes down from elevation 10 m, along the base of the mountains on the north site boundary. The majority of the site has a grade elevation in the range of 3 to 5 m which will be leveled to 8.00m according to Vietnamese national elevation.

The power plant will comprise of main equipment for two supercritical coal fired units and auxiliary systems. Each unit will comprise of one coal fired boiler, one steam condensing turbine and one generator. The plant will use a supercritical steam cycle, with steam parameters at main stop valve of 24.2 MPa and 566 °C and reheat at 4.26 MPa and 566 °C. The steam turbine will be a tandem-compound, three cylinder unit. It will have a combined high and medium pressure cylinder, and two double flow low-pressure cylinders.

The steam turbine-generator (STG) and its auxiliaries will be housed in a STG building. The steam generators will be located outdoors with weather protection on the roof.

The plant will be fueled by a mixture of imported bituminous and sub-bituminous coals, delivered to the site by Over-Panamax size ships (capacity up to 100,000 deadweight tonnage). Coal unloading from a ship will be at an average rate of 1200 t/h. The coal receiving port will have unloading cranes with grab buckets and a trestle, consisting of a concrete deck on steel piles, extending to the shore, on which the conveyor belts will be located to deliver the unloaded coal to on-shore coal storage. VA II will have two covered coal storage sheds with a combined capacity of 616,000 tones, which will be sufficient for 45 days operation at full load. The coals will be blended on the reclaimer conveyor belts from the storage sheds at the ratio of about 70% sub-bituminous to 30% bituminous coal.

Diesel oil will be used for boiler start-up and firing support when the boiler is operating at low load. Total annual consumption is about 14743 tons which is stored in steel tanks of 2000 m³ which is located in one separation area with surrounding concrete tank. Oil supply is imported via facilities of Petrolimex or domestic source (from Dung Quat or Nghi Son oil refinery plants).

Boiler flue gas will be cleaned with dust in the ESP then desulphurized in the Seawater – FGD, processed flue gas meeting requirements from TCVN and International finance company will be discharged to the atmosphere via one stack of 210m height. The stack will comprise of one external concrete casing with steel flue gas duct, each duct for one boiler.

Coal ash (including bottom ash and fly ash) will be collected in storage silos, then will be loaded down on trucks from the ash pond.

Fresh water for the plant demands (about 8,000m³/day) will be supplied from the fresh water supply system from Ha Tinh water investment and business joint-stock company. In addition, in order to ensure safety of fresh water, the plant will provide one standby small-sized reverse osmosis seawater

desalination system to produce freshwater from seawater for necessary demands.

Turbine exhaust steam will be condensed in cooling surface steam condenser using seawater. Seawater will be taken from deep water places using water inlets, then water will run into the pump pit. There, there will have seawater circulating pumps for condensers to air scrubbers of the FGD. Hot water will be collected at the canal and discharged to the sea at one location, then recirculation to the water inlets will be minimized.

Generator terminal voltage is 22kV and will be stepped upto 500kV in the unit step-up transformer. The plant auxiliary loads will be powered from unit auxiliary transformer, this transformer will reduce terminal voltage from 22kV to 6.6kV. One plant auxiliary transformer will be used during plant commissioning process and unit start-up.

Auxiliary power system will be designed at voltages of 6.6kV and 0.4kV. Two emergency diesel generators will be installed to supply power for emergency loads and lighting, when the normal power source is interrupted.

One 500kV switchyard will be constructed within the plant scope. The switchyard will have AIS or GIS types. There will be two outgoing circuits to Ha Tinh stations and two outgoing circuits to Da Nang stations. One 500kV/220kV auto transformer will be used to connect Vung Ang 2's 500kV switchyard with Vung Ang 1's 220kV switchyard.

The plant main controls will be monitored and operated by the DCS. The main systems for each unit to be controlled and monitored will comprise of: load control system and unit protection system, burner management system; turbine main and bypass steam system; coal supply system; seawater – FGD; ESP, fly ash and bottom ash system; water treatment system; cooling water system; closed-circuit cooling water system; condensate water system and feedwater degasing. Controls for coal unloading system and ash handling system will be shared by both units.

The plant will have all required auxiliary systems such as fire fighting system, water treatment system; compressed air system; closed-circuit cooling water system, hydrogen generation system; air ventilation, heating and conditioning system, lifting equipment, ash and fly ash collection and storage; start-up boiler; diesel oil storage and supply system; waste water collection and treatment system; administrative building; operator's living houses; and all other items and facilities required for the power plant operation.

The plant items comprise of:

1. Turbine/generator area
2. Boiler
3. ESP
4. Main transformers
5. Unit auxiliary transformers
6. Station start-up transformer
7. Excitation transformer
8. Switchgear building
9. 500kV switchyard
10. Coal yard dry storage shed
11. Coal supply and receiving system
12. Central control house
13. ESP control house
14. FGD control house
15. Fire fighting pumphouse
16. Workshop
17. Waste water treatment area
18. Diesel oil storage
19. Diesel oil transfer pumphouse
20. Lubrication oil storage
21. Administrative building
22. Car garage
23. Water treatment area
24. Main gate house
25. Hydrogen plant building

26. Emergency diesel generator
27. Dirt/clean lube oil storage
28. Coal/ash control house
29. DO unloading pumphouse
30. Coal crusher house
31. Other auxiliary systems ...

3. DESIGN OF VUNG ANG 2 TPP FIRE PROTECTION AND FIGHTING SYSTEM

3.1. General

Vung Ang 2 TPP will be designed to maintain a safe operation condition for plant operators and equipment. Plant equipment will be designed to minimize fighting risks with selection of suitable equipment and material. To carry out this purpose, the fire prevention and fighting system shall perform well the following works

- Early detection, warning and fire fighting.
- Prevent flare up flame or spread firing.
- Protect operators working in plant
- Minimize the firing damage .

Dangerous areas in the plant such as boiler area, steam turbine, airheater area, coal storage, coal conveyor, oil pumphouse, oil tank, hydrogen house, cable trench and storage places of combustible and explosive materials. Especially, diesel oil storage and reserve area for serving ignition and firing support when the power plant operation is at low load and hydrogen generation station for generator cooling has high possibility of fire and explosion.

3.2. Applicable standards and codes for design of fire fighting and protection system

Design basis for the power plant is to ensure safety for fire protection and fighting, apply and comply with current Vietnamese Standards, especially when those standards are higher than international standard.

Referring such contents not specified by Vietnamese standards, related international standards will be applied i.e NFPA (National Fire Protection Association) or others

i.e IEC 79-10 (International Electromechanical Commission). In such cases, applied standards will be approved by fire protection and fighting police department.

Fire fighting and protection system will be designed, constructed according to fire protection and fighting law, together with following decrees and instructions:

1. Decree No. 35/2003/NĐ-CP dated 04/ 04/ 2003 by the government specifying execution details of some fire protection and fighting law.
2. Circular No. 04/2004/TT-BCA dated 31/03/ 2004 by Ministry of Policeman guiding implementation of Decree No. 35/2003/NĐ-CP dated 4/4/2003 by the Government specifying implementation of some articles for fire protection and fighting.
3. Decree No.123/2005/NĐ-CP dated 05/10/2005 by the government specifying administrative punishment in fire fighting and protection.
4. Decree No.02/2006/TTG dated 23/01/2006 by the Prime Minister referring to strengthening of instruction and effective implementation of fire protection and fighting.
5. Decree No.10/CP dated 17/2/1993 by the government referring to issuance of safe protection for oil and petrol works.
6. Regulations for safe protection of oil and petrol works (accompanied by Decree No.10/CP dated 17/02/1993 by the Government)
7. Circular No.05/TT/TM-NV dated 01/10/1993 by Interministerial of Commerce and Trade, and Interior guiding execution of Decree No.10/CP dated 17/2/1993 by the government issuing stipulation and safe protection for petrol and oil works.
8. Interministerial Circular No.10/TT-LB dated 31/12/1994 by Ministry of Interior and Ministry of Construction guiding urban fire fighting supply.
9. Decree No.47/1999/NĐ-CP dated 06/07/1999 by the government referring to adjustment and addition for safe protection of petrol works, accompanied by the government's Decree No.10/CP dated 17/2/1993.
10. Decree No.08/2001/ND-CP dated 22/February/2001 by the Government specifying security conditions and orders for some sectors, business sector.
11. Joint circular No.15/2001/TTLT-BTM-BCA dated 10/05/2001 by Ministry of Commerce – Ministry of Policeman specifying provision and stipulations for fire fighting facilities for oil storage and oil products.

12. Combining with other related circulars, decrees and instructions

3.2.1. Vietnamese Standards

Relating to safe fire protection and fighting, following Vietnamese standards will be applied in design of fire fighting and protection system:

+ TCVN 2622: 1995	Fire prevention and protection for buildings and structures
+ TCVN 5760: 1993	Fire fighting system – Design and application requirements
+ TCVN 5738: 2000	Fire alarm system – Technical requirement
+ TCVN 5307: 1991	Oil and petrol storage and oil product
+ TCVN 6153: 1996	Safety for pressure vessels – Design, installation and utilization
+ TCVN 6100: 1996	CO ₂ fire protection fire extinguishing media – carbon dioxide
+ TCVN 4090: 1985	Oil pipe and oil products – Design standard
+ 20 TCN 46	Lightning in construction works
+ TCVN 6101: 1996	Fire protection and fighting – CO ₂ used fire fighting system equipment – Design and Installation
+ TCVN 5303: 1990	Fire safety – Terminology and definitions.
+ TCVN 3254:1989	Fire safety – General requirement
+ TCVN 4878:1989	Fire classification
+ TCVN 4879: 1989	Safety signs
+ TCVN 5040: 1990	Fire protection and fighting equipment - Symbols
+ TCVN 6103: 1996	Fire protection and fighting - Terminology – Smoke control.
+ TCVN 6305: 1997	Fire protection and fighting – Automatic sprinkler system
+ TCVN 7336: 2003	Fire protection and fighting – Design and installation requirements.
+ TCVN 7161: 2002	Air used fire fighting system – Physical property and system design.
+ TCVN 6379: 1998	Fire fighting equipment – Fire hydrant – Technical requirement.

- + TCVN 7435-1: 2004 Fire protection and fighting – Portable extinguishers and fire trolley
- + TCVN 5279: 1990 Explosion and fighting safety – General requirement.
- + TCVN 3255: 1986 Explosion safety – General requirement.
- + TCVN 4586: 1997 Industrial explosive material – Safe requirement for reserve, transport and utilization.
- + TCVN 6174: 1997 Industrial explosive material – Safety requirement for manufacturing, explosion testing and approval.
- + TCVN 5334: 1991 Electrical equipment for oil storage and oil products, safety specifications for design and installation.
- + TCVN 5684: 2003 Fire safety for oil works and products – General requirement
- + TCVN 3991: 1985 Fire protection standard in construction design – Terminology - Definition
- + TCVN 6160: 1996 Fire protection and fighting – Design requirement for high storeyed building – Design requirement.
- + TCVN 4317: 1986 Storage house – Basic principle for design.
- + TCVN 4513: 1988 Water supply within design standard
- + TCVN 6100: 1996 Fire fighting system - CO₂
- + TCVN 6102: 1995 Fire fighting system - Power
- + TCN 33: 1985 External and project's water supply network – Design standard.
- + TCN 25: 1991 Electrical line installation in living hoist and public work – Design standard.
- + TCN 18-24 Code of electrical installation – General regulation .
- + TCN 46-84 Lightning for public construction works – Design standard.
- + TCXD 218: 1998 Fire detection and alarm system – General regulation.
- + In addition, other related standards are also applied.

3.2.2. NFPA standards

- + NFPA 10 Standard for Portable Fire Extinguishers

+ NFPA 11	Standard for Low-Expansion Foam
+ NFPA 12	Standard on Carbon Dioxide Extinguishing Systems
+ NFPA 13	Standard for the Installation of Sprinkler Systems
+ NFPA 14	Standard for the Installation of Standpipe and Hose Systems
+ NFPA 15	Standard for Water Spray Fixed Systems for Fire Protection
+ NFPA 17	Standard for Dry Chemical Extinguishing Systems
+ NFPA 20	Standard for the Installation of Centrifugal Fire Pumps
+ NFPA 22	Standard for Water Tanks for Private Fire Protection
+ NFPA 24	Standard for the Installation of Private Fire Service Mains and Their Appurtenances
+ NFPA 25	Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems
+ NFPA 30	Flammable and Combustible Liquids Code
+ NFPA 70	National Electrical Code
+ NFPA 72	National Fire Alarm Code
+ NFPA 80	Standard for Fire Doors and Fire Windows
+ NFPA 204M	Guide for Smoke and Heat Venting
+ NFPA 497	Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas
+ NFPA 850	Recommended Practice for Fire Protection for Electric Generating Plants and High Voltage Direct Current Converter Stations
+ NFPA 120:	Standard for Fire Prevention and Control in Coal Mines.

In addition, other international standards will be considered to be applied.

3.3. Main design principle for fire protection and fighting system of Vung Ang 2 TPP

- 1) Complying with principle “Fire prevention first and Combined fire prevention and fighting”, strictly combining with standards and regulations of related fire protection and fighting. Each equipment, system following technology specific features shall comply with principle of equipment and material selecting and arrangement for fire prevention and extinguishing.
- 2) Important buildings and equipment will be provided with automatic fire alarm system.. In addition, locations with fire spreading possibility will be considered to prevent fire spreading to surrounding area.
- 3) Water fire fighting system will be the station main fire fighting system. This project will be provided with one independent fire water supply system to supply water for outdoor fire hydrants, automatic water spray system and spray system. The system will be calculated to be designed for one greatest fire in the plant area within a certain period.
- 4) Equipment and building will provide some conventional fire fighting systems, also some portable fire extinguishers and some special fire equipment as follows:
 - CO₂ fire fighting system will be applied for rooms with electronic equipment, tunnels and cable trench, central control house and system control house.
 - CO₂ fire fighting system will be installed for 6kV cable trenches in steam turbine house
 - Spray system/overflow spray system will be applied for main transformer, unit auxiliary transformer, station start-up transformer, oil ignition and turbine lube oil tank, areas with oil pipes, diesel generator...
 - Automatic water spray system will be applied for coal bunker area, power house, coal storage area, separate fire fighting facilities for each coal conveyor tower and connecting between the power house and coal storage area.
 - Foam fire fighting system will be provided for ignition oil tank
- 5) Installing the fire control, detection and alarm for the whole power plant
- 6) Fire water pumps will be provided in fire water pumphouse. Fire water reserve tank and one close-circuited fire water system will be installed for all the plant items.

3.4. Determining dangerous area

Dangerous areas will be provided with fire fighting equipment, fire detection and alarm system. Following areas in the power plant are determined as dangerous area:

- Main transformer area
- Oil storage area (ignition oil tank, lube oil tank, transformer oil tank...)
- Oil pumphouse
- Hydrogen generation house
- Hydrogen cooler (near the generator)

3.5. Requirements for building and structure

Classification of fire danger in building/structure and fire resistance is given in following table:

Item	Fire and explosive danger	Fire resistance
Power house (Turbine house, deaerator area, coal bunker area, boiler area)	D	II
IDF	D	
ESP	D	II
Stack	D	II
Main transformer	D	II
Unit auxiliary transformer	D	II
ESP control house	D	II
FGD control house	D	II
Station start-up transformer	D	II
Excitation transformer	D	II
Emergency diesel generator	D	II
Switchgear building	D	II
DO unloading pumphouse	C	II
Clean/dirty lube oil room	C	II
Coal conveyor	C	II
Transfer tower and coal crusher	C	II
Coal storage area	C	II
Ignition oil tank and pumphouse	C	II
Central control house and coal supply system control	D	II
Circulating cooling water pumphouse	E	II

Item	Fire and explosive danger	Fire resistance
500kV switchyard	D	II
Bottom ash discharge water pumphouse and clarification tank	E	II
Chemical dosing and mixing house	E	II
Chemical water treatment house and cooling water treatment house	E	III
Hydrogen generation station and storage tank	A	II
Auxiliary boiler area	D	II
Compressed air house (hydraulic is not used)	E	II
Air compressor area (hydraulic oil is used)	D	II
Metal testing, electrical and C&I equipment room	D	II
Maintenance workshop	E	II
Waste water treatment station	E	II
Warehouse	C	II
Storage yard	E	III
Dump truck house	D	II
Garage, carrier house	D	II
Fire truck house	D	II

(See TCVN 2622: Fire protection and fighting for house and structure – Design requirements)

Note: Excluding above buildings/structures, fire and explosive danger and fire resistance of other building will be designed suitably with regulations specified by current international standard “Fire protection and fighting standard for building”

3.6. Fire protection and fighting system for areas

Fire protection and fighting system for areas/buildings in the power plant will comprise of:

- 1) Power house
- 2) Transformer area
- 3) Switchyard area
- 4) Coal storage area
- 5) Oil tank area
- 6) Hydrogen station area

...

In addition to such areas/main buildings, there are also other areas/auxiliary buildings.

Fire fighting facilities are arranged for buildings, equipment and the whole power plant.

Building/area	Detection equipment	Operation	Fire extinguish- ed by	Combustibles	Fire classification
I. Central monitoring room/System monitoring room					
1. Chemical equipment monitoring room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical, cable wire	Normal
2. Emergency electrical distribution room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical, cable wire	Normal
3. Boiler, turbine monitoring room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical, cable wire	Normal
4. DC room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical, cable wire	Normal
5. UPS room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical, cable wire	Normal
6. Cable trench	Sensing cable	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical, cable wire	High
7. Electrical equipment room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical, cable wire	Normal
8. Operation engineer room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical wire, general combustible	Normal

9. Control room	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical wire, general combustible	Normal
10. ESP control house	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical wire, general combustible	Normal
11. FGD control house	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical wire, general combustible	Normal
12. Coal handling PDC/control building	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical wire, general combustible	Normal
13. Ash handling PDC/control building	Smoke, heat detection heads	Auto/Manual/Mechanical	Inert gas/CO2 spray	Electrical wire, general combustible	Normal

II. Turbine area

1. Turbine lube oil tank	Thermal sensing cable	Auto/Manual/Mechanical	Open typed water spray	Lube oil	High
2. Hydraulic electrical equipment room	Thermal sensing cable	Auto/Manual/Mechanical	Open typed water spray	Lube oil	High
3. Lube oil pipe area	Thermal sensing cable	Auto/Manual/Mechanical	Water spray, overflow	Lube oil	High
4. Steam turbine support	Thermal sensing cable	Auto/Manual/Mechanical	Fire foam tank	Lube oil	High
5. Turbine oil tank	Thermal sensing cable	Auto/Manual/Mechanical	Water spray, overflow	Lube oil	High
6. Cable	Smoke, heat	Auto/Ma	Inert gas/	Cable	High

trench and 6kV switchboard	detection heads	nual/ Mechanical	CO2 spray		
III. Boiler area					
1. Burner	Thermal sensing cable	Auto/Manual/ Mechanical	Open typed water spray	Ignition oil	Normal
2. Lube oil tank	Thermal sensing cable	Auto/Manual/ Mechanical	Open typed water spray	Lube oil	Normal
3. Air-heater area	Thermal detector	Manual	Fire hydrant	Non-combustible material	Normal
4. Coal unloading hopper	Thermal sensing cable	Auto/Manual/ Mechanical	CO2	Coal	Normal
IV. Transformer					
1. Main transformer	Thermal detector	Auto/Manual/ Mechanical	Open typed water spray	Oil	High
2. Unit auxiliary transformer	Thermal detector	Auto/Manual/ Mechanical	Open typed water spray	Oil	High
3. Station start-up transformer	Thermal detector	Auto/Manual/ Mechanical	Open typed water spray	Oil	High
V. Open and closed coal storage area					
1. Coal storage		Manual	Fire hydrant	Coal	Normal
2. Coal conveyor	Thermal detection cable	Auto/Manual/ Mechanical	Wet typed fire fighting	Coal	Normal
3. Coal mobilization	Thermal detector	Auto/Manual/	Wet typed fire	Coal	Normal

area		Mechanical	fighting		
4. Transfer tower	Thermal detector	Auto/Manual/Mechanical	Open typed water spray	Coal	Normal
5. Coal crushing area	Thermal detector	Auto/Manual/Mechanical	Open typed water spray	Coal	Normal
VI. Others					
1. Diesel generator	Smoke, heat detection heads	Auto/Manual/Mechanical	Open typed water spray	Diesel oil	Normal
2. Oil tank for diesel generator	Thermal detector	Manual	Foam	Diesel oil	Normal
3. Fire fighting pump	Smoke, heat detection heads	Auto/Manual/Mechanical	Water spray	Diesel oil	Normal
4. Diesel tank for fire pump	Thermal detector	Manual	Foam	Diesel oil	Normal
5. Station monitoring area	Smoke detector	Manual	Fire hydrant	Electrical wire and general combustible	Normal
6. Oil tank of the Diesel-Driven Fire Pump	Smoke, heat detection heads	Auto/Manual/Mechanical	Water spray	Diesel oil	Normal

3.6.1. Transformer

According to NFPA 850 for transformers with fire partition walls, transformer areas will be constructed with emergency oil pits.

According to NFPA 850, all main and secondary transformers will be provided with deluge spray system.

According to NFPA 15, fire water flow for transformer area will be designed as 10.2 l/min/ m² and about 6.1 l/min/ m² for area within the cofferdam.

Design and installation of emergency oil collection system for transformer area shall comply with standard No. 11TCN 21-84. Following matters will be considered:

Space within the collection pit shall sufficient for the whole stored oil inside the transformer.

Required time for oil collection and fire water shall be considered in the detail design.

Capacity of oil discharge collection pit will be designed following 11TCN 21-84.

3.6.2. Diesel oil tank

Diesel oil tank area of F.O 2x2000m³ serving fire ignition and fire support when units are at low load shall have temperature of under fire ignition. In order to avoid oil temperature increase to ignition temperature, oil tanks will be installed with cooling water spray system. According to Vietnamese standard 5307-2002, water spray flow is 30.0 l/min/m of the tank circumference.

In order to avoid fire spreading in the cofferdam area (or partition wall), oil tank will be provided with three foam fire hydrants with spray flow of 190 l/min (according to NFPA 11).

Foam nozzles (hoses) can be connected to foam hydrants surrounding the cofferdam or partition wall.

Each cofferdam of the oil tank will have one oil collection tank. On the pipe outside the cofferdam connection from the discharge oil pit to the oil separation pit will be installed with one isolation valve to prevent oil fire spreading.

3.6.3. Steam turbine area

Turbine lube oil storage area will be protected by automatic water spray system.

Fire danger inside the steam turbine area:

- Steam turbine lube oil storage area – (High danger area)
- H2 cooler area (High danger area)
- Steam turbine support area (Normal danger)

Turbine lube oil storage tank area and H2 cooler will be protected by automatic water spray system.

Turbine lube oil storage tanks will be arranged within the cofferdam by reinforced concrete with sufficient capacity to store the whole oil and fire water quantity with flow of 12 l/min/ m² within 10 minutes as specified in NFPA 850 standard.

Automatic water spray system permitting water spray in the tank will reduce stored oil temperature in the tank not permitting oil to achieve fire ignition temperature or extinguish the fire.

Oil discharge pipe leading to pits under the turbine area will be installed with quick open valves. Valves installed on the pipe outside the turbine area will be quick on-off type to avoid oil fire spreading inside oil storage area under case of fire occurrence.

Turbine supports will be protected by chemical foam fire fighting system and indoor fire hydrants.

All area considered as danger normally in the turbine area is protected by close circuit water supply system, fire water hydrants will be arranged adjacent to area of high danger inside the turbine area. In addition, portable fire extinguishers will be located near the walkway on the ground floor of the turbine area.

3.6.4. *Electrical equipment area and cable bed*

Electrical – electronic room and electric cable floor is considered as “High danger area”. Electrical – electronic room and cable tunnel is protected by CO₂ system, fire detectors will be provided with signals to activate the CO₂ used automatic fire fighting system.

Alarm signals (optical or voice) will be activated at least 30 seconds before operation of CO₂ system.

3.6.5. *LV, MV, DC electrical equipment storage room and cable floor*

LV, MV, DC equipment rooms and cable floor is evaluated as “High danger area”. Such rooms and cable tunnel will be protected by fixed CO₂ system, fire detectors will provide signs to activate CO₂ used automatic fire fighting system.

3.6.6. *Emergency diesel generator*

Emergency diesel generator will be protected by fixed CO₂ system, fire detectors will provide signs to activate CO₂ automatic fire fighting system.

In addition, emergency diesel generator will be protected by indoor fire hydrants. Water hydrants will be arranged near area of high fire risks.

3.6.7. *Boiler*

Boiler protected by outdoor fire hydrants and chemical portable extinguishers will be arranged near walkways on different floors.

At location of burners, fire alarm system and direct fire nozzles will be installed to fight fire under case of emergency.

In addition, mobile extinguishers such as fire truck will be used under emergency case.

3.6.8. *Electrical & switching equipment room*

Electrical and switching equipment room will be protected by indoor fire water system. In addition, there are also portable extinguishers to protect other electrical equipment and monitoring equipment.

Smoke detectors will be located at special locations to initiate fire alarm signs.

3.6.9. *Switchyard equipment room*

Switchyard house and DCS room will be protected by indoor fire fighting system. In addition, there are also portable extinguishers to protect electrical equipment and monitoring equipment.

Smoke detectors will be located at special locations to initiate alarm signs under fire occurrence.

Different floors of the central monitoring room are also protected by indoor fire fighting system.

3.6.10. *Administrative building, warehouse and garage*

Smoke detectors are arranged in each room to initiate alarm signals under fire occurrence.

Buildings will be protected by indoor fire hydrants. Such hydrants will be located at areas of high danger.

Dry chemical extinguishers will be installed near walkways.

3.6.11. *Canteen and workshop*

Smoke detectors will be arranged in each room to be able generate alarm signals under fire occurrence.

Buildings will be protected by indoor fire hydrants. Indoor fire hydrants will be arranged at areas of high danger.

Dry chemical foam extinguishers will be installed near walkways.

3.7. Fire protection and fighting service road

Plant internal road will be arranged as close loop circuit, combining between service road and fire fighting road. Power house and coal storage areas will have road routes surround. Main road route has width of 12m, other routes have widths of 4m, 6m, 8m depending on utilization purpose of each route. Road design load is based on standard No.20 – Applied for trucks. If the plant internal road run under access bridges, it shall ensure static clearance height of over 5.0m, smallest distance from road side to buildings/works shall be greater than 1.5m.

3.8. Configuration of fire fighting and fire water supply system

Fire water supply system is an independent system. This system is designed under loop circuit providing water for outdoor fire hydrants, spray system, wet spray system, wall plate fire fighting system. Fire water supply system will comprise of fire water tank, fire pump, jockey pump and indoor & outdoor fire water pipes.

In addition to fire water system, CO₂ fire fighting system, foam fire fighting system, dry chemical foam fire fighting system will be provided depending on features of each area and equipment.

3.9. Station fire protection and fighting system

Station fire protection and fighting system will be installed and provided with following systems:

- Fire water loop circuit with hydrants and hose box.
- Indoor and outdoor fire water system.
- Wet typed water spray system (Wet sprinkler system, wet spray system).
- Delude system
- Foam used fire fighting system
- Dry powder fire fighting system
- CO₂ fire fighting system
- Portable extinguishers
- Fire truck and accessories.

Main fire fighting facilities comprise of:

- Fire water pump.

- Fire water loop circuit (main distribution pipe).
- Outdoor fire hydrants and hose box.
- Sprinkler system
- Spray system
- Cooling water spray system for oil storage tank.
- Foam fire fighting system (Oil tank and oil pumphouse area) .
- Fire proof clothing and other special rescue tools.

3.10. Fire protection system

3.10.1. Fire detection system

Fire detection system together with fire fighting facilities will be designed, installed basing on current Vietnamese fire protection and fighting code and equivalent international codes. Fire detection equipment will be selected suitable with separate failures. Fire detection and alarm system will operate continuously 24 hours/day to ensure timely detection and alarm under case of failure.

The fire detection and alarm system shall meet following basic requirements:

- Timely fire detection
- Clear signal transmission

Fire detection, alarm system and fire fighting facilities will be designed, installed according to Vietnamese fire protection and fighting code.

Fire detection system will use fire detectors:

Smoke Detector:

Will be located on house ceilings, where smoke can be generated as initial signs of fire occurrence.

Smoke is generated from the fire source, rising high and intruded into sensing bulb of the smoke detector and generating alarm signals which will be transmitted to the fire alarm cubicle.

There are many types of smoke detectors: ionization smoke detectors, electro optical smoke detector, optical smoke detector, projection ray detector...

Heat Detector:

Will be located on house ceilings, where smoke can be generated as initial signs of fire occurrence.

Heat is generated from the fire source, activating heat sensor and alarm signals which will be transmitted to the fire alarm cubicle.

There are two types of heat detectors: Fix temperature heat detector and Rate-of-rise heat detector.

Fix temperature heat detector will activate alarm signals when its surrounding temperature increases to a pre-set level, for example 58°C , 68°C , 108°C .

Rate-of-rise heat detector will activate alarm signals when its surrounding temperature increase suddenly within a short period, for example, sudden increase of $6^{\circ}\text{C}/\text{min.}$, $8^{\circ}\text{C}/\text{min.}$

Flame Detector:

Detecting ultraviolet from the flame. It is sensitive with flame, normally used at sensitive environments, easy ignition, for example oil storage tank.

Very Early Smoke Detection Aspiration (VESDA)

The system can detect micro smoke elements, find early fire before appearing visible smoke or flame. Working principle is protection area air sample that is put to optical detectors through small pipe system arranging over or under protection area ceiling to detect smoke element, sensor will send the signal to controller, there, warning signal (if any) will be created with 04 warning levels. Small drilled holes on piping will create a point sample matrix. VESDA will provide many warning levels, and the warning levels can be established in a wide range, allowing early warning, so increasing to evacuate time.

Installation and maintenance system cost is low, the system can be used for both small and very large space up to 2000 m^2 with simple pipe system arrange on existing piping or hiding on ceiling, easily connecting with control board. At area difficult to access, detectors can be arranged on centre area wall as well as inside or outside

protection area. Expansion of the system when requiring protection for new area will be simply done by adding more piping.

Because of strict features and requirements for control cubicles at the central control room, then such location shall be designed and provided with special fire protection and fighting system. Control cubicles will be provided with VESDA.

Emergency button:

If above activation equipment are automatic activation equipment, emergency button will be manual type. In the other way, persons can actively initiate fire signals by impacting such equipment when witnessing fire prior to fire alarm equipment activation.

3.10.2. Fire alarm system

Bell/Siren/Loud speaker/Fire alarm light:

Fire failure will be informed to people in the affected area for early evacuation.

Fire alarm system shall be strictly combined with fire detection system and other fire fighting means to create safety during plant operation.

Similar to fire detection system, fire alarm system shall operate 24 continuous hours per day to timely alarm in all fire problem in plant.

Emergency alarm system shall ensure the main following functions

- Produce alarm signal and transfer signal.
- Easy recognize signal
- Automatic and continuous self – checking function.

The power supply of system is from UPS and battery backup source during fire failure in the power plant.

Backup battery capacity shall ensure at least 24 continuous hour operation of equipment, and at least 3 hours during fire.

Fire alarm centre shall be protectively earthed. Protective earthing shall satisfy existing earthing codes.

Fire alarm system for Vung Ang 2 TPP shall be designed following TCVN 5738-1993 standard on Fire alarm system – Technical requirement and NFPA 72: 1996.

3.11. Water fire fighting system

Water fire fighting system includes water source, fire fighting pump, water pipe and fire fighting water sprayers. The system shall ensure enough water quantity and necessary pressure for fire fighting.

3.11.1. Fire pumphouse

Fire pumphouse shall be installed near fire water source. The pumphouse shall be arranged four fire pumps: fire pump driven by electric motor, standby pump driven by diesel motor and 02 jockey boost up pumps to maintain pressure in piping.

Installing 02 jockey boost up pumps will have more advantages over 01 jockey boost up pump in operating system. When the piping is leaked or 01 jockey boost up pump is failed or maintained, pressure in piping shall not be kept at desired level; when pressure in piping reducing to active level of electric pump, this pump shall be automatically stopped. After some operation minutes, it will push up pressure in piping to required pressure and the pump shall automatically stop. This process will continuously repeat until repairing completely jockey pump. Using 02 jockey boost up pumps for the system will overcome above disadvantages.

Fire pumps will be controlled by pressure button to be automatically started or stopped when pressure is at pre-set level.

At time of fire failure, the fire fighting system is operated resulting in drop pipe pressure to permitted level, fire operation pump will be automatically started.

In case of failure of online fire pump (due to power source failure or motor failure), emergency diesel fire pumps will be automatically started when the pressure is down to permitted level.

Fire pumps can be controlled manually or remotely. In addition, auto control requirements for fire pumps shall be ensured as follows:

- Auto starting step-up pump
- Auto starting online pump
- Auto starting emergency diesel pump if the online fire pump can not be started due to technical reason.

When the online fire pump is interrupted due to failure, the standby pump shall be automatically open.

In order to protect pump and other system components, depression valves will be provided for each valve.

Calculation of fire water flow:

In Vung Ang 2 TPP, fire fighting water flow shall be optimally selected based on three (03) following calculation cases.

Case 1: (Oil store warehouse)	Calculation of water demand of foam system for Diesel oil tank + demand of oil tank cooling system + demand of two fire fighting hydrants.
Case 2:	Calculation of water demand of steam spray system of main transformer + demand of two fire fighting hydrant
Case 3:	Calculation of water demand of two fire fighting hydrant indoor at the highest in Boiler compartment + demand of two outdoor fire fighting hydrant.

The above calculation is to find the maximum water demand for fire fighting, so with the same equipments, the calculation shall be carried out with the equipment with higher fire fighting water demand.

Following is fire water flow table for three above cases:

Item	Unit	Value		
		Case 1	Case 2	Case 3
Foam spray for oil tank	l/min.	1202		
Cooling for oil tank	l/min.	1809		
Spray for transformer	l/min.		5411	
Two outdoor fire hydrants	l/min.	1893	1893	1893
Two boiler area fire hydrants	l/min.			300
Total	l/min.	4904	7304	2193
	m ³ /h	294	438	132

From above calculations, we found that case 2 requires highest fire water flow. Then, the pumphouse capacity is selected as 440 m³/h.

Description	Symbol	Formula	Unit	Result
Pumphouse capacity	Q_b	$Q = \max (\text{Case1, Case 2, Case 3})$	m ³ /h	440
Internal diameter of the transport pipe	d	$d = \sqrt[4]{4*Q/(\pi*v)}$	mm	322
Flow velocity	v	TCVN 33-7.14	m/s	1.5
Internal diameter of transport pipe	d	Following handbook	mm	348
External diameter	D	Following handbook	mm	355
Pressure of fire pump	H_b	$H_b = H_{dh} + H_{tt} + H_{td}$	mH ₂ O	112.5
Total loss on water pipe	H_{tt}	$H_{tt} = H_1 + H_2$	m	4.0
Pressure loss due to pipe length friction	H_1	$H_1 = A*Q^2*L$	m	3.7
Unit resistance (refer to table)	A	TCVN 4513 - 6.15		0.4078
Pipe length	L	Estimated	m	600
Local loss of water supply network	H_2	$H_2 = 10\%H_1$	m	0.37
Differential from furthest nozzle to fire pump	H_{dh}	Estimated	m	70
Free pressure loss inside pipe and hose	H_{td}	TCN 33-3.15; TCVN 4513-6.19 $H_{td} = 0.00385*q^2*l$	m	38
Flow via hose	q	NFPA 850.4-2.1(b) $Q = 1893 \text{ l/min.}$	l/s	15.8
Hose length	l	Selected	m	40
Fire pump motor capacity	N_b	$N_b = \frac{k*(Q*H_b*G_n)}{(102*H_s)}$	kW	202
Efficiency	H_s	Following handbook	%	80
Standby factor	k	-		1.2
Fire water ratio	G_n	-	kg/m ³	1000
Fire water reserve	V_{dt}	$V_{dt} = Q_b*t_{dt}$	m ³	1320
Fire water reserve time	t_{dt}	TCVN 2622-10.22	hour	3
Minimum flow for recovering reserved fire	Q_{ph}	$Q_{ph} = V_{dt}/t_{ph}$	m ³ /h	55

water				
Time for recovering reserved fire water	t_{ph}	TCVN 2622-10.23	hour	24

From above fire water flow calculation, pumphouse capacity will be selected to ensure minimum flow of 440 m³/h for fire demand (Case 2).

According to TCVN 2622: 1995, water quantity for fire reserve shall be calculated basing on the greatest fire water volume during three hours. Therefore, minimum fire water volume is 1320 m³.

According to TCVN 2622 – Item 2.3, time for reserved water recovery shall not exceed 24 hours. Therefore, we have make-up water pump capacity:

$$Q_{bs} = 1320/24 = 55 \text{ m}^3/\text{h}.$$

Pump basic parameters:

+ Main pump:

Quantity : 01

Type : Centrifugal, power motor

Capacity : 440 m³/h

Pressure : 112.5 mH₂O

Motor capacity : 202 kW

+ Standby pump:

Quantity : 01

Type : Centrifugal, diesel motor

Capacity : 440 m³/h

Pressure : 112.5 mH₂O

Motor capacity : Corresponding to main pump (according to TCVN 2622 Item 10.24).

+ Jockey pump:

Number : 02

Pressure : 120 mH₂O

+ Make-up water pump

Capacity : 55 m³/h

3.11.2. Sprinkler system

Sprinkler system is the most popular fire fighting system. This system will cool machines, equipment under case of over heat, fire and explosion proof, or extinguish fire with protected scope.

The fire will be extinguished by direct water spray into the fire area, where sprinklers are activated at pre-set temperature level.

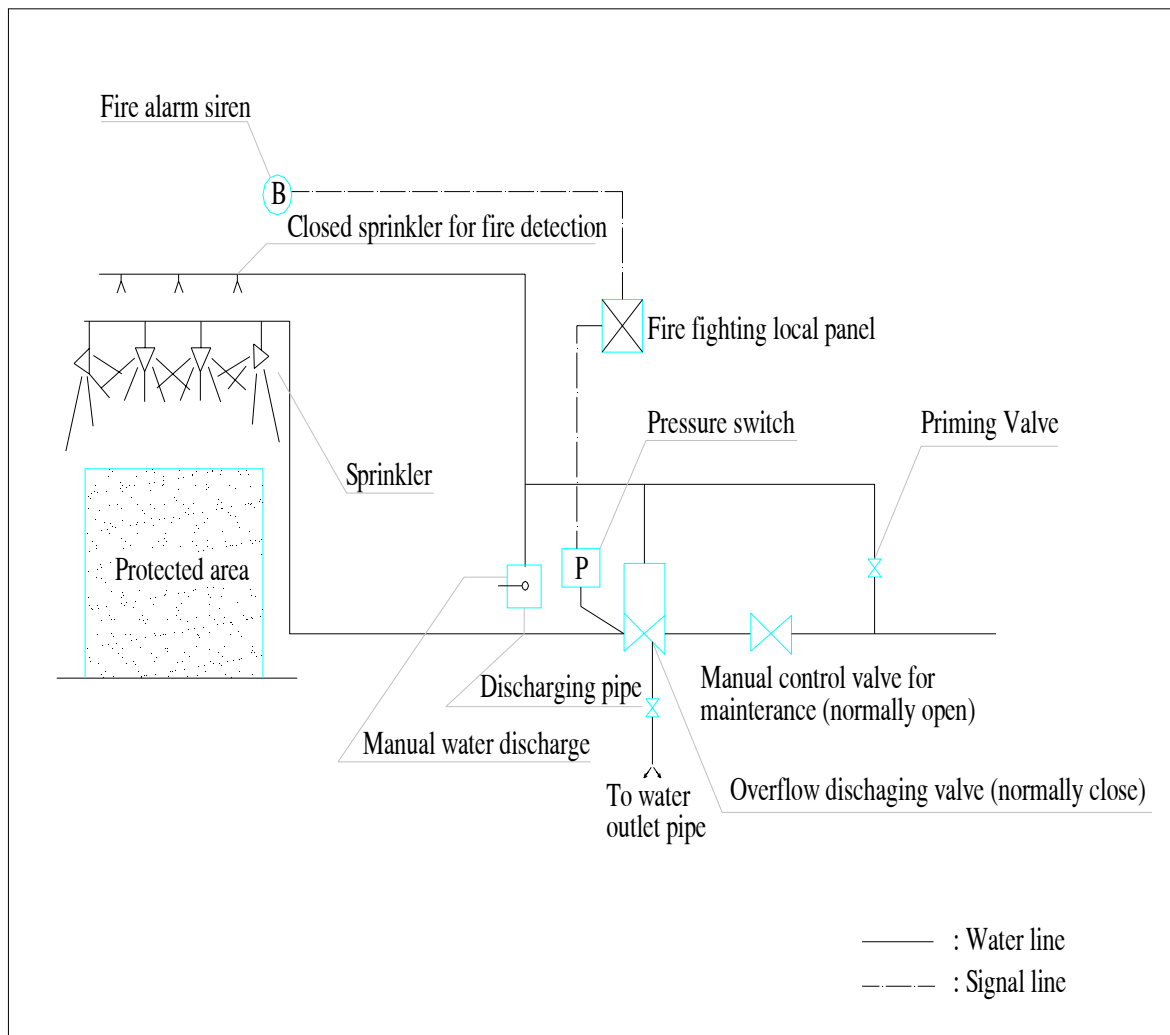
Sprinkler system is one system connecting underground and ground pipe and which is designed following fire fighting technology standards. Ground system is one pipe network which is designed following “hydraulic calculation” principle or “pipe calibration” principle or installed inside buildings, structures or one overhead places. On such pipes, sprinklers will be arranged so that when water is sprayed, it will cover one calculated space. Control valve for each system riser will be located on riser or its water supply pipe. Each riser for the sprinkler comprises of one alarm activation system when the system start fire fighting. Normally, the system is activated by heat generated from the fire, and it will spray water covering the firing area.

Spray system will be arranged at following areas:

- Boiler feed water pump area
- Motor driven feed pump (area having lube oil or hydraulic oil firing danger)
- Turbine driven water pump oil tank
- Emergency diesel generator oil tank area
- Substation

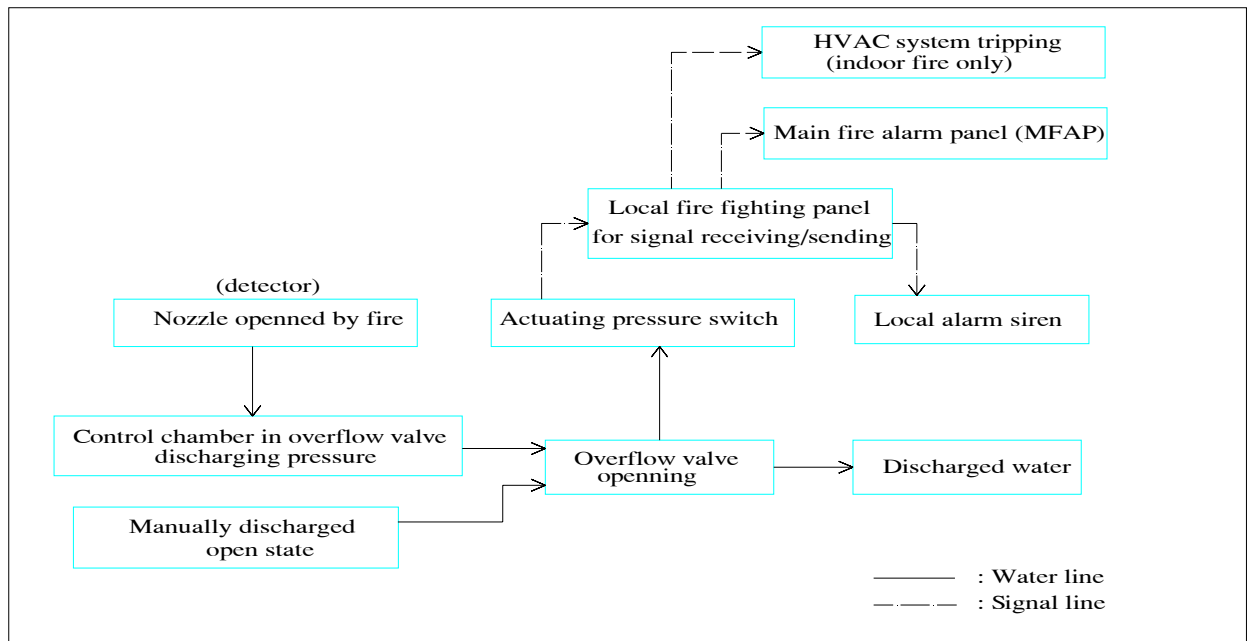
- Hydrogen generation house
- Cooling system for oil tanks
- Warehouse, garage
- Administrative building and canteen

System structure diagram:



Operation principle:

Under case of fire (detected by fire detectors), deluge valve will operate to activate the spray system (water is discharged via sprinklers). In addition, local fire fighting cubicle will receive and transmit signals to main fire fighting cubicle and local bells will be activated.



There are many types of sprinklers:

Wet Pipe System : Sprinkler system with automatic sprinklers attached to the system with water pipe system, and connected to such water source, then, water will be sprayed immediately, via sprinklers opened by fire heat.

Dry Pipe System : Sprinkler system with automatic sprinklers attached to the air or nitrogen pipe will be maintained by pressure, and when sprinklers are opened by heat from the fire, air inside the pipe will be exhaust, permitting water pressure to open dry pipe valve. Therefore, water will run into the pipe system and sprayed via open sprinklers.

Preaction System: Sprinkler system with automatic sprinklers attached to the air piping system may have or haven't pressure and have one make-up fire alarm system installed at locations with sprinklers. When the fire alarm system is activated, it will open valve and permitting water to run into the pipe system, the spraying water via open sprinklers.

Deluge System: Sprinkler system will use open sprinkles which are attached to one water source, via one valve which is opened by activation of one fire alarm system located at sprinkler places. When such valves are open, water will run into the pipe system, then sprayed via all installed sprinklers.

Deluge system will be designed to spray one thick, great water volume. It covers one large area, water is sprayed simultaneously by many hoses, and arranged in ranks depending on each site requirement.

The sites are classified with high hazards in oil transformer area which can extinguish fire by normal sprinkler system, using deluge or water spray systems.

Calculation of station spray system:

Density of design water shall not less than 10.2 litre/min./m² for spray system (according to NFPA 15.4 -4.3.4). Density $q = 14$ l/min./m² referring to NFPA 15.4 – 4.3.

Area	Protected dimension			Greatest protected area A
	Length(m)	Width(m)	Height(m)	m ²
Step-up transformer	8.6	7.5	10	387
Unit auxiliary transformer	7.5	4.5	8	226
Station auxiliary transformer	7.2	3.7	6.8	175
Turbine oil room	15	8		120
Lube oil tank	8	4	3.5	116
Furnace face	35	10		350
Diesel generator	14	8		112
Hydrogen generation house	24	12		288

Area	Water demand $Q=A*q$		Pipe (mm) $D=\sqrt{4*Q/\pi*v}$	
	l/ph	m ³ /h	Calculated	Selected
Step-up transformer	5411	325	277	300
Unit auxiliary transformer	3161	190	212	250
Station auxiliary transformer	2448	147	186	200

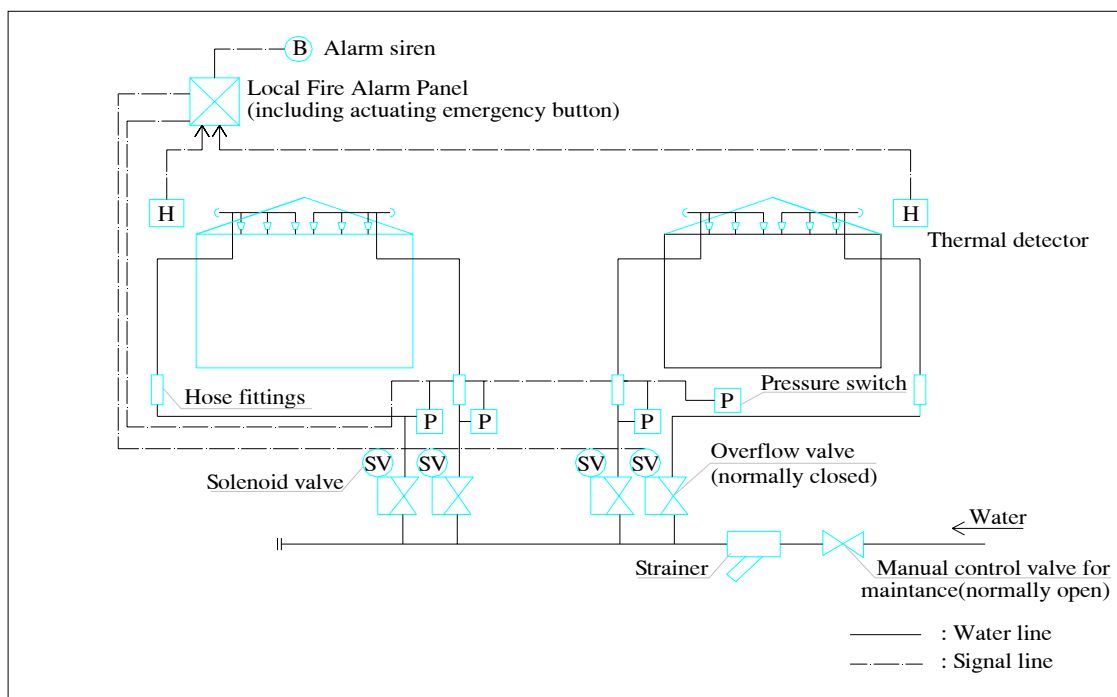
Turbine oil area	1680	101	154	150
Lube oil tank	1624	97	152	150
Furnace face	4900	294	263	250
Diesel generator house	1568	94	149	150
Hydrogen generation house	4032	242	239	250
<i>Greatest flow</i>	<i>5411</i>	<i>325</i>	<i>277</i>	<i>300</i>

Water flow supplied for spray system $Q = 5411 \text{ l/min.} = 325 \text{ m}^3/\text{h}$. Pipe diameter $d = 300\text{mm}$.

3.11.3. Oil tank cooling water system

Each oil tank will be cooled on the roof and surround the tank wall. Oil storage tank will be provided with consolidation bands inside the tank, therefore, in order to equally cool the wall tank, there will be two independent water spray loops for each tank, one on the top and one under the consolidation bands. Cooling water spray flow is 30 liters/min/circumferential following TCVN 5307-1991.

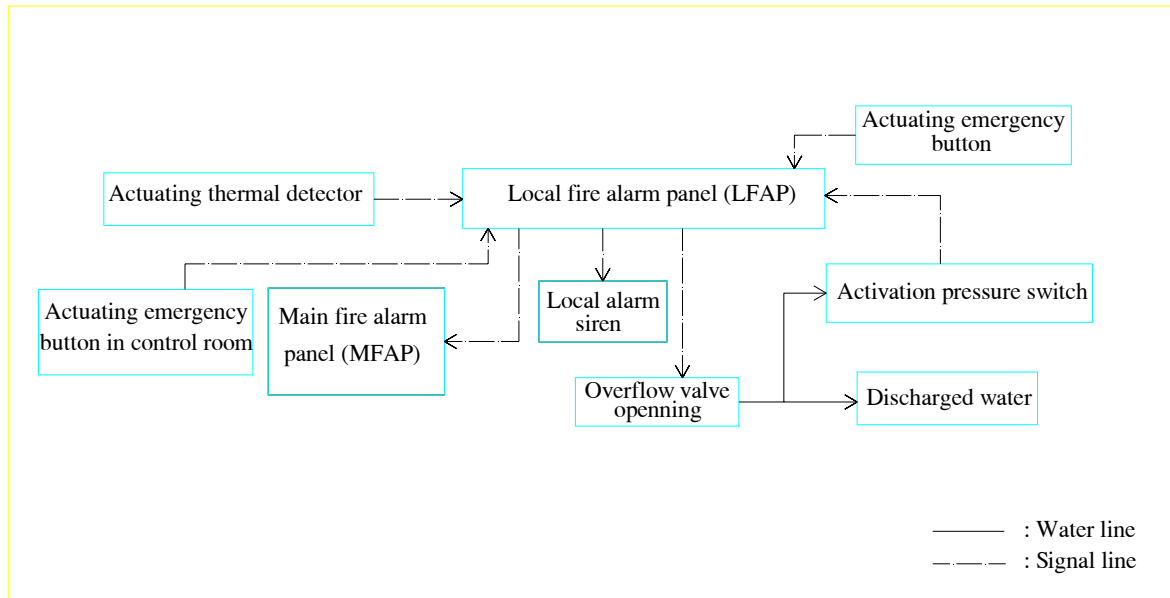
Oil tank cooling system diagram:



Operation principle:

Under case of fire (detected by fire detectors or manually activated by emergency buttons), local fire alarm central cubicles will receive and send signals to main fire

alarm central cubicles, local fire alarm bells will be activated and deluge valve will activate the oil tank cooling system (water is discharged via water sprinklers).



Cooling system will be provided for diesel oil tanks to avoid expansion on occurrence of fire.

Water cooling system will comprise of stationary water supply pipe, nozzles, and quick on – off valves which are operated manually and remotely on fire fighting control cubicle at oil pumphouse and at the central control room by opening & closing gate valves.

Loops for nozzles for one storage tank will be divided into two systems: one cooling water system and one foam fire fighting system.

Fire tank will be cooled with flow of 0.5 litre/second for one meter of tank circumferential.

The tank near the fire tank within distance of less than or equal to twice of the fire tank diameter will be cooled with flow of 0.2 litres/second on one circumferential meter of tank (only cooling haft of fire tank circumference TCVN 5307: 91, Item 2.10.8).

Calculation of oil tank cooling:

Oil tank will comprise of two adjacent tanks with diameters of 16m, height of 12m.

Applicable standard: TCVN 5307-91, Item 2-10.8

The firing tank will be cooled with flow of 0.5 l/s for one circumferential meter of tank. The tank adjacent to the firing tank will be cooled with flow of 0.2 l/s for one circumferential meter of tank.

Area	Density q	Water demand Q=A*q		Pipe diameter (mm) D=sqrt(4*Q/pi*v)	
	l/(ph.m)	l/ph	m³/h	Calculated	Selected
Firing tank	30	1507	90	160	200
Adjacent oil tank	12	301	18		
Total flow		1890	109		

3.11.4. Fire hydrant system

Fire hydrant system will be designed consisting of one network of ground and underground pipes connected with each other in loop circuit (ensuring continuous water supply). This system will supply pressurized water to throttle valves at indoor and outdoor hydrants. Hydrants will be arranged at suitable places in the power plant. Hose of suitable length will be installed with accessories such as by-pass pipe, hose, connection joint...will be located at metal boxes of fire hose near water hydrants. Soft hoses shall have length of over 20m and without joints.

Indoor hose shall be located at each floor in the control house, places in the turbine and other adjacent houses, walkways, on landing, corridor and conspicuous places, easy to use (according to TCVN 2622 Item 10.18). Centre of indoor fire hose will be located at height of 1.25m comparing with the floor.

Outdoor hydrants will be located at intervals among outdoor water hydrants as follows: around the power house: not higher than 80m, around the oil; around the oil tank: not higher than 30m, and other areas: not higher than 120m., ensuring sufficient fire water volume for each fire location (according to TCVN 2622). At each hydrants, there is one steel red painted box for weather proof.

3.12. Foam fire fighting system for diesel oil area

Ignition oil tank area will be provided with two oil tank of 2x 2000 m³ capacity, main oil pumphouse and 01 transfer oil pumphouse. The fire fighting system in this area will use a fixed foam fire fighting system. One foam-air mixer and two foam storage receiver of 2 x 2.65m³ will be provided in the foam house.

Pressurized water for the foam – air mixer will be supplied from the fire water supply system, foam solution using AFF 3%, after mixing following the foam water solution ratio will be transported to the foam generator in each oil tank and foam hydrants outside the fire partition wall by special foam mixing pipe.

Water source for fixed oil tank cooling system will be supplied from the main pipe system and transported to the water hose for cooling oil tank.

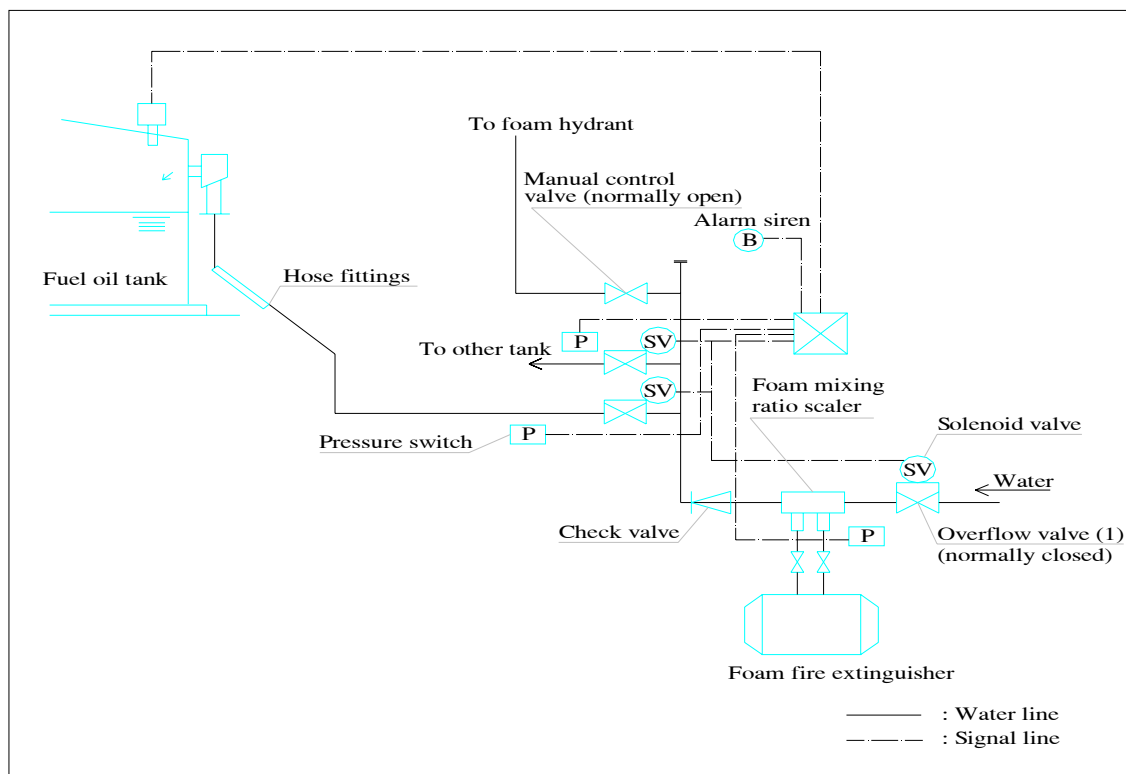
Under case of fire, foam fire fighting system will be manually started.

Popular foams:

AFF foam: is a foam which will cover the firing fuel by hydrocarbon.

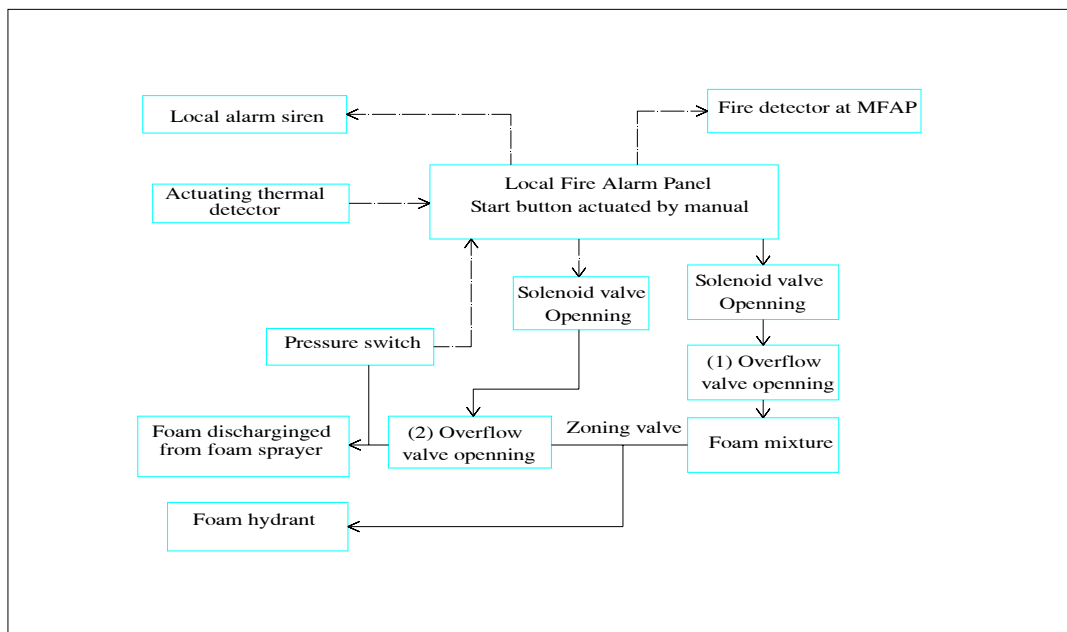
FOAM ARC (alcohol-resistant concentrate) is a foam which will cover the undissolved fuel plan.

Diagram of the fire foam system:



Operation philosophy:

Under case of heat detectors giving fire alarm, or emergency button is manually activated the local fire cubicle will receive and send signals to the central fire alarm cubicle, to the deluge valve via electronic valve (to activate foam mixer) and foam mixer will be transported to fixed foam hose and foam hydrant via fire selection valve (discharged foam mixer).



Foam fire fighting system, when activated, will spray one type of foam to cover the oil and petrol surface, combustible liquid is separated from the air and fire, then the flame is extinguished.

The fixed fire foam is installed stably consisting of fixed piping system connected with the fire water pipe, solid foam generator tank, foam generator, foam spray pipe and foam checker. The system will be provided with 3% solid foam generator named Fluoroprotein. Reserved quantity for fire fighting is three times of foam volume used for fire fighting (TCVN 5307: 91, Item 2-10,7).

Foam hydrants will be provided with two windings of 15m with hose diameter of 40mm located in steel box.

Cone shafe of diesel oil tank will be provided with fixed foam sprayer, each oil receiver will be provided with two foam chamber located symmetrically via the tank center. Foam hydrants will be located around the oil pumphouse area to avoid the fire spreading.

Pipe system connecting the foam supply and sprinklers via one control valve, this valve will be automatically controlled by fire detectors provided in the protected area. When the control valve is open, fire water and solid foam melts to the ratio generator to creat a fire foam solution. The foam solution will be supplied depending on fire scope until the control valves are closed.

Foam nozzles used shall be listed by authorized units or approved in discharge flow, effective protection scope and code, coverage area. The foam spray system will be manually controlled. Start of the foam system will be done by opening gate valve and the system will stall operation by closing such valves.

The foam system will be designed with design density following NFPA 11 standard. Foam density shall be not less than 4.1 litre/min./m² of surface area.

Calculation for fire foam system :

Requirement:

Reserved foam should be three times of required foam volume (TCVN 6307-91, Item 2-10.7). Designed according to NFPA 11.

Foam density shall not be less than 4.1 litre/min./m² for firing area.

Oil tank dimension:

Volume: 2 ones

Diameter: 16 m

Height: 12 m

Calculation:

Description	Symbol	Formula	Unit	Result
Flow for foam chambers	Q_A	$Q_A = A \cdot q$	l/min.	824
Area of fire fighting surface	A	$A = \pi \cdot d^2 / 4$	m ²	201
Density	q	NFPA 11.3-2.3.2.2	l/min./m ²	4.1
Flow for foam hose	Q_B	$Q_B = q_h \cdot N$	l/min.	378
Foam hose flow	q_h	NFPA 11.3 - 9.1	l/min.	189
Required smallest hose number	N	TCVN	hose	2
Total flow	Q_1	$Q_1 = Q_A + Q_B$	l/min.	1202
Solid foam demand	Q_c	$Q_c = Q_{ca} + Q_{cb} + Q_{cc}$	litre	1188
Select			litre	1190
Foam demand for liquid surface	Q_{ca}	$Q_{ca} = A \cdot q \cdot t \cdot S$	litre	742
Min. fire fighting time	t	NFPA 11.3-2.3.2.2	min.	30
Solid foam rate	S	NFPA 11.3-2.3.2.3		3%
Foam demand for hoses	Q_{cb}	$Q_{cb} = q_h \cdot S \cdot t \cdot N$	litre	340
Foam demand for pipe filling	Q_{cc}	$Q_{cc} = (\pi \cdot d^2 / 4) \cdot L \cdot S$	litre	106
Water supply pipe diameter for foam system	d	$d = \sqrt[4]{4 \cdot Q_1 / (\pi \cdot v)}$	mm	130
		Select	mm	150
Pipe length	L	Estimated	mm	200

Total water flow for foam system is 824 litre/min. Solid foam demand is 1190 litres. Solid foam demand is 1190 litres. Pipe diameter is 100mm.

3.13. CO₂ fire fighting system

Vung Ang 2 TPP will be provided with CO₂ fire fighting system for other areas following equipment specifications in each protected area and fire affect scope.

Protected areas of the deluge CO₂ fire fighting system comprise of: emergency power supply house in the central control building, PC boiler house, DC power supply room, UPS room, cable floor, electric equipment house, DC power room of the system control house, and cable floor of the turbine area's 6kV voltage distribution house.

Fixed CO₂ fire fighting system comprise of CO₂ receivers, distribution pipe system, hoses, activation and alarm equipment. The system is capable of automatically activating by thanking to fire detectors or manually activating the system by breaking protection glass covers and pushing the buttons.

Equipment for CO₂ fire stations and distribution system comprise of:

CO₂ receivers and supports are designed to sufficiently store total liquid CO₂ volume following design requirements. Each CO₂ receiver will be divided into each chamber following estimated number of receivers of about 12 receivers x 50 liters. Each chamber comprises of all required supports and equipment for keeping the receivers at best stable location.

Headers, junction pipes to each receiver, isolation valves, check valve, pressure indicator, drain valve and safety valves together with other required equipment.

Required CO₂ for fire fighting and reserve in the receiver will be designed according to TCVN 6101-1995.

CO₂ portable receivers, dry foam receivers and portable trolleys are designed following NFPA 10. Areas in the power plant is classified following danger groups as A, B or C (following NFPA 10 Item 1-5.1 to apply suitable fire extinguishing substances). Types of ABC foam are used for extinguish fire for A typed fire, B & C typed fires with solid combustibles.

In addition, design and installation of CO₂ system shall comply with TCVN 6101-1995, TCVN 6100-1996.

Portable extinguishers shall be sealed and marked outside with specified weights of containers and extinguishing matters. Foam quality is specified in TCVN 6102-1996 Fire protection and fighting – Fire foam. On installation, the receiver top shall not exceed height of 1.5m comparing with the floor, in case the receiver weight is higher than the above weight, the receiver top shall be lower than the height of 01m, the receiver bottom shall be 10cm apart from the floor (complying with NFPA-10 Item 1-6.10).

Maximum protected scope of popular extinguishers is about 1m², therefore, it is necessary to calculate the coverage area of protected equipment.

Furthest distance from portable extinguishers to protected locations is about 23m (NFPA standard).

For CO₂ extinguishing substance: As specified in Table 1 of TCVN 6100-1996 Fire protection and fighting – Carbon dioxide – CO₂ quality shall meet purity of 99.5% in volume respect, water content shall not exceed 0.015% in quantity respect.

Trolleyed mobile fire extinguishers are designed for greatest protection areas, requirements for fire extinguishing substance quantity, highest concentration density i.e in turbine area, warehouse and workshop...

3.14. Dry chemical foam fire fighting system

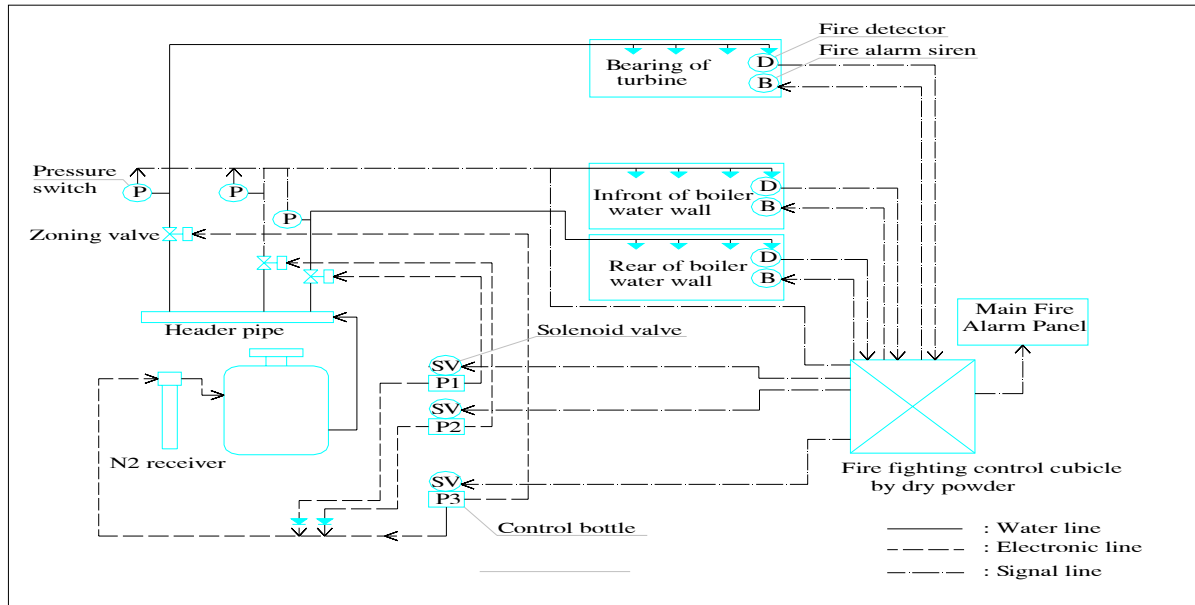
Dry chemical fire fighting system is provided to extinguish the fire caused by the fuel by discharging the dry foam on the diesel oil surface where foam and water fire fighting system is not effective.

Dry foam fire fighting system is designed for following areas:

Support of steam turbine generator

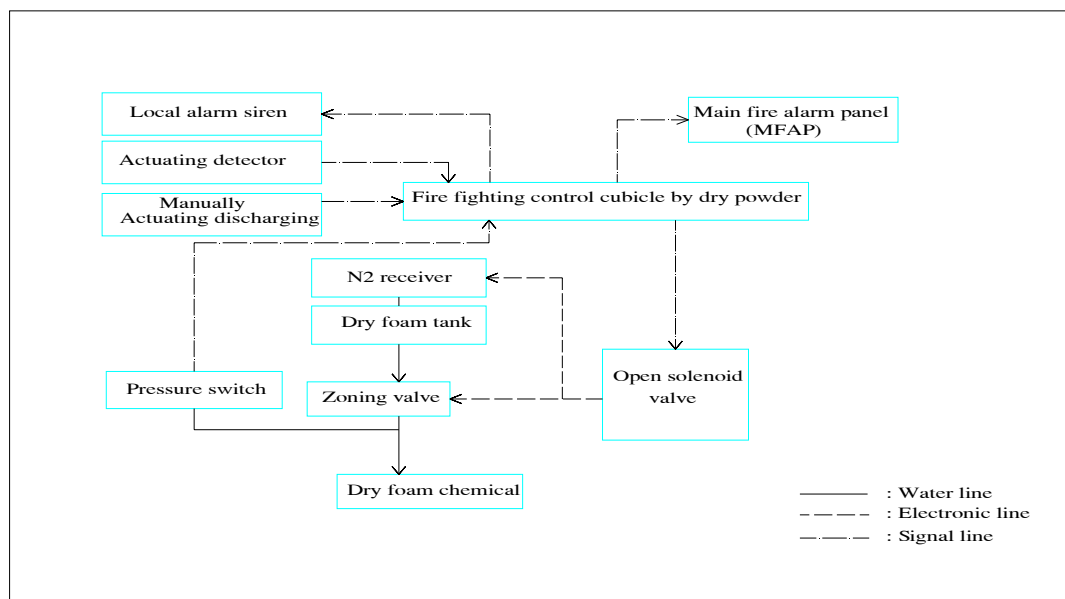
Oil tank for auxiliary boiler support firing at areas behind and in front of boiler furnace.

System structure:



Operation philosophy:

On occurrence of fire (detected by flame detectors or manually activated by gas discharging button), the control cubicle for the dry foam system will receive and send signals to the main fire alarm cubicle, to electronic valve (dry foam – control receivers) to activate the N2 receiver (control valve), dry foam chemical will be discharged at the firing area via select valve.



3.15. FM200 gas fire fighting system

FM200 is one fire fighting system using substance with chemical formula of CF_3CHF_2 of clean air, no colour, no smell. It will discharge at clean fire extinguishing substance named FM200 (Tyco, Kidde) or HFC227ea (Nittan). Such gas can effectively extinguish the fire and also balance required O_2 volume for temporary human respiration. Assets and material in the protected area shall not be destroyed for not creating a mist coverage, not affecting to the health, corresponding to covered areas with special property i.e Central control room.

FM200 system can start to extinguish the fire within 10 seconds or less on detection of the fire, prevent the fire source before causing damage. It is usually stored in pressured receivers under liquid mode then space is saved (similar protection scope, FM200 occupies a space seven times less than that of the CO_2 system).

3.16. Portable and trolley fire fighting equipment

Portable and trolley fire fighting equipment shall include portable extinguisher, portable and trolley foam container that are equipped for protection requirement of each area.

According to TCVN 7435-1: 2004 – Foam and gas fire fighting system - Design standard, to classify fire, the select extinguisher type.

The portable fire fighting equipment shall be provided for all buildings next to door.

All fire fighting equipment shall be complied with existing standard. Areas in plant shall be classified into A, B or C fire danger group (following TCVN 2622 standard) to use suitable fire fighting substance.

The extinguisher shall be sealed, and manufacturer trade stamp shall clearly mark shell weight and gas weight. When installing, extinguisher top shall not be 1.5m higher than floor.

Selecting fire fighting substance and extinguisher weight for each area shall ensure protection capability for construction structure of building and tool/device at this area.

The maximum protection scope of popular extinguisher is about 1m^2 , so when selecting extinguisher quantity, it shall be calculated following cover area for equipment and tool to be protected in the area.

The maximum distance from extinguisher positions to protection area shall be 23m (following NFPA- 10 item 3-2.1), at area with high fire risk equipment, this distance shall be 15m.

The mobile extinguisher with trolley shall be designed and installed for large protection area with high requirement on substance quantity and concentration.

3.17. Fire water source

Water supplied to the plant fire fighting system will be taken from three (03) filtered water tanks of 4000m³ in the water treatment area.

The fire fighting water system of plant shall ensure water requirement with flow of 440 m³/h. Fire fighting reserving water quantity shall satisfy the maximum fire fighting water quantity in 03 hours with the minimum reserving quantity of 1320 m³. Fire fighting water and other technology process water have the same source, so it is required a solution for maintain supplying fire fighting water in 3 hours and not be influenced by technology process demands. (Normal solution in the power plant: The source for technical purposes has priority over outlets to fire pump. In order to maintain water level for capacity of 1320m² for fire water reserve, the minimum water flow for fire water recovery is 55m³/h.

Besides, this water source shall ensure to reserve fire fighting water after firing in 24 hours (following TCVN 2622-2.3). Fire water reserve shall meet the highest fire water volume within 03 hours with minimum water volume of 1320m². The fire water source is shared with source of other technical procedures, then it should have alternatives to ensure fire water supply (3 hours) for not affecting to technical water supply (Normal alternative used in power plants: the source to technical purposes is higher than to the fire pump in order to maintain water level with capacity of 1320m² for fire water reserve). Minimum water quantity for recovering fire water is 55m³/h.

In addition, this water source shall ensure the reserved fire water to be recovered within 24 hours after the fire (According to TCVN 2622-2.3).

3.18. Fire water piping system

Fire water will be supplied from the fire water pumphouse and distributed to the fire water network to following items:

Indoor and outdoor fire water hoses

Indoor fire water hoses accompanied with hose cubicles

Sprinkler system

Spray system

Foam system

Oil tank cooling system

Water distribution system is designed under water loop circuit to following areas:

Steam turbine area

Transformer area

Emergency diesel generator station

Control house

Administrative building

Air dryer area

Garage

Warehouse

Workshop

ESP control house

FGD control house

Coal handling PDC/control building

Ash handling PDC/control building

Coal yard dry storage shed and conveyor system & transfer towers

Oil tank area

Oil pumphouse

Water treatment area and waste treatment area,...

Connecting to the coal port fire fighting piping and oil port/general port

Fire pumphouse will supply water for the system under two loop circuited piping routes to ensure that when one circuit is maintained the rest one still ensure water supply for fire fighting demand. The fire water system will be provided with outdoor fire water hoses with diameter of 100mm and divided into some independent sections isolated by throttle valves. When one valve is locked for maintenance, number of isolated fire fighting hoses shall not be higher than 5. Intervals among outdoor water hydrants as follows: around the power house: not higher than 80m, around the oil; around the oil tank: not higher than 30m, and other areas: not higher than 120m.

Indoor water supply system: According to design stipulations, indoor hydrants will be provided for almost plant buildings with diameter of about 65mm, at interval of ≤ 30 m intervals. Supplied water will be taken from water network for outdoor hydrants. If the main pipe route should be maintained at some certain locations, it is necessary to ensure water supply for indoor fire fighting hydrants.

Indoor and outdoor hydrants will be located at locations convenient for fire fighting. Indoor and outdoor hydrants will be provided to hose cubicles. Fire fighting soft hoses will have quick couples and kept in hose box.

Foam fire dydzant and hose box will be arranged near areas with capability of fire risk caused by oil and grease. All buildings will be provided with fire water drainage system.

According to 20TCN-33-85 outdoor water supply network and works – Design standard, it is recommended to use galvanized steel pipe meeting BS 1387, for pipe with diameter of 70mm upward. Black steel pipe meeting NFPA 5 will be used for pipes of diameter greater than 70mm. Ground pipes will be painted and completed by three layers (anti-risk layer, two red painted layers) and located above reinforced concrete or steel supports and fixed by bolts. Referring to underground pipes, pipe depth is necessary to comply with NFPA-24 Item 8-11 to level of over 1.2m comparing to road surface. Underground pipes on the ground are usually located at minimum depth of over 0.8m higher than the ground. When the pipes are arranged underground, the pipes shall be covered of two layers of asphalt for anti-erosion protection. In case that pipes cross traffic road, it is necessary to arrange pipe across reinforced concrete ditch. Concrete ditch structure shall ensure resistance of greatest weight classified following the road.

Throttle valves under case of pipe repair shall be arranged complying with Item 8.8.a and 8.9 in 20TCN-33-85. External water supply network and structures – Design standard.

3.19. Dry chemical powder system

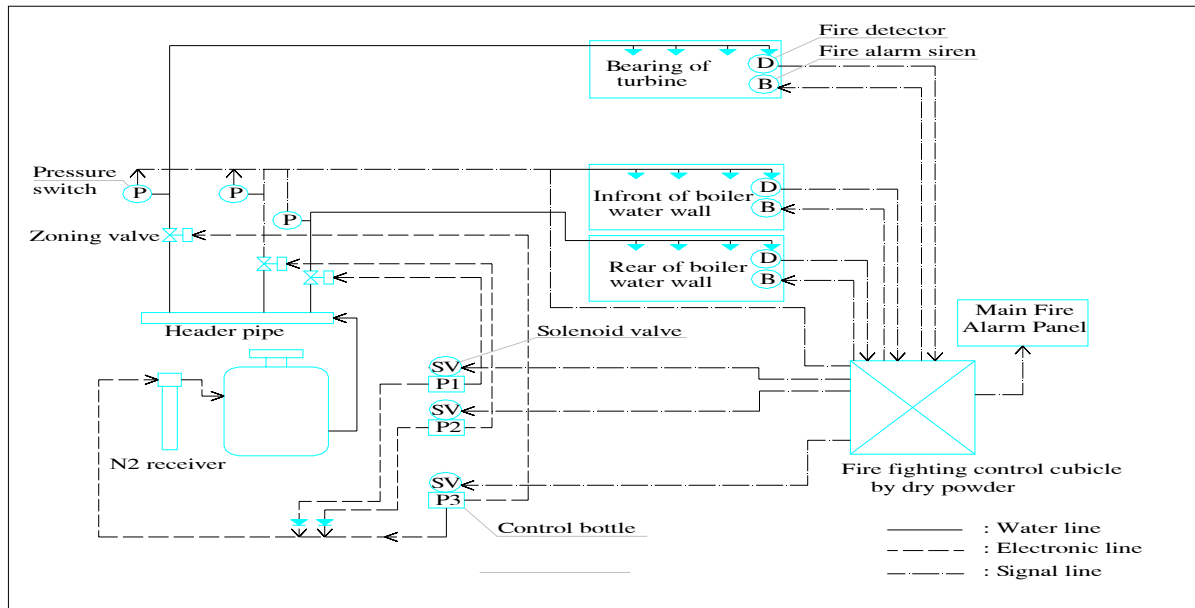
Dry powder fire fighting system will be provided for fire fighting for fires caused the fuel by discharging dry powder on the diesel oil liquid surface, where the power and water fire fighting system is not effective.

Dry power fire fighting system is designed for following areas:

Turbine generator support

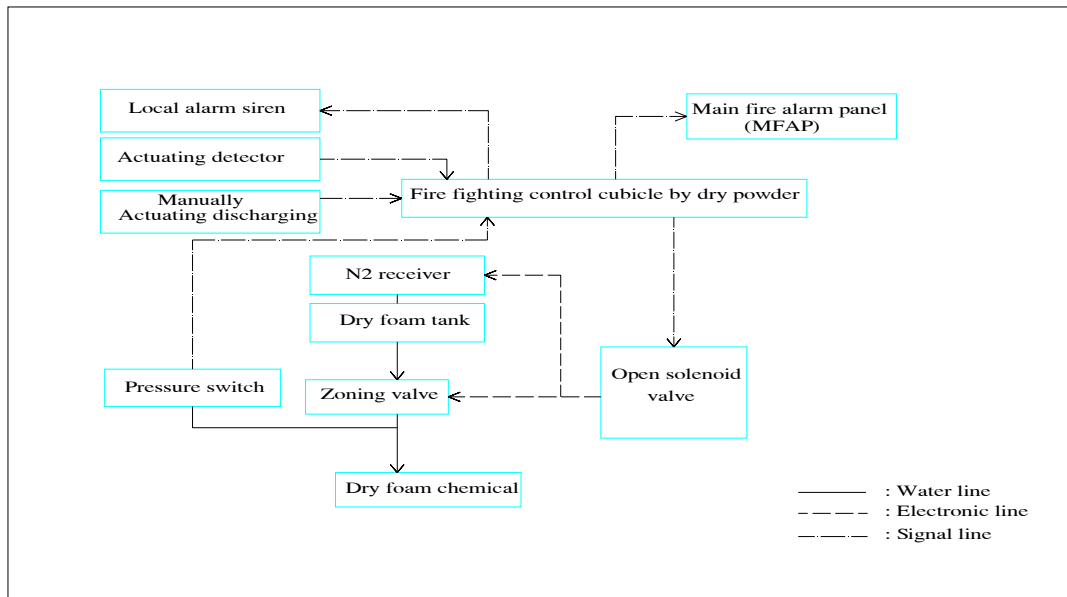
Boiler oil burner areas at the front and rear of the furnace

System diagram:



Operation philosophy:

On firing, control cubicle of the dry powder system will receive and send signals from the main fire alarm central cubicle to the electromagnetic valves (dry powder containers – Control container) for activating N2 container (control one), dry powder chemical will be discharged at firing area via select valve.



3.20. Outdoor hydrant system

Each hydrant will comprise of two hose couples

Fire fighting box will be galvanized steel

Each fire hose box comprise of:

- Fire hose 65Ax20m: 4 ones
- Fire hydrant: 2 ones
- Hydrant locking key: 1 one
- Connection locking key: 1 one

3.21. Indoor hydrants

In order to provide the fire fighting system for items in each buildings following fire protection and fighting requirements, each indoor fire hydrant will comprise of:

- Fire fighting valve: 40A
- Fire water pipe: 40Ax15m x 2 rolls
- Water hose: Spray hose

3.22. Fire water drainage system

On firing, it is necessary to use the wall partition fire fighting system, fire water will be collected by the ditch system and discharged externally following drainage culvert system on the floor. When the outdoor fire fighting system is utilized, fire water will be collected and discharged into the storm water drainage culvert system. On fire fighting for transformers, ignition oil tank, fire fighting water is discharged into the emergency oil pit or oil separation pit.

3.23. Fire brigade station and fire trucks

Fire station and fire trucks will have temporary dimensions of about 15mx30m located adjacent to the workshop (refer to Item No.56 on the general layout drawing), the fire trucks will be located in the station. Inside the station and fire trucks will be provided with panels of fire alarm signal indication generated from all areas armed with the plant fire protection and fighting equipment in order to inform the operators/reception staff about the failure; Such panesl will be connected to all fire protected areas. In addition, inside the fire station and fire trucks will be provided with internal telephone system for communication purpose between the central control room and operation staff.

3.24. Fire alarm system

Fire detection and alarm system will consist of MFAP and LFAP, voice and light fire alarm system, automatic detector and connection cables. - Fire alarm areas include: control house, electronic equipment house, technical room, relay house, cable trench, DC supply house and UPS room, battery room, power distribution house and relay house in the areas of 500kV switchyard, main transformer and auxiliary transformer, boiler burner area, air heating oil system, turbine oil system, power distribution house, diesel generator house, coal storage system and other specified required areas to be connected in loop circuit to the local fire alarm panel.

Then the LFAP will be connected to MFAP and central monitoring room.

Fire bell will be provided at all locations in the power plant and all office areas.

The fire detection system will be designed, provided and installed according to NFPA 72 standard.

3.24.1. Fire alarm areas

The plant will be provided with one monitoring computer for the fire alarm horn system which is located in the central control house. Fire alarm areas comprise of: control house, electronic equipment house, technical room, relay house, cable trench, DC power supply house and UPS room, battery room, power distribution house, relay house in 500kV switchyard area, main transformer, auxiliary transformer, boiler burner area, air drying oil system, turbine oil system, power distribution system, diesel generator station, coal storage system and other areas as specified.

- Areas in CO₂ automatic fire fighting system
- Areas applied for water (powder) automatic fire fighting system

3.24.2. Selection of equipment

Automatic fire alarm computer system comprise of LCD:

Components of the system will comprise of smoke & heat, infrared detectors, horns, temperature sensing cable and other interlock relays. This computer will also have role of display on the screen and also connection for printers and other computer for printable, exporting messages or files in a automatic way. Power for such computer system is AC220V electricity, under case of power failure, the emergency power system will be started for returning to the operation mode in an automatic way.

Interlock relays will be capable of executing control and display operation of the spray system and fire pump.

3.25. Power supply for the fire protection and fighting system

The fire protection and fighting system will comprise of two fire water pumps, (of which one online, one standby, one power and one diesel). Voltage supplied for power fire fighting pump is 6.6kV voltage level. In addition, there will be two jockey pumps (one online, one standby), power source for such pumps is taken from the water treatment area. Fire fighting pump will be automatically controlled, under case of online pump failure, the standby one will be automatically started. The fire pump system can be controlled from both central control room and fire water pumphouse.

3.26. Fire protection for the air conditioning, ventilation system

3.26.1. Fire protection alternative for the air conditioning and ventilation system

Following fire protection alternatives will be considered to be applied for the air ventilation and conditioning in each building:

- (1) Main building, turbine house will use air alternative taken in by natural method and discharged outside by forced method. In order to avoid fire and explosion from H₂ accumulation, ventilators will use anti-explosion motor which is located at highest locations on the building roof. Number of ventilators and specifications are specified in detail design stage.
- (2) Plastic fan with anti-explosion glass will be used for such working locations as water treatment area, warehouse where have gases capable of causing high fire and explosion and will meet requirement of air ventilation capacity.
- (3) Forced ventilators with indoor air variation factor of over 10 times/hour will be provided for oil treatment area and oil pumphouse. Ventilators will use anti-explosion motor. Indoor air shall not permit recirculation.
- (4) Hydrogen generation station will use natural air ventilation system with air ventilation sphere located at highest point on the roof.
- (5) Battery room also apply natural air supply system and forced air discharge system with plastic fan with anti-explosion glass fiber reinforcement. Indoor air will not permit re-circulation.
- (6) Emergency air ventilator with air variation factor in the room shall not be less than 10 times/h which will be provided in the power distribution

equipment room, when power auxiliary distribution equipment catch fire, power supplied for ventilators can be automatically interrupted.

(7) Dust filters will be provided for transfer tower and coal bunker area. Dust filter for transfer tower will be of water spray type or bag type. The coal bunker area will be installed with bag filter. Dust filter will be anti-explosion centrifugal fan.

(8) Air ventilation pipes will be steel galvanized or plastic incombustible glass fiber consolidated material, dust filtering pipe will be steel plate of 3mm thickness.

3.26.2. Fire protection alternative for air conditioning system

The central air conditioning system will be installed for the main building of the power plant, the whole design will meet fire protection and fighting standards.

The central control house, electronic equipment house, power distribution house, cable trench will be provided with independent emergency IDF. Under case of fire, when alarm equipment send signals outside, the air conditioner will be automatically tripped and send Off signals to the central control house of the fire protection and fighting system and tripping air ventilator and conditioner, after confirmation of fire ending, the IDF will be on for flue gas discharging outside.

Damper will be provided for flue gas supply and discharge line of the air ventilation system and conditioning system for one of following cases (working temperature is 72⁰C).

- (1) The main air supply and recovery line cross partition walls and floor of the turbine area
- (2) The main air supply and recovery line cross main equipment or partition walls and floor of high ignitability
- (3) At the cross section of the pipe connecting the air supply and recovery pipe.

The air and water supply pipes of the conditioning system will be made of non-combustible, galvanized or insulated glass plastic. Insulation material will use compound rubber material of B1 insulation level.

3.27. The fire protection and fighting system for coal port

The coal port will be designed with the fire protection and fighting system using outdoor hydrants. The water source is connected from the fire fighting loop circuit of Vung Ang 2 TPP. The pipe and fire fighting hose cubicle is in the scope of supply of the port project.

4. GROUNDING SYSTEM AND LIGHTNING PROTECTION SYSTEM

4.1. Grounding system

The grounding and lightning protection system is required for being installed for the whole plant area. This system is installed to protect the whole power plant, all systems, structure and buildings.

The grounding and lightning system will comprise of earth bars and bare copper cables. Grounding works will be done for all power equipment installed in the boiler/turbine area and 500kV switchyard system for grounding connection between all equipment and station grounding system such as: plant structures, outdoor metal pillars or other uncharged metals.

The grounding system will be connected to reduce step voltage and touch voltage between human and charged equipment. Calculated value is designed within safe limit under normal and abnormal cases.

The grounding and lightning protection system for the plant items will be designed following Vietnamese standard (TCVN), IEEE80, NFPA 780) or other stricter standard. All calculated parts of the earthing system will be applied following IEEE80 standard, calculation results will be referred to Vietnamese standard.

The earthing system will be designed so that following conditions are achieved:

- Touch voltage will be limited for not causing danger for human and equipment under all normal and abnormal cases.
- Ensuring relay protection, communication, control.. are safe and reliable
- Safety for all operation equipment
- Quickly dispersing the stroke current and short-circuited current

The main grounding system will comprise of one system of grounding wires and bars. Design of the grounding system is implemented after survey of grounding resistivity. Minimum requirement for each building is earthed at least having one copper bar with one grounding wire for the building.

The grounding system will comprise of the grounding system and lightning bar according to IEC-1024 standard referring to grounding and lightning protection for equipment and building.

Each building will be provided with at least one grounding bar connected to grounding wire of the building.

The lightning and grounding system is designed complying with technical requirements of IEC-1024 referring to lightning protection and earthing for equipment and building.

4.2. Lightning protection system

The lightning protection system is designed as follows:

Buildings, metal structures and equipment areas will be designed under mode of lightning needle.

The oil tank area will be designed under mode of double protection system for both conventional and modern lightning.

Grounding for the lightning system shall be connected to the grounding network and implemented via junction box as follows:

Wire connecting lightning needle and down conductor will be suitable with Vietnamese standards or equivalent and/or higher international standards.

One supplementary grounding pile

Grounding bar is galvanized copper

Connecting to the system

Down conductor will be drilled for bolting to the grounding bar

Wire connecting to the grounding grid is at least wire copper wire of 120mm² section which is compressed at terminal and bolted to the earthing bar.

Wire connecting to the supplementary earthing bar is at least wire copper bar of 120mm² section which is compressed at terminal and bolted to the earthing bar.

5. SAFETY AND FIRE PROTECTION & FIGHTING DURING CONSTRUCTION

Labour safety during construction shall strictly comply with current labor safety standards and regularly examined by professional safety groups.

Referring to earth work, the slope is paid duly attention to ensure safety during construction, and avoiding sliding to the workers.

Prior to construction, the workers shall be trained via a labor safety course for avoiding accidents during construction and sufficiently armed with labor protection tools such as helmets, boots and cloths etc..

Referring to the formwork, reinforcement assembly and concrete pouring shall be regularly examined with stability of the scaffolding system. The scaffolds shall have stable handrail for not falling. During overhead working, it is necessary to have lifeline worn.

Derusting workers shall wear glasses. When reinforcement is installed in the formwork, the worker shall stand on the adjacent work floor and not stand on the wall lining.

During transport of formwork, the overhead reinforcement shall be carefully tightened.

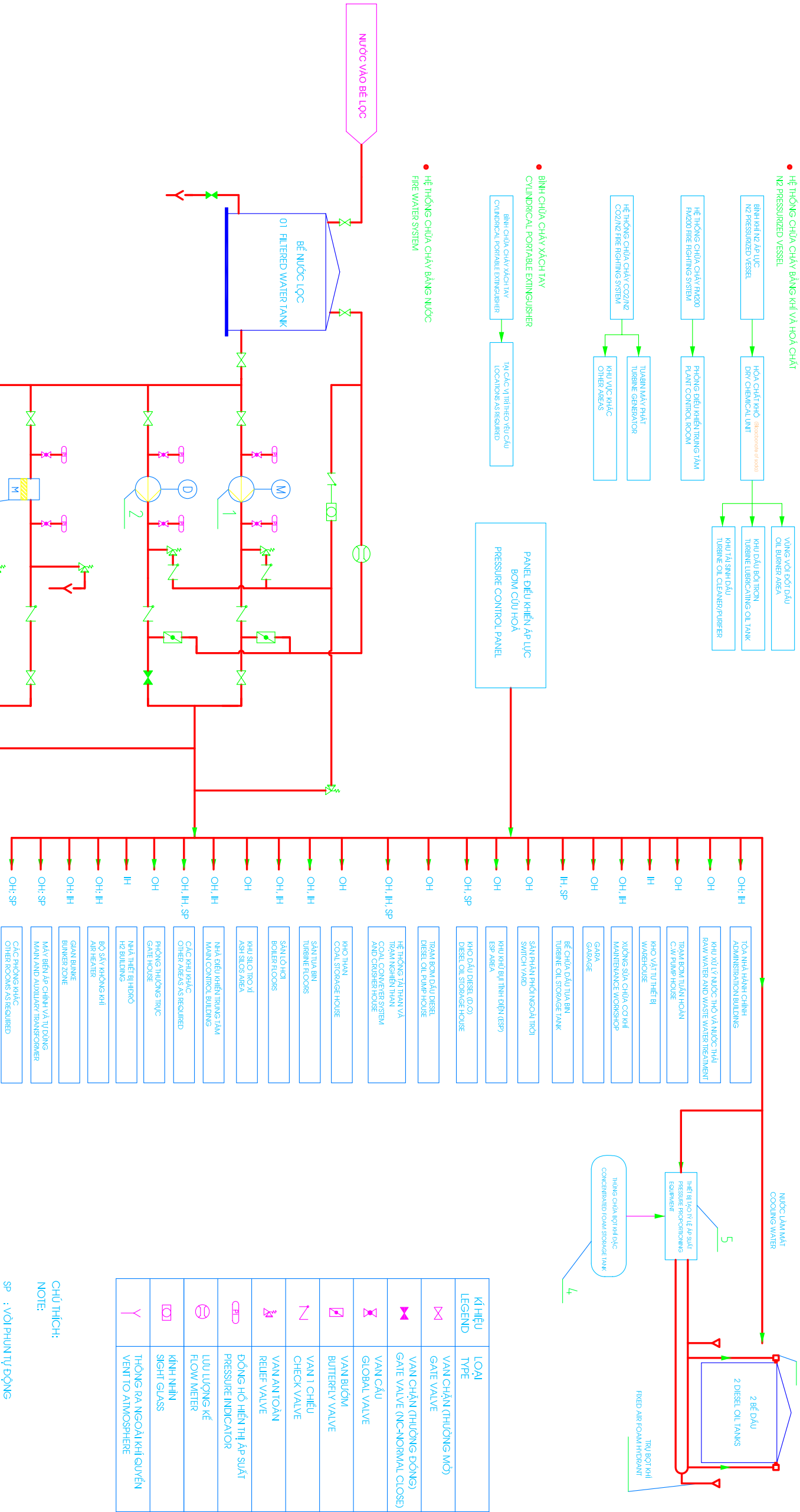
Construction work shall have suitable layout, the scaffold shall be safe and workers shall be carefully insured. During construction of basement wall, it is necessary to stop traveling under to avoid material falling.

Construction electricity shall be suitable arranged and comply with regulations of power safety. It is necessary to comply with traffic safety regulation and fire protection and fighting in the plant during construction process according to Vietnamese standard.

During construction of oil pipe and related works, it is necessary to pay attention fire protection and fighting system, there shall be safety equipment for fire and explosion. Contractors shall have slogan of labor safety.

There shall be labor safety examination group.

Portable fire extinguishers shall be provided on site at suitable locations for serving local fire fighting.



KÍ HIỆU LEGEND	LOẠI TYPE
	VAN CHẶN (THƯỜNG MỞ) GATE VALVE
	VAN CHẶN (THƯỜNG ĐÓNG) GATE VALVE (NC-NORMAL CLOSE)
	VAN CẦU GLOBAL VALVE
	VAN BƯỚM BUTTERFLY VALVE
	VAN T CHIEU CHECK VALVE
	VAN AN TOÀN RELIEF VALVE
	ĐỒNG HỒ HIỂN THỊ ÁP SUẤT PRESSURE INDICATOR
	LƯU LƯỢNG KẾ FLOW METER
	KÍNH NHÌN SIGHT GLASS
	THÔNG RA NGOÀI KHÍ QUYỂN VENT TO ATMOSPHERE

CHÚ THÍCH:
NOTE:

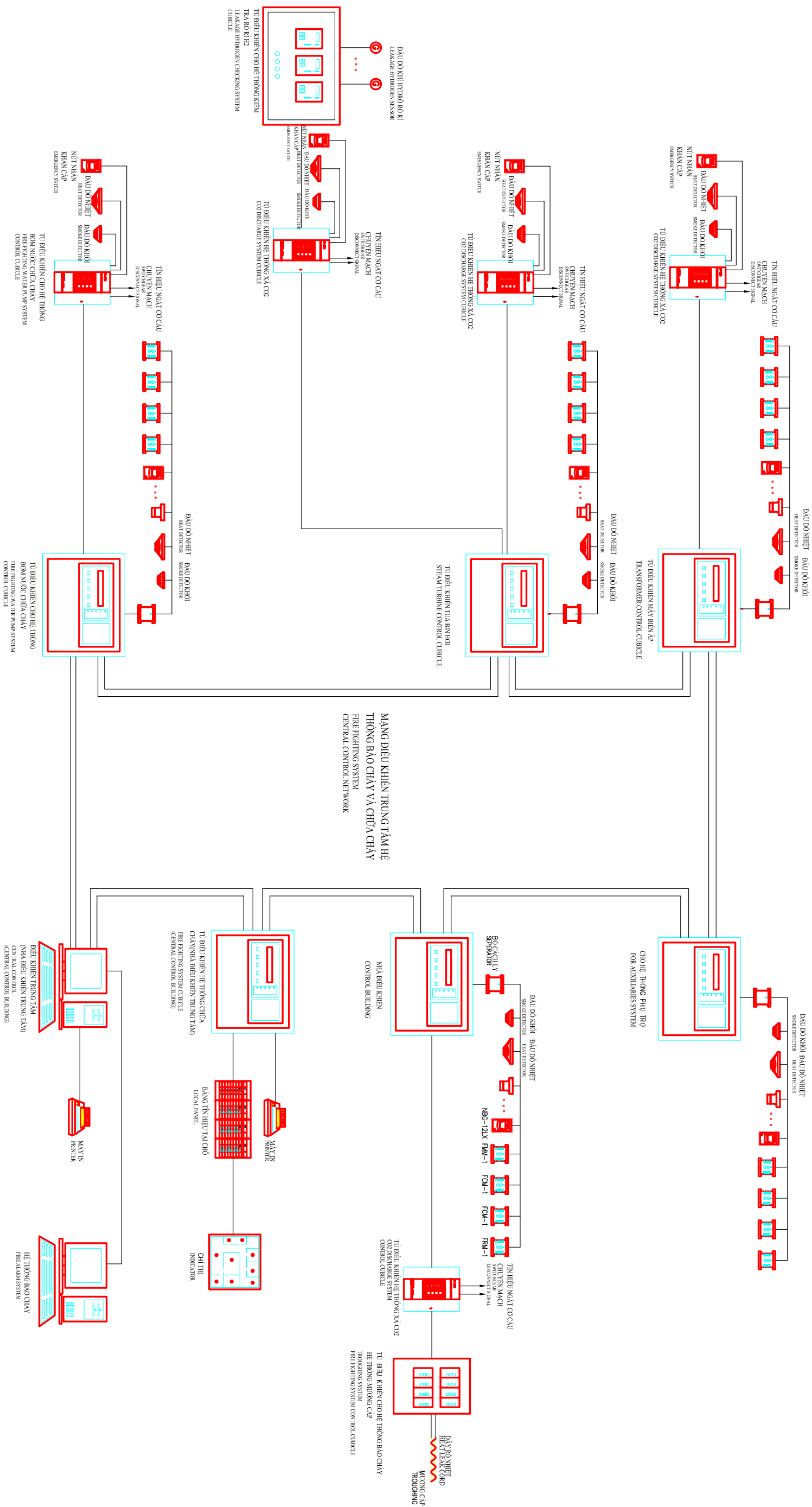
SP : VỎI PHUN TỰ ĐỘNG
SPRINKLER/SPRAY
OH : TRỤ NƯỚC CỨU HOÀ NGOÀI TRỜI
OUTDOOR HYDRANT
IH : TRỤ NƯỚC CỨU HOÀ TRONG NHÀ
INDOOR HYDRANT

5	THIẾT BỊ TẠO TỶ LỆ ÁP SUẤT PRESSURE PROPORTIONING EQUIPMENT		CÁI SET	1	
4	THÙNG CHỨA BƠI KHÍ ĐẶC CONCENTRATED FOAM STORAGE TANK	V = 740 LT	CÁI SET	1	
3	BƠM TẠO ÁP LỰC JOCKEY PUMP	H = 20 m H ₂ O	CÁI SET	2	
2	BƠM CỨU HOÀ DẪN ĐỘNG DIESEL DIESEL MOTOR DRIVEN FIRE PUMP	Q = 440 m ³ /h; H = 112,5 m H ₂ O P = 202 kW	CÁI SET	1	DỰ PHÒNG FOR SPARE
1	BƠM CỨU HOÀ DẪN ĐỘNG ĐIỆN ELECTRIC MOTOR DRIVEN FIRE PUMP	Q = 440 m ³ /h; H = 112,5 m H ₂ O P = 202 kW	CÁI SET	1	
STT NO.	TÊN GỌI DESCRIPTION	ĐẶC TÍNH KỸ THUẬT SPECIFICATION	ĐƠN VỊ UNIT	SỐ LƯỢNG QUANTITY	GHI CHÚ NOTE

BẢNG KÊ THIẾT BỊ CHÍNH

LIST OF MAIN EQUIPMENTS

CÔNG TY CỔ PHẦN TƯ VẤN XÂY DỰNG ĐIỆN 1 (PECC1)			PHẦN CƠ NHIỆT / MECHANICAL		
NHÀ MÁY NHIỆT ĐIỆN VÙNG ÁNG 2 2x660 MW			HỆ THỐNG PCCC SƠ ĐỒ NGUYÊN LÝ		
2x660 MW VÙNG ÁNG 2 THERMAL POWER PLANT			FIRE FIGHTING SYSTEM FLOW DIAGRAM		
Trưởng phòng Chịu nhiệm			TKCS Tỷ lệ		
Bành Đức Vỹ Phạm Ngọc Thôi			9/2/2009 60.09-V A2-ME-25		
Kiểm tra Thiết kế					
Hoàng Mạnh Hà Vũ Ngọc Sơn					



CÔNG TY CỔ PHẦN TƯ VẤN XÂY DỰNG ĐIỆN 1 (PECC1)			PHẦN CƠ NHIỆT / MECHANICAL		
NHÀ MÁY NHIỆT ĐIỆN VÙNG ÁNG 2 2x660 MW			SƠ ĐỒ NGUYÊN LÝ HỆ THỐNG BẢO CHẤY		
2x660 MW VÙNG ÁNG 2 THERMAL POWER PLANT			FIRE DETECTIONS ALARM SYSTEM FLOW DIAGRAM		
Trường phòng	Bành Đức Vỹ				
Chủ nhiệm	Phạm Ngọc Thôi				
Kiểm tra	Hoàng Mạnh Hà		TKCS	9/2009	60.09-VA2-ME-26
Thiết kế	Vũ Ngọc Sơn		Tỷ lệ		

PMGC Guidelines for Contractors

Execution and Operation Phases

Introduction

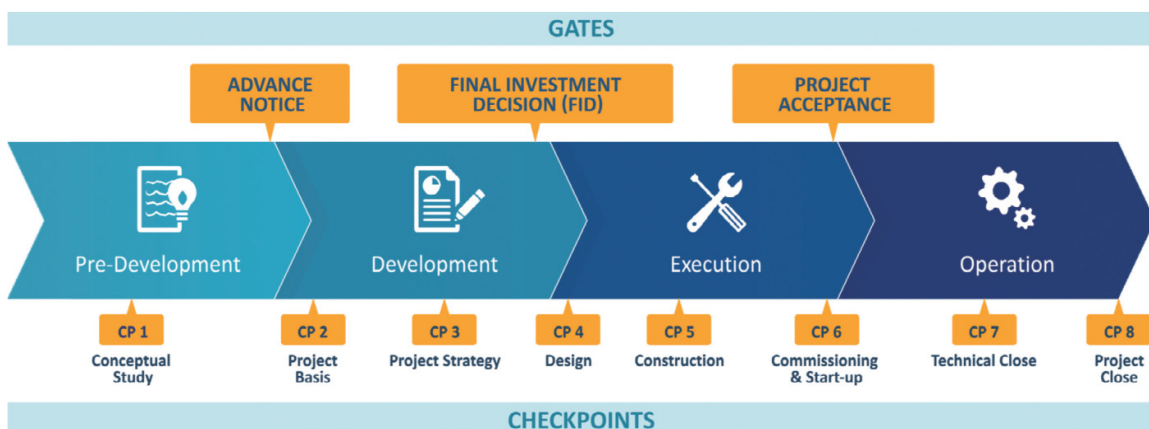
The Project Management Governance Checkpoints (PMGC) govern the implementation of VA2 project, it is intended to facilitate a consistently safe, timely and cost effective delivery of high quality project. This helps in ensuring clear accountability and responsibility for project management and governance at all levels of the organisation and also aids in helping the Project Team ensure compliance with all applicable codes, standards, laws, and regulations.

The PMGC is based on the best international project management practices, and defines the minimum requirements for the development, execution, and delivery of the project. The PMGC covers all the phases of the project lifecycle, including Pre-Development, Development, Execution, and Operation, and sets the minimum requirements for project management. It is based on the four-pillar project basics of scope, cost, schedule, and quality supported by a strong HSSE foundation.

This document is prepared to help our Contractors understand the deliverables that are expected by VAPCO at each stage of project implementation to allow better collaboration and delivery of project.

The PMGC Project Lifecycle is divided into four phases:

- Pre-Development
- Development
- Execution
- Operation



The deliverables required at Checkpoints related to **Execution and Operation phases** of the Project life cycle are mentioned in the following sections.

1. Checkpoint 4 – Design

1.1 Purpose

The purpose of this checkpoint is to ensure that the Project Team has properly staffed-up and organised itself, finalised the execution plan with due consideration to possible risks and ensured that all key design has been completed.

1.2 Deliverables

Given below are the expected deliverables with a detailed description of each:

PMGS Deliverables	PMGC Deliverable Description	Contractor's Deliverables
Project Management Team Organisation Chart	The Project Director should form the Project Team. The team members should have the skills and qualifications necessary to accomplish the project goals. Team members should be aware of each other's roles and responsibilities. For each team member to achieve project goals, all should agree on project goals and coordinate their efforts to meet project goals. A project organization chart should be published to document the organisation.	The Contractor should form a dedicated Project Team. The Contractor's Project Team should include sufficient members with suitable skills and qualifications necessary to accomplish the project goals. A Project Team Organisation Chart should be published to document the organisation and submitted to VAPCO.

PMGS Deliverables	PMGC Deliverable Description	Contractor's Deliverables
Final Project Execution Plan	The Project Team should finalise a Project Execution Plan (PEP) which sets out the strategy for managing the project from start to completion. The PEP is a tool to aid the Project Team to consider how to best execute the project. The plan also serves as the main communication vehicle to ensure that everyone is aware and knowledgeable of project objectives and how they will be accomplished. The PEP should provide firm information on project scope, execution strategy, cost, schedule and risks in implementing the project. The PEP is a living document throughout the whole project cycle and should be updated from time to time, if required, to describe current and future processes and procedures based on the discretion of the Project Team.	The Contractor is expected to provide necessary information to VAPCO for completion of the PEP and for periodic updates from time to time, if required.
Project Procedures	The Project Team should issue Project Procedures which provide direction on the required project processes, a description of key roles and responsibilities and contains applicable documents, forms, policies and templates. The list of required procedures in the PMGC Standard is a minimum requirement and each project should develop the appropriate number and type of procedures to fit their specific needs. • HSSE • Quality (Equipment & Construction) • Management of Change • Regulatory Compliance • Project Controls • Public Affairs / Community Social Responsibility	The Contractor should issue Project Procedures specific to project requirements in accordance with requirements in the Owner's Technical Requirements.

PMGS Deliverables	PMGC Deliverable Description	Contractor's Deliverables
Project Risk Register	The Project Team should formally re-review the risks at this stage of the job and consider in detail the critical risks to ensure that they have been effectively mitigated. This is essentially a refreshed look at the risk review conducted earlier and modified against the backdrop of the project actually in execution.	The Contractor is expected to provide necessary information to VAPCO for updating the Project Risk Register and for periodic updates from time to time, if required.
Key Construction Drawings	Although the construction work is not yet started in this stage, it is important that some key construction drawings should be reviewed, issued and ready for use. Generally, design institutes, consultants or contractors actually issue the drawings but the Project Team is responsible to make sure it happens. Key construction drawing could include General Arrangement (GA) drawings, process schematics, mechanical flow diagrams, structural and civil drawing (civil project) etc. The Project Team should also ensure that a system is in place to distribute and control the drawings.	The Contractor is responsible for ensuring that drawings are submitted to VAPCO for review, and should also ensure that a system is in place to distribute and control the drawings.

2. Checkpoint 5 – Construction

2.1 Purpose

The purpose of this checkpoint is to ensure that the Project Team is well prepared for the construction commencement. The main feature of this checkpoint is readiness: meaning that the methodologies are reviewed and in place, the people have been well trained, and all regulatory requirements are met before the construction starts.

2.2 Deliverables

Given below are the expected deliverables with a detailed description of each.

Deliverables	Deliverable Description (as applicable)	Contractor's Deliverables
Contractors' Method Statements	The Project Team is to ensure that the Contractors' Method Statements have been submitted and reviewed and is understood by those performing the work. Construction shall be done according to the design and method statements.	The Contractor is to ensure that the Contractors' Method Statements have been submitted to VAPCO for review and is understood by those performing the work. Construction shall be done according to the design and method statements.
Conduct HSSE Induction Training	The Project Team shall ensure that Health, Safety, Security and Environment (HSSE) induction training has been completed for all workers. Most often this will require a customized training program for the project. The expectation is that everyone working on the project should be familiar with VAPCO's HSSE requirements and committed to the goal of no accidents, no harm to people and no damage to the environment. . The Project Team will also probably need to spend more time monitoring and policing the contractor in the beginning or when there are significant changes in project personnel such as new work crews arriving. Project staffing should be considered to provide adequate safety supervision especially at the beginning of a project.	The Contractor shall ensure that Health, Safety, Security and Environment (HSSE) induction training has been completed by all its workers. The expectation is that everyone working on the project should be familiar with VAPCO's HSSE requirements and committed to the goal of no accidents, no harm to people and no damage to the environment. The Contractor should ensure adequate project staffing for safety supervision especially at the beginning of a project.

Deliverables	Deliverable Description (as applicable)	Contractor's Deliverables
Check all regulatory requirements and permits	The Project Team is to ensure that all regulatory requirements and permits have been obtained before commencing work in accordance with local law and regulations. Evidence of this may be provided with a register or other equivalent methods.	The Contractor is responsible for ensuring that regulatory requirements and permits that Contractor is responsible for have been obtained before commencing work in accordance with local law and regulations, and should provide evidence of this to VAPCO for record.

3. Checkpoint 6 – Commissioning & Start-up

3.1 Purpose

The purpose of this checkpoint is to ensure the project is ready for commissioning and start-up, and it is important to ensure a comprehensive commissioning & start up procedure is available and closely followed. VAPCO will be heavily involved in the hot commissioning activities to energise the system, and then to run it up to its design capacity.

3.2 Deliverables

Given below are the expected deliverables with a detailed description of each.

Deliverables	Deliverable Description (As applicable)	Contractor's Deliverables
Commissioning & Start-up Team organisation chart O&M Team organization chart	By this stage, both the Commissioning & Start-Up and Operations & Maintenance (O&M) Teams should be established.	The Contractor should provide its Commissioning & Start-up Team Organisation Chart to VAPCO for record. The O&M Team organisation is not applicable (Part of O&M Contract)
Operating and Maintenance Procedures	Operating and Maintenance Procedures may be provided by Original Equipment Manufacturer (OEM), prepared by a specialist consultant, done in-house or any combination of the above. In any case, the Project Team is responsible to see that these are completed and reviewed by the O&M team for comments.	The Contractor is responsible for providing Operations and Maintenance Procedures to VAPCO for all systems and equipment that has been installed as part of the Project.
O&M training records	O&M training should be completed and documented. The training often will be provided by the OEM or contractor, but in some cases may be done by the O&M Team itself. The Project team should liaison with the training provider as well as the O&M team to ensure the training is completed.	The Contractor should discuss with VAPCO on specific arrangements for O&M training and ensure that arrangements are made as necessary for timely completion of such training as provided by the Contract.

Deliverables	Deliverable Description (As applicable)	Contractor's Deliverables
Handover Plan	The Project Team should prepare and develop a Handover Plan. The plan should be reviewed by both the Commissioning & Start-up Team and O&M Team. The key components of the Handover Plan should include a list of project documentation, O&M manuals and drawings, and the handover schedule. The plan should also include responsible parties for each item.	The Contractor is expected to provide necessary information to VAPCO for completing the Handover Plan as provided by the Contract.
Regulatory Requirements and Permits Register	The Project Team is to ensure that all regulatory requirements and permits have been obtained before commissioning and start-up in accordance with local law and regulations. Evidence of this may be provided with a register or other equivalent document.	The Contractor is responsible for ensuring that regulatory requirements and permits that the Contractor is responsible for have been obtained before commissioning and start-up in accordance with local law and regulations, and should provide evidence of this to VAPCO for record.
Updated Risk Register	The Project Team should re-review the risk register especially with respect to those items related to commissioning and start-up to ensure that those required mitigation measures have been completed. This is also an excellent time to make the Commissioning & Start-up Team, O&M Team familiar with the Risk Register and prepare for the transition to operations.	The Contractor is expected to provide necessary information to VAPCO for updating the Project Risk Register and for periodic updates from time to time, if required.
Workshop Established List of Special tools, Equipment and Commissioning Spare parts on-site	The Project Team shall take the lead to check that all the required support facilities and parts are on-site and available. In some cases, the O&M Team may be responsible for the workshop and even the purchase of the spare parts, but it is still important that the Project Team follow-up on these items to ensure a smooth commissioning & start-up. The Project Team should work closely with the Commissioning & Start-up Team, O&M Team to coordinate this activity.	Any workshop facilities, special tools, equipment and commissioning spare parts required as part of the Contract should be provided by this stage. The Contractor is expected to demonstrate to VAPCO that any such requirements have been met.

Deliverables	Deliverable Description (As applicable)	Contractor's Deliverables
Acceptance Criteria Identified	Acceptance Criteria are the requirements that have to be met for a plant or equipment to be assessed as complete. The Project Team shall ensure that the Acceptance Criteria for handover has been identified and agreed with the O&M Team prior to the Commissioning & Start-up. The Acceptance Criteria should already been documented in the PEP and in the key project contracts and this step is just a reminder for all parties to focus on the Acceptance Criteria in preparation for the Project Acceptance Gate.	The Contractor should familiarise itself with the Acceptance Criteria if it has not done so already.
Commissioning & Start-up procedure (incl. isolation requirements) reviewed and issued	The Commissioning & Start-up Procedures are written routines / instructions that describe the logical sequence of activities required to perform Commissioning & Start-up and the specific actions to perform each activity. If there are no written procedures, there is no basis for monitoring performance of Commissioning & Start-up. The establishment of Commissioning & Start-up Procedures allow more time and mental energy to deal with the unexpected, which always happen during commissioning. The Commissioning and Start-up Teams is responsible to write the procedures while the Project Team shall ensure that the Commissioning & Start-up procedures have been reviewed and distributed to involved parties.	The Contractor is expected to submit written Commissioning and Start-up procedures to VAPCO for review, and to distribute procedures as necessary.

Deliverables	Deliverable Description (As applicable)	Contractor's Deliverables
Pre Start-up Readiness Review Checklist	With brand new greenfield projects, and in view of the inexperience with a new process and time pressure to complete, start-up is considered as one of the most hazardous stages in a project. The Project Team should coordinate and lead a Pre Start-up Readiness Review to be conducted by an independent party. This review typically consists of a desktop review of documentation including key drawings and procedures as well as walk-downs of the installation to ensure readiness. All relevant parties should be involved including the Project Team, Contractor, Commissioning & Start-up Team and O&M Team etc.	The Contractor is expected to attend the Pre Start-up Readiness Review, and to provide necessary information to VAPCO for completion of the Pre Start-up Readiness Review Checklist.

4. Gate – Project Acceptance

4.1 Purpose

This gate often involves contractual requirements and is usually about the same time as the Commercial Operation Date. The majority of the deliverables from this stage are documents which formalise the completion of the project.

4.2 Deliverables

Given below are the expected deliverables with a detailed description of each:

Deliverables	Deliverable Description (as applicable)	Contractor's Deliverables
Punch List	Before the issue of the Substantial Completion Certificate to the Contractor, the Project Team must ensure that a defect list (or commonly known as a Punch List) is agreed with the contractor, together with the proposed completion dates of corrective actions. The Punch List should be reviewed with the O&M Team as well as other parties as delegated by the VAPCO. It is important that the Project Team diligently follow-up and produce a thorough Punch List to ensure proper follow-up of the project.	The Contractor should submit a defect list (or commonly known as a Punch List), to be reviewed by and agreed with VAPCO, with proposed completion dates of corrective items.
Substantial Completion Certificate	Once satisfied that the project is complete, the Project Team should issue a Substantial Completion Certificate (or similar type document) to the contractor. At this point, the Defects Liability Period begins and the project insurance coverage stops. It is a major milestone as it signifies the completion of the majority of the project and the contractor is often entitled to substantial payments.	N/A (For reference only)

Deliverables	Deliverable Description (as applicable)	Contractor's Deliverables
Taking-over Certificate Issued	At the time of substantial completion, the Project Team should handover the operation responsibility of the project from the Project Team to O&M Team. The Project Team shall prepare a Taking-over Certificate for the O&M Team to sign to confirm acceptance of the project. This is the formal documentation of the Gate and is an important document.	The Contractor should note that at the time of substantial completion, the VAPCO Project Team may hand over the operation responsibility of the project to O&M Team.
Permit to Work System in effect	Having taken over the facility, the O&M Team shall put a Permit to Work (PTW) System in place to ensure safe operation and maintenance of the plant. This basically means that the project is now an operating facility and any changes or work done on the facilities must be coordinated and approved by the O&M Team.	The Contractor should note that it will need to work within the requirements of the PTW System when completing any outstanding activities in the Punch List.
IT Systems, Business Systems	It is typically required to set-up new IT and Business Systems to support the operation of the project.	N/A
Management Procedures in Place	The Management Procedures should be written and published. The O&M Team shall be responsible to develop Management Procedures and to ensure all staff understands the procedures.	N/A

5. Checkpoint 7 – Technical Close

5.1 Purpose

The purpose of this checkpoint is to establish a systematic approach for project technical close to ensure all project technical close-out activities are performed in a coordinated and effective manner, which in turn will satisfy project requirements.

5.2 Deliverables

Given below are the expected deliverables with a detailed description of each.

Deliverable	Deliverable Description (as applicable)	Contractor's Deliverables
Punch List cleared	When the project is completed and handed over to the O&M Team, a punch list shall be maintained to keep track of the remedial actions for outstanding issues and defects. Together with the Contractor, the Project Team should resolve the issues in the punch list as soon as possible and agree on deadlines for completion. The Project Team should obtain sign-off acceptance of completed items from the O&M Team.	The Contractor is expected to work with VAPCO to ensure that the Punch List items are cleared as per the agreed completion dates.
Risk Register closed	The Project Team should conduct a review of the risk register and ensure that all items required for completion by the Contractor or Project Team have been completed. It is recommended that the updated risk register be formally transferred from the Project Team to the O&M Team and note made of any remaining open items that the O&M Team will assume the responsibility for completing.	The Contractor is expected to work with VAPCO to ensure that outstanding risks associated with the Contract, if any, are closed.

Deliverable	Deliverable Description (as applicable)	Contractor's Deliverables
Receipt of essential spare parts, special tools and equipment	The Project Team shall ensure that the O&M Team has received: • All the required essential spare parts, special tools and equipment as stipulated in the contract • Names, addresses and contacts (email addresses and phone numbers) of repair and services companies for each major system / equipment installed • All equipment warranty statements and guarantees	VAPCO will work with the Contractor to ensure that all necessary items in the Contract have been provided, including all items listed on the left.
Document Register	The Project Team shall ensure that all final documents are received, including: • Final documents such as catalogue data sheet for equipment installed including performance curves, rating data sheets, etc. wherever applicable • Testing and commissioning records / reports • As-built drawings • Operating & maintenance manuals, etc. As a result of modifications to complete the punch list items, the final documentation (including as-built drawings, O&M manuals, etc.) must be updated and modified to reflect such modifications. If necessary, the Project Team shall formally re-issue the as-built documentation to the O&M Team.	The Contractor is expected to provide VAPCO with all necessary information / documentation for completion of the final documents. The Contractor may need to update the final documentation (including as-built drawings, O&M manuals, etc.) to reflect any modifications made when completing the Punch List items.
Performance Monitoring and Reporting in place	Finally, the O&M Team shall establish the plant Performance Monitoring and Reporting, with reporting templates to be agreed with respective business units of Sponsors. These business units can provide input and support on this process if required.	N/A

6. Checkpoint 8 – Project Close

6.1 Purpose

The purpose of this checkpoint is to verify acceptance of the final project deliverables, providing final closure for the project, and to document and share lessons learned, success factors and best practices captured during the Post Implementation Review (PIR) for use on future projects.

6.2 Tasks / Deliverables

Given below are the tasks and associated deliverables with a detailed description of each.

Deliverable	Deliverable Description (as applicable)	Contractor's Deliverables
Post Implementation Review (PIR) Report	The Post Implementation Review shall include reviews of the project development and execution along with operating performance data. It serves as an important document in the project lifecycle. Any scope, cost, or schedule changes should be captured and the cause for these changes identified. Likewise, any HSSE or quality issues should be high-lighted along with the lessons learned. It is important to highlight both positive and negative items with the key lessons learned being recorded and available for future projects. The PIR may start with a survey designed to solicit feedback on the project from the Project Team and stakeholders. Once feedback has been collected and evaluated, a project assessment meeting should be conducted to capture success factors and best practices and formulate lessons learned for future projects. Having verified acceptance of final project deliverables, the lessons learned from a project should be documented, which can be used as reference for future projects. All applicable stakeholders should be involved in the process to ensure full capture of lessons learned.	As a stakeholder, the Contractor is expected to provide input to VAPCO for the PIR Report with regards to HSSE, quality, lessons learned etc. through the execution of the Project.

Deliverable	Deliverable Description (as applicable)	Contractor's Deliverables
Completed Defect Liability Period	The project should not be closed before the Defect Liability Period (DLP) has expired. At the end of the DLP, the Project Team shall check and confirm with other parties (Finance, Commercial, Legal and Insurance etc.) that no issues (technical or commercial) are outstanding with the Contractor. All payments and claims have to be settled between the contracting parties involved. Any retention should also be released in full to the Contractor.	The Contractor is expected to work with VAPCO in clearing any outstanding issues.

Deliverable	Deliverable Description (as applicable)	Contractor's Deliverables
Project Close Out Form	The project should not be closed before the Defect Liability Period (DLP) has expired. At the end of the DLP, the Project Team shall check and confirm with other parties (Finance, Commercial, Legal and Insurance etc.) that no issues (technical or commercial) are outstanding with the Contractor. There is no specific deliverable associated with this other than for the Project Team to keep track of the date and then to ensure that there are no outstanding items. For the Final Account Settlement and Contract Closing, all payments and claims have to be settled between the contracting parties involved. Any retention money should also be released in full to the Contractor. Contract closure involves verification that all work has been completed correctly and satisfactorily, updating of contract records to reflect final results, and archiving information for future use. Among other activities contract closure includes: Confirming the project has addressed the terms and conditions of the contract Confirming completion of exit criteria for contract closure Formally closing out all contracts associated with the completed project Finally, normal commercial documents, as required, such as the Final Account Settlement / Contract Closing should be issued by the party responsible for contract close-out.	The Contractor is expected to work with VAPCO in clearing any outstanding issues.
Performance Certificate	At the end of the Defect Liability Period, the Project Team shall issue a Performance Certificate (or similar) to the Contractor to formally receive the project in full and release the retention money accordingly.	N/A (For reference only)